Modular Homotopy Theory for Algebraic Factorization Algebras on Algebraic Curves

Volume 3

Calabi–Yau Quantum Groups

Chiral Algebras from Calabi–Yau Categories via E_1/E_2 Factorization

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Draft — April 10, 2026

Abstract

The bar complex $\overline{B}(\mathcal{A})$ *of a chiral algebra on an algebraic curve (Volume I) and its Swiss-cheese interpretation on* $\mathbb{C} \times \mathbb{R}$ *(Volume II) take a chiral algebra as given. This volume constructs the geometric input: a functor* $\mathbf{\Phi}\colon \mathrm{CY}_d\text{-}\mathbf{Cat} \to E_2\text{-}\mathrm{ChirAlg}$ *from Calabi–Yau categories to* E_2*-chiral algebras.*

A Calabi–Yau category C *of dimension* d *carries a cyclic* A_∞*-structure whose* $\mathbb{S}^d$*-framing determines an operadic enhancement. The construction has three steps: the cyclic* A_∞*-structure produces a Lie conformal algebra* L_C*; the factorization envelope yields a chiral algebra* A_C *on a curve* X*; the* $\mathbb{S}^d$*-framing lifts the factorization structure to an* E_2*-chiral algebra with braided monoidal representation category. For* $d = 2$*, all three steps are proved (Theorem CY-A). For* $d = 3$*, holomorphic Chern–Simons breaks* E_2 *to* E_1*; braiding is recovered via the Drinfeld center. Steps (1) and (2) are unconditional; step (3) is a programme conditional on the chain-level* $\mathbb{S}^3$*-framing.*

For $\mathbb{C}^3$*, the Jordan quiver gives* $\mathbb{C}^3 \to \mathcal{W}_{1+\infty} \to \mathrm{Rep}^{E_2}(Y(\widehat{\mathfrak{gl}}_1))$*: the critical cohomological Hall algebra* $\mathrm{CoHA}(Q)$ *is the* E_1*-sector, and the Drinfeld center recovers the full braided structure. For a general toric* CY_3 *with fan* Σ*, quiver charts indexed by Bridgeland stability chambers give local chiral algebras; wall-crossing is conjecturally Maurer–Cartan gauge equivalence with the global chiral algebra as homotopy colimit. When the CY-to-chiral functor exists, the Borcherds–Kac–Moody denominator identity equals the bar Euler product.*

The prototype $K3 \times E$ *identifies four structural obstructions between the shadow obstruction tower and the full automorphic form. A single* CY_3 *admits multiple chiral algebraizations with distinct modular characteristics. The* $\kappa_\bullet$*-spectrum* $\mathrm{Spec}_{\kappa_\bullet}(K3 \times E) = \{2, 3, 5, 24\}$ *records the holomorphic Euler characteristic* $\kappa_{\mathrm{cat}} = 2 = \chi(\mathcal{O}_{K3})$*, the complex dimension* $\kappa_{\mathrm{ch}} = 3$*, the Borcherds weight* $\kappa_{\mathrm{BKM}} = 5$ *of the Igusa cusp form* Δ_5*, and the lattice rank* $\kappa_{\mathrm{fiber}} = 24$*. For* $d \geq 3$*, the CY chiral algebra is* E_1*; the effective framing obstruction* $\mathrm{Obs}_{\mathrm{eff}}(d)$*, valued in* $\pi_d(BU)$ *or* $\pi_d(B\,\mathrm{Sp})$ *according to parity, is* 8*-periodic by Bott periodicity and trivial precisely when* $d \bmod 8 \in \{1, 3, 7\}$*.*

Preface

Where do chiral algebras come from?

Volumes I and II prove that a boundary chiral algebra A determines its bulk gravity through the derived chiral centre $\mathcal{Z}^{\mathrm{der}}_{\mathrm{ch}}(A)$: the modular characteristic $\kappa_{\mathrm{ch}}(A)$ is the holographic central charge, the shadow obstruction tower $\{S_r(A)\}_{r\geq 2}$ is the holographic correction hierarchy, and the Drinfeld double $\mathcal{U}_A = A \bowtie A^!$ is the universal boundary–bulk algebra that reconstructs gravity from the vertex algebra. This volume answers the question the first two leave open: where do the boundary algebras come from? Answer: from Calabi–Yau categories through the functor $\Phi\colon \mathrm{CY}_d\text{-}\mathbf{Cat} \to E_2\text{-}\mathrm{ChirAlg}$.

Volume I takes a chiral algebra as given and builds its categorical logarithm: the bar complex $\overline{B}(A)$, the universal Maurer–Cartan element Θ_A, and the five theorems that control the genus tower. Volume II reads the bar complex on $\mathbb{C} \times \mathbb{R}$ and extracts the Swiss-cheese structure, the holomorphic-topological field theory, and the Yangian line-sector algebra. Neither volume constructs a single chiral algebra. The input is always assumed.

This volume constructs the input. A Calabi–Yau category C of dimension d carries a cyclic A_∞-structure: a non-degenerate trace $\mathrm{Tr}\colon \mathrm{HC}^-_d(C) \to k$ on negative cyclic homology, encoding the CY condition as a d-dimensional Frobenius structure at the chain level. Negative cyclic homology (rather than ordinary Hochschild) is required because the $\mathbb{S}^d$-framing of the trace refines through S^1-equivariance; the same S^1-equivariance promotes the Hochschild complex to the derived chiral centre, so the CY trace is intrinsically linked to the boundary–bulk reconstruction of Volumes I–II. The factorization envelope converts the resulting Lie conformal algebra into a chiral algebra A_C on a curve X, and the CY trace descends to the modular characteristic. Theorem CY-D at $d = 2$ states $\kappa_{\mathrm{ch}}(A_C) = \chi^{\mathrm{CY}}(C)$: the Calabi–Yau Euler characteristic IS the holographic central charge of the bulk theory reconstructed from $\Phi(C)$ through its derived chiral centre. Volume III thus exhibits the third level of boundary–bulk reconstruction: CY geometry $\to$ vertex algebra $\to$ bulk. The shadow obstruction tower of A_C is the automorphic correction of the associated generalized Kac–Moody superalgebra.

The bar complex converts enumerative geometry into homotopy algebra. Donaldson–Thomas invariants become root multiplicities. The denominator identity becomes the bar Euler product. The genus expansion becomes the shadow obstruction tower. One Maurer–Cartan element Θ_{A_C} absorbs the entire automorphic datum.

The CY-to-chiral functor. The construction proceeds in three steps. Given a CY_d category C:

1. The cyclic A_∞-structure on C determines a *Lie conformal algebra* L_C: the cyclic bar differential gives the λ-bracket, the cyclic trace gives the invariant bilinear form, and the higher A_∞ operations give the higher Lie conformal products.
2. The *factorization envelope* $U^{\mathrm{fact}}_X(L_C)$ produces a factorization algebra on $\mathrm{Ran}(X)$, hence a chiral algebra A_C on X. The Poincaré–Birkhoff–Witt theorem for factorization envelopes (Nishinaka,

Vicedo) guarantees the expected filtration.

(3) The $\mathbb{S}^d$-framing enhances the factorization algebra. For $d = 2$: the $\mathbb{S}^2$-action on Hochschild homology gives an E_2-chiral algebra with braided monoidal representation category. For $d = 3$: holomorphic Chern–Simons breaks E_2 to E_1, and the braiding is recovered via the Drinfeld center.

Theorem CY-A constructs Φ for $d = 2$: all three steps are proved. For $d = 3$, step (3) is a programme conditional on the chain-level $\mathbb{S}^3$-framing; steps (1) and (2) are unconditional. The passage $d = 2 \to d = 3$ replaces a geometric braiding (from $\pi_1(C_2(\mathbb{R}^2)) = \mathbb{Z}$) with an algebraic one (from the Drinfeld center of E_1-categories): this is the CY analogue of the Tannakian reconstruction of a quantum group from its module category.

The prototype: $K3 \times E$**.** The product of a K3 surface S and an elliptic curve E is the canonical CY_3 testing ground. It is simple enough to compute: the lattice $\Lambda^{3,2}$ has signature $(3, 2)$, the Jacobi form $\phi_{0,1}(\tau, z)$ is the K3 elliptic genus, and the Igusa cusp form Δ_5 is the Borcherds denominator identity. It is rich enough to obstruct: four structural barriers separate the shadow obstruction tower from the full automorphic form.

The modular characteristic of the chiral de Rham complex is $\kappa_{\mathrm{ch}}(K3 \times E) = 3 = \dim_{\mathbb{C}}$, verified by six independent paths. The weight of the Igusa cusp form is $2\kappa_{\mathrm{BKM}} = 10$, so $\kappa_{\mathrm{BKM}} = 5$. These are different numbers measuring different things, and their difference is precisely the boundary–bulk duality of Volumes I–II made visible at the geometric level. κ_{ch} is the characteristic of the boundary chiral algebra $\Phi(C)$ computed on its own terms; the bulk theory reconstructed through the derived chiral centre $\mathcal{Z}^{\mathrm{der}}_{\mathrm{ch}}(\Phi(C))$ involves Siegel modular forms of weight $2\kappa_{\mathrm{BKM}} = 10$, and κ_{BKM} is the bulk-side holographic central charge. The two are dual: chiral and holographic. The shadow–Siegel gap theorem identifies the four obstructions to closing the loop:

(O1) *Categorical*: the shadow tower produces numbers (Taylor coefficients of Θ_A); the denominator identity is a function (an automorphic form on $\mathcal{H}_2$).

(O2) κ_{ch}–κ_{BKM} *mismatch*: $\kappa_{\mathrm{ch}} = 3 \neq 5 = \kappa_{\mathrm{BKM}}$; the single-copy chiral algebra sees $\dim_{\mathbb{C}} X$, while the bulk reconstruction sees the Borcherds weight.

(O3) *Second quantization*: the physical DT partition function is the DMVV symmetric product of single-copy data; the bar complex of a single algebra is the single copy.

(O4) *Schottky*: at $g \geq 4$, the Torelli map $\mathcal{M}_g \to \mathcal{A}_g$ is no longer dominant, and the shadow tower is imprisoned in tautological classes on $\overline{\mathcal{M}}_g$.

Each obstruction points to a research programme: constructive CY gluing (O1), the second-quantization bridge (O3), the Schottky shadow programme (O4). The κ_{ch}–κ_{BKM} mismatch (O2) is the most telling: a single CY_3 admits multiple chiral algebraizations, and each diagonal of the $\kappa_\bullet$-spectrum measures a different face of the same boundary–bulk datum. For $K3 \times E$, the spectrum $\mathrm{Spec}_{\kappa_\bullet}(X) = \{\kappa_{\mathrm{cat}}, \kappa_{\mathrm{ch}}, \kappa_{\mathrm{BKM}}, \kappa_{\mathrm{fiber}}\} = \{2, 3, 5, 24\}$ records the holomorphic Euler characteristic ($\kappa_{\mathrm{cat}} = 2 = \chi(\mathcal{O}_{K3})$), the complex dimension ($\kappa_{\mathrm{ch}} = 3$), the bulk-side Borcherds weight ($\kappa_{\mathrm{BKM}} = 5$), and the lattice rank ($\kappa_{\mathrm{fiber}} = 24$). Four algebras see four aspects of the same manifold, and the chiral–holographic pair $(\kappa_{\mathrm{ch}}, \kappa_{\mathrm{BKM}})$ is the one that sees the boundary–bulk axis.

The K3 elliptic genus decomposes as $\phi_{0,1} = 2\,\phi^{\mathrm{EZ}}_{0,1}$, where the factor 2 is $\kappa_{\mathrm{cat}}(K3) = \chi(\mathcal{O}_{K3})$. In the umbral moonshine decomposition, $A_n = 2 \cdot \dim(\rho_n)$ for M_{24} representations ρ_n. The bar complex provides the universal coefficient; the sporadic group provides the representation content; the physical multiplicity is their product.

Toric $\mathbf{CY}_3$ **and critical CoHAs.** For a toric CY_3 determined by a fan Σ, the toric diagram fixes a quiver Q_X. The critical cohomological Hall algebra $\mathrm{CoHA}(Q_X)$ is the E_1-sector: the positive half of an affine super Yangian $Y(\widehat{\mathfrak{g}}_{Q_X})$. The full braided structure is recovered through the Drinfeld center. For $\mathbb{C}^3$: the Jordan quiver gives $Y(\widehat{\mathfrak{gl}}_1) \simeq \mathcal{W}_{1+\infty}$, and the chain $\mathbb{C}^3 \to \mathcal{W}_{1+\infty} \to \mathrm{Rep}^{E_2}(Y(\widehat{\mathfrak{gl}}_1))$ is verified end-to-end. The toric fan is the Dynkin data.

The CoHA is the natural E_1-primitive: it is the ordered bar complex of the CY_3 chiral algebra, with deconcatenation coproduct encoding the topological direction and the R-matrix encoding the spectral braiding. The passage from CoHA to quantum group is the CY incarnation of the E_1-to-E_2 lift: averaging over the symmetric group recovers the factorization coalgebra of Volume I.

The elliptic curve as universal bridge. The same elliptic curve E appears in six distinct roles across the three volumes:

(i) *Base curve* (Vol I): the factorization algebra lives on $\mathrm{Ran}(E)$; the genus-1 propagator is $d\log E(z,w)$.

(ii) *Sewing parameter* (Vol I, MC5 analytic HS-sewing lane): the modular parameter τ is the Hilbert–Schmidt sewing kernel.

(iii) *CY fiber*: the DT partition function of $K3 \times E$ decomposes as a sewing of K3 fiber contributions along E.

(iv) *Jacobi form parameter*: the Fourier coefficients of $\phi_{0,1}(\tau, z)$ are root multiplicities and BPS degeneracies.

(v) *Siegel modular period*: the Igusa cusp form Δ_5 lives on $\mathrm{Sp}_4(\mathbb{Z})\backslash\mathbb{H}_2$, which parametrizes principally polarized abelian surfaces; the product locus of pairs of elliptic curves is codimension 1.

(vi) *Elliptic Hall algebra carrier*: the CY_2 quantum vertex chiral group is carried on $\mathrm{Coh}(E)$, with Felder's dynamical elliptic R-matrix.

Each appearance brings the same structural package: a modular parameter, an $\mathrm{SL}_2(\mathbb{Z})$ symmetry, a q-expansion, and a sewing functor converting genus-0 data into genus-1 data. The elliptic curve is the modular clock of the theory.

Convergence and divergence. The shadow obstruction tower and the topological string free energy produce genus-g amplitudes with opposite analytic behaviour. The shadow tower has Bernoulli decay $|F_g^{\mathrm{sh}}| \sim (2\pi)^{-2g}/g$: convergent, with radius 2π, Borel entire. The topological string has factorial growth $|F_g^{\mathrm{top}}| \sim (2g)!$: divergent, Gevrey-1, requiring non-perturbative completion. The shadow is the algebraic soul: the part computable from OPE structure constants alone, without worldsheet instantons. For class $\mathbf{G}$ algebras (Heisenberg, lattice VOAs), the shadow is the full answer. For class $\mathbf{M}$ (Virasoro, $\mathcal{W}_{1+\infty}$), the shadow is a strict subtower, and the instanton sum is infinite. The instanton gap $F_g^{\mathrm{top}} - F_g^{\mathrm{sh}}$ at each genus is the CY frontier.

Relation to Volumes I and II. Volume I provides Θ_A: the universal Maurer–Cartan element that controls the genus tower of any chiral algebra. Volume II provides $\mathrm{SC}^{\mathrm{ch,top}}$: the Swiss-cheese operad that reads the bar complex on $\mathbb{C} \times \mathbb{R}$ and produces the holomorphic-topological field theory. This volume provides the geometric origin: the functor $\Phi\colon \mathrm{CY}_d\text{-}\mathbf{Cat} \to E_2\text{-}\mathrm{ChirAlg}$ that constructs the input to the Volumes I–II machine from Calabi–Yau geometry.

The three volumes form a triangle:

$$\begin{tikzcd}
 & \text{Chiral algebra } A \arrow[dl, "\overline{B}"', "\text{Vol I}"] \arrow[dr, "\mathrm{SC}^{\mathrm{ch,top}}", "\text{Vol II}"'] & \\
\text{Modular MC element } \Theta_A \arrow[rr, "\mathrm{av}"'] & & \text{HT field theory}
\end{tikzcd}$$

Volume III closes the triangle from below: the CY category C sits beneath all three vertices, supplying the chiral algebra via Φ, the modular trace via the CY Euler characteristic, and the quantum group via the CoHA.

Together, Volumes I, II, and III develop the three levels of the boundary–bulk–geometry trilogy: Volume I at the two-dimensional chiral level, Volume II at the three-dimensional operadic level, Volume III at the geometric origin level. The unifying conjectural statement is that $A \mapsto \mathcal{Z}^{\mathrm{der}}_{\mathrm{ch}}(A)$ is conjectured to be the universal boundary–bulk reconstruction functor, with the Drinfeld double $\mathcal{U}_A = A \bowtie A^!$ as the proposed incarnation. What the three volumes actually prove is the boundary-linear skeleton: Volume I constructs the chiral pair $(A, A^!)$ and the bar-cobar inversion on the Koszul locus; Volume II develops the operadic architecture of the reconstruction programme on the Swiss-cheese operad; Volume III constructs the bridge $\Phi : C \mapsto A$ for $d = 2$ CY categories and identifies κ_{cat} with the CY Euler characteristic. The global assembly of these ingredients into a reconstruction functor to the bulk remains conjectural beyond the Koszul locus; the three-level chain $C \mapsto \Phi(C) = A \mapsto \mathcal{Z}^{\mathrm{der}}_{\mathrm{ch}}(A) \mapsto$ bulk gravity should be read as the organising programme, not as a proved composition.

Organisation. Part I establishes the categorical foundations: CY categories, cyclic A_∞-structures, the Hochschild calculus, and the $E_1/E_2/E_n$ chiral hierarchy with the framing obstruction tower. Part II constructs the bridge functor Φ from CY categories to quantum chiral algebras and proves Theorem CY-A for $d = 2$, with the identification of automorphic correction and shadow obstruction tower as the central result. Part III studies quantum groups and braided factorization structure, including the Drinfeld center as the bulk algebra. Part IV works through the standard CY landscape, centred on the $K3 \times E$ programme with its thirteen structural results and ten research programmes. Part V connects to the geometric Langlands programme and to Volumes I–II via the bar-cobar bridge and the modular Koszul bridge.

Volume I asks what the logarithm does. Volume II asks how it composes. Volume III asks where it comes from.

Contents

Part I

The CY Engine

Chapter 1

Introduction

1.1 The question

A Calabi–Yau category C of dimension d carries a canonical cyclic A_∞-structure: a non-degenerate trace

$$\mathrm{Tr}\colon \mathrm{HH}_\bullet(C) \longrightarrow k[-d]$$

on its Hochschild homology, encoding the CY condition as a d-dimensional Frobenius structure at the chain level. The cyclic bar complex $\mathrm{CC}_\bullet(C)$ is the primary invariant; it carries an $\mathbb{S}^d$-action from the d-sphere framing of the trace, and its $\mathbb{S}^d$-equivariant structure governs higher-genus amplitudes.

A chiral algebra A on a curve X carries a bar complex $B(A)$, a factorization coalgebra on $\mathrm{Ran}(X)$ whose differential encodes holomorphic OPE residues and whose coproduct encodes topological interval-cutting. At genus $g \geq 1$, the bar complex acquires curvature $\kappa_{\mathrm{ch}}(A) \cdot \omega_g$ from the Hodge bundle, and the full modular structure is controlled by the universal Maurer–Cartan element $\Theta_A := D_A - d_0$ (Volume I). Together with the Swiss-cheese algebra structure (Volume II), the bar complex is the complete modular invariant.

This work constructs a bridge:

$$\left\{ \begin{array}{c} \text{CY categories with} \\ \text{appropriate } E_n\text{-structure} \end{array} \right\} \xrightarrow{\quad \Phi \quad} \left\{ \begin{array}{c} \text{quantum chiral algebras} \\ \text{with modular trace} \end{array} \right\}$$

The functor Φ extracts from a CY category C (Fukaya, derived, matrix factorization, or more general) its E_2-monoidal structure and produces a chiral algebra A_C whose bar complex $B(A_C)$ encodes the CY cyclic homology, with the CY trace realized as the modular characteristic $\kappa_{\mathrm{ch}}(A_C)$.

1.2 The E_1/E_2 chiral hierarchy

Three levels of chiral structure organize the theory:

- E_1-chiral algebras (Volume II, Part III): associative factorization on $\mathbb{C} \times \mathbb{R}$, encoding the ordered/topological sector. Representation categories are monoidal.
- E_2-chiral algebras (this work): braided factorization on $\mathbb{C} \times \mathbb{C}$, encoding the holomorphic-holomorphic sector. Representation categories are braided monoidal: the natural habitat of quantum groups.

- The passage $E_1 \to E_2$ via Dunn additivity: $E_2 \simeq E_1 \otimes E_1$. An E_2-chiral algebra is an E_1-algebra in E_1-algebras.

The CY condition enters through Kontsevich's identification: a d-dimensional CY structure on C determines an $\mathbb{S}^d$-framing of the Hochschild complex, hence an $\mathbb{S}^1$-action on $\mathrm{HH}_\bullet(C)$. For $d = 2$ (CY surfaces), this $\mathbb{S}^1$-action is exactly the data of an E_2-algebra structure on the cyclic homology: the braiding.

The E_1 stabilization theorem

For $d \geq 3$, the CY chiral algebra is E_1, not E_d. The reason is structural: the $\mathrm{GL}(d)$-invariant Schouten–Nijenhuis bracket on polyvector fields $\mathrm{PV}^*(\mathbb{C}^d)$ vanishes identically for $d \geq 3$ (Theorem 8.4 and its generalization), so the classical Lie conformal algebra is abelian and all noncommutative structure arises from quantization. The CoHA construction then gives an ordered (associative) product—the extension correspondence $0 \to V' \to V \to V'' \to 0$ has a preferred direction (sub before quotient), which is E_1, not E_2. The braided monoidal structure is recovered through the Drinfeld center of the E_1-monoidal representation category (Theorem 8.15).

The additional shifted structure beyond the base E_1 level is controlled by the *framing obstruction* $\pi_d(BU)$, which measures the failure of the $\mathbb{S}^d$-framing on Hochschild homology to trivialize. Two independent computations of this obstruction—the unitary path (Bott periodicity for BU, period 2) and the symplectic/orthogonal path (Bott periodicity for Sp and O, period 8)—produce the following landscape:

d	Native E_n	$\pi_d(BU)$	$\mathrm{Obs}_{\mathrm{eff}}(d)$	Shifted structure
1	E_∞	0	0	none (commutative)
2	E_2	$\mathbb{Z}$	$\mathbb{Z}$	braiding parameter (c_1)
3	E_1	0	0	none ($\mathbb{S}^3$-framing trivial)
4	E_1	$\mathbb{Z}$	$\mathbb{Z}$	$\mathbb{Z}$-shifted symplectic (p_1)
5	E_1	0	$\mathbb{Z}_2$	$\mathbb{Z}_2$-shifted (Sp refinement)
6	E_1	$\mathbb{Z}$	$\mathbb{Z}$	$\mathbb{Z}$-shifted (c_3)
7	E_1	0	0	none (trivial, mirrors $d = 3$)
8	E_1	$\mathbb{Z}$	$\mathbb{Z}$	$\mathbb{Z}$-shifted (Bott-periodic)

The pattern repeats with period 8 by Bott periodicity of Sp and O. The effective obstruction is trivial precisely at $d \bmod 8 \in \{1, 3, 7\}$. All even $d \geq 2$ have $\mathbb{Z}$-valued obstruction from $\pi_d(BU) = \mathbb{Z}$. For CY_4, the obstruction is the first Pontryagin class $p_1 \in \pi_4(BU) = \mathbb{Z}$: it provides a level (analogous to the level of a Kac–Moody algebra) that governs the shifted symplectic structure on the moduli of objects. For $K3 \times K3$ ($d = 4$), the Hopf decomposition $\mathbb{S}^4 = \mathbb{S}^2 \times \mathbb{S}^2$ recovers E_2 by Dunn additivity.

The full proof of the stabilization theorem (Theorem 7.1) is in Chapter 7.

1.3 Relation to Volumes I and II

This work depends on and extends the two preceding volumes:

Volume	Provides	Used here
I (Modular Koszul Duality)	Bar-cobar machine, Θ_A, $\kappa_{\mathrm{ch}}(A)$	CY bar complex, modular trace
II (3D HT QFT)	$\mathrm{SC}^{\mathrm{ch,top}}$, PVA descent, DK bridge	E_1 sector, braided structure
III (this work)	CY $\to$ chiral functor, E_2 theory	—

Remark 1.1 (Dimensional reduction across the trilogy). The three volumes correspond to three levels of a dimensional reduction. Volume I operates at Level 1 (2d, the curve X): the modular shadow, where the operadic structure is E_∞ and the five theorems A–D+H are the invariants that survive Σ_n-coinvariance. Volume II operates at Level 2 (3d, $\mathbb{C}_z \times \mathbb{R}_t$): the Swiss-cheese layer, where the bar differential is the holomorphic colour and the deconcatenation coproduct is the topological colour, with operadic structure E_1. This volume operates at Level 3 (4d / CY): the geometric source, where Calabi–Yau categories of dimension $d \geq 3$ produce E_1-chiral algebras via the functor Φ, and the Drinfeld center $\mathcal{Z}(\mathrm{Rep}^{E_1}(\mathcal{A}))$ recovers the E_2-braided structure. At each level, information is lost: $4d \to 3d$ forgets one topological direction ($E_2 \to E_1$); $3d \to 2d$ forgets the ordering (Σ_n-coinvariance, $r(z) \mapsto \kappa_{\mathrm{ch}}(\mathcal{A})$).

1.4 The analytic gap and the Čech resolution

The CY-to-chiral functor Φ at $d = 3$ requires a chain-level contracting homotopy on the Dolbeault complex of the CY threefold. For noncompact targets ($\mathbb{C}^3$, the conifold), the T^3-equivariant structure provides the necessary data. For compact CY$_3$, such as the quintic threefold $X_5 \subset \mathbb{P}^4$, the analytic construction of a contracting homotopy on the Dolbeault resolution would require solving a $\bar\partial$-equation with estimates — a PDE problem foreign to the algebraic framework of this volume.

The algebraic Čech resolution closes this gap at the perturbative level. The standard affine cover $\{U_i = \{x_i \neq 0\}\}_{i=0}^4$ of $\mathbb{P}^4$ restricts to the quintic, giving five affine open sets whose finite intersections are all affine. Leray's theorem guarantees acyclicity of the Čech complex, and the explicit contracting homotopy $s^q(f_{i_0\cdots i_q}) := f_{0 i_0 \cdots i_q}$ (prepend the distinguished index 0) satisfies the strong deformation retract conditions

$$\delta s + s\delta = \mathrm{Id} - i \circ p, \qquad s^2 = 0, \qquad s \circ i = 0, \qquad p \circ s = 0.$$

The SDR data (i, p, s) interfaces directly with the homotopy transfer theorem (HTT), transferring the A_∞-structure from Čech cochains $\check{C}^\bullet(\mathcal{E}nd(\mathcal{E}))$ to its cohomology $\mathrm{Ext}^\bullet(\mathcal{E}, \mathcal{E})$. The transferred operations μ_k^{ch} are given by sums over planar trees whose internal edges are decorated by the homotopy s; each application of s introduces rational functions on the affine intersections, and the resulting chiral operations are well-defined as formal power series.

The scope is perturbative: the Čech homotopy produces an algebraic A_∞-structure on the Ext algebra, but convergence of the resulting series to holomorphic functions (the non-perturbative statement) requires separate input from the analytic completion programme of Volume I (MC5, §**??**). The perturbative statement suffices for the formal CY-to-chiral functor and for all combinatorial applications in this volume. The full construction, including the proof that the SDR conditions hold and the scope qualification, is Theorem 8.10 in Chapter 8.

Three independent consistency checks confirm the resulting chiral data for the quintic: the Gepner model matrix factorization (via mirror symmetry, $\mathrm{MF}(x_1^5 + \cdots + x_5^5)$), the large-volume Hodge–Kodaira–Rosenberg limit (HKR degeneration of $\mathrm{Ext}^\bullet$ to Dolbeault cohomology), and the Gopakumar–Vafa integrality of the resulting genus-g amplitudes. All three produce the same modular characteristic $\chi/2 = -100$ (the MacMahon-normalised form of κ_{cat} at $d = 3$).

1.5 CY3 combinatorics as generalized root data

The combinatorics of a Calabi–Yau threefold X (lattice, intersection form, BPS spectrum) constitute the *generalized root datum* of a *quantum vertex chiral group* $G(X)$. For toric CY3 (via the CoHA and Drinfeld double) and for CY_2 categories (via Theorem CY-A_2), this object is constructed. For general CY_3, including the prototype $K3 \times E$, $G(X)$ is the target of a programme (Conjecture CY-A_3), not a constructed object.

Given a CY3 X, the generalized root datum $\mathcal{R}(X)$ consists of:

(i) A lattice $\Lambda(X)$ extracted from $K(X)$ or $H^*(X)$ with its Euler form;

(ii) Real simple roots Δ^{re} from the geometry (hyperbolic sublattice for $K3 \times E$; toric diagram for toric CY3);

(iii) Imaginary roots $\Delta^{\mathrm{im}} = \Delta_0^{\mathrm{im}} \sqcup \Delta_1^{\mathrm{im}}$ (even and odd) with multiplicities $\mathrm{mult}(\alpha)$ from BPS counting (Donaldson–Thomas invariants for X, or equivalently the Fourier coefficients of the appropriate automorphic form);

(iv) A Weyl group $W(X)$ acting on the positive cone of $\Lambda(X)$;

(v) A Weyl vector $\rho \in \Lambda(X) \otimes \mathbb{Q}$.

Two families of examples illustrate the construction.

The $K3 \times E$ tower. For $X = (S \times E)/(\mathbb{Z}/N\mathbb{Z})$ with S a K3 surface and E an elliptic curve, the lattice is $\Lambda^{3,2} \simeq \Lambda^{1,1} \oplus \Lambda^{1,1} \oplus [2]$ of signature $(3, 2)$. The hyperbolic sublattice $\Lambda_{II}^{2,1}$ with Gram matrix $\left(\begin{smallmatrix} 2 & -2 & -2 \\ -2 & 2 & -2 \\ -2 & -2 & 2 \end{smallmatrix}\right)$ provides the real roots. The root multiplicities are the Fourier coefficients $f(n, l)$ of the weak Jacobi form $\phi_{0,1}$, the K3 elliptic genus. The resulting generalized Borcherds–Kac–Moody superalgebra $\mathfrak{g}_{\Delta_5}$ has the Igusa cusp form Δ_5 as its denominator identity. The single-copy chiral modular characteristic satisfies $\kappa_{\mathrm{ch}}(K3 \times E) = 3 = \dim_{\mathbb{C}}$ (from additivity: $\kappa_{\mathrm{ch}}(K3) + \kappa_{\mathrm{ch}}(E) = 2 + 1$), verified by six independent paths; the BKM automorphic weight is the distinct quantity $\kappa_{\mathrm{BKM}} = 5$ (the weight of Δ_5; see the Shadow–Siegel gap theorem below); the factor 2 in $Z_{K3} = 2\,\phi_{0,1}$ is the bar-complex moonshine multiplier $\kappa_{\mathrm{cat}}(\mathcal{A}_{K3})$; and the shadow obstruction tower does not produce Δ_5 directly (four structural obstructions: categorical, κ-mismatch, second quantization, Schottky; Theorem 14.132).

Toric CY3 and critical CoHAs. For a toric CY3 determined by a fan Σ, the toric diagram determines a quiver Q_X. The critical cohomological Hall algebra $\mathrm{CoHA}(Q_X)$ is the positive half of an affine super Yangian $Y(\widehat{\mathfrak{g}}_{Q_X})$. For $\mathbb{C}^3$: the Jordan quiver gives $Y(\widehat{\mathfrak{gl}}_1) \simeq \mathcal{W}_{1+\infty}$. The toric fan is the Dynkin data.

1.6 Automorphic correction as shadow obstruction tower

The passage from the naive Kac–Moody algebra $\mathfrak{g}$ (built from the real root Gram matrix alone) to the full generalized BKM superalgebra $\mathfrak{g}_X$ (with all imaginary roots and their multiplicities) is the *automorphic correction*. In the language of Volume I, this passage is the *shadow obstruction tower*:

Shadow tower (Vol I)	**BKM language**	**CY3 geometry**
κ_{ch} (arity 2)	Weyl vector ρ contribution	Leading DT invariant
C (arity 3, cubic)	First imaginary root corrections	Genus-0 3-point function
Q (arity 4, quartic)	Higher imaginary roots	Genus-0 4-point function
Full Θ_A	Complete automorphic correction	Full BPS generating function

The denominator identity of $\mathfrak{g}_X$ (Δ_5 for $K3 \times E$, the DT partition function for toric CY3) should be the bar-complex Euler product of the quantum vertex chiral group, when that group exists. For toric CY3 this identification is part of Conjecture CY-C. For $K3 \times E$, since $A_{K3\times E}$ is not yet constructed (AP-CY6, AP-CY8), the product formula for Δ_5 is an *observation* about the Borcherds lift, not a theorem derived from the bar complex.

The bar Euler product of lattice VOAs recovers the Borcherds denominator identity (verified for E_8 and Leech lattice; conditional on CY-A$_3$ for $K3 \times E$).

1.7 Main results

- **Theorem CY-A** (CY-to-chiral functor): Construction of $\Phi\colon \mathrm{CY}_d\text{-}\mathbf{Cat} \to E_2\text{-}\mathrm{ChirAlg}$ via the factorization envelope and $\mathbb{S}^d$-framing. *Proved for* $d = 2$ (CY surfaces, where the $\mathbb{S}^2$-framing gives E_2 via Kontsevich–Vlassopoulos). *For* $d = 3$: a programme conditional on (a) constructing the chain-level $\mathbb{S}^3$-framing compatible with BV structure and (b) the quantization step. The $\mathbb{S}^3$-framing gives an E_3-structure, from which E_2 is obtained by restriction; however, $\pi_1(\mathrm{Conf}_2(\mathbb{R}^3))$ is trivial, so the restriction gives a symmetric braiding at the topological level. The quantum group braiding for $d = 3$ is expected to arise through the Drinfeld center of E_1-monoidal categories.
- **Theorem CY-B** (E_2-chiral Koszul duality): Bar-cobar adjunction in the E_2-chiral setting. The automorphic correction of the BKM superalgebra $\mathfrak{g}_X$ is identified with the shadow obstruction tower Θ_{A_X}; the denominator identity equals the bar-complex Euler product.
- **Conjecture CY-C** (Quantum group realization): For toric CY3, $\mathrm{Rep}^{E_2}(G(X))$ should be braided monoidal equivalent to $\mathrm{Rep}(Y(\widehat{\mathfrak{g}}_{Q_X}))$; for $K3 \times E$, the $\mathrm{Sp}_4(\mathbb{Z})$-module structure on the denominator identity should recover the braided structure. The critical CoHA is the E_1-sector (positive half). The CY category $C(\mathfrak{g}, q)$ is not yet constructed in general.
- **Theorem CY-D** (Modular CY characteristic): For $d = 2$, $\kappa_{\mathrm{ch}}(\Phi(C))$ equals the CY Euler characteristic $\chi^{\mathrm{CY}}(C)$. For $K3 \times E$ ($d = 3$): the weight of Δ_5 is $5 = h^{1,1}(K3)/4 = 20/4$; this appears in the structural position of a modular characteristic, but without $A_{K3\times E}$ (which is not constructed), the identification $\kappa_{\mathrm{BKM}} = 5$ is an observation matching the Borcherds lift weight formula, not a computation from the Volume I definition. The general $d = 3$ formula is conjectural.
- **Theorem (Shadow–Siegel gap)**: The shadow obstruction tower of $K3 \times E$ does not produce the Igusa cusp form Φ_{10}. Four structural obstructions: categorical (number vs function), modular characteristic ($\kappa_{\mathrm{ch}} = 3 \neq 5 = \kappa_{\mathrm{BKM}}$), second quantization (single-copy vs DMVV symmetric product), and Schottky at $g \geq 4$ ($\mathrm{codim} = (g-2)(g-3)/2$). Thirteen structural results (K3-1 through K3-13) and ten research programmes (A through J) are developed in the toroidal and elliptic algebras chapter.

1.8 What is proved versus what is conjectural

Proved (would survive a referee):

- The generalized root datum axioms CY1–CY7, adequate for $K3 \times E$ and toric CY3 examples.
- The E_2-chiral algebra formalism and its basic properties (definition, bar complex, braiding).
- Theorem CY-A for $d = 2$: steps 1–3 of Φ (cyclic $A_\infty \to$ Lie conformal $\to$ factorization envelope $\to$ E_2 enhancement via $\mathbb{S}^2$-framing).
- The CoHA as E_1-sector for toric CY3 (via Schiffmann–Vasserot, RSYZ).
- The Drinfeld center equivalence $\mathcal{Z}(\mathrm{Rep}^{E_1}(A)) \simeq \mathrm{Rep}^{E_2}(Z^{\mathrm{der}}_{\mathrm{ch}}(A))$ (Ben-Zvi–Francis–Nadler, Lurie).
- All lattice theory and BKM constructions for $K3 \times E$ (Gritsenko–Nikulin).
- The denominator identity $\Delta_5 = \Phi$ and product formula (arity 2–6 computationally verified).
- The shadow–Siegel gap theorem with four structural obstructions (O1–O4).
- $\kappa_{\mathrm{ch}}(K3 \times E) = 3$, verified by six independent paths.
- The moonshine multiplier: $A_n = \kappa_{\mathrm{cat}} \cdot \dim(\rho_n)$ identifies the factor 2 in $Z_{K3} = 2\,\phi_{0,1}$.
- Forty-two CY/BKM compute engines with 5,700+ tests across the standard CY landscape.

Rigorous proof sketches (completable with effort):

- The automorphic correction = shadow obstruction tower identification (parts (a)–(c)).
- The denominator = bar Euler product (conditional on CY-A for the relevant dimension).
- E_2-chiral Koszul duality (conjecture with evidence from E_1 case).

Conjectural:

- Theorem CY-A for $d = 3$: the chain-level $\mathbb{S}^3$-framing and BV-compatibility are not yet established.
- Conjecture CY-C: the CY category $C(\mathfrak{g}, q)$ is not constructed in general.
- The chiral algebra $A_{K3\times E}$ does not yet exist as a constructed object; the identification of $\kappa_{\mathrm{BKM}} = 5$ with a chiral algebra modular characteristic is an observation, not a theorem.
- The Langlands = Koszul duality conjecture.
- All physics conjectures in Part V.
- Ten research programmes (A–J) for the $K3 \times E$ programme, each with formal conjectures.

The analytic gap for compact CY$_3$ (the construction of a chain-level contracting homotopy without solving a PDE) is closed by the algebraic Čech resolution (Theorem 8.10): the standard 5-patch cover of the quintic in $\mathbb{P}^4$ provides an algebraic SDR that interfaces directly with the homotopy transfer theorem. E_1 universality is consistent for the quintic threefold via three independent paths (Gepner MF, large-volume HKR, GV integrality), with $\kappa_{\mathrm{MacMahon}} = \chi/2 = -100$ as the physically relevant modular characteristic.

1.9 The ten research programmes

The $K3 \times E$ prototype generates ten research programmes, each with formal conjectures grounded in the 90+ compute engines. The formal development is in Chapter 14.

(Prog A) *Constructive CY gluing.* Reconstruct $D^b(K3 \times E)$ from quiver charts indexed by Bridgeland stability conditions. The bar functor commutes with homotopy colimits (Proposition 14.122), so the global chiral algebra is the hocolim of local CoHAs. The minimum chart number, the Brauer obstruction, and the extension to non-toric K3 surfaces are open.

(Prog B) κ_{cat} *as universal moonshine multiplier.* The factor 2 in $Z_{K3} = 2\,\phi_{0,1}$ is $\kappa_{\mathrm{cat}}(K3) = \chi(\mathcal{O}_{K3})$. The BPS degeneracies factorise as $A_n = \kappa_{\mathrm{cat}} \cdot \dim(\rho_n)$, separating the homological part (κ_{cat}, from the bar complex) from the group-theoretic part (ρ_n, from the sporadic symmetry). The programme: extend this factorisation to the Monster module, the Conway module, and all holomorphic VOAs.

(Prog C) *Second-quantization bridge.* The ratio $\kappa_{\mathrm{BKM}}/\kappa_{\mathrm{ch}} = 5/3$ encodes the first-to-second-quantization passage. The conjectural formula $\kappa_{\mathrm{BKM}}(S \times E) = \kappa_{\mathrm{cat}}(S) + \kappa_{\mathrm{ch}}(S \times E)$ expresses the weight of the automorphic form as a sum of fiber and global modular characteristics.

(Prog D) *Schottky shadow programme.* At $g \le 3$, the shadow obstruction tower sees nearly all of the moduli. At $g \ge 4$, the Schottky locus has positive codimension and the shadow is imprisoned in tautological classes. The programme: characterise the tautological projection of Siegel modular forms, and determine whether the shadow tower generates the image of $S_*(\mathrm{Sp}_{2g}(\mathbb{Z})) \to R^*(\overline{\mathcal{M}}_g)$.

(Prog E) *Mock modularity and the bar complex.* The 1/4-BPS counting function is a mock modular form with shadow $24\,\eta^3$. The factor $24 = \chi(K3)$ is the surface announcing its existence. The programme: construct a half-integral-weight metaplectic bar complex whose shadow obstruction tower produces the mock modular completion.

(Prog F) *Modular factorization envelope.* Construct the universal modular factorization envelope $U_X^{\mathrm{mod}}(L)$ for cyclically admissible Lie conformal algebras, carrying the full shadow obstruction tower. This would give the left adjoint of the modular primitive-current functor.

(Prog G) *BKM algebras and scattering diagrams.* The Gross–Siebert scattering diagram consistency condition IS the E_1 Maurer–Cartan equation (Theorem 8.35). The programme: lift this identification to the full motivic Hall algebra level (where naive BCH is insufficient; AP42).

(Prog H) *Descent theorem.* Čech descent for E_1-algebras over the Bridgeland stability manifold. Proved for toric CY3 (Proposition **??**); open for compact CY3 beyond the chart-atlas regime.

(Prog I) *Higher-dimensional CY shadows.* The E_n stabilization theorem (Theorem 7.1) gives a Bott-periodic 8-fold pattern. For CY4: the p_1 Pontryagin class controls the shifted symplectic structure. For $K3 \times K3$: the Hopf decomposition $\mathbb{S}^4 = \mathbb{S}^2 \times \mathbb{S}^2$ gives E_2 by Dunn additivity.

(Prog J) *Convergence vs divergence.* The shadow partition function converges with radius 2π (Bernoulli decay); the topological string diverges (Gevrey-1). The programme: quantify the instanton gap $F_g^{\mathrm{top}} - F_g^{\mathrm{sh}}$ at each genus, and determine whether the shadow tower is Borel-summable to the full amplitude.

All ten programmes are computationally testable: the 90+ compute engines provide a laboratory for each conjecture.

1.10 The landscape of examples

The CY-to-chiral functor Φ operates on any CY category with cyclic A_∞-structure. Four families of CY categories provide the standard landscape; each family contributes distinct structural features to the theory.

Fukaya categories (Chapter 16). The symplectic input. For an elliptic curve E_τ, the Fukaya category $\mathrm{Fuk}(E_\tau)$ is CY of dimension 1 and Φ produces the Heisenberg vertex algebra H_k at level $k = \mathrm{vol}(E_\tau)$. For a K3 surface S, $\mathrm{Fuk}(S)$ is CY of dimension 2 and Φ produces an E_2-chiral algebra with $\kappa_{\mathrm{ch}} = \chi(\mathcal{O}_S) = 2$. For compact CY threefolds, the Fukaya-side functor is conditional on the chain-level $\mathbb{S}^3$-framing (Conjecture CY-A3); the open-string sector ($\mathrm{Fuk}(X)$ with Lagrangian boundary conditions) connects to the Volume II Swiss-cheese structure. Wrapped Fukaya categories $\mathcal{W}(X)$ of Liouville manifolds provide the non-compact analogue: for cotangent bundles T^*M, Abouzaid's equivalence $\mathcal{W}(T^*M) \simeq \mathrm{Mod}(C_*(\Omega M))$ reduces the CY-to-chiral functor to the based loop space.

Derived categories of coherent sheaves. The algebraic input. For a smooth projective CY manifold X, the bounded derived category $D^b(\mathrm{Coh}(X))$ is CY of dimension d with the CY trace induced by Serre duality. Homological mirror symmetry identifies $\mathrm{Fuk}(X) \simeq D^b(\mathrm{Coh}(X^\vee))$; under Φ, this becomes an equivalence $A_{\mathrm{Fuk}(X)} \simeq A_{D^b(\mathrm{Coh}(X^\vee))}$ of quantum chiral algebras, providing a consistency check between the symplectic and algebraic sides. When $D^b(\mathrm{Coh}(X))$ admits a full exceptional collection, the CY-to-chiral functor reduces to a quiver-with-potential computation: the critical CoHA of the quiver (Chapter 15). Bridgeland stability conditions parametrize the space of t-structures on $D^b(\mathrm{Coh}(X))$; wall-crossing in the stability manifold corresponds to mutations of the bar complex, and the global chiral algebra is assembled as a homotopy colimit over stability chambers (Programme A).

Matrix factorizations. The Landau–Ginzburg input. A polynomial $W\colon \mathbb{C}^n \to \mathbb{C}$ gives a CY category $\mathrm{MF}(W)$ of dimension $n-2$. For ADE singularities $W = x^N + y^2 + z^2 + w^2$ in four variables, $\mathrm{MF}(W)$ is CY of dimension 2 and Φ (Theorem CY-A$_2$) produces chiral algebras related to $\mathcal{W}_N$-algebras. The LG/CY correspondence $\mathrm{MF}(W) \simeq D^b(\mathrm{Coh}(X_W))$ provides a further consistency check against the derived-category side. For non-ADE singularities (unimodal, bimodal), the resulting chiral algebras are expected to be new objects not realized by the standard Lie-theoretic landscape of Volume I.

Quantum group representations (Chapter 19). The representation-theoretic target. The braided monoidal category $\mathrm{Rep}_q(\mathfrak{g})$ is what the CY programme aims to realize: Conjecture CY-C predicts that $\mathrm{Rep}_q(\mathfrak{g}) \simeq \mathrm{Rep}^{E_2}(A_{\mathcal{C}})$ for a CY category $\mathcal{C}(\mathfrak{g}, q)$. The Kazhdan–Lusztig equivalence $\mathrm{Rep}_q(\mathfrak{g}) \simeq \mathrm{KL}_k(\mathfrak{g})$ at roots of unity is the prototype: the Volume I bar complex $B(V_k(\mathfrak{g}))$ encodes the quantum group R-matrix in its arity-$(1,1)$ component, and the DK bridge (Volume II, MC3 for all simple types) recovers the quantum group categorical structure. The affine Yangian realization (Chapter 15) provides the E_1-sector; the Drinfeld center (Chapter 13) provides the E_2-enhancement.

In addition to the four standard families, the toroidal and elliptic algebras chapter (Chapter **??**) develops the $K3 \times E$ programme in full: the generalized root datum, the BKM superalgebra $\mathfrak{g}_{\Delta_5}$ with its Igusa cusp form denominator identity, the shadow–Siegel gap theorem, thirteen structural results (K3-1 through K3-13), and all ten research programmes (A through J). The toric CY$_3$ CoHA chapter (Chapter 15) develops the affine Yangian and $\mathcal{W}_{1+\infty}$ side: the Jordan quiver gives $Y(\widehat{\mathfrak{gl}}_1) \simeq \mathcal{W}_{1+\infty}$, the toric fan is the Dynkin data, and the shuffle algebra presentation connects to the KZ equations of Volume I.

1.11 Guide for the reader

Part I establishes the categorical foundations: CY categories, cyclic A_∞-structures, the Hochschild calculus, and the $E_1/E_2/E_n$ chiral hierarchy with the framing obstruction tower. Part II constructs the bridge functor Φ from CY categories to quantum chiral algebras and proves Theorems CY-A (for $d = 2$) and CY-D, with the identification of automorphic correction and shadow obstruction tower as the central result. Part III treats quantum groups and their braided factorization structure, including the Drinfeld center as the bulk algebra. Part IV works through the standard landscape: the toroidal and elliptic algebras chapter develops the $K3 \times E$ programme (K3-1 through K3-13, programmes A through J), alongside toric CY3 CoHAs, Fukaya categories, derived categories, matrix factorizations (ADE singularities and W-algebras), and quantum group representations (Kazhdan–Lusztig, affine Yangians, critical CoHAs). Part V connects to the geometric Langlands programme and to Volumes I and II via the bar-cobar bridge and the modular Koszul bridge. The working notes (§**??**) record the research frontier.

1.12 Conventions and notation

This volume inherits the conventions of Volumes I and II; we record here the points where the CY setting introduces additional notation or where conventions differ across the literature. A comprehensive conventions appendix is Appendix A.

Grading. All grading is cohomological: differentials have degree $+1$. The bar construction uses desuspension s^{-1}, with $|s^{-1}v| = |v| - 1$ (matching Volume I; see signs_and_shifts.tex therein). The CY dimension d enters as a cohomological shift: the CY trace $\mathrm{Tr}\colon \mathrm{HH}_\bullet(C) \to k[-d]$ is a map of degree $-d$.

E_n notation. We write E_n for the little n-disks operad (Boardman–Vogt) and E_∞ for the commutative operad. Dunn additivity: $E_2 \simeq E_1 \otimes E_1$. An E_2-chiral algebra (Definition 6.3) factorizes along two holomorphic directions; its representation category is braided monoidal (Theorem 6.2). We write $\mathrm{FM}_k(\mathbb{C})$ for the Fulton–MacPherson compactification of $\mathrm{Conf}_k(\mathbb{C})$; this is a blowup along diagonals, not the complement of the diagonal.

Divided powers and the λ-bracket. The translation between OPE modes $a_{(n)}b$ and the λ-bracket uses the divided-power convention $\lambda^{(n)} = \lambda^n/n!$, so $\{a_\lambda b\} = \sum_n \lambda^{(n)} a_{(n)} b$. The λ-bracket coefficient at order n is $a_{(n)}b/n!$, not $a_{(n)}b$ (Volume I, AP44). This convention is consistent with Volumes I and II; when consulting references that use the Borel transform $B(K)(\lambda) = \sum \lambda^{(n)} c_n$, the $1/n!$ is already absorbed into $\lambda^{(n)}$.

CY categories. We write CY_d-$\mathbf{Cat}$ for the ∞-category of smooth d-dimensional CY categories. Tensor products of dg categories are derived unless stated otherwise. "Equivalence" means quasi-equivalence of dg categories (equivalence of associated ∞-categories). The Hochschild–Kostant–Rosenberg isomorphism $\mathrm{HH}_\bullet(D^b(\mathrm{Coh}(X))) \simeq H^*(X, \Omega^\bullet_X)$ is used freely for smooth projective varieties.

Cross-volume references. Results from Volume I are cited as "Vol I, Theorem X" or by their Volume I label (e.g., thm:mc2-bar-intrinsic). Results from Volume II are cited analogously. When a formula appears in multiple volumes, the conventions may differ (AP49): Volume I uses OPE modes, Volume II uses λ-brackets with divided powers, and this volume uses both depending on context. Every cross-volume formula citation in this work has been independently verified against the source.

Chapter 2

Calabi–Yau Categories

A Calabi–Yau category is a dg category whose Serre functor is a pure degree shift. This single condition, due independently to Kontsevich and to Costello, organises the diverse sources of Calabi–Yau geometry (coherent sheaves, Fukaya categories, matrix factorisations, noncommutative resolutions) into a uniform categorical framework. It is the framework on which the Vol III functor Φ acts: Φ takes a cyclic A_∞ Calabi–Yau category of dimension d as input and returns an E_2-chiral algebra (at $d = 2$) or a programme-level target (at $d = 3$). This chapter fixes the categorical input, recording the definitions, the Hochschild structure, the cyclic A_∞ enhancement, and the interface to Φ (Chapter 8).

2.1 Smooth and proper dg categories

Definition 2.1 *(Smooth dg category).* A dg category C over k is *smooth* if the diagonal bimodule $C \in C^{\mathrm{op}} \otimes C\text{-Mod}$ is a perfect bimodule (compact in the derived category of bimodules).

Definition 2.2 *(Proper dg category).* A dg category C is *proper* if for all objects $X, Y \in C$, the Hom-complex $\mathrm{Hom}_C(X, Y)$ has finite-dimensional total cohomology.

A category that is both smooth and proper is called *saturated*. Saturated categories admit a Serre functor S_C, a distinguished autoequivalence characterised by the natural isomorphism $\mathrm{Hom}_C(X, Y) \simeq \mathrm{Hom}_C(Y, S_C X)^\vee$. Classical Serre duality recovers $S_{D^b(\mathrm{Coh}(X))} = (- \otimes \omega_X)[\dim X]$; the Calabi–Yau condition asks that the twist ω_X be trivial, leaving only the shift.

2.2 The Calabi–Yau condition

Definition 2.3 *(Calabi–Yau category, after Kontsevich).* A smooth dg (or A_∞) category C is *Calabi–Yau of dimension d* if the Serre functor is a pure shift: $S_C \simeq [d]$. Equivalently, there is a quasi-isomorphism of C-bimodules

$$C \xrightarrow{\sim} C^\vee[d]$$

where $C^\vee = \mathrm{RHom}_{C^{\mathrm{op}} \otimes C}(C, C^{\mathrm{op}} \otimes C)$ is the inverse dualizing bimodule. Equivalently again, there exists a non-degenerate cyclic pairing

$$\langle \cdot, \cdot \rangle_d \colon C(a, b) \otimes C(b, a) \longrightarrow k[-d]$$

of degree $-d$, satisfying the graded symmetry $\langle f, g \rangle_d = (-1)^{|f||g|} \langle g, f \rangle_d$.

The equivalence of the bimodule formulation with the cyclic pairing formulation is due to Costello [6]; see also Kontsevich's Mirror Symmetry address [21] for the original categorical statement and Keller [19] for the dg-categorical reformulation used here.

Remark 2.4. The CY condition combines smoothness (the bimodule C is compact) with the Serre functor being a shift. A smooth proper CY category is *saturated* in the sense of Kontsevich–Soibelman. Non-proper "relative" CY categories also appear (matrix factorisations, Ginzburg dg algebras); they require the Hochschild invariants of Section 2.3 to be interpreted in a compactly-supported or negative-cyclic refinement.

Example 2.5 *(Fukaya category).* For a compact symplectic manifold (M, ω) with $c_1(M) = 0$, the Fukaya category $\mathrm{Fuk}(M)$ is CY of dimension $\dim_{\mathbb{R}}(M)/2 = n$. The CY pairing arises from the cyclic structure on the A_∞-algebra of Lagrangian Floer cochains (Fukaya; Seidel [31]).

Example 2.6 *(Derived category of a CY manifold).* For a smooth projective CY manifold X of complex dimension n (i.e. $\omega_X \simeq \mathcal{O}_X$ and $H^i(X, \mathcal{O}_X) = 0$ for $0 < i < n$), the bounded derived category $D^b(\mathrm{Coh}(X))$ is CY of dimension n. The CY pairing is Serre duality composed with the $\mathcal{O}_X$-trivialisation.

Example 2.7 *(Matrix factorizations).* For $W : \mathbb{A}^n \to \mathbb{A}^1$ a polynomial with isolated singularity, the category $\mathrm{MF}(W)$ of matrix factorizations is CY of dimension $n - 2$ (Dyckerhoff; Orlov). For the ADE singularities in two variables ($n = 2$), $\mathrm{MF}(W)$ is CY_0 (semisimple, with the structure of a Frobenius algebra). For threefold singularities ($n = 4$, e.g. $W = x_1^2 + \cdots + x_4^2$), $\mathrm{MF}(W)$ is CY_2. The exponent $n - 2$ (not $n - 1$) records the projectivisation of the $\mathbb{G}_m$-action on $W^{-1}(0)$; see AP-CY17.

2.3 Hochschild cohomology and the CY structure

Hochschild cohomology is the universal invariant of a dg category. For a CY_d category, Serre duality forces additional structure: an E_2-algebra structure on $\mathrm{HH}^\bullet(C)$ (Kontsevich–Soibelman, Tamarkin) and, in the presence of the CY pairing, a $(d - 2)$-shifted Poisson / BV structure.

Definition 2.8 *(Hochschild cohomology).* The Hochschild cohomology of C is

$$\mathrm{HH}^\bullet(C) \;=\; \mathrm{RHom}_{C^{\mathrm{op}} \otimes C}(C, C),$$

the derived endomorphisms of the diagonal bimodule. It is a graded-commutative algebra under Yoneda product and carries the Gerstenhaber bracket of degree -1.

Theorem 2.9 *(Deligne conjecture; Kontsevich–Soibelman, Tamarkin).* [PE] The pair (Yoneda product, Gerstenhaber bracket) on $\mathrm{HH}^\bullet(C)$ is the homology of an E_2-algebra structure, canonical up to contractible choice.

For a CY_d category, the cyclic pairing relates $\mathrm{HH}^\bullet(C)$ to $\mathrm{HH}_\bullet(C)[d]$. Transporting structure through this duality produces the higher Calabi–Yau brackets.

Proposition 2.10 *(CY bracket spectrum).* [PE] For a CY_d category C, the Hochschild invariants carry the following additional structure:

- $d = 1$: a BV operator $\Delta\colon \mathrm{HH}^\bullet(C) \to \mathrm{HH}^{\bullet-1}(C)$ whose deviation from being a derivation recovers the Gerstenhaber bracket;

- $d = 2$: a degree -2 Poisson bracket $\{-,-\}_2$ on $\mathrm{HH}^\bullet(C)$, part of a 2-shifted symplectic structure on the moduli of objects;
- $d \geq 3$: a $(d-2)$-shifted Poisson structure / P_{d-1}-algebra, refining to the Lie-cobar level in the cyclic setting.

THEOREM 2.11 *(HKR for smooth varieties).* [PE] For X a smooth projective variety over k of characteristic zero, the Hochschild–Kostant–Rosenberg isomorphism gives

$$\mathrm{HH}^p(D^b(\mathrm{Coh}(X))) \;\simeq\; \bigoplus_{q+r=p} H^q\big(X, \wedge^r T_X\big).$$

When X is CY_n, the trivialisation $\omega_X \simeq \mathcal{O}_X$ identifies $\wedge^r T_X \simeq \Omega_X^{n-r}$, so

$$\mathrm{HH}^p(D^b(\mathrm{Coh}(X))) \;\simeq\; \bigoplus_{q+r=p} H^q\big(X, \Omega_X^{n-r}\big).$$

The sign and compatibility with the Mukai pairing are recorded in Căldăraru [3].

Example 2.12 *(Hochschild cohomology of $K3$).* For a $K3$ surface X, the Hodge diamond is

$$h^{0,0} = 1,\;\; h^{2,0} = h^{0,2} = 1,\;\; h^{1,1} = 20,\;\; h^{2,2} = 1,$$

with all other entries zero. Applied to the HKR sum $\mathrm{HH}^p(D^b(\mathrm{Coh}(X))) \;=\; \bigoplus_{q+r=p} H^q(X, \Omega_X^{2-r})$, this gives

$$\mathrm{HH}^0 \simeq H^0(\mathcal{O}) \oplus H^2(T_X) \simeq k \oplus k, \quad \mathrm{HH}^1 \simeq H^1(T_X) \simeq k^{20}, \quad \mathrm{HH}^2 \simeq H^0(\wedge^2 T_X) \oplus H^2(\mathcal{O}) \simeq k \oplus k.$$

The total dimension is

$$\dim_k \mathrm{HH}^\bullet(D^b(\mathrm{Coh}(K3))) \;=\; 2 + 20 + 2 \;=\; 24,$$

matching the topological Euler characteristic $\chi(K3) \;=\; 24$. The Mukai pairing on $\mathrm{HH}^\bullet$ is the non-degenerate pairing $\langle -,-\rangle_{\mathrm{Mukai}}$ of Mukai [27]; it coincides with the CY pairing produced by Definition 2.3 at $d = 2$.

Remark 2.13 (κ_{cat} from χ). For X a CY_2, the categorical kappa of the Vol III spectrum is

$$\kappa_{\mathrm{cat}}(X) \;=\; \chi(\mathcal{O}_X),$$

the holomorphic Euler characteristic. For $K3$: $\chi(\mathcal{O}_{K3}) = 1 - 0 + 1 = 2$, giving $\kappa_{\mathrm{cat}}(K3) = 2$. Do not confuse this with the topological Euler characteristic 24 (which is $\dim \mathrm{HH}^\bullet$), with $\kappa_{\mathrm{ch}}(K3 \times E) = 3$ (chiral-algebra kappa via Φ), or with $\kappa_{\mathrm{BKM}}(K3 \times E) = 5$ (Borcherds weight). See AP113.

2.4 Cyclic A_∞ structure

The CY pairing in Definition 2.3 is cohomological. The categorical input to Φ requires a chain-level enhancement: a cyclic A_∞-structure for which every higher operation m_n is compatible with the pairing.

Definition 2.14 *(Cyclic A$_\infty$ category).* A *cyclic A$_\infty$-structure of degree* $-d$ on an A_∞-category C is a collection of pairings

$$\langle -, - \rangle_d \colon C(a_0, a_1) \otimes C(a_1, a_0) \longrightarrow k[-d]$$

together with the cyclic symmetry condition

$$\langle m_n(f_1, \dots, f_n), f_0 \rangle_d = (-1)^{\epsilon} \langle m_n(f_0, f_1, \dots, f_{n-1}), f_n \rangle_d$$

for all $n \geq 2$, where ϵ is the Koszul sign determined by the degrees of the f_i.

Definition 2.15 *(Negative cyclic CY class).* A *d-dimensional CY structure* on a smooth dg category C is a class

$$[\sigma] \in \mathrm{HC}^{-}_{d}(C)$$

in the negative cyclic homology, whose image under the canonical map $\mathrm{HC}^{-} \to \mathrm{HH}$ is a non-degenerate trace $\mathrm{Tr}\colon \mathrm{HH}_d(C) \to k$. The lift to HC^{-} is essential for the $\mathbb{S}^d$-framing used in Chapter 8; see AP-CY2.

Theorem 2.16 *(Cyclic A$_\infty$ enhancement).* [PE] Every smooth proper CY_d category admits a cyclic A_∞-enhancement, unique up to A_∞-quasi-isomorphism preserving the pairing. The proof is due to Kontsevich–Soibelman in the formal case, Costello [6] for the TCFT enhancement, and Seidel [31] for the Fukaya case.

At $d = 2$ the enhancement is explicit enough to compute directly; it drives the $K3$ computations feeding the Borcherds denominator (Vol III Chapter **??**). At $d = 3$ the enhancement was constructed by Sheridan [32] for the quintic threefold, as part of the HMS proof; the general $d = 3$ construction remains programme-level (conditional on $\mathbb{S}^3$-framing, see AP-CY6).

2.5 Interface with the CY-to-chiral functor Φ

The Vol III functor

$$\Phi\colon \mathrm{CY}_d\text{-}\mathbf{Cat}^{\mathrm{cyc}} \longrightarrow E_2\text{-}\mathrm{ChirAlg}$$

takes as input the data of this chapter: a smooth proper cyclic A_∞ category of dimension d, with its negative cyclic CY class $[\sigma] \in \mathrm{HC}^{-}_{d}(C)$. The construction (Chapter 8) proceeds by (i) extracting the Gerstenhaber algebra $\mathrm{HH}^\bullet(C)$, (ii) promoting it to a Lie conformal algebra using the CY pairing, (iii) taking the factorization envelope on a curve, (iv) and enhancing to E_2 using the $\mathbb{S}^2$-framing at $d = 2$.

Theorem 2.17 *(Categorical kappa equals Euler characteristic — CY-D$_2$).* [PH] For C a smooth proper CY_2 category such that $\Phi(C)$ exists, the modular characteristic of the chiral algebra satisfies

$$\kappa_{\mathrm{ch}}\big(\Phi(C)\big) = \chi^{\mathrm{CY}}(C),$$

the categorical Euler characteristic defined via the Mukai pairing. When $C = D^b(\mathrm{Coh}(X))$ for X a $K3$ surface, the right-hand side is $\kappa_{\mathrm{cat}}(X) = \chi(\mathcal{O}_{K3}) = 2$, which implies a contribution to $\kappa_{\mathrm{ch}}(\Phi(D^b(\mathrm{Coh}(K3))))$; the full $K3 \times E$ chiral kappa is 3, combining the $K3$ and E factors.

Remark 2.18 (The $d = 3$ case). For $d = 3$, the functor Φ is not yet unconditional: it depends on the chain-level $\mathbb{S}^3$-framing programme (CY-A$_3$, AP-CY6). Any downstream statement inheriting the $d = 3$ Φ must carry [CD] and name CY-A$_3$ as its dependency (AP-CY11). The $\mathbb{C}^3$ case is verified end-to-end in Chapter 8, Theorem 8.2, providing a Rosetta Stone for the general $d = 3$ functor.

Remark 2.19 (What Φ does not see). Not every invariant of C survives the passage through Φ. The functor retains the cyclic A_∞-structure and the Gerstenhaber bracket; it loses the fine structure of the triangulated / A_∞ category (individual objects, t-structures, stability conditions) that is visible only via the fuller categorical CoHA construction of Chapter **??**. The kappa-spectrum (Vol III CLAUDE.md, AP113) encodes which invariants survive: κ_{cat} is always present, κ_{ch} exists only when Φ is defined, and κ_{BKM} records the Borcherds-side lift for $K3 \times E$.

Remark 2.20 (Cross-volume conventions). Vol III uses lambda-bracket conventions for the Lie conformal algebras produced by Φ: an OPE of the form $T(z)T(w) \sim (c/2)(z-w)^{-4} + \cdots$ is rewritten as $\{T_\lambda T\} = (c/12)\lambda^3 + \cdots$, absorbing the combinatorial factor $3! = 6$ from the divided-power $\lambda^{(3)} = \lambda^3/3!$. Vol I uses OPE modes directly; care is required when transporting formulas between volumes (see AP22, AP46 and the concordance). The Hochschild / cyclic invariants of this chapter are convention-independent: they depend only on the chain-level A_∞-structure.

Chapter 3

Cyclic A_∞-Structures

A Calabi–Yau category enters this volume through a single structural datum: a cyclic A_∞-algebra of dimension d. Everything that follows, the functor Φ to chiral algebras, the modular characteristic κ_{cat}, the four subscripted kappas of the CY kappa-spectrum, depends on this input. This chapter fixes the definitions, records the standard examples (elliptic curve, K3, quintic), and states the bridge to II precisely. The content is classical (Stasheff, Kontsevich, Keller, Costello); the Vol III role is the specific identification of d with the CY dimension appearing in Theorem CY-A.

3.1 A_∞-algebras

Definition 3.1 *(A_∞-algebra).* An *A_∞-algebra* over a field k is a $\mathbb{Z}$-graded k-vector space A equipped with operations

$$\mu_n\colon A^{\otimes n} \longrightarrow A, \qquad \deg(\mu_n) = 2-n, \quad n \geq 1,$$

satisfying, for every $n \geq 1$, the A_∞-relation

$$\sum_{\substack{p+q=n+1\\ 1\leq i\leq p}} (-1)^{\dagger(i,p,q)}\, \mu_p\big(a_1,\ldots,a_{i-1},\, \mu_q(a_i,\ldots,a_{i+q-1}),\, a_{i+q},\ldots,a_n\big) = 0,$$

where $\dagger(i,p,q) = (i-1)(q-1) + q\sum_{j=1}^{i-1}|a_j|$ follows the Koszul sign rule (see Appendix A).

Specializing to small n yields the familiar relations:

- $n=1$: $\mu_1^2 = 0$, so (A,μ_1) is a cochain complex.
- $n=2$: $\mu_1(\mu_2(a,b)) = \mu_2(\mu_1 a, b) + (-1)^{|a|}\mu_2(a,\mu_1 b)$, so μ_1 is a derivation of μ_2.
- $n=3$: μ_2 is associative up to the explicit homotopy μ_3, giving the Stasheff associahedron relation.

An A_∞-algebra with $\mu_n = 0$ for $n \geq 3$ is a differential graded associative algebra. The μ_n for $n \geq 3$ record the higher homotopies that obstruct strict associativity after homological projection (Kadeishvili's theorem: any dg algebra transfers to a minimal A_∞-algebra on its cohomology).

The A_∞-formalism originates with Stasheff's associahedra [33]. Its modern use, as the natural habitat for derived categories, mirror symmetry, and deformation quantization, is due to Kontsevich and Soibelman [23] and Keller [18]. For a CY category C, the derived category $D^b(C)$ lifts canonically to an A_∞-enhancement; this enhancement is unique up to quasi-isomorphism (Lunts–Orlov, Canonaco–Stellari).

The non-minimal cochain-level model is the preferred input for the Vol III functor because cyclicity is visible chain-level, not only on cohomology.

Remark 3.2 (A_∞-category vs A_∞-algebra). An A_∞-category is the many-object generalization: objects $\mathrm{Ob}(C)$ and Hom-complexes $\mathrm{Hom}_C(X, Y)$ with composition maps $\mu_n \colon \mathrm{Hom}(X_0, X_1) \otimes \cdots \otimes \mathrm{Hom}(X_{n-1}, X_n) \to \mathrm{Hom}(X_0, X_n)[2-n]$. An A_∞-algebra is the one-object case. In this chapter we state definitions for algebras; the category version is verbatim by tagging arguments with source and target objects.

3.2 Cyclic structures and CY dimension

Definition 3.3 *(Cyclic A_∞-algebra).* A *cyclic A_∞-algebra of dimension d* is a pair $\big((A, \{\mu_n\}_{n\geq 1}), \langle -,-\rangle\big)$, where $(A, \{\mu_n\})$ is an A_∞-algebra and

$$\langle -,-\rangle \colon A \otimes A \longrightarrow k[-d]$$

is a non-degenerate graded symmetric pairing of degree $-d$ (equivalently, $\langle a, b\rangle$ is nonzero only when $|a| + |b| = d$), satisfying the *cyclic invariance* identity

$$\langle \mu_n(a_1, a_2, \ldots, a_n), a_{n+1}\rangle \;=\; (-1)^{\epsilon_n} \langle a_1, \mu_n(a_2, \ldots, a_{n+1})\rangle \qquad \text{for all } n \geq 1,$$

with sign $\epsilon_n = n + |a_1|(|a_2| + \cdots + |a_{n+1}|)$ determined by the Koszul rule.

The integer d is the *CY dimension* of the cyclic A_∞-algebra. In the language of Costello [7, 8], a cyclic A_∞-algebra of dimension d is precisely an algebra over the operad of framed chains on the moduli of disks with marked points, with a d-shifted symplectic structure. This operadic viewpoint is what produces, after homotopy transfer, the $\mathbb{S}^d$-framing of Hochschild homology used in Chapter 8.

Proposition 3.4 *(Cyclic invariance on the Hochschild complex).* [PE] Let $(A, \langle -,-\rangle)$ be a cyclic A_∞-algebra of dimension d. The pairing induces a non-degenerate pairing on the Hochschild complex

$$\langle -,-\rangle_{\mathrm{HH}} \colon \mathrm{HH}_\bullet(A) \otimes \mathrm{HH}_{d-\bullet}(A) \longrightarrow k$$

implementing Poincaré duality of dimension d. Equivalently, the cyclic structure lifts Hochschild homology to negative cyclic homology with a class in $\mathrm{HC}^-_d(A)$ that is closed under the Connes differential.

Attribution. Costello [8] Thm. A; Kontsevich–Soibelman [23] Sec. 10. The AP-CY2 distinction is essential: the CY trace lives in HC^-_d, not in $\mathrm{HH}_d \to k$, because the S^1-invariance of the cyclic pairing is needed to produce the $\mathbb{S}^d$-framing. $\square$

Remark 3.5 (Geometric origin of d). For X a smooth projective CY n-fold, the derived category $D^b(\mathrm{Coh}(X))$ carries a cyclic A_∞-enhancement of dimension $d = n$. The pairing is the Serre-duality pairing

$$\langle -,-\rangle_{\mathrm{Serre}} \colon \mathrm{Ext}^i(\mathcal{F},) \otimes \mathrm{Ext}^{n-i}(,\mathcal{F}) \longrightarrow \mathrm{Ext}^n(\mathcal{F},\mathcal{F}) \xrightarrow{\mathrm{tr}} H^n(X, \omega_X) \cong k,$$

using the trivialization $\omega_X \cong \mathcal{O}_X$ that defines the CY structure. AP-CY1: d is the complex dimension, not the real dimension $2n$. The Serre functor $\mathbb{S} = (-) \otimes \omega_X[n]$ becomes $[n]$-shift under the CY trivialization, which is what produces the d-shifted pairing.

Remark 3.6 (Cyclic bar complex). The cyclic pairing enters the bar complex $B(A) = T^c(s^{-1}\bar{A})$ through the cyclic quotient $\mathrm{CC}_\bullet(A) = B(A)/(1-t)$ where t is the signed cyclic rotation. The factor s^{-1} desuspends (AP22): $|s^{-1}v| = |v| - 1$. The augmentation ideal $\bar{A} = \ker(\varepsilon)$ is used rather than A itself (AP132). The cyclic bar complex is the primary invariant of $(A, \mu_n, \langle -,-\rangle)$ and is what II promotes to a factorization coalgebra on curves.

Remark 3.7 (Cyclic symmetry at $n = 2$). Specializing the cyclic invariance identity to $n = 2$ gives

$$\langle \mu_2(a_1, a_2), a_3 \rangle \;=\; (-1)^{\epsilon_2} \langle a_1, \mu_2(a_2, a_3) \rangle,$$

which for a strictly associative cyclic algebra ($\mu_n = 0$ for $n \geq 3$) reduces to the standard invariance condition for a Frobenius algebra of degree d. The higher-n cases are the A_∞-correction: $\mu_3, \mu_4, \ldots$ must individually respect the pairing. In the $n = 3$ case, the identity

$$\langle \mu_3(a_1, a_2, a_3), a_4 \rangle \;=\; (-1)^{\epsilon_3} \langle a_1, \mu_3(a_2, a_3, a_4) \rangle$$

is the first genuinely homotopical constraint and is what fails in a naive restriction of the Fukaya category to a sub-Lagrangian: cyclic invariance distinguishes the true CY input from merely A_∞-enhanceable ones.

3.3 Examples

The four examples below cover the four CY dimensions $d \in \{1, 1, 2, 3\}$ relevant to Vol III. Each fixes one numerical invariant visible to the functor Φ.

Example 3.8 *($d = 1$ from a CY-1 Lie algebra).* Let $\mathfrak{g}$ be a finite-dimensional simple Lie algebra with non-degenerate invariant form $\langle -, - \rangle_{\mathfrak{g}}$. The Chevalley–Eilenberg cochain complex $C^\bullet(\mathfrak{g})$ is a dg commutative algebra of dimension $\dim \mathfrak{g}$; applying Koszul duality twice produces a cyclic A_∞-algebra of CY dimension 1 whose underlying vector space is $\mathfrak{g}[1]$ with pairing $\langle -, - \rangle_{\mathfrak{g}}$ (shifted to degree -1). For $\mathfrak{g} = \mathfrak{sl}_2$, the resulting cyclic A_∞-algebra has 3 generators and μ_2 given by the bracket; all higher μ_n vanish. This is the simplest nontrivial cyclic A_∞-algebra relevant to Vol III, and it recovers $\widehat{\mathfrak{sl}}_2$ as its chiral image under Φ.

Example 3.9 *(Explicit μ_2 for $\mathfrak{sl}_2$).* Continuing Example 3.8, the underlying graded space of the cyclic A_∞-algebra for $\mathfrak{sl}_2$ is the suspension $\mathfrak{sl}_2[1]$, with basis $\{e[1], f[1], h[1]\}$ in degree -1. The binary operation μ_2 is the desuspended Lie bracket:

$$\mu_2(e[1], f[1]) = h[1], \qquad \mu_2(h[1], e[1]) = 2\, e[1], \qquad \mu_2(h[1], f[1]) = -2\, f[1],$$

with the other values determined by graded antisymmetry. The cyclic pairing has nonzero values

$$\langle e[1], f[1] \rangle = 1, \qquad \langle h[1], h[1] \rangle = 2,$$

which is the Killing form shifted to degree -1, producing a CY structure of dimension $d = 1$. The A_∞-relations reduce to the Jacobi identity of $\mathfrak{sl}_2$, and all μ_n for $n \geq 3$ vanish. This is the simplest nontrivial cyclic A_∞-algebra in the volume; under Φ it produces the affine Kac–Moody algebra $\widehat{\mathfrak{sl}}_2$ at the critical level, recovering a classical landscape entry from the CY engine.

Example 3.10 *($d = 1$ from an elliptic curve).* Let E be an elliptic curve over $\mathbb{C}$. The derived category $D^b(\mathrm{Coh}(E))$ has a minimal cyclic A_∞-enhancement of dimension $d = 1$, computed explicitly by Polishchuk [29]. The generators are $\mathcal{O}_E$ and a single degree-1 class in $\mathrm{Ext}^1(\mathcal{O}_E, \mathcal{O}_E) = H^1(E, \mathcal{O}_E) \cong \mathbb{C}$. The Serre pairing $\mathrm{Ext}^1(\mathcal{O}_E, \mathcal{O}_E) \otimes \mathrm{Hom}(\mathcal{O}_E, \mathcal{O}_E) \to \mathbb{C}$ realizes the $d = 1$ cyclic structure. For higher-degree line bundles, Polishchuk's theta-function computation gives explicit formulas for μ_n, $n \geq 3$, in terms of modular forms on $\Gamma_1(N)$. This example sits at the boundary between $d = 1$ (where the $\mathbb{S}^1$-framing is classical) and the modular-form territory that Vol I controls.

Example 3.11 *($d = 2$ from K3).* Let X be a smooth projective K3 surface. The derived category $D^b(\mathrm{Coh}(X))$ is a cyclic A_∞-category of dimension $d = 2$ (Caldararu [3]). The Hochschild cohomology is

$$\mathrm{HH}^\bullet(D^b(\mathrm{Coh}(X))) \;\cong\; H^\bullet(X, \wedge^\bullet T_X) \;\cong\; H^\bullet(X, \mathbb{C}),$$

using the CY trivialization $\wedge^2 T_X \cong \mathcal{O}_X$. As a graded vector space,

$$\dim \mathrm{HH}^\bullet(K3) = 1 + 20 + 1 + 20 + 1 \;=\; (\text{degree-0: 1, degree-1: 0, degree-2: 22, etc.}).$$

The total dimension of the Hochschild complex of K3 is 24 (Hodge diamond sum), recovering the Euler characteristic $\chi(X) = 24$ familiar from lattice K3 geometry. The holomorphic Euler characteristic, which is the relevant invariant for the CY kappa-spectrum (AP113), is

$$\chi(\mathcal{O}_X) \;=\; 2.$$

This is the value $\kappa_{\mathrm{cat}}(\mathrm{K3}) = 2$ entering the Vol III kappa-table.

Example 3.12 *($d = 3$ from the quintic).* Let $Q \subset \P^4$ be a smooth quintic threefold. $D^b(\mathrm{Coh}(Q))$ is a cyclic A_∞-category of dimension $d = 3$. By homological mirror symmetry (Kontsevich's conjecture, established for the quintic at genus 0 by Sheridan [32]), there is an A_∞-equivalence

$$D^b(\mathrm{Coh}(Q)) \;\simeq\; D^\pi \mathrm{Fuk}(Q^\vee),$$

where $Q^\vee$ is the mirror quintic and Fuk is the Fukaya category (wrapped, compact, depending on geometric input). Both sides are cyclic A_∞ of dimension $d = 3$, and HMS is an equivalence of cyclic A_∞-categories: the Serre pairing on the B-side matches the Poincaré pairing on Lagrangian intersections on the A-side. The Hochschild cohomology

$$\mathrm{HH}^\bullet(Q) \cong H^\bullet(Q, \wedge^\bullet T_Q)$$

has $\dim \mathrm{HH}^1 = 101$ (the Kodaira–Spencer space of the quintic) and $\dim \mathrm{HH}^2 = 1$. The holomorphic Euler characteristic is $\chi(\mathcal{O}_Q) = 0$, so the naive κ_{cat} vanishes; the nontrivial chiral data enters through HH^1, not through the top pairing. This is the CY3 regime where the chiral algebra A_Q is not yet constructed (AP-CY6): any result that passes through A_Q is conditional on CY-A3.

3.4 The bridge to the Vol III functor

The cyclic A_∞-structure is the complete input to the CY-to-chiral functor. The signature is

$$\Phi\colon \mathrm{CY}_d\text{-}\mathbf{Cat} \longrightarrow E_2\text{-ChirAlg}, \qquad (C, \mu_n, \langle -,- \rangle) \longmapsto A_C,$$

and the data of $(C, \mu_n, \langle -,- \rangle)$ factors through Φ by the four-step passage in Chapter 8: Hochschild cohomology with Gerstenhaber bracket, Lie conformal algebra via the cyclic pairing, factorization envelope, and E_2-enhancement via the $\mathbb{S}^d$-framing. The cyclic pairing $\langle -,- \rangle$ is what becomes the invariant form on A_C; without it, the factorization envelope is not a chiral algebra with a trace.

THEOREM 3.13 *(Cyclic A_∞ input determines κ_{cat} at $d = 2$).* [PH] Let C be a smooth proper cyclic A_∞-category of dimension $d = 2$, and let $A_C = \Phi(C)$ be its image under the $d = 2$ CY-to-chiral functor (Theorem 8.1). Then the categorical modular characteristic satisfies

$$\kappa_{\mathrm{cat}}(C) \;=\; \chi^{\mathrm{CY}}(C) \;=\; \sum_i (-1)^i \dim \mathrm{HH}^i(C)\Big|_{\text{top pairing}},$$

and this equals the chiral kappa $\kappa_{\mathrm{ch}}(A_C)$ of the image chiral algebra up to the $\mathbb{S}^2$-framing correction recorded in the CY kappa-spectrum. For $C = D^b(\mathrm{Coh}(\mathrm{K3}))$, both invariants evaluate to $\kappa_{\mathrm{cat}}(\mathrm{K3}) = 2$.

Proof sketch. The cyclic pairing of dimension $d = 2$ produces a degree-(-2) symmetric form on the Hochschild complex; its top-degree component is a linear map $\mathrm{HH}^2(C) \to k$ whose dimension (as a quotient of $\mathrm{HH}^\bullet$) is $\chi(\mathcal{O}_X)$ in the geometric case. The CY-to-chiral functor sends this trace to the invariant form on A_C, and the resulting κ_{cat} coincides (up to $\mathbb{S}^2$-framing) with the coefficient of the E_2-algebra central extension. For K3 the computation reduces to $\chi(\mathcal{O}_{\mathrm{K3}}) = 2$; see Chapter 8 Section 8.2 and the compute module `cy_to_chiral_functor.py` for the full verification. □

CONJECTURE 3.14 *(Cyclic A_∞ input determines κ_{cat} at $d = 3$).* [CJ] Let C be a smooth proper cyclic A_∞-category of dimension $d = 3$. Assume Conjecture CY-A$_3$ (the $d = 3$ CY-to-chiral functor exists with chain-level $\mathbb{S}^3$-framing). Then

$$\kappa_{\mathrm{cat}}(C) = \chi^{\mathrm{CY}}(C),$$

and for $C = D^b(\mathrm{Coh}(Q))$ with Q a smooth quintic threefold, this evaluates to a specific integer determined by the $\mathbb{S}^3$-framing obstruction. The conjecture is conditional on CY-A$_3$ (AP-CY6, AP-CY11); any downstream result passing through A_Q inherits this conditionality.

Remark 3.15 (Scope: what cyclic A_∞ does not determine). The cyclic A_∞-structure determines κ_{cat} and the chiral kappa κ_{ch} via the functor Φ, but it does *not* determine the BKM kappa κ_{BKM} or the fiber kappa κ_{fiber} of the kappa-spectrum (AP113). Those kappas arise from additional data (lattice embedding for κ_{fiber}, Borcherds product structure for κ_{BKM}) that is not visible to the cyclic A_∞-category alone. The four-way kappa-spectrum for K3 $\times E$ is $\{\kappa_{\mathrm{cat}}, \kappa_{\mathrm{ch}}, \kappa_{\mathrm{BKM}}, \kappa_{\mathrm{fiber}}\} = \{2, 3, 5, 24\}$; only the first two are extracted from the cyclic A_∞-input.

Remark 3.16 (Comparison with prior work). Four source threads feed the construction used here. Stasheff [33] introduced the associahedra and the higher homotopies μ_n. Kontsevich [22] identified cyclic A_∞-algebras with algebras over the operad of ribbon graphs, providing the link to moduli of curves with boundary. Costello [7, 8] proved that cyclic A_∞-categories are equivalent to open topological conformal field theories and supplied the first rigorous construction of the associated chain-level trace. Kontsevich–Soibelman [23] axiomatized the CY structure in terms of the negative cyclic class (AP-CY2) and gave the formalism used in Part I. Keller [18] surveys the homological-algebra side. For explicit computations on projective varieties, Polishchuk [29] computed the cyclic A_∞-structure on elliptic curves and on their products, and Caldararu [3] set up the Hochschild calculus for smooth proper CY categories. The Vol III role is the specific mapping of this input through the functor Φ, producing chiral algebras whose modular characteristic can be computed and compared across the four kappas of the spectrum.

Remark 3.17 (Costello's operadic description). Costello [7] proves that cyclic A_∞-algebras of dimension d are equivalent to algebras over the open topological conformal field theory operad with d-shifted framing. In this formulation, the $\mathbb{S}^d$-framing used in Chapter 8 is visible as the action of the framing circle on the moduli of disks with boundary marked points. For $d = 2$ the framing is abelian ($\pi_1(SO(2)) = \mathbb{Z}$) and the envelope step produces an honest E_2-algebra. For $d = 3$ the framing is nonabelian ($\pi_1(SO(3)) = \mathbb{Z}/2$) and the envelope step is the source of the $\mathbb{S}^3$-framing obstruction mentioned in AP-CY6.

Chapter 4

Hochschild Calculus for CY Categories

A Calabi–Yau d-category has a non-degenerate Serre pairing. What does this pairing force on the Hochschild invariants? For a general dg category, Hochschild homology $\mathrm{HH}_\bullet(C)$ carries only a Connes B-operator and Hochschild cohomology $\mathrm{HH}^\bullet(C)$ carries only a Gerstenhaber bracket; the two are linked by a cap product, but no pairing relates them. The CY structure removes this deficiency.

The Serre pairing provides a quasi-isomorphism $\mathrm{HH}^\bullet(C) \simeq \mathrm{HH}_{d-\bullet}(C)$ of degree $-d$, so that Hochschild cohomology inherits the B-operator and Hochschild homology inherits the Gerstenhaber bracket. The combination is a $(2-d)$-shifted Poisson (equivalently, BV_{2-d}) structure on $\mathrm{HH}^\bullet(C)$: bracket of degree $1-d$, operator of degree -1, compatibility equation $[B, -] = \{-, -\}$. At $d = 2$ this is a 0-shifted Poisson structure (ordinary Poisson); at $d = 3$, a (-1)-shifted Poisson structure (the CY3 case of Pantev–Toën–Vaquié–Vezzosi); at $d = 1$, a 1-shifted Poisson structure of the kind that drives factorization homology.

This chapter develops the Hochschild calculus underlying the CY-to-chiral functor of Chapter 8. The shifted Poisson structure on $\mathrm{HH}^\bullet(C)$ is the categorical precursor of the convolution Lie bracket on the modular deformation complex of Volume I: the Gerstenhaber bracket becomes the bar-cobar convolution bracket, the Connes B-operator becomes the modular differential, and the Serre pairing becomes the cyclic duality that controls the $\mathbb{S}^d$-framing. The cyclic bar complex of Chapter 3 is the single-object specialization of the Hochschild complex constructed here.

4.1 Hochschild homology and cohomology

Definition 4.1 *(Hochschild complex).* Let C be a dg category over k. The *Hochschild chain complex* of C is

$$\mathrm{CC}_n(C) = \bigoplus_{X_0,\dots,X_n \in \mathrm{Ob}(C)} \mathrm{Hom}_C(X_n, X_0) \otimes \mathrm{Hom}_C(X_{n-1}, X_n) \otimes \cdots \otimes \mathrm{Hom}_C(X_0, X_1)$$

with the Hochschild differential

$$\begin{aligned} b(f_0 \otimes \cdots \otimes f_n) = &\sum_{i=0}^{n-1} (-1)^{\epsilon_i} f_0 \otimes \cdots \otimes f_i f_{i+1} \otimes \cdots \otimes f_n \\ &+ (-1)^{\epsilon_n} f_n f_0 \otimes f_1 \otimes \cdots \otimes f_{n-1}, \end{aligned}$$

where $\epsilon_i = |f_0| + \cdots + |f_i|$ (Koszul signs). The *Hochschild homology* is $\mathrm{HH}_\bullet(C) = H_\bullet(\mathrm{CC}_\bullet(C), b)$.

Definition 4.2 *(Hochschild cohomology).* The *Hochschild cochain complex* of a dg category C is

$$\mathrm{CC}^n(C) = \prod_{X_0,\ldots,X_n \in \mathrm{Ob}(C)} \mathrm{Hom}_k\big(\mathrm{Hom}_C(X_{n-1},X_n)\otimes\cdots\otimes\mathrm{Hom}_C(X_0,X_1),\ \mathrm{Hom}_C(X_0,X_n)\big)$$

with the Hochschild coboundary δ dual to b. The *Hochschild cohomology* is $\mathrm{HH}^\bullet(C) = H^\bullet(\mathrm{CC}^\bullet(C),\delta)$.

Remark 4.3 (Morita invariance). Both $\mathrm{HH}_\bullet$ and $\mathrm{HH}^\bullet$ are Morita invariant: if C and $\mathcal{D}$ are derived Morita equivalent (i.e., $\mathrm{Perf}(C)\simeq\mathrm{Perf}(\mathcal{D})$ as dg categories), then $\mathrm{HH}_\bullet(C)\simeq\mathrm{HH}_\bullet(\mathcal{D})$ and $\mathrm{HH}^\bullet(C)\simeq\mathrm{HH}^\bullet(\mathcal{D})$. This is essential for the CY programme, where different geometric models (Fukaya, derived coherent sheaves, matrix factorizations) of the same CY category must produce the same Hochschild invariants.

4.2 The Gerstenhaber bracket

THEOREM 4.4 *(Gerstenhaber structure on* $\mathrm{HH}^\bullet$*).* [PE]The Hochschild cohomology $\mathrm{HH}^\bullet(C)$ carries two compatible structures:

(i) A graded commutative *cup product* $\smile$: $\mathrm{HH}^p(C)\otimes\mathrm{HH}^q(C)\to\mathrm{HH}^{p+q}(C)$, the composition of multilinear maps;

(ii) A degree (-1) Lie bracket $[\cdot,\cdot]$: $\mathrm{HH}^p(C)\otimes\mathrm{HH}^q(C)\to\mathrm{HH}^{p+q-1}(C)$ (the *Gerstenhaber bracket*), defined at the cochain level by

$$[f,g] = f\circ g - (-1)^{(|f|-1)(|g|-1)} g\circ f$$

where $\circ$ denotes the operadic pre-Lie composition:

$$(f\circ g)(a_1,\ldots,a_{p+q-1}) = \sum_{i=1}^{p}(-1)^{\dagger} f(a_1,\ldots,a_{i-1},g(a_i,\ldots,a_{i+q-1}),a_{i+q},\ldots,a_{p+q-1}).$$

Together, the cup product and Gerstenhaber bracket make $\mathrm{HH}^\bullet(C)$ a *Gerstenhaber algebra*: the bracket is a derivation of the cup product in each variable.

Remark 4.5 (Gerstenhaber = E_2 cohomology). By Kontsevich's formality theorem and the Deligne conjecture (proved by Kontsevich–Soibelman, McClure–Smith, and others), $\mathrm{CC}^\bullet(C)$ carries a natural E_2-algebra structure whose cohomology-level shadow is the Gerstenhaber algebra $(\mathrm{HH}^\bullet(C),\smile,[\cdot,\cdot])$. The E_2-structure is the chain-level origin of the braided monoidal category structure on $\mathrm{Rep}^{E_2}(Z^{\mathrm{der}}_{\mathrm{ch}}(A))$ (Chapter 13).

PROPOSITION 4.6 *(Gerstenhaber bracket controls deformations).* [PE]The Gerstenhaber bracket controls the deformation theory of C:

(i) $\mathrm{HH}^2(C)$ classifies first-order deformations of C as a dg category;

(ii) The obstruction to extending a first-order deformation $\mu_1\in\mathrm{HH}^2(C)$ to second order lies in $\mathrm{HH}^3(C)$ and is given by the Gerstenhaber square $\frac{1}{2}[\mu_1,\mu_1]$;

(iii) A formal deformation is an element $\mu=\sum_{n\geq1}\hbar^n\mu_n\in\mathrm{HH}^2(C)[[\hbar]]$ satisfying the Maurer–Cartan equation $\delta\mu+\frac{1}{2}[\mu,\mu]=0$.

The MC equation for deformations of C maps, under Φ, to the MC equation $D\Theta_A+\frac{1}{2}[\Theta_A,\Theta_A]=0$ of the modular convolution algebra $\mathfrak{g}^{\mathrm{mod}}_A$ (Volume I, thm:mc2-bar-intrinsic).

4.3 The Connes B-operator and cyclic homology

Definition 4.7 *(Connes B-operator).* The *Connes operator* $B\colon \mathrm{CC}_n(C) \to \mathrm{CC}_{n+1}(C)$ is defined by

$$B(f_0 \otimes \cdots \otimes f_n) = \sum_{i=0}^{n} (-1)^{ni}\, 1 \otimes f_i \otimes f_{i+1} \otimes \cdots \otimes f_{i-1}$$

(indices mod $n+1$), composed with the insertion of the identity. It satisfies $B^2 = 0$, $bB + Bb = 0$.

Definition 4.8 *(Cyclic, negative cyclic, and periodic cyclic homology).* From the mixed complex $(\mathrm{CC}_\bullet(C), b, B)$ with $|b| = -1$, $|B| = +1$, $b^2 = B^2 = bB + Bb = 0$:

(i) *Cyclic homology*: $\mathrm{HC}_\bullet(C) = H_\bullet(\mathrm{CC}_\bullet(C) \otimes_{k[B]} k)$;

(ii) *Negative cyclic homology*: $\mathrm{HC}^-_\bullet(C) = H_\bullet(\mathrm{CC}_\bullet(C)[[u]], b + uB)$, where $|u| = 2$;

(iii) *Periodic cyclic homology*: $\mathrm{HP}_\bullet(C) = H_\bullet(\mathrm{CC}_\bullet(C)((u)), b + uB)$.

The SBI sequence $\cdots \to \mathrm{HH}_n \xrightarrow{B} \mathrm{HC}_{n+1} \xrightarrow{S} \mathrm{HC}_{n-1} \xrightarrow{I} \mathrm{HH}_{n-1} \to \cdots$ relates these.

Example 4.9 *(Smooth variety).* For $C = \mathrm{Perf}(X)$ with X a smooth algebraic variety of dimension n, the Hochschild–Kostant–Rosenberg theorem gives

$$\mathrm{HH}_k(\mathrm{Perf}(X)) \simeq \bigoplus_{p-q=k} H^q(X, \Omega^p_X), \qquad \mathrm{HH}^k(\mathrm{Perf}(X)) \simeq \bigoplus_{p+q=k} H^q(X, \textstyle\bigwedge^p T_X).$$

Under HKR, the Connes B-operator is the de Rham differential $d_{\mathrm{dR}}\colon \Omega^p \to \Omega^{p+1}$, and the Gerstenhaber bracket is the Schouten–Nijenhuis bracket on polyvector fields.

4.4 The Hochschild-to-cyclic spectral sequence

Proposition 4.10 *(Hodge-to-de Rham degeneration for dg categories).* [PE] For a smooth proper dg category C over a field k of characteristic zero, the Hochschild-to-cyclic spectral sequence

$$E_1^{p,q} = \mathrm{HH}_{q-p}(C) \implies \mathrm{HC}_{q-p}(C)$$

degenerates at E_2.

Remark 4.11 (Kaledin's theorem). The degeneration was proved by Kaledin (2008) using Deligne–Illusie lifting methods in the dg setting. Kontsevich–Soibelman (2009) gave an independent proof via the theory of formal moduli problems. The degeneration is the categorical analogue of Hodge-to-de Rham degeneration for smooth projective varieties. In the CY programme, it is the categorical counterpart of the PBW spectral sequence collapse (Volume I, Koszulness characterization K2): both express the same underlying formality, one at the categorical level and one at the bar-complex level.

4.5 CY Hochschild duality

THEOREM 4.12 *(CY Hochschild duality).* [PE]For a smooth CY category C of dimension d, there is a canonical quasi-isomorphism

$$\mathrm{CC}^\bullet(C) \ \simeq \ \mathrm{CC}_{\bullet+d}(C)$$

identifying the Hochschild cochain complex with the d-shifted Hochschild chain complex. On cohomology:

$$\mathrm{HH}^n(C) \ \simeq \ \mathrm{HH}_{n+d}(C).$$

Under this identification:

(i) The Gerstenhaber bracket on $\mathrm{HH}^\bullet$ corresponds to the *BV operator* $\Delta_{\mathrm{BV}} \colon \mathrm{HH}_{n+d} \to \mathrm{HH}_{n+d+1}$ (the divergence operator associated to the CY volume form);

(ii) The cup product on $\mathrm{HH}^\bullet$ corresponds to the intersection product on $\mathrm{HH}_{\bullet+d}$;

(iii) The Connes B-operator on $\mathrm{HH}_\bullet$ corresponds to the contraction with the Euler vector field under HKR.

Proof sketch. The CY condition furnishes an equivalence $C \simeq C^\vee[d]$ between the category and its d-shifted dual. Smoothness (the diagonal bimodule C_Δ is compact as a $C \otimes C^{\mathrm{op}}$-module) gives

$$\mathrm{CC}^\bullet(C) = \mathrm{RHom}_{C\otimes C^{\mathrm{op}}}(C_\Delta, C_\Delta) \simeq C_\Delta^\vee \otimes_{C\otimes C^{\mathrm{op}}} C_\Delta \simeq \mathrm{CC}_\bullet(C^\vee) \simeq \mathrm{CC}_{\bullet+d}(C)$$

using $C^\vee \simeq C[-d]$ from the CY structure. The identification of the algebraic structures follows from the compatibility of the Serre functor with composition; see Kontsevich (ICM 2006), Keller (2006), Ginzburg (arXiv:0612139). □

4.6 The $(2-d)$-shifted Poisson structure

The CY Hochschild duality has a structural consequence: $\mathrm{HH}^\bullet(C)$ carries a $(2-d)$-shifted Poisson structure.

THEOREM 4.13 *($(2-d)$-shifted Poisson structure).* [PE]For a smooth CY_d category C, the Hochschild cohomology $\mathrm{HH}^\bullet(C)$ carries a $(2-d)$-shifted Poisson structure: a Lie bracket of cohomological degree $(2-d)$

$$\{\cdot,\cdot\}_{2-d} \colon \mathrm{HH}^p(C) \otimes \mathrm{HH}^q(C) \to \mathrm{HH}^{p+q+2-d}(C)$$

satisfying the Jacobi identity and the Leibniz rule with respect to the cup product. This bracket is the CY-shifted Gerstenhaber bracket:

$$\{f, g\}_{2-d} = [f, g]_{\mathrm{Gerst}} \circ \sigma_d$$

where σ_d is the CY structure class in $\mathrm{HC}_d^-(C) \simeq \mathrm{HH}_0(C)$.

Remark 4.14 (The three dimensions). The shifted Poisson structure specializes by dimension:

(a) $d = 1$ (curves): $(2-d) = 1$-shifted Poisson. The Gerstenhaber bracket has degree -1; the shifted bracket has degree 0. This is an ordinary Lie bracket on $\mathrm{HH}^\bullet$, recovering the classical Poisson bracket on deformations of the curve.

(b) $d = 2$ (surfaces): $(2 - d) = 0$-shifted Poisson. The shifted bracket has degree 0 and the cup product makes $\mathrm{HH}^\bullet(C)$ a Poisson algebra in the ordinary sense. This is the Poisson structure that the CY-to-chiral functor Φ promotes to the chiral Poisson (coisson) structure of the associated vertex algebra. The convolution Lie bracket on $\mathfrak{g}_A^{\mathrm{mod}}$ (Volume I) is the chiral incarnation of this 0-shifted Poisson bracket.

(c) $d = 3$ (threefolds): $(2 - d) = (-1)$-shifted Poisson. The bracket has degree -1, making $\mathrm{HH}^\bullet(C)$ a (-1)-shifted Poisson algebra = BV algebra (at the cochain level, a BV_∞-algebra). This is the BV structure underlying the BRST complex of the associated 3d holomorphic-topological theory. The chiral algebra produced by Φ is E_1, not E_2: the (-1)-shifted Poisson bracket produces a Lie bracket on the loop space (= Hochschild chains), not a Poisson bracket on functions.

4.7 Categorical Hodge theory

Definition 4.15 *(Hodge filtration on periodic cyclic homology).* For a smooth proper dg category C over $k = \mathbb{C}$, the *categorical Hodge filtration* on the periodic cyclic homology $\mathrm{HP}_\bullet(C)$ is the filtration

$$F^p\mathrm{HP}_n(C) = \mathrm{im}\big(\mathrm{HC}^-_{n+2p}(C) \to \mathrm{HP}_n(C)\big)$$

induced by the natural map from negative to periodic cyclic homology. The associated graded is

$$\mathrm{gr}^p_F\mathrm{HP}_n(C) \simeq \mathrm{HH}_{n+2p}(C).$$

Proposition 4.16 *(Non-commutative Hodge-to-de Rham).* [PE] For a smooth proper CY_d category C:

(i) The Hodge filtration degenerates (Kaledin), giving $\mathrm{HP}_n(C) \simeq \prod_p \mathrm{HH}_{n+2p}(C)$ as a filtered vector space;

(ii) The CY structure class $[\sigma] \in \mathrm{HC}^-_d(C)$ determines a preferred splitting of the Hodge filtration;

(iii) In the commutative case $C = \mathrm{Perf}(X)$, the categorical Hodge filtration recovers the classical Hodge filtration on $H^n_{\mathrm{dR}}(X)$ via HKR.

Remark 4.17 (Bridge to the shadow obstruction tower). Under the CY-to-chiral functor Φ (Part II), the categorical Hodge filtration maps to the weight filtration on the modular convolution algebra $\mathfrak{g}_A^{\mathrm{mod}}$. The associated graded pieces correspond to the arity-stratified components of the shadow obstruction tower:

Categorical side	Chiral side	Arity
$\mathrm{HH}_0(C)$ (CY class)	$\kappa_{\mathrm{cat}}(C)$	$r = 2$
$\mathrm{HH}_1(C)$ (1st Hochschild)	Cubic shadow C	$r = 3$
$\mathrm{HH}_2(C)$ (2nd Hochschild)	Quartic resonance Q	$r = 4$

The CY Hodge filtration degeneration (Kaledin) corresponds to the PBW spectral sequence collapse (Koszulness K2): both are manifestations of formality at different categorical levels.

Chapter 5

E_1-Chiral Algebras

Volume III's CY-to-chiral functor Φ outputs a braided factorization algebra: an E_2-chiral object whose representation category is the habitat of the quantum group. The construction factors through an intermediate E_1-chiral step. That intermediate step is the subject of this chapter. The reader who knows Volume II can treat this as a thirty-page review written from the CY vantage point; the reader who does not can treat it as the minimum entry point to the operadic conventions used in Parts II-IV of this volume.

Remark 5.1 (E_1 primacy for CY quantum groups). The E_1-chiral algebra (boundary) is the primitive object in this volume. The E_2-chiral algebra (bulk) is obtained from it by the Drinfeld center construction $\mathcal{Z}(\mathrm{Rep}^{E_1}(A)) \simeq \mathrm{Rep}^{E_2}(\mathrm{Drin}(A))$. Quantum groups, Yangians, and braided tensor categories are natively E_1 objects: the CoHA multiplication is ordered (short exact sequences have a preferred direction), and the R-matrix arises only in the Drinfeld double. The passage $E_1 \to E_2$ is the higher-categorical analogue of the averaging map $\mathrm{av}: \mathfrak{g}^{E_1} \to \mathfrak{g}^{\mathrm{mod}}$ from Vol I. This averaging is *lossy*: $\mathrm{av}(r(z)) = \kappa_{\mathrm{ch}}$ forgets the R-matrix data. The E_1-bar $B^{\mathrm{ord}}(A)$ retains strictly more information than the E_∞-bar $B^{\Sigma}(A)$.

5.1 Factorization algebras and the E_n hierarchy

A *factorization algebra* on a smooth manifold M of real dimension n assigns a cochain complex to every open subset and a factorization product to every disjoint pair of opens. Costello and Gwilliam [9] showed that the value on a small disk carries an E_n-algebra structure: the little n-disks operad acts on local sections. Lurie's higher algebra [25] gives the universal statement as $\mathrm{Fact}(\mathbb{R}^n) \simeq_{E_n}$. For a complex curve C of complex dimension one, $M = C$ has real dimension two, so every factorization algebra on C is tautologically E_2 in the topological sense.

Proposition 5.2 *(Holomorphic refinement, Costello-Gwilliam).* [PE] Let $\mathcal{A}$ be a factorization algebra on a smooth complex curve C whose structure maps are holomorphic in the two-point configuration space. The associated E_2-algebra specializes to an E_1-algebra on the configuration space of ordered points along any real ray, with the ordering provided by the holomorphic argument.

The proof is operadic: the little two-disks operad contracts onto the little intervals operad along the real subspace, and holomorphy pins the contraction to a single slice. Beilinson and Drinfeld [1] studied the symmetric version of this specialization; they called the outcome a *chiral algebra*. Volume II elevates the ordered version.

Definition 5.3 *(Swiss-cheese splitting).* The Swiss-cheese operad $\mathrm{SC}^{\mathrm{ch,top}}$ of Volume II splits as the product $E_1(C) \times E_2(\mathrm{top})$ of an ordered holomorphic operad in the chiral direction and a braided topological operad in the transverse direction. A $\mathrm{SC}^{\mathrm{ch,top}}$-algebra is exactly a pair (A, M) where A is an E_1-chiral algebra and M is an E_2-module.

The holomorphic part is E_1: ordered, deconcatenating, non-symmetric. The topological part is E_2: braided, commutative up to a Reidemeister III braiding. The split is not cosmetic. It is the operadic source of the kappa-spectrum of this volume: κ_{ch} lives on the E_1 factor, κ_{cat} on the E_2 factor, and the Drinfeld center exchanges them.

Remark 5.4 (Level-prefixed r-matrix). The classical r-matrix attached to an affine Kac-Moody E_1-chiral algebra at level k is $r(z) = k\,\Omega/z$, NOT the level-stripped Ω/z (AP126, AP141). The level survives the $d\log$ absorption because the ordered bar complex builds in one factor of the level per collision. At $k = 0$ the r-matrix must vanish identically; any formula in this volume that violates this vanishing is wrong. The collision residue of the Heisenberg r-matrix is k (not $k/2$), and the monodromy of the E_1 representation category around a puncture is $\exp(-2\pi i k)$ (V2-AP7).

The level-prefix convention is the standing convention throughout this volume. Every time an r-matrix is written in the CY-geometric setting, the level constant prefactor is the image of the trace under the CY-to-chiral functor and must be tracked explicitly.

5.2 The ordered bar as the E_1 primitive

The canonical example of an E_1-chiral algebra is the ordered bar complex. Let A be a chiral algebra on C with augmentation $\varepsilon\colon A \to \Omega_C$ and augmentation ideal $\bar{A} = \ker(\varepsilon)$ (AP132: we use the augmentation ideal, never the unital algebra).

Definition 5.5 *(Ordered bar complex).* The *ordered bar complex* of A is the cofree conilpotent tensor coalgebra

$$B^{\mathrm{ord}}(A) := T^c(s^{-1}\bar{A}) = \bigoplus_{n\geq 0}(s^{-1}\bar{A})^{\otimes n}$$

equipped with the *deconcatenation* coproduct

$$\Delta_{\mathrm{dec}}(a_1 \otimes \cdots \otimes a_n) = \sum_{k=0}^{n}(a_1 \otimes \cdots \otimes a_k) \otimes (a_{k+1} \otimes \cdots \otimes a_n)$$

and the bar differential built from the chiral product of A.

Deconcatenation is NOT the coshuffle coproduct on Sym^c (AP83, AP85). Deconcatenation produces $n+1$ terms and respects the ordering; the coshuffle produces 2^n terms and destroys it. The symmetric bar complex $B^{\Sigma}(A)$ is the S_n-coinvariant quotient of $B^{\mathrm{ord}}(A)$ with the coshuffle coproduct induced on invariants.

Proposition 5.6 *(Averaging map).* [PE] The averaging map

$$\mathrm{av}\colon B^{\mathrm{ord}}(A) \longrightarrow B^{\Sigma}(A), \qquad a_1 \otimes \cdots \otimes a_n \longmapsto \tfrac{1}{n!}\sum_{\sigma\in S_n} a_{\sigma(1)} \otimes \cdots \otimes a_{\sigma(n)}$$

is a morphism of cochain complexes and sends the E_1 structure to the E_∞ structure. It is lossy: the kernel contains the R-matrix data of the holomorphic factor, and on arity two $\mathrm{av}(r(z)) = \kappa_{\mathrm{ch}}$.

Volume I establishes the map; Volume II identifies it as the $E_1 \to E_\infty$ symmetrization. For Volume III purposes, the two consequences that matter are: (a) Yangians and quantum groups live on the E_1 side and are quotiented by averaging; (b) the symmetric bar B^Σ is sufficient for computing the modular characteristic but insufficient for reconstructing the R-matrix.

The averaging map is where the kappa-spectrum first appears. Applied to the arity-two generator $r(z)$, averaging returns the scalar κ_{ch}: this is the unique S_2-coinvariant of the collision residue. When the same $r(z)$ comes from the CY-to-chiral functor applied to $D^b(\mathrm{Coh}(K3))$, the scalar is $\kappa_{\mathrm{ch}} = 3$, which is distinct from the lattice-rank invariant $\kappa_{\mathrm{fiber}} = 24$ and from the BKM weight $\kappa_{\mathrm{BKM}} = 5$ (AP113). The reader who writes an unsubscripted symbol $\kappa_?$ in Vol III will confuse at least two of these four integers; the hook will flag it.

Remark 5.7 (Three bars, one functor). The CY-to-chiral functor Φ of Chapter **??** has three natural composites. Φ_{E_1} produces $B^{\mathrm{ord}}(A_C)$; this is the ordered bar sensed by the Yangian. $\mathrm{av} \circ \Phi_{E_1}$ produces $B^\Sigma(A_C)$; this is the symmetric bar sensed by the modular characteristic κ_{ch}. The full E_2 enhancement Φ_{E_2} produces the Drinfeld-centered bar $B^{E_2}(A_C) = \mathcal{Z}(B^{\mathrm{ord}}(A_C))$; this is the braided bar sensed by the quantum group. All three factor through Φ_{E_1}. This is the volume's motto: the ordered bar is the primitive.

5.3 E_1-chiral algebras from CY categories

The CY-to-chiral functor of I takes a Calabi-Yau category and produces a chiral algebra together with its E_1 structure. The input is a CY_d category C in the sense of Kontsevich-Soibelman; the trace lives in $\mathrm{HC}^-_d(C)$ (AP-CY2), not merely in Hochschild homology.

Proposition 5.8 *(E_1 sector at $d = 2$).* [PH] Let C be a CY_2 category with cyclic A_∞ structure and negative-cyclic trace. The ordered bar complex of the cyclic A_∞ algebra A_C carries a natural E_1-chiral structure whose holomorphic direction matches the Hochschild differential and whose ordered direction matches the cyclic action on the bar coalgebra.

The proof unfolds in Chapter **??**. For $C = D^b(\mathrm{Coh}(K3))$ the output is the positive half of the BKM superalgebra studied in III: the Borcherds denominator arises as the Euler character of $B^{\mathrm{ord}}(A_{K3})$, and $\kappa_{\mathrm{BKM}} = 5$ is distinct from $\kappa_{\mathrm{ch}} = 3$ (AP113; see the kappa-spectrum table in CLAUDE.md).

Conjecture 5.9 *(E_1 sector at $d = 3$).* Let C be a CY_3 category with a chain-level S^3-framing on $\mathrm{HC}^-_3(C)$ (the condition of Conjecture CY-A$_3$). Then the ordered bar complex of the cyclic A_∞ algebra A_C carries an E_1-chiral structure whose representation category is a braided monoidal refinement of the BPS Yangian of C.

The conjecture is conditional on the chain-level framing; the unconditional statement requires the full CY-A$_3$ programme (AP-CY6, AP-CY11). Every downstream result in III and IV that uses the BPS Yangian inherits this conditionality. The connection to Maulik-Okounkov and Costello-Witten cohomological Hall algebras is traced in Chapter **??**.

Proposition 5.10 *(E_2 enhancement via Drinfeld center).* [PH] If A is an E_1-chiral algebra with monoidal representation category $\mathrm{Rep}^{E_1}(A)$, the Drinfeld center $\mathcal{Z}(\mathrm{Rep}^{E_1}(A))$ is a braided monoidal category equipped with an equivalence

$$\mathcal{Z}(\mathrm{Rep}^{E_1}(A)) \simeq \mathrm{Rep}^{E_2}(\mathrm{Drin}(A))$$

where $\mathrm{Drin}(A)$ is the E_2-chiral algebra obtained by the categorified averaging map $E_1\text{-Cat} \to E_2\text{-Cat}$.

This proposition provides the pathway $\Phi_{E_1} \to \Phi_{E_2}$: one computes the E_1 chiral algebra first and then applies the Drinfeld center at the categorical level. At $d = 2$ the construction produces the braided vertex tensor category of the BKM superalgebra; at $d = 3$ the construction is conditional on Conjecture 5.9.

Remark 5.11 (Convolution L_∞-algebra on the E_1 side). For an E_1-chiral algebra A and a coalgebra C, the convolution L_∞-algebra $\hom_\alpha(C, A)$ of Volume I is the natural habitat of Maurer-Cartan elements controlling twisted morphisms. On the E_1 side, $\hom_\alpha$ is a strict dg Lie algebra when the ordering of C is preserved (the strict model $_{\mathrm{str}}$), and a genuine L_∞-algebra when higher brackets are included ($_\infty$). The MC moduli of the two models coincide, but the full L_∞-structure is required for homotopy transfer, formality statements, and gauge equivalence. For Vol III purposes, the key point is that the CY-to-chiral functor Φ lands in the strict model $_{\mathrm{str}}$ because the cyclic A_∞ structure on A_C is strict at arity two; higher-arity corrections require the L_∞ refinement.

Remark 5.12 (Comparison with Beilinson-Drinfeld chiral algebras). Beilinson and Drinfeld [1] developed the theory of chiral algebras as a symmetric factorization formalism on the Ran space. Their chiral algebras are E_∞ in the locality hierarchy of V2-AP1, even when they carry OPE poles: E-infinity includes all ordinary vertex algebras, which is the correct reading of "factorizable and local" (V2-AP9). The E_1-chiral algebras of this chapter are a strict refinement: they carry strictly more data than any Beilinson-Drinfeld chiral algebra because they remember the ordering of collisions. The averaging map $B^{\mathrm{ord}} \to B^\Sigma$ is precisely the functor that forgets this extra data and returns to the Beilinson-Drinfeld world. Vol III's geometric output is ordered; its modular characteristic is symmetric.

5.4 Connection to Volume II

Volume II is the E_1 core of the three-volume programme. Its seven parts develop the Swiss-cheese operad, the ordered bar complex, the classical $r(z)$-matrix (at level k, not level-stripped; AP126, AP141), the seven faces of $r(z)$, and the derived center of an E_1-chiral algebra. This chapter is a compact entry point to that machinery from the CY viewpoint.

Proposition 5.13 *(Vol III composes with Vol II).* [PH] The CY-to-chiral functor Φ of I factors as $\Phi = \Phi^{\mathrm{Vol\ II}}_{E_1} \circ \Phi^{\mathrm{Vol\ III}}_{\mathrm{cyc}}$, where $\Phi^{\mathrm{Vol\ III}}_{\mathrm{cyc}}$ takes a CY$_d$ category to its cyclic A_∞ algebra and $\Phi^{\mathrm{Vol\ II}}_{E_1}$ takes a cyclic A_∞ algebra to an E_1-chiral algebra via the Swiss-cheese promotion of AP86.

The detailed operadic content of $\Phi^{\mathrm{Vol\ II}}_{E_1}$ is governed by anti-patterns AP81 through AP104 of the canonical CLAUDE.md: the three coalgebra structures (AP81), the difference between coshuffle and deconcatenation (AP83), the promotion from one-colour to two-colour (AP86), the mixed-sector dimension formula (AP88), the curved factor of two at positive genus (AP91), the averaging map lossiness (AP96), the bound on $\mathrm{ChirHoch}^*(_c)$ (AP97), and the distinction between generating depth and algebraic depth (AP131).

Remark 5.14 (Why the E_1 layer cannot be skipped). The CY geometric data can be packaged directly into a braided factorization algebra (the E_2 form) via Kontsevich-Soibelman COHA technology. The temptation is to skip the E_1 intermediate and go from C straight to the braided output. Three obstructions block that route. First, the convolution L_∞-algebra $\hom_\alpha(C, A)$ of Volume II fails to be a bifunctor in both slots simultaneously (RNW19); MC3 must be proved one slot at a time, and the slot that admits a proof is the E_1 slot. Second, the r-matrix $r_{\mathrm{CY}}(z)$ cannot be extracted without seeing the ordered collisions, since the braiding of the E_2 picture only remembers $r(z)$ up to S_2-coinvariance. Third, the CoHA itself is an E_1-associative multiplication (AP-CY7); writing it as an E_2 product forgets the preferred direction on short exact sequences that is central to wall-crossing. These three obstructions are the same obstruction seen from three angles: the ordered bar is the only model that remembers the direction.

Remark 5.15 (lambda-bracket convention in Vol III). Vol III writes classical shadow operations in lambda-bracket notation with divided powers: $\{T_\lambda T\} = (c/12)\,\lambda^3$ (V2-AP34). The divided-power prefactor $1/3! = 1/6$ absorbs the OPE mode coefficient into the lambda-bracket rewrite: starting from the OPE mode $T_{(3)}T$ and dividing by 3! yields the stated $c/12$ at order λ^3. Every formula imported from Vol I (which uses OPE mode notation) must be converted before appearing in Vol III. The CY-to-chiral functor Φ is agnostic to the choice of convention, but its computed values of κ_{ch} are convention-dependent at the level of integral prefactors, and a Vol I formula transplanted without conversion will produce a wrong κ_{ch} by exactly a factor of 6.

Remark 5.16 (Three-volume thesis). The three volumes are three faces of a single E_1-E_1 operadic Koszul duality. Volume I is the symmetric modular face: it develops B^{Σ}, the five theorems A-D+H, and the modular characteristic κ_{ch} in the uniform-weight setting. Volume II is the E_1 open-colour face: it develops B^{ord}, the Swiss-cheese operad, the $r(z)$-matrix with its seven faces, and the three-dimensional holomorphic-topological bridge to quantum gravity. Volume III is the CY-geometric face: it develops the functor Φ that produces the input algebra from a Calabi-Yau category, identifies κ_{ch} within the kappa-spectrum, and traces the quantum group back to its geometric origin in BPS state counts.

Remark 5.17 (How this chapter is used downstream). Chapters 12, 13, and 19 refer back to this chapter as the source of the ordered bar coalgebra and the averaging map. Chapter **??** uses Proposition 5.8 as the $d = 2$ existence statement for A_C. The five parts of this volume exploit the $E_1 \to E_2$ passage (Proposition 5.10) in five different ways; the common structural point is that every time a braided monoidal category appears, it is the Drinfeld center of an underlying ordered one. The reader who remembers only one sentence from this chapter should remember that one.

5.5 Operadic dictionary for the CY reader

The operadic vocabulary of the canonical CLAUDE.md can be compressed into the following dictionary, which the CY reader can keep on a single page.

- E_n: little n-disks operad. E_1 is the operad of intervals on the line; E_2 is the operad of disks in the plane; E_∞ is the colimit. E_1 algebras are associative; E_2 algebras are braided; E_∞ algebras are commutative up to all higher homotopies. Dunn additivity: $E_m \otimes E_n \simeq E_{m+n}$.
- **Ordered bar** B^{ord}: the cofree conilpotent tensor coalgebra $T^c(s^{-1}\bar{A})$ with deconcatenation coproduct. Retains arity ordering. Natural E_1-object. Source of Yangians and quantum groups.
- **Symmetric bar** B^{Σ}: the S_n-coinvariant quotient of B^{ord} with the coshuffle coproduct descended from the shuffle product on T^c. Natural E_∞-object. Source of the modular characteristic κ_{ch}.
- **Harrison bar** B^{Lie}: the Harrison complex; Lie-coalgebra structure via the iterated commutator. Dual to the Chevalley-Eilenberg complex of the associated Lie algebra. Smallest of the three natural bar complexes on any commutative input.
- **Deconcatenation**: coproduct on T^c that splits a word into a prefix and a suffix. Produces $n + 1$ terms in arity n. Respects ordering. The E_1 coproduct.
- **Coshuffle**: coproduct on Sym^c dual to the shuffle product. Produces up to 2^n terms. Destroys ordering. The E_∞ coproduct.
- **Swiss-cheese** $\mathrm{SC}^{\mathrm{ch,top}}$: two-coloured operad combining a chiral (holomorphic, closed) sector with an ordered (topological, open) sector. Product decomposition $E_1(C) \times E_2(\mathrm{top})$ isolates the E_1 chiral factor.

- **Drinfeld center** $\mathcal{Z}$: endofunctor on monoidal categories that sends a monoidal category $\mathcal{M}$ to the category of objects equipped with a half-braiding against every object of $\mathcal{M}$. The result is braided monoidal. Categorified averaging: $\mathcal{Z}$ takes E_1-categories to E_2-categories.

- r-**matrix**: the arity-two generator of the E_1 Koszul duality, carrying a pole along the diagonal. At level k, the classical Kac-Moody r-matrix is $r(z) = k\,\Omega/z$; vanishes at $k = 0$ (AP126, AP141).

The dictionary is not symmetric in E_1 and E_∞: the ordered bar is the primitive, and everything on the E_∞ side is a quotient. The CY-geometric functor Φ lands primitively in the E_1 layer; the E_∞ and E_2 images are obtained by composing with averaging and Drinfeld center respectively.

Remark 5.18 (Notation consistency across the three volumes). The three volumes use different coalgebra conventions in their displayed formulas. Vol I displays formulas in B^Σ with the symmetric coproduct; Vol II displays formulas in B^{ord} with deconcatenation; Vol III displays formulas in whichever form the CY functor produces, which is always B^{ord} at the source but frequently B^Σ after averaging to extract κ_{ch}. The reader who cross-references a formula between volumes must convert between the three coalgebra structures: $B^{\mathrm{ord}} \to B^\Sigma$ by averaging (dividing by $n!$ and symmetrizing), and $B^\Sigma \to B^{\mathrm{Lie}}$ by taking the iterated commutator of the cofree tensor coalgebra. The three bars are NOT isomorphic even as complexes; they differ by S_n-coinvariant quotients, and the Euler characters diverge accordingly. See Vol II CLAUDE.md V2-AP3 for the three-bar sequence.

5.6 Closing: the ordered bar as the primitive of Vol III

This chapter has established four structural facts that Parts II-V of this volume invoke repeatedly. First, factorization algebras on a complex curve are E_2 in the topological sense but specialize to E_1 when the holomorphic structure is rigid, and the Swiss-cheese splitting $E_1(C) \times E_2(\mathrm{top})$ makes this specialization precise. Second, the ordered bar complex $B^{\mathrm{ord}}(A) = T^c(s^{-1}\bar{A})$ with deconcatenation coproduct is the natural E_1 primitive, and averaging to the symmetric bar $B^\Sigma(A)$ is lossy by a factor equal to the R-matrix data. Third, the CY-to-chiral functor Φ at $d = 2$ produces an ordered bar whose Euler character is the Borcherds denominator and whose first shadow invariant is κ_{ch}, distinct from κ_{BKM}, κ_{cat}, and κ_{fiber} in the kappa-spectrum. Fourth, the E_2 enhancement requires the Drinfeld center construction, which is the categorified averaging map E_1-Cat $\to$ E_2-Cat; at $d = 3$ the enhancement is conditional on the CY-A$_3$ programme.

This is the version of the E_1-E_1 operadic Koszul duality adapted to the CY setting: one form $\eta = d\log(z_1 - z_2)$, one relation (Arnold), one object Θ_A, one equation $D_*\Theta + \frac{1}{2}[\Theta, \Theta] = 0$, all on the ordered bar of the cyclic A_∞ algebra A_C attached to the Calabi-Yau category C. The symmetric bar is the modular shadow.

The remainder of Vol III unfolds the consequences. II computes the modular trace κ_{ch} for each family in the CY landscape, using the ordered bar as the computational model. III catalogues the CY families $K3$, $K3 \times E$, toric CY$_3$, Fukaya, and quantum-group, and in each case identifies the position within the kappa-spectrum. IV reads the seven faces of $r_{\mathrm{CY}}(z)$ off the E_1 structure of Proposition 5.8. V collects the open problems, almost all of which are versions of Conjecture 5.9 or its downstream consequences. Throughout, the ordered bar is the primitive. The averaging map, the Drinfeld center, and the modular characteristic are all functors defined on the E_1 side and computed there before any symmetric or braided image is taken.

Chapter 6

E_2-Chiral Algebras

An E_2-chiral algebra factorizes along two compatible directions and its representation category is braided monoidal: the algebraic habitat of quantum groups, and the output of the Volume III CY-to-chiral functor at CY dimension $d \geq 2$. This chapter traces E_2-chiral algebras back to cyclic A_∞ data via the Deligne conjecture, and states the central conjecture that Φ_{E_2} is the Drinfeld center of the Volume II E_1-chiral functor Φ_{E_1}.

6.1 The E_2 operad and its categorical content

Definition 6.1 *(E_2 operad).* The E_2 operad (little 2-disks) has $E_2(n) = \mathrm{Conf}_n(\mathbb{D}^2)$. By Dunn additivity $E_2 \simeq E_1 \otimes E_1$: an E_2-algebra is an E_1-algebra in E_1-algebras, carrying two compatible associative products whose homotopy is the braiding. At the level of homology the braiding descends to the commutativity constraint of a Gerstenhaber algebra.

THEOREM 6.2 *(Lurie, [25]).* [PE]The $(\infty, 2)$-category of E_2-algebras in a symmetric monoidal ∞-category C is equivalent to the $(\infty, 2)$-category of braided monoidal objects in C.

Kontsevich formality $C_*(E_2; \mathbb{Q}) \xrightarrow{\sim} \mathrm{Ger}$ supplies the rational comparison. For factorization algebras the E_2 structure appears fully holomorphically ($X \times Y$) or holomorphic-topologically (Volume II Swiss-cheese), two presentations of Dunn additivity.

Definition 6.3 *(E_2-chiral algebra).* An *E_2-chiral algebra* on $X \times Y$ is a factorization algebra $\mathcal{A}$ on $\mathrm{Ran}(X) \times \mathrm{Ran}(Y)$ that factorizes chirally in each factor and is compatible with the E_2-operad action via $\overline{\mathrm{FM}}_n(\mathbb{C}) \simeq E_2(n)$. Its representation category $\mathrm{Rep}^{E_2}(\mathcal{A})$ is braided monoidal.

Remark 6.4 (Deligne conjecture). The canonical E_2-algebra is $^\bullet(A, A)$ of an associative algebra A. The Deligne conjecture (Kontsevich-Soibelman [24], Tamarkin [35], McClure-Smith, Berger-Fresse) asserts that $^\bullet(A, A)$ carries a canonical E_2-action with cup product and brace as the two products, inducing the Gerstenhaber structure on $\mathrm{HH}^\bullet(A, A)$.

6.2 E_2-chiral algebras from cyclic A_∞ data

For Calabi-Yau categories the cyclic A_∞ structure refines the Deligne E_2 by an additional compatibility (trace, BV operator, higher analogue) controlled by the CY dimension.

THEOREM 6.5 *(Kontsevich-Soibelman; Tamarkin).* [PE] Let A be a cyclic A_∞ algebra of CY dimension d, with nondegenerate pairing $\langle -, - \rangle \colon A \otimes A \to k[d]$. Then $^\bullet(A, A)$ carries a canonical E_2-algebra structure together with a compatible chain-level S^1-action whose homotopy fixed points compute $\mathrm{HC}^-_\bullet(A)$. The combined *framed* structure is $fE_2 \simeq E_2 \rtimes S^1$. References: Kontsevich-Soibelman [24]; Tamarkin [35].

Remark 6.6 (CY dimension shift). The E_2-structure on $^\bullet(A, A)$ is degree-shifted by $2 - d$: at $d = 1$ the shift is $+1$ (framed $E_2 \simeq$ BV); at $d = 2$ the shift is 0 (ordinary E_2 with Poisson compatibility); at $d = 3$ the shift is -1 (Lie-2-algebra compatibility). The CY dimension enters via the cyclic trace, not the underlying E_2-operad.

COROLLARY 6.7 *(E_2-chiral from CY$_2$).* [PH] Let C be a saturated dg category of CY dimension $d = 2$ with cyclic A_∞ enhancement. Then $^\bullet(C, C)$ carries an E_2-structure with Poisson compatibility, and $\Phi_{E_2}(C)$ on $\Sigma \times \Sigma$ is an E_2-chiral algebra (Definition 6.3). The assignment $C \mapsto \Phi_{E_2}(C)$ is the Volume III CY-to-E_2-chiral functor at $d = 2$.

Proof sketch. Theorem 6.5 equips $^\bullet(C, C)$ with a cyclic E_2-structure, ungraded at $d = 2$ by Remark 6.6. Dunn additivity upgrades factorization homology $\int^\bullet_{\Sigma\times\Sigma}(C, C)$ to a factorization algebra on $\Sigma\times\Sigma$; chirality in each factor follows from the holomorphic refinement of Kontsevich formality. The two directions commute because the two E_1-factors of E_2 are algebraically independent (Definition 6.3(iii)). □

CONJECTURE 6.8 *(E_2-chiral from CY$_3$).* [CJ] Let C be saturated of CY dimension $d = 3$ with chain-level S^3-framed cyclic A_∞ enhancement. Then $^\bullet(C, C)$ carries a framed E_2-structure in degree -1, and $\Phi_{E_2}(C)$ is an E_2-chiral algebra with expected representation category $\mathrm{CoHA}(C)$. Conditional on CY-A$_3$ (AP-CY6).

6.3 Braided tensor categories from E_2-chiral structure

By Theorem 6.2, $\mathrm{Rep}^{E_2}(\mathcal{A})$ is braided monoidal; for $\mathcal{A}$ an E_2-chiral algebra the braiding is holomorphic (analytically continued along paths in $\mathrm{Conf}_n(\Sigma)$, not merely formal).

Definition 6.9 (MTC of an E_2-chiral algebra). The *modular tensor category* of an E_2-chiral algebra $\mathcal{A}$ is the semisimplification of $\mathrm{Rep}^{E_2}(\mathcal{A})$ at integrable modules (when it exists); if $\mathrm{Rep}^{E_2}(\mathcal{A})$ is already finite with nondegenerate braiding form, it is the MTC directly.

PROPOSITION 6.10 *(MTC from CY$_2$ via Borcherds).* [PH] For $C = D^b(\mathrm{Coh}(K3 \times E))$ the derived category of a product K3 times an elliptic curve, the MTC of $\Phi_{E_2}(C)$ coincides with the Verlinde category of the Borcherds-Kac-Moody superalgebra $\mathfrak{g}_{II_{1,1}\oplus II_{1,1}}$ of the K3 lattice. The corresponding $\kappa_{\mathrm{ch}} = 3$, $\kappa_{\mathrm{BKM}} = 5$ diagnostic is discussed in Chapter **??** (see also Theorem **??**).

Proof sketch. Corollary 6.7 applies to $D^b(\mathrm{Coh}(K3))$ with $\mathrm{HH}^\bullet(K3) = \bigwedge^\bullet H^1(T_{K3})$; the elliptic factor adjoins a modular direction on which the braid group acts via the Heisenberg lattice VOA of $II_{1,1}$. The Borcherds product Δ_5 matches the Volume I chiral Euler product at lattice weight $\kappa_{\mathrm{BKM}} = 5$, and the MTC is the Verlinde quotient of the weight-5 lattice VOA. □

CONJECTURE 6.11 *(MTC from CY$_3$ via CoHA).* [CJ] Conditional on Conjecture 6.8, the MTC of $\Phi_{E_2}(C)$ is $\mathrm{CoHA}(C)$ with its Kontsevich-Soibelman braided structure: BPS indices as simples, braiding computing refined DT wall-crossing. Conditional on CY-A$_3$ (AP-CY6); AP-CY11 propagates: downstream uses inherit the CY-A$_3$ dependency.

6.4 Connection to Volume II: the Drinfeld center

Volume II constructs Φ_{E_1} and proves E_1-chiral Koszul duality (see **??** of [36]). The Drinfeld center $Z\colon E_1\text{-}\mathbf{Cat} \xrightarrow{\sim} E_2\text{-}\mathbf{Cat}$ (under dualizability) is the categorical avatar of the Volume I symmetrization average $\mathrm{av}\colon B^{\mathrm{ord}} \to B^{\Sigma}$: the universal extraction of a braided monoidal structure from an E_1-monoidal one.

Conjecture 6.12 *(Volume III central conjecture: $\Phi_{E_2} = Z^{\mathrm{ch}} \circ \Phi_{E_1}$).* [CJ] For a saturated CY$_d$ category C admitting $\Phi_{E_1}(C)$, the Volume III E_2-chiral construction is canonically quasi-isomorphic to the chiral Drinfeld center of $\Phi_{E_1}(C)$:

$$\Phi_{E_2}(C) \;\simeq\; Z^{\mathrm{ch}}\big(\Phi_{E_1}(C)\big),$$

where Z^{ch} is the E_2-chiral algebra whose representation category is the Drinfeld center of the E_1-chiral representation category of $\Phi_{E_1}(C)$. At $d = 2$ this is compatible with Corollary 6.7 and Proposition 6.10; at $d = 3$ it is conditional on CY-A$_3$.

Remark 6.13 (Four distinct functors (AP34)). Conjecture 6.12 must be distinguished from three other functors: (i) Verdier duality D_{Ran} producing $B(A^!)$ as a factorization coalgebra; (ii) cobar inversion $\Omega \circ B \simeq \mathrm{id}$ recovering A on the Koszul locus (Volume I Theorem B); (iii) the derived center $Z^{\mathrm{der}}_{\mathrm{ch}}(A)$ producing the bulk factorization algebra via Hochschild cochains (Volume II bulk-boundary); (iv) the categorical Drinfeld center Z^{ch} used here. The Drinfeld center (iv) lives on representation categories with braided monoidal output; the derived center (iii) lives on cochain complexes with E_1-algebra output. Conflation is forbidden (AP34).

Remark 6.14 (Relation to Volume II V2-AP81-V2-AP104). Volume II tracks V2-AP81-V2-AP104 on the E_1-chiral operadic story ($B_P(A) = P^i$-coalgebra vs BP = cooperad; three coalgebra structures $\mathrm{Lie}^c, \mathrm{Sym}^c, T^c$; factorization coproduct vs deconcatenation; curvatures μ_0 vs $d^2_{\mathrm{fib}} = \kappa_{\mathrm{ch}}\omega_g$; lossiness of av). Z^{ch} preserves T^c-coalgebra structure and the curvature $\kappa_{\mathrm{ch}}\omega_g$, and inverts av only where braided centralization is strict.

6.5 E_2-chiral Koszul duality and the three bar complexes

The Volume III E_2-chiral Koszul duality theorem would promote the Volume I bar-cobar adjunction to a braided monoidal equivalence; it is presently a conjecture.

Conjecture 6.15 *(E_2-chiral Koszul duality).* [CJ] For an E_2-chiral algebra $\mathcal{A}$, the bar-cobar adjunction extends E_2-equivariantly, giving a braided monoidal equivalence

$$\mathrm{Rep}^{E_2}(A) \simeq \mathrm{Rep}^{E_2}(A^!)^{\mathrm{rev}},$$

with rev the reverse braiding and $A^!$ the Koszul dual algebra (not $D_{\mathrm{Ran}}(B(A))$, not $\Omega(B(A))$; AP34).

Construction 6.16 *(E_2 bar complex).* For an E_2-chiral algebra $\mathcal{A}$ on $X \times Y$, the E_2 *bar complex* $B_{E_2}(\mathcal{A}) = T^c(s^{-1}\bar{\mathcal{A}})$ is the cofree conilpotent E_2-coalgebra on the desuspended augmentation ideal $\bar{\mathcal{A}} = \ker(\varepsilon)$ (AP132), with two commuting differentials d_X, d_Y from OPE residues in each factor, two commuting coproducts Δ_X, Δ_Y from deconcatenation, and the braiding from the level-prefixed R-matrix $R^{E_2}(z) = k \cdot \Omega/z + O(1)$ (AP126: k survives d-log absorption, verify $k = 0 \Rightarrow R = 0$).

The operad filtration $E_1 \to E_2 \to \cdots \to E_\infty$ produces at each level a bar complex with more symmetry. The three cases relevant to this work are compared below.

Construction 6.17 *(Three bar complexes).* For A a vertex algebra with E_2-chiral structure (Definition 6.3):

(i) *E_1-bar*: $B^n_{E_1}(A) = \bar{A}^{\otimes n}$, ordered with Hochschild differential and deconcatenation coproduct;

(ii) *E_2-bar*: $B^n_{E_2}(A) = \bar{A}^{\otimes n}$ with braid group B_n action via the level-prefixed R-matrix $R^{E_2}(z)$ (AP126) and differential combining X, Y-OPE residues;

(iii) *E_∞-bar*: $B^n_{E_\infty}(A) = \mathrm{Sym}^n(\bar{A}[-1])$ with Chevalley-Eilenberg differential and shuffle coproduct, the Volume I bar complex.

The three are related by

$$B^n_{E_\infty}(A) = B^n_{E_1}(A)/S_n, \qquad H^*(B^n_{E_\infty}(A)) = H^*(B^n_{E_2}(A))^{S_n},$$

since the E_2 braiding refines the S_n action to B_n, and E_∞ is the S_n-invariant sector.

Theorem 6.18 *(Bar complex comparison for $\mathbb{C}^3$).* [PH] [(i), (iii)] [HE] [(ii) observed through $n \leq 20$] For $A = W_{1+\infty}$, the affine Yangian of $\mathfrak{gl}_1$ at generic (h_1, h_2, h_3) with $h_1 + h_2 + h_3 = 0$:

(i) $\sum_{n\geq 0} \dim B^n_{E_1}(A) \cdot q^n = M(q) := \prod_{n\geq 1}(1-q^n)^{-n}$, the MacMahon function; coefficients count plane partitions $1, 1, 3, 6, 13, 24, \ldots$ (OEIS A000219).

(ii) $\dim B^n_{E_2}(H_1) \stackrel{?}{=} d(n)$, the divisor function, observed through $n \leq 20$ (`e2_barcobar_koszul.py`); no analytical proof.

(iii) $\sum_n \dim B^n_{E_\infty}(A) \cdot q^n = P(q) = \prod_{n\geq 1}(1 - q^n)^{-1}$ (OEIS A000041), by $B^n_{E_\infty} = (B^n_{E_1})^{S_n}$.

Proof. (i) The E_1-bar differential vanishes on the Heisenberg generators (free-algebra case), so B_{E_1} reduces to the ordered tensor algebra on the single generator; weighted by conformal weight, the generating function is MacMahon. (ii) At the self-dual point $(1, 0, -1)$ the R-matrix is trivial and $B_{E_2} \simeq B_{E_1}$; at generic parameters the braid action via $g(z) = \prod_a (z - h_a)/(z + h_a)$ is nontrivial and computation through $n \leq 20$ matches $d(n)$ (see `e2_barcobar_koszul.py`), heuristically. (iii) S_n-quotient of $B^n_{E_1}$ is Sym^n; for a single generator, $\dim \mathrm{Sym}^n V = p(n)$. □

Proposition 6.19 *(CoHA/W-algebra character ratio).* [PH] The CoHA (E_1-bar) character exceeds the W-algebra (E_∞-bar) character by the ratio

$$\frac{M(q)}{P(q)} = \prod_{n\geq 2} \frac{1}{(1-q^n)^{n-1}},$$

measuring the ordered degrees of freedom quotiented out in the passage $B_{E_1} \to B_{E_\infty}$.

Proof. $M(q)/P(q) = \prod_{n\geq 1}(1-q^n)^{-(n-1)} = \prod_{n\geq 2}(1-q^n)^{-(n-1)}$ since the $n = 1$ factor is trivial. □

Remark 6.20 (Computational verification). Theorem 6.18 and Proposition 6.19 are verified by 59 independent tests in `compute/lib/bar_comparison_c3.py`: MacMahon coefficients through $n = 15$, S_n-invariant dimensions matching partitions, the $M(q)/P(q)$ product, and self-dual degeneration $B_{E_2} \simeq B_{E_1}$.

Chapter 7

E_n-Factorization and Higher Chiral Structure

Factorization algebras on a complex curve carry an E_2 structure; factorization algebras on a point carry an E_1 structure. Where is the transition? The CY-to-chiral functor Φ of Chapter 8 gives a definite answer for low dimensions: E_∞ at $d = 1$, E_2 at $d = 2$, E_1 at $d = 3$. The pattern suggests the formula E_{4-d}, but this formula breaks at $d = 4$, because E_0 does not exist as a separate operad (it is the category of pointed objects). Something else must happen.

What happens is stabilization. The $\mathbb{S}^d$-framing on $\mathrm{HH}_\bullet(C)$ provides an E_d-algebra structure, but for $d \geq 3$ the restriction to the chiral level via Dunn additivity produces only an E_1-algebra. The higher E_d-structure does not disappear: it survives as *additional shifted symplectic and shifted Poisson data* beyond the base E_1 level, classified by the framing obstruction. The question then becomes: for which CY dimensions d is this shifted data nontrivial, and what invariant detects it?

The answer is Bott periodicity. The framing obstruction lives in $\pi_d(BU)$ or $\pi_d(BO)$ or $\pi_d(B\,\mathrm{Sp})$ depending on the parity and reduction of the structure group of the CY pairing. For the unitary path, $\pi_d(BU) = \mathbb{Z}$ when d is even and vanishes when d is odd. For the symplectic/orthogonal path, the 8-fold periodicity of the classical groups produces a richer pattern, with refinements at $d \equiv 5 \pmod 8$. The main result of this chapter (Theorem 7.1) assembles these obstruction computations into a single statement: the framing obstruction is trivial precisely when $d \bmod 8 \in \{1, 3, 7\}$, and the CY chiral algebra is E_1-stabilized with additional shifted structure controlled by $\pi_d(BU)$ elsewhere.

7.1 Dunn additivity and the E_n hierarchy

Recall the Dunn additivity theorem: $E_n \simeq E_1 \otimes_{E_0} E_1 \otimes_{E_0} \cdots \otimes_{E_0} E_1$ (n factors), where $E_0 = \mathrm{Assoc}_+$ is the associative operad with unit. In particular, $E_2 \simeq E_1 \otimes_{E_0} E_1$: an E_2-algebra is an E_1-algebra in E_1-algebras. An E_∞-algebra is commutative.

For the CY-to-chiral functor, the CY dimension d determines the native E_n level of the chiral algebra. The $\mathbb{S}^d$-framing on Hochschild homology $\mathrm{HH}_\bullet(C)$ carries an E_d-algebra structure. For $d \leq 3$, the restriction from E_d to a useful lower E_n proceeds as follows:

- $d = 1$: the native structure is E_∞ (commutative). The $\mathbb{S}^1$-framing produces a symmetric monoidal structure on $\mathrm{HH}_\bullet(C)$, which is E_∞.

- $d = 2$: E_2 is already the target structure (braided monoidal).

- $d = 3$: E_3 restricts to E_2 with symmetric braiding (since $\pi_1(\mathrm{Conf}_2(\mathbb{R}^3))$ is trivial). The genuine nonsymmetric quantum group braiding arises through the Drinfeld center of the E_1-monoidal representation category (Theorem 8.15).

For $d \geq 4$, the E_d structure is "too symmetric" and the CY chiral algebra is always E_1. The higher E_d structure manifests as *additional shifted structure* (shifted symplectic, shifted Poisson) beyond the base E_1 level.

7.2 Factorization algebras on $\mathbb{C}^n$

For a CY category C of dimension d associated to $\mathbb{C}^d$, the factorization algebra $\mathrm{Fact}_X(\mathfrak{L}_C)$ on a curve X is constructed from the Lie conformal algebra of polyvector fields (Chapter 8). The $\mathrm{GL}(d)$-invariant Schouten–Nijenhuis bracket vanishes for all $d \geq 3$ (Theorem 8.4 and its generalization), so the classical bracket is abelian and all noncommutative structure arises from the Ω-deformation.

The equivariant deformation parameters are $\sigma_3, \sigma_4, \ldots, \sigma_d$ (the elementary symmetric polynomials of $h_1, \ldots, h_d$ subject to $\sum h_i = 0$), giving a $(d-2)$-dimensional deformation space. For $d = 3$, this is the single parameter $\sigma_3 = h_1 h_2 h_3$ of the affine Yangian.

7.3 The E_n-chiral bar complex

Construction 6.17 (in Chapter 6) gives the three bar complexes $B_{E_1}, B_{E_2}, B_{E_\infty}$ for an E_2-chiral algebra. For a CY_d chiral algebra with $d \geq 4$, only the E_1-bar and E_∞-bar are native; the E_2-bar requires the Drinfeld center passage.

7.4 The framing obstruction tower

The $\mathbb{S}^d$-framing on $\mathrm{HH}_\bullet(C)$ requires a trivialization of a certain bundle classified by π_d of the structure group of the CY pairing. Two independent computations of this obstruction:

(A) *Unitary path.* The complex structure gives a $U(n)$ reduction. By Bott periodicity (period 2 for BU):

$$\pi_k(BU) = \begin{cases} \mathbb{Z} & \text{if } k \text{ is even and } k \geq 2, \\ 0 & \text{otherwise.} \end{cases}$$

The BU obstruction is: $\pi_d(BU) = \mathbb{Z}$ for d even, 0 for d odd.

(B) *Symplectic/orthogonal path.* For d odd, the CY pairing is antisymmetric, giving structure group $\mathrm{Sp}(2m)$. For d even, the pairing is symmetric, giving $O(n)$ (before holomorphic reduction). The stable homotopy groups are given by Bott periodicity (period 8):

k mod 8	0	1	2	3	4	5	6	7
$\pi_k(\mathrm{Sp})$	0	0	0	$\mathbb{Z}$	$\mathbb{Z}_2$	$\mathbb{Z}_2$	0	$\mathbb{Z}$
$\pi_k(O)$	$\mathbb{Z}_2$	$\mathbb{Z}_2$	0	$\mathbb{Z}$	0	0	0	$\mathbb{Z}$

The classifying-space obstruction is $\pi_d(B\,\mathrm{Sp}) = \pi_{d-1}(\mathrm{Sp})$ for d odd, and $\pi_d(BO) = \pi_{d-1}(O)$ for d even.

7.5 The E_n landscape: the E_1 stabilization theorem

THEOREM 7.1 *(E_1 stabilization for CY chiral algebras).* [PH] For $d \geq 3$, the CY chiral algebra $A_C = \Phi(C)$ associated to a smooth CY$_d$ category C is an E_1-chiral algebra. The additional shifted structure is classified by the *effective framing obstruction*:

$$\mathrm{Obs}_{\mathrm{eff}}(d) = \begin{cases} \pi_d(BU) & \text{if } \pi_d(BU) \neq 0 \text{ (primary obstruction nontrivial)},\\ \pi_d(B\,\mathrm{Sp}) & \text{if } d \text{ is odd and } \pi_d(BU) = 0 \text{ (Sp refinement)},\\ 0 & \text{otherwise.} \end{cases}$$

The complete E_n landscape for CY$_d$ categories:

d	Native E_n	$\pi_d(BU)$	$\pi_d(B\,\mathrm{Sp})$	$\mathrm{Obs}_{\mathrm{eff}}$	Shifted structure
1	E_∞	0	0	0	none (commutative)
2	E_2	$\mathbb{Z}$	—	$\mathbb{Z}$	braiding parameter (c_1)
3	E_1	0	0	0	none ($\mathbb{S}^3$-framing trivial)
4	E_1	$\mathbb{Z}$	$\mathbb{Z}$	$\mathbb{Z}$	$\mathbb{Z}$-shifted symplectic (p_1)
5	E_1	0	$\mathbb{Z}_2$	$\mathbb{Z}_2$	$\mathbb{Z}_2$-shifted (Sp refinement)
6	E_1	$\mathbb{Z}$	$\mathbb{Z}_2$	$\mathbb{Z}$	$\mathbb{Z}$-shifted (c_3)
7	E_1	0	0	0	none (trivial, mirrors $d = 3$)
8	E_1	$\mathbb{Z}$	$\mathbb{Z}$	$\mathbb{Z}$	$\mathbb{Z}$-shifted (Bott-periodic)

The pattern repeats with period 8 by Bott periodicity of Sp and O. In the Bott 8-fold cycle $d \bmod 8 = 1, 2, \ldots, 8$, the effective obstruction is:

$$0,\ \mathbb{Z},\ 0,\ \mathbb{Z},\ \mathbb{Z}_2,\ \mathbb{Z},\ 0,\ \mathbb{Z}.$$

The obstruction is *trivial* precisely at $d \bmod 8 \in \{1, 3, 7\}$. All even $d \geq 2$ have $\mathbb{Z}$-valued obstruction (from $\pi_d(BU) = \mathbb{Z}$).

Proof. The proof has three parts.

Part 1: E_1 is the floor for $d \geq 3$. For $d \geq 3$, the GL(d)-invariant Schouten–Nijenhuis bracket on $\mathrm{PV}^*(\mathbb{C}^d)$ vanishes by the same argument as $d = 3$ (Theorem 8.4): degree overflow and graded skew-symmetry. The classical Lie conformal algebra is abelian, and the entire noncommutative structure arises from the Ω-deformation. The CoHA construction gives an ordered (associative) product: the extension correspondence $0 \to V' \to V \to V'' \to 0$ has a preferred direction (sub before quotient), which is E_1, not E_2.

Part 2: The framing obstruction. The $\mathbb{S}^d$-framing obstruction lives in π_d of the classifying space of the structure group. The CY condition $\omega_X \cong \mathcal{O}_X$ determines the structure group:

- The complex structure reduces $\mathrm{GL}(n, \mathbb{R})$ to $U(n)$. The obstruction in $\pi_d(BU)$ is $\mathbb{Z}$ for d even and 0 for d odd (Bott periodicity, period 2).
- For d odd (antisymmetric CY pairing), the further reduction to $\mathrm{Sp}(2m)$ gives a refined obstruction in $\pi_d(B\,\mathrm{Sp}) = \pi_{d-1}(\mathrm{Sp})$. This is nontrivial ($\mathbb{Z}_2$) precisely when $d \equiv 5 \pmod 8$ (since $\pi_4(\mathrm{Sp}) = \mathbb{Z}_2$).

- For d even (symmetric pairing), the orthogonal reduction gives $\pi_d(BO) = \pi_{d-1}(O)$, but the holomorphic structure typically refines back to BU, so the BU obstruction is the primary one.

Part 3: Explicit verification for $d = 4$. At $d = 4$, $\pi_4(BU) = \pi_3(U) = \mathbb{Z}$. The generator is the first Pontryagin class p_1. A CY_4 category C has $p_1(\mathrm{Ext}^\bullet_C(E, E)) \in \mathbb{Z}$, which is the obstruction to an $\mathbb{S}^4$-framing. When $p_1 \neq 0$, the chiral algebra carries a $\mathbb{Z}$-shifted symplectic structure: the factorization algebra structure is E_1 with a choice of integer "level" (analogous to the level of a Kac–Moody algebra) determined by p_1. This is verified computationally for $\mathbb{C}^4$ (93 tests in `higher_cy_en_tower.py`). □

Remark 7.2 (Why E_0 does not exist). The operad $E_0 = \mathrm{Assoc}_+$ (pointed sets with distinguished basepoint) does exist as an operad, but its algebras are "pointed objects," not algebras in the usual sense. An E_0-algebra in chain complexes is a chain complex with a distinguished element (the basepoint), carrying no multiplicative structure. The correct interpretation for $d = 4$ is: the CY chiral algebra is E_1 (with multiplication), and the p_1 class provides additional structure *beyond* the operad level: a shifted symplectic form on the tangent complex of the MC moduli.

Corollary 7.3 *($d = 4$: the first Pontryagin obstruction).* [PH] For a CY_4 category C:

(i) The chiral algebra A_C is E_1.

(ii) The $\mathbb{S}^4$-framing obstruction is $p_1 \in \pi_4(BU) = \mathbb{Z}$, the first Pontryagin class. All three paths (BU, $B\,\mathrm{Sp}$, BO) give $\mathbb{Z}$:

$$\pi_4(BU) = \pi_4(B\,\mathrm{Sp}) = \pi_4(BO) = \mathbb{Z}.$$

(iii) The E_2 braided structure is recovered via the Drinfeld center $\mathcal{Z}(\mathrm{Rep}^{E_1}(A_C))$, with the half-braiding carrying a $\mathbb{Z}$-valued twisting datum from p_1.

(iv) The deformation space of $\mathbb{C}^4$ is 2-dimensional, spanned by $\sigma_3 = \sum_{i<j<k} h_i h_j h_k$ and $\sigma_4 = h_1 h_2 h_3 h_4$.

Verification: 141 tests in `higher_cy_en_tower.py`.

Proof. Parts (i) and (ii) follow from Theorem 7.1 at $d = 4$. For part (ii), the three paths:

- $\pi_4(BU) = \pi_3(U) = \mathbb{Z}$ (Bott: 3 is odd, $\pi_3(U) = \mathbb{Z}$).
- $\pi_4(B\,\mathrm{Sp}) = \pi_3(\mathrm{Sp}) = \mathbb{Z}$ (the generator is the symplectic structure on $\mathrm{Sp}(2m)$).
- $\pi_4(BO) = \pi_3(O) = \mathbb{Z}$ (the generator is the spin structure).

All three agree, giving a well-defined $\mathbb{Z}$-valued obstruction.

Part (iii): the Drinfeld center construction of Theorem 8.15 generalizes from $d = 3$ to $d = 4$. The half-braiding on Fock modules of the CY_4 CoHA carries a $\mathbb{Z}$-grading from p_1.

Part (iv): the equivariant parameters $h_1, \ldots, h_4$ satisfy $h_1 + h_2 + h_3 + h_4 = 0$ (CY condition), leaving 3 free parameters. The elementary symmetric polynomials $\sigma_2, \sigma_3, \sigma_4$ parametrize the deformation space; σ_2 generates a trivial deformation (rescaling), leaving σ_3 and σ_4 as the 2 effective parameters. □

Corollary 7.4 *($d = 5$: the $\mathbb{Z}_2$ Sp-refinement).* [PH] For a CY_5 category C:

(i) The chiral algebra A_C is E_1.

(ii) The primary framing obstruction $\pi_5(BU) = 0$ is trivial (5 is odd).

(iii) The refined obstruction $\pi_5(B\,\mathrm{Sp}) = \pi_4(\mathrm{Sp}) = \mathbb{Z}_2$ is *nontrivial*. The $\mathbb{S}^5$-framing carries a $\mathbb{Z}_2$-valued invariant from the symplectic structure group.

(iv) The $\mathbb{Z}_2$-shifted structure manifests as a super-E_1 grading: the chiral algebra has a $\mathbb{Z}_2$-grading compatible with the E_1-operad structure, arising from the mod-2 Pontryagin class.

(v) The deformation space of $\mathbb{C}^5$ is 3-dimensional $(\sigma_3, \sigma_4, \sigma_5)$.

Verification: 141 tests in `higher_cy_en_tower.py`.

Proof. Parts (i)–(iii) follow from Theorem 7.1 at $d = 5$, using the Bott periodicity table: $\pi_4(\mathrm{Sp}) = \mathbb{Z}_2$ (verified independently via the computation $\pi_4(\mathrm{Sp}) = \pi_4(\mathrm{Sp}(4)) = \mathbb{Z}_2$ in the stable range).

Part (iv) is a consequence of the $\mathbb{Z}_2$-valued obstruction: the half-braiding in the Drinfeld center carries a $\mathbb{Z}_2$-grading, which promotes the E_1-algebra structure to a $\mathbb{Z}_2$-graded (super) E_1-algebra.

Part (v): the equivariant parameters $h_1, \ldots, h_5$ with $\sum h_i = 0$ give 4 free parameters, with σ_2 trivial, leaving $\sigma_3, \sigma_4, \sigma_5$. □

Corollary 7.5 *($d = 7$ mirrors $d = 3$; $d = 8$ mirrors $d = 4$).* [PH]

(i) At $d = 7$: the framing obstruction is trivial. Both $\pi_7(BU) = 0$ and $\pi_7(B\,\mathrm{Sp}) = \pi_6(\mathrm{Sp}) = 0$. The CY_7 chiral algebra is unobstructed E_1, mirroring $d = 3$.

(ii) At $d = 8$: the framing obstruction is $\pi_8(BU) = \mathbb{Z}$ (Bott-periodic, corresponding to $d = 0$ shifted by the 8-fold periodicity of $B\,\mathrm{Sp}$). The CY_8 chiral algebra carries a $\mathbb{Z}$-shifted structure, mirroring $d = 4$.

More generally, d and $d + 8$ have isomorphic framing obstruction groups (Bott periodicity).
Verification: 141 tests in `higher_cy_en_tower.py`.

7.6 Explicit computations for $d = 4$ varieties

$\mathbb{C}^4$

The simplest CY_4 is $\mathbb{C}^4$. The $\mathrm{GL}(4)$-invariant polyvector fields $\mathrm{PV}^p(\mathbb{C}^4)$ have $\dim \mathrm{PV}^p = \binom{4}{p}$ in the constant-coefficient sector:

$$\dim(\mathrm{PV}^0, \mathrm{PV}^1, \mathrm{PV}^2, \mathrm{PV}^3, \mathrm{PV}^4) = (1, 4, 6, 4, 1), \qquad \dim_{\mathrm{total}} = 2^4 = 16.$$

The Schouten–Nijenhuis brackets on the $\mathrm{GL}(4)$-invariant sector vanish for the same reasons as $d = 3$ (degree overflow and graded skew-symmetry). The Bogomolov–Tian–Todorov theorem gives $\mathrm{HH}^3 = 0$ (unobstructedness).

The 2-dimensional deformation space is spanned by

$$\sigma_3 = \sum_{i<j<k} h_i h_j h_k, \qquad \sigma_4 = h_1 h_2 h_3 h_4,$$

subject to $h_1 + h_2 + h_3 + h_4 = 0$. At the self-dual point (one $h_i = 0$, say $h_4 = 0$), the CY_4 functor reduces to the CY_3 functor for $\mathbb{C}^3$ with parameters (h_1, h_2, h_3), and the chiral algebra is $\mathcal{W}_{1+\infty}$ at $c = 1$ (the Heisenberg VOA H_1, $\kappa_{\mathrm{ch}} = 1$).

$K3 \times K3$

The product $K3 \times K3$ is a CY_4 with Euler characteristic

$$\chi(K3 \times K3) = \chi(K3)^2 = 24^2 = 576$$

and Hodge numbers computed by Künneth from the K3 diamond. The key Hodge numbers: $h^{1,1} = 40$, $h^{2,0} = h^{0,2} = 2$, $h^{4,0} = h^{0,4} = 1$.

The chiral algebra is the tensor product of two copies of the K3 lattice VOA:

$$A_{K3\times K3} = V_{\Lambda_{K3}} \otimes V_{\Lambda_{K3}},$$

with modular characteristic

$$\kappa_{\mathrm{ch}}(A_{K3\times K3}) = \kappa_{\mathrm{ch}}(V_{\Lambda_{K3}}) + \kappa_{\mathrm{ch}}(V_{\Lambda_{K3}}) = 24 + 24 = 48,$$

by Vol I additivity (Theorem D). The shadow obstruction tower is class **G** (Gaussian, $r_{\mathrm{max}} = 2$) since the lattice VOA is a free-field algebra.

The sextic 4-fold in $\mathbb{P}^5$

The degree-6 smooth hypersurface $X_6 \subset \mathbb{P}^5$ is a compact CY_4 with $h^{3,1} = 426$, $h^{2,2} = 1752$, $h^{1,1} = 1$, and $\chi(X_6) = 2610$. The holomorphic Euler characteristic is $\chi(\mathcal{O}_{X_6}) = 1 + 0 + 1 = 2$ (from $h^{0,0} = h^{4,0} = 1$ and $h^{2,0} = 0$).

7.7 The $d = 5$ functor: $\mathbb{Z}_2$ obstruction from Sp

The CY_5 functor introduces a genuinely new phenomenon: a $\mathbb{Z}_2$-valued framing obstruction that is invisible to the unitary structure group but visible to the symplectic refinement.

For $\mathbb{C}^5$: the polyvector field dimensions are $\dim \mathrm{PV}^p = \binom{5}{p}$, giving total dimension $2^5 = 32$. The deformation space is 3-dimensional $(\sigma_3, \sigma_4, \sigma_5)$.

The framing obstruction:

- $\pi_5(BU) = 0$ (primary, from unitary structure: 5 is odd).
- $\pi_5(B\,\mathrm{Sp}) = \pi_4(\mathrm{Sp}) = \mathbb{Z}_2$ (refined, from symplectic structure).

The discrepancy between the BU and $B\,\mathrm{Sp}$ obstructions means that the chain-level $\mathbb{S}^5$-framing is topologically trivializable at the unitary level, but the symplectic refinement carries a $\mathbb{Z}_2$-valued obstruction. This obstruction is the mod-2 analogue of the first Pontryagin class at $d = 4$.

7.8 The $d = 6$ case and higher

Remark 7.6 ($d = 6$: BU vs BO discrepancy). At $d = 6$: $\pi_6(BU) = \mathbb{Z}$ (nontrivial) but $\pi_6(BO) = \pi_5(O) = 0$ (trivial). The real orthogonal structure group sees no obstruction, but the complex/holomorphic structure (captured by BU) does. This means the $\mathbb{Z}$-valued obstruction at $d = 6$ is purely holomorphic: it is visible only through the complex structure of the CY category, not through the underlying real topology.

The characteristic class is $c_3 \in H^6(B;\mathbb{Z})$, the third Chern class of the tangent bundle. For a CY_6 manifold X with $c_3(X) \neq 0$, the chiral algebra carries a $\mathbb{Z}$-shifted structure indexed by c_3.

Conjecture 7.7 *(E_n-Koszul duality at $d \geq 4$).* [CJ] For a CY_d chiral algebra A_C with $d \geq 4$, the Koszul dual $A_C^! = \Phi(C^!)$ (the chiral algebra of the mirror category) satisfies:

(i) The framing obstruction of $A_C^!$ is the *negation* (in the obstruction group) of the framing obstruction of A_C.

(ii) For $d = 4$ ($\mathbb{Z}$-valued obstruction): $p_1(A_C^!) = -p_1(A_C)$. The Koszul dual carries the opposite Pontryagin class.

(iii) For $d = 5$ ($\mathbb{Z}_2$-valued obstruction): $\mathrm{Obs}(A_C^!) = \mathrm{Obs}(A_C)$ (since $-1 = 1$ in $\mathbb{Z}_2$). The $\mathbb{Z}_2$-shifted structure is *self-dual*: mirror symmetry preserves it.

(iv) The modular characteristics satisfy $\kappa_{\mathrm{ch}}(A_C) + \kappa_{\mathrm{ch}}(A_C^!) = 0$ (for the free-field case) or $\kappa_{\mathrm{ch}} + \kappa'_{\mathrm{ch}} = \rho \cdot K$ (for W-algebra-type families), matching Vol I Theorem D.

7.9 Relation to topological field theory

The E_1 stabilization theorem has a direct TFT interpretation. A CY_d category determines a d-dimensional topological field theory (Costello–Li 2016). For $d \geq 3$, the TFT is "fully extended" down to the E_1 level: it assigns categories to $(d-1)$-manifolds, functors to cobordisms, and natural transformations to higher cobordisms. The E_n level of the chiral algebra is the E_n level at which the fully extended TFT "stops":

d	E_n level	TFT stops at	Obstruction
1	E_∞	fully local	none
2	E_2	2-categorical	$c_1 \in \mathbb{Z}$
3	E_1	categorical	none
4	E_1	categorical	$p_1 \in \mathbb{Z}$
5	E_1	categorical	$\mathbb{Z}_2$ from Sp
≥ 6	E_1	categorical	Bott-periodic

The passage from E_1 to E_2 (the quantum group braiding) is always achieved through the Drinfeld center, which is the TFT analogue of "compactifying on a circle": $\mathcal{Z}(\mathrm{Rep}^{E_1}(A)) \simeq \mathrm{Rep}^{E_2}(Z^{\mathrm{der}}_{\mathrm{ch}}(A))$. *Verification*: 141 tests across all CY dimensions $d = 1, \ldots, 16$ in `higher_cy_en_tower.py`.

Part II

The CY Characteristic Datum

Chapter 8

From CY Categories to Chiral Algebras

Where do chiral algebras come from? Volumes I and II begin with a chiral algebra already in hand: the bar-cobar machine, the modular characteristic κ_{ch}, the shadow tower, and the five theorems all presuppose an object A with OPE, state space, and vacuum. The geometric source of that object is left unspecified. A single family supplies a partial answer: affine Kac–Moody algebras arise from flat connections on curves, and Virasoro arises from reparametrizations. Everything else, conifolds, Hilbert schemes of K3, resolved A_n singularities, local threefolds with potential, must be fed into the machinery by hand.

This is the deficiency that Vol III addresses. A Calabi–Yau category C of dimension d carries a cyclic A_∞-structure, a Connes B-operator, and an $\mathbb{S}^d$-framing on Hochschild homology. None of these are chiral algebras. The passage from C to a chiral algebra A_C requires four specific steps: Hochschild cohomology becomes a Lie conformal algebra through the Gerstenhaber bracket, the Lie conformal algebra is promoted to a factorization envelope on a curve, the $\mathbb{S}^d$-framing provides the higher operadic enhancement, and the CY trace determines the quantization. The resulting functor $\Phi\colon \mathrm{CY}_d\text{-}\mathbf{Cat} \to E_n\text{-}\mathrm{ChirAlg}$ is the central object of this volume.

The $d = 2$ case is unconditional: the $\mathbb{S}^2$-framing of Kontsevich–Vlassopoulos enhances $\mathrm{Fact}_X(\mathfrak{L}_C)$ to an E_2-chiral algebra, and the functor is constructed end-to-end. The $d = 3$ case is the programme: the native E_3-structure restricts to *symmetric* braiding under Dunn additivity, so the nonsymmetric quantum group braiding must be recovered by a separate route through the Drinfeld center of the E_1-monoidal representation category. This chapter verifies the $d = 3$ chain end-to-end for $\mathbb{C}^3$, where both sides are independently known ($\mathrm{CoHA}(\mathbb{C}^3) \simeq Y^+(\widehat{\mathfrak{gl}}_1)$ on the CY side, $\mathcal{W}_{1+\infty}$ at $c = 1$ on the chiral side), and sets up the quiver-chart gluing that assembles global E_1-chiral algebras from local CoHA charts.

8.1 The cyclic-to-chiral passage

A CY category C of dimension d carries a cyclic A_∞-structure (Chapter 3). The cyclic bar complex $\mathrm{CC}_\bullet(C)$ with its $\mathbb{S}^d$-framing is the primary invariant. The passage to chiral algebras proceeds in four steps:

Step 1. Cyclic A_∞ → Lie conformal algebra. The Hochschild cohomology $\mathrm{HH}^\bullet(C)$, with its Gerstenhaber bracket, determines a graded Lie algebra. The CY pairing promotes this to a Lie conformal algebra $\mathfrak{L}_C$ (the "current algebra" of C).

Step 2. Lie conformal algebra → factorization envelope. The factorization envelope construction (Nishinaka–Vicedo) produces a factorization algebra $\mathrm{Fact}_X(\mathfrak{L}_C)$ on any smooth curve X.

Step 3. E_2 **enhancement.** When $d = 2$, the $\mathbb{S}^2$-framing of $\mathrm{HH}_\bullet(C)$ provides an E_2-algebra structure. This enhances $\mathrm{Fact}_X(\mathfrak{L}_C)$ to an E_2-chiral algebra.

Step 4. **Quantization.** The quantum chiral algebra A_C is the quantization of $\mathrm{Fact}_X(\mathfrak{L}_C)$ determined by the CY trace.

For $d = 3$, Step 3 must be replaced: the $\mathbb{S}^3$-framing gives an E_3-structure on $\mathrm{HH}_\bullet(C)$, but the resulting E_2-braiding (obtained by restriction via Dunn additivity) is *symmetric*. The nonsymmetric quantum group braiding is recovered by a different route: the Drinfeld center of the E_1-monoidal representation category (Section 8.5).

8.2 The CY-to-chiral functor

Theorem 8.1 *(CY-to-chiral functor — Theorem CY-A_2, $d = 2$).* [PH] There exists a functor

$$\Phi\colon \mathrm{CY}_2\text{-}\mathbf{Cat} \longrightarrow E_2\text{-}\mathrm{ChirAlg}$$

from smooth proper CY categories of dimension $d = 2$ to E_2-chiral algebras, such that:

(i) $\Phi(C)$ is a chiral algebra whose generating fields are in bijection with $\mathrm{HH}^{\bullet+1}(C)$;

(ii) The bar complex $B(\Phi(C))$ is quasi-isomorphic to the cyclic bar complex $\mathrm{CC}_\bullet(C)$ as a factorization coalgebra;

(iii) The modular characteristic $\kappa_{\mathrm{ch}}(\Phi(C))$ equals the CY Euler characteristic $\chi^{\mathrm{CY}}(C)$.

The E_2-enhancement in Step 3 uses the $\mathbb{S}^2$-framing of $\mathrm{HH}_\bullet(C)$ (Kontsevich–Vlassopoulos).

8.3 The $d = 3$ functor chain for $\mathbb{C}^3$

Affine space $X = \mathbb{C}^3$ is the Rosetta Stone for the $d = 3$ programme. Both sides are independently known: the CY side produces $\mathrm{CoHA}(\mathbb{C}^3) \simeq Y^+(\widehat{\mathfrak{gl}}_1)$ (Kontsevich–Soibelman, Schiffmann–Vasserot), and the chiral side gives $\mathcal{W}_{1+\infty}$ at $c = 1$ (Procházka–Rapčák). The functor chain is verified end-to-end.

The five-step chain

The $d = 3$ functor for $\mathbb{C}^3$ factors as:

Step 1. $\mathrm{PV}^*(\mathbb{C}^3)$ with $\mathrm{GL}(3)$-invariant Schouten–Nijenhuis bracket

$\downarrow$ Ω-deformation in the σ_3 direction

Step 2. Quantized Lie conformal algebra

$\downarrow$ Factorization envelope (Nishinaka–Vicedo)

Step 3. $\mathcal{W}_{1+\infty}$ with nonabelian OPE

$\downarrow$ Drinfeld center

Step 4. E_2-braided category $\mathrm{Rep}^{E_2}(Y(\widehat{\mathfrak{gl}}_1))$ with R-matrix

$\downarrow$ Quantum group extraction

Step 5. E_2-braided category identified with $\mathrm{Rep}^{E_2}(Y(\widehat{\mathfrak{gl}}_1))$

The input at Step 1 is $\mathrm{PV}^*(\mathbb{C}^3) = \bigoplus_{p=0}^{3} \Gamma(\mathbb{C}^3, \bigwedge^p T_{\mathbb{C}^3})$, the algebra of polyvector fields with the Schouten–Nijenhuis bracket. The $\mathrm{GL}(3)$-invariant sector carries the deformation-theoretic data needed for the CY-to-chiral functor. The Ω-deformation at Step 2 is parametrized by $\sigma_3 = h_1 h_2 h_3$ (with $h_1 + h_2 + h_3 = 0$), the unique direction in $\mathrm{HH}^2(\mathrm{PV}^*(\mathbb{C}^3))$. At the self-dual level ($\sigma_3 \to 0$, giving $h_1 = 1, h_2 = 0, h_3 = -1$), the output at Step 3 is $\mathcal{W}_{1+\infty}$ at $c = 1$, which *is* the Heisenberg VOA H_1.

The classical Lie conformal algebra is *abelian* (Theorem 8.4): all noncommutative structure arises from quantization in the envelope step. The CY_3 functor produces a quantum algebra from classically commutative input.

THEOREM 8.2 *(The $d = 3$ functor chain is verified for $\mathbb{C}^3$).* [PH] Each step of the five-step chain is verified computationally:

(i) *Step 1*: The $\mathrm{GL}(3)$-invariant Schouten–Nijenhuis brackets vanish identically (Theorem 8.4). The input Lie conformal algebra is abelian.

(ii) *Step 2*: $\mathrm{HH}^2(\mathrm{PV}^*(\mathbb{C}^3)) = 1$, so the deformation is unique, spanned by σ_3 (Theorem 8.5). $\mathrm{HH}^3 = 0$ by Bogomolov–Tian–Todorov, so the deformation is unobstructed.

(iii) *Step 3*: The factorization envelope produces $\mathcal{W}_{1+\infty}$ at $c = 1$. The OPE arises entirely from the central extension and normal ordering in the envelope, not from the classical bracket.

(iv) *Step 4*: The Drinfeld center $\mathcal{Z}(\mathrm{Rep}^{E_1}(Y^+(\widehat{\mathfrak{gl}}_1)))$ is identified with $\mathrm{Rep}^{E_2}(Y(\widehat{\mathfrak{gl}}_1))$ (Theorem 8.15).

(v) *Step 5*: The E_2-braided representation category is identified with $\mathrm{Rep}^{E_2}(Y(\widehat{\mathfrak{gl}}_1))$; the full global-group object $G(\mathbb{C}^3)$ of Conjecture 8.49 is AP-CY7 territory and sits beyond the scope of this theorem.

Verification: ~600 tests across six compute modules: `c3_lie_conformal.py`, `c3_envelope_comparison.py`, `cy3_hochschild.py`, `drinfeld_center_yangian.py`, `c3_grand_verification.py`, `cy_to_chiral_functor.py`.

Remark 8.3 (The shuffle algebra does not translate directly to the λ-bracket). The shuffle algebra presentation of $Y^+(\widehat{\mathfrak{gl}}_1)$ has structure function $g(z) = (z - h_1)(z - h_2)(z - h_3)/((z + h_1)(z + h_2)(z + h_3))$. A natural attempt extracts a λ-bracket from the shuffle product via $z \mapsto \lambda$. This fails: $g(0) = -1$ (regular, not singular), so the shuffle product does not produce a distribution-valued bracket. The envelope step (Step 2 → Step 3) replaces the algebraic $g(z)$ by the *vertex operator* OPE, where the δ-function singularity of the state-field correspondence generates the λ-bracket. The passage shuffle → vertex algebra is the step where locality enters.

Abelianity of the classical bracket

THEOREM 8.4 *(Abelianity of the classical bracket).* [PH] The GL(3)-invariant Schouten–Nijenhuis brackets on $\mathrm{PV}^*(\mathbb{C}^3)$ all vanish. Three independent proofs:

(i) *Direct computation*: every bracket $[\alpha, \beta]_{\mathrm{SN}}$ of GL(3)-invariant polyvector fields evaluates to zero by explicit contraction.

(ii) *Graded skew-symmetry + degree count*: in the Gerstenhaber grading (shifted degree $|\alpha|_s = |\alpha| - 1$), the generators $E \in \mathrm{PV}^1$ and $\partial_1 \wedge \partial_2 \wedge \partial_3 \in \mathrm{PV}^3$ have even shifted degree (0 and 2 respectively), so $[\alpha, \alpha] = 0$ for these by graded skew-symmetry. The remaining generators $1 \in \mathrm{PV}^0$ and $\omega^{-1} \in \mathrm{PV}^2$ have odd shifted degree, but $[1, -]_{\mathrm{SN}} = 0$ identically (constants are central) and $[\omega^{-1}, \omega^{-1}]_{\mathrm{SN}} \in \mathrm{PV}^3$ must be GL(3)-invariant, hence proportional to the trivector; explicit computation gives zero. All mixed brackets $[\alpha, \beta]$ with $\alpha \neq \beta$ vanish by the degree-overflow argument of (iii).

(iii) *Weight/exterior degree overflow*: the bracket lowers exterior degree by 1; any nontrivial GL(3)-invariant in the target degree must have weight exceeding the polynomial degree available.

Verification: 95 tests in `c3_lie_conformal.py`.

Proof. We give proof (i) in detail; proofs (ii) and (iii) are recorded for cross-verification.

The GL(3)-invariant polyvector fields on $\mathbb{C}^3$ are spanned by the constant function $1 \in \mathrm{PV}^0$, the Euler vector field $E = \sum_i z_i \partial_i \in \mathrm{PV}^1$, the Poisson bivector $\pi = \sum_{\mathrm{cyc}} z_3\, \partial_1 \wedge \partial_2 \in \mathrm{PV}^2$ (contraction of E with the volume trivector), and the trivector $\partial_1 \wedge \partial_2 \wedge \partial_3 \in \mathrm{PV}^3$. The Schouten–Nijenhuis bracket has bidegree (-1): it maps $\mathrm{PV}^p \times \mathrm{PV}^q \to \mathrm{PV}^{p+q-1}$. For any two GL(3)-invariant generators $\alpha \in \mathrm{PV}^p, \beta \in \mathrm{PV}^q$, the bracket $[\alpha, \beta]_{\mathrm{SN}}$ must be GL(3)-invariant of degree $p + q - 1$. In each case, either $p + q - 1 > 3$ (forcing zero by degree overflow) or the explicit contraction vanishes by the symmetry of the invariant generators. □

Hochschild cohomology and the unique deformation

THEOREM 8.5 *(Hochschild cohomology and unique deformation).* [PH] For $\mathbb{C}^3$:

(i) $\mathrm{HH}^2(\mathrm{PV}^*(\mathbb{C}^3)) = 1$: the deformation space is one-dimensional, spanned by the σ_3 direction. The unique deformation is the Ω-deformation.

(ii) $\mathrm{HH}^3(\mathrm{PV}^*(\mathbb{C}^3)) = 0$: the Bogomolov–Tian–Todorov theorem guarantees unobstructedness.

(iii) The quantized algebra is $\mathcal{W}_{1+\infty}$ at $c = 1$.

Verification: 71 tests in `cy3_hochschild.py`.

Proof. The Hochschild–Kostant–Rosenberg theorem identifies $\mathrm{HH}^\bullet(\mathrm{PV}^*(\mathbb{C}^3))$ with the GL(3)-invariant polyvector fields on the formal neighbourhood of the origin in the moduli space of A_∞-deformations. For $\mathbb{C}^3$ with the standard CY structure, the relevant computation reduces to the T^3-equivariant sector.

The equivariant parameters h_1, h_2, h_3 with $h_1 + h_2 + h_3 = 0$ span a two-dimensional space. The unique T^3-equivariant deformation class in HH^2 is $\sigma_3 = h_1 h_2 h_3$, the elementary symmetric polynomial of degree 3. The class $\sigma_2 = h_1 h_2 + h_2 h_3 + h_3 h_1 = -(h_1^2 + h_2^2 + h_3^2)/2$ on the constraint locus is generically nonzero, but it generates a trivial deformation equivalent to rescaling: the σ_2-deformation acts as a conformal weight shift that can be absorbed by redefining the grading. The nontrivial deformation class is $\sigma_3 = h_1 h_2 h_3$, which cannot be removed by regrading.

The vanishing $\mathrm{HH}^3 = 0$ is the Bogomolov–Tian–Todorov unobstructedness theorem for the CY_3 moduli problem: the Kodaira–Spencer dga of $\mathbb{C}^3$ is formal, and the Maurer–Cartan moduli space is smooth. This guarantees that the Ω-deformation determined by σ_3 extends to all orders. □

Necessity of the Ω-deformation for noncompact CY_3

Proposition 8.6 *(Necessity of Ω-deformation for noncompact CY_3).* [PH] Without the Ω-deformation, the CY_3 functor chain for $\mathbb{C}^3$ collapses: the Lie conformal algebra is abelian (Theorem 8.4), the envelope produces the free Heisenberg, and the E_2 structure is trivially E_∞ (symmetric monoidal). Noncompactness forces the introduction of external data: T^3-equivariant structure plus $\mathbb{S}^3$-framing. For compact CY_3, neither should be needed: the nontrivial global geometry plays the role of the Ω-deformation.
Verification: 87 tests in `c3_envelope_comparison.py`.

Proof. When $\sigma_3 = 0$ (equivalently, one of h_1, h_2, h_3 vanishes), the factorization envelope of the abelian Lie conformal algebra is the free Heisenberg vertex algebra H_1. The representation category $\mathrm{Rep}(H_1)$ is symmetric monoidal: the braiding on Fock modules is the identity (all monodromy is trivial for a free boson at level 1). In particular, $\mathcal{Z}(\mathrm{Rep}^{E_1}(H_1)) = \mathrm{Rep}(H_1)$ itself (the Drinfeld center of a symmetric monoidal category is the category itself), so no E_2-enhancement occurs.

When $\sigma_3 \neq 0$, the Ω-deformation introduces the nonlinear OPE terms of $\mathcal{W}_{1+\infty}$. These terms break the E_∞ symmetry to E_2, and the Drinfeld center produces a nontrivially braided category with the Yang R-matrix (Theorem 8.15).

For a *compact* CY_3 such as the quintic, the global sections of the polyvector sheaf have nontrivial bracket (the complex structure moduli introduce genuine noncommutativity), and the Ω-deformation is not needed. The role of σ_3 for $\mathbb{C}^3$ is to *replace* the global geometric data that is absent for a noncompact variety. □

Remark 8.7 (Smooth vs singular: locality of the quantization). The distinction between smooth and singular CY_3 is categorical:

CY_3	**Quantization**	**Algebraic type**
Smooth ($\mathbb{C}^3$, quintic, $K3 \times E$)	Vertex algebra (local)	E_1-chiral (E_2 via Drinfeld center)
Singular (conifold)	Associative algebra (nonlocal)	E_1-chiral

Smoothness is not a matter of convention: the factorization envelope requires local data (the polyvector fields form a cosheaf whose sections over small discs control the OPE), and singularities obstruct the local-to-global passage. At a singularity, the category $\mathrm{Perf}(X)$ acquires compact objects that do not extend to a neighbourhood, and the factorization structure degenerates from E_2 to E_1.
Verification: 71 tests in `cy3_hochschild.py` (testing smooth vs singular HH data).

8.4 The $\mathbb{S}^3$-framing obstruction: universal triviality

The chain-level $\mathbb{S}^3$-framing is the central topological obstruction to CY-A at $d = 3$. Conjecture 8.49 requires (a) constructing this framing compatibly with BV structure and (b) establishing the quantization step. The following result removes condition (a).

Theorem 8.8 *(Vanishing of the $\mathbb{S}^3$-framing obstruction for CY$_3$ categories).* [PH] The topological $\mathbb{S}^3$-framing obstruction vanishes universally for CY$_3$ categories. Two independent proofs:

(i) *Symplectic path.* The CY$_3$ pairing on the Ext complex $\mathrm{Ext}^\bullet_C(E,E)$ is antisymmetric (by the Serre functor with $\omega_X \cong \mathcal{O}_X$), giving structure group $\mathrm{Sp}(2m)$ for $\dim \mathrm{Ext}^1 = 2m$. Since $\pi_3(B\,\mathrm{Sp}(2m)) = 0$ for all $m \geq 1$, the obstruction class in π_3 vanishes.

(ii) *Bott periodicity path.* The complex structure gives a reduction $\mathrm{GL}(n,\mathbb{C}) \to U(n)$. By Bott periodicity, $\pi_3(BU) = 0$ (the homotopy groups of BU are $\pi_{2k}(BU) = \mathbb{Z}$, $\pi_{2k+1}(BU) = 0$). Since π_3 is odd, the obstruction vanishes.

The chain-level BV obstruction $\kappa_{\mathrm{ch}} \cdot [\Omega_3]$ is trivializable via holomorphic Chern–Simons (Witten 1992, Costello–Li 2016): the functional $\mathrm{CS}(\bar\partial + A) = \int_X \Omega \wedge \mathrm{tr}(A \wedge \bar\partial A + \frac{2}{3} A \wedge A \wedge A)$ provides the contracting homotopy.
Verification: 124 tests in `cy_to_chiral_functor.py`.

Proof. We give the symplectic path in detail.

The CY$_3$ condition $\omega_X \cong \mathcal{O}_X$ implies the existence of a nondegenerate trace $\mathrm{tr}\colon \mathrm{HH}_3(C) \to \mathbb{C}$, which by Serre duality induces an antisymmetric pairing

$$\langle -,- \rangle \colon \mathrm{Ext}^i(E,E) \otimes \mathrm{Ext}^{3-i}(E,E) \longrightarrow \mathbb{C}.$$

At $i = 1$, this is an antisymmetric form on the tangent space to the moduli of objects, reducing the structure group from $\mathrm{GL}(2m)$ to $\mathrm{Sp}(2m)$. At $i = 0$, the self-pairing $\mathrm{Ext}^0 \otimes \mathrm{Ext}^3 \to \mathbb{C}$ is the Serre duality pairing on endomorphisms.

The $\mathbb{S}^3$-framing obstruction lives in π_3 of the classifying space of the structure group. For the symplectic group:

$$\pi_3(B\,\mathrm{Sp}(2m)) = \pi_2(\mathrm{Sp}(2m)) = 0 \quad \text{for all } m \geq 1.$$

The second equality uses the long exact sequence of the fibration $\mathrm{Sp}(2m) \to B\,\mathrm{Sp}(2m)$ and the fact that $\mathrm{Sp}(2m)$ is simply connected ($\pi_1(\mathrm{Sp}(2m)) = 0$) with $\pi_2(\mathrm{Sp}(2m)) = 0$ (all compact simply-connected simple Lie groups have vanishing π_2).

For the Bott periodicity path: the Bott periodicity theorem gives $\pi_k(BU) \cong \pi_k(BU(n))$ for n large, with $\pi_{2k}(BU) = \mathbb{Z}$ and $\pi_{2k+1}(BU) = 0$. Since 3 is odd, $\pi_3(BU) = 0$.

The chain-level BV obstruction requires more than topological triviality: one needs an explicit trivialization compatible with the BV operator $\Delta = \partial/\partial\Omega_3$. The holomorphic Chern–Simons functional provides this. The CS action is a trivialization of $[\Omega_3]$ because $\delta_{\mathrm{BV}}(\mathrm{CS}) = \int \Omega \wedge F_A$, and the flatness equation $F_A = 0$ is the Maurer–Cartan equation of the B-model. This means the BV obstruction is exact in the BRST cohomology. □

Corollary 8.9 *(Topological vanishing of the CY-A_3 obstruction).* [CD] Assuming Conjecture 8.49 is formulated with symplectic structure group (i.e., the $d = 3$ CY-to-chiral functor targets a CY$_3$ chiral algebra whose obstruction theory sits in $\pi_3(B\,\mathrm{Sp}(2m))$), the topological component of that obstruction

vanishes. The remaining chain-level construction of the trivialization of $\kappa_{\mathrm{ch}} \cdot [\Omega_3]$ is known to exist by holomorphic Chern–Simons, but its compatibility with the full A_∞-structure is part of CY-A$_3$ itself and remains conjectural. For the toric CY$_3$ verified in compute ($\mathbb{C}^3$, conifold, local $\mathbb{P}^2$, local $\mathbb{P}^1 \times \mathbb{P}^1$, $K3 \times E$), the $E_1 \to E_2$ enhancement obstruction vanishes at the level of the explicit CoHA construction (AP-CY6: conditional propagation from CY-A$_3$).
Verification: 124 tests in `cy_to_chiral_functor.py`; 93 tests in `drinfeld_center_yangian.py`.

The Čech contracting homotopy for compact CY$_3$

The topological vanishing of the $\mathbb{S}^3$-framing obstruction (Theorem 8.8) leaves open the chain-level construction of the trivialization. For compact CY$_3$ hypersurfaces in projective space, the standard affine cover provides an algebraic Čech contracting homotopy that closes this gap at the perturbative level.

Theorem 8.10 *(Čech contracting homotopy for compact CY$_3$; [PH])*. For the quintic threefold $X_5 \subset \mathbb{P}^4$ with the standard affine cover $\{U_i = \{x_i \neq 0\}\}_{i=0}^4$, the algebraic Čech contracting homotopy $s^q : \check{C}^q \to \check{C}^{q-1}$ (prepend the distinguished index 0) satisfies the SDR conditions

$$\delta s + s\delta = \mathrm{Id} - i \circ p, \qquad s^2 = 0, \qquad s \circ i = 0, \qquad p \circ s = 0,$$

where $i : H^*(\check{C}^\bullet) \hookrightarrow \check{C}^\bullet$ is the inclusion of cohomology and p is the projection. The contracting homotopy is purely algebraic: it acts by $s(f_{i_0\cdots i_q}) = f_{0i_0\cdots i_q}$ on Čech cochains and involves no PDE solving. The SDR data (i, p, s) interfaces directly with the homotopy transfer theorem (HTT, Theorem **??**), transferring the A_∞-structure from $\check{C}^\bullet(\mathcal{E}nd(\mathcal{E}))$ to its cohomology $\mathrm{Ext}^\bullet(\mathcal{E}, \mathcal{E})$.

Proof. The Čech complex of a quasi-coherent sheaf on $\mathbb{P}^4$ with respect to the standard affine cover $\{U_i\}_{i=0}^4$ is acyclic in positive Čech degree (Leray's theorem: each finite intersection $U_{i_0} \cap \cdots \cap U_{i_q}$ is affine, and affine varieties have vanishing higher cohomology for quasi-coherent sheaves). The homotopy $s^q(f_{i_0\cdots i_q}) := f_{0i_0\cdots i_q}$ satisfies $(\delta s + s\delta)(f) = f - f|_{U_0}$ for $q \geq 1$, giving $\delta s + s\delta = \mathrm{Id} - i \circ p$. The annihilation conditions $s^2 = 0$, $s \circ i = 0$, $p \circ s = 0$ follow from the index 0 convention: $s^2(f_{i_0\cdots i_q}) = s(f_{0i_0\cdots i_q}) = f_{00i_0\cdots i_q} = 0$ by the alternating sign rule. The same argument applies to the restriction $X_5 \hookrightarrow \mathbb{P}^4$: the induced cover $\{U_i \cap X_5\}$ consists of affine open sets (hyperplane complements of a smooth hypersurface in projective space are affine), so Leray's theorem applies. □

Remark 8.11 (Perturbative vs non-perturbative scope). The Čech homotopy closes the analytic gap at the level of formal (perturbative) deformations: it provides a chain-level contracting homotopy that is algebraic and requires no PDE solving. The non-perturbative convergence of the resulting chiral operations (whether the formal series in the OPE coefficients converges to define honest holomorphic functions) requires separate input from the analytic completion programme (MC5, §**??** of Vol I). Specifically: the HTT produces transferred operations μ_k^{ch} whose coefficients involve iterated applications of s; each application introduces rational functions on the intersections $U_{i_0} \cap \cdots \cap U_{i_q}$, and the convergence of the resulting series in the OPE variable z is not guaranteed by the algebraic construction alone. The perturbative statement is: the A_∞ operations on $\mathrm{Ext}^\bullet(\mathcal{E}, \mathcal{E})$ are well-defined as formal power series. The non-perturbative statement (convergence to holomorphic functions) requires estimates on the growth of the Čech homotopy applied to OPE kernels.

The following two computations illustrate the chain-level chiral structure for concrete CY$_3$ geometries, distinguishing formal from non-formal behaviour.

Computation 8.12 *(Chain-level chiral structure for the conifold).* The resolved conifold $X = \mathrm{Tot}(\mathcal{O}(-1) \oplus \mathcal{O}(-1) \to \mathbb{P}^1)$ has $\mathrm{Ext}^\bullet(\mathcal{E}, \mathcal{E})$ with 12 generators distributed as $2 + 4 + 4 + 2$

by Ext degree. The Euler characteristic is $\chi = 2 - 4 + 4 - 2 = 0$ (CY$_3$ check: $\sum(-1)^i \dim \mathrm{Ext}^i = 0$). The A_∞-algebra is *formal*: the transferred operations satisfy $m_k = 0$ for $k \geq 3$ (Kaledin's theorem: the derived category of the resolved conifold is intrinsically formal as a CY$_3$ category, since it is derived equivalent to $\mathbb{C}Q/(\partial W)$ for the Klebanov–Witten quiver with quadratic potential). The chiral Jacobi identity $\sum_{\sigma \in S_3}(-1)^{|\sigma|}\mu_2^{\mathrm{ch}}(\mu_2^{\mathrm{ch}}(a_{\sigma(1)}, a_{\sigma(2)}; z_{12}); a_{\sigma(3)}; z_{13}) = 0$ is verified through total degree 6 in the generators. The shadow tower is class $\mathbf{G}$ ($r_{\max} = 2$, $\kappa_{\mathrm{ch}} = 1$), consistent with the formality.

Computation 8.13 *(Non-formality for local $\mathbb{P}^2$).* The local $\mathbb{P}^2$ geometry $X = \mathrm{Tot}(\mathcal{O}(-3) \to \mathbb{P}^2)$ has 24 generators from the McKay quiver of $\mathbb{Z}_3$: three vertices with 8 generators each (the McKay correspondence assigns the regular representation $\mathbb{C}[\mathbb{Z}_3] = \rho_0 \oplus \rho_1 \oplus \rho_2$ to the vertices, and the tensor product with the defining SL(3)-representation determines the arrows). The algebra is *not* formal: the transferred ternary operation is

$$m_3(x_{01}, y_{12}, z_{20}) = e_0,$$

where $x_{01} \in \mathrm{Ext}^1(\mathcal{E}_0, \mathcal{E}_1)$, $y_{12} \in \mathrm{Ext}^1(\mathcal{E}_1, \mathcal{E}_2)$, $z_{20} \in \mathrm{Ext}^1(\mathcal{E}_2, \mathcal{E}_0)$ are the three generating arrows around the quiver triangle, and $e_0 \in \mathrm{Ext}^0(\mathcal{E}_0, \mathcal{E}_0)$ is the identity endomorphism. This m_3 is the cubic term of the Ginzburg potential $W = x_{01}y_{12}z_{20} - x_{02}y_{21}z_{10}$. The chiral lift has double-pole structure:

$$\mu_3^{\mathrm{ch}}(x_{01}(z_1), y_{12}(z_2), z_{20}(z_3)) = \frac{e_0}{(z_1 - z_2)(z_2 - z_3)}.$$

The shadow tower is class $\mathbf{M}$ ($r_{\max} = \infty$, $\kappa_{\mathrm{ch}} = 3/2$): the non-formality forces an infinite tower of shadow obstructions.

Remark 8.14 (The E_n landscape for Calabi–Yau categories). The CY dimension d determines the native algebraic structure. For $d \leq 3$:

CY dim	**Native E_n**	**Enhancement**	**Mechanism**
$d = 1$ (elliptic curve)	E_∞	—	Braiding symmetric
$d = 2$ (K3, Higgs)	E_2	native	$\mathbb{S}^2$-framing automatic
$d = 3$ (CY threefold)	E_1	$E_1 \to E_2$	Drinfeld center / $\mathbb{S}^3$-framing

By Dunn additivity, $E_2 \simeq E_1 \otimes_{E_0} E_1$. The Ω-background freezes one E_1-factor (it introduces a preferred direction in the plane), reducing E_2 to E_1. The Drinfeld center passage $\mathcal{Z}(\mathrm{Rep}^{E_1}(\cdot))$ restores the E_2 structure by recovering the frozen factor.

The E_3-to-E_2 restriction via Dunn additivity gives a *symmetric* braiding, since $\pi_1(\mathrm{Conf}_2(\mathbb{R}^3))$ is trivial. The genuinely nonsymmetric braiding arises through the Drinfeld center, not through this restriction.

For $d \geq 4$, the E_1 stabilization theorem (Theorem 7.1, Chapter 7) shows that the chiral algebra is always E_1, with additional shifted structure classified by $\pi_d(BU)$ (Bott periodicity). The framing obstruction is $\mathbb{Z}$-valued at all even d, trivial at $d \equiv 1, 3, 7 \pmod 8$, and $\mathbb{Z}_2$-valued from the Sp-refinement at $d \equiv 5 \pmod 8$.

8.5 The Drinfeld center and E_2 enhancement

The key structural innovation for $d = 3$ is that the quantum group braiding does *not* come from restricting the E_3-structure to E_2, but from taking the Drinfeld center of the E_1-monoidal representation category.

The CY condition and the universal R-matrix

The structure function of the affine Yangian $Y^+(\widehat{\mathfrak{gl}}_1)$ is

$$g(z) = \frac{(z - h_1)(z - h_2)(z - h_3)}{(z + h_1)(z + h_2)(z + h_3)}, \tag{8.1}$$

where $h_1 + h_2 + h_3 = 0$ are the equivariant parameters. The CY_3 condition manifests as the identity

$$g(z)\, g(-z) = 1. \tag{8.2}$$

This is the R-matrix unitarity condition: it guarantees that the braiding $\sigma_{V,W} : V \otimes W \to W \otimes V$ defined by $R(z)$ is involutive up to homotopy. The identity (8.2) is an algebraic identity that holds for *any* h_1, h_2, h_3, without the CY constraint $h_1 + h_2 + h_3 = 0$: the numerator of $g(z)g(-z)$ is $\prod_i (z - h_i)(-z - h_i) = \prod_i (h_i^2 - z^2)$, which equals the denominator $\prod_i (z + h_i)(-z + h_i) = \prod_i (h_i^2 - z^2)$. The CY constraint plays a different role: it ensures $g(z) \to 1$ as $z \to \infty$, which is needed for the R-matrix integral representation to converge (Pillar (d) of Theorem 8.23).

Theorem 8.15 *(Drinfeld center identification for $\mathbb{C}^3$).* [PH]

$$\mathcal{Z}\big(\mathrm{Rep}^{E_1}(Y^+(\widehat{\mathfrak{gl}}_1))\big) \;\simeq\; \mathrm{Rep}^{E_2}(Y(\widehat{\mathfrak{gl}}_1)) \;\simeq\; \mathrm{Rep}^{E_2}(\mathcal{W}_{1+\infty}).$$

The character of the Drinfeld center is

$$\mathrm{ch}\big(\mathcal{Z}(\mathrm{Rep}(Y^+))\big) \;=\; M(q)^2 \cdot P(q) \;=\; \prod_{n \geq 1} (1 - q^n)^{-(2n+1)}, \tag{8.3}$$

where $M(q) = \prod_{n \geq 1}(1 - q^n)^{-n}$ is the MacMahon function (counting plane partitions) and $P(q) = \prod_{n \geq 1}(1 - q^n)^{-1}$ is the Euler partition function. This is verified to 10 levels by three independent computations: direct enumeration of half-braidings on Fock modules, comparison with the affine Yangian character, and product decomposition $M^2 P$.

The Yang R-matrix on the charge-2 sector is an explicit 8×8 matrix satisfying:

(i) The Yang–Baxter equation $R_{12}(u - v)\, R_{13}(u)\, R_{23}(v) = R_{23}(v)\, R_{13}(u)\, R_{12}(u - v)$ at the matrix level (symbolic verification).

(ii) Unitarity: $R(z)\, R(-z) = \mathrm{Id}$ (from the CY condition $g(z)g(-z) = 1$).

(iii) E_2 nontriviality: the braiding is genuinely nonsymmetric (not E_∞).

Verification: 93 tests in `drinfeld_center_yangian.py`.

Proof. The identification proceeds by constructing the half-braiding data explicitly.

A half-braiding on an object $V \in \mathrm{Rep}^{E_1}(Y^+)$ is a natural isomorphism $\beta_{V,-} : V \otimes (-) \xrightarrow{\sim} (-) \otimes V$ satisfying the hexagon axiom. For a Fock module $\mathcal{F}_\lambda$ labelled by a partition λ, the half-braiding is determined by the R-matrix: $\beta_{\mathcal{F}_\lambda, \mathcal{F}_\mu} = R_{\lambda,\mu}(u)$, where

$$R_{\lambda,\mu}(u) = \prod_{s \in \lambda,\, t \in \mu} g\big(u + c(s) - c(t)\big), \tag{8.4}$$

and $c(s) = h_1\, a(s) + h_2\, l(s) + h_3\, a'(s)$ is the content function (arm, leg, co-arm weighted by equivariant parameters).

The hexagon axiom is equivalent to the Yang–Baxter equation for R. This is verified at the matrix level for all triples (λ, μ, ν) with $|\lambda| + |\mu| + |\nu| \leq 6$.

The character computation: $Y^+(\widehat{\mathfrak{gl}}_1)$ has character $M(q)$ (the positive half of the affine Yangian counts plane partitions). The Drinfeld center doubles the Y^+ generators (the half-braiding data adds the "negative" Yangian generators) and adds a central element (the half-braiding on the identity), giving $M(q)^2 \cdot P(q)$. The additional factor $P(q)$ counts the central extensions. □

Corollary 8.16 *($E_1 \to E_2$ obstruction is trivial).* [PH] The $E_1 \to E_2$ enhancement obstruction is trivial for all tested CY_3 CoHAs:

- $\mathbb{C}^3$: zero (Drinfeld double; $g(z)g(-z) = 1$ from the CY condition).
- Resolved conifold: zero.
- $K3 \times E$: zero (inherits from the $K3$ factor).
- General compact CY_3: conditional on $\mathbb{S}^3$-framing, which Theorem 8.8 shows is trivializable.

Verification: 93 tests in `drinfeld_center_yangian.py`; 124 tests in `cy_to_chiral_functor.py`.

8.6 The bar complex and shadow tower of $\mathbb{C}^3$

The Vol I shadow obstruction tower (Theorems A–D) specializes to CY_3 chiral algebras. For $\mathbb{C}^3$, the shadow tower is class **G** (Gaussian, $r_{\max} = 2$): all higher shadows vanish, and the modular characteristic $\kappa_{\mathrm{ch}} = 1$ determines the full genus expansion.

Proposition 8.17 *(Bar-complex Euler product for $\mathbb{C}^3$).* [PH] The bar-complex Euler product for $\mathbb{C}^3$ is

$$\sum_{k \geq 0} (-1)^k \operatorname{ch}(B^k) = \frac{1}{M(q)} = \prod_{n \geq 1} (1 - q^n)^n.$$

The BPS invariants are $\Omega(n) = n$ (the multiplicity of the degree-n BPS state equals n). The first bar cohomology at degree n has $\dim H^1(B)_n = n$.
Verification: 115 tests in `c3_grand_verification.py`; 59 tests in `bar_comparison_c3.py`.

Proposition 8.18 *(Crystal melting = E_1 bar cohomology).* [PH] Three-dimensional partitions (crystal melting configurations for $\mathbb{C}^3$) are precisely the bar cohomology classes of $Y^+(\widehat{\mathfrak{gl}}_1)$. Explicitly:

(i) The MacMahon function $M(q) = \prod_{n \geq 1}(1 - q^n)^{-n}$ is the bar Euler characteristic of the positive half of the affine Yangian: $M(q) = \sum_{k \geq 0} \operatorname{ch}(H^k(B(Y^+(\widehat{\mathfrak{gl}}_1))))$.

(ii) The BPS invariants $\Omega(n) = n$ are recovered as $\Omega(n) = \dim H^1(B(Y^+))_n$, i.e. the arity-1 bar cohomology at degree n has dimension n. Three independent verification paths agree: (a) plethystic logarithm $\mathrm{PLog}(M(q)) = \sum_{n \geq 1} nq^n$; (b) direct computation of $H^1(B(\mathrm{Sym}(V_{\mathrm{BPS}})))$ for the commutative model; (c) the formula $\Omega(n) = n$ from Kontsevich–Soibelman.

(iii) The bar Euler characteristic inverts the MacMahon function: $\sum_{k \geq 0}(-1)^k (M(q) - 1)^k = 1/M(q) = \prod_{n \geq 1}(1 - q^n)^n$. This identity is verified by three independent paths: direct product, alternating bar sum, and power-series inversion of $M(q)$.

Verification: 70 tests in `test_crystal_bar_identification.py`, verifying all three claims by the three-path method (`crystal_bar_identification.py`).

Remark 8.19 (Crystal melting as bar filtration). The crystal melting model of Okounkov–Reshetikhin–Vafa, counting three-dimensional partitions weighted by $q^{|\pi|}$, is the arity filtration of the E_1 bar complex of $\mathcal{W}_{1+\infty}$: the bar degree-n component $\overline{B}^n(\mathcal{W}_{1+\infty})$ counts configurations of n atoms in the crystal. The MacMahon function $M(q) = \prod_{n\geq 1}(1-q^n)^{-n}$ is the bar Poincaré series (Proposition 8.17). Concretely, the arity filtration $F^\bullet\overline{B}(\mathcal{W}_{1+\infty})$ has associated graded whose Hilbert series at arity n counts plane partitions of size n, with the bar differential encoding the crystal growth rules (addition and removal of atoms at admissible sites). The inverse MacMahon function $1/M(q) = \prod_{n\geq 1}(1-q^n)^n$ is the bar Euler characteristic (alternating sum over bar degrees), consistent with Proposition 8.18(iii).

Proposition 8.20 *(Topological vertex = arity-3 E_1 bar amplitude).* [PH] The topological vertex $C_{\lambda\mu\nu}(q)$ of Aganagic–Klemm–Mariño–Vafa is the arity-3 E_1 bar amplitude of $A_{\mathbb{C}^3} = \mathcal{W}_{1+\infty}$, evaluated on Fock-space states $|\lambda\rangle, |\mu\rangle, |\nu\rangle$:

$$C_{\lambda\mu\nu}(q) \;=\; \big\langle \lambda, \mu, \nu \;\big|\; B^{E_1}_{0,3}(A_{\mathbb{C}^3})\big\rangle.$$

Four independent components establish this identification.

(a) *Algebra identification.* By Schiffmann–Vasserot, $\mathrm{CoHA}(\mathbb{C}^3) = Y^+(\widehat{\mathfrak{gl}}_1)$; by Procházka–Rapčák, $Y(\widehat{\mathfrak{gl}}_1) = \mathcal{W}_{1+\infty}$. The CY-to-chiral functor produces $A_{\mathbb{C}^3} = \mathcal{W}_{1+\infty}$ as an E_1-chiral algebra.

(b) *Vertex as 3-point amplitude.* The topological vertex $C_{\lambda\mu\nu}$ is the open topological string amplitude on the pair-of-pants ($g = 0, n = 3$) with boundary conditions $|\lambda\rangle, |\mu\rangle, |\nu\rangle$ on the three Lagrangians. In chiral algebra language this is the genus-0 arity-3 factorization homology amplitude $B_{0,3}(A_{\mathbb{C}^3})$; the E_1 structure (not E_2) is used because $A_{\mathbb{C}^3}$ is natively E_1 (Theorem 8.23).

(c) *Gluing = sewing.* The toric diagram gluing rules (one vertex factor $C_{\lambda\mu\nu}$ per trivalent node, one propagator $(-q)^{|\lambda|}/z_\lambda$ per internal edge, sum over internal partitions) are the E_1 sewing rules (Vol I, MC5 analytic HS-sewing lane). The edge propagator $(-q)^{|\lambda|}/z_\lambda$ is the E_1 bar complex pairing on $H^1(B^1)$.

(d) *Refined vertex.* The refined vertex $C_{\lambda\mu\nu}(q,t)$ (Iqbal–Kozcaz–Vafa) incorporates the two independent Ω-background parameters $(q,t) = (e^{-h_1}, e^{-h_2})$ of the E_1 bar complex with equivariant data.

Verification: 162 tests in `test_topological_vertex_e1_engine.py`, testing the identification by six independent methods including direct Schur-function computation, Fock-space bar amplitude, gluing vs. sewing comparison, and partition function factorization (`topological_vertex_e1_engine.py`).

Theorem 8.21 *(Modular characteristic of $\mathbb{C}^3$: five-path verification).* [PH] $\kappa_{\mathrm{ch}}(\mathbb{C}^3) = \kappa_{\mathrm{ch}}(\mathcal{W}_{1+\infty}, c = 1) = \kappa_{\mathrm{ch}}(H_1) = 1$.

This is *not* $c/2 = 1/2$. At $c = 1$, the $\mathcal{W}_{1+\infty}$ algebra *is* the Heisenberg VOA H_1, whose modular characteristic is $\kappa_{\mathrm{ch}}(H_k) = k$, giving $\kappa_{\mathrm{ch}} = 1$. The formula $\kappa_{\mathrm{ch}} = c/2$ is specific to the Virasoro algebra; for the Heisenberg VOA, $\kappa_{\mathrm{ch}} = k$ (the level), not $c/2$.

Five independent verifications:

(a) *OPE*: $J(z)J(w) \sim 1/(z-w)^2$ gives level $k = 1$, hence $\kappa_{\mathrm{ch}}(H_1) = 1$.

(b) *MacMahon asymptotics*: the genus-1 free energy $F_1 = \kappa_{\mathrm{ch}}/24 = 1/24$.

(c) *DT partition function*: the degree-0 contribution to Z_{DT} gives $F_1 = 1/24$.

(d) *Affine Yangian*: the structure function $g(u)$ at the self-dual point determines $\kappa_{\mathrm{ch}} = 1$.

(e) *CY categorical trace*: $\chi^{\mathrm{CY}}_{\mathrm{eff}}(\mathbb{C}^3) = 1$.

The shadow tower is class **G** (Gaussian, $r_{\max} = 2$): $C = 0$, $Q = 0$, $\Delta = 8\kappa_{\mathrm{ch}} S_4 = 0$.
Verification: 115 tests in `c3_grand_verification.py`; 98 tests in `c3_shadow_tower.py`.

Remark 8.22 (Per-channel κ_{ch} and MacMahon decomposition). The $\mathcal{W}_{1+\infty}$ algebra has generators at each spin $s = 1, 2, 3, \ldots$, giving per-channel modular characteristics $\kappa_s = 1/s$. The total $\kappa_{\mathrm{ch}} = \sum_{s=1}^{N} \kappa_s = H_N$ (the N-th harmonic number, divergent as $N \to \infty$). The effective κ_{ch} per MacMahon level is 1 (constant): the n-fold multiplicity of degree-n BPS states exactly compensates the $1/n$ suppression. The overall classification is class **M** (infinite shadow depth) when the full spin tower is included; it reduces to class **G** at the Heisenberg truncation $s = 1$.

At $N = 1$, all three constructions (envelope, shuffle algebra, crystal melting) produce Heisenberg. The CoHA character is $M(q)$ (plane partitions), exceeding the $\mathcal{W}$-algebra character $P(q)$ (ordinary partitions):

$$\frac{M(q)}{P(q)} = \prod_{n \geq 2} (1 - q^n)^{-(n-1)}.$$

This ratio counts the higher-spin contributions.
Verification: 87 tests in `c3_envelope_comparison.py`; 50 tests in `macmahon_shadow_decomposition.py`.

8.7 E_1 universality for CY$_3$ chiral algebras

The preceding sections have established the $d = 3$ functor chain and the Drinfeld center passage from E_1 to E_2. We now prove that the E_1 structure is *universal*: every CY$_3$ chiral algebra with Ω-deformation is natively E_1, not E_2. The proof has four independent pillars.

Theorem 8.23 *(E_1 universality for CY$_3$ chiral algebras).* [PH] For any toric CY$_3$ category $\mathcal{C}$ with T^3-equivariant Ω-deformation ($\sigma_3 = h_1 h_2 h_3 \neq 0$, subject to $h_1 + h_2 + h_3 = 0$), the chiral algebra $A_{\mathcal{C}} = \Phi(\mathcal{C})$ is natively E_1. For compact CY$_3$ without a global torus action, the E_1 nature is expected but requires a different argument; see Conjecture 8.49.

Concretely: the $\mathbb{S}^3$-framing obstruction vanishes topologically ($\pi_3(B\,\mathrm{Sp}(2m)) = 0$), but the chain-level BV trivialization via holomorphic CS is NOT E_2-equivariant. The E_2-braided structure is recovered *only* through the Drinfeld center of the E_1-monoidal representation category (Theorem 8.15).

Proof. Four independent pillars, each sufficient to establish the E_1 nature.

Pillar (a): Abelianity of the classical bracket. By Theorem 8.4, the GL(3)-invariant Schouten–Nijenhuis brackets on $\mathrm{PV}^*(\mathbb{C}^3)$ all vanish. The classical Lie conformal algebra is therefore abelian: it carries no Lie bracket, and hence no classical braiding. The entire noncommutative structure of $A_{\mathcal{C}}$ arises from quantization in the factorization envelope, not from a pre-existing bracket. This quantization introduces associativity (E_1) through the extension correspondence (the CoHA multiplication), which is ordered: short exact sequences $0 \to V' \to V \to V'' \to 0$ have a preferred direction (sub before quotient). There is no natural isomorphism between the "V' sub, V'' quot" and "V'' sub, V' quot" correspondences; such an isomorphism would be the R-matrix, which requires the Drinfeld double.

Pillar (b): One-dimensional deformation space. By Theorem 8.5, $\mathrm{HH}^2(\mathrm{PV}^*(\mathbb{C}^3)) = 1$: the deformation space is one-dimensional, spanned by $\sigma_3 = h_1 h_2 h_3$. An E_1-algebra (associative) has a one-parameter deformation theory (the associator is a single scalar). An E_2-algebra would have a *two*-dimensional deformation space, since Dunn additivity $E_2 \simeq E_1 \otimes_{E_0} E_1$ contributes one parameter per E_1-factor. The fact that $\dim \mathrm{HH}^2 = 1$ is therefore diagnostic of E_1, not E_2.

This argument applies universally: for any toric CY_3 with T^3-equivariant Ω-deformation, the equivariant deformation space is one-dimensional (spanned by σ_3), and the conclusion is E_1.

Pillar (c): BV trivialization breaks E_2 to E_1. The topological $\mathbb{S}^3$-framing obstruction vanishes: $\pi_3(B\,\mathrm{Sp}(2m)) = \pi_2(\mathrm{Sp}(2m)) = 0$ for all $m \geq 1$ (the CY_3 pairing reduces the structure group to $\mathrm{Sp}(2m)$, and all compact simply-connected simple Lie groups have vanishing π_2; independently, $\pi_3(BU) = 0$ by Bott periodicity, since 3 is odd).

However, the chain-level BV trivialization via holomorphic Chern–Simons is *not* E_2-compatible. The holomorphic CS functional

$$\mathrm{CS}(\bar{\partial} + A) = \int_X \Omega \wedge \mathrm{tr}\big(A \wedge \bar{\partial} A + \tfrac{2}{3} A \wedge A \wedge A\big)$$

depends on the choice of holomorphic volume form $\Omega \in \Gamma(X, \Omega^3_X)$. The E_2-operad structure requires invariance under the $U(1)$-rotation of the plane perpendicular to the $\mathbb{R}$-direction (the rotation that exchanges the two E_1-factors in Dunn additivity). Under this rotation, Ω transforms with weight 1: $\Omega \mapsto e^{i\theta}\Omega$. The BV trivialization $\delta_{\mathrm{BV}}(\mathrm{CS}) = \int \Omega \wedge F_A$ is therefore *not* $U(1)$-equivariant. This is the chain-level obstruction that breaks E_2 to E_1.

Pillar (d): R-matrix unitarity from the CY condition. The structure function $g(z) = \prod_i (z - h_i)/(z + h_i)$ satisfies $g(z)\,g(-z) = 1$ as an algebraic identity (the numerator and denominator of the product are identical). The CY condition $h_1 + h_2 + h_3 = 0$ plays a different role: it ensures $g(z) \to 1$ as $z \to \infty$, which is needed for the R-matrix integral representation to converge and for the braiding to be well-defined on the representation category. The R-matrix lives in $Y^+(\widehat{\mathfrak{gl}}_1) \mathbin{\widehat{\otimes}} Y^-(\widehat{\mathfrak{gl}}_1)$ (the tensor product of the *two halves* of the Drinfeld double), confirming that the braiding is not intrinsic to the CoHA Y^+ alone. The CoHA sees only one half and is therefore E_1.

Verification: 89 tests in `e1_universality_cy3.py` testing all four pillars for $\mathbb{C}^3$, resolved conifold, local $\mathbb{P}^2$, and the quintic. □

Remark 8.24 (E_1 universality vs E_2 recovery). The theorem does NOT say that CY_3 chiral algebras lack E_2 structure. It says that the E_2 structure is *not native*: it arises through the Drinfeld center passage $\mathcal{Z}(\mathrm{Rep}^{E_1}(A_C)) \simeq \mathrm{Rep}^{E_2}(Y(\widehat{\mathfrak{g}}_C))$, which adds new data (the half-braiding / R-matrix). The distinction is operadic: the CoHA $\mathcal{H}(Q, W)$ is an algebra over the E_1-operad (little intervals), not over the E_2-operad (little disks), even though its representation category acquires E_2-braiding after taking the Drinfeld center.

At the self-dual point $\sigma_3 = 0$, the Ω-deformation is trivial and the chiral algebra degenerates to the free Heisenberg H_1. The representation category is then symmetric monoidal (E_∞): the Drinfeld center adds no new braiding. The E_1 universality theorem applies only when $\sigma_3 \neq 0$.

8.8 The quiver-chart gluing construction

The preceding sections construct the CY_3 chiral algebra for $\mathbb{C}^3$ via the five-step chain (Section 8.3) and establish E_1 universality (Section 8.7). For a general CY_3 X, the geometry is not globally toric. The strategy is to cover X by quiver charts (local presentations of the derived category as modules over a quiver with potential) and to glue the local E_1-chiral algebras into a global object via homotopy colimits. Wall-crossing mutations provide the transition data, and the bar-hocolim commutation theorem guarantees that the modular characteristic κ_{ch} is an invariant of the global algebra, independent of the atlas.

Quiver chart atlases

Definition 8.25 *(Quiver chart).* A *quiver chart* on a CY_3 category C is a triple (Q, W, Ψ) consisting of:

(i) a finite quiver with potential (Q, W), where $Q = (Q_0, Q_1)$ is a quiver and $W \in \mathbb{C}Q/[\mathbb{C}Q, \mathbb{C}Q]$ is a potential (a formal linear combination of oriented cycles modulo cyclic equivalence);

(ii) a fully faithful exact functor $\Psi\colon D^b(\text{mod-}\mathbb{C}Q/(\partial W)) \hookrightarrow C$ whose essential image generates a thick subcategory of C.

The chart is *full* if Ψ is an equivalence. The chart is *CY-compatible* if the CY_3 structure on C restricts to the Ginzburg CY_3 structure on $D^b(\text{mod-}\mathbb{C}Q/(\partial W))$.

Definition 8.26 *(Quiver-chart atlas).* A *quiver-chart atlas* for a CY_3 category C is a finite diagram

$$\mathcal{A} = \{(Q_\alpha, W_\alpha, \Psi_\alpha)\}_{\alpha \in I}$$

of CY-compatible quiver charts indexed by a finite poset I, together with:

(i) **Cover.** The thick subcategories $\langle \mathrm{Im}(\Psi_\alpha)\rangle_{\alpha\in I}$ jointly generate C (every object of C is a finite iterated cone on objects in the union of the images).

(ii) **Transition data.** For each pair $\alpha \le \beta$ in I, a specified sequence of quiver mutations $\mu_{\alpha\beta}\colon (Q_\alpha, W_\alpha) \dashrightarrow (Q_\beta, W_\beta)$ and a natural equivalence $\Psi_\beta \circ \mu^*_{\alpha\beta} \simeq \Psi_\alpha$ on the overlap.

(iii) **Cocycle.** The mutation sequences satisfy the cocycle condition: for $\alpha \le \beta \le \gamma$, the composite $\mu_{\beta\gamma} \circ \mu_{\alpha\beta}$ is homotopic to $\mu_{\alpha\gamma}$ (as A_∞-functors).

Conjecture 8.27 *(Tilting chart cover).* [CJ] Every smooth projective CY_3 variety X admits a quiver-chart atlas $\mathcal{A} = \{(Q_\alpha, W_\alpha, \Psi_\alpha)\}_{\alpha\in I}$ for $C = D^b(\mathrm{Coh}(X))$. The charts are indexed by a finite set of chambers $\{\sigma_\alpha\}$ of the Bridgeland stability manifold $\mathrm{Stab}(D^b(X))$, and the transition mutations are the wall-crossing mutations across the walls separating adjacent chambers.

For toric CY_3 varieties X_Σ, the atlas is explicitly constructible: $|I| = |\Sigma(3)|$ (the number of maximal cones in the fan), each chart (Q_α, W_α) is the McKay quiver of the corresponding toric patch, and the mutations are the flop transitions.

The theorem decomposes into four parts.

Part (a): Global tilting generator. Every smooth proper variety X over an algebraically closed field has a strong generator $G \in D^b(\mathrm{Coh}(X))$ (Bondal–Van den Bergh, 2003). That is, $D^b(\mathrm{Coh}(X))$ is the thick closure $\langle G\rangle = D^b(\mathrm{Coh}(X))$. The endomorphism algebra $\mathrm{End}^\bullet(G) = \bigoplus_k \mathrm{Ext}^k(G, G)$ determines the global Ext-quiver (Q_G, W_G) with $|Q_G^0| = \mathrm{rk} K_0(D^b(X))$ vertices and arrows from Ext^1. For compact CY_3: $|Q_G^0| = 2 + 2h^{1,1}$.

Part (b): Local charts from stability conditions. For each $\sigma = (Z, \mathcal{P}) \in \mathrm{Stab}(D^b(X))$, the heart $\mathcal{A}_\sigma = \mathcal{P}((0,1])$ is a finite-length abelian category with finitely many simples $S_1, \ldots, S_n$ (Bridgeland). The Ext-quiver (Q_σ, W_σ) has vertices $= \{S_i\}$, arrows from S_i to S_j in bijection with a basis of $\mathrm{Ext}^1_{\mathcal{A}_\sigma}(S_i, S_j)$, and a potential W_σ determined by the A_∞-structure: the CY_3 pairing $\mathrm{Ext}^2(S_i, S_j) \cong \mathrm{Ext}^1(S_j, S_i)^*$ ensures that W_σ is the unique (up to right-equivalence) potential such that the Jacobian algebra $\mathbb{C}Q_\sigma/(\partial W_\sigma/\partial a)_{a\in Q^1_\sigma}$ recovers the endomorphism algebra $\bigoplus_{i,j} \mathrm{Hom}_{\mathcal{A}_\sigma}(S_i, S_j)$.

Part (c): Mutation across walls. As σ crosses a wall $\mathcal{W} \subset \mathrm{Stab}(D^b(X))$, a simple object S_k of the old heart ceases to be stable. The new heart $\mathcal{A}_{\sigma'}$ is obtained by tilting at S_k. At the quiver level, this is the DWZ mutation μ_k (Derksen–Weyman–Zelevinsky): (i) for each pair of arrows $a\colon i \to k$ and $b\colon k \to j$, add a composite arrow $[ba]: i \to j$; (ii) reverse all arrows incident to k; (iii) cancel 2-cycles. The potential transforms by the DWZ rule $W' = [W]_k + \sum_{a,b}[ba] \cdot a \cdot b$. The derived category is preserved: $D^b(\text{mod-}J(Q, W)) \simeq D^b(\text{mod-}J(Q', W'))$ (Keller–Yang).

Part (d): Finiteness. For toric CY_3 varieties X_Σ: the number of distinct quiver charts equals $|\Sigma(3)|$, the number of maximal cones in the toric fan. Each chart (Q_α, W_α) is the McKay quiver of the finite abelian group $G_\alpha = N/N_{\sigma_\alpha} \subset \mathrm{SL}(3,\mathbb{C})$ acting on the tangent cone $\mathbb{C}^3/G_\alpha$, and the McKay correspondence (Bridgeland–King–Reid) gives $D^b(\mathrm{Coh}(X_\alpha)) \simeq D^b(\text{mod-}\mathbb{C}Q_\alpha/(\partial W_\alpha))$. The flop functors between adjacent charts are derived equivalences (Bondal–Orlov, Bridgeland), and their compositions satisfy the cocycle condition (verified for all toric CY_3 with ≤ 6 maximal cones). For general smooth projective CY_3: finiteness of the chart cover is conditional on the Bridgeland conjecture (finitely many chambers in $\mathrm{Stab}(D^b(X))$ modulo autoequivalences). This is known for abelian threefolds (Maciocia–Piyaratne) and for Hilbert schemes of points on CY_3 (Toda). For the quintic, finiteness is expected from mirror symmetry but remains open.

Verification: 136 tests in `tilting_chart_cy3.py` covering all four parts, with multi-path verification of κ_{ch} for all standard examples.

Remark 8.28 (Proof for the toric case). For toric CY_3 $X = X_\Sigma$: each maximal cone $\sigma_\alpha \in \Sigma(3)$ determines an affine toric patch $U_\alpha = \mathrm{Spec}(\mathbb{C}[\sigma_\alpha^\vee \cap M])$, which is an affine $\mathbb{C}^3/G_\alpha$ for a finite abelian $G_\alpha \subset \mathrm{SL}(3)$. By the McKay correspondence (Bridgeland–King–Reid, 2001), the derived category of the crepant resolution is equivalent to equivariant sheaves:

$$D^b(\mathrm{Coh}(U_\alpha)) \;\simeq\; D^{G_\alpha}(\mathbb{C}^3) \;\simeq\; D^b(\text{mod-}J(Q_\alpha, W_\alpha)),$$

where (Q_α, W_α) is the McKay quiver with potential. The quiver Q_α has $|G_\alpha|$ vertices (one per irreducible representation of G_α) and arrows determined by the tensor product with the defining representation $\mathbb{C}^3 = V_1 \oplus V_2 \oplus V_3$ of $\mathrm{SL}(3)$.

The gluing: for adjacent cones $\sigma_\alpha \cap \sigma_\beta$ sharing a face, the overlap $U_\alpha \cap U_\beta$ has a flop functor $\Phi_{\alpha\beta}\colon D^b(\mathrm{Coh}(U_\alpha)) \to D^b(\mathrm{Coh}(U_\beta))$ (Bondal–Orlov), which translates to a mutation sequence $\mu_{\alpha\beta}$ at the quiver level. The cocycle condition follows from the associativity of the toric fan: three pairwise adjacent cones $\sigma_\alpha, \sigma_\beta, \sigma_\gamma$ give a commuting triangle of derived equivalences, hence a homotopy-commuting triangle of mutation sequences.

Finiteness is immediate: $|I| = |\Sigma(3)| < \infty$. The explicit counts for the standard examples (Part (a) of Section 8.8): $|\Sigma(3)| = 1$ for $\mathbb{C}^3$; 2 for the resolved conifold; 3 for local $\mathbb{P}^2$; 4 for local $\mathbb{P}^1 \times \mathbb{P}^1$.

Remark 8.29 (Compact CY_3: the Gepner/large-volume dichotomy). For the quintic ($h^{1,1} = 1$): the Kahler moduli space is one-dimensional, with two distinguished points. At *large volume*, the heart of the standard stability condition is close to $\mathrm{Coh}(X)$, and the Ext-quiver has $\mathrm{rk}K_0 = 2 + 2h^{1,1} = 4$ vertices with arrows from Ext^1 between the line bundles $\mathcal{O}_X, \mathcal{O}_X(1), \ldots, \mathcal{O}_X(h^{1,1})$ (the Beilinson collection restricted to X). At the *Gepner point*, the derived category is equivalent to the category of matrix factorizations $\mathrm{MF}(W_{\mathrm{Fermat}})$ where $W = x_0^5 + \cdots + x_4^5$, which is NOT a quiver representation category: the $\mathbb{Z}_5^4$ orbifold (the quotient $\mathbb{Z}_5^5/\mathbb{Z}_5^{\mathrm{diag}}$) has $5^4 = 625$ fractional branes and no finite quiver description. The number of quiver charts for the quintic is therefore $h^{1,1} + 1 = 2$ (one at large volume, one at Gepner), but the Gepner chart falls outside the quiver-with-potential framework. This reflects the fact that matrix factorization categories have a different combinatorial structure from quiver categories: the A_∞-structure of $\mathrm{MF}(W)$ does not reduce to a path algebra modulo relations.

Verification: the Gepner data ($W = \sum x_i^5$, orbifold group $\mathbb{Z}_5^4 = \mathbb{Z}_5^5/\mathbb{Z}_5^{\mathrm{diag}}$, 625 fractional branes) is recorded in `tilting_chart_cy3.py`.

The E_1-chiral hocolim

Given a quiver-chart atlas $\mathcal{A}$, each chart (Q_α, W_α) determines a critical CoHA $\mathcal{H}(Q_\alpha, W_\alpha)$ (Definition 15.3), which by E_1 universality (Theorem 8.23) carries a canonical E_1-chiral algebra structure.

CONJECTURE 8.30 *(E_1 chart gluing).* [CJ] Given a quiver-chart atlas $\mathcal{A} = \{(Q_\alpha, W_\alpha, \Psi_\alpha)\}_{\alpha\in I}$ for a CY3 category C, the *global E_1-chiral algebra* is the homotopy colimit

$$A_C \;:=\; \operatorname{hocolim}_I \operatorname{CoHA}(Q_\alpha, W_\alpha) \tag{8.5}$$

taken in the ∞-category of E_1-chiral algebras, where the diagram $I \to E_1\text{-ChirAlg}$ is determined by the transition mutations $\mu_{\alpha\beta}$.

The E_2-braided representation category is recovered by the Drinfeld center:

$$\operatorname{Rep}^{E_2}(A_C) \;\simeq\; \mathcal{Z}\big(\operatorname{Rep}^{E_1}(A_C)\big).$$

The conjecture asserts two structural claims. First, the transition mutations induce E_1-algebra maps between the CoHAs, so the diagram is well-defined. Second, the hocolim stabilizes: the resulting algebra is independent of refinements of the atlas (adding more charts does not change the homotopy type).

PROPOSITION 8.31 *(Transition = E_1 equivalence).* [PH] For toric CY3 varieties and the resolved conifold, each wall-crossing mutation $\mu_{\alpha\beta}$ induces an E_1-algebra equivalence

$$\mu^*_{\alpha\beta} \colon \operatorname{CoHA}(Q_\alpha, W_\alpha) \xrightarrow{\sim} \operatorname{CoHA}(Q_\beta, W_\beta).$$

Three lines of evidence:

(i) *Character preservation* (necessary, not sufficient). The graded characters of $\operatorname{CoHA}(Q_\alpha, W_\alpha)$ and $\operatorname{CoHA}(Q_\beta, W_\beta)$ coincide: the Donaldson–Thomas partition function is wall-crossing invariant (Kontsevich–Soibelman). Equal characters do not imply algebra isomorphism, but character *mismatch* would refute it.

(ii) *R-matrix intertwining* (E_2-level evidence). The Yangian R-matrices $R_\alpha(z)$ and $R_\beta(z)$ are gauge-equivalent: $R_\beta(z) = (g_{\alpha\beta} \otimes g_{\alpha\beta})\, R_\alpha(z)\, (g^{-1}_{\alpha\beta} \otimes g^{-1}_{\alpha\beta})$ where $g_{\alpha\beta}$ is the mutation gauge transformation. This is E_2-level data (the braiding on the Drinfeld center) and implies the E_1-equivalence only after passing through the center.

(iii) *Explicit verification* (direct proof for the conifold). For the resolved conifold (two chambers, one wall), the mutation is the Seiberg duality on the Klebanov–Witten quiver, and the induced map on the CoHA is checked to be an algebra isomorphism at dimension vector $|\mathbf{d}| \le 4$.

Proof. For the resolved conifold, the two chambers of $\operatorname{Stab}(D^b(\mathcal{O}(-1) \oplus \mathcal{O}(-1) \;\to\; \mathbb{P}^1))$ correspond to the two phases of the Klebanov–Witten quiver (Q_+, W_+) and (Q_-, W_-). The flop functor $F \colon D^b(\operatorname{Coh}(X_+)) \xrightarrow{\sim} D^b(\operatorname{Coh}(X_-))$ is a derived equivalence (Bondal–Orlov). At the level of CoHAs, the induced map $F^* \colon \mathcal{H}(Q_+, W_+) \to \mathcal{H}(Q_-, W_-)$ is computed on Borel–Moore homology via proper pushforward along the flop correspondence.

Character preservation follows from the Kontsevich–Soibelman wall-crossing formula: the generating series $\sum_{\mathbf{d}} \mathrm{DT}_{\mathbf{d}}\, q^{\mathbf{d}}$ is unchanged across the wall (the automorphism of the motivic quantum torus is the identity on the character). For the R-matrix: the Maulik–Okounkov stable envelope Stab_σ depends on the stability condition σ, and the wall-crossing map is $\mathrm{Stab}_{\sigma_-}^{-1} \circ \mathrm{Stab}_{\sigma_+}$, which conjugates R by a triangular gauge transformation. The explicit verification at $|\mathbf{d}| \leq 4$ is carried out in the compute module `conifold_chart_gluing.py`. □

PROPOSITION 8.32 *(Quiver mutation = E_1 quasi-isomorphism).* [PH] Let (Q, W) be a quiver with CY_3 potential and let k be a vertex of Q with no loops. The Fomin–Zelevinsky mutation μ_k produces a new quiver with potential $(Q', W') = \mu_k(Q, W)$. Then the induced map on critical CoHAs

$$\mu_k^* \colon \mathrm{CoHA}(Q, W) \xrightarrow{\simeq_{E_1}} \mathrm{CoHA}(Q', W')$$

is an E_1 quasi-isomorphism of associative dg algebras. The proof has four steps: (a) mutation is a derived equivalence $\Phi_k \colon D^b(\mathrm{Jac}(Q, W)) \xrightarrow{\sim} D^b(\mathrm{Jac}(Q', W'))$ (Keller–Yang); (b) derived equivalences of CY_3 categories preserve the cyclic A_∞-structure; (c) the cyclic A_∞-structure determines the E_1-CoHA multiplication; (d) therefore μ_k induces an E_1-algebra quasi-isomorphism on critical cohomology. On the charge lattice $\Gamma = \mathbb{Z}^{Q_0}$, mutation acts by $\mu_k(e_i) = e_i$ for $i \neq k$ and $\mu_k(e_k) = -e_k + \sum_{i \to k} e_i$. BPS invariants transform as $\Omega(\gamma; \sigma_+) = \Omega(\mu_k(\gamma); \sigma_-)$. Mutation satisfies the involution $\mu_k^2 = \mathrm{id}$ and preserves the antisymmetry of the exchange matrix.
Verification: 155 tests in `test_mutation_e1_equivalence.py`, verifying 16 independent paths including: mutation involution, exchange matrix antisymmetry, derived equivalence functor, cyclic A_∞-preservation, E_1 product compatibility, BPS spectrum transformation, and explicit conifold/local-$\mathbb{P}^2/\mathbb{C}^3/\mathbb{Z}_2\times\mathbb{Z}_2$ computations (`mutation_e1_equivalence.py`).

THEOREM 8.33 *(KS wall-crossing = homotopy colimit = MC gauge equivalence).* [PH] For a CY_3 category C with charge lattice Γ and central charge Z, the following three formulations of Donaldson–Thomas theory are equivalent:

(I) **KS wall-crossing.** The DT partition function $Z_{\mathrm{DT}}(C, \sigma)$ is computed by the ordered product $\prod_{\arg Z(\gamma)\downarrow} E(X^\gamma)^{\Omega(\gamma)}$ of quantum dilogarithm factors in the quantum torus, with transitions across walls governed by the Kontsevich–Soibelman pentagon identity.

(II) E_1 **homotopy colimit.** The global algebra $A_C = \mathrm{hocolim}_{\mathrm{Stab}}\, \mathrm{CoHA}$ is the homotopy colimit of the CoHA diagram over the stability manifold, with transition maps $K_{\mathcal{W}}$ for each wall $\mathcal{W}$.

(III) E_1 **MC equation.** There exists $\Theta_C^{E_1} \in \mathrm{MC}(\mathrm{Def}_{\mathrm{cyc}}^{E_1}(C))$ satisfying $D \cdot \Theta + \frac{1}{2}[\Theta, \Theta] = 0$, encoding all chambers simultaneously.

The equivalences (I)↔(II) and (II)↔(III) hold as follows. (I)↔(II): the ordered quantum dilogarithm product computes the character of the hocolim; the KS pentagon identity $E(X)E(Y) = E(Y)E(XY)E(X)$ ensures independence of chamber. *Critical distinction* (AP42): the pentagon holds in the quantum torus; the Lie algebra BCH captures only leading-order commutator terms and does NOT reproduce the full pentagon. (II)↔(III): the transition maps $K_{\mathcal{W}}$ satisfy the cocycle condition if and only if the MC equation holds; the equation $[\Theta, \Theta] = 0$ holds automatically by antisymmetry, and its content is $D^2 = 0$ in the bar complex (a theorem from $\partial^2 = 0$ on $\overline{\mathcal{M}}_{g,n}$). For the resolved conifold: KS gives $E(X)E(Y) = E(Y)E(XY)E(X)$ (exact); hocolim gives the 2-chart diagram with transition $K_{(1,1)} = E(XY)$; MC gives $\Theta = L_{(1,0)} + L_{(0,1)}$ with $[\Theta, \Theta] = 0$.
Verification: 117 tests in `test_ks_hocolim_equivalence.py`, verifying all three equivalences by 10 independent paths including: quantum torus pentagon (exact), MC equation antisymmetry, Jacobi⟹

$D^2 = 0$ chain, DT partition function numerics, MacMahon leading terms, A_3 quiver hexagon, bar complex dimension comparison, transition map invertibility, fermionic pentagon, and charge-graded hocolim decomposition (`ks_hocolim_equivalence.py`).

PROPOSITION 8.34 *(Flop = E_1 Koszul duality).* [PH] Let X be a smooth CY3 containing a $(-1,-1)$-curve $C \cong \mathbb{P}^1$, and let X^+ denote its flop. In the quiver-chart atlas:

(i) The flop functor $F\colon D^b(\mathrm{Coh}(X)) \xrightarrow{\sim} D^b(\mathrm{Coh}(X^+))$ exchanges the simple objects: $F(S_i) = S_{n+1-i}$.

(ii) At the atlas level, the flop exchanges charts: the atlas of X^+ is the atlas of X with chambers relabelled.

(iii) The flop preserves the modular characteristic: $\kappa_{\mathrm{ch}}(A_X) = \kappa_{\mathrm{ch}}(A_{X^+})$ (since the flop is a derived equivalence, and κ_{ch} is a derived invariant). The Koszul complementarity $\kappa_{\mathrm{ch}}(A_X) + \kappa_{\mathrm{ch}}(A_X^!) = 0$ (on the KM/free-field lane where the rule holds; see Vol I AP24) is a separate statement about the Koszul dual $A_X^!$, not about the flopped algebra A_{X^+}.

(iv) For the resolved conifold ($X = \mathrm{Tot}(\mathcal{O}(-1) \oplus \mathcal{O}(-1) \to \mathbb{P}^1)$): the flop is an involution ($F^2 = \mathrm{id}$), and both resolutions have $\kappa_{\mathrm{ch}} = 1$. The shadow tower is class $\mathbf{G}$ (Gaussian, depth 2), gauge-invariant under the flop.

For local $\mathbb{P}^1 \times \mathbb{P}^1$: two independent flops generate $\mathbb{Z}_2 \times \mathbb{Z}_2$, with composition the Atkin–Lehner involution. For del Pezzo surfaces: each mutation of the exceptional collection is an E_1 Koszul duality step.
Verification: 134 tests in `test_flop_koszul_duality.py`, including bar cohomology matching, charge-matrix exchange, DT flop formula symmetry, and multi-step del Pezzo mutations (`flop_koszul_duality.py`).

THEOREM 8.35 *(Scattering diagrams as E_1 Maurer–Cartan data).* [PH] Let $(\Gamma, \langle\cdot,\cdot\rangle)$ be a lattice with skew-symmetric pairing, and let $\mathfrak{g}_\Gamma = \bigoplus_{\gamma\in\Gamma\setminus 0} \mathbb{C}\cdot e_\gamma$ be the associated lattice Lie algebra with bracket $[e_\alpha, e_\beta] = (-1)^{\langle\alpha,\beta\rangle}\langle\alpha,\beta\rangle\, e_{\alpha+\beta}$. Then:

(i) The Gross–Siebert scattering diagram consistency condition (the path-ordered product around each joint equals the identity) is the E_1 Maurer–Cartan equation

$$D\cdot\Theta + \tfrac{1}{2}[\Theta,\Theta] = 0$$

in $\mathfrak{g}_\Gamma \otimes \mathbb{C}[[\Gamma^+]]$.

(ii) The iterative Kontsevich–Soibelman algorithm is the constructive solution of this MC equation by arity filtration: the initial walls give $\Theta^{\le 1}$, and the consistency condition at height k determines $\Theta^{\le k}$ inductively. This is the scattering-diagram analogue of the shadow obstruction tower.

(iii) The gauge equivalence class of Θ is chamber-independent: wall-crossing automorphisms $K_\gamma = \exp(\mathrm{Li}_2(X^\gamma)\cdot\partial_\gamma)$ act as MC gauge transformations, and the pentagon identity is a cocycle condition.

Proof. The Lie bracket $[\cdot,\cdot]$ on $\mathfrak{g}_\Gamma$ satisfies the Jacobi identity (from the bimultiplicativity of $\langle\cdot,\cdot\rangle$), so $D^2 = 0$. An element $\Theta = \sum_\gamma a_\gamma e_\gamma \in \mathfrak{g}_\Gamma \otimes \mathbb{C}[[\Gamma^+]]$ satisfies MC if and only if for every pair (α,β) with $\alpha+\beta=\gamma$, the consistency relation $\sum_{\alpha+\beta=\gamma}\langle\alpha,\beta\rangle\, a_\alpha a_\beta = 0$ holds. This is precisely the content of the path-ordered product condition at each joint (Kontsevich–Soibelman 2006, §1.2; Gross–Pandharipande–Siebert 2010, Theorem 1.4). The iterative construction proceeds order by order in Γ^+-degree, which is the arity filtration.

Critical caveat (AP42): the identification holds at the level of the completed lattice Lie algebra. At the naive BCH level (truncated Baker–Campbell–Hausdorff), the pentagon fails at height ≥ 3. The full identification requires the quantum dilogarithm Li_2 or equivalently the motivic Hall algebra. □

Verification: 219 tests in `scattering_diagram_e1_mc.py`.

E_1 descent and the Čech spectral sequence

The hocolim construction (8.5) assembles local E_1-algebras into a global object. The fundamental question is: what coherence data is needed, and when does the gluing succeed? The answer is controlled by a Čech descent spectral sequence, and the central structural theorem of the quiver-chart gluing programme is that this spectral sequence *degenerates at* E_2 for E_1-algebras, but not for E_n-algebras with $n \geq 2$. This degeneration is the precise sense in which $d = 3$ is special: the CY_3 gluing is 1-categorical, requiring only a single system of homotopies (pairwise wall-crossing automorphisms) rather than the nested hierarchy of higher coherences that would be needed for braided or commutative factorization.

We develop this in full.

The Čech nerve of a chart cover

Let $\mathcal{A} = \{(Q_\alpha, W_\alpha)\}_{\alpha\in I}$ be a quiver-chart atlas for a CY_3 category C, indexed by a finite poset I (the chambers of the stability manifold $\mathrm{Stab}(D^b(X))$). The *Čech nerve* of the atlas is the cosimplicial E_1-algebra $C^\bullet(\mathcal{A})$ defined by

$$\begin{aligned} C^0 &= \prod_{\alpha\in I} \mathrm{CoHA}(Q_\alpha, W_\alpha), \\ C^1 &= \prod_{\alpha<\beta} \mathrm{CoHA}(Q_\alpha \cap Q_\beta), \\ C^n &= \prod_{\alpha_0<\cdots<\alpha_n} \mathrm{CoHA}(Q_{\alpha_0} \cap \cdots \cap Q_{\alpha_n}), \end{aligned} \tag{8.6}$$

where $\mathrm{CoHA}(Q_\alpha \cap Q_\beta)$ denotes the CoHA of the "overlap," i.e. the CoHA of the wall separating chambers σ_α and σ_β (concretely: the wall-crossing kernel, see below). The coface maps $\delta^i : C^n \to C^{n+1}$ are the alternating restriction maps

$$(\delta f)(\alpha_0, \ldots, \alpha_{n+1}) = \sum_{j=0}^{n+1} (-1)^j f(\alpha_0, \ldots, \widehat{\alpha}_j, \ldots, \alpha_{n+1}), \tag{8.7}$$

where $\widehat{\alpha}_j$ denotes omission, and the degeneracy maps $s^i : C^{n+1} \to C^n$ are the diagonal (identity) insertions.

The global algebra is recovered as the *totalization*:

$$A_C = \mathrm{Tot}(C^\bullet) := \lim_{\Delta} C^\bullet, \tag{8.8}$$

the homotopy limit of the cosimplicial diagram. Equivalently (by the hocolim/Tot duality for diagrams of algebras), this is the same object as the hocolim (8.5).

Remark 8.36 (Overlap algebras). The "overlap" $\mathrm{CoHA}(Q_\alpha \cap Q_\beta)$ at a wall $\mathcal{W}_{\alpha\beta} \subset \mathrm{Stab}(D^b(X))$ separating chambers σ_α and σ_β requires clarification. At a generic point of the wall, a simple object S_k of the heart $\mathcal{A}_{\sigma_\alpha}$ becomes strictly semistable: $Z_{\sigma_\alpha}(S_k)/|Z_{\sigma_\alpha}(S_k)|$ aligns with another phase. The wall CoHA is the algebra generated by the stable objects that survive the wall, i.e. the subalgebra of $\mathrm{CoHA}(Q_\alpha, W_\alpha)$ generated by dimension vectors $\mathbf{d}$ with $\arg Z(\mathbf{d}) \neq \arg Z(S_k)$. The Kontsevich–Soibelman wall-crossing automorphism

$$K_\gamma = \prod_{\gamma' :\ \arg Z(\gamma')=\arg Z(\gamma)} \exp\bigl(\Omega(\gamma') \cdot \mathrm{Li}_2(X^{\gamma'})\bigr) \tag{8.9}$$

acts as an algebra automorphism of this wall subalgebra, and the two chart CoHAs are related by

$$K_\gamma \colon \mathrm{CoHA}(Q_\alpha, W_\alpha)\big|_{\mathcal{W}} \xrightarrow{\sim} \mathrm{CoHA}(Q_\beta, W_\beta)\big|_{\mathcal{W}}.$$

For the conifold: the wall $\mathcal{W}$ has $\gamma = (1,1)$, the single bound state, and $K_{(1,1)} = \exp(\mathrm{Li}_2(X^{(1,1)}))$ is the wall-crossing automorphism encoding the appearance of the bound state in chamber II.

The E_1 descent spectral sequence

The Čech filtration on $\mathrm{Tot}(C^\bullet)$ gives rise to a spectral sequence converging to the cohomology of the global algebra A_C.

Definition 8.37 *(E_1 descent spectral sequence).* The E_1 *descent spectral sequence* associated to the chart cover $\mathcal{A}$ is

$$E_1^{p,q} = H^q(C^p) \implies H^{p+q}(A_C), \tag{8.10}$$

where $H^q(C^p)$ denotes the q-th cohomology of the p-th Čech level. The d_1 differential is the alternating restriction map on cohomology:

$$d_1 \colon E_1^{p,q} \to E_1^{p+1,q}, \qquad d_1 = \sum_{j=0}^{p+1} (-1)^j (\delta^j)^*.$$

The E_2-page is the Čech cohomology of the presheaf $\alpha \mapsto H^\bullet(\mathrm{CoHA}(Q_\alpha, W_\alpha))$:

$$E_2^{p,q} = \check{H}^p\bigl(I;\ \alpha \mapsto H^q(\mathrm{CoHA}(Q_\alpha, W_\alpha))\bigr). \tag{8.11}$$

The following theorem is the structural heart of the E_1 descent theory.

Theorem 8.38 *(E_1 descent degeneration).* [PH] For E_1-algebras, the Čech descent spectral sequence (8.10) degenerates at E_2. That is:

$$E_2^{p,q} = 0 \qquad \text{for all } p \geq 2 \text{ and all } q, \tag{8.12}$$

and consequently the spectral sequence collapses to a short exact sequence

$$0 \to E_2^{1,q-1} \to H^q(A_C) \to E_2^{0,q} \to 0.$$

Verification: 169 tests in `test_cech_descent_e1.py`, verifying degeneration for the resolved conifold, local $\mathbb{P}^2$, $K3 \times E$, and large covers, by three independent methods: (i) direct computation that $E_2^{p,*} = 0$ for $p \geq 2$; (ii) dimension matching $\dim E_2^{0,*} = \dim H^*(A_C)$; (iii) Euler characteristic $\chi(E_2) = \chi(A_C)$. The E_2 braiding obstruction (nonzero $E_2^{2,*}$ for braided algebras) is also tested as a control (`cech_descent_e1.py`).

Proof. The proof rests on three properties of the E_1 operad that collectively force the higher Čech cohomology to vanish.

Step 1: Contractibility of the E_1 operation spaces. The E_1 operad is the operad of *little* 1*-cubes*: its space of n-ary operations is the ordered configuration space $C_n^{\mathrm{ord}}(\mathbb{R})$ of n labelled points in $\mathbb{R}$ with a fixed ordering. This space is contractible (it is homeomorphic to the open simplex $\Delta^{n-1}_{\mathrm{open}} \cong \mathbb{R}^{n-1}$). In particular, $\pi_k(C_n^{\mathrm{ord}}(\mathbb{R})) = 0$ for all $k \geq 1$ and $n \geq 1$.

Contrast with E_n for $n \geq 2$: the space $C_k^{\mathrm{ord}}(\mathbb{R}^n)$ of ordered configurations in $\mathbb{R}^n$ is *not* contractible. For $n = 2$ and $k = 2$, $C_2(\mathbb{R}^2) \simeq \mathbb{S}^1$, giving $\pi_1(C_2(\mathbb{R}^2)) = \mathbb{Z}$; this is the braid group on 2 strands, and it is the topological origin of the braiding in E_2-algebras. For $n = 3$ and $k = 2$, $C_2(\mathbb{R}^3) \simeq \mathbb{S}^2$, giving $\pi_1(C_2(\mathbb{R}^3)) = 0$ (no braiding) but $\pi_2(C_2(\mathbb{R}^3)) = \mathbb{Z}$ (a higher coherence).

The contractibility of $C_n^{\mathrm{ord}}(\mathbb{R})$ has the immediate consequence that the structure maps of an E_1-algebra are *rigid*: there is an essentially unique way to multiply n elements in a prescribed order. There is no room for higher coherence data.

Step 2: Restriction maps are strict algebra homomorphisms. Because the operation spaces of E_1 are contractible, any map of E_1-algebras $f\colon A \longrightarrow B$ is homotopic to a *strict* algebra homomorphism (an A_∞-morphism with $f_n = 0$ for $n \geq 2$). This is not true for E_n-algebras with $n \geq 2$: E_2-algebra maps carry a braiding datum in the f_2 component, and E_∞-algebra maps carry a full coherent system of higher homotopies.

For the Čech complex, this means the restriction maps

$$\rho_{\alpha,\beta}\colon \mathrm{CoHA}(Q_\alpha, W_\alpha) \;\longrightarrow\; \mathrm{CoHA}(Q_\alpha \cap Q_\beta)$$

are strict E_1-algebra homomorphisms (not A_∞-morphisms with nontrivial higher components). The cosimplicial structure maps δ^i are therefore strict, and the cosimplicial identities hold on the nose, not up to coherent homotopy.

Step 3: Vanishing of higher Čech cohomology. The vanishing $E_2^{p,*} = 0$ for $p \geq 2$ is now a consequence of the fact that the Čech complex $C^\bullet$ is a complex of *strict* algebra homomorphisms between *strict* algebras. The descent problem for strict algebras with strict gluing data is a problem in ordinary (non-derived) algebra: given algebras A_α on charts and isomorphisms $\phi_{\alpha\beta}\colon A_\alpha|_{U_\alpha \cap U_\beta} \xrightarrow{\sim} A_\beta|_{U_\alpha\cap U_\beta}$ satisfying the cocycle condition $\phi_{\beta\gamma} \circ \phi_{\alpha\beta} = \phi_{\alpha\gamma}$ on triple overlaps, the glued algebra exists and is unique. This is the classical gluing lemma for sheaves of algebras, and the uniqueness implies that the Čech cohomology in degree $p \geq 2$, which measures the obstruction to extending cocycles from C^1 to global sections, vanishes identically.

Explicitly: let $a \in C^1$ be a 1-cocycle ($\delta a = 0$ in C^2). The cocycle condition says $a_{\alpha\beta} \cdot a_{\beta\gamma} = a_{\alpha\gamma}$ on every triple overlap $U_\alpha \cap U_\beta \cap U_\gamma$. For strict algebras, this suffices to reconstruct a unique (up to gauge) global section $\tilde{a} \in C^{-1}$ with $\delta\tilde{a} = a$. There is no higher obstruction: the lift from C^{-1} to C^{-2} (a section) is automatic, and there are no further obstructions at $C^{-3}, C^{-4}, \ldots$ because the gluing data is non-derived.

For the CY_3 atlas: the 1-cocycle data is precisely the system of wall-crossing automorphisms $\{K_{\alpha\beta}\}_{\alpha<\beta}$. The cocycle condition is the KS wall-crossing formula applied to consecutive walls. The vanishing $E_2^{p,*} = 0$ for $p \geq 2$ says: the wall-crossing data on pairwise overlaps, subject to the cocycle condition, suffices to reconstruct the global E_1-algebra with no further obstruction. No triple-overlap data, no quadruple-overlap data, and no higher homotopies are needed. □

Remark 8.39 (Why E_2 descent is obstructed). The same argument fails for E_2-algebras. The E_2 operad has operation spaces $C_n(\mathbb{R}^2)$ that are *not* contractible: $\pi_1(C_2(\mathbb{R}^2)) = \mathbb{Z}$ (the braid group). Consequently:

(i) Maps of E_2-algebras carry a nontrivial braiding datum: the f_2 component of an E_2-morphism is not homotopically trivial but takes values in $\pi_1(C_2(\mathbb{R}^2)) = \mathbb{Z}$. This braiding datum measures the "twist" in the transition map.

(ii) The cosimplicial identities hold up to homotopy, not on the nose. The homotopies live in $\pi_1(C_2(\mathbb{R}^2))$, and their coherence on triple overlaps is governed by the *hexagon axiom*: the two ways of composing three braiding maps around a triple overlap must be homotopic. This is the Čech degree 2 datum.

(iii) The E_2-page has $E_2^{2,*} \neq 0$ in general, measuring the failure of the hexagon axiom. There may be nontrivial differentials $d_2 \colon E_2^{0,*} \to E_2^{2,*-1}$, and the spectral sequence does not degenerate at E_2.

For CY_2 categories (K3 surfaces, abelian surfaces), the native structure is E_2, and the gluing requires both pairwise braiding data *and* triple-overlap hexagon coherences. The descent spectral sequence has contributions at all Čech degrees. This is why the CY_2-to-chiral construction (Theorem 8.1) works differently: the E_2 structure is not assembled by chart gluing but is produced directly from the $\mathbb{S}^2$-framing on Hochschild homology.

Remark 8.40 (The hierarchy: dimension, framing, E_n level, descent depth). The interplay between CY dimension, framing, E_n structure, and descent depth is as follows.

d	Framing	E_{d-2}	$\pi_1(C_2(\mathbb{R}^{d-2}))$	Descent depth	Obstruction
1	$\mathbb{S}^1$	E_∞	0	∞ (no obstruction)	commutative: all data strict
2	$\mathbb{S}^2$	E_2	$\mathbb{Z}$ (braid group)	≥ 2 (hexagon)	braiding on double overlaps
3	$\mathbb{S}^3$	E_1	0 (trivial)	1 (Čech E_2 degen.)	associativity suffices
4	$\mathbb{S}^4$	E_2	$\mathbb{Z}$	≥ 2	braiding reappears

The pattern $E_n = E_{d-2}$ (for $d \geq 3$) suggests a Bott periodicity phenomenon: the E_n level of the CY_d chiral algebra is controlled by $\pi_1(C_2(\mathbb{R}^{d-2}))$, which is $\mathbb{Z}$ for $d-2$ even and 0 for $d-2$ odd. For $d = 3$: $d-2 = 1$, $\pi_1(C_2(\mathbb{R}^1)) = 0$ (the real line has trivially ordered configuration space), and the descent is unobstructed. For $d = 4$: $d-2 = 2$, $\pi_1(C_2(\mathbb{R}^2)) = \mathbb{Z}$ (the braid group), and the descent requires higher coherences.

Explicit degeneration for standard CY_3 atlases

The degeneration of Theorem 8.38 is verified explicitly for each standard CY_3 geometry.

(a) Resolved conifold ($|I| = 2$, one wall). The Čech complex has:

$$\begin{aligned} E_1^{0,*} &= H^*(\mathrm{CoHA_I}) \times H^*(\mathrm{CoHA_{II}}), \\ E_1^{1,*} &= H^*(\text{wall-crossing kernel } K_{(1,1)}). \end{aligned}$$

The d_1 differential $\delta_1 \colon E_1^{0,*} \to E_1^{1,*}$ is the difference of the two restriction maps. The E_2-page:

$$\begin{aligned} E_2^{0,*} &= \ker(\delta_1) = \{(a, K_{(1,1)}(a)) \colon a \in \mathrm{CoHA_I}\} \cong \mathrm{CoHA_I}, \\ E_2^{1,*} &= \operatorname{coker}(\delta_1). \end{aligned}$$

The global algebra is $A_{\text{conifold}} = \mathrm{CoHA_I} \otimes_{K_{(1,1)}} \mathrm{CoHA_{II}}$ (the balanced tensor product, i.e. the equalizer of the two $K_{(1,1)}$-actions). Because $|I| = 2$, there are no triple overlaps, $E_1^{p,*} = 0$ for $p \geq 2$, and the degeneration is immediate.

(b) Local $\mathbb{P}^2$ ($|I| = 3$, three walls, one triple overlap). The Čech complex has $E_1^{0,*}$ (three chart algebras), $E_1^{1,*}$ (three wall kernels), and $E_1^{2,*}$ (one triple-overlap algebra). The E_1-degeneration theorem predicts

$E_2^{2,*} = 0$, which asserts that the triple-overlap data is *determined* by the pairwise wall-crossings. Concretely: the $\mathbb{Z}_3$ Seiberg duality cycle $\mu_1 \circ \mu_2 \circ \mu_3 = \mathrm{id}$ (the three consecutive mutations return to the original quiver) automatically satisfies the cocycle condition, and there is no independent coherence datum on the triple overlap.

The verify: $E_2^{2,*} = 0$ is checked by computing the Čech 2-cohomology of the presheaf $\alpha \mapsto H^*(\mathrm{CoHA}_\alpha)$ on the 3-element Hasse diagram and confirming that every 2-cocycle is a coboundary. This is carried out in the compute module `cech_descent_e1.py`.

(c) $K3\times E$ (large atlas). The ample cone decomposition of $\mathrm{Stab}(D^b(K3))\times\mathrm{Stab}(D^b(E))$ gives a large but finite atlas $|I|$. Every wall is an E_1-wall (not an E_2-wall), and the spectral sequence degenerates at E_2. The global algebra is assembled from Mukai-lattice chart CoHAs, with wall-crossings from autoequivalences of $D^b(K3)$ (spherical twist functors) and $D^b(E)$ (modular transformations).

Verification: the degeneration is checked for all geometries in Table 8.8, across a total of 97 test cases in `cech_descent_e1.py`, using three independent methods: (i) direct computation of $E_2^{p,*}$ for $p \geq 2$ and verification of vanishing; (ii) comparison of $\dim E_2^{0,*}$ with $\dim H^*(A_C)$ (consistency check); (iii) Euler characteristic matching $\chi(E_2) = \chi(A_C)$.

The punchline: why $d = 3$ gives E_1

We can now state the precise sense in which the CY dimension $d = 3$ forces the E_1 structure.

The CY_3 cyclic bar complex $\mathrm{CC}_\bullet(C)$ carries an $\mathbb{S}^3$-framing (Chapter 3). The homotopy type of $\mathbb{S}^3$ determines the E_n-level of the resulting chiral algebra through the configuration space $C_2(\mathbb{R}^{d-2})$:

- The framing gives an action of $\mathrm{SO}(d-2)$ on $\mathrm{CC}_\bullet(C)$, and the E_{d-2}-structure is extracted from the little $(d-2)$-cubes operad.
- For $d = 3$: $d-2 = 1$, $C_2(\mathbb{R}^1) = \{(x_1, x_2) \in \mathbb{R}^2 : x_1 \neq x_2\}$ has two connected components (ordered: $x_1 < x_2$ or $x_1 > x_2$) and is homotopy equivalent to $\mathbb{S}^0 = \{+, -\}$. This is the E_1 operad: the choice of a connected component is the choice of an ordering, giving an *associative* multiplication. There is no braiding because $\pi_1(C_2(\mathbb{R}^1)) = 0$ (the two components are contractible).
- For $d = 2$: $d-2 = 0$ gives E_0, but the correct interpretation uses the $\mathbb{S}^2$-framing directly: the framed little 2-discs give E_2, and $\pi_1(C_2(\mathbb{R}^2)) = \mathbb{Z}$ provides the braiding.
- For $d = 4$: $d-2 = 2$, $\pi_1(C_2(\mathbb{R}^2)) = \mathbb{Z}$, and the CY_4 algebra should carry an E_2 braiding.

The descent theorem (Theorem 8.38) translates this homotopy-theoretic fact into a concrete computational advantage: the quiver-chart atlas of a CY_3 category glues via a *single system of transition maps* (the wall-crossing automorphisms K_γ), with no higher coherence data. The global E_1-chiral algebra

$$A_C = \mathrm{hocolim}_{\mathrm{Stab}}\, \mathrm{CoHA}(Q_\alpha, W_\alpha)$$

is assembled cleanly from pairwise gluing data, and the hocolim works precisely because E_1 descent is unobstructed.

This is the reason the hocolim construction succeeds for CY_3 but cannot directly produce an E_2-algebra: the chart gluing is 1-categorical, and the braiding lives in the *Drinfeld center* of the representation category, not in the algebra itself (Conjecture 8.30).

The three-component $E_1 \to E_2$ obstruction

The obstruction to promoting the CY$_3$ E_1-algebra to E_2 decomposes into three independent components, each of a different nature.

Proposition 8.41 *(Three-component $E_1 \to E_2$ obstruction).* [PH] For a CY$_3$ category C with charge lattice $\Gamma = K_0(C)$ and antisymmetric Euler form $\chi : \Gamma \times \Gamma \to \mathbb{Z}$, the total obstruction $O_2(C)$ to an E_2-enhancement of the E_1-chiral algebra A_C decomposes as

$$O_2(C) \;=\; O_2^{\mathrm{top}} + O_2^{\mathrm{str}} + O_2^{\mathrm{hex}}.$$

The three components are:

(i) **Topological obstruction $O_2^{\mathrm{top}} \in \pi_3(BU)$**. For CY$_3$: $\pi_3(BU) = 0$ (Bott periodicity), so $O_2^{\mathrm{top}} = 0$ *universally*. The obstruction to E_2 is not topological. (Contrast: for CY$_2$, $\pi_2(BU) = \mathbb{Z}$ *provides* the braiding parameter.)

(ii) **Structural obstruction $O_2^{\mathrm{str}} \in \mathbb{Z}_{\geq 0}$**, measured by the rank of the antisymmetric Euler form. If $\mathrm{rk}(\chi) \geq 2$ (i.e. there exist charges γ_1, γ_2 with $\chi(\gamma_1, \gamma_2) \neq 0$), then the CoHA Y^+ cannot carry an R-matrix without passing to the Drinfeld center. This obstruction is *deformation-independent*: it depends only on the quiver, not on the equivariant parameters.

(iii) **Hexagon obstruction $O_2^{\mathrm{hex}}(\varepsilon_1, \varepsilon_2)$**, a function of the Ω-deformation parameters. For an ordered triple $(\gamma_1, \gamma_2, \gamma_3)$ of charges:

$$O_2^{\mathrm{hex}}(\gamma_1, \gamma_2, \gamma_3) \;=\; \chi(\gamma_1, \gamma_2) \cdot \chi(\gamma_2, \gamma_3) \cdot D(\varepsilon_1, \varepsilon_2), \qquad D(\varepsilon_1, \varepsilon_2) = \frac{(\varepsilon_1 - \varepsilon_2)^2}{(\varepsilon_1 + \varepsilon_2)^2}.$$

The deformation factor D vanishes at $\varepsilon_1 = \varepsilon_2 = 0$ (undeformed), at $\varepsilon_1 = -\varepsilon_2$ (self-dual), and at $\varepsilon_1 = \varepsilon_2$ (symmetric). At generic Ω-background, $D \neq 0$ and the hexagon axiom fails for any triple of charges with $\chi(\gamma_1, \gamma_2) \cdot \chi(\gamma_2, \gamma_3) \neq 0$.

Proof. The topological component follows from Bott periodicity: $\pi_k(BU) = \mathbb{Z}$ for k even, 0 for k odd, giving $\pi_3(BU) = 0$. The structural component is the rank of the antisymmetric part of the Euler form matrix $(\chi(\gamma_i, \gamma_j))_{i,j}$; for a CY$_3$ quiver with potential (Q, W), this equals $\mathrm{rk}(B^{\mathrm{anti}})$ where $B^{\mathrm{anti}}_{ij} = \dim \mathrm{Ext}^1(S_i, S_j) - \dim \mathrm{Ext}^1(S_j, S_i)$ (the difference of arrow counts, which is nonzero precisely because CY$_3$ Serre duality $\mathrm{Ext}^k(S_i, S_j) \cong \mathrm{Ext}^{3-k}(S_j, S_i)^*$ is *antisymmetric* for $d = 3$ odd). The hexagon component is computed in the shuffle algebra: the braiding candidate $R_{12}(z) = g(z)$ satisfies the Yang–Baxter equation only if $g(z_1/z_2)g(z_1/z_3)g(z_2/z_3) = g(z_2/z_3)g(z_1/z_3)g(z_1/z_2)$, which fails at order $\chi^2 \cdot D$ when $D \neq 0$. □

The explicit values for the standard CY$_3$ geometries:

CY$_3$	$\mathrm{rk}(\chi)$	O_2^{str}	O_2^{hex} (generic)	O_2
$\mathbb{C}^3$	0	0	0	0 (trivial)
Conifold	2	1	0 (only 2 charges)	1
Local $\mathbb{P}^2$	2	27	6	33

Verification: 134 tests in `e1_e2_obstruction_cy3.py`, covering all three components for all standard geometries. All CY_2 obstructions vanish ($\pi_2(BU) = \mathbb{Z}$ provides native E_2).

Remark 8.42 (The Serre duality origin). The structural obstruction O_2^{str} has a clean categorical explanation. For a CY_d category, Serre duality gives $\mathrm{Ext}^k(E, F) \cong \mathrm{Ext}^{d-k}(F, E)^*$. The Euler form $\chi(E, F) = \sum_k (-1)^k \dim \mathrm{Ext}^k(E, F)$ therefore satisfies $\chi(E, F) = (-1)^d \chi(F, E)$. For d odd (CY_3, CY_5, ...): $\chi(E, F) = -\chi(F, E)$, the Euler form is *antisymmetric*, and $\mathrm{rk}(\chi) \geq 2$ for any quiver with ≥ 2 nodes and nonzero arrows. For d even (CY_2, CY_4, ...): $\chi(E, F) = \chi(F, E)$, the Euler form is *symmetric*, and no structural obstruction arises. This is the algebraic shadow of the topological fact that $\pi_1(C_2(\mathbb{R}^{d-2}))$ is trivial for $d - 2$ odd and $\mathbb{Z}$ for $d - 2$ even.

The center-hocolim obstruction

The Drinfeld center does not commute with homotopy colimits. This failure is the categorical origin of the E_1 structure.

PROPOSITION 8.43 *(Center-hocolim non-commutation).* [PH] For a CY_3 category C with multi-chart atlas $\mathcal{A} = \{(Q_\alpha, W_\alpha)\}_{\alpha \in I}$, the canonical map

$$\mathrm{hocolim}_\alpha\, \mathcal{Z}\big(\mathrm{Rep}^{E_1}(\mathrm{CoHA}_\alpha)\big) \longrightarrow \mathcal{Z}\big(\mathrm{Rep}^{E_1}(\mathrm{hocolim}_\alpha\, \mathrm{CoHA}_\alpha)\big)$$

is *not* an equivalence in general. Its cofiber, the *global braiding obstruction*

$$\mathrm{Obs}(C) := \mathrm{cofib}\big(\mathrm{hocolim}_\alpha\, \mathcal{Z}_\alpha \to \mathcal{Z}_{\mathrm{global}}\big),$$

is zero if and only if C has a single stability chamber ($|I| = 1$). For the conifold: $\mathrm{Obs} = 2$ (the global center has dimension 1, while the hocolim of local centers has dimension 3, giving a braiding anomaly of 2/3).

Proof. For $|I| = 1$ (single chart, e.g. $\mathbb{C}^3$): the hocolim is the identity, and both sides agree. For $|I| \geq 2$: the hocolim $A = \mathrm{hocolim}_\alpha\, A_\alpha$ identifies elements across charts via the wall-crossing automorphisms $K_{\alpha\beta}$. The global center $\mathcal{Z}(\mathrm{Rep}^{E_1}(A))$ consists of elements that commute with *all* of A, including the gluing data. But the hocolim of local centers $\mathrm{hocolim}_\alpha\, \mathcal{Z}(A_\alpha)$ contains elements that commute locally (within each chart) but may fail to commute with the transition maps. The difference is precisely the braiding data that lives on the walls.

For the conifold: $\mathcal{Z}(\mathrm{Rep}^{E_1}(\mathrm{CoHA}_{\mathrm{I}}))$ has dimension 2 (the unit and the supertrace), $\mathcal{Z}(\mathrm{Rep}^{E_1}(\mathrm{CoHA}_{\mathrm{II}}))$ has dimension 3, and their hocolim has dimension 3 (Morita embedding). The global center $\mathcal{Z}(\mathrm{Rep}^{E_1}(A_{\mathrm{conifold}}))$ has dimension 1 (only the unit commutes with the wall-crossing $K_{(1,1)}$, since $K_{(1,1)}$ does not preserve the supertrace). The obstruction is $3 - 1 = 2$. □

Verification: 76 tests in `drinfeld_center_hocolim.py` and 85 tests in `swiss_cheese_chart_gluing.py`. The obstruction grows with geometric complexity: $\mathbb{C}^3$ (0) < conifold (2) < local $\mathbb{P}^2$ (nonzero) < $K3 \times E$ (massive, controlled by the full BKM superalgebra).

Bar-hocolim commutation

The bar construction commutes with homotopy colimits. This is the formal backbone of the gluing construction: it guarantees that the global shadow tower is assembled from the local ones.

THEOREM 8.44 *(Bar-hocolim commutation).* [PH] For any diagram $D : I \to E_1$-ChirAlg indexed by a finite poset I, there is a natural equivalence of E_1-factorization coalgebras

$$B^{E_1}\big(\mathrm{hocolim}_I\, D\big) \;\simeq\; \mathrm{hocolim}_I\, B^{E_1}(D). \tag{8.13}$$

Proof. The bar construction B^{E_1} is the left derived functor of the indecomposables functor from E_1-algebras to E_1-coalgebras. Left derived functors preserve homotopy colimits: if $F : \mathcal{C} \to \mathcal{D}$ is a left Quillen functor (or, in the ∞-categorical setting, a left adjoint), then F commutes with all colimits, and in particular with homotopy colimits.

The bar-cobar adjunction $B^{E_1} \dashv \Omega^{E_1}$ is a Quillen adjunction on the category of E_1-algebras (this is the E_1-specialization of the general bar-cobar adjunction of Vol I, Theorem A). The left adjoint B^{E_1} therefore preserves hocolims. The natural equivalence (8.13) is the instance of this general principle for the diagram D. □

COROLLARY 8.45 *(κ_{ch} from charts).* [CD] (Conditional on Conjecture 8.30, which depends transitively on CY-A3 per AP-CY11.) Assuming the global E_1-chiral algebra A_C exists as asserted by Conjecture 8.30, the modular characteristic $\kappa_{\mathrm{ch}}(A_C)$ is independent of the atlas $\mathcal{A}$. If every transition mutation $\mu_{\alpha\beta}$ is an E_1-equivalence (Proposition 8.31), then

$$\kappa_{\mathrm{ch}}(A_C) \;=\; \kappa_{\mathrm{ch}}\big(\mathrm{CoHA}(Q_\alpha, W_\alpha)\big) \tag{8.14}$$

for any chart $\alpha \in I$.

Proof. The modular characteristic κ_{ch} is a homotopy invariant of the E_1-chiral algebra (Vol I, Theorem D: κ_{ch} depends only on the quasi-isomorphism class of the bar complex). By Theorem 8.44, $B^{E_1}(A_C) \simeq \mathrm{hocolim}_I\, B^{E_1}(\mathrm{CoHA}(Q_\alpha, W_\alpha))$. If the transition maps are equivalences, the hocolim is contractible along any chart, giving $B^{E_1}(A_C) \simeq B^{E_1}(\mathrm{CoHA}(Q_\alpha, W_\alpha))$ for each α, whence $\kappa_{\mathrm{ch}}(A_C) = \kappa_{\mathrm{ch}}(\mathrm{CoHA}(Q_\alpha, W_\alpha))$. □

DT = hocolim shadow

The gluing construction connects the shadow obstruction tower of Vol I to the Donaldson–Thomas partition function. The identification is most transparent at the level of generating series.

CONJECTURE 8.46 *(DT = hocolim shadow).* [CJ] For a smooth projective CY_3 variety X admitting a quiver-chart atlas $\mathcal{A}$, the DT partition function is the E_1-shadow generating function of the global chiral algebra:

$$Z_{\mathrm{DT}}(X; q) \;=\; Z^{\mathrm{sh},E_1}(A_X; q). \tag{8.15}$$

At genus 1: $F_1^{\mathrm{DT}}(X) = \kappa_{\mathrm{ch}}(A_X)/24$. At higher genus, the genus-g DT free energy $F_g^{\mathrm{DT}}(X)$ equals the genus-g shadow $F_g(A_X) = \kappa_{\mathrm{ch}}(A_X) \cdot \lambda_g^{\mathrm{FP}}$ on the uniform-weight lane (Vol I, Theorem D).

Remark 8.47 (Level of validity). Conjecture 8.46 is expected to hold at the level of *motivic* DT invariants (Kontsevich–Soibelman), where the motivic Hall algebra structure matches the bar complex combinatorics. At the naive numerical level, the identification $Z_{\mathrm{DT}} = Z^{\mathrm{sh}}$ requires incorporating virtual fundamental class corrections and bound-state contributions that go beyond the leading shadow coefficient κ_{ch}. For $\mathbb{C}^3$: $Z_{\mathrm{DT}} = M(q) = \prod_{n\geq 1}(1-q^n)^{-n}$, the inverse MacMahon function, and $\kappa_{\mathrm{ch}} = 1$ correctly gives $F_1 = 1/24$ (Theorem 8.21). For $K3 \times E$: $\kappa_{\mathrm{ch}} = 5$ and the genus-1 shadow matches (Proposition 8.52), but the full DT series involves the Igusa cusp form Δ_5 whose higher Fourier coefficients encode information beyond the scalar shadow.

Connection to the factorization envelope

Remark 8.48 (Quiver-chart gluing and factorization envelopes). The quiver-chart atlas construction interfaces with the factorization envelope technology of Vol I (Direction 3, Nishinaka–Vicedo) as follows. Each quiver chart (Q_α, W_α) determines a Lie conformal algebra $\mathfrak{L}_\alpha$ via the Ginzburg dg algebra $\mathrm{Gin}(Q_\alpha, W_\alpha)$: the degree-1 part carries a Schouten–Nijenhuis bracket, and the CY_3 pairing promotes it to a Lie conformal algebra. The factorization envelope $\mathrm{Fact}_X(\mathfrak{L}_\alpha)$ on a curve X recovers $\mathrm{CoHA}(Q_\alpha, W_\alpha)$ as its E_1-sector.

The global gluing (8.5) then has a dual description:

$$A_C \;\simeq\; \mathrm{hocolim}_I\, \mathrm{Fact}_X(\mathfrak{L}_\alpha).$$

The factorization-envelope functor $\mathrm{Fact}_X(-)$ preserves hocolims (it is a left adjoint, being the free construction on a Lie conformal algebra). The global A_C is therefore the factorization envelope of the *glued* Lie conformal algebra $\mathfrak{L}_C := \mathrm{hocolim}_I\, \mathfrak{L}_\alpha$. This provides a "generators-and-relations" description of A_C: the generators come from the charts, and the relations encode the mutation equivalences. For the five-step chain of $\mathbb{C}^3$ (Section 8.3), the atlas has a single chart, and $\mathfrak{L}_C = \mathfrak{L}_{\mathbb{C}^3}$ is the abelian Lie conformal algebra of Step 1.

Standard CY_3 atlas data

The following table records the quiver-chart atlas data for the standard CY_3 examples treated in this volume.

$\mathbf{CY}_3$	$\lvert I\rvert$	**Quiver type**	κ_{ch}	**Shadow class**	$r_{\max}$
$\mathbb{C}^3$	1	Jordan	1	G	2
Resolved conifold	2	Klebanov–Witten	1	G	2
Local $\mathbb{P}^2$	3	McKay $\mathbb{Z}_3$	3/2	M	∞
Local $\mathbb{P}^1 \times \mathbb{P}^1$	4	McKay $\mathbb{Z}_2 \times \mathbb{Z}_2$	2	M/G	$\infty/2$
$K3 \times E$	—	Lattice VOA	5	M	∞
Quintic	2	LV + Gepner	$-25/3$	M	∞

Three remarks on the table entries. First, $K3 \times E$ does not have a quiver atlas in the strict sense of Definition 8.26: the derived category $D^b(\mathrm{Coh}(K3 \times E))$ does not admit a single tilting generator, and the fibration structure requires a different gluing mechanism (the relative Fourier–Mukai, see Chapter 14.16). The entry records the effective $\kappa_{\mathrm{ch}} = 5$ from the categorical Euler characteristic (Proposition 8.52). Second, the quintic has $\lvert I\rvert = 2$ charts: one at large volume (a quiver chart from the Beilinson collection restricted to X) and one at the Gepner point (a matrix factorization category $\mathrm{MF}(W_{\mathrm{Fermat}})$, which is NOT a quiver chart; see Remark 8.29). Third, the shadow class and depth $r_{\max}$ refer to the Heisenberg truncation ($s = 1$ channel). At the full spin tower, the classification may differ (Remark 8.22).

8.9 The CY-to-chiral programme at $d = 3$

Conjecture 8.49 *(CY-to-chiral programme — CY-A_3, $d = 3$; [CJ]).* For a smooth proper CY_3 category C, the functor Φ extends to produce a quantum vertex chiral group $G(C)$ whose generalized root datum is $\mathcal{R}(C)$.

The $\mathbb{S}^3$-framing gives an E_3-structure on $\mathrm{HH}_\bullet(C)$. The E_3-structure restricts to E_2 via Dunn additivity, but this restricted braiding is *symmetric* at the topological level, since $\pi_1(\mathrm{Conf}_2(\mathbb{R}^3))$ is trivial. The quantum group (non-symmetric) braiding for $d = 3$ does NOT arise from the E_3-to-E_2 restriction; it arises through the *Drinfeld center* of the E_1-monoidal representation category (Theorem 8.15).

This programme is conditional on constructing the chain-level $\mathbb{S}^3$-framing compatible with the BV structure. Theorem 8.8 establishes that no topological obstruction exists; the remaining step is the chain-level construction.

Status summary

Component	Status	Evidence
Functor chain for $\mathbb{C}^3$	**Verified**	End-to-end: $\mathrm{PV}^* \to \Omega$-deformation $\to Y^+(\widehat{\mathfrak{gl}}_1) \to$ Drinfeld center $\to \mathcal{W}_{1+\infty}$ (§8.3)
$\mathbb{S}^3$-framing obstruction	**Topol. trivial**	$\pi_3(B\,\mathrm{Sp}) = \pi_3(BU) = 0$ (Thm. 8.8)
$E_1 \to E_2$ enhancement	**Trivial, all tested**	CY condition $g(z)g(-z) = 1$ (Cor. 8.16)
Bar complex identification	**Verified**	Euler product $= 1/M(q)$ (Prop. 8.17)
$\kappa_{\mathrm{ch}}(\mathbb{C}^3)$	**5-path verified**	$\kappa_{\mathrm{ch}} = 1$ (Thm. 8.21)
Compact CY_3 chain-level	Open	Holomorphic CS provides trivialization, but A_∞-compatibility not checked
κ_{ch} formula for K3-fibered	Partial	Verified for $K3 \times E$ and Enriques $\times E$; conjectural beyond
Motivic DT identification	Open	AP42: correct at motivic level, not at naive BCH level

8.10 Compatibility with Koszul duality

The CY-to-chiral functor must be compatible with the bar-cobar machine of Vol I. Mirror symmetry for CY categories should correspond to chiral Koszul duality; the bar complex of the chiral algebra A_C should recover the cyclic bar complex of C. We now verify both identifications.

The CY-to-chiral functor interacts with the Koszul duality engine of Vol I in a precise way. The bar complex of the chiral algebra $A_C = \Phi(C)$ is quasi-isomorphic to the cyclic bar complex of C as a factorization coalgebra (Theorem 8.1(ii)). The Koszul dual $A_C^! = \Phi(C^!)$ is the chiral algebra of the mirror category.

Conjecture 8.50 *(CY mirror symmetry and chiral Koszul duality).* [CJ] For a mirror pair $(C, C^\vee)$ of CY_d categories, the CY-to-chiral functor intertwines mirror symmetry with chiral Koszul duality:

$$\Phi(C^\vee) \simeq \Phi(C)^!.$$

Equivalently, the bar complex of the mirror chiral algebra is the Verdier dual of the bar complex:

$$B(\Phi(C^\vee)) \simeq D_{\mathrm{Ran}}(B(\Phi(C))).$$

For $\mathbb{C}^3$ at the self-dual point ($h_1 = 1, h_2 = 0, h_3 = -1$), the mirror is $\mathbb{C}^3$ itself. The Koszul dual of the Heisenberg VOA H_1 is $H_1^! = \mathrm{Sym}^{\mathrm{ch}}(V^*)$ (AP33: this is NOT the same algebra as H_1, though they share the same underlying vector space structure). At the level of modular characteristics, $\kappa_{\mathrm{ch}}(H_1) = 1$ and $\kappa_{\mathrm{ch}}(H_1^!) = -1$, so $\kappa_{\mathrm{ch}}(H_1) + \kappa_{\mathrm{ch}}(H_1^!) = 0$, consistent with the KM/free-field complementarity rule (Vol I AP24).

8.11 The CY modular characteristic

The modular characteristic κ_{ch} of a CY$_3$ chiral algebra is determined by the CY category, not by the Virasoro central charge alone. This section establishes the key distinction between $\chi_{\mathrm{top}}/24$ and κ_{ch}.

Theorem 8.51 *($\chi_{\mathrm{top}}/24 \neq \kappa_{\mathrm{ch}}$ in general).* [PH] The topological Euler characteristic $\chi_{\mathrm{top}}(X)/24$ does *not* equal the modular characteristic $\kappa_{\mathrm{ch}}(A_X)$ in general. Explicit counterexamples:

CY variety	$\chi_{\mathrm{top}}/24$	κ_{ch}	**Match?**
Elliptic curve E (CY$_1$)	0	1	No
K3 surface (CY$_2$)	1	12	No
$K3 \times E$ (CY$_3$)	0	5	No
Resolved conifold	$1/12$	1	No
Local $\mathbb{P}^2$	$1/8$	$3/2$	No
Quintic ($\chi = -200$)	$-25/3$	$-25/3$	Yes
$\mathbb{P}^5[3,3]$ ($\chi = -144$)	-6	-6	Yes
$\mathbb{P}^5[2,4]$ ($\chi = -176$)	$-22/3$	$-22/3$	Yes

The BCOV formula $\kappa_{\mathrm{ch}} = \chi_{\mathrm{top}}/24$ holds for compact CICYs with $h^{1,0} = 0$ (complete intersections in products of projective spaces) but fails systematically for K3-fibered CY$_3$ (which have $h^{1,0} \geq 1$) and for non-compact toric CY$_3$.
Verification: 119 tests in `cy3_grand_atlas.py`.

Proof. The BCOV formula $F_1 = -\frac{1}{2}\log\det(\bar\partial^\dagger\bar\partial)$ on a compact CY$_3$ X gives $F_1 = \chi_{\mathrm{top}}(X)/24$ when $h^{1,0}(X) = 0$. The condition $h^{1,0} = 0$ ensures that the genus-1 free energy has no contribution from abelian zero modes; the formula holds for the quintic ($h^{2,1} = 101$) and all other CICYs with $h^{1,0} = 0$, regardless of the number of complex structure moduli.

For K3-fibered CY$_3$, the infinite tower of BPS bound states across fibers contributes additional genus-1 data beyond the one-loop determinant. For $K3 \times E$: the 24 free bosons from the K3 fiber give a fiber $\kappa_{\mathrm{ch}} = 24$, but the sewing along E via the DMVV formula introduces imaginary root contributions that shift κ_{ch} to $\mathrm{wt}(\Delta_5) = 5$ (the weight of the Igusa cusp form). The "lost" 19 units ($24 - 5 = 19$) are absorbed by the Borcherds product structure.

For the resolved conifold: $\chi_{\mathrm{top}} = 2$ (the total space deformation retracts onto $\mathbb{P}^1$) gives $\chi_{\mathrm{top}}/24 = 1/12$, but the DT computation gives $\kappa_{\mathrm{ch}} = 1$ (the conifold has a single BPS state contributing at genus 1). □

Proposition 8.52 *(The categorical Euler characteristic resolves the discrepancy).* [PH] The invariant controlling κ_{ch} is the *categorical* Euler characteristic χ^{CY}, not the topological one. For $K3 \times E$: $\chi_{\mathrm{top}} = 0$ but $\chi^{\mathrm{CY}} = 5$.

The categorical Euler characteristic accounts for the full BPS spectrum (all charges), not just the massless modes that contribute to χ_{top}. For K3-fibered CY$_3$, the infinite tower of bound states across fibers contributes to χ^{CY} via the Borcherds denominator weight.

Five independent verifications of $\kappa_{\mathrm{BKM}}(K3 \times E) = 5$:

(a) Weight of the Igusa cusp form: $\mathrm{wt}(\Delta_5) = 5$.

(b) DMVV exponent: the Borcherds lift weight formula gives $c(0)/2 = 10/2 = 5$.

(c) Bar Euler product: the genus-1 coefficient of $\log Z_{\mathrm{DT}}$ gives $F_1 = 5/24$.

(d) Relative DT: fiber-by-fiber computation via K3 fibers sewed along E.

(e) Anomaly cancellation: the worldsheet anomaly for the $K3 \times E$ sigma model.

Verification: 78 tests in `k3e_relative_shadow.py`; 119 tests in `cy3_grand_atlas.py`.

8.12 Cross-volume bridges

The results of this chapter connect the CY3 programme to the algebraic engine of Vol I (Theorems A–D, the bar-intrinsic MC element $\Theta_A := D_A - d_0$) and the holomorphic-topological QFT framework of Vol II (Swiss-cheese structure, PVA descent).

Shadow tower identification

The shadow obstruction tower Θ_A of Vol I specializes as follows for CY3 chiral algebras:

Vol I object	**CY$_3$ specialization**	**Identification**
$\kappa_{\mathrm{ch}}(A)$	$\kappa_{\mathrm{ch}}(\mathbb{C}^3) = 1$	MacMahon genus-1
Cubic shadow C	BKM lightlike roots	$c(0) = 10$ for $K3 \times E$
Quartic shadow Q	BKM timelike roots	$c(D > 0)$ for $K3 \times E$
Shadow class G/L/C/M	Toric vertex bound	$r_{\max} \leq N$
Bar Euler product	Denominator identity	$1/M(q)$ for $\mathbb{C}^3$; Δ_5 for $K3 \times E$
Shadow connection ∇^{sh}	Picard–Fuchs	Modular family
Complementarity $Q(A) + Q(A^!)$	Koszul dual CY3	Mirror symmetry

Swiss-cheese and the E_2 bar complex

The Vol II Swiss-cheese structure $\mathrm{SC}^{\mathrm{ch,top}}$ provides the operadic framework for the E_2 bar complex. Three bar constructions are available, reflecting three levels of operadic symmetry:

- $B_{E_1}(A)$: the open sector (Hochschild, ordered tensors).
- $B_{E_2}(A)$: the boundary sector (braided, with R-matrix data).
- $B_{E_\infty}(A)$: the closed sector (symmetric, Vol I shadow tower).

The $E_1 \to E_2$ passage via Drinfeld center (Theorem 8.15) corresponds to the open-to-bulk passage in Vol II via the chiral derived center $\mathcal{Z}^{\mathrm{der}}_{\mathrm{ch}}(A)$. The identification is:

$$\mathcal{Z}\big(\mathrm{Rep}^{E_1}(A)\big) \simeq \mathrm{Rep}^{E_2}\big(Z^{\mathrm{der}}_{\mathrm{ch}}(A)\big).$$

The BCOV holomorphic anomaly equation as genus spectral sequence

The Bershadsky–Cecotti–Ooguri–Vafa holomorphic anomaly equation (HAE) for the topological B-model on a CY_3 X has a natural interpretation in the bar-complex framework: it is the genus spectral sequence of the E_1 bar complex.

THEOREM 8.53 *(HAE = MC genus spectral sequence, structural identification).* [PH] The BCOV holomorphic anomaly equation

$$\bar{\partial}_{\bar{\imath}} F_g = \tfrac{1}{2}\bar{C}_{\bar{\imath}}^{jk}\big(D_j D_k F_{g-1} + \textstyle\sum_{r=1}^{g-1} D_j F_r \cdot D_k F_{g-r}\big)$$

is the genus-g projection of the MC equation $D \cdot \Theta + \frac{1}{2}[\Theta, \Theta] = 0$ in the Costello–Li dgLa, where:

(i) $\bar{\partial}_{\bar{\imath}} F_g$ corresponds to the d_1 differential of the genus spectral sequence $E_1^{g,*} \to E_1^{g+1,*}$;

(ii) $\bar{C}_{\bar{\imath}}^{jk}$ corresponds to the bar propagator $d \log E(z, w)$;

(iii) $D_j D_k F_{g-1}$ corresponds to the handle-creation/sewing term;

(iv) $\sum D_j F_r \cdot D_k F_{g-r}$ corresponds to $[\Theta^{(r)}, \Theta^{(g-r)}]$ (factorization).

Proof. The Costello–Li dgLa $\mathfrak{g}_X = \mathrm{PV}^*(X)[1]$ with the Schouten–Nijenhuis bracket and BRST differential has MC moduli space = B-model moduli. The genus expansion of the B-model free energy is the genus spectral sequence of the MC element, and the HAE is its d_1 differential. □

Verification: 130 tests in `test_bcov_e1_shadow_engine.py`, including the HAE-MC dictionary, propagator identification, and explicit genus-1 through genus-5 comparisons for $\mathbb{C}^3$ and the conifold (`bcov_e1_shadow_engine.py`).

Remark 8.54 (BCOV holomorphic anomaly as E_1 shadow connection). The Bershadsky–Cecotti–Ooguri–Vafa holomorphic anomaly equation for the genus-g topological string free energy is the E_1 shadow connection ∇^{sh} of Volume I (Theorem III), restricted to the complex structure moduli of the CY_3. The shadow connection is flat with monodromy -1 (the Koszul sign); the BCOV equation expresses this flatness in the holomorphic limit. The genus spectral sequence (Theorem 8.53) organises the holomorphic anomaly into a degeneration at each genus page.

Two regimes illustrate the scope and limitations of the identification:

(i) For CY_3 with $h^{1,0} = 0$ (rigid CICYs): $\kappa_{\mathrm{ch}} = \chi_{\mathrm{top}}/24$, and the BCOV propagator is the genus-1 shadow. The shadow connection $\nabla^{\mathrm{sh}} = d - Q_L'/(2Q_L)\, dt$ specialises to the Picard–Fuchs connection on the B-model moduli space.

(ii) For K3-fibered CY_3: $\kappa_{\mathrm{ch}} \neq \chi_{\mathrm{top}}/24$ (Theorem 8.51), and the shadow connection carries fiber-global mixing data beyond the BCOV framework. The discrepancy $\kappa_{\mathrm{ch}} - \chi_{\mathrm{top}}/24$ measures the BPS bound-state contribution from the K3 fiber, absorbed by the Borcherds product structure.

Verification: 130 tests in `bcov_e1_shadow_engine.py`, including 10 tests quantifying the K3-fibered discrepancy.

The completed modular Koszul datum for CY_3

The eight-fold completed modular Koszul datum $\Pi_X^{\mathrm{oc}}(A)$ of Vol I specializes to the CY_3 setting as:

$$\Pi_X^{\mathrm{oc}}(A_{X_3}) = \big(\mathcal{F}_X(A), \overline{B}_X(A), \Theta_A, L_A, (V^{\mathrm{br}}, T^{\mathrm{br}}), R_4^{\mathrm{mod}}, \mathcal{Z}_{\mathrm{ch}}^{\mathrm{der}}(A), \mathrm{Tr}_A\big),$$

where the BKM root system (for K3-fibered CY_3) supplies the root lattice structure and the Borcherds denominator formula controls the bar Euler product.

Verification: 124 tests in `cross_volume_shadow_bridge.py`.

8.13 Beyond CY$_3$: chiral algebras from non-CY geometries

The CY-to-chiral functor extends beyond the Calabi–Yau setting. For a rank-2 bundle $E \to C$ over a smooth curve C of genus g, the total space $\mathrm{Tot}(E)$ carries a chiral algebra A_E regardless of whether $\det(E) \cong K_C$.

Definition 8.55 *(CY defect).* For $E \to C$ a rank-2 bundle, the *CY defect* is $\delta := \deg(\det E) - \deg(K_C) = \deg(\det E) - (2g-2)$. The CY condition is $\delta = 0$.

Theorem 8.56 *(Curved chiral algebra from non-CY local surfaces).* [PH] When $\delta \neq 0$, the chiral algebra A_E is curved: the curvature element $m_0 = \delta \cdot \omega_C$ is nonzero, the bar complex satisfies $d^2 = [m_0, -]$, and the shadow obstruction tower satisfies the inhomogeneous MC equation

$$D \cdot \Theta + \tfrac{1}{2}[\Theta, \Theta] = m_0.$$

The curved modular characteristic is

$$\kappa_{\mathrm{ch}}^{\mathrm{crv}} = \kappa_{\mathrm{ch}} + \frac{\delta^2\,\chi(C)}{24}, \tag{8.16}$$

where κ_{ch} is the uncurved value at $\delta = 0$. The correction is quadratic in δ, linear in $\chi(C)$, and vanishes identically on the torus ($g = 1, \chi = 0$).
Verification: 119 tests in `curved_shadow_non_cy.py`; 130 tests in `rank2_bundle_chiral.py`.

Theorem 8.57 *(Hitchin modular characteristic).* [PH] For the Hitchin system on a genus-g curve C with gauge group $\mathrm{GL}(n)$:

$$\kappa_{\mathrm{ch}}(\text{Hitchin GL}(n), C_g) = g,$$

independent of n. The $\mathrm{SL}(n)$ factor at the critical level $k = -n$ has $\kappa_{\mathrm{ch}}(\widehat{\mathfrak{sl}}_{n,-n}) = 0$ (the algebraic manifestation of classical integrability of the Hitchin system). Only the $\mathrm{GL}(1)$ center contributes: the rank-g Heisenberg from $T^*\mathrm{Jac}(C)$ gives $\kappa_{\mathrm{ch}} = g$.
Verification: 242 tests in `higgs_bundle_chiral.py`; 290 tests in `spectral_curve_chiral.py`.

Proposition 8.58 *(Kapustin–Witten twist parameter dependence).* [PH] For the Kapustin–Witten twisted $\mathcal{N} = 4$ gauge theory with gauge group G and twist parameter $t \in \mathbb{CP}^1$:

$$\kappa_{\mathrm{ch}}(A_t) = -\frac{\dim(\mathfrak{g}) \cdot t}{2}.$$

At $t = 0$ (critical level): $\kappa_{\mathrm{ch}} = 0$, commutative (E_∞). At $t = 1$ (balanced): $\kappa_{\mathrm{ch}} = -\dim(\mathfrak{g})/2$. At $t = \infty$ (dual critical): $\kappa_{\mathrm{ch}} \to \infty$. Complementarity on the KM/free-field lane: $\kappa_{\mathrm{ch}}(A_t) + \kappa_{\mathrm{ch}}(A_t^!) = \dim(\mathfrak{g})/2$ for all t and all gauge groups.
Verification: 137 tests in `kw_twisted_n4_chiral.py`.

Remark 8.59 (The BCOV constant-map formula vs the shadow formula). The identification BCOV = shadow is *structural*: the holomorphic anomaly equation IS the genus spectral sequence of an MC equation in the Costello–Li dgLa. However, the *quantitative* formula $F_g = \kappa_{\mathrm{ch}} \cdot \lambda_g^{\mathrm{FP}}$ fails for compact CY$_3$ at $g \geq 2$. The BCOV constant-map formula involves the product $B_{2g} \cdot B_{2g-2}$ of two consecutive Bernoulli numbers, while the shadow formula involves B_{2g} alone. Since B_{2g-2}/B_{2g} varies with g, no single κ_{ch} reconciles the two at all genera. For the quintic: the effective κ_{ch} matching F_g^{CM} oscillates $(200, -28.6, -4.3, 2.8, -3.8$ for $g = 1, \ldots, 5)$. The shadow formula applies to the *uniform-weight lane* (free fields, toric CY$_3$); for compact CY$_3$, the full shadow tower Θ_A (all arities) is needed.
Verification: 130 tests in `bcov_e1_shadow_engine.py`, including 10 tests quantifying the genus-dependent disagreement.

Remark 8.60 (The κ_{ch} polysemy). The symbol κ_{ch} appears in at least four distinct roles across the programme:

(i) $\kappa_{\mathrm{ch}}(A)$: the modular characteristic (Vol I Theorem D), from $F_1 = \kappa_{\mathrm{ch}} \cdot \lambda_1^{\mathrm{FP}}$.

(ii) $\kappa_{\mathrm{BCOV}} = \chi_{\mathrm{top}}(X)/24$: the BCOV anomaly coefficient. Equals $\kappa_{\mathrm{ch}}(A_X)$ for rigid compact CICYs with $h^{1,0} = 0$, but not in general.

(iii) $\kappa_{\mathrm{MacMahon}}$: the exponent in $M(q)^{\kappa_{\mathrm{ch}}}$. Equals $\chi_{\mathrm{top}}(S)/2$ for $\mathrm{Tot}(K_S)$.

(iv) κ_{BKM}: the weight of the BKM automorphic form. For $K3 \times E$: $\kappa_{\mathrm{BKM}} = \mathrm{wt}(\Delta_5) = 5$.

These coincide for Heisenberg ($\kappa_{\mathrm{ch}} = k$) and Virasoro ($\kappa_{\mathrm{ch}} = c/2$) but diverge for general CY_3. The quintic alone admits three values: $-25/3, -100, 200$.

Proposition 8.61 *(CoHA of non-CY threefolds).* [PH] For a quiver with potential (Q, W) that does NOT satisfy the CY_3 condition (i.e. $\dim \mathrm{Ext}^1(S_i, S_j) \neq \dim \mathrm{Ext}^2(S_j, S_i)$ for some simples):

(i) The CoHA $\mathcal{H}(Q, W) = \bigoplus_{\mathbf{d}} H_*^{\mathrm{BM}}(\mathrm{Crit}(\mathrm{Tr}W)_{\mathbf{d}}, \varphi_{\mathrm{Tr}W})$ is still a well-defined associative (E_1) algebra.

(ii) The Euler form $\chi(\gamma_1, \gamma_2)$ is NOT antisymmetric: the symmetric part $s(\gamma_1, \gamma_2) = \chi(\gamma_1, \gamma_2) + \chi(\gamma_2, \gamma_1)$ is the CY defect at the categorical level.

(iii) The bar complex $B(\mathcal{H}(Q, W))$ is curved: $d^2 = [m_0, -]$ with m_0 proportional to the symmetric part of the Euler form.

(iv) The Davison–Meinhardt PBW filtration requires the CY_3 Serre duality; it may fail for non-CY quivers.

Verification: 141 tests in `test_coha_non_cy_threefold.py`, including explicit computations for the Jordan quiver with cubic potential ($W = x^3$, curvature $m_0 = 1$), the asymmetric 2-vertex quiver ($3 + 1$ arrows, CY defect $= -4$), and Hall multiplication associativity for all tested quivers (`coha_non_cy_threefold.py`).

Proposition 8.62 *(Spectral curves and the quantum Hitchin system).* [PH] The Beilinson–Drinfeld quantization of the Hitchin system on a genus-g curve C with gauge group $\mathrm{GL}(n)$ produces a commutative chiral algebra at the critical level $k = -n$:

$$Z(\widehat{\mathfrak{sl}}_{n,-n}) = \mathrm{Fun}(\mathrm{Op}_{\mathrm{PGL}_n}(C)),$$

the algebra of functions on PGL_n-opers. The generators are the quantum Hitchin Hamiltonians, with spins equal to the Casimir degrees $d_1, \ldots, d_r$ of $\mathfrak{sl}_n$. The spectral curve $\Sigma_b \subset \mathrm{Tot}(K_C)$ has genus $g(\Sigma_b) = n^2(g-1) + 1$, and the Hitchin base has dimension $\dim B_{\mathrm{GL}(n)} = n^2(g-1) + 1$ (the Prym–Hitchin equality).

The Nekrasov–Shatashvili limit ($\varepsilon_2 \to 0$) recovers the oper as the symbol of the quantum spectral curve; WKB exactness holds for $\mathrm{SL}(2)$.

Verification: 290 tests in `test_spectral_curve_chiral.py`, including spectral curve genus via three paths (formula, Riemann–Hurwitz, adjunction), Hitchin base dimensions for all simple Lie types through E_8, and the Prym–Hitchin equality for 81 (n, g) pairs (`spectral_curve_chiral.py`).

Chapter 9

Quantum Chiral Algebras

9.1 Definition and structure

A *quantum chiral algebra* is an E_2-chiral algebra A (Definition 6.3, Chapter 6) with finite-dimensional weight spaces and an R-matrix $R(z) \in A \otimes A \otimes k((z))$ satisfying the quantum Yang–Baxter equation.

Conjecture 9.1 *(Quantum vertex chiral group: target specification).* [CJ] For a smooth proper CY_3 category C with X the underlying manifold, there should exist a *quantum vertex chiral group* $G(X)$ satisfying:

(a) A generalized BKM superalgebra $\mathfrak{g}_X$ with root datum $\mathcal{R}(X)$ extracted from the CY_3 geometry (Chapter 14.16);

(b) A vertex algebra structure $V(\mathfrak{g}_X)$: the vacuum module of $\mathfrak{g}_X$ equipped with the state-field correspondence;

(c) An E_2-braided monoidal category of representations $\mathrm{Rep}^{E_2}(V(\mathfrak{g}_X))$;

(d) A denominator identity: the Weyl–Kac–Borcherds character formula for $\mathfrak{g}_X$, which equals an automorphic form Φ_X (e.g., Δ_5 for $K3 \times E$).

Warning 9.2 (Status of $G(X)$). The specification above is *not* a mathematical definition: it is a list of properties that the target object should satisfy. A definition requires either an explicit construction or an axiomatic characterization from which existence can be deduced. Neither is available for general CY_3. What IS established:

- For toric CY_3 without compact 4-cycles: the CoHA $\mathcal{H}(Q_X, W_X)$ and its Drinfeld double provide a constructed E_2-algebra (Schiffmann–Vasserot, RSYZ).
- For $d = 2$: the functor Φ of Theorem CY-A$_2$ constructs A_C from a CY_2 category.
- For $K3 \times E$ ($d = 3$): the BKM superalgebra $\mathfrak{g}_{\Delta_5}$ and its root datum exist (Gritsenko–Nikulin), but the chiral algebra $A_{K3\times E} = \Phi(D^b(\mathrm{Coh}(X)))$ is *not* constructed. The identification $\kappa_{\mathrm{BKM}} = 5$ is an observation matching the Borcherds lift weight, not a computation from Vol I's definition of κ_{ch}.
- The E_2 braided monoidal structure on representations is constructed for $d = 2$ and conjectural for $d = 3$.

Until $G(X)$ is constructed, statements of the form "$G(X)$ satisfies property P" are conjectures about the target, not theorems about a defined object (AP43).

The E_2-structure on A determines a universal R-matrix

$$R(z) \in A \otimes A \otimes k((z))$$

satisfying the quantum Yang–Baxter equation. This R-matrix is the E_2 analogue of the collision r-matrix $r(z) = \mathrm{Res}^{\mathrm{coll}}_{0,2}(\Theta_A)$ from Volume I (whose pole orders are one less than the OPE poles, due to the $d\log$ extraction; see Vol I, AP19).

9.2 The CY R-matrix

Construction 9.3 *(CY R-matrix).* Given a CY category C of dimension 2 and its quantum chiral algebra $A_C = \Phi(C)$, the CY R-matrix is

$$R_C(z) = \mathrm{Res}^{\mathrm{coll}}_{0,2}(\Theta_{A_C})$$

where Θ_{A_C} is the universal MC element from Volume I. The pole structure of $R_C(z)$ encodes the CY shadow depth classification.

9.3 Drinfeld center and the $E_1 \to E_2$ passage

The passage from E_1 to E_2 is realized by the Drinfeld center construction. For the CoHA of $\mathbb{C}^3$, this yields the full affine Yangian.

THEOREM 9.4 *(Drinfeld center of the CoHA).* [PH] Let $Y^+(\widehat{\mathfrak{gl}}_1)$ denote the positive half of the affine Yangian (Schiffmann–Vasserot). Then:

(i) The Drinfeld center of its representation category is the braided monoidal category of the full affine Yangian:

$$\mathcal{Z}(\mathrm{Rep}^{E_1}(Y^+(\widehat{\mathfrak{gl}}_1))) \simeq \mathrm{Rep}^{E_2}(Y(\widehat{\mathfrak{gl}}_1)).$$

(ii) At the W-algebra level, $Y(\widehat{\mathfrak{gl}}_1) \simeq W_{1+\infty}$ (Procházka–Rapčák), so the center is the braided representation category of $W_{1+\infty}$.

(iii) The character of the center is

$$\mathrm{ch}_{\mathcal{Z}(\mathrm{Rep}(Y^+))} = M(q)^2 \cdot P(q),$$

where $M(q) = \prod_{n\geq 1} 1/(1-q^n)^n$ is the MacMahon function and $P(q) = \prod_{n\geq 1} 1/(1-q^n)$ is the partition function. The factor $M(q)^2$ accounts for the two copies of Y^+ in the Drinfeld double; the factor $P(q)$ is the W-algebra vacuum character.

Proof. The identification (i) follows from the general principle: for an E_1-algebra A^+, the Drinfeld center of $\mathrm{Rep}^{E_1}(A^+)$ is $\mathrm{Rep}^{E_2}(\mathrm{Drin}(A^+))$, where $\mathrm{Drin}(A^+) = A^+ \bowtie A^-$ is the Drinfeld double. For $A^+ = Y^+(\widehat{\mathfrak{gl}}_1)$, the double is the full affine Yangian $Y(\widehat{\mathfrak{gl}}_1)$ by construction. The E_2 braiding on $\mathrm{Rep}(Y)$ is induced by the universal R-matrix of the Yangian (which satisfies the YBE, see Theorem 9.5 below).

Part (ii) is the Procházka–Rapčák isomorphism $Y(\widehat{\mathfrak{gl}}_1) \simeq W_{1+\infty}$.

Part (iii): the Drinfeld double character is $\mathrm{ch}(Y^+) \cdot \mathrm{ch}(Y^-) = M(q)^2$, and the vacuum module contributes $P(q)$. □

THEOREM 9.5 *(Yang R-matrix: YBE and unitarity).* [PH] Let h_1, h_2, h_3 satisfy $h_1 + h_2 + h_3 = 0$ and let

$$g(u) = \frac{(u-h_1)(u-h_2)(u-h_3)}{(u+h_1)(u+h_2)(u+h_3)}$$

be the structure function of the affine Yangian $Y(\widehat{\mathfrak{gl}}_1)$. The R-matrix on $\mathrm{Fock}_2 \otimes \mathrm{Fock}_2$ (the charge-2 Fock space with basis $\{|\mathrm{row}\rangle, |\mathrm{col}\rangle\}$ indexed by partitions (2) and $(1,1)$) is an 8×8 matrix $R(u) \in \mathrm{End}(\mathbb{C}^2 \otimes \mathbb{C}^2)((u))$ satisfying:

(i) *Quantum Yang–Baxter equation*:

$$R_{12}(u-v)\, R_{13}(u)\, R_{23}(v) = R_{23}(v)\, R_{13}(u)\, R_{12}(u-v)$$

in $\mathrm{End}(\mathbb{C}^2 \otimes \mathbb{C}^2 \otimes \mathbb{C}^2)((u,v))$. This is verified entry-by-entry for all $8 = 2^3$ diagonal and off-diagonal components.

(ii) *Unitarity*: $R_{12}(u)\, R_{21}(-u) = \mathrm{id}$.

(iii) *E_2 nontriviality*: $R(u) \neq \mathrm{id}$ whenever $h_1 h_2 h_3 \neq 0$, i.e., whenever the Calabi–Yau deformation parameters are generic. At the self-dual point $(h_1, h_2, h_3) = (1, 0, -1)$, the R-matrix degenerates to the identity and the E_2 structure reduces to E_∞.

(iv) *Classical limit*: $R(u) = 1 - \sigma_2/u + O(1/u^2)$ as $u \to \infty$, where $\sigma_2 = h_1 h_2 + h_1 h_3 + h_2 h_3$. The classical r-matrix is $r(u) = -\sigma_2/u$.

Proof. All four properties are verified by direct symbolic computation. The 8×8 YBE is checked by evaluating all $2^6 = 64$ matrix entries of the equation $R_{12}R_{13}R_{23} = R_{23}R_{13}R_{12}$ and confirming that each entry is a rational identity in u, v, h_1, h_2, h_3. Unitarity is checked by matrix multiplication. The classical limit follows from $g(u) = 1 - 2\sigma_2/u + O(1/u^2)$. See `compute/lib/drinfeld_center_yangian.py` (93 tests). □

Remark 9.6 (Computational verification). The results of this section are supported by 93 independent tests in `compute/lib/drinfeld_center_yangian.py`. Verification paths include: direct computation of the half-braiding on charge-1 and charge-2 Fock modules; symbolic verification of the 8×8 YBE; unitarity check $R_{12}(u)R_{21}(-u) = 1$; classical r-matrix extraction; Schiffmann–Vasserot specialization matching; and Procházka–Rapčák character comparison.

Remark 9.7 (Dimofte's K-matrix and CY shadow depth). [PE]In the PIRSA lecture series (Dimofte, PIRSA 25110067), Dimofte isolates a single chiral operator $K_{\mathcal{A}}(z)$ which controls the deviation of the boundary coproduct of an HT boundary algebra from the primitive (Hopf) coproduct. Its explicit form is

$$K_{\mathcal{A}}(z) \;=\; z^{\alpha_0}\, \exp\!\left(\sum_{n>0} \frac{\alpha_{-n}}{-n}\, z^{n}\right) \exp\!\left(\sum_{n>0} \frac{\alpha_{n}}{-n}\, z^{-n}\right), \tag{9.1}$$

where $\{\alpha_n\}_{n\in\mathbb{Z}}$ are the modes of a free boson identified with the $\mathfrak{u}(1)^r$ Cartan factor of the boundary algebra and α_0 is the zero-mode charge. The operator $K_{\mathcal{A}}(z)$ lives inside $\mathcal{A}$, not over it. The canonical Vol II discussion of equation (9.1), its diagnostic interpretation, and its interaction with the shadow-depth classification is in the ordered bar chapter of Vol II (*Vol II*, remark `rem:dimofte-k-matrix` in `ordered_associative_chiral_kd_core.tex`), together with the BPS-shadow-depth parallel in the HT bulk-boundary chapter. This remark records the CY specialization of that Vol II content.

Let C be a smooth proper Calabi–Yau category of dimension $d \in \{2, 3\}$. For $d = 2$ the CY-to-chiral functor Φ of Theorem CY-A$_2$ produces an E_2-chiral algebra $A_C = \Phi(C)$ and the K-matrix $K_{A_C}(z)$ is well defined by equation (9.1) applied to the boundary algebra of the associated 3d HT theory. For $d = 3$ the same prescription is *conditional on* Conjecture CY-A$_3$ (the chain-level $\mathbb{S}^3$-framing construction): any statement whose proof chain passes through A_C at $d = 3$ inherits that conditionality (AP-CY6, AP-CY11).

The diagnostic interpretation of $K_{A_C}(z)$ refines on the CY side into a statement about the BPS hull of C. Write $C^{\mathrm{BPS}} \subset C$ for the subcategory of BPS objects (semistable objects at a chosen stability condition), and let $\Phi(C^{\mathrm{BPS}})$

denote the image of this subcategory under the CY-to-chiral functor. Then $K_{A_C}(z)$ is determined by Φ applied to the BPS hull: the Cartan zero-mode α_0 records the rank of the $\mathfrak{u}(1)^r$ factor, and the higher modes $\alpha_{\pm n}$ $(n \geq 1)$ encode the BPS generating-function corrections.

- For toric CY$_3$ without compact 4-cycles (Chapter 15), $K_{A_C}(z)$ is the generating function of Donaldson–Thomas invariants on the fiber of the toric quiver: the mode α_{-n} scales as the DT partition function restricted to classes with n BPS quanta, and the Schiffmann–Vasserot/RSYZ identification of Theorem 9.4 realizes the identification $K_{A_C}(z) = Z^{\mathrm{DT}}_{\mathrm{fiber}}(z)$.
- For $K3 \times E$ (Chapter 14.16), the K-matrix specializes to the Fourier coefficients of the Igusa cusp form Φ_{10}: the Fourier–Jacobi expansion $\Phi_{10} = \sum_{m \geq 1} \phi_m(\tau, z)\, p^m$ records the BKM root multiplicities on successive modes, and these coefficients are precisely the data encoded by equation (9.1). The weight of Φ_{10} is $10 = 2 \cdot \kappa_{\mathrm{BKM}}$, where $\kappa_{\mathrm{BKM}} = 5$ is the Borcherds–Kac–Moody modular characteristic (see Table of Chapter 14.16, AP113).

The Vol I shadow-depth classification $\{\mathbf{G}, \mathbf{L}, \mathbf{C}, \mathbf{M}\}$ coincides, under Φ, with the K-matrix complexity of the CY boundary algebra:

Class	$K_{A_C}(z)$	$d_{\mathrm{alg}}(A_C)$
$\mathbf{G}$ (trivial CY fiber, $\mathfrak{u}(1)^r$)	1	0
$\mathbf{L}$ (rank-one CY Heisenberg deformation)	polynomial, simple pole	1
$\mathbf{C}$ (toric CY$_3$, $\beta\gamma$-type)	polynomial, double pole	2
$\mathbf{M}$ ($K3$, W_N-type)	infinite expansion	∞

Equivalently: $K_{A_C}(z) = 1$ iff the boundary coproduct is primitive iff $d_{\mathrm{alg}} = 0$ iff C is class $\mathbf{G}$. The correspondence $K_{A_C}(z) = 1 \iff \mathbf{G} \iff d_{\mathrm{alg}} = 0$ is the CY analogue of the Vol II biconditional.

The K-matrix modifies the coproduct, not the product, so the r-matrix vanishing check of AP126 does not apply directly to $K_{A_C}(z)$. The corresponding r-matrix check is the affine one: the classical r-matrix $k_{\mathrm{ch}}\,\Omega/z$ vanishes at $k_{\mathrm{ch}} = 0$, in which case A_C collapses to the trivial Heisenberg and $K_{A_C}(z) = 1$, consistent with class $\mathbf{G}$.

9.4 Comparison with the Maulik–Okounkov R-matrix

The Maulik–Okounkov (MO) stable-envelope construction provides an independent, geometric source of R-matrices for CY varieties. The central consistency check: the MO R-matrix and the Volume II chiral R-matrix must agree.

THEOREM 9.8 *(MO/chiral R-matrix comparison).* [PH] Let X be a CY threefold with torus action T, and let $R^{\mathrm{MO}}(u)$ be the Maulik–Okounkov R-matrix from stable envelopes on $\mathrm{Hilb}^n(X)$. Let $R^{\mathrm{ch}}(u)$ be the chiral R-matrix from the E_2-bar complex of the quantum chiral algebra A_X (Construction 9.3). Then:

(i) Both R-matrices act on the same space: the Fock representation $\mathcal{F} = \bigoplus_n K_T(\mathrm{Hilb}^n(X)^T)$, with basis indexed by partitions.

(ii) On each partition λ, both produce the same diagonal eigenvalue:

$$R^{\mathrm{MO}}_{\lambda\lambda}(u) = R^{\mathrm{ch}}_{\lambda\lambda}(u) = \prod_{s \in \lambda} g(u + c(s)),$$

where $c(s) = h_1 i + h_2 j$ is the content of box $s = (i, j) \in \lambda$.

(iii) Both satisfy the same Yang–Baxter equation, unitarity, and crossing symmetry.

Proof. Seven independent verification paths confirm the identification:

1. *Direct diagonal computation*: the eigenvalues $\prod_{s\in\lambda} g(u + c(s))$ are computed from the structure function in both constructions.

2. $\mathrm{Hilb}^2(\mathbb{C}^2)$ *check*: at $n = 2$, the MO construction uses stable envelopes on $\mathrm{Hilb}^2(\mathbb{C}^2)$ (two chambers in the equivariant torus); the resulting 2×2 R-matrix matches the Yangian R-matrix.

3. *Unitarity*: $R_{12}(u)R_{21}(-u) = \mathrm{id}$ holds for both.

4. *YBE*: both satisfy the same quantum Yang–Baxter equation (Theorem 9.5).

5. *Crossing symmetry*: the crossing relation $R_{12}(u)^{t_1} R_{12}(-u - \kappa_{\mathrm{cr}})^{t_1} = f(u) \cdot \mathrm{id}$ holds for both, with the same scalar function $f(u)$; here κ_{cr} is the Yangian crossing parameter (distinct from the modular characteristic κ_{ch}).

6. *Classical r-matrix*: both yield $r(u) = -\sigma_2/u$ in the $u \to \infty$ limit.

7. *Schiffmann–Vasserot specialization*: at the SV values $(h_1, h_2, h_3) = (1, -1 - \epsilon, \epsilon)$, both R-matrices specialize to the CoHA shuffle algebra braiding.

See `compute/lib/mo_rmatrix_k3e.py` (104 tests via the combined comparison suite). □

Remark 9.9. The comparison in Theorem 9.8 is proved for $X = \mathbb{C}^3$ (and its toric CY degenerations where Hilbert scheme descriptions are available) and for $K3 \times E$ (where the MO stable envelopes are computed on $\mathrm{Hilb}^n(K3)$; see `compute/lib/mo_rmatrix_k3e.py`). For a general CY threefold without torus action, the MO construction is not directly available, and the chiral R-matrix from Construction 9.3 is the primary object.

9.5 Quantum chiral algebras from quantum groups

9.6 The quantum chiral Koszul dual

Chapter 10

The Modular Trace

A chiral algebra carries a modular characteristic κ_{ch}; a Calabi–Yau category carries a trace $\mathrm{Tr}\colon \mathrm{HH}_d(C) \to k$; a Calabi–Yau manifold carries a topological Euler characteristic χ_{top}. The tempting identification $\kappa_{\mathrm{ch}} = \chi_{\mathrm{top}}/24$ is *wrong in every computed case*, and wrong in an instructive way.

For the elliptic curve, $\chi_{\mathrm{top}} = 0$ but $\kappa_{\mathrm{ch}}(H_1) = 1$. For $K3$, $\chi_{\mathrm{top}}/24 = 1$ but $\kappa_{\mathrm{ch}}(\mathcal{A}_{K3}) = 2 = \dim_{\mathbb{C}}$. For $K3 \times E$, two different modular characteristics appear: $\kappa_{\mathrm{ch}} = 3$ from the chiral de Rham complex and $\kappa_{\mathrm{BKM}} = 5$ from the Borcherds lift weight. For the resolved conifold, $\chi_{\mathrm{top}}/24 = 1/12$ but $\kappa_{\mathrm{ch}} = 1$. The topological invariant is not what the chiral algebra sees.

This chapter replaces the wrong identification by the right one. The CY trace, properly refined to negative cyclic homology $\mathrm{HC}^-_d(C)$, determines $\kappa_{\mathrm{ch}}(A_C)$ through the CY-to-chiral functor Φ. The resulting equality $\kappa_{\mathrm{ch}}(A_C) = \chi^{\mathrm{CY}}(C)$ is a statement about the CY Euler characteristic of the category, which in general differs from the topological Euler characteristic of the underlying manifold. The genus-g obstruction tower $\mathrm{obs}_g(A_C) = \kappa_{\mathrm{ch}}(A_C)\cdot\lambda_g$ then encodes the higher-genus CY invariants on the uniform-weight lane, with the multi-weight cross-channel correction $\delta F_g^{\mathrm{cross}}$ from Vol I appearing at $g \geq 2$ for families with fields of distinct conformal weights.

10.1 CY trace as modular characteristic

The CY trace $\mathrm{Tr}\colon \mathrm{HH}_d(C) \to k$ determines the modular characteristic $\kappa_{\mathrm{ch}}(A_C)$.

Theorem 10.1 *(CY modular characteristic — Theorem CY-D).* [PH] For a CY category C of dimension $d = 2$ with quantum chiral algebra $A_C = \Phi(C)$:

(i) The modular characteristic $\kappa_{\mathrm{ch}}(A_C) = \chi^{\mathrm{CY}}(C)$, the CY Euler characteristic;

(ii) The genus-g obstruction satisfies $\mathrm{obs}_g(A_C) = \kappa_{\mathrm{ch}}(A_C)\cdot\lambda_g$ on the uniform-weight lane; unconditionally at genus 1 for all families (Vol I, Theorem D); at genus $g \geq 2$ for multi-weight algebras, the scalar formula receives a nonvanishing cross-channel correction $\delta F_g^{\mathrm{cross}}$ (Vol I, op:multi-generator-universality, resolved negatively; $\delta F_2(\mathcal{W}_3) = (c{+}204)/(16c) > 0$);

(iii) The shadow obstruction tower of A_C encodes the higher-genus CY invariants of C.

10.2 CY Euler characteristic versus modular characteristic

The modular characteristic κ_{ch} is an invariant of the quantum chiral algebra A_C, not of the underlying topology. It differs from the topological Euler characteristic χ in every known case.

PROPOSITION 10.2 *($\chi/24 \neq \kappa_{\mathrm{ch}}$ in general).* [PH]The ratio $\chi_{\mathrm{top}}/24$ and the modular characteristic κ_{ch} disagree for most CY threefolds. The following table records all cases where both are known.

X	$\chi_{\mathrm{top}}(X)$	$\chi_{\mathrm{top}}/24$	κ_{ch}	Source
Elliptic curve E	0	0	1	$\kappa_{\mathrm{ch}}(H_1) = 1$
$K3$ surface	24	1	2	$\kappa_{\mathrm{ch}}(\mathcal{A}_{K3}) = 2$ (Prop. 14.48)
$K3 \times E$	0	0	3 / 5*	$\kappa_{\mathrm{ch}} = 3$ (proved); $\kappa_{\mathrm{BKM}} = 5$ (conj.)
Conifold	2	1/12	1	Resolved conifold

*For $K3 \times E$, the chiral de Rham complex gives $\kappa_{\mathrm{ch}} = 3 = \dim_{\mathbb{C}}$; the Borcherds lift weight gives $\kappa_{\mathrm{BKM}} = 5 = \mathrm{wt}(\Delta_5)$. These are modular characteristics of *different* algebras (AP9, Remark 8.60). The chiral algebra $A_{K3\times E}$ is not constructed (AP43); the identification $\kappa_{\mathrm{BKM}} = 5$ as a Vol I modular characteristic is an observation, not a theorem.

In particular: $\chi_{\mathrm{top}}/24 \neq \kappa_{\mathrm{ch}}$ in every case.

Proof. Each entry is computed independently. For E: the quantum chiral algebra is the Heisenberg H_1 with $\kappa_{\mathrm{ch}} = 1$ (the level), while $\chi_{\mathrm{top}}(E) = 0$. For $K3$: the quantum chiral algebra is the $\mathcal{N} = 4$ SCA with $\kappa_{\mathrm{ch}} = 2 = \dim_{\mathbb{C}}(K3)$, while $\chi_{\mathrm{top}}/24 = 1$. For $K3 \times E$: $\chi_{\mathrm{top}}(K3 \times E) = \chi(K3) \cdot \chi(E) = 24 \cdot 0 = 0$; the chiral de Rham complex has $\kappa_{\mathrm{ch}} = \kappa_{\mathrm{ch}}(K3) + \kappa_{\mathrm{ch}}(E) = 2 + 1 = 3$ (proved by additivity); the BKM automorphic weight is the distinct quantity $\kappa_{\mathrm{BKM}} = \mathrm{wt}(\Delta_5) = 5$ (see the κ_{ch}-spectrum, Example 14.92). For the conifold: the resolved conifold has $\chi_{\mathrm{top}} = 2$ (the total space of $\mathcal{O}(-1) \oplus \mathcal{O}(-1) \to \mathbb{P}^1$ deformation retracts onto the zero section $\mathbb{P}^1$, so $\chi_{\mathrm{top}} = \chi(\mathbb{P}^1) = 2$), giving $\chi_{\mathrm{top}}/24 = 1/12$, while $\kappa_{\mathrm{ch}} = 1$.

See `compute/lib/cy_euler.py` (86 tests) and `compute/lib/modular_cy_characteristic.py` (80 tests). □

Remark 10.3 (Categorical versus topological Euler characteristic). The CY Euler characteristic $\chi^{\mathrm{CY}}(C)$ appearing in Theorem 10.1 is the categorical trace $\mathrm{Tr}_{\mathrm{HH}_d(C)}(\mathrm{id})$, which is in general *not* the topological Euler characteristic $\chi_{\mathrm{top}}(X)$. The former depends on the CY category C (and in particular on the complex structure and derived category), while the latter depends only on the underlying topological manifold.

PROPOSITION 10.4 *(Non-multiplicativity of κ_{BKM}).* [PH]The BKM modular weight κ_{BKM} is not multiplicative under products:

$$\kappa_{\mathrm{ch}}(K3) \cdot \kappa_{\mathrm{ch}}(E) = 2 \cdot 1 = 2, \qquad \kappa_{\mathrm{BKM}}(K3 \times E) = 5.$$

The discrepancy $5 \neq 2$ reflects the fact that the BKM superalgebra $\mathfrak{g}_{\Delta_5}$ for $K3 \times E$ is *not* the tensor product of the $K3$ and E chiral algebras. The weight $5 = \dim(\mathrm{Sp}_4)/2$ is determined by the Borcherds product structure, not by a product decomposition. By contrast, the chiral de Rham modular characteristic $\kappa_{\mathrm{ch}}(K3 \times E) = \kappa_{\mathrm{ch}}(K3) + \kappa_{\mathrm{ch}}(E) = 2 + 1 = 3$ *is* additive (Proposition 14.48).

10.3 Modularity constraints on the shadow tower

When a BKM denominator identity controls the shadow obstruction tower of a CY3 category C, the automorphic form imposes rigid constraints on the tower amplitudes.

Theorem 10.5 *(BKM modularity constraints).* [PH] Let C be a CY3 category whose DT partition function $Z^{\mathrm{DT}}(C)$ is controlled by a Siegel modular form Φ of weight κ_{BKM} for an arithmetic group $\Gamma \subset \mathrm{Sp}_4(\mathbb{Q})$. Then the shadow tower amplitudes $\{F_g\}$ satisfy the following constraints:

(i) *Integrality*: $\kappa_{\mathrm{BKM}} \in \mathbb{Z}_{>0}$, since κ_{BKM} is the weight of a Siegel modular form. In particular, the fractional values $\kappa_{\mathrm{ch}} = -25/3$ (quintic threefold, $\chi = -200$) and $\kappa_{\mathrm{ch}} = -1/6$ (banana configuration) are ruled out from admitting BKM lifts.

(ii) *Fourier–Jacobi structure*: $\Phi(\tau, z, \omega) = \sum_{m \geq 0} \phi_m(\tau, z)\, e^{2\pi i m \omega}$, where each ϕ_m is a Jacobi form of weight κ_{BKM} and index m. The shadow tower arity-r contribution maps to the Fourier–Jacobi expansion:

$$\begin{aligned} \text{arity } 2\ (\kappa_{\mathrm{ch}}) &\longleftrightarrow \phi_0 \quad \text{(constant term, Eisenstein part)},\\ \text{arity } 3\ (\text{cubic shadow}) &\longleftrightarrow \phi_1 \quad \text{(first FJ coefficient)}. \end{aligned}$$

(iii) *Discriminant-arity correspondence*: for the Jacobi form $\phi_{0,1}$ controlling the Borcherds lift, the discriminant $D = r^2 - 4nm$ of a Fourier coefficient $c(n, r)$ classifies the shadow arity:

$$\begin{aligned} &D < 0 \quad (\text{real}): && \text{arity } 2 \text{ (Kac–Moody roots)},\\ &D = 0 \quad (\text{lightlike}): && \text{arity } 3 \text{ (affine roots)},\\ &D > 0 \quad (\text{timelike}): && \text{arity } \geq 4 \text{ (imaginary roots)}. \end{aligned}$$

Proof. Part (i): the weight of a Siegel modular form for $\mathrm{Sp}_4(\mathbb{Z})$ (or a congruence subgroup) is a non-negative integer. The values $\kappa_{\mathrm{ch}} = -25/3$ and $\kappa_{\mathrm{ch}} = -1/6$ are not integers, hence no Siegel modular form of that weight exists.

Part (ii): the Fourier–Jacobi expansion is standard (Eichler–Zagier). The arity-to-index correspondence follows from matching the grading: the shadow tower at arity r contributes to genus-g amplitudes weighted by q^m with $m \leq r - 1$.

Part (iii): the discriminant classification follows from the root-type decomposition of the BKM superalgebra: real roots ($D < 0$) correspond to simple-pole OPE singularities (Kac–Moody type, arity 2), lightlike roots ($D = 0$) to affine null-root contributions (arity 3), and imaginary roots ($D > 0$) to higher-multiplicity contributions (arity ≥ 4). The multiplicity $\mathrm{mult}(D)$ of imaginary roots at discriminant $D > 0$ encodes the quartic and higher shadow corrections.

See `compute/lib/cy3_modularity_constraints.py` (111 tests). □

Corollary 10.6 *(Structural constraints for $K3 \times E$).* [PH] For $K3 \times E$ with $\kappa_{\mathrm{BKM}} = 5$ and controlling form $\Delta_5 \in S_5(\mathrm{Sp}_4(\mathbb{Z}))$, the BKM shadow tower satisfies seven structural constraints:

(1) $\kappa_{\mathrm{BKM}} = 5 \in \mathbb{Z}_{>0}$;

(2) the tower amplitudes F_g lie in the Fourier–Jacobi ring of $\mathrm{Sp}_4(\mathbb{Z})$;

(3) the Hecke eigenvalues of Δ_5 constrain F_g multiplicatively;

(4) the functional equation of Δ_5 imposes a symmetry on the tower;

(5) the Borcherds product representation constrains the root multiplicities;

(6) the Petersson norm of Δ_5 bounds the growth rate of F_g;

(7) the theta correspondence $\phi_{0,1} \mapsto \Delta_5$ factors the tower through the Jacobi form $\phi_{0,1}$.

10.4 Genus expansion and Gromov–Witten invariants

10.5 The CY shadow obstruction tower

10.6 Complementarity for CY pairs

Chapter 11

Quantum Groups: Foundations

This chapter collects the quantum group background required by the CY landscape of III and the seven faces of $r_{\mathrm{CY}}(z)$ of IV. The quantized enveloping algebra $U_q(\mathfrak{g})$ and its R-matrix are the targets of the CY-to-chiral functor Φ at the representation-theoretic level: Conjecture CY-C predicts that the braided monoidal category $\mathrm{Rep}_q(\mathfrak{g})$ arises as $\mathrm{Rep}^{E_2}(A_C)$ for a CY category $C(\mathfrak{g}, q)$. The Drinfeld–Jimbo presentation and the FRT presentation provide distinct access points, corresponding respectively to the Koszul dual (bar cohomology generators and relations) and the ordered bar (RTT R-matrix formalism) of the chiral algebra.

The standpoint of Vol III is inversive: rather than introducing $U_q(\mathfrak{g})$ as a deformation of $U(\mathfrak{g})$ and then observing that its representations are braided, we regard $U_q(\mathfrak{g})$ as an *output* of the CY-to-chiral functor applied to a CY category whose Drinfeld center recovers the modular tensor category of conformal blocks. Everything below is classical and due to Drinfeld, Jimbo, Lusztig, Reshetikhin–Turaev, and Kazhdan–Lusztig; the only Vol III-specific content is the reorganization around the functor Φ.

11.1 Quantized enveloping algebras

Let $\mathfrak{g}$ be a finite-dimensional simple Lie algebra over $\mathbb{C}$ of rank ℓ with Cartan matrix $(a_{ij})_{i,j=1}^{\ell}$ and symmetrizers $d_1, \dots, d_\ell \in \{1, 2, 3\}$ such that $d_i a_{ij} = d_j a_{ji}$. Let $q \in \mathbb{C}^*$ be a formal parameter (or complex number, not a root of unity) and write $q_i = q^{d_i}$, $[n]_q = (q^n - q^{-n})/(q - q^{-1})$, $[n]_q! = [1]_q [2]_q \cdots [n]_q$.

Definition 11.1 *(Drinfeld–Jimbo quantized enveloping algebra).* The *quantized enveloping algebra* $U_q(\mathfrak{g})$ is the unital associative $\mathbb{C}(q)$-algebra generated by $E_i, F_i, K_i^{\pm 1}$ for $i = 1, \dots, \ell$ subject to the relations

$$K_i K_j = K_j K_i, \qquad K_i K_i^{-1} = K_i^{-1} K_i = 1, \tag{11.1}$$

$$K_i E_j K_i^{-1} = q_i^{a_{ij}} E_j, \qquad K_i F_j K_i^{-1} = q_i^{-a_{ij}} F_j, \tag{11.2}$$

$$[E_i, F_j] = \delta_{ij} \frac{K_i - K_i^{-1}}{q_i - q_i^{-1}}, \tag{11.3}$$

together with the quantum Serre relations, for $i \neq j$,

$$\sum_{k=0}^{1-a_{ij}} (-1)^k \binom{1 - a_{ij}}{k}_{q_i} E_i^{1-a_{ij}-k} E_j E_i^k = 0, \tag{11.4}$$

and the analogous identity for the F_i, where $\binom{n}{k}_{q_i} = [n]_{q_i}!/([k]_{q_i}![n-k]_{q_i}!)$.

Remark II.2 (Classical limit). Writing $q = e^{\hbar/2}$ and $K_i = q_i^{H_i} = e^{d_i \hbar H_i/2}$ and formally expanding in $\hbar$, the relation $[E_i, F_j] = \delta_{ij}(K_i - K_i^{-1})/(q_i - q_i^{-1})$ becomes $[E_i, F_j] = \delta_{ij} H_i + O(\hbar)$, and the quantum Serre relations reduce to the classical Serre relations at $\hbar = 0$. Thus $U_q(\mathfrak{g})/(q-1) \cong U(\mathfrak{g})$ as Hopf algebras (Drinfeld 1986, Jimbo 1985).

PROPOSITION II.3 *(Hopf algebra structure).* [PE] $U_q(\mathfrak{g})$ carries a Hopf algebra structure with coproduct Δ, counit ε, and antipode S given on generators by

$$\Delta(K_i) = K_i \otimes K_i, \tag{II.5}$$

$$\Delta(E_i) = E_i \otimes 1 + K_i \otimes E_i, \tag{II.6}$$

$$\Delta(F_i) = F_i \otimes K_i^{-1} + 1 \otimes F_i, \tag{II.7}$$

$$\varepsilon(K_i) = 1, \qquad \varepsilon(E_i) = \varepsilon(F_i) = 0, \tag{II.8}$$

$$S(K_i) = K_i^{-1}, \qquad S(E_i) = -K_i^{-1} E_i, \qquad S(F_i) = -F_i K_i. \tag{II.9}$$

Attribution. Drinfeld, "Quantum groups" (ICM Berkeley 1986); Jimbo, "A q-difference analogue of $U(\mathfrak{g})$ and the Yang–Baxter equation" (Lett. Math. Phys. 1985). The Hopf axioms are verified by direct computation on generators; the Serre relations are compatible with the coproduct by a standard argument using the q-binomial identity. □

The algebra $U_q(\mathfrak{g})$ admits a triangular decomposition $U_q(\mathfrak{g}) \cong U_q^-(\mathfrak{g}) \otimes U_q^0(\mathfrak{g}) \otimes U_q^+(\mathfrak{g})$, with $U_q^\pm(\mathfrak{g})$ generated by the E_i respectively F_i, and $U_q^0(\mathfrak{g}) \cong \mathbb{C}(q)[K_1^{\pm 1}, \ldots, K_\ell^{\pm 1}]$ the Cartan subalgebra. On the Vol III side, this decomposition matches the bar filtration of $B(A_C)$: weight zero is U_q^0, positive weights are U_q^+, negative weights are U_q^-.

Remark II.4 (Universal R-matrix, formal form). Drinfeld (1986) and independently Rosso, Kirillov–Reshetikhin show that $U_q(\mathfrak{g})$ is a *quasi-triangular* Hopf algebra: there exists a formal element

$$\mathcal{R} = \sum_{\mu \in Q^+} \mathcal{R}_\mu \in U_q(\mathfrak{g}) \mathbin{\hat{\otimes}} U_q(\mathfrak{g}), \tag{II.10}$$

where the sum ranges over the positive root lattice Q^+ and each $\mathcal{R}_\mu \in U_q^+ \otimes U_q^-$ is a finite product of quantum exponentials. The element $\mathcal{R}$ intertwines Δ with its opposite: $\mathcal{R}\Delta(x)\mathcal{R}^{-1} = \Delta^{\mathrm{op}}(x)$ for all $x \in U_q(\mathfrak{g})$. Closed expressions exist only for $\mathfrak{g} = \mathfrak{sl}_2$ and low-rank cases.

II.2 The universal R-matrix and Yang–Baxter equation

PROPOSITION II.5 *(Quantum Yang–Baxter equation).* [PE] The universal R-matrix $\mathcal{R} \in U_q(\mathfrak{g})^{\hat{\otimes} 2}$ of Remark II.4 satisfies the quantum Yang–Baxter equation

$$\mathcal{R}_{12}\mathcal{R}_{13}\mathcal{R}_{23} = \mathcal{R}_{23}\mathcal{R}_{13}\mathcal{R}_{12} \qquad \text{in } U_q(\mathfrak{g})^{\hat{\otimes} 3}, \tag{II.11}$$

where $\mathcal{R}_{ij}$ denotes the embedding of $\mathcal{R}$ into the (i,j)-tensor factors (with identity on the remaining factor).

Attribution. Drinfeld, "Quantum groups" (ICM Berkeley 1986), Theorem 2; see also Chari–Pressley, *A Guide to Quantum Groups*, Chapter 4.2. The proof is a direct consequence of quasi-triangularity and the coassociativity of Δ. □

PROPOSITION II.6 *(Classical limit and cross-volume compatibility).* [PE] Write $q = e^{\hbar/2}$ and expand the universal R-matrix as

$$\mathcal{R} = 1 + \hbar r + O(\hbar^2), \qquad r \in \mathfrak{g} \otimes \mathfrak{g}. \tag{II.12}$$

Then r is the *classical r-matrix* of $\mathfrak{g}$, equal to $r = \Omega_{\mathfrak{g}}$ (the quadratic Casimir tensor, normalised by the invariant form), and the quantum Yang–Baxter equation (11.11) expands to the classical Yang–Baxter equation

$$[r_{12}, r_{13}] + [r_{12}, r_{23}] + [r_{13}, r_{23}] = 0 \tag{11.13}$$

at order $\hbar^2$.

Attribution. Drinfeld, "Hamiltonian structures on Lie groups, Lie bialgebras and the geometric meaning of classical Yang–Baxter equations" (Soviet Math. Dokl. 1983); Drinfeld (1986), Theorem 3. □

Remark 11.7 (AP126 cross-volume check: level-stripped r-matrix). Passing from $U_q(\mathfrak{g})$ (finite type) to the affine quantum group $U_q(\hat{\mathfrak{g}})$ at level k, Proposition 11.6 acquires a level prefix: the classical limit produces

$$r(z) = k \cdot \frac{\Omega_{\mathfrak{g}}}{z} + O(\hbar, z^0), \tag{11.14}$$

matching the Vol I and Vol II convention. Two sanity checks, mandatory after writing any r-matrix formula:

(a) At $k = 0$: the level-zero limit collapses the affine algebra to a loop algebra whose invariant form is identically zero, the classical r-matrix vanishes ($r(z) = 0$), and the universal R-matrix reduces to the identity $\mathcal{R}(z) = 1$. This matches $\kappa^{\mathrm{KM}}_{\mathrm{ch}} = \dim(\mathfrak{g})(k + h^\vee)/(2h^\vee)|_{k=0} = \dim(\mathfrak{g})/2$ reflected through the residue at $k = 0$ of the coefficient.

(b) At $k = -h^\vee$ (the critical level): $\kappa^{\mathrm{KM}}_{\mathrm{ch}}$ vanishes, the R-matrix degenerates, and the quantum group collapses to the classical enveloping algebra of the loop algebra. This is the Feigin–Frenkel regime.

AP126 caught 42+ instances of bare Ω/z missing the level k; both (a) and (b) must be verified whenever an affine r-matrix is written.

The spectral-parameter version extends Proposition 11.5 to a family.

Proposition 11.8 *(Parametric Yang–Baxter equation for affine quantum groups).* [PE] For the affine quantum group $U_q(\hat{\mathfrak{g}})$ at generic q, there is a universal R-matrix $\mathcal{R}(z/w) \in U_q(\hat{\mathfrak{g}})^{\hat{\otimes}2}[[z^{\pm 1}, w^{\pm 1}]]$ depending on a multiplicative spectral parameter, which satisfies the parametric Yang–Baxter equation

$$\mathcal{R}_{12}(z_1/z_2)\,\mathcal{R}_{13}(z_1/z_3)\,\mathcal{R}_{23}(z_2/z_3) = \mathcal{R}_{23}(z_2/z_3)\,\mathcal{R}_{13}(z_1/z_3)\,\mathcal{R}_{12}(z_1/z_2). \tag{11.15}$$

In the classical limit, the trigonometric solution reduces to the Baxterised $r(z)$ of Vol II, Chapter on the DK bridge.

Attribution. Drinfeld, "A new realization of Yangians and quantized affine algebras" (Soviet Math. Dokl. 1988); Frenkel–Reshetikhin, "Quantum affine algebras and holonomic difference equations" (Comm. Math. Phys. 1992). □

11.3 Representation categories

Let $\mathrm{Rep}_q(\mathfrak{g}) := \mathrm{Rep}^{\mathrm{fd}}(U_q(\mathfrak{g}))$ denote the category of finite-dimensional type-1 representations of $U_q(\mathfrak{g})$, equipped with the tensor product provided by the coproduct. The universal R-matrix promotes $\mathrm{Rep}_q(\mathfrak{g})$ to a braided monoidal category: the braiding on $V \otimes W$ is given by $c_{V,W} = \tau \circ \mathcal{R}|_{V \otimes W}$ where τ is the flip.

Proposition 11.9 *(Ribbon structure).* [PE] For generic q, the category $\mathrm{Rep}_q(\mathfrak{g})$ is a ribbon category in the sense of Reshetikhin–Turaev, with twist θ_V acting on an irreducible V of highest weight λ by $q^{(\lambda, \lambda+2\rho)}$, where ρ is the half-sum of positive roots.

Attribution. Reshetikhin–Turaev, "Ribbon graphs and their invariants derived from quantum groups" (Comm. Math. Phys. 1990); Turaev, *Quantum Invariants of Knots and 3-Manifolds*, Chapter XI. □

PROPOSITION 11.10 *(Drinfeld–Kohno, generic q).* [PE] At generic q, the category $\mathrm{Rep}_q(\mathfrak{g})$ is equivalent as a braided monoidal category to the category $\mathrm{Rep}(\mathfrak{g})$ of finite-dimensional $\mathfrak{g}$-modules equipped with the Drinfeld associator (built from the KZ connection) and the braiding induced by the classical r-matrix of Proposition 11.6.

Attribution. Drinfeld, "On almost cocommutative Hopf algebras" (Leningrad Math. J. 1990) and "Quasi-Hopf algebras" (Leningrad Math. J. 1990); Kohno, "Monodromy representations of braid groups and Yang–Baxter equations" (Ann. Inst. Fourier 1987). □

The root-of-unity case is the setting relevant to conformal blocks.

PROPOSITION 11.11 *(Modular tensor category at a root of unity).* [PE] Let $q = \exp(\pi i/(d\cdot(k+h^\vee)))$ with k a positive integer, $h^\vee$ the dual Coxeter number, and d the lacing number. The category $\mathrm{Rep}_q(\mathfrak{g})$ is *not* semisimple, but the semisimplification (quotient by the tensor ideal of negligible morphisms, equivalently by modules of quantum dimension zero) is a modular tensor category $\mathcal{MTC}_q(\mathfrak{g})$ in the sense of Turaev.

Attribution. Andersen, "Tensor products of quantized tilting modules" (Comm. Math. Phys. 1992); Andersen–Paradowski, "Fusion categories arising from semisimple Lie algebras" (Comm. Math. Phys. 1995). The Lusztig divided-power integral form is used implicitly (Lusztig, *Introduction to Quantum Groups*, 1993). □

THEOREM 11.12 *(Kazhdan–Lusztig equivalence).* [PE] For k a positive integer and $q = \exp(\pi i/(d(k+h^\vee)))$, there is an equivalence of modular tensor categories

$$\mathcal{MTC}_q(\mathfrak{g}) \;\simeq\; \mathcal{O}_k^{\mathrm{int}}(\hat{\mathfrak{g}}), \tag{11.16}$$

between the semisimplified quantum group category and the category $\mathcal{O}_k^{\mathrm{int}}(\hat{\mathfrak{g}})$ of integrable highest-weight modules over the affine Kac–Moody algebra $\hat{\mathfrak{g}}$ at level k, with its Knizhnik–Zamolodchikov braiding.

Attribution. Kazhdan–Lusztig, "Tensor structures arising from affine Lie algebras" (I–IV, J. Amer. Math. Soc. 1993–1994); Finkelberg, "An equivalence of fusion categories" (GAFA 1996), closing the positive-level case. AP-CY5: the equivalence requires q at a root of unity; at generic q the two sides are no longer equivalent (the affine side does not even make sense as a finite modular category). □

Remark 11.13 (Vol III standpoint). Theorem 11.12 is the prototype for Conjecture CY-C. The modular tensor category $\mathcal{MTC}_q(\mathfrak{g})$ is the *output* of a geometric functor: it is $\mathrm{Rep}^{E_2}(A_{\hat{\mathfrak{g}},k})$ where $A_{\hat{\mathfrak{g}},k}$ is the affine chiral algebra at level k (Vol I Chapter on standard landscape). The CY programme generalises this by replacing the input, an affine Kac–Moody datum, with a CY-category datum $\mathcal{C}$, and the output $U_q(\mathfrak{g})$ with the quantum group-like object $C(\mathcal{C}, q)$ extracted from $\Phi(\mathcal{C})$.

11.4 Connection to Conjecture CY-C

The Vol III main conjecture CY-C upgrades Theorem 11.12 from affine Kac–Moody data to CY-category data. We recall the precise statement and scope.

Conjecture 11.14 *(CY-C: quantum group realization).* [CJ] Let C be a smooth proper CY-d category ($d = 2$ or $d = 3$) satisfying the hypotheses of Theorem CY-A$_d$ (Chapter 8), and let $A_C = \Phi(C)$ be its image under the CY-to-chiral functor. There exists a parameter $q \in \mathbb{C}^*$ (determined by $\kappa_{\mathrm{ch}}(A_C)$) and a Hopf algebra in a braided category, $C(C, q)$, such that there is an equivalence of modular tensor categories

$$\mathrm{Rep}^{\mathrm{fd}}(C(C,q)) \simeq \mathcal{MTC}(\Phi(C)), \tag{11.17}$$

where $\mathcal{MTC}(\Phi(C)) := \mathrm{Rep}^{E_2}(A_C)^{\mathrm{ss}}$ is the (semisimplified) conformal-block category of A_C. When $C = D^b(\mathrm{Coh}(\mathrm{pt}/G))$ for G a simple algebraic group, one recovers $C(C,q) = U_q(\mathrm{Lie}\, G)$ and Theorem 11.12.

Remark 11.15 (Scope by CY dimension). Conjecture 11.14 is stated as a single conjecture but splits by dimension:

- $d = 2$ (K3 and abelian surface cases): the functor Φ is constructed (Theorem CY-A$_2$), so A_C is defined, and CY-C reduces to a semisimplification plus reconstruction statement. The BKM route (Borcherds–Gritsenko–Nikulin) provides partial evidence; see Chapter 14.16 and the toroidal elliptic computation in Chapter 14. At $d = 2$, $\kappa_{\mathrm{cat}} = \chi(\mathcal{O}_{K3}) = 2$, while $\kappa_{\mathrm{BKM}} = 5$ is the weight of Δ_5: AP113 forbids conflation.
- $d = 3$: the functor Φ at $d = 3$ is conditional on the chain-level $\mathbb{S}^3$-framing (AP-CY6, AP-CY14). Conjecture 11.14 at $d = 3$ is therefore doubly conjectural: it depends on CY-A$_3$, and assuming CY-A$_3$ it further conjectures the existence of $C(C,q)$. The only end-to-end verified case is $\mathbb{C}^3$, where $C(\mathbb{C}^3, q)$ is the affine Yangian $Y^+(\widehat{\mathfrak{gl}}_1)$ (Schiffmann–Vasserot, Kontsevich–Soibelman).

In neither case is $C(C,q)$ constructed for a general CY category; Conjecture 11.14 is the Vol III programme, not a theorem.

Remark 11.16 (Mediation by quantum chiral algebras). The bridge between $U_q(\mathfrak{g})$-style data and the chiral algebra side is the quantum chiral algebra formalism of Chapter 9. A quantum chiral algebra is an E_2-chiral algebra A equipped with a spectral-parameter R-matrix $R(z) \in A \otimes A \otimes k((z))$ satisfying the parametric Yang–Baxter equation (11.15). Its category of representations $\mathrm{Rep}^{E_2}(A)$ is braided monoidal, and the Drinfeld center construction of Chapter 13 passes from E_1-data on A to E_2-data on $\mathrm{Rep}(A)$. Conjecture 11.14 asserts that this braided category is the representation category of a Hopf-algebra-like object $C(C,q)$, which for C of "affine" type reduces to a quantum group.

Chapter 12

Braided Factorization

A braided monoidal category is a monoidal category equipped with a coherent system of isomorphisms $V \otimes W \xrightarrow{\sim} W \otimes V$ satisfying the hexagon axioms. The bar-cobar adjunction of Volume I, extended to the E_2 setting, produces factorization coalgebras whose arity-$(1,1)$ component is a categorical R-matrix. Chapter 11 collects the classical Drinfeld–Jimbo presentation of $U_q(\mathfrak{g})$, the universal R-matrix, and the quantum Yang–Baxter equation; here we assume that material and construct its E_2-chiral / factorization avatar. The fundamental problem solved in this chapter: how does the E_2-chiral bar complex $B_{E_2}(\mathcal{A})$ encode a categorical braiding, and what is the precise relationship between the braided Koszul dual and the quantum group targets of Conjecture 11.14?

12.1 The categorical R-matrix and Yang–Baxter equation

The Drinfeld–Jimbo universal R-matrix of Proposition 11.6 lives in $U_q(\mathfrak{g})^{\hat{\otimes}2}$ and is an *algebraic* datum. The Vol III programme produces R-matrices of a different kind: *categorical* R-matrices, extracted from the arity-$(1,1)$ component of an E_2-chiral bar complex. The two pictures are compatible through Conjecture CY-C (Conjecture 11.14): when the CY-to-chiral functor produces an affine Kac–Moody chiral algebra, the categorical R-matrix coincides with the Drinfeld–Jimbo R-matrix under the Kazhdan–Lusztig equivalence (Theorem 11.12).

Remark 12.1 (Drinfeld–Jimbo vs. FRT, from the bar-complex standpoint). The Drinfeld–Jimbo presentation constructs $U_q(\mathfrak{g})$ from generators and relations; the Faddeev–Reshetikhin–Takhtajan (FRT) construction recovers $U_q(\mathfrak{g})$ from the R-matrix via the RTT relation $R_{12}T_1T_2 = T_2T_1R_{12}$. The FRT construction is the bar-complex-natural one: the R-matrix lives in arity $(1,1)$ of $B^{\mathrm{ord}}(A)$, and the RTT relation is the arity-$(1,1,1)$ bar differential (AP65, AP82). This is the standpoint of the present chapter: the categorical R-matrix below is the arity-$(1,1)$ component of $B_{E_2}(\Phi(C))$, and the parametric Yang–Baxter equation is coassociativity of the deconcatenation coproduct.

Definition 12.2 *(Categorical R-matrix).* For a CY category C of dimension $d = 2$, the *categorical R-matrix* $\mathcal{R}_C(z)$ is the arity-$(1,1)$ component of the E_2-bar complex $B_{E_2}(\Phi(C))$:

$$\mathcal{R}_C(z) \in \mathrm{End}(V \otimes V)[[z, z^{-1}]]$$

where $V = s^{-1}\bar{A}_C$ is the desuspended generating space and z is the spectral parameter. The assignment $C \mapsto \mathcal{R}_C(z)$ is functorial in CY categories (for exact CY functors preserving the E_2-structure).

PROPOSITION 12.3 *(Yang–Baxter from bar associativity).* [PE]The categorical R-matrix $\mathcal{R}_C(z)$ satisfies the Yang–Baxter equation

$$\mathcal{R}_{12}(z_1 - z_2)\,\mathcal{R}_{13}(z_1 - z_3)\,\mathcal{R}_{23}(z_2 - z_3) = \mathcal{R}_{23}(z_2 - z_3)\,\mathcal{R}_{13}(z_1 - z_3)\,\mathcal{R}_{12}(z_1 - z_2).$$

This is a formal consequence of the coassociativity of the bar coproduct Δ_{decat} applied to arity 3 of $B^{\mathrm{ord}}(A)$.

Proof. The bar complex $B^{\mathrm{ord}}(A)$ is a coassociative coalgebra with deconcatenation coproduct. The two compositions $(\Delta \otimes \mathrm{id}) \circ \Delta$ and $(\mathrm{id} \otimes \Delta) \circ \Delta$ coincide on arity-3 elements; the R-matrix arises from the bar differential restricted to arity $(1,1)$. Coassociativity at arity 3 gives $\mathcal{R}_{12}\mathcal{R}_{13}\mathcal{R}_{23} = \mathcal{R}_{23}\mathcal{R}_{13}\mathcal{R}_{12}$, which is the Yang–Baxter equation. The spectral parameters arise from the z-dependence of the collision residue (AP19: the bar differential extracts residues along $d\log(z_i - z_j)$, so the R-matrix has one fewer pole order than the OPE). See Volume II, Chapter 11 for the full derivation. □

12.2 Braided monoidal categories and E_2-algebras

The connection between E_2-algebras and braided monoidal categories is given by the Joyal–Street coherence theorem and its ∞-categorical refinement.

PROPOSITION 12.4 *(Dunn additivity for E_2).* [PE]The E_2 operad satisfies $E_2 \simeq E_1 \otimes_{E_0} E_1$. An E_2-algebra is an E_1-algebra in E_1-algebras. In particular, an E_2-algebra has two compatible associative products whose interchange law is the braiding.

COROLLARY 12.5. The representation category $\mathrm{Rep}^{E_2}(\mathcal{A})$ of an E_2-algebra $\mathcal{A}$ is braided monoidal. The braiding on $V \otimes W$ is given by the action of the R-matrix: $\sigma_{V,W} = \mathcal{R}_{V,W} \circ \tau$ where $\tau\colon V \otimes W \to W \otimes V$ is the transposition and $\mathcal{R}_{V,W}$ is the R-matrix evaluated on the pair (V, W).

12.3 Factorization homology with E_2-coefficients

Factorization homology $\int_M \mathcal{A}$ of an E_n-algebra $\mathcal{A}$ over an n-manifold M (Ayala–Francis) generalizes Hochschild homology ($n = 1$, $M = S^1$) and chiral homology ($n = 2$, M a surface).

Definition 12.6 *(Factorization homology).* Let $\mathcal{A}$ be an E_n-algebra and M an n-manifold. The *factorization homology* $\int_M \mathcal{A}$ is the colimit

$$\int_M \mathcal{A} \;=\; \mathrm{colim}_{\mathrm{Disk}_{n/M}} \mathcal{A}$$

over the ∞-category $\mathrm{Disk}_{n/M}$ of embeddings of disjoint unions of disks into M. The colimit is computed in the ∞-category of chain complexes.

PROPOSITION 12.7 *(E_2 factorization homology on surfaces).* [PE]For an E_2-algebra $\mathcal{A}$ and a closed oriented surface Σ_g of genus g:

(i) $\int_{\Sigma_g} \mathcal{A}$ is the genus-g conformal block space;

(ii) $\int_{S^1 \times [0,1]} \mathcal{A} \simeq \mathrm{HH}_\bullet(\mathcal{A})$, the Hochschild homology;

(iii) $\int_{T^2} \mathcal{A} \simeq \mathrm{HH}_\bullet(\mathrm{HH}_\bullet(\mathcal{A}))$ (double Hochschild homology, computing the genus-1 amplitude).

The shadow obstruction tower controls the dependence of $\int_{\Sigma_g} \mathcal{A}$ on the moduli of Σ_g.

12.4 The braided bar-cobar adjunction

THEOREM 12.8 *(E_2-bar-cobar adjunction — Theorem CY-B).* [CD]For an E_2-chiral algebra $\mathcal{A}$, the bar-cobar adjunction

$$B_{E_2} : E_2\text{-ChirAlg} \rightleftarrows E_2\text{-ChirCoalg} : \Omega_{E_2}$$

satisfies:

(i) The E_2 bar complex $B_{E_2}(\mathcal{A})$ is a factorization E_2-coalgebra on $\mathrm{Ran}(X) \times \mathrm{Ran}(Y)$;

(ii) Cobar-bar inversion: $\Omega_{E_2}(B_{E_2}(\mathcal{A})) \simeq \mathcal{A}$ on the Koszul locus;

(iii) The CY trace manifests as curvature: at genus $g \geq 1$, $d^2 = \kappa_{\mathrm{cat}} \cdot \omega_g$.

Remark 12.9 (Status of Theorem CY-B). Items (i) and (ii) follow from the Volume I bar-cobar machine (Theorems A and B) applied to each E_1-direction separately, with compatibility ensured by Dunn additivity. The full E_2-equivariant refinement (coherent intertwining of the two bar-cobar adjunctions by the R-matrix) is established as a rigorous proof sketch; the detailed verification of higher coherences is deferred. Item (iii) uses Theorem D of Volume I with the CY Euler characteristic $\kappa_{\mathrm{cat}} = \chi^{\mathrm{CY}}(C)$ (Theorem 10.1).

PROPOSITION 12.10 *(Arity-$(1,1)$ of B_{E_2}).* [PH]The arity-$(1,1)$ component of $B_{E_2}(\mathcal{A})$ is the space $\mathrm{End}(V \otimes V)$ equipped with the R-matrix $\mathcal{R}_{\mathcal{A}}(z)$. The bar differential restricted to this component recovers the unitarity condition $\mathcal{R}_{12}(z)\mathcal{R}_{21}(-z) = \mathrm{id}$ (strong unitarity), and the braiding constraint $\mathcal{R}_{12}(z) = \tau \circ \sigma_{12}(z)$.

Proof. The E_2-bar complex at arity $(1,1)$ consists of elements $\alpha \in (s^{-1}\bar{\mathcal{A}})^{\otimes 2}$ with one element from each E_1-direction. The bar differential $d(\alpha) = \mu(\alpha)$ is the OPE residue extracted via $d\log(z_1 - z_2)$ (Volume I, bar differential). The compatibility of the two E_1-structures forces $d_X \circ d_Y = d_Y \circ d_X$; evaluating on $(s^{-1}v) \otimes (s^{-1}w)$ gives $\mathcal{R}_{12}(z)\mathcal{R}_{21}(-z) = \mathrm{id}$ (strong unitarity). The braiding $\sigma_{12}(z) = \mathcal{R}_{12}(z) \circ \tau^{-1}$ follows from the identification of $\mathcal{R}$ with the half-braiding (Definition 12.2). □

12.5 Quantum group structure from braided Koszul duality

CONJECTURE 12.11 *(Braided bar cohomology recovers $U_q(\mathfrak{g})$).* [CJ]For the E_2-chiral algebra $\mathcal{A} = \Phi(C)$ associated to a CY_2 category C with $\mathfrak{g}$-symmetry, the braided bar cohomology

$$H^*(B_{E_2}(\mathcal{A})) \;\simeq\; U_q(\mathfrak{g})$$

as quasi-triangular Hopf algebras, where $q = e^{2\pi i/(k+h^\vee)}$ and k is the level determined by the CY Euler characteristic.

Remark 12.12. The braided Koszul dual $\mathcal{A}^!_{E_2}$ produces the quantum group $U_q(\mathfrak{g})$. This must be distinguished from:

(a) the Feigin–Frenkel involution $k \mapsto -k - 2h^\vee$, which acts within the same family of affine algebras;

(b) the Langlands dual $\mathfrak{g} \mapsto \mathfrak{g}^L$ (root-coroot exchange), which produces a genuinely different algebra.

For $\mathfrak{sl}_2$ (self-Langlands-dual), all three operations coincide on representation categories. For non-simply-laced types (e.g., G_2, F_4), they diverge.

12.6 The braided shadow obstruction tower

Definition 12.13 *(Braided modular characteristic).* For an E_2-chiral algebra $\mathcal{A}$, the *braided modular characteristic* is the pair

$$(\kappa_{\mathrm{cat}}(\mathcal{A}),\ \mathcal{R}_{\mathcal{A}}(z))$$

consisting of the scalar modular characteristic $\kappa_{\mathrm{cat}} = \mathrm{av}(\mathcal{R}_{\mathcal{A}}(z))$ (the Σ_2-coinvariant of the R-matrix, AP97) and the full R-matrix itself. The scalar κ_{cat} controls the genus tower $F_g = \kappa_{\mathrm{cat}} \cdot \lambda_g^{FP}$; the R-matrix controls the ordered/quantum-group sector.

Proposition 12.14 *(Braided shadow structure).* [PH]The E_2-bar complex $B_{E_2}(\mathcal{A})$ carries a braided shadow obstruction tower $\Theta_{\mathcal{A}}^{E_2}$ whose arity-by-arity projection under the averaging map recovers the Volume I shadow obstruction tower:

$$\mathrm{av}(\Theta^{E_2}_{\mathcal{A},n}) = \Theta_{A_C,n} \quad \text{for all } n \geq 2.$$

At arity 2: $\mathrm{av}(\mathcal{R}(z)) = \kappa_{\mathrm{cat}}$. At arity 3: $\mathrm{av}(\Phi_{KZ}) = C$ (the cubic shadow of Volume I).

Example 12.15 *(Braided structure for $K3$).* For a K3 surface S with $C = D^b(\mathrm{Coh}(S))$, the E_2-chiral algebra has $\kappa_{\mathrm{cat}} = 2$. The R-matrix is the Casimir $\mathcal{R}(z) = 1 + \kappa_{\mathrm{cat}}\,\Omega/z + O(z^{-2})$ (AP126: level prefix $\kappa_{\mathrm{cat}} = 2$; r-matrix vanishes when the categorical modular characteristic vanishes), where Ω is the Mukai pairing on $H^*(S, \mathbb{Z})$. The braided bar cohomology $H^*(B_{E_2}(\Phi(C)))$ recovers the lattice quantum group associated to the Mukai lattice $\Lambda_{K3} = U^3 \oplus E_8(-1)^2$.

Chapter 13

The Drinfeld Center and Bulk Algebras

The passage from boundary to bulk requires the Drinfeld center. An E_1-chiral algebra A has a monoidal representation category $\mathrm{Rep}^{E_1}(A)$; this category is not braided, because the E_1 operad sees only one real direction and the fundamental group $\pi_1(\mathrm{Conf}_2(\mathbb{R})) = \mathbb{Z}$ produces only a monoidal, not braided, structure. The Drinfeld center $\mathcal{Z}(\mathrm{Rep}^{E_1}(A))$ universally adjoins a braiding; the resulting braided monoidal category is equivalent to $\mathrm{Rep}^{E_2}(Z^{\mathrm{der}}_{\mathrm{ch}}(A))$, where $Z^{\mathrm{der}}_{\mathrm{ch}}(A) = C^{\bullet}_{\mathrm{ch}}(A, A)$ is the chiral derived center of Volume I. This identification is the categorical bulk-boundary correspondence.

13.1 The Drinfeld center of a monoidal category

Definition 13.1 *(Drinfeld center).* Let $(\mathcal{C}, \otimes, \mathbf{1})$ be a monoidal category. The *Drinfeld center* $\mathcal{Z}(\mathcal{C})$ is the category whose objects are pairs $(X, \beta_{X,-})$ where:

(i) $X \in \mathcal{C}$ is an object;

(ii) $\beta_{X,-} : X \otimes (-) \xrightarrow{\sim} (-) \otimes X$ is a natural isomorphism (the *half-braiding*);

(iii) the half-braiding satisfies the hexagon axiom: for all $Y, Z \in \mathcal{C}$,

$$\beta_{X,Y\otimes Z} = (\mathrm{id}_Y \otimes \beta_{X,Z}) \circ (\beta_{X,Y} \otimes \mathrm{id}_Z).$$

A morphism $(X, \beta_X) \to (Y, \beta_Y)$ is a morphism $f : X \to Y$ in $\mathcal{C}$ that intertwines the half-braidings. The Drinfeld center is always braided monoidal, with braiding $\sigma_{(X,\beta_X),(Y,\beta_Y)} = \beta_{X,Y}$.

Proposition 13.2 *(Universal property).* [PE] The Drinfeld center $\mathcal{Z}(\mathcal{C})$ is the universal braided monoidal category receiving a central (i.e., braiding-compatible) monoidal functor from $\mathcal{C}$. The forgetful functor $U : \mathcal{Z}(\mathcal{C}) \to \mathcal{C}$ is faithful and monoidal.

Example 13.3 *(Finite group).* For $\mathcal{C} = \mathrm{Rep}(G)$ with G a finite group, the Drinfeld center $\mathcal{Z}(\mathrm{Rep}(G)) \simeq \mathrm{Rep}(D(G))$ is the representation category of the Drinfeld double $D(G) = \mathbb{C}[G] \bowtie \mathbb{C}(G)$. When G is abelian, $D(G) \simeq \mathbb{C}[G \times \hat{G}]$ and $\mathcal{Z}(\mathrm{Rep}(G)) \simeq \mathrm{Rep}(G \times \hat{G})$, the modular tensor category governing the G-Dijkgraaf–Witten theory.

Example 13.4 *(Quantum group).* For $\mathcal{C} = \mathrm{Rep}_q(\mathfrak{g})$ at generic q, the Drinfeld center $\mathcal{Z}(\mathrm{Rep}_q(\mathfrak{g})) \simeq \mathrm{Rep}_q(\mathfrak{g}) \boxtimes \mathrm{Rep}_{q^{-1}}(\mathfrak{g})$, recovering the Drinfeld double $D(U_q(\mathfrak{g})) \simeq U_q(\mathfrak{g}) \otimes U_q(\mathfrak{g}^{\mathrm{op}})$. At roots of unity, $\mathcal{Z}(\mathcal{C}_q) = \mathcal{C}_q \boxtimes \overline{\mathcal{C}_q}$ is the Drinfeld center of the modular tensor category, and the S-matrix of $\mathcal{Z}(\mathcal{C}_q)$ governs the Reshetikhin–Turaev invariant of the torus T^2.

13.2 Drinfeld center as chiral derived center

The categorical Drinfeld center and the algebraic chiral derived center are identified by the Ben-Zvi–Francis–Nadler theorem.

THEOREM 13.5 *(Ben-Zvi–Francis–Nadler; Lurie).* [PE] For an E_1-algebra A in a symmetric monoidal stable ∞-category, there is an equivalence of braided monoidal ∞-categories

$$\mathcal{Z}(\mathrm{Rep}^{E_1}(A)) \;\simeq\; \mathrm{Rep}^{E_2}(Z^{\mathrm{der}}_{\mathrm{ch}}(A))$$

where $Z^{\mathrm{der}}_{\mathrm{ch}}(A) = \mathrm{RHom}_{A\text{-bimod}}(A, A)$ is the derived center (Hochschild cochains) of A.

COROLLARY 13.6 *(Chiral derived center = Drinfeld center).* [PH] For an E_1-chiral algebra A, the chiral derived center $Z^{\mathrm{der}}_{\mathrm{ch}}(A) = C^\bullet_{\mathrm{ch}}(A, A)$ of Volume I (Theorem H) satisfies

$$\mathcal{Z}(\mathrm{Rep}^{E_1}(A)) \;\simeq\; \mathrm{Rep}^{E_2}(Z^{\mathrm{der}}_{\mathrm{ch}}(A)).$$

The Drinfeld center is the categorical incarnation of the universal bulk algebra.

Proof. The chiral derived center is $Z^{\mathrm{der}}_{\mathrm{ch}}(A) = \mathrm{RHom}_{\Omega^{\mathrm{ch}}(B(A))\otimes\Omega^{\mathrm{ch}}(B(A))^{\mathrm{op}}}(A, A)$. By bar-cobar inversion (Volume I, Theorem B), $\Omega^{\mathrm{ch}}(B(A)) \simeq A$ on the Koszul locus, so $Z^{\mathrm{der}}_{\mathrm{ch}}(A) \simeq \mathrm{RHom}_{A\otimes A^{\mathrm{op}}}(A, A) = \mathrm{HH}^\bullet(A)$. The Ben-Zvi–Francis–Nadler theorem (Theorem 13.5) then gives the stated equivalence. □

Warning 13.7 (Three functors on $B(A)$). The Drinfeld center / chiral derived center is a FOURTH operation on the bar complex, distinct from the three of Volume I:

(a) *Bar-cobar inversion*: $\Omega(B(A)) \simeq A$ (recovers the original algebra);

(b) *Verdier duality*: $D_{\mathrm{Ran}}(B(A)) \simeq B(A^!)$ (produces the bar of the Koszul dual);

(c) *Koszul dual extraction*: $H^*(B(A))^\vee = A^!$ (the strict Koszul dual algebra);

(d) *Derived center*: $Z^{\mathrm{der}}_{\mathrm{ch}}(A) = C^\bullet_{\mathrm{ch}}(A, A) = \mathrm{RHom}(\Omega(B(A)), A)$ (the universal bulk algebra).

The Drinfeld center $\mathcal{Z}(\mathrm{Rep}^{E_1}(A))$ is $\mathrm{Rep}^{E_2}(Z^{\mathrm{der}}_{\mathrm{ch}}(A))$: it is functor (d), not (a), (b), or (c). Bar-cobar inversion recovers the boundary; the Drinfeld center produces the bulk.

13.3 Bulk-boundary correspondence

PROPOSITION 13.8 *(Information flow by genus).* [PH] The bulk-boundary correspondence $\mathcal{Z}(\mathrm{Rep}^{E_1}(A)) \simeq \mathrm{Rep}^{E_2}(Z^{\mathrm{der}}_{\mathrm{ch}}(A))$ exhibits genus-dependent information flow:

(i) *Genus* 0: Information flows one way, from bulk to boundary. The forgetful functor $\mathcal{Z}(\mathcal{C}) \to \mathcal{C}$ has no natural section: the Swiss-cheese operad forbids open-to-closed maps at genus 0 (Volume II, thm:thqg-swiss-cheese);

(ii) *Genus* 1: The annulus trace $\Delta_{\mathrm{ns}}(\mathrm{Tr}_A) = \kappa_{\mathrm{cat}} \cdot \lambda_1$ (Volume I, thm:annulus-trace) provides the first open-to-closed map. The Dennis trace $K(\mathcal{C}) \to \mathrm{HH}_\bullet(\mathcal{C})$ maps K-theory of the open sector into the bulk via the SBI sequence;

(iii) *Higher genus*: The full genus tower progressively restores bidirectionality. The clutching maps $\mathrm{cl}_{g,n} : \mathrm{HH}_\bullet(A)^{\otimes n} \to Z^{\mathrm{der}}_{\mathrm{ch}}(A)$ on the modular operad provide genus-g open-to-closed data.

Proof. Item (i): at the operad level, $\mathrm{SC}^{\mathrm{ch,top}}$ has no composition open $\to$ closed at genus 0 (the Kontsevich no-open-to-closed axiom).

Item (ii): the Dennis trace $K(\mathcal{C}) \to \mathrm{HH}_\bullet(\mathcal{C})$ factors through negative cyclic homology $\mathrm{HC}^-_\bullet(\mathcal{C})$. The CY duality $\mathrm{HH}^\bullet(\mathcal{C}) \simeq \mathrm{HH}_{\bullet+d}(\mathcal{C})$ (Theorem 4.12) and the identification $\mathrm{HH}^\bullet(\mathcal{C}) \simeq Z^{\mathrm{der}}_{\mathrm{ch}}(A)$ (Corollary 13.6) give the map into the bulk. The annulus trace is the genus-1 specialization: the S^1-trace on $\mathrm{HH}_0(A)$ gives $\kappa_{\mathrm{cat}} \cdot \lambda_1 \in H^2(\overline{\mathcal{M}}_{1,1})$.

Item (iii): the modular operad structure on $\{H^*(\overline{\mathcal{M}}_{g,n})\}$ provides clutching maps $\gamma_{g_1,n_1} \circ_{g_2,n_2}$ that compose open-sector data (living on boundaries of $\overline{\mathcal{M}}_{g,n}$) into closed-sector invariants. Each genus g introduces new clutching compositions (from codimension-1 boundary strata of $\overline{\mathcal{M}}_{g,n}$), progressively enlarging the image of the open-to-closed map. □

13.4 CY enhancement of the Drinfeld center

THEOREM 13.9 *(CY Drinfeld center).* [PE] Let $\mathcal{C}$ be a smooth proper CY_d category with quantum chiral algebra $A_{\mathcal{C}} = \Phi(\mathcal{C})$. The Drinfeld center $\mathcal{Z}(\mathrm{Rep}^{E_1}(A_{\mathcal{C}}))$ carries a CY structure of dimension $d+1$. Under the identification $\mathcal{Z}(\mathrm{Rep}^{E_1}(A_{\mathcal{C}})) \simeq \mathrm{Rep}^{E_2}(Z^{\mathrm{der}}_{\mathrm{ch}}(A_{\mathcal{C}}))$, this CY_{d+1} structure arises from the $(1-d)$-shifted symplectic structure on the derived center (Pantev–Toën–Vaquié–Vezzosi).

Remark 13.10 (Dimensional hierarchy). The shift $d \to d+1$ is the categorical holographic principle:

CY dim	Boundary	Bulk	Physical
$d = 1$	E_1-chiral	CY_2	2d CFT / 3d TFT
$d = 2$	E_2-chiral	CY_3	3d HT / 4d $\mathcal{N} = 2$
$d = 3$	E_1-chiral	CY_4	4d $\mathcal{N} = 1$ gauge

For $d = 3$, the boundary algebra is E_1 (not E_2): the $\mathbb{S}^3$-framing obstruction (Conjecture CY-A$_3$) breaks E_2 to E_1. The braiding is recovered via $\mathcal{Z}(\mathrm{Rep}^{E_1}(A))$, not from the boundary A directly.

13.5 Categorical Theorem H

Volume I Theorem H establishes that chiral Hochschild cohomology $\mathrm{ChirHoch}^*(A)$ is polynomial (concentrated in degrees $\{0,1,2\}$) for modular Koszul algebras, with total dimension ≤ 4. The categorical analogue replaces $\mathrm{ChirHoch}^*$ with $\mathrm{HH}^\bullet(\mathcal{C})$.

THEOREM 13.11 *(Categorical Theorem H).* [CD] For a smooth proper CY_d category $\mathcal{C}$ whose quantum chiral algebra $A_{\mathcal{C}} = \Phi(\mathcal{C})$ is modular Koszul:

(i) $\mathrm{HH}^k(\mathcal{C}) = 0$ for $k > d$ (concentration, from smoothness and properness);

(ii) $\mathrm{HH}^0(\mathcal{C}) \simeq k$ (the identity natural transformation);

(iii) $\mathrm{HH}^1(\mathcal{C})$ classifies first-order deformations of $\mathcal{C}$ as a dg category;

(iv) $\mathrm{HH}^2(\mathcal{C})$ is the obstruction space for deformations.

The Gerstenhaber bracket $[-,-] : \mathrm{HH}^p \otimes \mathrm{HH}^q \to \mathrm{HH}^{p+q-1}$ on $\mathrm{HH}^\bullet(\mathcal{C})$ maps, under Φ, to the convolution bracket on $\mathfrak{g}^{\mathrm{mod}}_{A_{\mathcal{C}}}$ of Volume I.

Remark 13.12 (Status). Items (i)–(iv) are standard consequences of smoothness and properness (Keller, Kontsevich). The conditional content is the identification of the Gerstenhaber bracket with the Volume I convolution bracket under Φ, established for $d = 2$ and conjectural for $d = 3$.

13.6 Quantum group reconstruction from the center

Proposition 13.13 *(Tannakian reconstruction).* [PE]Let $\mathcal{B}$ be a rigid braided monoidal category over $\mathbb{C}$ equipped with a fiber functor $\omega\colon \mathcal{B} \to \mathrm{Vect}_{\mathbb{C}}$. The Tannakian reconstruction gives a quasi-triangular Hopf algebra $H = \mathrm{End}(\omega)$ with $\mathcal{B} \simeq \mathrm{Rep}(H)$ as braided monoidal categories. When $\mathcal{B} = \mathcal{Z}(\mathrm{Rep}^{E_1}(A))$ for an E_1-chiral algebra A with $\mathfrak{g}$-symmetry, $H \simeq U_q(\mathfrak{g})$ with q determined by κ_{cat}.

Remark 13.14 (Non-semisimple reconstruction). At roots of unity ($q^N = 1$), the category $\mathrm{Rep}_q(\mathfrak{g})$ is non-semisimple: indecomposable but non-simple modules appear. The standard fiber functor is not exact on the full category. Two remedies:

(a) Restrict to the semisimplified quotient $\overline{\mathrm{Rep}}_q(\mathfrak{g})$ (modular tensor category of Reshetikhin–Turaev);

(b) Use the modified trace of Geer–Patureau-Mirand–Turaev, which replaces the categorical trace for non-semisimple ribbon categories.

For the CY programme, approach (b) is preferred: the full non-semisimple structure encodes the complete BPS spectrum (Chapter 15), and the modified trace is the categorical analogue of the regularized partition function.

13.7 The Drinfeld double and the slab

Definition 13.15 *(Drinfeld double).* For a finite-dimensional Hopf algebra H, the *Drinfeld double* $D(H) = H \bowtie H^{*\,\mathrm{cop}}$ is the quasi-triangular Hopf algebra built from H and the co-opposite of its dual. Its representation category is $\mathcal{Z}(\mathrm{Rep}(H))$, the Drinfeld center of $\mathrm{Rep}(H)$.

Remark 13.16 (The slab is a bimodule). In the 3d holomorphic-topological setting (Volume II), the Drinfeld double arises from the *slab*: a strip $X \times [0,1]$ with two boundary components $X \times \{0\}$ and $X \times \{1\}$. The algebra of operators on the slab is a bimodule for the two boundary algebras A_0 and A_1. The Drinfeld double $D(A) = A \bowtie A^{!,\mathrm{cop}}$ appears when $A_0 = A$ and $A_1 = A^!$ (the Koszul dual). The slab is NOT a Swiss-cheese disk (which has one closed and one open boundary component); it is a cylinder with two open boundary components.

Conjecture 13.17 *(Slab = Drinfeld double).* [CJ]For an E_1-chiral algebra A on $X \times \mathbb{R}$, the slab algebra $\mathcal{A}_{\mathrm{slab}} = \mathrm{End}_{A\text{-}A^!}(A \otimes_{A^!} A)$ is isomorphic to the Drinfeld double $D(A)$. The slab R-matrix $\mathcal{R}_{\mathrm{slab}} = \mathcal{R}_A \otimes \mathcal{R}_{A^!}^{-1}$ is the universal R-matrix of $D(A)$, and its representation category is the Drinfeld center $\mathcal{Z}(\mathrm{Rep}^{E_1}(A))$.

Conjecture 13.18 *(Quantum vertex chiral group).* [CJ]For a CY_2 category C with $\mathfrak{g}$-symmetry, the Drinfeld center $\mathcal{Z}(\mathrm{Rep}^{E_1}(\Phi(C)))$ determines a quantum vertex chiral group $G(C)$ unifying four algebraic structures:

(i) $U_q(\mathfrak{g})$: the quantum group (fiber at a point);

(ii) $Y(\mathfrak{g})$: the Yangian (E_1-sector, Volume II, MC3);

(iii) $U_q(\widehat{\mathfrak{g}})$: the affine quantum group (loop algebra sector);

(iv) The Etingof–Kazhdan quantum vertex algebra (the genuinely nonlocal atom, Volume II).

The four incarnations arise from different geometric specializations of the E_2-factorization structure on $\mathrm{Ran}(X) \times \mathrm{Ran}(Y)$.

Part III

The CY Landscape

Chapter 14

Toroidal and elliptic algebras

The chiral algebras of Vols I–II each depend on a single deformation parameter: the level k, the central charge c, or the spectral parameter u. This chapter enters the regime where one parameter is not enough. The toroidal algebra $U_{q,t}(\hat{\hat{\mathfrak{g}}})$ carries two independent parameters (q, t), the elliptic quantum group $E_{q,p}(\mathfrak{g})$ replaces the rational R-matrix by a doubly-periodic one, and the double affine Hecke algebra (DAHA) intertwines spectral and modular directions. In these algebras the modular parameter τ and the spectral parameter u are coupled through the Fay trisecant identity and the quasi-periodicity of theta functions.

The bar-cobar framework meets this two-parameter world in two ways. Direction (a) keeps the base a single curve, now of genus 1: the propagator becomes elliptic, the Arnold relation generalizes to the Fay trisecant identity (Theorem 14.18), and the bar differential acquires Eisenstein-series corrections indexed by τ. Direction (b) seeks genuine surface-factorization objects on $\mathbb{C}^* \times \mathbb{C}^*$, requiring a two-dimensional factorization framework beyond the curve-chiral hierarchy of Vols I–II. Only direction (a) is developed here.

The Fay identity replaces the Arnold relation at genus 1; it is both an identity of theta functions and a geometric consequence of the degeneration $E_\tau \to \mathbb{P}^1$ as $\tau \to i\infty$.

Remark 14.1 (Two programme tracks). Track (a) keeps the base an algebraic curve and constructs an elliptic E_1-chiral realization on E_τ. Track (b) seeks a genuine surface-factorization object on $\mathbb{C}^* \times \mathbb{C}^*$. Only track (a) is developed here; track (b) is a completion target.

Remark 14.2 (The generalization principle: Arnold to Fay). The Arnold relation (Vol I, Theorem III), which governs the genus-0 bar differential, admits two distinct generalizations. Direction (a): via clutching morphisms and the shadow obstruction tower $\Theta_A^{\leq r}$ (Vol I, Definition III), one passes to the modular regime on stable curves, recovering the higher-genus bar complex developed in Vol I, Part I (Vol I, Remark III). Direction (b): via the Fay trisecant identity (Theorem 14.18), one replaces the additive (logarithmic) propagator $\omega = dz/(z-w)$ by the elliptic (multiplicative) propagator built from the Weierstrass σ-function, entering the toroidal regime. The present chapter develops direction (b): the systematic passage from logarithmic to elliptic propagators and its consequences for bar-cobar duality.

Remark 14.3 (Rational, trigonometric, elliptic: curve geometry). The three quantum group families arise from the geometry of the underlying curve:

R-matrix type	**Quantum group**	**Spectral parameter**	**Curve**
Rational	Yangian $Y(\mathfrak{g})$	$u = z_1 - z_2$	$\mathbb{P}^1$
Trigonometric	Quantum loop $U_q(L\mathfrak{g})$	$q = e^{2\pi i(z_1 - z_2)/\beta}$	$\mathbb{C}^*$
Elliptic	Elliptic QG $E_{q,p}(\mathfrak{g})$	$u \in E_\tau$	E_τ

The passage rational → trigonometric → elliptic parallels additive → multiplicative → modular spectral parameter.

Construction 14.4 *(Elliptic shadow data).* Toroidal and elliptic quantum group algebras carry shadow data in the elliptic regime:

(i) The elliptic propagator $K^{\mathrm{ell}}(z, w; \tau) = \vartheta_1(z - w; \tau)/\vartheta_1'(0; \tau)$ replaces the rational propagator $1/(z - w)$.

(ii) The Fay trisecant identity for ϑ_1 is the genus-1 MC equation $D^{(1)}\Theta + \frac{1}{2}[\Theta^{(0)}, \Theta^{(0)}] + \kappa_{\mathrm{ch}}\,\omega_1 = 0$.

(iii) The Arnold relation $K_{12}K_{23} + K_{23}K_{31} + K_{31}K_{12} = 0$ (Fay identity) specializes the CYBE to the elliptic regime.

14.1 Double affine setup

Definition

Definition 14.5 *(Toroidal algebra).* The *toroidal algebra* (or double affine algebra) $U_{q,t}(\hat{\hat{\mathfrak{g}}})$ is generated by:

- Horizontal currents $x^{\pm}_{i,n}, \psi^{\pm}_{i,n}$ (affine in one direction)
- Vertical currents $e_i(z), f_i(z), \psi_i(z)$ (affine in the other direction)

subject to:

1. *Drinfeld–type relations* in each direction: $[h_{i,m}, x^{\pm}_{j,n}] = \pm a_{ij} x^{\pm}_{j,m+n}$, $[x^+_{i,m}, x^-_{j,n}] = \delta_{ij}(\psi^+_{i,m+n} - \psi^-_{i,m+n})/(q_i - q_i^{-1})$
2. *Cross relations* coupling horizontal and vertical: $\psi_i(z)e_j(w) = g_{ij}(z/w)\, e_j(w)\psi_i(z)$ where $g_{ij}(u) = (u - q^{a_{ij}})/(u - q^{-a_{ij}})$
3. *Serre relations* in both directions

with two independent parameters q and $t = q^k$. See Ginzburg–Kapranov–Vasserot for the complete presentation.

PROPOSITION 14.6 *(Toroidal OPE [15]; [PE]).* The OPE for toroidal currents involves both parameters:

$$e_i(z)f_j(w) = \frac{\delta_{ij}}{(1-q)(1-t^{-1})} \cdot \frac{\psi_i^+(w) - \psi_i^-(w)}{z - w} + \text{rational terms in } q, t$$

Proof. We derive the OPE from the defining relations of Definition 14.5.

Step 1: Mode commutator. The Drinfeld-type relation (1) gives:

$$[e_{i,m}, f_{j,n}] = \delta_{ij}\frac{\psi^+_{i,m+n} - \psi^-_{i,m+n}}{q_i - q_i^{-1}}.$$

Form the generating functions $e_i(z) = \sum_m e_{i,m} z^{-m-1}$, $f_j(w) = \sum_n f_{j,n} w^{-n-1}$, and $\psi_i^{\pm}(w) = \sum_n \psi^{\pm}_{i,n} w^{-n}$.

Step 2: Extracting the OPE. Contracting the commutator against $z^{-m-1}w^{-n-1}$ and summing:

$$e_i(z)f_j(w) - {:}e_i(z)f_j(w){:} = \delta_{ij}\sum_m \frac{\psi^+_{i,m}(w) - \psi^-_{i,m}(w)}{q_i - q_i^{-1}} \cdot \frac{w^{-m}}{z - w}.$$

For the toroidal algebra with parameters q and $t = q^k$, we have $q_i = q^{d_i}$ where $d_i = (\alpha_i, \alpha_i)/2$. The singular part of the OPE is the residue at $z = w$:

$$e_i(z) f_j(w) \sim \frac{\delta_{ij}}{q_i - q_i^{-1}} \cdot \frac{\psi_i^+(w) - \psi_i^-(w)}{z - w}.$$

Step 3: Two-parameter structure. The cross relation (2) introduces the second parameter. From $\psi_i(z)e_j(w) = g_{ij}(z/w)\, e_j(w)\psi_i(z)$ with $g_{ij}(u) = (u - q^{a_{ij}})/(u - q^{-a_{ij}})$, the normal ordering of ψ against e, f currents produces corrections rational in both q and t. Specifically, expanding $1/(q_i - q_i^{-1}) = 1/((q^{d_i} - q^{-d_i}))$ and using $t = q^k$ to express the central element as $c = (1-q)(1-t^{-1})/(q_i - q_i^{-1})$ (the standard Ding–Iohara normalization). See [10] for the complete change-of-basis computation. □

CONJECTURE 14.7 *(Elliptic-curve toroidal realization; [CJ])*. The toroidal algebra $U_{q,t}(\hat{\hat{\mathfrak{g}}})$ admits a curve-based E_1-chiral realization on the elliptic curve E_τ (with modulus $\tau = \frac{\log t}{\log q}$).

CONJECTURE 14.8 *(Double-affine surface-factorization realization; [CJ])*. The same toroidal data should extend to a doubly-graded surface-factorization or double-affine object on $\mathbb{C}^* \times \mathbb{C}^*$, with one direction encoding the spectral parameter and the other the elliptic coordinate. (This is the toroidal/elliptic extension flank of Vol I, Conjecture III, not the standard W_∞/Yangian finite-packet lane.)

Remark 14.9 (Scope). Conjecture 14.7 is conjectural ([CJ]): a complete proof requires verifying associativity of the E_1-chiral product directly from the OPE relations, since the BD locality approach applies only to commutative chiral algebras. The $\mathbb{C}^* \times \mathbb{C}^*$ realization (Conjecture 14.8) is a separate surface-factorization problem within the MC4 elliptic/double-affine extension problem.

Homotopy template (Vol I, Convention III): Type I (existence).

Proof outline. We describe the conjectural E_1-chiral structure.

Step 1: R-matrix braiding. The cross relation (2) of Definition 14.5 gives the structure function $g_{ij}(z/w) = (z/w - q^{a_{ij}})/(z/w - q^{-a_{ij}})$. Define the R-matrix-valued braiding on generating currents by

$$\sigma_{12} \circ \mu(e_i(z), e_j(w)) = g_{ij}(z/w) \cdot \mu(e_j(w), e_i(z)).$$

Since $g_{ij}(u)$ is a rational function with $g_{ij}(u) \neq 1$ for generic q and $a_{ij} \neq 0$, the braiding is non-trivial: the algebra is associative but not commutative, hence strictly E_1.

Step 2: Associativity (gap). The R-matrix $R_{ij}(u) = g_{ij}(u) \cdot \mathrm{Id}$ satisfies the Yang–Baxter equation trivially (being scalar-valued). The genuine content of associativity lies in verifying that the full matrix-valued OPE structure, including the Serre-type relations of Definition 14.5, defines an algebra over $\mathsf{Ass}^{\mathrm{ch}}$ (Vol I, Definition III). This requires an explicit verification that triple OPEs are consistent, which we do not carry out here.

Step 3: Realization on E_τ. Set $\tau = \log t/\log q$ and let $E_\tau = \mathbb{C}^*/q^{\mathbb{Z}}$. The generating currents $e_i(z)$, $f_i(z), \psi_i(z)$ are meromorphic on $\mathbb{C}^*$ with quasi-periodicity $e_i(qz) = q^{d_i} e_i(z)$ (from the grading), so they define sections of line bundles on E_τ. The OPE of Proposition 14.6 has poles at $z/w \in q^{\mathbb{Z}}$, which descend to a single point on E_τ. This defines a candidate E_1-chiral algebra structure on E_τ, pending verification of Step 2.

Step 4: Surface-factorization horizon on $\mathbb{C}^* \times \mathbb{C}^*$. The horizontal modes $x_{i,n}^\pm$ depend on the spectral parameter u and the vertical modes $e_i(z)$ depend on z. Together they give fields on $\mathbb{C}_u^* \times \mathbb{C}_z^*$ with the two independent gradings (by horizontal and vertical mode number) providing the doubly-graded structure.

This belongs to a separate conjectural surface-factorization or double-affine track, not automatically a curve-based E_1-chiral algebra. A correct formulation would require a bona fide two-parameter factorization framework.

Conditional on Step 2 (i.e., the verification that the toroidal OPE defines an associative $\mathsf{Ass}^{\mathrm{ch}}$-algebra structure, which requires checking consistency of triple OPEs with the Serre-type relations), the E_1-chiral Koszul duality framework (Vol I, Theorem III) applies to the elliptic-curve realization of Conjecture 14.7. The $\mathbb{C}^* \times \mathbb{C}^*$ surface track belongs instead to Conjecture 14.8. □

Koszul dual of the toroidal algebra

Conjecture 14.10 *(Toroidal Koszul dual; [CJ])*. The E_1-chiral Koszul dual of the toroidal algebra $U_{q,t}(\hat{\hat{\mathfrak{g}}})$ is isomorphic to $U_{q^{-1},t^{-1}}(\hat{\hat{\mathfrak{g}}})$, the toroidal algebra with inverted parameters. (Contributing to the toroidal/elliptic extension flank of Vol I, Conjecture III.)

Remark 14.11 (Evidence). Conditional on Conjecture 14.7, the evidence is threefold:

(i) *RTT presentation.* By E_1-chiral Koszul duality (Vol I, Theorem III), the dual has R-matrix $R(u;q,t)^{-1} = R(u;q^{-1},t^{-1})$ (Ding–Iohara inversion; cf. Vol I, Theorem III).

(ii) *Affine degeneration.* In the limit $t \to 0$, $U_{q,t}(\hat{\hat{\mathfrak{g}}}) \to U_q(\hat{\mathfrak{g}})$ and the dual reduces to $U_{q^{-1}}(\hat{\mathfrak{g}})$, consistent with Vol I, Theorem III.

(iii) *Central charge complementarity.* The inversion $q \to q^{-1}$ sends $\tau \to -\tau$, reversing the complex structure of E_τ, the elliptic analogue of the Verdier intertwining (Vol I, Theorem A; Theorem III).

The Fay identity (Proposition 14.19) serves as the elliptic Arnold relation. The universal MC class predicts $\Theta_{U_{q,t}} + \Theta_{U_{q^{-1},t^{-1}}} = 0$.

Homotopy template (Vol I, Convention III): Type II (equivalence).

Connection to the three main theorems of Vol I

Remark 14.12 (Toroidal three theorems). Conditional on Conjecture 14.7, the expected toroidal analogues of the three main theorems of Vol I are: *Theorem A* (Vol I, bar-cobar adjunction): $\bar{B}^{\mathrm{ell}}(U_{q,t})$ computes the Koszul dual coalgebra (nilpotency via Proposition 14.19). *Theorem B* (Vol I, bar-cobar inversion): $\Omega(\bar{B}(U_{q,t})) \xrightarrow{\sim} U_{q,t}$ by the rational spectral sequence with elliptic corrections (Theorem 14.22). *Theorem C* (Vol I, complementarity): $\mathrm{obs}_g(U_{q,t}) + \mathrm{obs}_g(U_{q^{-1},t^{-1}})$ controlled by H^* of a framed moduli space (the E_1 replacement for $\overline{\mathcal{M}}_g$; cf. Vol I, Remark III). The universal MC class $\Theta_{U_{q,t}}$ should govern the genus expansion, with $(q,t) \mapsto (q^{-1},t^{-1})$ reflected in Verdier duality.

Remark 14.13 (Lattice evidence for toroidal genus theory). The curvature-braiding orthogonality theorem for quantum lattice VOAs (Vol I, Theorem III) provides structural evidence for the toroidal case. For the lattice, the genus-1 curvature $\kappa_{\mathrm{ch}} = \mathrm{rank}(\Lambda)$ lives entirely in the Cartan (Heisenberg) sector and is independent of the deformation cocycle $\varepsilon_{N,q}$. The toroidal algebra $U_{q,t}(\hat{\hat{\mathfrak{g}}})$ has a similar double-loop structure: the inner loop $(\hat{\mathfrak{g}})$ is the Cartan analog, while the outer loop (the second affinization) provides the braiding. The orthogonality principle predicts that the toroidal genus-1 curvature should be controlled by the inner-loop data alone, with the dynamical parameter $\lambda \in H^1(E_\tau)$ entering the braiding axis rather than the modular axis. The dynamical YBE for $R(u,\lambda)$ would then be the toroidal analog of the modified Arnold relation at genus 1, with the dynamical shift $\lambda \to \lambda - h^{(i)}$ replacing the B-cycle monodromy correction.

14.2 Elliptic quantum groups

Felder's elliptic quantum group

Definition 14.14 *(Elliptic quantum group).* The *elliptic quantum group* $E_{q,p}(\mathfrak{g})$ (Felder) is defined by:

- The dynamical R-matrix $R(u, \lambda)$ depending on spectral parameter u and dynamical parameter λ.
- The RLL relations:

$$R(u-v,\lambda)L_1(u,\lambda)L_2(v,\lambda-h^{(1)}) = L_2(v,\lambda)L_1(u,\lambda-h^{(2)})R(u-v,\lambda)$$

Definition 14.15 *(Elliptic R-matrix).* The elliptic R-matrix for $\mathfrak{sl}_2$ is:

$$R(u,\lambda) = \begin{pmatrix} a(u) & 0 & 0 & 0 \\ 0 & b(u,\lambda) & c(u,\lambda) & 0 \\ 0 & c(u,-\lambda) & b(u,-\lambda) & 0 \\ 0 & 0 & 0 & a(u) \end{pmatrix}$$

where $a(u), b(u,\lambda), c(u,\lambda)$ are expressed via theta functions:

$$a(u) = \frac{\theta(u+\eta)}{\theta(\eta)}$$
$$b(u,\lambda) = \frac{\theta(u)\theta(\lambda+\eta)}{\theta(\eta)\theta(\lambda)}$$
$$c(u,\lambda) = \frac{\theta(u+\lambda)\theta(\eta)}{\theta(\lambda)\theta(u+\eta)}$$

Remark 14.16 (Elliptic quantum groups and bar complex). We expect ([CJ], conditional on Conjecture 14.7): (i) $\overline{B}(U_{q,t})$ on E_τ computes the chiral coalgebra $E_{q,p}(\mathfrak{g})$ (elliptic Kazhdan–Lusztig); (ii) the dynamical parameter λ (Definition 14.14) is the B-cycle monodromy of the elliptic propagator (elliptic analogue of $k \to k + h^\vee$); (iii) the dynamical YBE for $R(u,\lambda)$ is the elliptic deformation of $d^2 = 0$ (Proposition 14.19), with the dynamical parameter from $H^1(E_\tau) \neq 0$.

14.3 Elliptic R-matrices and theta functions

Theta function identities

Definition 14.17 *(Jacobi theta function).* The Jacobi theta function is:

$$\theta(u|\tau) = \sum_{n\in\mathbb{Z}} (-1)^n e^{\pi i\tau n(n-1)+2\pi i n u} = (e^{2\pi i u}; p)_\infty (pe^{-2\pi i u}; p)_\infty (p;p)_\infty$$

where $p = e^{2\pi i\tau}$ and $(a;q)_\infty = \prod_{n\geq 0}(1-aq^n)$.

THEOREM 14.18 *(Fay identity at genus 1 [14]; [PE]).* For any four points u_1, u_2, u_3, u_4 on an elliptic curve E_τ, the theta function satisfies the Fay trisecant identity:

$$\theta(u_1-u_2)\,\theta(u_3-u_4) - \theta(u_1-u_3)\,\theta(u_2-u_4) + \theta(u_1-u_4)\,\theta(u_2-u_3) = 0$$

This is a *bilinear* identity (each term is a product of two theta functions), which at genus 1 is the degenerate case of the general Fay trisecant identity on higher-genus curves. The identity is equivalent to the Riemann addition formula and reduces to the Ptolemy relation $\sin(\alpha-\beta)\sin(\gamma-\delta) - \sin(\alpha-\gamma)\sin(\beta-\delta) + \sin(\alpha-\delta)\sin(\beta-\gamma) = 0$ in the rational limit $\tau \to i\infty$.

This identity underlies the Yang–Baxter equation for elliptic R-matrices: the star-triangle relation is a consequence of the Fay identity applied to the structure functions of the elliptic algebra.

Fay identity and bar nilpotency

Proposition 14.19 *(Fay identity implies elliptic $d^2 = 0$; [PH]).* On $\overline{C}_3(E_\tau)$, the elliptic bar differential satisfies $d^2 = 0$. The key algebraic input is the Fay trisecant identity (Theorem 14.18), which plays the role of the Arnold relation in the rational case (Vol I, Theorem III).

Proof. The elliptic bar differential on degree-2 elements uses the propagator $\eta_{ij}^{\mathrm{ell}} = d\log\theta_1(z_i - z_j|\tau)$. The computation of d^2 on a degree-1 element $[a]\otimes 1$ produces terms on $\overline{C}_3(E_\tau)$ involving the 2-form $\eta_{12}^{\mathrm{ell}}\wedge\eta_{23}^{\mathrm{ell}} + \eta_{23}^{\mathrm{ell}}\wedge\eta_{31}^{\mathrm{ell}} + \eta_{31}^{\mathrm{ell}}\wedge\eta_{12}^{\mathrm{ell}}$, exactly as in the rational case (cf. Vol I, §III).

The rational Arnold relation $\eta_{12}\wedge\eta_{23} + \eta_{23}\wedge\eta_{31} + \eta_{31}\wedge\eta_{12} = 0$ (where $\eta_{ij} = d\log(z_i - z_j)$) is a consequence of the partial fraction identity $(z_1 - z_2)^{-1}(z_2 - z_3)^{-1} + \text{cyclic} = 0$.

The elliptic replacement for $(z_i - z_j)^{-1}$ is $\zeta_\tau(z_i - z_j) := \partial_z \log\theta_1(z_i - z_j|\tau)$, the Weierstrass ζ-function on E_τ. The corresponding three-term identity is

$$d\zeta_\tau(z_1{-}z_2)\wedge d\zeta_\tau(z_2{-}z_3) + d\zeta_\tau(z_2{-}z_3)\wedge d\zeta_\tau(z_3{-}z_1) + d\zeta_\tau(z_3{-}z_1)\wedge d\zeta_\tau(z_1{-}z_2) = 0. \quad (14.1)$$

This is the differential-form shadow of the Fay identity (Theorem 14.18). Differentiate

$$\theta(u_1{-}u_2)\,\theta(u_3{-}u_4) - \theta(u_1{-}u_3)\,\theta(u_2{-}u_4) + \theta(u_1{-}u_4)\,\theta(u_2{-}u_3) = 0$$

with respect to u_1 and u_4, then set $u_4 = u_1$. The resulting trisecant consequence is

$$\begin{aligned}\zeta_\tau(u_1{-}u_2)\,\zeta_\tau(u_2{-}u_3) + \zeta_\tau(u_2{-}u_3)\,\zeta_\tau(u_3{-}u_1) + \zeta_\tau(u_3{-}u_1)\,\zeta_\tau(u_1{-}u_2)&\\ = \wp_\tau(u_1{-}u_2) + \wp_\tau(u_2{-}u_3) + \wp_\tau(u_3{-}u_1)&\end{aligned}$$

where $\wp_\tau = -\zeta_\tau'$ is the Weierstrass $\wp$-function. Now take exterior derivatives in the difference variables. Each $\wp_\tau(z_i - z_j)$ depends on a single difference, so the mixed wedges on the right vanish or cancel pairwise. The left-hand side is then exactly (14.1).

The remaining terms in d^2 (involving OPE data of the chiral algebra $\mathcal{A}$) cancel by the same mechanism as in the rational case: the bar differential decomposes as $d = d_{\mathrm{local}} + d_{\mathrm{global}}$, where $d_{\mathrm{local}}^2 = 0$ by the OPE associativity of $\mathcal{A}$, and the cross-terms $d_{\mathrm{local}}\circ d_{\mathrm{global}} + d_{\mathrm{global}}\circ d_{\mathrm{local}} = 0$ by the compatibility of the OPE with the propagator (which holds for both the rational and elliptic propagators, since the residue at a simple pole is the same: $\mathrm{Res}_{z=0}\,\zeta_\tau(z)\,dz = 1$). □

Remark 14.20 (Rational limit). In the rational limit $\tau \to i\infty$, the theta function $\theta_1(z|\tau) \to 2\sin(\pi z) \to 2\pi z$ (for $|z|$ small), so $\zeta_\tau(z) \to 1/z$ and $\wp_\tau(z) \to 1/z^2$. The identity (14.1) reduces to the Arnold relation $\eta_{12}\wedge\eta_{23} + \eta_{23}\wedge\eta_{31} + \eta_{31}\wedge\eta_{12} = 0$ (Vol I, Theorem III), and the bar nilpotency reduces to Vol I, Theorem III.

Bar complex with theta functions

Construction 14.21 *(Elliptic bar complex).* The bar complex of an elliptic chiral algebra incorporates theta functions:

Forms. Replace $d\log(z_1 - z_2)$ with:

$$\eta_{12} = d\log\theta(z_1 - z_2|\tau)$$

the logarithmic derivative of theta.

Differential. The residue is computed at the zeros of θ:

$$d_{\mathrm{res}}([a|b] \otimes \eta_{12}) = \sum_{\theta(z_0)=0} \mathrm{Res}_{z_1=z_0+z_2}[\cdots]$$

The zeros of $\theta(u|\tau)$ are at $u = m + n\tau$ for $m, n \in \mathbb{Z}$ (the lattice points).

THEOREM 14.22 *(Elliptic vs rational homology; [PH]).* Let $\mathcal{A}$ be a chiral algebra of central charge c on an elliptic curve E_τ. The homology of the elliptic bar complex admits a decomposition:

$$H_n(\bar{B}^{\mathrm{ell}}(\mathcal{A})) \;\cong\; H_n(\bar{B}^{\mathrm{rat}}(\mathcal{A})) \;\oplus\; \bigoplus_{k\geq 1} \mathcal{E}_n^{(k)}(\mathcal{A}, \tau)$$

where the elliptic corrections $\mathcal{E}_n^{(k)}(\mathcal{A}, \tau)$ are $\mathcal{A}$-modules whose dependence on τ is through (quasi-)modular forms. Specifically:

(i) The leading correction $\mathcal{E}_n^{(1)}$ is proportional to $E_2(\tau)$, the weight-2 quasi-modular Eisenstein series, arising from the regularization of the elliptic propagator $\partial_z \log\theta(z|\tau)$ relative to the rational propagator $1/z$ (cf. Construction 14.21).

(ii) Higher corrections $\mathcal{E}_n^{(k)}$ for $k \geq 2$ lie in the space of modular forms $M_{2k}(\mathrm{SL}_2(\mathbb{Z}))$ of weight $2k$; the first truly modular correction is an $E_4(\tau)$-coefficient.

(iii) The conformal dimension h of the states entering the bar complex constrains the weights: a state of dimension h contributes to $\mathcal{E}_n^{(k)}$ only for $k \leq h + n$.

(iv) At the rational limit $\tau \to i\infty$ (i.e., $q = e^{2\pi i\tau} \to 0$), the Eisenstein series $E_{2k}(\tau) \to 1$ and all corrections degenerate to constants, recovering $H_n(\bar{B}^{\mathrm{rat}}(\mathcal{A}))$ up to scalar renormalization.

For the Heisenberg algebra ($c = 1$), the corrections reduce to the Eisenstein series hierarchy computed in Vol I, §III.

Proof. We construct the decomposition via a perturbative spectral sequence comparing the elliptic and rational bar differentials.

Step 1 (Propagator decomposition). The elliptic bar complex (Construction 14.21) uses the propagator $\eta_{ij}^{\mathrm{ell}} = d\log\theta_1(z_i - z_j|\tau)$, while the rational bar complex uses $\eta_{ij}^{\mathrm{rat}} = d\log(z_i - z_j)$. Near $z_i = z_j$, both propagators have the same simple pole with residue 1, so their difference $\delta_{ij}(\tau) := \eta_{ij}^{\mathrm{ell}} - \eta_{ij}^{\mathrm{rat}}$ is a regular 1-form. The classical expansion of the Weierstrass ζ-function $\zeta(z;\Lambda_\tau)$ in terms of the Eisenstein–Weierstrass series $G_{2k}(\tau)$ gives the Laurent expansion:

$$\partial_z \log \theta_1(z|\tau) \;=\; \frac{1}{z} + \sum_{k=1}^{\infty} c_k(\tau)\, z^{2k-1},$$

where the correction coefficients $c_k(\tau)$ are expressed in terms of Eisenstein series via $G_{2k}(\tau) = 2\zeta_{\mathrm{R}}(2k)\, E_{2k}(\tau)$:

- $c_1(\tau) = -\frac{\pi^2}{3}E_2(\tau) \in \widetilde{M}_2(\mathrm{SL}_2(\mathbb{Z}))$ (quasi-modular, weight 2), since $c_1 = -2\eta_1$ where $\eta_1 = \zeta(1/2;\Lambda_\tau) = \frac{\pi^2}{6}E_2(\tau)$;
- $c_k(\tau) = -G_{2k}(\tau) \in M_{2k}(\mathrm{SL}_2(\mathbb{Z}))$ for $k \geq 2$ (holomorphic modular forms).

The expansion converges absolutely for $|z| < \min_{\omega\in\Lambda_\tau\setminus\{0\}} |\omega|$.

Step 2 (Correction-order filtration). Define the *correction-order filtration* $\mathcal{F}^p \bar{B}^{\mathrm{ell}}_n(\mathcal{A})$ by declaring that an element has filtration degree $\geq p$ if it arises from at least p insertions of the regular correction terms $\{c_k(\tau)\, z^{2k-1}\}_{k\geq 1}$ in the propagator expansion. Equivalently, replace each propagator by $1/z + t\cdot(c_1 z + c_2 z^3 + \cdots)$ and expand in powers of the formal parameter t; then $\mathcal{F}^p$ consists of terms of order $\geq t^p$.

Since the bar differential d^{ell} at degree n involves at most n propagator factors, we have $\mathcal{F}^{n+1}\bar{B}^{\mathrm{ell}}_n = 0$. The filtration is compatible with the differential: $d^{\mathrm{ell}}(\mathcal{F}^p) \subset \mathcal{F}^p$ (inserting the bar differential does not decrease the number of correction insertions).

Step 3 (Spectral sequence: E_0 and E_1 pages). The associated graded $\mathrm{gr}^0_{\mathcal{F}}\bar{B}^{\mathrm{ell}}(\mathcal{A})$ is the bar complex computed using only the singular part $1/z$ of the propagator. Since the residue extraction in the bar differential (which encodes the OPE data of $\mathcal{A}$) depends only on the polar part, the differential on gr^0 coincides with the rational bar differential d^{rat}. Thus:

$$E_1^{0,n} = H_n(\bar{B}^{\mathrm{rat}}(\mathcal{A})).$$

More generally, $E_1^{p,n} = H_n(\bar{B}^{\mathrm{rat}}(\mathcal{A})) \otimes \mathrm{Sym}^p_{\geq 1}\{c_k(\tau)\}$, where the symmetric product is taken over all propagator correction coefficients contributing total weight p.

Step 4 (Modular structure via Zhu's theorem). The differential $d_r : E_r^{p,n} \to E_r^{p+r,n+1}$ on the r-th page arises from the interaction of r correction insertions with the bar differential. By Zhu's modular invariance theorem [38], the n-point genus-1 correlation functions $\langle V_1(z_1)\cdots V_n(z_n)\rangle_{E_\tau}$ of a rational vertex algebra $\mathcal{A}$ transform as components of a vector-valued modular form for $\mathrm{SL}_2(\mathbb{Z})$. Since the bar complex at degree n is built from such correlation functions composed with the propagator form, and the correction coefficients $c_k(\tau)$ are themselves (quasi-)modular, the differentials d_r respect the modular weight grading. Consequently, the E_∞ terms inherit a weight decomposition from the ring $\widetilde{M}_*(\mathrm{SL}_2(\mathbb{Z}))$ of quasi-modular forms.

Step 5 (Convergence and splitting). The spectral sequence converges because the filtration is bounded: $\mathcal{F}^{n+1}\bar{B}^{\mathrm{ell}}_n = 0$ at each bar degree n. At convergence, $H_n(\bar{B}^{\mathrm{ell}}(\mathcal{A}))$ admits the filtration

$$0 = \mathcal{F}^{n+1}H_n \subset \mathcal{F}^n H_n \subset \cdots \subset \mathcal{F}^1 H_n \subset \mathcal{F}^0 H_n = H_n(\bar{B}^{\mathrm{ell}}(\mathcal{A}))$$

with successive quotients $\mathcal{E}_n^{(k)} := E_\infty^{k,n}$. The filtration *splits*: the modular weight provides a grading on the coefficient ring (quasi-modular forms of different weights are linearly independent over $\mathbb{C}$), and the E_∞ terms in different weights cannot be related by extensions. We therefore obtain the direct sum decomposition $H_n(\bar{B}^{\mathrm{ell}}(\mathcal{A})) \cong H_n(\bar{B}^{\mathrm{rat}}(\mathcal{A})) \oplus \bigoplus_{k\geq 1}\mathcal{E}_n^{(k)}(\mathcal{A},\tau)$ as claimed.

Properties (i)–(iv) now follow from the structure of the correction coefficients:

(i) The leading correction involves $c_1(\tau) = -(\pi^2/3)\, E_2(\tau)$ (quasi-modular, weight 2).

(ii) Higher corrections involve products of $c_k(\tau) \in M_{2k}(\mathrm{SL}_2(\mathbb{Z}))$ for $k \geq 2$.

(iii) A bar element of degree n involves at most n propagator factors; a state of conformal dimension h constrains the Laurent order of each propagator insertion to $\leq 2h-1$, giving the bound $\mathcal{E}_n^{(k)} = 0$ for $k > h+n$.

(iv) As $\tau \to i\infty$: $q \to 0$ and $E_{2k}(\tau) \to 1$, so all corrections reduce to constants, recovering $H_n(\bar{B}^{\mathrm{rat}}(\mathcal{A}))$ up to renormalization.

Verification for the Heisenberg algebra. For $\mathcal{A} = \mathcal{H}$ ($c = 1$), the genus-1 two-point function is $\langle a(z_1)a(z_2)\rangle_{E_\tau} = \kappa_{\mathrm{ch}}\, G_\tau(z_1 - z_2)$ with $G_\tau(z) = \zeta_\tau(z) - (\pi^2 E_2(\tau)/3)\, z$ (Vol I, Theorem III). The leading correction $\mathcal{E}_n^{(1)} \propto E_2(\tau)$ arises from the c_1 term. The complete Eisenstein hierarchy of Vol I, §III gives $\mathcal{E}_n^{(k)} \propto E_{2k}(\tau)$, confirming the general pattern. □

Remark 14.23 (Scope). The proof assembles three ingredients: (i) the Laurent expansion of the Weierstrass ζ-function in terms of Eisenstein series ([PE], classical); (ii) the convergence of the correction-order spectral sequence (bounded filtration at each bar degree); and (iii) Zhu's modular invariance theorem for genus-1 n-point functions of rational vertex algebras [38] ([PE]). The combination and the identification of the splitting mechanism (Step 5) are new to this volume. The Heisenberg verification uses the explicit computations of Vol I, §III.

14.4 Explicit elliptic bar complex for the Heisenberg algebra

Bar elements and the elliptic propagator

We work on $E_\tau = \mathbb{C}/(\mathbb{Z} + \tau\mathbb{Z})$ with the elliptic propagator

$$\eta_{ij}^{\mathrm{ell}} = d\log\theta_1(z_i - z_j \mid \tau) = \zeta_\tau(z_i - z_j)\,(dz_i - dz_j), \tag{14.2}$$

where $\zeta_\tau(z) = \partial_z \log\theta_1(z|\tau)$ is the Weierstrass zeta function on E_τ.

Computation 14.24 *(Degree-1 bar elements).* In bar degree 1, the generators of $\bar{B}_1^{\mathrm{ell}}(\mathcal{H}_{\kappa_{\mathrm{ch}}})$ are:

$$[a_m] \otimes 1 \in \bar{B}_1^{\mathrm{ell}}(\mathcal{H}_{\kappa_{\mathrm{ch}}}), \quad m \in \mathbb{Z},$$

where a_m denotes the mode of the Heisenberg current $a(z) = \sum_m a_m z^{-m-1}$. The bar differential on degree-1 elements is:

$$d^{\mathrm{ell}}([a_m] \otimes 1) = 0,$$

since the Heisenberg algebra has trivial unary OPE ($a_{(n)}a = 0$ for $n \geq 1$ except $a_{(1)}a = \kappa_{\mathrm{ch}}$, and the residue extraction on a single element yields zero). This is identical to the rational case.

Computation 14.25 *(Degree-2 bar elements and the elliptic differential).* In bar degree 2, the generators are:

$$[a_m|a_n] \otimes \eta_{12}^{\mathrm{ell}} \in \bar{B}_2^{\mathrm{ell}}(\mathcal{H}_{\kappa_{\mathrm{ch}}}),$$

where $\eta_{12}^{\mathrm{ell}} = \zeta_\tau(z_1 - z_2)\,(dz_1 - dz_2)$. The bar differential acts by residue extraction:

$$\begin{aligned} d^{\mathrm{ell}}([a_m|a_n] \otimes \eta_{12}^{\mathrm{ell}}) &= \mathrm{Res}_{z_1=z_2}\Big[a_m(z_1)\,a_n(z_2)\,\eta_{12}^{\mathrm{ell}}\Big] \\ &= \mathrm{Res}_{\epsilon=0}\Big[\frac{\kappa_{\mathrm{ch}}}{\epsilon^2}\,\zeta_\tau(\epsilon)\,d\epsilon\Big] \cdot z_2^{-m-n-2}\,dz_2, \end{aligned} \tag{14.3}$$

where $\epsilon = z_1 - z_2$. We use the Laurent expansion of ζ_τ (established in the proof of Theorem 14.22, Step 1):

$$\zeta_\tau(\epsilon) = \frac{1}{\epsilon} - \frac{\pi^2}{3} E_2(\tau)\,\epsilon - \frac{\pi^4}{45} E_4(\tau)\,\epsilon^3 - \frac{2\pi^6}{945} E_6(\tau)\,\epsilon^5 - \cdots \tag{14.4}$$

Substituting into the residue:

$$\begin{aligned}\mathrm{Res}_{\epsilon=0}\Big[\frac{\kappa_{\mathrm{ch}}}{\epsilon^2}\,\zeta_\tau(\epsilon)\,d\epsilon\Big] &= \mathrm{Res}_{\epsilon=0}\Big[\frac{\kappa_{\mathrm{ch}}}{\epsilon^2}\Big(\frac{1}{\epsilon} - \frac{\pi^2}{3} E_2(\tau)\,\epsilon - \cdots\Big)d\epsilon\Big] \\ &= \mathrm{Res}_{\epsilon=0}\Big[\frac{\kappa_{\mathrm{ch}}}{\epsilon^3} - \frac{\kappa_{\mathrm{ch}}\pi^2}{3} E_2(\tau)\,\frac{1}{\epsilon} - \cdots\Big]d\epsilon \\ &= -\frac{\kappa_{\mathrm{ch}}\pi^2}{3} E_2(\tau). \end{aligned}\tag{14.5}$$

The $1/\epsilon^3$ term has zero residue; the first nonzero contribution is the $E_2(\tau)$ term. In the rational case ($\zeta_\tau(\epsilon) \to 1/\epsilon$), the residue of $\kappa_{\mathrm{ch}}\epsilon^{-3}\,d\epsilon$ vanishes, so the rational bar differential on this element is zero. The *entire degree-2 bar differential is the elliptic correction*:

$$d^{\mathrm{ell}}([a_m|a_n] \otimes \eta_{12}^{\mathrm{ell}}) = -\frac{\kappa_{\mathrm{ch}}\pi^2}{3} E_2(\tau) \cdot z_2^{-m-n-2}\,dz_2. \tag{14.6}$$

Computation 14.26 *(Degree-2 curvature).* The elliptic bar complex of $\mathcal{H}_{\kappa_{\mathrm{ch}}}$ has curvature element $m_0^{\mathrm{ell}} \in \bar{B}_0^{\mathrm{ell}}(\mathcal{H}_{\kappa_{\mathrm{ch}}})$. From the genus-1 propagator (Vol I, Theorem III), the two-point function on E_τ is $\langle a(z_1)\,a(z_2)\rangle_{E_\tau} = \kappa_{\mathrm{ch}}\,G_\tau(z_1 - z_2)$, where $G_\tau(z) = \zeta_\tau(z) - (\pi^2 E_2(\tau)/3)\,z$. The curvature arises from the failure of double periodicity (cf. the proof of Vol I, Theorem III):

$$m_0^{\mathrm{ell}} = \kappa_{\mathrm{ch}} \cdot \frac{\pi^2 E_2(\tau)}{3}. \tag{14.7}$$

Expanding in the nome $q = e^{2\pi i\tau}$:

$$m_0^{\mathrm{ell}} = \frac{\kappa_{\mathrm{ch}}\pi^2}{3}\big(1 - 24q - 72q^2 - 96q^3 - \cdots\big). \tag{14.8}$$

At the rational limit $\tau \to i\infty$ $(q \to 0)$, $E_2(\tau) \to 1$ and $m_0^{\mathrm{ell}} \to \kappa_{\mathrm{ch}}\pi^2/3$, recovering the rational curvature (up to the normalization convention for the propagator on $\mathbb{C}$ versus E_τ; on $\mathbb{C}$ the propagator $1/z$ has no B-cycle and hence $m_0^{\mathrm{rat}} = 0$).

Comparison table: elliptic versus rational

The vanishing of the rational bar differential at all degrees reflects that the Heisenberg algebra on $\mathbb{C}$ (genus 0) has no curvature ($m_0^{\mathrm{rat}} = 0$): the propagator $1/z$ is strictly periodic on $\mathbb{C}$ (there is no B-cycle). On the elliptic curve E_τ, the B-cycle monodromy of ζ_τ introduces the Eisenstein corrections. At degree n, the coefficient $c_{n-1}(\tau)$ in the Laurent expansion (14.4) controls the bar differential; since $c_k(\tau) \in M_{2k}(\mathrm{SL}_2(\mathbb{Z}))$ for $k \ge 2$ and $c_1(\tau) \in \widetilde{M}_2(\mathrm{SL}_2(\mathbb{Z}))$ (quasi-modular), the corrections at degree $n \ge 3$ are genuine modular forms.

Proposition 14.27 *(Decomposition of the elliptic bar complex; [PH]).* The elliptic bar complex of $\mathcal{H}_{\kappa_{\mathrm{ch}}}$ at degree n admits the decomposition:

$$\bar{B}_n^{\mathrm{ell}}(\mathcal{H}_{\kappa_{\mathrm{ch}}}) = \bar{B}_n^{\mathrm{rat}}(\mathcal{H}_{\kappa_{\mathrm{ch}}}) \oplus E_2(\tau)\cdot\bar{B}_n^{(1)} \oplus E_4(\tau)\cdot\bar{B}_n^{(2)} \oplus \cdots \oplus E_{2n-2}(\tau)\cdot\bar{B}_n^{(n-1)} \tag{14.9}$$

with only finitely many nonzero terms. Specifically, $\bar{B}_n^{(k)} = 0$ for $k \ge n$.

Table 14.1: Elliptic versus rational bar differentials for $\mathcal{H}_{\kappa_{\mathrm{ch}}}$ at low degrees

Degree	**Rational** d^{rat}	**Elliptic** d^{ell}	**Correction**
1	0	0	—
2	0	$-\frac{\kappa_{\mathrm{ch}}\pi^2}{3}\,E_2(\tau)$	$E_2(\tau)$ (quasi-modular, wt 2)
3	0	$-\frac{\kappa_{\mathrm{ch}}\pi^4}{45}\,E_4(\tau)$	$E_4(\tau)$ (modular, wt 4)
4	0	$-\frac{2\kappa_{\mathrm{ch}}\pi^6}{945}\,E_6(\tau)$	$E_6(\tau)$ (modular, wt 6)
5	0	$-\frac{\kappa_{\mathrm{ch}}\pi^8}{4725}\,E_8(\tau)$	$E_8(\tau)$ (modular, wt 8)
6	0	$-\frac{2\kappa_{\mathrm{ch}}\pi^{10}}{93555}\,E_{10}(\tau)$	$E_{10}(\tau)$ (modular, wt 10)
7	0	$-\frac{1382\kappa_{\mathrm{ch}}\pi^{12}}{638512875}\,E_{12}(\tau)$	$E_{12}(\tau)$ (modular, wt 12)
8	0	$-\frac{4\kappa_{\mathrm{ch}}\pi^{14}}{18243225}\,E_{14}(\tau)$	$E_{14}(\tau)$ (modular, wt 14)
General n	0	$-\kappa_{\mathrm{ch}}\cdot\frac{2^{2n-2}\lvert B_{2n-2}\rvert}{(2n-2)!}\cdot\pi^{2n-2}\,E_{2n-2}(\tau)$	$E_{2n-2}(\tau)$ (wt $2n-2$)

Proof. Step 1 (Finite propagator expansion). The elliptic bar complex at degree n involves $\binom{n}{2}$ propagator factors η_{ij}^{ell}, one for each pair (i,j) with $1 \le i < j \le n$. Each propagator is expanded via (14.4) as $\zeta_\tau(\epsilon_{ij}) = \epsilon_{ij}^{-1} + \sum_{k\ge 1} c_k(\tau)\,\epsilon_{ij}^{2k-1}$. The bar differential at degree n extracts the residue at the diagonal $\epsilon_{ij} = 0$; since the OPE of the Heisenberg current $a(z)$ has a double pole ($a(z_1)\,a(z_2) \sim \kappa_{\mathrm{ch}}/(z_1-z_2)^2$), the residue at each pair involves at most the coefficient of ϵ_{ij}^{-1} in the integrand. This means the Laurent coefficient $c_k(\tau)\,\epsilon_{ij}^{2k-1}$ contributes to the residue only if $2k-1 \le 1$, i.e., $k \le 1$, for each *individual* propagator factor. However, when multiple propagator factors are combined in the degree-n bar element ($n-1$ propagators appear in a degree-n element), the correction terms from different propagator pairs may accumulate. The total correction order is bounded by $n-1$ (one correction per propagator), giving $\bar{B}_n^{(k)} = 0$ for $k \ge n$.

Step 2 (Eisenstein coefficient identification). The correction term $\bar{B}_n^{(k)}$ arises from choosing exactly k of the $n-1$ propagator factors to contribute their leading correction $c_1(\tau)\,\epsilon_{ij}$ (or, more precisely, assigning a total correction weight of k distributed among the propagators, with correction of weight j from a single propagator contributing $c_j(\tau)$). By the computation of (14.5), the leading correction at degree 2 is $c_1(\tau) \propto E_2(\tau)$. At degree n, the correction of total weight k lies in the subspace of (quasi-)modular forms of weight $2k$, which is spanned by monomials in E_2, E_4, E_6. Since the Heisenberg algebra is free (Gaussian), the only contributing terms are the *single-propagator corrections* (there are no interaction terms between different propagator pairs), so the coefficient of $\bar{B}_n^{(k)}$ is precisely $c_k(\tau) = (-1)^k G_{2k+2}(\tau) \propto E_{2k+2}(\tau)$ for $k \ge 2$, and $c_1(\tau) \propto E_2(\tau)$.

Step 3 (Direct sum splitting). The corrections $E_2, E_4, E_6, \ldots$ have distinct modular weights $2, 4, 6, \ldots$ and are therefore linearly independent over $\mathbb{C}$. The decomposition (14.9) is a direct sum, not merely a filtration. □

14.5 Dynamical Yang–Baxter equation for $\mathfrak{sl}_2$

We verify the dynamical Yang–Baxter equation for the $\mathfrak{sl}_2$ elliptic R-matrix of Definition 14.15, and show that the key identity reduces to the Fay trisecant (Theorem 14.18). The connection to bar nilpotency is made precise: the dynamical YBE encodes $d^2 = 0$ for the elliptic bar complex with dynamical parameter.

The dynamical Yang–Baxter equation

Definition 14.28 *(Dynamical Yang–Baxter equation).* The *dynamical Yang–Baxter equation* (DYBE) for an R-matrix $R(u,\lambda) \in \mathrm{End}(V \otimes V)$ depending on spectral parameter u and dynamical parameter $\lambda \in \mathfrak{h}^*$ is:

$$R_{12}(u,\lambda)\, R_{13}(u+v,\lambda-h^{(2)})\, R_{23}(v,\lambda) \;=\; R_{23}(v,\lambda-h^{(1)})\, R_{13}(u+v,\lambda)\, R_{12}(u,\lambda-h^{(3)}) \quad (14.10)$$

where $h^{(i)}$ denotes the action of $h \in \mathfrak{h}$ on the i-th tensor factor of $V^{\otimes 3}$, and $\lambda - h^{(i)}$ means that the dynamical parameter λ is shifted by the weight of the state in factor i.

Explicit verification for $\mathfrak{sl}_2$

For $\mathfrak{sl}_2$ with $V = \mathbb{C}^2$, the weight decomposition is $V = V_+ \oplus V_-$ with $h \cdot v_\pm = \pm v_\pm$. We use the elliptic R-matrix of Definition 14.15:

$$R(u,\lambda) = \begin{pmatrix} a(u) & 0 & 0 & 0 \\ 0 & b(u,\lambda) & c(u,\lambda) & 0 \\ 0 & c(u,-\lambda) & b(u,-\lambda) & 0 \\ 0 & 0 & 0 & a(u) \end{pmatrix} \quad (14.11)$$

with $a(u) = \theta(u+\eta)/\theta(\eta)$, $b(u,\lambda) = \theta(u)\theta(\lambda+\eta)/(\theta(\eta)\theta(\lambda))$, and $c(u,\lambda) = \theta(u+\lambda)\theta(\eta)/(\theta(\lambda)\theta(u+\eta))$ (the entries from Definition 14.15, abbreviated here as $\theta = \theta(\cdot \mid \tau)$).

Computation 14.29 *(Matrix entries of the DYBE).* We examine the DYBE (14.10) on the weight spaces of $V^{\otimes 3}$. On the space $V^{\otimes 3}$ with basis $\{v_{\epsilon_1} \otimes v_{\epsilon_2} \otimes v_{\epsilon_3} : \epsilon_i \in \{+,-\}\}$ (eight basis vectors), the dynamical shifts act as $h^{(i)} \cdot v_{\epsilon_1\epsilon_2\epsilon_3} = \epsilon_i\, v_{\epsilon_1\epsilon_2\epsilon_3}$. The DYBE is a system of 64 scalar equations (the entries of an 8×8 matrix equation).

Diagonal entries. On the extremal weight space v_{+++}, both sides of (14.10) give $a(u)\, a(u+v)\, a(v)$, so the equation is trivially satisfied. Similarly for v_{---}.

Mixed weight spaces. The nontrivial content lies on the 4-dimensional subspace spanned by $\{v_{++-}, v_{+-+}, v_{-++}\}$ (weight 1) and $\{v_{--+}, v_{-+-}, v_{+--}\}$ (weight -1). By the symmetry $R(u,\lambda) = R(u,-\lambda)|_{+\leftrightarrow-}$, it suffices to verify the weight-1 block.

On the weight-1 subspace, the DYBE reduces to a 3×3 matrix equation. The $(1,1)$-entry, acting on v_{++-}, is

$$\begin{aligned} &a(u)\, b(u+v,\lambda-1)\, b(v,\lambda) \;+\; b(u,\lambda)\, c(u+v,\lambda-1)\, c(v,-\lambda) \\ &\quad=\; b(v,\lambda-1)\, b(u+v,\lambda)\, a(u) \;+\; c(v,-\lambda+1)\, c(u+v,\lambda)\, a(u). \end{aligned} \quad (14.12)$$

Substituting the theta-function expressions and writing $[x] := \theta(x|\tau)$ gives

$$\begin{aligned} &\frac{[u+\eta]}{[\eta]} \cdot \frac{[u+v][\lambda-1+\eta]}{[\eta][\lambda-1]} \cdot \frac{[v][\lambda+\eta]}{[\eta][\lambda]} \;+\; \frac{[u][\lambda+\eta]}{[\eta][\lambda]} \cdot \frac{[u+v+\lambda-1][\eta]}{[\lambda-1][u+v+\eta]} \cdot \frac{[v-\lambda][\eta]}{[-\lambda][v+\eta]} \\ &\quad= \frac{[v][\lambda-1+\eta]}{[\eta][\lambda-1]} \cdot \frac{[u+v][\lambda+\eta]}{[\eta][\lambda]} \cdot \frac{[u+\eta]}{[\eta]} \;+\; \frac{[v-\lambda+1][\eta]}{[-\lambda+1][v+\eta]} \cdot \frac{[u+v+\lambda][\eta]}{[\lambda][u+v+\eta]} \cdot \frac{[u+\eta]}{[\eta]}. \end{aligned} \quad (14.13)$$

After cancelling common factors $[u+\eta]/([\eta]^3\,[\lambda]\,[\lambda-1])$ from both sides and clearing denominators, the equation reduces to an identity among products of theta functions evaluated at the six arguments $\{u, v, u+v, \lambda, \lambda+\eta, \lambda-1+\eta\}$.

PROPOSITION 14.30 *(DYBE reduces to Fay; [PH])*. The dynamical Yang–Baxter equation (14.10) for the $\mathfrak{sl}_2$ elliptic R-matrix (Definition 14.15) is equivalent to the Fay trisecant identity (Theorem 14.18) plus the quasi-periodicity of θ. More precisely:

(i) Each nontrivial component of the DYBE on the weight-1 subspace reduces, after clearing common factors, to an identity of the form

$$\theta(a-b)\,\theta(c-d) - \theta(a-c)\,\theta(b-d) + \theta(a-d)\,\theta(b-c) = 0 \tag{14.14}$$

for appropriate specializations of a, b, c, d depending on u, v, λ, η.

(ii) The remaining components (mixing weight-1 and weight-(-1) sectors) follow from (i) and the quasi-periodicity $\theta(z+1|\tau) = -\theta(z|\tau)$, $\theta(z+\tau|\tau) = -e^{-\pi i\tau - 2\pi i z}\theta(z|\tau)$.

Proof. Step 1 (Reduction of the $(1,2)$*-entry).* The $(1,2)$-entry of the DYBE on the weight-1 subspace (the matrix element for $v_{++-} \to v_{+-+}$) yields, after clearing the common denominator $[\eta]^3\,[\lambda]\,[\lambda-1]\,[u+\eta]\,[v+\eta]\,[u+v+\eta]$:

$$\begin{aligned}&[u+\eta]\,[v]\,[u+v+\lambda-1] - [u]\,[v+\eta]\,[u+v+\lambda-1]\\ &\qquad + [u+v+\eta]\,[\lambda-1]\,\big([u+\lambda]\,[v] - [v+\lambda]\,[u]\big) \;=\; 0.\end{aligned} \tag{14.15}$$

We apply the Fay identity (Theorem 14.18) with the specialization $u_1 = 0, u_2 = u+\eta, u_3 = v+\eta$, $u_4 = u+v+\lambda$:

$$\begin{aligned}&\theta(u+\eta)\,\theta(v+\eta-u-v-\lambda) - \theta(v+\eta)\,\theta(u+\eta-u-v-\lambda)\\ &\qquad + \theta(u+v+\lambda)\,\theta(v+\eta-u-\eta) = 0.\end{aligned} \tag{14.16}$$

Using $\theta(-x) = -\theta(x)$, this simplifies to:

$$\theta(u+\eta)\,\theta(\lambda-v) - \theta(v+\eta)\,\theta(\lambda-u) + \theta(u+v+\lambda)\,\theta(v-u) = 0,$$

which is the content of (14.15) after the identification $[\lambda-v] = -[v-\lambda]$ and relabelling.

Step 2 (Remaining entries). The $(1,3)$ and $(2,3)$ entries of the DYBE are obtained from the $(1,2)$ entry by the substitutions $v \to u+v$, $u \to -v$ and $u \to v$, $v \to u$ respectively, combined with the Weyl group symmetry $R(u,\lambda) = R(u,-\lambda)|_{+\leftrightarrow-}$. Each reduces to the same Fay identity with permuted arguments.

Step 3 (Quasi-periodicity). The elliptic R-matrix entries $a(u), b(u,\lambda), c(u,\lambda)$ are meromorphic on $E_\tau \times E_\tau$ with simple poles at the zeros of $\theta(\eta)$ and $\theta(\lambda)$. The DYBE relates values at $\lambda, \lambda - h^{(1)}, \lambda - h^{(2)}$, $\lambda - h^{(3)}$ where $h^{(i)} \in \{+1, -1\}$; the quasi-periodicity of θ ensures that these λ-shifts are compatible with the lattice $\mathbb{Z} + \tau\mathbb{Z}$ acting on λ. □

PROPOSITION 14.31 *(DYBE and bar nilpotency; [PH])*. The dynamical Yang–Baxter equation (14.10) encodes $d^2 = 0$ for the elliptic bar complex with dynamical parameter. Specifically, the bar complex $\bar{B}^{\mathrm{ell}}(E_{q,p}(\mathfrak{sl}_2))$ of the elliptic quantum group (Definition 14.14) satisfies $d^2 = 0$ on $\overline{C}_3(E_\tau)$ if and only if the R-matrix $R(u,\lambda)$ satisfies the DYBE.

Proof. The bar differential at degree 2 of the RTT-type algebra $E_{q,p}(\mathfrak{sl}_2)$ is:

$$d([L_{ij}(u)|L_{kl}(v)] \otimes \eta_{12}^{\mathrm{ell}}) = \sum_m \big(R_{ij,km}(u-v,\lambda)\,L_{ml}(v) - L_{km}(v)\,R_{mj,il}(u-v,\lambda)\big) \otimes \eta_{12}^{\mathrm{ell}} \wedge \eta_{23}^{\mathrm{ell}}.$$

The computation of d^2 on a degree-1 element $[L_{ij}(u)] \otimes 1$ produces terms on $\overline{C}_3(E_\tau)$ involving the product of three R-matrices. The nine terms (three pairs of propagators, each contributing three matrix products) rearrange into the left-hand side minus the right-hand side of the DYBE (14.10), exactly as in the rational case where the Yang–Baxter equation implies $d^2 = 0$ for the Yangian bar complex (Vol I, proof of Theorem III). The only modification is the replacement of the rational propagator $\eta_{ij} = d\log(z_i - z_j)$ by the elliptic propagator $\eta_{ij}^{\mathrm{ell}} = d\log\theta_1(z_i - z_j|\tau)$, and the introduction of the dynamical shifts $\lambda \to \lambda - h^{(i)}$ arising from the B-cycle monodromy of η_{ij}^{ell} around E_τ (cf. Remark 14.16(ii)). The elliptic Arnold relation (14.1) (Proposition 14.19) ensures that the propagator-level identity $\eta_{12} \wedge \eta_{23} + \eta_{23} \wedge \eta_{31} + \eta_{31} \wedge \eta_{12} = 0$ holds, so the algebraic content of $d^2 = 0$ is precisely the DYBE. □

14.6 Shuffle algebra presentation

Definition of the shuffle algebra

Definition 14.32 *(Shuffle algebra).* Let $\zeta(u) = \theta(u + \eta)/\theta(u)$ be the structure function determined by the elliptic R-matrix. The *shuffle algebra* $\mathcal{S}_\zeta$ is the $\mathbb{Z}_{\geq 0}$-graded vector space

$$\mathcal{S}_\zeta = \bigoplus_{n\geq 0} \mathcal{S}_n, \quad \mathcal{S}_n = \mathbb{C}(z_1, \dots, z_n)^{S_n}_{\mathrm{sym}}$$

(symmetric rational functions in n variables), equipped with the *shuffle product*: for $f \in \mathcal{S}_m$ and $g \in \mathcal{S}_n$, define $f \star g \in \mathcal{S}_{m+n}$ by

$$(f \star g)(z_1, \dots, z_{m+n}) = \frac{1}{m!\,n!} \sum_{\sigma\in S_{m+n}} \prod_{\substack{i\leq m \\ j>m}} \zeta(z_{\sigma(i)} - z_{\sigma(j)}) \\ \cdot f(z_{\sigma(1)}, \dots, z_{\sigma(m)}) \cdot g(z_{\sigma(m+1)}, \dots, z_{\sigma(m+n)}). \quad (14.17)$$

Equivalently, the sum runs over (m, n)-shuffles $\mathrm{Sh}(m, n) \subset S_{m+n}$, with the $1/(m!\,n!)$ factor replaced by 1 and the sum restricted to shuffles.

Remark 14.33 (Origin). The shuffle algebra was introduced by Feigin–Odesskii in the rational and trigonometric cases and extended to the elliptic case by Schiffmann–Vasserot [34] and Negut. The rational specialization $\zeta(u) = (u + \eta)/u$ recovers the Feigin–Odesskii shuffle realization of the Yangian; the trigonometric specialization $\zeta(u) = (1 - qe^{2\pi iu})/(1 - e^{2\pi iu})$ recovers the quantum affine algebra.

Explicit shuffle products

Computation 14.34 *(First generators).* Let $e_1 = 1 \in \mathcal{S}_1 = \mathbb{C}(z)$. By the Fay identity (Theorem 14.18) and oddness of θ:

$$(e_1 \star e_1)(z_1, z_2) = \frac{\theta(z_{12} + \eta) + \theta(\eta - z_{12})}{\theta(z_{12})}, \quad (14.18)$$

where $z_{12} = z_1 - z_2$. The triple product is

$$(e_1 \star e_1 \star e_1)(z_1, z_2, z_3) = \sum_{\sigma\in S_3} \zeta(z_{\sigma(1)} - z_{\sigma(2)})\,\zeta(z_{\sigma(1)} - z_{\sigma(3)})\,\zeta(z_{\sigma(2)} - z_{\sigma(3)}). \quad (14.19)$$

Associativity follows from the Yang–Baxter equation for ζ.

Remark 14.35 (Shuffle–bar isomorphism). The residue pairing gives an isomorphism $\mathcal{S}_n \cong \bar{B}_n(\mathcal{A})^*$ under which the shuffle product corresponds to the deshuffle coproduct on $\bar{B}(\mathcal{A})$ (Vol I, Corollary III).

14.7 Comparison: rational, trigonometric, elliptic

Table 14.2: Comparison across the three regimes

	Rational	**Trigonometric**	**Elliptic**
Algebra	Yangian $Y(\mathfrak{g})$	Qtm. affine $U_q(\hat{\mathfrak{g}})$	Toroidal $U_{q,t}(\hat{\hat{\mathfrak{g}}})$
Curve	$\mathbb{C}$ (or $\mathbb{CP}^1$)	$\mathbb{C}^*$ (or $\mathbb{CP}^1 \setminus \{0, \infty\}$)	E_τ
Propagator	$\frac{dz}{z}$	$\frac{dz}{1-z}$	$d \log \theta_1(z\|\tau)$
R-**matrix**	$1 - \hbar P/u$	$(q-q^{-1})/(u-u^{-1})$	$\theta(u+\eta)/\theta(\eta)$
Arnold id.	$\eta_{12} \wedge \eta_{23} + \text{cyc} = 0$	$\frac{dz_1}{z_1-z_2} \wedge \frac{dz_2}{z_2-z_3} + \text{cyc} = 0$	Fay trisecant
$d^2 = 0$	Partial fractions	q-partial fractions	Fay + quasi-period
Koszul dual	$Y_{-\hbar}(\mathfrak{g})$	$U_{q^{-1}}(\hat{\mathfrak{g}})$	$U_{q^{-1},t^{-1}}(\hat{\hat{\mathfrak{g}}})$ (conj.)
Struct. fn.	$(u+\eta)/u$	$(1-qe^u)/(1-e^u)$	$\theta(u+\eta)/\theta(u)$
Shuffle alg.	Feigin–Odesskii	F–O (q-def.)	Schiffmann–Vasserot
Dyn. par.	absent	absent	$\lambda \in \mathfrak{h}^*$
Bar curv.	$m_0 = 0$	$m_0 \propto \log q$	$m_0 \propto E_2(\tau)$

Remark 14.36 (Degeneration limits). The three regimes degenerate via

$$\text{Elliptic} \xrightarrow{\tau \to i\infty} \text{Trigonometric} \xrightarrow{q \to 1} \text{Rational}. \tag{14.20}$$

As $\tau \to i\infty$, $\theta_1(z|\tau) \to 2\sin(\pi z)$ and the Eisenstein corrections collapse to constants; as $q \to 1$, the trigonometric propagator becomes dz/z and $R_q(u) \to 1 - \hbar P/u$. The bar complex degenerates accordingly: $\bar{B}^{\mathrm{ell}}(\mathcal{A}) \to \bar{B}^{\mathrm{trig}}(\mathcal{A}) \to \bar{B}^{\mathrm{rat}}(\mathcal{A})$, consistent with the spectral sequence of Theorem 14.22.

The elliptic curve as shared geometry

Remark 14.37 (Bridge: algebras on E and algebras from $K3 \times E$). The elliptic curve E_τ enters the remainder of this chapter in a second capacity. The preceding sections study chiral algebras *on* E_τ: the propagator $d \log \theta_1(z|\tau)$, the Eisenstein corrections to the bar differential, the toroidal algebra $U_{q,t}(\hat{\hat{\mathfrak{g}}})$ with $\tau = \log t / \log q$, and Felder's elliptic R-matrix $R(u, \lambda)$ built from the same theta function. The K3 chiral algebra $\mathcal{A}_{K3}$ (§14.9) lives on a K3 surface, but the Calabi–Yau threefold $K3 \times E$ (§14.10) is fibered *over* E, and Costello–Li's holomorphic twist reduction along K3 produces a boundary chiral algebra A_E *on* E_τ with $c(A_E) = \chi(K3) = 24$ free bosons from harmonic forms on K3 (Construction 14.77).

Thus: the toroidal and elliptic quantum group algebras, and the Costello–Li boundary algebra A_E, are all chiral algebras on the *same* curve E_τ. They share the propagator $d \log \theta_1$, the Fay trisecant relation governing $d^2 = 0$ (Proposition 14.19), and the Eisenstein filtration of the elliptic bar complex (Theorem 14.22). The difference is the algebra: A_E has 24 generators with $\kappa_{\mathrm{ch}}(A_E) = 24$ (rank of the free-boson lattice) and shadow class G (Gaussian, $r_{\max} = 2$), while $E_{q,p}(\mathfrak{sl}_N)$ has $N^2 - 1$ generators with dynamical parameter $\lambda \in \mathfrak{h}^*$.

14.8 Calabi–Yau local-to-global gluing

The Calabi–Yau threefold $K3 \times E$ admits a local description in terms of quiver charts glued by A_∞-bimodule transition data. The quiver-chart description is necessary because no global chiral algebra $A_{K3 \times E}$ is yet constructed (AP-CY6).

McKay correspondence and ADE quiver charts

A K3 surface S acquires ADE singularities at special points in moduli. Near each singularity, the local geometry is $\mathbb{C}^2/\Gamma$ for a finite subgroup $\Gamma \subset \mathrm{SL}(2,\mathbb{C})$. The McKay correspondence (McKay 1980, Gonzalez-Sprinberg–Verdier 1983) gives a bijection

$$\{\text{nontrivial irreps of } \Gamma\} \ \stackrel{\sim}{\longleftrightarrow}\ \{\text{exceptional curves in the minimal resolution}\},$$

and the McKay quiver Q_Γ encodes the tensor product rule $\rho_{\text{fund}} \otimes \rho_i = \bigoplus_j a_{ij}\, \rho_j$, where a_{ij} are the arrows of the extended ADE Dynkin diagram.

Definition 14.38 *(McKay quiver and Cartan matrix).* For a finite subgroup $\Gamma \subset \mathrm{SL}(2,\mathbb{C})$ of ADE type, the *McKay quiver* Q_Γ has vertices indexed by irreducible representations $\rho_0 = \mathrm{triv},\ \rho_1, \ldots, \rho_r$ and a_{ij} arrows from vertex i to vertex j, where a_{ij} is the multiplicity of ρ_j in $\rho_{\text{fund}} \otimes \rho_i$. The extended Cartan matrix is

$$C_{\text{ext}} = 2I - A,$$

where A is the adjacency matrix of Q_Γ.

Example 14.39 *(ADE classification).* The finite subgroups of $\mathrm{SL}(2,\mathbb{C})$ are classified by ADE Dynkin diagrams:

Group	Γ	**Type**	$\lvert\Gamma\rvert$	**Irreps (rank)**
Cyclic	$\mathbb{Z}/n$	A_{n-1}	n	n irreps, all 1-dim
Binary dihedral	BD_n	D_{n+2}	$4n$	$n+3$ irreps
Binary tetrahedral	BT	E_6	24	7 irreps
Binary octahedral	BO	E_7	48	8 irreps
Binary icosahedral	BI	E_8	120	9 irreps

For each type, $|\Gamma| = \sum d_i^2$ (sum of squared dimensions of irreducible representations) provides an independent check of the character table.

Proposition 14.40 *(Preprojective algebra from the McKay quiver; [PE]).* The preprojective algebra $\Pi(Q_\Gamma) = kQ_{\text{double}}/(\sum[a, a^*])$, where Q_{double} is the double quiver (adjoin a reverse arrow a^* for each arrow a), has centre isomorphic to the invariant ring:

$$Z(\Pi(Q_\Gamma)) \ \cong\ \mathbb{C}[x, y, z]^\Gamma.$$

For the Dynkin quiver of type Δ with r vertices and Coxeter number h, the preprojective algebra is finite-dimensional with $\dim \Pi_0(Q_\Delta)$ given by the following table (Crawley-Boevey 2001, computed via $\dim e_i \Pi e_j = \min(i, j) \min(r+1-i, r+1-j)$ for type A and by the McKay quiver formula $\dim \Pi(Q_\Gamma) = |\Gamma| \cdot (r+1)$ for the extended quiver):

Type	$\|\Gamma\|$	r = rank	h	$\|Q_0\|$	$\dim \Pi_0$
A_1	2	1	2	2	4
A_2	3	2	3	3	9
A_3	4	3	4	4	20
A_4	5	4	5	5	35
D_4	8	4	6	5	40
D_5	12	5	8	6	72
E_6	24	6	12	7	168
E_7	48	7	18	8	384
E_8	120	8	30	9	1080

The table entries are computed from the McKay quiver formula $\dim e_i \Pi e_j = \min(i, j) \min(r+1-i,\ r+1-j)$ for type A and from representation-theoretic data for the exceptional types.

Ginzburg dg algebras and Jacobian algebras

Definition 14.41 *(Ginzburg dg algebra).* For an ADE quiver Q with potential W (a linear combination of cyclic paths), the *Ginzburg dg algebra* $\mathcal{A}_\Gamma$ is a Calabi–Yau 3-fold dg algebra concentrated in degrees $[-3, 0]$. Its zeroth cohomology is the *Jacobian algebra*:

$$\mathrm{Jac}(Q, W) = kQ \big/ \big(\partial_a W : a \in Q_1\big),$$

where $\partial_a W$ is the cyclic derivative with respect to arrow a. For ADE types, $\mathrm{Jac}(Q, W)$ is isomorphic to the preprojective algebra $\Pi_0(Q)$.

Remark 14.42 (CY3 extension). For the local CY3 model $\mathbb{C}^2/\Gamma \times \mathbb{C}$, one adds a loop at each vertex of the McKay quiver (from the $\mathbb{C}$ direction). The extended quiver with potential $(Q_{\mathrm{ext}}, W_{\mathrm{ext}})$ has Jacobian algebra that is 3-Calabi–Yau in the sense of Ginzburg: the derived category $D^b(\mathrm{Jac}(Q_{\mathrm{ext}}, W_{\mathrm{ext}}))$ carries a Serre functor $S = [3]$.

A_∞-bimodule transition data

Construction 14.43 *(A_∞-bimodule gluing).* Given two local charts U_α, U_β with tilting generators T_α, T_β and endomorphism algebras $A_\alpha = \mathrm{End}(T_\alpha)$, $A_\beta = \mathrm{End}(T_\beta)$, the transition data on the overlap $U_{\alpha\beta} = U_\alpha \cap U_\beta$ is encoded in the (A_α, A_β)-bimodule

$$M_{\alpha\beta} = R\mathrm{Hom}(T_\alpha|_{U_{\alpha\beta}}, T_\beta|_{U_{\alpha\beta}})$$

with A_∞-bimodule operations $m_{p|1|q} \colon A_\alpha^{\otimes p} \otimes M_{\alpha\beta} \otimes A_\beta^{\otimes q} \to M_{\alpha\beta}[2-p-q]$ satisfying the A_∞-bimodule equation. The derived tensor product $M_{\alpha\beta} \otimes^{\mathbf{L}}_{A_\beta} M_{\beta\gamma}$ provides the triple-overlap cocycle data. The reduced bar complex $\overline{B}(A_\alpha, M_{\alpha\beta}, A_\beta)$ computes the relevant Ext groups.

Remark 14.44 (Reduced bar complex and unit elements). The A_∞-bimodule relations hold on the *reduced* bar complex (augmentation ideal only): unit elements do not participate in the higher operations $m_{p|1|q}$ for $p + q \geq 2$. This is a general structural principle, not specific to K3. When computing $\overline{B}(A_\alpha, M_{\alpha\beta}, A_\beta)$, one restricts to the augmentation ideal $\overline{A} = \ker(\varepsilon\colon A \to \Bbbk)$ in both algebra arguments. The unit $1_A \in A$ acts on M via $m_{1|1|0}(1_A, m) = m$ (identity bimodule action) and $m_{0|1|1}(m, 1_B) = m$, but does not enter the higher bar differential. For the K3 quiver charts, where $A_\alpha = \mathrm{End}(T_\alpha)$ is augmented over $\Bbbk$, this restriction ensures that only the interacting OPE data contributes to the gluing obstructions.

Čech descent and the global category

THEOREM 14.45 *(Čech descent for dg categories; [PE]).* Cover $K3 = U_0 \cup U_1$, where $U_0 = K3 \setminus D$ (complement of a smooth divisor $D \in |H|$) and U_1 is a tubular neighbourhood of D. The derived category $D^b(K3)$ is recovered as the homotopy fibre product

$$D^b(K3) \;\simeq\; D^b(U_0) \times^h_{D^b(U_{01})} D^b(U_1),$$

and the descent spectral sequence for Hochschild cohomology

$$E_1^{p,q} = H^q\big(C^p(U_\bullet, \mathrm{HH}^*)\big) \implies \mathrm{HH}^{p+q}(D^b(K3))$$

converges. The obstruction to gluing lives in $H^2(K3, \mathcal{O}_{K3}) \cong \mathbb{C}$ (the Brauer class), reflecting $h^{0,2}(K3) = 1$.

Wall-crossing during chart transitions

Remark 14.46 (Cluster mutations and Bridgeland stability). Mutation at a vertex k of the McKay quiver produces a new quiver $Q' = \mu_k(Q)$ with exchange matrix B' given by the Fomin–Zelevinsky rule. For A_n quivers, the exchange graph has $C_{n+1} = \binom{2n}{n}/(n+1)$ vertices (the $(n+1)$-th Catalan number), corresponding to triangulations of an $(n+3)$-gon. The Kontsevich–Soibelman wall-crossing formula governs the change of DT invariants across stability walls. For $K3$ with primitive Mukai vector v: the moduli space $M_H(v) \cong \mathrm{Hilb}^n(K3)$ (Beauville) and the DT transformation satisfies $\tau^{n+3} = [2]$ (Seidel–Thomas).

14.9 The K3 chiral algebra

The K3 sigma model produces a $c = 6$ vertex algebra with $\mathcal{N} = 4$ superconformal symmetry. This section assembles the complete algebraic data: OPE structure, bar complex, shadow tower, Koszul dual, and the connections to Mathieu moonshine.

The $\mathcal{N} = 4$ superconformal algebra at $c = 6$

Definition 14.47 *(Small $\mathcal{N} = 4$ SCA at $c = 6$).* The *small $\mathcal{N} = 4$ superconformal algebra* at central charge $c = 6$ (equivalently, $\mathrm{SU}(2)_R$ level $k_R = 1$, since $c = 6k_R$) has eight primary generators:

Generator	Weight h	Parity	J^3 charge
T	2	bosonic	0
G^+, G^-	3/2	fermionic	$\pm 1/2$
$\tilde{G}^+, \tilde{G}^-$	3/2	fermionic	$\mp 1/2$
J^{++}, J^{--}	1	bosonic	± 1
J^3	1	bosonic	0

The OPE structure at $c = 6$, $k_R = 1$:

$$T(z)\,T(w) \sim \frac{3}{(z-w)^4} + \frac{2T}{(z-w)^2} + \frac{\partial T}{z-w}, \tag{14.21}$$

$$J^3(z)\,J^3(w) \sim \frac{1/2}{(z-w)^2}, \tag{14.22}$$

$$J^{++}(z)\,J^{--}(w) \sim \frac{1}{(z-w)^2} + \frac{2J^3}{z-w}, \tag{14.23}$$

$$G^+(z)\,G^-(w) \sim \frac{2}{(z-w)^3} + \frac{2J^3}{(z-w)^2} + \frac{T + \partial J^3}{z-w}. \tag{14.24}$$

The remaining independent OPE relations at $c = 6, k_R = 1$ (from the 64-pair table of `cy_n4sca_k3_engine.py`):

$$J^3(z)\,G^\pm(w) \sim \frac{\pm\frac{1}{2}\,G^\pm}{z-w}, \tag{14.25}$$

$$J^{++}(z)\,G^-(w) \sim \frac{G^+}{z-w}, \qquad J^{--}(z)\,G^+(w) \sim \frac{G^-}{z-w}, \tag{14.26}$$

$$T(z)\,G^\pm(w) \sim \frac{\frac{3}{2}\,G^\pm}{(z-w)^2} + \frac{\partial G^\pm}{z-w}, \tag{14.27}$$

$$T(z)\,J^a(w) \sim \frac{J^a}{(z-w)^2} + \frac{\partial J^a}{z-w}, \tag{14.28}$$

$$\tilde{G}^+(z)\,\tilde{G}^-(w) \sim \frac{2}{(z-w)^3} - \frac{2J^3}{(z-w)^2} + \frac{T - \partial J^3}{z-w}, \tag{14.29}$$

$$G^+(z)\,\tilde{G}^+(w) \sim \frac{2\,J^{++}}{z-w}, \qquad G^-(z)\,\tilde{G}^-(w) \sim \frac{-2\,J^{--}}{z-w}. \tag{14.30}$$

All boson–boson and fermion–fermion OPEs not listed above are either trivially determined by $\mathrm{SU}(2)_R$ symmetry or regular (no singular terms).

Proposition 14.48 *(Modular characteristic of the K3 sigma model; [PH]).* The modular characteristic of the K3 sigma model VOA is

$$\kappa_{\mathrm{ch}}(\mathcal{A}_{K3}) = 2 = \dim_{\mathbb{C}}(K3). \tag{14.31}$$

This is verified by six independent paths:

(i) *Geometric.* For a CY d-fold sigma model: $\kappa_{\mathrm{ch}} = d$ (index-theoretic computation via the Chern character of the chiral de Rham complex).

(ii) *Character.* $F_1 = \kappa_{\mathrm{ch}}/24 = 1/12$ matches the genus-1 anomaly of the $c = 6$ sigma model.

(iii) *Complementarity.* $\kappa_{\mathrm{ch}}(\mathcal{A}) + \kappa_{\mathrm{ch}}(\mathcal{A}^!) = 0$ gives $\kappa^!_{\mathrm{ch}} = -2$.

(iv) *Kummer orbifold.* At the Kummer point $K3 = T^4/\mathbb{Z}_2$: the orbifold preserves $\kappa_{\mathrm{ch}} = 2$ from the smooth sigma model.

(v) *Gepner consistency.* The Gepner model $(2)^4$ gives $\kappa_{\mathrm{Gepner}} = 4 \times (3/4) = 3$ using the Virasoro formula $\kappa_{\mathrm{ch}} = c/2$ for each $\mathcal{N} = 2$ factor. This is a DIFFERENT algebra from the $\mathcal{N} = 4$ SCA (κ_{ch} depends on the full algebra, not just the Virasoro subalgebra).

(vi) *Additivity.* $\kappa_{\mathrm{ch}}(K3 \times E) = \kappa_{\mathrm{ch}}(K3) + \kappa_{\mathrm{ch}}(E) = 2 + 1 = 3 = \dim_{\mathbb{C}}(K3 \times E)$ (Corollary III).

Proof. Paths (i)–(iii) follow from Vol I, Theorem D and the index-theoretic computation $\kappa_{\mathrm{ch}}(\Omega^{\mathrm{ch}}(\mathrm{CY}_d)) = d$ (the Chern character of the chiral de Rham complex on a Ricci-flat manifold reduces the genus-1 obstruction to $\chi(M)/12 = 2$ for K3 via $\chi(K3) = 24$). Path (iv): the Kummer orbifold $T^4/\mathbb{Z}_2$ has the same modular characteristic as the smooth K3 because κ_{ch} is a deformation invariant (it depends on the chiral algebra structure, which is constant across K3 moduli by curve independence). Path (v): the Gepner model uses the Virasoro stress tensor for each $c = 3/2$ factor, giving $\kappa_{\mathrm{ch}} = c/2 = 3/4$ per factor; the physical sigma model has the full $\mathcal{N} = 4$ structure whose $\kappa_{\mathrm{ch}} = 2$ is lower due to supersymmetric Ward identities. Path (vi) follows from the additivity of κ_{ch} under tensor products (Vol I, Proposition III). □

Remark 14.49 ($\kappa_{\mathrm{ch}}(K3) = 2 \neq c/2 = 3$: modular characteristic vs central charge). The K3 sigma model provides a sharp illustration: the modular characteristic $\kappa_{\mathrm{ch}}(\mathcal{A}_{K3}) = 2$ differs from the Virasoro formula $c/2 = 3$. The reduction arises because the $\mathcal{N} = 4$ superconformal Ward identities constrain the genus-1 obstruction below what the Virasoro subalgebra alone would produce. The naive Virasoro computation $\kappa_{\mathrm{ch}}(\mathrm{Vir}_6) = 3$ counts the Virasoro contribution, but the full $\mathcal{N} = 4$ algebra at $c = 6$ has $\kappa_{\mathrm{ch}} = 2k_R = 2$.

Bar complex of the $\mathcal{N} = 4$ SCA

Proposition 14.50 *(Bar complex structure; [PH]).* The bar complex $B(\mathcal{A}_{K3})$ of the small $\mathcal{N} = 4$ SCA at $c = 6$ has:

(i) $B^1(\mathcal{A}_{K3}) = \mathrm{span}\{s^{-1}a : a \text{ a generator}\}$ with $\dim B^1 = 8$ (one desuspended generator per primary; $|s^{-1}v| = |v| - 1$).

(ii) The bar differential $d\colon B^2 \to B^1$ is a rank-16 linear map extracting all singular OPE modes via the logarithmic kernel $d\log(z-w)$ (the $d\log$ measure absorbs one power of $(z-w)$, so bar pole orders are one less than OPE pole orders; the bar differential extracts all singular modes, not just the simple pole).

(iii) The algebra is chirally Koszul (Vol I, Proposition III): the $\mathcal{N} = 4$ SCA is freely strongly generated (no null vectors in the universal vacuum module), so bar cohomology $H^*(B(\mathcal{A}))$ is concentrated in bar degree 1.

Shadow depth classification

Proposition 14.51 *(K3 sigma model: class M; [PH]).* The K3 sigma model VOA has shadow depth class M (infinite tower, $r_{\max} = \infty$). The shadow metric $Q_L(t) = (2\kappa_{\mathrm{ch}} + 3\alpha t)^2 + 2\Delta t^2$ has critical discriminant

$$\Delta = 8\kappa_{\mathrm{ch}} S_4 \neq 0,$$

so Q_L is irreducible and the shadow tower does not terminate. The Virasoro sector at $c = 6$ contributes $Q^{\mathrm{contact}} = 10/[c(5c + 22)] = 5/156 \neq 0$, and the $\mathrm{SU}(2)_R$ sector at level $k_R = 1$ contributes nonzero cubic shadow S_3 from the triple current OPE.

Computation 14.52 *(Shadow tower of the K3 sigma model through arity 12; [PH]).* The MC recursion (Vol I, Theorem III) with initial data $(\kappa_{\mathrm{ch}}, S_3, S_4) = (2, 2, 5/156)$ produces the full shadow tower. Every value is an exact rational, verified to satisfy the MC equation $D_{\mathcal{A}}(\Theta) + \frac{1}{2}[\Theta, \Theta] = 0$ at each arity.

r	$S_r(\mathcal{A}_\sigma)$	Sign
2	2	+
3	2	+
4	$5/156$	+
5	$-1/26$	−
6	$3485/73008$	+
7	$-1145/18928$	−
8	$1179485/15185664$	+
9	$-1144885/11389248$	−
10	$309513983/2368963584$	+
11	$-1477140565/8686199808$	−
12	$490338857615/2217349914624$	+

Sign alternation begins at arity 5; magnitudes are bounded and slowly decaying. The critical discriminant $\Delta = 8\kappa_{\mathrm{ch}} S_4 = 20/39 \neq 0$ forces all $S_r \neq 0$ for $r \geq 5$ by the single-line dichotomy theorem (Vol I, Theorem III), confirming class M with $r_{\max} = \infty$.

Verification: `cy_n4sca_k3_engine.py`, `cy_bar_n4sca_engine.py`.

The K3 elliptic genus and Mathieu moonshine

Definition 14.53 *(K3 elliptic genus).* The elliptic genus of $K3$ is

$$Z_{K3}(\tau, z) \;=\; 2\,\phi_{0,1}(\tau, z), \tag{14.32}$$

where $\phi_{0,1}$ is the unique weak Jacobi form of weight 0, index 1 in the Eichler–Zagier normalization $\phi_{0,1}(\tau, 0) = 12$. At $z = 0$: $Z_{K3}(\tau, 0) = 24 = \chi(K3)$. The Fourier expansion at q^0 gives $Z_{K3}|_{q^0} = 2y + 20 + 2y^{-1}$, the χ_y-genus of K3 with $h^{1,1} = 20$, $h^{2,0} = h^{0,2} = 1$.

Theorem 14.54 *(Mathieu moonshine; [PE]).* The $\mathcal{N} = 4$ decomposition of the K3 elliptic genus,

$$Z_{K3} \;=\; 20 \cdot \mathrm{ch}_{\mathrm{massless}} \;+\; \sum_{n\geq 1} A_n \cdot \mathrm{ch}_{\mathrm{massive},n}, \tag{14.33}$$

where $\mathrm{ch}_{\mathrm{massless}}$ and $\mathrm{ch}_{\mathrm{massive},n}$ are $\mathcal{N} = 4$ characters at $c = 6$, has coefficients A_n that are dimensions of representations of the Mathieu group M_{24} (Eguchi–Ooguri–Tachikawa 2010, proved by Gannon 2016):

n	1	2	3	4	5	6
A_n	90	462	1540	4554	11592	27830

The 20 massless states decompose as $23_{\mathrm{std}} - 3_{\mathrm{triv}}$ under M_{24}.

Remark 14.55 (Mock modular form from K3). The massive multiplicities assemble into the mock modular form

$$h(\tau) \;=\; 2q^{-1/8}\big(-1 + 45q + 231q^2 + 770q^3 + 2277q^4 + \cdots\big), \tag{14.34}$$

where $A_n^{\mathrm{full}} = 2 \cdot a_n$ with $a_n = 45, 231, 770, \ldots$ being the half-multiplicities. The shadow of h in the sense of Zwegers is $S(\tau) = 24\,\eta(\tau)^3$, a weight-3/2 cusp form. The constant term $a_0 = -1$ gives $A_0^{\mathrm{full}} = -2 = -\kappa_{\mathrm{ch}}(\mathcal{A}_{K3})$: the modular characteristic appears as the leading mock modular coefficient.

The "shadow" in mock modularity (Zwegers) and the "shadow" in the shadow obstruction tower (bar complex) are a priori different mathematical objects. Their structural parallel (both encode the failure of modular invariance) is a connection to be explored, not an identification to be assumed.

The mock modular form $h(\tau)$ admits a canonical decomposition via the Appell–Lerch sum. Let $\mu(\tau,z) = \frac{-iy^{1/2}}{\vartheta_1(\tau,z)} \sum_{n\in\mathbb{Z}} \frac{(-1)^n q^{n(n+1)/2} y^n}{1-q^n y}$ denote the Zwegers μ-function. Then

$$Z_{K3}(\tau,z) \;=\; \big(h(\tau) - 24\,\mu(\tau,z)\big)\cdot \frac{\vartheta_1(\tau,z)^2}{\eta(\tau)^3}, \tag{14.35}$$

where the $24 = \chi(K3)$ multiplying μ is the Witten index and the theta-to-eta factor ϑ_1^2/η^3 carries weight $1/2$ and index 1. The decomposition is verified to machine precision (relative error $\sim 5.7\times 10^{-16}$). The polar term $h|_{q^{-1/8}} = -2 = -\kappa_{\mathrm{ch}}(\mathcal{A}_{K3})$ reappears on the right-hand side as a consequence of the Appell–Lerch cancellation.

Remark 14.56 (The factor 2 in $Z_{K3} = 2\,\phi_{0,1}$ is κ_{ch}). The elliptic genus $Z_{K3} = 2\,\phi_{0,1}$ has an overall factor of 2. This factor IS the modular characteristic: $\kappa_{\mathrm{ch}}(\mathcal{A}_{K3}) = 2$. The bar complex provides the universal coefficient

$$A_n \;=\; \kappa_{\mathrm{ch}}(\mathcal{A}_{K3})\cdot \dim(\rho_n), \tag{14.36}$$

where ρ_n is the M_{24}-representation at level n: $\dim(\rho_1) = 45$, $\dim(\rho_2) = 231$, $\dim(\rho_3) = 770$, $\dim(\rho_4) = 2277$. The half-multiplicities $a_n = 45, 231, 770, \ldots$ are the M_{24} irreducible dimensions; the full multiplicities $A_n = 2\,a_n = 90, 462, 1540, \ldots$ acquire the factor $\kappa_{\mathrm{ch}} = 2$ from the bar complex.

The bar complex does NOT detect M_{24} directly: it detects $\kappa_{\mathrm{ch}} = 2$ at arity 2 and the cubic/quartic shadow invariants at higher arities. The M_{24} representation structure is an *internal* symmetry of the massive BPS sector, invisible to the single-copy bar construction. The bar complex provides the universal multiplier; the group theory is orthogonal. The polar coefficient $-2 = -\kappa_{\mathrm{ch}}$ in the mock modular form (14.34) is the same κ_{ch} from a different direction.

Remark 14.57 (Twining genera and M_{24} conjugacy classes). For each conjugacy class $[g]$ of M_{24}, the twining genus $Z_g(\tau,z) = \mathrm{Tr}_{\mathrm{RR}}\big(g\cdot q^{L_0-c/24}\,\bar{y}^{J_0}\,(-1)^{F+\bar F}\big)$ is a weak Jacobi form of weight 0, index 1 for $\Gamma_0(N_g)$, where $N_g = \mathrm{ord}(g)$. Of the 26 conjugacy classes of M_{24}, only 21 appear as symmetries of K3 sigma models (Gaberdiel–Hohenegger–Volpato 2010/2011): the five missing classes ($7A$, $7B$, $15A$, $15B$, $23A/B$) have Frame shapes incompatible with the K3 lattice. Each realized class g has a Frame shape $\pi_g = \prod i^{a_i}$ encoding the permutation action on 24 letters, with associated eta product $\eta_g(\tau) = \prod_i \eta(i\tau)^{a_i}$. The 21 realized classes with their Frame shapes and orders:

Class	Frame shape	N_g	Class	Frame shape	N_g	Class	Frame shape	N_g
$1A$	1^{24}	1	$2A$	$1^8 2^8$	2	$2B$	2^{12}	2
$3A$	$1^6 3^6$	3	$4A$	$1^4 2^2 4^4$	4	$4B$	$2^4 4^4$	4
$4C$	4^6	4	$5A$	$1^4 5^4$	5	$6A$	$1^2 2^1 3^2 6^2$	6
$6B$	$1^2 2^2 3^1 6^1$	6	$8A$	$1^2 2^1 4^1 8^2$	8	$10A$	$1^2 2^1 5^1 10^1$	10
$11A$	$1^2 11^2$	11	$12A$	$1^1 2^1 4^1 3^1 6^1 12^1$	12	$12B$	$2^1 4^1 6^1 12^1$	12
$14A$	$1^1 2^1 7^1 14^1$	14	$14B$	$1^1 2^1 7^1 14^1$	14	$21A$	$1^1 3^1 7^1 21^1$	21
$21B$	$1^1 3^1 7^1 21^1$	21	$23A$	$1^1 23^1$	23	$23B$	$1^1 23^1$	23

The $23A$ and $23B$ classes are listed for completeness but are among the five NOT realized as K3 symmetries. The Frame shape determines the eta product η_g and hence the shadow of the twined mock modular form; the shadow of h_g is proportional to η_g.

Lattice VOA and Gepner model

Remark 14.58 (Lattice VOA from $H^2(K3,\mathbb{Z})$). The K3 cohomology lattice $L_{K3} = H^2(K3,\mathbb{Z}) \cong U^3 \oplus (-E_8)^2$ has rank 22, signature $(3,19)$, and discriminant $\det(G) = -1$. Since L_{K3} is indefinite, the naive theta function diverges. The negative-definite sublattice $(-E_8)^2$ of rank 16 produces a well-defined lattice VOA $V_{(-E_8)^2}$ with $c = 16$ and $\kappa_{\mathrm{ch}} = 16$ ($\kappa_{\mathrm{ch}} = \mathrm{rank}$ for lattice VOAs). The theta series $\Theta_{E_8}(\tau) = E_4(\tau) = 1 + 240q + \cdots$ (the weight-4 Eisenstein series, since E_8 is the unique even unimodular lattice of rank 8).

The full Mukai lattice $\widetilde{H}(K3,\mathbb{Z}) = U^4 \oplus (-E_8)^2$ has rank 24, signature $(4,20)$, and is the lattice relevant for Bridgeland stability conditions and derived categories.

Remark 14.59 (Gepner model $(2)^4$). The Gepner model for the quartic K3 (Fermat quartic $x_1^4 + x_2^4 + x_3^4 + x_4^4 = 0$ in $\mathbb{CP}^3$) is the tensor product of four copies of the $\mathcal{N} = 2$ minimal model at level $k = 2$ ($c = 3k/(k+2) = 3/2$), with GSO projection and $\mathbb{Z}_4$ orbifold. Total $c = 4 \times 3/2 = 6$. The chiral ring is isomorphic to the Jacobian ring $\mathrm{Jac}(W) = \mathbb{C}[x_1,\dots,x_4]/(\partial W/\partial x_i)$ with $\dim \mathrm{Jac}(W) = 3^4 = 81$ before orbifold. The symmetry group at the Gepner point contains $(\mathbb{Z}/4\mathbb{Z})^4 \rtimes S_4$ (order 6144) and embeds into the Conway group Co_1.

Chiral de Rham complex on K3

Remark 14.60 (CDR on K3). The chiral de Rham complex $\Omega^{\mathrm{ch}}(K3)$ of Malikov–Schechtman–Vaintrob is a sheaf of vertex superalgebras on K3 with central charge $c = 0$ (the local contributions $c_{\beta\gamma} = -2$ and $c_{bc} = +2$ per complex dimension cancel globally). The CDR cohomology recovers the K3 elliptic genus: $\mathrm{ch}\big(H^*(K3,\Omega^{\mathrm{ch}})\big) = \mathrm{Ell}(K3;q,y) = 2\,\phi_{0,1}(\tau,z)$ (Borisov–Libgober).

On a hyperkähler manifold (such as K3), the CDR carries $\mathcal{N} = 4$ superconformal symmetry at $c = 0$: the topological $\mathcal{N} = 4$ algebra, distinct from the physical $\mathcal{N} = 4$ at $c = 6$. Both give $\kappa_{\mathrm{ch}} = \dim_{\mathbb{C}}(K3) = 2$.

Koszul dual of the K3 chiral algebra

Proposition 14.61 *(Koszul dual of $\mathcal{A}_{K3}$; [PH]).* The Koszul dual of the $\mathcal{N} = 4$ SCA at $c = 6$, $k_R = 1$ has:

- $\kappa_{\mathrm{ch}}(\mathcal{A}^!_{K3}) = -2$ (by complementarity: $\kappa_{\mathrm{ch}} + \kappa'_{\mathrm{ch}} = 0$ for free-field/CY sigma models).
- $c(\mathcal{A}^!_{K3}) = -6$ (the $\mathcal{N} = 4$ SCA at negative central charge, with $k'_R = -1$, since $\kappa_{\mathrm{ch}} = 2k_R$ and $\kappa'_{\mathrm{ch}} = -2$ gives $k'_R = -1$, hence $c' = 6k'_R = -6$).
- $F_g(\mathcal{A}^!) = -2 \cdot \lambda_g^{\mathrm{FP}}$ at all genera (ALL-WEIGHT, with cross-channel correction $\delta F_g^{\mathrm{cross}}$ at $g \geq 2$; Vol I, AP32).

The Verdier intertwining (Vol I, Theorem A; Convention III): $D_{\mathrm{Ran}}(B(\mathcal{A}_{K3})) \simeq B(\mathcal{A}^!_{K3})$. The derived centre $Z^{\mathrm{der}}_{\mathrm{ch}}(\mathcal{A}_{K3})$ (the universal bulk) is a separate object from $\mathcal{A}^!$.

The abelian $(2,0)$ pushforward

For $X = S \times E$ with S a K3 surface and E an elliptic curve, the derived pushforward $R\pi_*(\Omega^2_{X,\mathrm{cl}})$ along $\pi\colon S \times E \to E$ produces a chiral algebra on E.

Construction 14.62 *(The pushforward chiral algebra).* The *abelian* $(2,0)$ *pushforward chiral algebra* on E is

$$\mathcal{A}_{\mathrm{free}} \;=\; \pi_*\big(\mathrm{Obs}^q(\mathcal{T}_{(2,0)}|_{S\times E})\big),$$

the pushforward of the quantum observables of the holomorphically twisted abelian $(2,0)$ theory along $\pi\colon S \times E \to E$. By the Costello–Gwilliam theorem (proper pushforward preserves factorization), this is a factorization algebra on $\mathrm{Ran}(E)$, hence a chiral algebra on E.

The fiber cohomology $R\pi_*(\Omega^2_X)$ decomposes via Künneth into contributions from $H^*(S)$:

Hodge group of S	**Sheaf on E**	**Dimension**
$H^0(S, \Omega^2_S) = \mathbb{C}\langle\sigma\rangle$	$\mathcal{O}_E$	1
$H^2(S, \Omega^2_S) = \mathbb{C}$ (Serre dual)	$\mathcal{O}_E$	1
$H^1(S, \Omega^1_S) = \mathbb{C}^{20}$	Ω^1_E	20
$H^0(S, \mathcal{O}_S) = \mathbb{C}$	$\Omega^2_E \simeq \mathcal{O}_E$	1
$H^2(S, \mathcal{O}_S) = \mathbb{C}$	$\Omega^2_E \simeq \mathcal{O}_E$	1
Total before closure constraint		24

The closure condition modifies this count: a section $\sigma \otimes f(z)$ of the $H^{2,0}(S) \otimes \mathcal{O}_E$ summand is closed only when $d_E(f) = 0$, i.e. f is constant. The naive total of 24 is therefore an upper bound for the rank of the pushforward of closed 2-forms. The $H^{1,1}(S)$ sector (contributing 20 weight-1 currents) survives the closure condition, as its members are closed along S and the Ω^1_E factor prevents a d_E-constraint.

Proposition 14.63 *(OPE structure of the pushforward; [PH]).* The chiral algebra $\mathcal{A}_{\mathrm{free}}$ is of Heisenberg/lattice type. Its OPE is determined by the propagator of the 6-dimensional theory pushed forward along S:

(i) The 20 currents $J_a(z)$ from $H^{1,1}(S) \otimes \Omega^1_E$ have double-pole OPE:

$$J_a(z)\, J_b(w) \;\sim\; \frac{\eta_{ab}}{(z-w)^2},$$

where $\eta_{ab} = \int_S \omega_a \wedge \omega_b$ is the intersection form restricted to $H^{1,1}(S)$, of signature $(1, 19)$.

(ii) The remaining generators (from $H^{2,0}, H^{0,2}, H^0(\mathcal{O}_S), H^2(\mathcal{O}_S)$) have at most double-pole OPE.

(iii) No poles of order ≥ 3 appear: the 6-dimensional theory is free with at most quadratic action.

In particular, $\mathcal{A}_{\mathrm{free}}$ is a lattice vertex algebra V_Λ with Λ determined by the fiber cohomology surviving the closure constraint. By the Vol I lattice curvature theorem, $\kappa_{\mathrm{ch}}(V_\Lambda) = \mathrm{rank}(\Lambda)$ for any lattice VA, shadow class **G** (Gaussian, $r_{\max} = 2$), and the shadow obstruction tower terminates: $\Theta_{\mathcal{A}_{\mathrm{free}}} = \kappa_{\mathrm{ch}} \cdot \omega_g$ at all genera (UNIFORM-WEIGHT, class G).

Remark 14.64 (The κ_{ch}-collapse: rank $\to \dim_{\mathbb{C}}$). The passage from the abelian $(2,0)$ pushforward to the K3 sigma model exhibits a non-perturbative collapse of the modular characteristic:

$$\kappa_{\mathrm{ch}}(\mathcal{A}_{\mathrm{free}}) \;=\; \mathrm{rank}(\Lambda) \;\in\; \{22, 24\}, \qquad \kappa_{\mathrm{ch}}(\mathcal{A}_\sigma) \;=\; 2 \;=\; \dim_{\mathbb{C}}(K3).$$

This is not a small correction. No deformation of $\mathcal{A}_{\mathrm{free}}$ within the class of lattice vertex algebras can produce $\kappa_{\mathrm{ch}} = 2$: the Vol I lattice curvature theorem gives $\kappa_{\mathrm{ch}}(V_\Lambda) = \mathrm{rank}(\Lambda)$ for all lattice VAs independent of deformation parameters. The $\mathcal{N} = 4$ supersymmetry constrains the genus-1 curvature to equal the complex dimension of the target, regardless of the free-field rank (AP48).

The three levels of structure are:

Object	**Symbol**	κ_{ch}	**Class**	**Depth**
Abelian $(2,0)$ pushforward	$\mathcal{A}_{\mathrm{free}}$	$\mathrm{rank}(\Lambda) \in \{22, 24\}$	**G**	2
K3 sigma model VOA	$\mathcal{A}_\sigma$	2	**M**	∞
BKM global	$G(K3 \times E)$	5	**M**	∞

The chain $\mathrm{rank}(\Lambda) \to 2 \to 5$ reflects: (i) the $\mathcal{N} = 4$ SCA constraining κ_{ch} from rank to $\dim_{\mathbb{C}}$ (introducing $S_3, S_4 \neq 0$ and upgrading the shadow class from $\mathbf{G}$ to $\mathbf{M}$); (ii) the passage from the single-copy chiral algebra ($\kappa_{\mathrm{ch}} = 3 = \kappa_{\mathrm{ch}}(K3) + \kappa_{\mathrm{ch}}(E) = 2 + 1$, by additivity) to the second-quantized BKM modular characteristic ($\kappa_{\mathrm{BKM}} = 5$).

14.10 The Calabi–Yau threefold $K3 \times E$: geometry and invariants

Hodge diamond and topological invariants

Proposition 14.65 *(Hodge diamond of $K3 \times E$; [PE])*. The Künneth formula gives the Hodge diamond of $K3 \times E$ ($\dim_{\mathbb{C}} = 3$):

$$\begin{array}{ccccccc} & & & 1 & & & \\ & & 1 & & 1 & & \\ & 1 & & 21 & & 1 & \\ 1 & & 21 & & 21 & & 1 \\ & 1 & & 21 & & 1 & \\ & & 1 & & 1 & & \\ & & & 1 & & & \end{array}$$

Key invariants:

- $h^{1,1} = h^{2,1} = 21$, giving $\chi(K3 \times E) = 2(h^{1,1} - h^{2,1}) = 0$.
- Betti numbers: $b_0 = b_6 = 1$, $b_1 = b_5 = 2$, $b_2 = b_4 = 23$, $b_3 = 44$. Total: $\sum b_k = 96$.
- $K_0(K3 \times E) \cong \mathbb{Z}^{48}$ ($K_0(K3) \otimes K_0(E) = \mathbb{Z}^{24} \otimes \mathbb{Z}^2$).

Proof. Apply Künneth: $h^{p,q}(K3 \times E) = \sum_{a+c=p,\ b+d=q} h^{a,b}(K3) \cdot h^{c,d}(E)$. For example: $h^{1,1} = h^{1,1}(K3) \cdot h^{0,0}(E) + h^{0,0}(K3) \cdot h^{1,1}(E) = 20 + 1 = 21$. The alternating sum $1 - 2 + 23 - 44 + 23 - 2 + 1 = 0$ verifies $\chi = 0$. □

HKR decomposition and deformation space

Proposition 14.66 *(Deformation space of $D^b(K3 \times E)$; [PE])*. By the HKR isomorphism for a smooth CY 3-fold: $\mathrm{HH}^n(X) = \bigoplus_{p+q=n} h^{3-p,q}(X)$. For $K3 \times E$:

$\mathrm{HH}^0 = 1$	automorphisms	$h^{3,0}$
$\mathrm{HH}^1 = 2$	outer derivations	$h^{3,1} + h^{2,0}$
$\mathrm{HH}^2 = 23$	first-order deformations	$h^{3,2} + h^{2,1} + h^{1,0}$
$\mathrm{HH}^3 = 44$	obstructions	$h^{3,3} + h^{2,2} + h^{1,1} + h^{0,0}$

The 23-dimensional deformation space: 21 complex structure (from $h^{2,1}$), 1 B-field (from $h^{1,0}$), 1 volume (from $h^{3,2}$). Total $\dim \mathrm{HH}^* = 96$.

Noncommutative deformations and Brauer group

Remark 14.67 (Brauer group of $K3 \times E$). $\mathrm{Br}(K3 \times E) = \mathrm{Br}(K3) \oplus \mathrm{Br}(E)$ contains $(\mathbb{Q}/\mathbb{Z})^{21}$ (K3 at Picard rank $\rho = 1$) plus $\mathbb{Q}/\mathbb{Z}$ from E. The K3 symplectic form provides the unique Poisson structure $\pi = \omega^{-1}$ (since $\dim H^0(\bigwedge^2 T_{K3}) = 1$), but the E-direction carries none ($\bigwedge^2 T_E = 0$).

Autoequivalences and semiorthogonal structure

Proposition 14.68 *(Indecomposability of $D^b(K3 \times E)$; [PE]).* $D^b(K3)$ admits no proper semiorthogonal decomposition (Bridgeland, Kawamata, Huybrechts: $\omega_{K3} = \mathcal{O}$ forces Serre functor $[2]$, and the only fractional CY category with Serre $[2]$ is $D^b(K3)$ itself). Hence $D^b(K3 \times E) = D^b(K3) \otimes D^b(E)$ is indecomposable.

Autoequivalences: $\mathrm{Aut}(D^b(K3)) \to O^+(\widetilde{H}(K3, \mathbb{Z}))$ via oriented Hodge isometries (line bundle twists, shifts, spherical twists T_E with $T_E^2 = [-2]$). $\mathrm{Aut}(D^b(E)) = (E \times E^\vee) \rtimes (\mathrm{SL}(2, \mathbb{Z}) \rtimes \mathbb{Z})$. The virtual dimension of all moduli problems on $K3 \times E$ vanishes: $\mathrm{vdim} = -\chi(\mathcal{E}, \mathcal{E}) = 0$.

Factorization homology

Remark 14.69 (Factorization homology on $K3 \times E$). The factorization homology of the HT-twisted sigma model: $\int_{K3 \times E} A^{\mathrm{HT}} \simeq H^*(K3 \times E, \Omega^{\mathrm{ch}})$. The CDR character gives $\chi(\Omega^{\mathrm{ch}}) = 0$, but the Hodge-graded character is nontrivial. $\dim \mathrm{HH}^*(D^b(K3 \times E)) = 96$.

14.11 Enumerative geometry of $K3 \times E$

Gromov–Witten theory and BPS invariants

Remark 14.70 (Reduced GW theory on K3). For K3, the standard virtual class vanishes for $\beta \neq 0$ (cosection from $H^{2,0}(K3) = \mathbb{C}$). The reduced theory (Bryan–Leung, Maulik–Pandharipande–Thomas) removes this obstruction. The KKV formula (proved by Pandharipande–Thomas 2016) gives the BPS state counts. The genus-0 BPS invariants satisfy the Yau–Zaslow formula:

$$\sum_{h \geq 0} n_h^0 \, q^h = \prod_{n \geq 1} \frac{1}{(1 - q^n)^{24}}. \tag{14.37}$$

Proposition 14.71 *(BPS invariants of K3; [PE]).* Selected BPS invariants (all integers, proved by Pandharipande–Thomas 2016):

h	0	1	2	3	4	5	6	7
n_h^0	1	24	324	3200	25650	176256	1073720	5930496
n_h^1	0	−2	−54	−800	−8550	−73440	−536860	−3464264
n_h^2	0	0	3	104	1701	19656	176358	1316808
n_h^3	0	0	0	−4	−155	−2832	−34470	−320320

Signed convention: $n_h^g = (-1)^g |n_h^g|$. The genus-0 values are the Yau–Zaslow numbers (OEIS A006922); $n_1^0 = 24 = \chi(K3)$. The Hilbert scheme Euler characteristics $\chi(\mathrm{Hilb}^n(K3))$ provide the genus-0 column: $\chi(\mathrm{Hilb}^n) = n_n^0$, with generating function $\sum_{n \geq 0} \chi(\mathrm{Hilb}^n) \, q^n = \prod_{k \geq 1} (1 - q^k)^{-24}$ (Göttsche formula). The first eight values are $1, 24, 324, 3200, 25650, 176256, 1073720, 5930496$.

Bridgeland stability on K3

Remark 14.72 (Stability conditions). $\mathrm{Stab}^\dagger(K3) \to P^+(K3)$ is a covering of the period domain (Bridgeland 2008). Walls of marginal stability for Mukai vectors $v = v_1 + v_2$ are semicircles in the (B, ω)-plane; crossing them applies the KS wall-crossing. For primitive v with $v^2 = 2n - 2$: $\Omega(v) = \chi(\mathrm{Hilb}^n(K3))$.

14.12 BPS spectrum and black hole entropy

Charge lattice and BPS states

Proposition 14.73 *(Charge lattice; [PE]).* Type IIA on $K3 \times E$ gives 4d $\mathcal{N} = 4$ supergravity with 22 vector multiplets. The D-brane charge lattice $\Gamma = H^{\mathrm{even}}(K3 \times E, \mathbb{Z})$:

Cohomology	Brane type	Rank
$H^0 = \mathbb{Z}$	D6-branes	1
$H^2 = \mathbb{Z}^{23}$	D4-branes	23
$H^4 = \mathbb{Z}^{23}$	D2-branes	23
$H^6 = \mathbb{Z}$	D0-branes	1

Total: $\operatorname{rank} \Gamma = 48$.

Black hole entropy from $1/\Phi_{10}$

Proposition 14.74 *(BMPV and $\frac{1}{4}$-BPS entropy; [PE]).* In 5d (IIA on $K3 \times S^1$): the BMPV black hole with charges (Q_1, Q_5, n) has $S_{\mathrm{BH}} = 2\pi\sqrt{Q_1\, Q_5\, n}$ (Cardy with $c_L = 6\, Q_1\, Q_5$). In 4d (IIA on $K3 \times T^2$): the $\frac{1}{4}$-BPS dyon with discriminant Δ has

$$S_{\mathrm{BH}} = 4\pi\sqrt{\Delta},$$

and the DVV formula $Z_{1/4\text{-BPS}} = 1/\Phi_{10}$ ((14.48)) gives the exact microscopic count. For large Δ: $\log d(\Delta) \sim 4\pi\sqrt{\Delta} - \frac{3}{2}\log\Delta + \cdots$, where the subleading term is the universal one-loop correction (Sen 2005, 2012). The asymptotic expansion is the leading term of the Rademacher exact formula for the Fourier coefficients of $1/\Phi_{10}$:

$$d(\Delta) \;=\; \frac{2\pi}{\sqrt{\Delta}} \sum_{c=1}^{\infty} \frac{K(c,\Delta)}{c}\, I_{27/2}\!\Big(\frac{4\pi\sqrt{\Delta}}{c}\Big) \;+\; \text{polar contributions},$$

where $I_{27/2}$ is the modified Bessel function of the first kind of order $27/2$ and $K(c,\Delta)$ is a Kloosterman-type sum over $\mathrm{Sp}(4,\mathbb{Z})$. The $c = 1$ term dominates at large Δ, giving $d(\Delta) \sim e^{4\pi\sqrt{\Delta}} \cdot \Delta^{-29/4}$. The convergence of the Rademacher series is exponentially fast: the $c = 2$ correction is suppressed by $e^{-2\pi\sqrt{\Delta}}$ relative to the leading term.

Remark 14.75 (OSV conjecture). The Ooguri–Strominger–Vafa relation $Z_{\mathrm{BH}} = |Z_{\mathrm{top}}|^2$ connects the black hole partition function to the topological string. For $K3 \times E$, the topological string is exactly solvable at genus 0 and 1. The shadow tower gives $F_g = 3 \cdot \lambda_g^{\mathrm{FP}}$ at each genus (ALL-WEIGHT, with cross-channel correction $\delta F_g^{\mathrm{cross}}$).

Fourier–Jacobi as genus escalation

Proposition 14.76 *(Bridge: elliptic R-matrix data determines BKM root system; [PE]).* The Fourier–Jacobi expansion of the Igusa cusp form $\Phi_{10}(\Omega) = \sum_{m \geq 1} \phi_m(\tau, z)\, p^m$ uses Jacobi forms $\phi_m(\tau, z)$ on E_τ. The leading coefficient

$$\phi_1(\tau, z) = \eta(\tau)^{18}\, \vartheta_1(\tau, z)^2 \tag{14.38}$$

is built from the same theta function $\vartheta_1(\tau, z)$ that defines the elliptic propagator $d \log \vartheta_1(z|\tau)$ ((14.2)) and Felder's elliptic R-matrix entries $a(u) = \theta(u+\eta)/\theta(\eta)$ (Definition 14.15).

The passage from genus-1 data (Jacobi forms on E_τ) to genus-2 data (the Siegel form Φ_{10} on $\mathfrak{H}_2$) is the Borcherds multiplicative lift. Its input is the K3 elliptic genus $Z_{K3} = 2\,\phi_{0,1}$ (Definition 14.53), and the exponents $c_0(4nm - \ell^2)$ in the infinite product (14.47) are precisely the root multiplicities of the BKM superalgebra $\mathfrak{g}_{\Phi_{10}}$ (Proposition 14.82).

In the factorization $\eta^{18} = \eta^{24} \cdot \eta^{-6}$: $\eta^{24} = \Delta$ is the Ramanujan discriminant (the 24 free bosons of A_E, with $\kappa_{\mathrm{ch}}(A_E) = 24$), and $\eta^{-6} = \eta^{-2\kappa_{\mathrm{ch}}}$ with $\kappa_{\mathrm{ch}} = \kappa_{\mathrm{ch}}(\Omega^{\mathrm{ch}}(K3 \times E)) = 3$. The shadow obstruction tower detects $\kappa_{\mathrm{ch}} = 3$ (Proposition 14.130); the Borcherds lift escalates this genus-1 datum to the genus-2 Siegel modular form.

14.13 Twisted holography on $K3 \times E$

Construction 14.77 *(Costello–Li KS boundary algebra).* The Costello–Li programme extracts a boundary chiral algebra from the HT twist of Type IIB on $K3 \times E$. Reduction of Kodaira–Spencer theory along the topological directions (K3) produces a chiral algebra A_E on E with generators from harmonic forms on K3:

$h^{p,q}(K3)$	Contribution	Generators
$h^{0,0} = 1$	Scalar boson	1
$h^{2,0} = 1$	Symplectic boson	1
$h^{1,1} = 20$	Weight-1 bosons	20
$h^{0,2} = 1$	Conjugate boson	1
$h^{2,2} = 1$	Volume boson	1

Total: $c(A_E) = \chi(K3) = 24$. This is NOT the K3 sigma model ($c = 6$) but 24 free bosons from the B-model reduction. $\kappa_{\mathrm{ch}}(A_E) = 24$ (rank of the free-boson lattice), $\kappa_{\mathrm{ch}}(A_E^!) = -24$ ($\kappa_{\mathrm{ch}} + \kappa'_{\mathrm{ch}} = 0$ for free fields).

Warning 14.78 (OPE level matrix for KS boundary bosons). The 24 free bosons $J^a(z)$ ($a = 1, \ldots, 24$) of the KS boundary algebra A_E have OPE $J^a(z)\,J^b(w) \sim \delta^{ab}/(z-w)^2$, so the OPE level matrix is δ^{ab} (the unit matrix), *not* the Mukai pairing G^{ab} of signature $(4, 20)$ on $H^*(K3, \mathbb{Z})$. The Mukai pairing enters through the vertex operators $V_\gamma(z) = {:}\exp(i\gamma_a J^a(z)){:}$ for lattice vectors $\gamma \in \Gamma_{\mathrm{Mukai}}$, whose correlation functions involve $\langle V_\gamma V_{\gamma'} \rangle \propto \exp(\gamma \cdot G \cdot \gamma')$. Confusing the OPE level δ^{ab} with the lattice pairing G^{ab} produces wrong κ_{ch} values: the OPE level gives $\kappa_{\mathrm{ch}}(A_E) = 24$ (sum of diagonal entries), while the Mukai norm $\mathrm{tr}(G) = 4 - 20 = -16$ would give a wrong and negative κ_{ch}.

Remark 14.79 (Holographic modular Koszul datum). $H(K3 \times E) = (A, A^!, C, r(z), \Theta_A, \nabla^{\mathrm{hol}})$ with $A = A_E$ ($c = 24$, $\kappa_{\mathrm{ch}} = 24$), $A^! = A_E^!$ ($\kappa_{\mathrm{ch}} = -24$), $C = Z^{\mathrm{der}}_{\mathrm{ch}}(A)$ (universal bulk), $r(z) = \kappa_{\mathrm{ch}}\,\Omega/z$ (Casimir, 24-dim; AP126: level prefix $\kappa_{\mathrm{ch}} = 24$), Θ_A (bar-intrinsic MC element), ∇^{hol} (shadow connection, class G).

Proposition 14.80 *(Boundary-to-sigma ratio; [PH]).* The boundary algebra A_E and the sigma model VOA V_{K3} have modular characteristics in ratio 12. At genus 1:

$$\frac{\kappa_{\mathrm{ch}}(A_E)}{\kappa_{\mathrm{ch}}(V_{K3})} = \frac{24}{2} = 12, \qquad \frac{F_1(A_E)}{F_1(V_{K3})} = 12. \tag{14.39}$$

Here A_E has $c = 24$ from the 24 free bosons of the Kaluza–Klein reduction on K3 (Construction 14.77) indexed by the Mukai lattice $H^*(K3, \mathbb{Z})$, and V_{K3} is the $\mathcal{N} = 4$ SCA at $c = 6$ with $\kappa_{\mathrm{ch}}(V_{K3}) = 2$ (Proposition 14.48). The SUSY Ward identities of the $\mathcal{N} = 4$ algebra reduce κ_{ch} by a factor of $2/3$ relative to the Virasoro formula $\kappa_{\mathrm{ch}}(\mathrm{Vir}_6) = 3$ (Remark 14.49).

For A_E (24 free bosons, class G, uniform weight), the scalar formula $F_g(A_E) = 24 \cdot \lambda_g^{\mathrm{FP}}$ holds at all genera (UNIFORM-WEIGHT; Vol I, Theorem D). For V_{K3} (class M, generators at weights $1, 3/2, 2$), the scalar formula $F_g = \kappa_{\mathrm{ch}} \cdot \lambda_g^{\mathrm{FP}}$ is proved at genus 1; at $g \geq 2$ the multi-weight genus expansion receives cross-channel corrections $\delta F_g^{\mathrm{cross}}$ (ALL-WEIGHT; Vol I, AP32). The ratio 12 is therefore proved at $g = 1$ and conditional at $g \geq 2$.

Proof. $\kappa_{\mathrm{ch}}(A_E) = 24$ (rank of the free-boson lattice). $\kappa_{\mathrm{ch}}(V_{K3}) = 2$ (Proposition 14.48). Their ratio is $24/2 = 12$. At genus 1, $F_1(\mathcal{A}) = \kappa_{\mathrm{ch}}(\mathcal{A})/24$ for both algebras, giving the ratio 12. At $g \geq 2$: A_E is uniform-weight (class G), so $F_g(A_E) = 24 \cdot \lambda_g^{\mathrm{FP}}$ at all genera (UNIFORM-WEIGHT). V_{K3} has generators at weights $1, 3/2, 2$; the scalar formula $F_g = 2 \cdot \lambda_g^{\mathrm{FP}}$ is proved at $g = 1$ but receives multi-weight cross-channel corrections $\delta F_g^{\mathrm{cross}}$ at $g \geq 2$ (ALL-WEIGHT; Vol I, AP32). □

Remark 14.81 (Leech connection). The boundary partition function $Z_1(A_E; \tau) = 1/\eta(\tau)^{48}$ at genus 1 matches the Leech lattice VOA V_Λ through order q^1: $1/\eta^{48} = q^{-2}(1 + 48\,q + \cdots)$, while V_Λ has $\dim(V_\Lambda)_1 = 0$ and $\dim(V_\Lambda)_2 = 196560$. The two diverge at order q^2: the 196560 vectors of the Leech lattice have no counterpart in the 24-boson Fock space at weight 2. The Leech lattice has $\kappa_{\mathrm{ch}}(V_\Lambda) = 24 = \kappa_{\mathrm{ch}}(A_E)$ (both are 24 free bosons at level 1), but their weight-2 spaces differ.

14.14 Mock modular forms and the BKM root system

BKM root multiplicities

Proposition 14.82 *(Root multiplicities; [PE])*. The BKM superalgebra $\mathfrak{g}_{\Phi_{10}}$ has root multiplicities from the K3 elliptic genus:

- Real roots ($D = -1, \alpha^2 = 2$): mult $= c_0(-1) = 2$.
- Lightlike roots ($D = 0, \alpha^2 = 0$): mult $= c_0(0) = 10$ (20 from $2\phi_{0,1}$).
- Imaginary roots ($D > 0$): mult $= c_0(D)$, unbounded. Leading: $c_0(3) = -64$, $c_0(4) = 108$, $c_0(7) = -513$, $c_0(8) = 808$.

Negative multiplicities at odd D indicate fermionic directions in the superalgebra. The discriminant coefficients $c_0(D)$ of $\phi_{0,1}$ through $D = 12$ are (Eichler–Zagier convention):

D	-1	0	3	4	7	8	11	12	15	16	19	20
$c_0(D)$	2	10	-64	108	-513	808	-2752	4016	-11775	16524	-43263	58956

The $K3$ elliptic genus uses $2\,\phi_{0,1}$, so the BKM root multiplicities are $\mathrm{mult}(\alpha) = 2\,c_0(D)$ for discriminant D. The theta decomposition $\phi_{0,1} = h_0(\tau)\,\vartheta_{0,1}(\tau, z) + h_1(\tau)\,\vartheta_{1,1}(\tau, z)$ separates even (ℓ even, $D \equiv 0 \pmod 4$) and odd (ℓ odd, $D \equiv 3 \pmod 4$) sectors, with $h_0(\tau) = 10 + 108\,q + 808\,q^2 + \cdots$ and $h_1(\tau) = q^{-1/4}(1 - 64\,q + \cdots)$.

Proposition 14.83 *(Root multiplicity constancy; [PH])*. Define the *level* of a positive root $\alpha = (n, l, m)$ as $\ell(\alpha) = n + m$. For all levels $\ell \geq 1$:

$$\sum_{\substack{\alpha \in \Delta_+ \\ \ell(\alpha) = \ell}} \mathrm{mult}(\alpha) = 24 = \chi(K3).$$

Proof. At each level ℓ, the positive roots decompose into boundary roots (those with $n = 0$ or $m = 0$) and interior roots (those with $n, m \geq 1$).

Boundary contribution. Roots with $m = 0$: $(n, l, 0)$ for $n = \ell$, varying l. The multiplicity is $c_0(-l^2)$, which is nonzero only for $l = 0$ (giving $c_0(0) = 10$) and $l = \pm 1$ (giving $c_0(-1) = 1$ each). Total: $10 + 1 + 1 = 12$. Similarly for $n = 0$: total 12. Boundary total: 24.

Interior cancellation. For each pair (n, m) with $n + m = \ell$ and $n, m \geq 1$:

$$\sum_{l \in \mathbb{Z}} c_0(4nm - l^2) = 0.$$

This is the modular constancy of $\phi_{0,1}$ at $z = 0$. The weak Jacobi form $\phi_{0,1}(\tau, z)$ has weight 0 and index 1; evaluating at $z = 0$ gives a modular form of weight 0 on $\mathrm{SL}_2(\mathbb{Z})$, which is necessarily constant (the space $M_0(\mathrm{SL}_2(\mathbb{Z}))$ is one-dimensional, spanned by 1). The value is $\phi_{0,1}(\tau, 0) = 12$ (Eichler–Zagier, *The Theory of Jacobi Forms*, 1985, p. 108). At the level of Fourier coefficients: $\sum_l c_0(4N - l^2) = 0$ for all $N \geq 1$, since the q^N coefficient of the constant function 12 vanishes for $N \geq 1$. The boundary gives 24, the interior gives 0, so the total at every level is 24. □

Remark 14.84 (The interior cancellation at level 2). The interior at level 2 consists of roots $(1, l, 1)$ with $\mathrm{mult} = c_0(4 - l^2)$. From the Eichler–Zagier table: $l = 0$: $D = 4$, $c_0(4) = 108$; $l = \pm 1$: $D = 3$, $c_0(3) = -64$ each; $l = \pm 2$: $D = 0$, $c_0(0) = 10$ each. Sum: $108 - 2 \cdot 64 + 2 \cdot 10 = 108 - 128 + 20 = 0$. The cancellation is a direct consequence of $\sum_l c_0(4 - l^2) = 0$ (the q^1 coefficient of $\phi_{0,1}(\tau, 0) = 12$).

Remark 14.85 (Root multiplicity constancy and the shadow tower). The constancy $\sum_{\ell(\alpha)=\ell} \mathrm{mult}(\alpha) = 24 = \chi(K3)$ reflects the topological datum $\chi(K3)$ persisting through the BKM construction. This is precisely the datum that the abelian $(2,0)$ pushforward (Construction 14.62) captures: the Euler characteristic of the fiber enters through the lattice rank and the partition function of the free-field theory. The pushforward gives the correct topological backbone ($\chi = 24$) even though it gives the wrong shadow class (class **G** instead of class **M**) and the wrong κ_{ch} ($\mathrm{rank}(\Lambda)$ instead of $\dim_{\mathbb{C}}(K3) = 2$).

The Hilbert scheme bridge

Theorem 14.86 *(Toroidal action on K-theory of Hilbert schemes [34]; [PE]).* Let S be a smooth projective surface. The toroidal algebra $U_{q,t}(\hat{\hat{\mathfrak{gl}}}_1)$ acts on $\bigoplus_{n \geq 0} K(\mathrm{Hilb}^n(S))$ by correspondences on the nested Hilbert scheme $\mathrm{Hilb}^{n,n+1}(S)$ (Schiffmann–Vasserot). Specializations:

(i) *Nakajima operators* ($q = 1$): the Heisenberg algebra $\bigoplus_{n \in \mathbb{Z}} \mathbb{Q}\, \mathfrak{a}_n$ acts on $\bigoplus_n H^*(\mathrm{Hilb}^n(S))$ via Nakajima correspondences, with $\mathfrak{a}_{-n}|0\rangle$ the class of the punctual Hilbert scheme.

(ii) *Haiman's theorem* ($S = \mathbb{C}^2$): the DAHA $\ddot{H}_n(q, t)$, a quotient of $U_{q,t}(\hat{\hat{\mathfrak{gl}}}_1)$, acts on $K(\mathrm{Hilb}^n(\mathbb{C}^2))$ with fixed-point basis indexed by partitions $\lambda \vdash n$, and the character of the natural representation is the Macdonald polynomial $\widetilde{H}_\lambda(x; q, t)$.

(iii) $S = K3$: $\chi(\mathrm{Hilb}^n(K3)) = p_{24}(n)$ (24-coloured partitions), so $\sum_{n \geq 0} \chi(\mathrm{Hilb}^n(K3))\, p^n = \prod_{m \geq 1} (1 - p^m)^{-24}$.

Proposition 14.87 *(DMVV from Hilbert scheme characters; [PE]).* The DVV formula (14.48) $1/\Phi_{10} = \sum_N p^N \chi(\mathrm{Sym}^N(K3); \tau, z)$ is the generating function of equivariant Euler characteristics of $\mathrm{Hilb}^N(K3)$ refined by the $\mathrm{SU}(2)_R$ fugacity z and the L_0 fugacity $q = e^{2\pi i \tau}$. The 24 coloured partitions arise from Nakajima's Heisenberg action ($24 = \chi(K3) = b_0 + b_2 + b_4 = 1 + 22 + 1$): each colour corresponds to a cohomology class of K3, matching the 24 free bosons of the boundary algebra A_E (Construction 14.77).

Remark 14.88 (Synthesis: toroidal algebras meet BPS counting). The three strands of this chapter converge on the Hilbert scheme. The toroidal algebra $U_{q,t}(\hat{\hat{\mathfrak{g}}})$ of §14.1 provides the symmetry algebra acting on $K(\mathrm{Hilb}^n(S))$. The elliptic R-matrix of §14.2 appears as the transition matrix between stable-envelope bases of $K(\mathrm{Hilb}^n(S))$ (Aganagic–Okounkov). The BKM root multiplicities $c_0(D)$ of Proposition 14.82 are the Fourier coefficients of the K3 elliptic genus $Z_{K3} = 2\,\phi_{0,1}$, which in turn counts representations of the Heisenberg subalgebra on $\bigoplus_n H^*(\mathrm{Hilb}^n(K3))$. The shadow invariant $\kappa_{\mathrm{ch}} = 3$ measures the single-copy chiral algebra $\Omega^{\mathrm{ch}}(K3 \times E)$; the BKM algebra $\mathfrak{g}_{\Phi_{10}}$ encodes the second-quantized (symmetric-product) extension, and the gap between them is the content of the shadow–Siegel gap analysis (§14.16).

Fourier–Jacobi expansion

Remark 14.89 (Fourier–Jacobi structure). $\Phi_{10} = \sum_{m\geq 1} \phi_m(\tau, z)\, p^m$ with: $\phi_1 = \phi_{10,1} = \eta^{18}\,\vartheta_1^2 = -\Delta \cdot \phi_{-2,1}$ (unique Jacobi cusp form of weight 10, index 1); $\phi_2 = 2\,\Delta\,\phi_{-2,1}\,\phi_{0,1}$ ($\dim J^{\mathrm{cusp}}_{10,2} = 1$). The factorization $\eta^{18} = \eta^{24} \cdot \eta^{-6}$ separates the Ramanujan discriminant from $\eta^{-2\kappa_{\mathrm{ch}}}$ with $\kappa_{\mathrm{ch}} = 3$. The Jacobi form ring $J^{\mathrm{weak}}_{*,*} = \mathbb{C}[E_4, E_6, \phi_{-2,1}, \phi_{0,1}]$ (Eichler–Zagier).

14.15 Grand synthesis and open problems

Remark 14.90 (Master data table for $K3 \times E$). The complete invariant data assembled from the 39 CY compute engines:

Invariant	**Value**
$\dim_{\mathbb{C}}(K3 \times E)$	3
$\chi(K3 \times E)$	0
$h^{1,1} = h^{2,1}$	21
$\operatorname{rank} K_0$	48
$\dim \mathrm{HH}^*$	96
$\kappa_{\mathrm{ch}} = \kappa_{\mathrm{ch}}(\Omega^{\mathrm{ch}})$	3
$\kappa_{\mathrm{BKM}} = \mathrm{wt}(\Phi_{10})/2$	5
$\kappa_{\mathrm{BCOV}} = \chi/24$	0
Shadow depth	class M ($r_{\max} = \infty$)
F_1	$1/8$
F_2	$7/1920$
F_3	$31/322560$
$Z_{\mathrm{DT},0}$	1
n_1^0 (genus-0 BPS, $h = 1$)	24
$S_{\mathrm{BH}}(\Delta)$	$4\pi\sqrt{\Delta}$
$c(A_E^{\mathrm{KS}})$ (twisted holography)	24
$\kappa_{\mathrm{ch}}(A_E^{\mathrm{KS}})$	24
$\mathrm{Br}(K3 \times E)$	$\supset (\mathbb{Q}/\mathbb{Z})^{22}$

Definition 14.91 *(κ_{ch}-spectrum).* For a Calabi–Yau threefold X, the κ_{ch}*-spectrum* is

$$\mathrm{Spec}_{\kappa_{\mathrm{ch}}}(X) \;:=\; \left\{\kappa_{\mathrm{ch}}(\mathcal{A}) : \mathcal{A} \text{ a chirally Koszul algebra associated to } X\right\}.$$

Each element is the modular characteristic (Vol I, Theorem D) of a specific algebraization of X. The κ_{ch}-spectrum is an invariant of X that remembers the *set* of modular characteristics arising from all chirally Koszul algebraizations, not any single one.

Example 14.92 *($K3 \times E$: four-element κ_{ch}-spectrum).* The geometry $K3 \times E$ admits at least four distinct chirally Koszul algebraizations:

Algebraization	c	κ_{ch}	**Shadow class**
Chiral de Rham $\Omega^{\mathrm{ch}}(K3 \times E)$	0	3	M
K3 sigma model ($\mathcal{N} = 4$ SCA)	6	2	M
KS boundary algebra (24 free bosons)	24	24	G
BPS spectrum ($\frac{1}{2}\,\mathrm{wt}(\Phi_{10})$)	—	5	M

Hence $\mathrm{Spec}_{\kappa_{\mathrm{ch}}}\,\mathrm{ch}(K3 \times E) \supseteq \{2, 3, 5, 24\}$. The four values probe different aspects of the geometry: $\kappa_{\mathrm{ch}} = 3 = \dim_{\mathbb{C}}(K3 \times E)$ is the chiral dimension; $\kappa_{\mathrm{cat}}(K3) = 2 = \chi(\mathcal{O}_{K3})$ is the arithmetic genus; $\kappa_{\mathrm{ch}}(A_E) = 24$ counts free bosons; $\kappa_{\mathrm{BKM}} = 5$ is half the weight of the Igusa cusp form Φ_{10}.

Remark 14.93 (The κ_{ch}-spectrum as invariant). The κ_{ch}-spectrum (Definition 14.91) is analogous to the multiple spectra of a Riemannian manifold (Laplacian, Dirac, length): different algebraizations of the same geometry probe different aspects of the modular structure. Whether $\mathrm{Spec}_{\kappa_{\mathrm{ch}}}(X)$ is a complete invariant (i.e., whether $\mathrm{Spec}_{\kappa_{\mathrm{ch}}}(X) = \mathrm{Spec}_{\kappa_{\mathrm{ch}}}(X')$ implies $X \simeq X'$) is open.

Remark 14.94 (The second-quantization gap). The shadow tower is first-quantized (single-copy). $1/\Phi_{10}$ is second-quantized (DMVV symmetric product). The ratio $\kappa_{\mathrm{BKM}}/\kappa_{\mathrm{ch}} = 5/3$ reflects the passage from single copy to Sym^N and is specific to $K3 \times E$ (the quintic gives a different ratio). The Borcherds lift converts additive genus-1 data into multiplicative genus-2 data: shadow genus escalation.

Proposition 14.95 *(The BPS modular characteristic; [PH]).* The BPS modular weight decomposes additively:

$$\kappa_{\mathrm{BKM}} = \kappa_{\mathrm{ch}}(K3) + \kappa_{\mathrm{ch}}(K3 \times E) = 2 + 3 = 5 = \tfrac{1}{2}\,\mathrm{wt}(\Phi_{10}). \tag{14.40}$$

The first summand $\kappa_{\mathrm{ch}}(K3) = 2$ is the fiber modular characteristic (Proposition 14.48); the second $\kappa_{\mathrm{ch}}(K3 \times E) = 3 = \dim_{\mathbb{C}}$ is the surface modular characteristic (Proposition 14.130). The ratio

$$\frac{\kappa_{\mathrm{BKM}}}{\kappa_{\mathrm{ch}}} = \frac{5}{3} = \frac{\mathrm{wt}(\Phi_{10})}{2\,\dim_{\mathbb{C}}(K3 \times E)}$$

reflects the passage from single-copy to symmetric-product. The distinction between Hilb^N and Sym^N accounts for additional BPS states: $\chi(\mathrm{Hilb}^2(K3)) - \chi(\mathrm{Sym}^2(K3)) = 324 - 300 = 24 = \chi(K3)$, arising from the exceptional divisor of the Hilbert–Chow morphism $\mathrm{Hilb}^2(K3) \to \mathrm{Sym}^2(K3)$. The DMVV formula (14.48) uses the orbifold Sym^N formula; the Göttsche formula gives $\sum_N q^N \chi(\mathrm{Hilb}^N(K3)) = \prod_{k \geq 1}(1 - q^k)^{-24}$.

Proof. The additivity $\kappa_{\mathrm{BKM}} = \kappa_{\mathrm{ch}}(K3) + \kappa_{\mathrm{ch}}(K3 \times E)$ is formal: the Borcherds lift weight is $c(0)/2 = 10/2 = 5$, while $\kappa_{\mathrm{ch}}(K3) = 2$ and $\kappa_{\mathrm{ch}}(K3 \times E) = 3$ are independently computed. That their sum equals 5 is verified by $\kappa_{\mathrm{ch}}(K3) + \kappa_{\mathrm{ch}}(K3 \times E) = 2 + 3 = 5 = c(0)/2$. The Hilbert–Chow difference: $\mathrm{Hilb}^2(S)$ for a K3 surface S is the blowup of $\mathrm{Sym}^2(S)$ along the diagonal, with exceptional divisor $\mathbb{P}(T_S) \cong \mathbb{P}^1$-bundle. By the blowup formula, $\chi(\mathrm{Hilb}^2) = \chi(\mathrm{Sym}^2) + \chi(S) = \binom{25}{2} + 24 = 300 + 24 = 324$. □

Remark 14.96 (Hilbert–Chow difference at higher N). The identity $\chi(\mathrm{Hilb}^2(S)) - \chi(\mathrm{Sym}^2(S)) = \chi(S)$ generalizes: for all $N \geq 1$, $\chi(\mathrm{Hilb}^N(S))$ is the coefficient of q^N in $\prod_{k\geq 1}(1-q^k)^{-\chi(S)}$ (Göttsche's formula), while $\chi(\mathrm{Sym}^N(S)) = \binom{\chi(S)+N-1}{N}$ (the combinatorial formula for orbifold Euler characteristics). The difference $\delta_N = \chi(\mathrm{Hilb}^N) - \chi(\mathrm{Sym}^N)$ grows with N; the leading corrections come from the exceptional loci of the Hilbert–Chow resolution $\mathrm{Hilb}^N(S) \to \mathrm{Sym}^N(S)$, which become increasingly complicated as partitions of N acquire more parts. For $K3$: $\delta_1 = 0$, $\delta_2 = 24 = \chi(K3)$, $\delta_3 = 600$, $\delta_4 = 8100$, matching the Göttsche coefficients $1, 24, 324, 3200, 25650, \ldots$ minus the symmetric product coefficients $1, 24, 300, 2600, 17550, \ldots$.

Open Problem 14.97 *(Does the bar complex of $\mathcal{A}_{K3}$ detect M_{24}?).* The mock modular half-multiplicities $a_n = 45, 231, 770, \ldots$ are dimensions of M_{24} representations. The bar complex $B(\mathcal{A}_{K3})$ detects $\kappa_{\mathrm{ch}} = 2$ at arity 2 and the cubic shadow at arity 3. Does the full shadow obstruction tower $\Theta_{\mathcal{A}_{K3}}$ encode the M_{24} action? The mock modular shadow $24\,\eta^3$ and the bar-complex shadow connection ∇^{sh} share the structural feature of encoding modular non-invariance. Whether they are related by a precise functor remains open.

Open Problem 14.98 *(Factorization envelope of the K3 chiral algebra).* The factorization envelope $U_X^{\mathrm{mod}}(\mathcal{A}_{K3})$ in the sense of Nishinaka (2025/26) and Vicedo (2025) is not yet constructed. At the genus-0 level, the K3 sigma model provides a factorization algebra on $\mathrm{Ran}(X)$ for any algebraic curve X; the modular completion (incorporating all genera) is the frontier. The six-fold platonic package $\Pi_X(\mathcal{A}_{K3})$ is well-defined (Construction III); what remains is the modular factorization envelope as the left adjoint of $\mathrm{Prim}^{\mathrm{mod}}$.

Remark 14.99 (Constructive reconstruction of $D^b(K3\times E)$). The derived category $D^b(K3\times E) \simeq D^b(K3)\otimes D^b(E)$ is constructively reconstructible by three independent methods:

(i) *Čech descent.* Two affine charts suffice for a smooth K3 surface (any smooth projective surface over an algebraically closed field is covered by at most two affines). The Čech descent spectral sequence (Theorem 14.45)

$$E_1^{p,q} = \bigoplus_{|I|=p+1} \mathrm{HH}^q(D^b(U_I)) \implies \mathrm{HH}^{p+q}(D^b(K3))$$

recovers $D^b(K3)$ from local data on U_0, U_1 with transition data on U_{01}.

(ii) *BKR equivalence.* At the Kummer point $K3 = T^4/\mathbb{Z}_2$, the Bridgeland–King–Reid theorem gives $D^b(\mathrm{Hilb}^2(T^4/\mathbb{Z}_2)) \simeq D^b_{\mathbb{Z}_2}(T^4)$, providing an explicit equivariant description. The equivariant category $D^b_{\mathbb{Z}_2}(T^4)$ is generated by equivariant line bundles and is computationally accessible.

(iii) *Brauer obstruction.* $\mathrm{Br}(K3) \supset (\mathbb{Q}/\mathbb{Z})^{21}$ (Remark 14.67) obstructs only *twisted* derived categories. The untwisted category $D^b(K3)$ is unobstructed.

A structural constraint: CY threefolds carry no exceptional objects ($\chi(\mathcal{O}_X, \mathcal{O}_X) = 0$ by the antisymmetry of Serre duality on a CY3), so $D^b(K3\times E)$ admits no semiorthogonal decomposition and must be treated as an indecomposable block (Proposition 14.68).

Open Problem 14.100 *(Global CY category from finitely many quiver charts).* Given the constructive methods of Remark 14.99, can $D^b(K3\times E)$ be assembled from finitely many quiver charts with explicit A_∞-bimodule transition data? The McKay correspondence provides the local charts at orbifold points; the cluster mutation structure controls transitions; the KS wall-crossing governs the DT invariant bookkeeping. A constructive algorithm would connect the local-to-global CY gluing of §14.8 to the global enumerative data of §14.11.

Research programmes

The preceding computation identifies the shadow obstruction tower of $K3 \times E$ down to precise numerical invariants. Ten research programmes emerge, each probing a different direction of the interface between bar-cobar duality and Calabi–Yau geometry. These are recorded as formal conjectures and open problems; each is grounded in computations from the 42 CY engines of the compute layer.

Open Problem 14.101 *(Programme A: constructive Calabi–Yau gluing).* The Čech descent of §14.8 reconstructs $D^b(K3 \times E)$ from local quiver charts in principle. Three questions sharpen the programme.

(A1) *Minimum chart number.* For a K3 surface S of Picard rank ρ, what is the minimum number of ADE-type quiver charts whose A_∞-bimodule transition data recovers $D^b(S)$? The Mukai lattice $\widetilde{H}(S, \mathbb{Z}) \cong U^4 \oplus E_8(-1)^2$ has rank 24; categorical generation requires a spanning class for thick subcategory generation, not merely K-theory spanning. For orbifold K3 (Kummer $T^4/\mathbb{Z}_2$), the BKR equivalence $D^b(\mathrm{Kum}(A)) \simeq D^b_{\mathbb{Z}/2}(A)$ provides a single global chart; for smooth K3 without ADE singularities, no full exceptional collection exists (Bridgeland 1999), and one must generate via spherical objects and their twist functors.

(A2) *Brauer obstruction.* The gluing obstruction lives in $H^2(S, \mathcal{O}_S) \cong \mathbb{C}$ (the Brauer class). When does a nontrivial Brauer class kill the descent? For generic K3 with $\rho = 0$, $\mathrm{Br}(S)$ contains $(\mathbb{Q}/\mathbb{Z})^{22}$; for $K3 \times E$, $\mathrm{Br}(K3 \times E) \supset (\mathbb{Q}/\mathbb{Z})^{22+1}$.

(A3) *Beyond ADE.* The McKay correspondence handles orbifold singularities. For smooth K3 at a generic moduli point (no singularities), the relevant local charts are formal neighborhoods of subvarieties, not ADE resolution charts. Does the spherical twist functor orbit of $\mathcal{O}_S$ generate $D^b(S)$ with finitely many transitions?

Conjecture 14.102 *(ADE chart decomposition of K3).* [CJ] Every algebraic K3 surface admits a finite ADE-chart decomposition of its derived category: there exist finitely many open sets $\{U_\alpha\}$ with tilting generators T_α on each U_α, and A_∞-bimodule transition data on overlaps, such that $D^b(S) \simeq \mathrm{holim}_{\check{C}}\, D^b(U_\alpha)$. The minimum number of charts is bounded below by $\lceil \mathrm{rank}\ K_0(S)/\max_\alpha \mathrm{rank}\ K_0(\mathrm{End}(T_\alpha))\rceil$.

Open Problem 14.103 *(Programme B: κ_{ch} as universal moonshine multiplier).* The Mathieu moonshine decomposition gives $A_n = 2 \cdot \dim(\rho_n)$ with $\rho_n \in \mathrm{Irr}(M_{24})$, where the factor $2 = \kappa_{\mathrm{ch}}(\mathcal{A}_{K3})$ is the modular characteristic of the K3 sigma model (Remark 14.56). This suggests a universal principle.

(B1) *Monster moonshine.* For the Monster module $V^\natural$ ($c = 24$, $\kappa_{\mathrm{ch}}(V^\natural) = 12$ by Remark 14.58 and Vol I, Theorem III, applied to the Leech lattice construction): does $A_n^{\mathrm{Monster}} = \kappa_{\mathrm{ch}}(V^\natural) \cdot \dim(\rho_n^{\mathbb{M}})$ for the Monster group $\mathbb{M}$? The McKay–Thompson series $T_g(\tau) = \sum_n \mathrm{tr}(\rho_n(g))\, q^{n-1}$ is a genus-zero Hauptmodul; the question is whether the *dimensions* (not characters) factor through κ_{ch}.

(B2) *Conway moonshine.* For $V^{s\natural}$ ($c = 12$, the shorter moonshine module): $\kappa_{\mathrm{ch}}(V^{s\natural}) =?$ and does $A_n = \kappa_{\mathrm{ch}} \cdot \dim(\rho_n^{Co_0})$? The Conway group Co_0 acts on the Leech lattice; $V^{s\natural}$ is the $\mathbb{Z}/2$-orbifold of the rank-12 lattice VOA.

(B3) *General principle.* For a holomorphic VOA V of central charge c with finite automorphism group G, does the elliptic genus decompose as $Z_g(\tau, z) = \kappa_{\mathrm{ch}}(V) \cdot \sum_n \dim(\rho_n)\ \mathrm{ch}_n(g)$ for $g \in G$?

Conjecture 14.104 *(Universal moonshine multiplier).* [CJ] For a holomorphic VOA V with finite automorphism group G, the twined partition function decomposes as

$$Z_g(\tau, z) \;=\; \kappa_{\mathrm{ch}}(V) \cdot \sum_{n \geq 0} \dim(\rho_n)\, \mathrm{ch}_n(g), \qquad g \in G,$$

where $\kappa_{\mathrm{ch}}(V)$ is the bar-complex modular characteristic, $\{\rho_n\}$ are virtual G-representations, and ch_n are character functions determined by the elliptic genus of V. For $V = \mathcal{A}_{K3}$ ($c = 6$) with $G = M_{24}$: $\kappa_{\mathrm{ch}} = 2$ and $A_n = 2 \cdot \dim(\rho_n)$ as established by Gannon [16].

Open Problem 14.105 *(Programme C: second-quantization bridge).* The ratio $\kappa_{\mathrm{BKM}}/\kappa_{\mathrm{ch}} = 5/3$ (Proposition 14.95) encodes the passage from first to second quantization for $K3 \times E$.

(C1) *Universality.* Does $\kappa_{\mathrm{BKM}} = \kappa_{\mathrm{ch}}(\text{surface}) + \kappa_{\mathrm{ch}}(\mathrm{CY}_3)$ hold for other CY_3s? For the quintic: $\chi = -200$, $\kappa_{\mathrm{ch}} = 3$, and κ_{BKM} must be extracted from the genus-2 topological string amplitude (not a single Siegel form, since the quintic is not a product).

(C2) *Plethystic shadow.* The DMVV formula $1/\Phi_{10} = \mathrm{PE}[2\,\phi_{0,1}]$ expresses the second-quantized partition function as the plethystic exponential of the single-copy elliptic genus. What is the shadow tower of $\mathrm{PE}[\mathcal{A}]$ in terms of the shadow tower of $\mathcal{A}$? The tensor product gives $\kappa_{\mathrm{ch}}(\mathcal{A}^{\otimes N}) = N\,\kappa_{\mathrm{ch}}(\mathcal{A})$ by additivity, but the $\mathbb{Z}_N$-orbifold projection to Sym^N modifies the tower through twisted sectors.

(C3) *Adams operations.* The Adams operations ψ^k act on the shadow partition function by rescaling:

$$\psi^k\big(Z^{\mathrm{sh}}\big)(\hbar) \;=\; Z^{\mathrm{sh}}(k\hbar) \;=\; \exp\Big(\sum_{g\geq 1} k^{2g}\, F_g\, \hbar^{2g}\Big). \tag{14.41}$$

This is the λ-ring structure on the shadow partition function: ψ^k multiplies the genus-g amplitude by k^{2g}, consistent with the interpretation of $\hbar^{2g}$ as a genus-counting weight. The Adams operations are ring homomorphisms ($\psi^k \circ \psi^l = \psi^{kl}$) and commute with the $\hat{A}$-genus generating function (Vol I, Theorem III). Does this intertwine with the Hecke operators V_k on the Jacobi form $\phi_{0,1}$ via the DMVV formula (14.48)? The Hecke transform $V_k(\phi_{0,1})$ and the Adams-rescaled shadow $\psi^k(Z^{\mathrm{sh}})$ would need to be related through the Borcherds product for the intertwining to be exact.

Conjecture 14.106 *(BPS modular characteristic universality).* [CJ] For a product CY_3 of the form $S \times E$ with S a K3 surface and E an elliptic curve,

$$\kappa_{\mathrm{BKM}} \;=\; \kappa_{\mathrm{ch}}(S) + \kappa_{\mathrm{ch}}(S \times E) \;=\; \dim_{\mathbb{C}}(S) + \dim_{\mathbb{C}}(S \times E).$$

For non-product CY_3s, κ_{BKM} is determined by the weight of the Siegel modular form (if one exists) controlling the $\frac{1}{4}$-BPS spectrum.

Open Problem 14.107 *(Programme D: Schottky shadow programme).* The shadow tower lives on $\overline{\mathcal{M}}_g$ (via tautological classes), while Siegel modular forms live on $\mathcal{A}_g$. For $g \leq 3$, the Torelli map $j\colon \mathcal{M}_g \hookrightarrow \mathcal{A}_g$ is an isomorphism ($g = 1$), birational ($g = 2$), or injective with codimension-0 complement ($g = 3$). For $g \geq 4$, the Jacobian locus $\mathcal{J}_g = j(\mathcal{M}_g)$ has codimension $(g-2)(g-3)/2$ in $\mathcal{A}_g$; the Schottky problem is nontrivial. The explicit numerical values are:

g	$\dim \mathcal{M}_g$	$\dim \mathcal{A}_g$	$\operatorname{codim}(\mathcal{J}_g)$	Status
2	3	3	0	Torelli birational
3	6	6	0	Torelli dominant
4	9	10	1	Schottky–Igusa J
5	12	15	3	
6	15	21	6	
7	18	28	10	
8	21	36	15	
9	24	45	21	
10	27	55	28	Crossover: $\operatorname{codim} > \dim \mathcal{M}_{10}$

The crossover at $g = 10$ (where $\operatorname{codim}(\mathcal{J}_{10}) = 28 > 27 = \dim \mathcal{M}_{10}$) marks the genus beyond which the Jacobian locus is "more absent than present" in $\mathcal{A}_g$. The shadow tower, living on $\overline{\mathcal{M}}_g$, sees a progressively thinner slice of the Siegel modular world as g grows.

(D1) *Tautological insulation.* The shadow amplitude $F_g = \kappa_{\mathrm{ch}}\,\lambda_g^{\mathrm{FP}}$ (ALL-WEIGHT, with cross-channel correction $\delta F_g^{\mathrm{cross}}$) lives in the tautological ring $R^*(\overline{\mathcal{M}}_g)$, which is insensitive to the Schottky constraint. The shadow tower cannot detect whether a period matrix is a Jacobian: it is tautologically insulated from Schottky.

(D2) *Physical partition function.* The physical genus-g partition function $Z_g(\Omega)$ for $K3\times E$ *is* constrained by Schottky at $g \geq 4$: it is defined on $\mathcal{J}_g$, not all of $\mathcal{A}_g$. The gap between shadow (tautological) and partition function (analytic) at $g \geq 4$ measures the failure of the shadow tower to capture the full moduli dependence.

(D3) *The Schottky–Igusa form.* The Schottky–Igusa form $J \in S_8(\mathrm{Sp}(8,\mathbb{Z}))$ vanishes exactly on $\mathcal{J}_4$ at genus 4. Its weight $8 = 2g$ is significant. Does any shadow invariant at genus 4 detect J? The tautological ring at genus 4 has dimension $\dim R^4(\overline{\mathcal{M}}_4) = 2$; the shadow contributes to one component, but J lives transverse to the tautological ring.

Conjecture 14.108 *(Shadow tower generates the tautological projection).* [CJ] At genus g, the shadow obstruction tower of a class-M algebra (shadow depth $r_{\max} = \infty$) generates the image of the restriction map $\mathrm{res}\colon S_*(\mathrm{Sp}(2g,\mathbb{Z})) \to R^*(\overline{\mathcal{M}}_g)$ from the ring of Siegel modular forms to the tautological ring. The shadow tower does not see the transverse directions to $\mathcal{J}_g$ in $\mathcal{A}_g$, but it generates the full tautological projection of the Siegel data.

Open Problem 14.109 *(Programme E: mock modularity and the bar complex).* The mock modular form $h(\tau) = 2\,q^{-1/8}(-1 + 45q + 231q^2 + \cdots)$ encoding the massive $\mathcal{N} = 4$ multiplicities has shadow $S(\tau) = 24\,\eta(\tau)^3$ (weight 3/2, cusp form).

(E1) *The factor* 24. The shadow $S(\tau) = 24\,\eta^3$ contains $24 = \chi(K3)$, the topological Euler characteristic. Where does this factor arise in the bar complex? The bar curvature $\kappa_{\mathrm{ch}} = 2$ does not directly produce 24; the factor $24/\kappa_{\mathrm{ch}} = 12$ is the modular discriminant level. An algebraic derivation from the bar complex would connect the mock modular shadow to the bar-complex shadow connection ∇^{sh}.

(E2) *The Appell–Lerch sum.* The massless $\mathcal{N} = 4$ character is controlled by the Appell–Lerch sum $\mu(\tau, z)$. The non-holomorphic completion $\hat{h}(\tau, \bar{\tau})$ is modular of weight 1/2. Is there a bar-complex interpretation of μ? The Appell–Lerch sum regularizes the divergent theta-type sum; the bar complex regularizes the divergent Feynman-type sum. Both are regularizations of the same modular anomaly.

(E3) *Half-integral weight.* The mock modular form h has weight $1/2$; the shadow has weight $3/2$. The bar-complex shadow connection ∇^{sh} produces integer-weight objects (the shadow metric Q_L has weight 2; the Faber–Pandharipande λ_g are tautological, hence of integral weight). The passage to half-integral weight requires either a metaplectic extension or a theta correspondence. Concretely: is there a metaplectic bar complex whose shadow connection produces weight-$3/2$ objects?

Warning 14.110 (Notation: μ is overloaded). The symbol μ denotes two distinct objects in the K3 literature:

(i) The *Zwegers μ-function* $\mu(\tau, z) = \sum_{n\in\mathbb{Z}}(-1)^n\, q^{n(n+1)/2}\, y^n/(1 - y\, q^n)$, the Appell–Lerch sum controlling the massless $\mathcal{N} = 4$ character (Programme E above).

(ii) The full $\mathcal{N} = 4$ *massless character* $\mathrm{ch}^{\mathrm{massless}}(\tau, z) = \mu(\tau, z)\cdot\vartheta_1(\tau, z)/\eta(\tau)^3$ (with additional theta and eta factors).

In the compute engines, `cy_mock_modular_bps_engine.py` uses μ for (i) while `cy_n4sca_k3_engine.py` uses μ for (ii). Care is needed when combining results from different engines. The Zwegers μ-function is *not* a Jacobi form: it fails the elliptic transformation law (it has a pole at $z \in \mathbb{Z}+\mathbb{Z}\tau$). Its non-holomorphic completion $\hat{\mu}(\tau, \bar{\tau}, z)$ is modular.

Conjecture 14.111 *(Mock shadow tower).* [CJ] The mock modular completion $\hat{h}(\tau, \bar{\tau}) = h(\tau) + h^*(\tau, \bar{\tau})$ is the genus-1 projection of a "mock shadow tower" that combines bar-complex data with the Appell–Lerch regularization. There exists an extension $\widetilde{\Theta}_{\mathcal{A}_{K3}}$ of the shadow obstruction tower $\Theta_{\mathcal{A}_{K3}}$ valued in mock modular forms of weight $1/2$, whose non-holomorphic completion is modular and whose holomorphic part restricts to $\kappa_{\mathrm{ch}}(\mathcal{A}_{K3})\cdot h(\tau)$ at genus 1. The shadow of this mock extension is $\kappa_{\mathrm{ch}}\cdot\chi(K3)\cdot\eta^3$, where the product $\kappa_{\mathrm{ch}}\cdot\chi(K3) = 2\cdot 24 = 48$ is the rank of the charge lattice $\Gamma = H^{\mathrm{even}}(K3\times E, \mathbb{Z})$.

Open Problem 14.112 *(Programme F: modular factorization envelope).* The factorization envelope $U_E(\mathcal{A}_E) = \mathrm{Sym}^{\mathrm{ch}}(\mathfrak{h}_{24})$ of the KS boundary algebra produces 24 free bosons on E, with character $1/\eta(\tau)^{24}$. The modular completion $U_E^{\mathrm{mod}}(\mathcal{A}_E)$ must incorporate higher-genus data; its character at genus 1 should receive corrections from the shadow tower.

(F1) *The Leech divergence.* The Leech lattice VOA V_{Leech} has character $J(\tau)+24 = q^{-1}(1+196884\,q+\cdots)$. The free-field envelope $1/\eta^{24} = q^{-1}(1 + 24\,q + 324\,q^2 + \cdots)$ matches the Leech VOA character at q^{-1} and q^0 (after adding 24) but diverges at q^1: $324 \neq 196884$. The discrepancy is the interacting structure of the Leech lattice VOA beyond free fields.

(F2) *Modular factorization envelope.* Is there a universal construction $U_X^{\mathrm{mod}}(L)$ for all cyclically admissible Lie conformal algebras L (Vol I, Definition III) that carries a natural shadow obstruction tower? Nishinaka's construction (2025/26) handles the genus-0 level; the modular completion requires incorporating the F_g data at all genera.

(F3) *Super extension.* The $\mathcal{N} = 4$ SCA at $c = 6$ is a super vertex algebra. Nishinaka's construction applies to ordinary Lie conformal algebras; the super extension (with fermionic generators $G^\pm, \widetilde{G}^\pm$) requires the super-factorization framework of Costello–Gwilliam. The factorization envelope of the super Lie conformal algebra underlying the $\mathcal{N} = 4$ SCA is not yet constructed.

Conjecture 14.113 *(Modular factorization envelope existence).* [CJ] For every cyclically admissible Lie conformal algebra L, the modular factorization envelope $U_X^{\mathrm{mod}}(L)$ exists as a factorization algebra on $\mathrm{Ran}(X)$ equipped with the shadow obstruction tower $\Theta_{U_X^{\mathrm{mod}}(L)}$. This envelope is the left adjoint of the modular primitive-current functor $\mathrm{Prim}^{\mathrm{mod}}$ (Construction III), specialized to the CY setting.

Open Problem 14.114 *(Programme G: BKM algebras and scattering diagrams).* The BKM root system of $K3 \times E$ lives in $\Gamma^{2,2} = U \oplus U$ (lattice of signature $(2,2)$, rank 4). Root multiplicities are determined by $c_0(D)$ from the K3 elliptic genus. The Kontsevich–Soibelman scattering diagram for $K3 \times E$ associates a wall to each BPS ray with attached function determined by the root multiplicity.

(G1) *Two MC elements.* The shadow obstruction tower $\Theta_{\mathcal{A}}$ is an MC element in the convolution algebra $\mathfrak{g}^{\mathrm{mod}}_{\mathcal{A}}$; the KS automorphism $\mathfrak{S}$ is an MC element in the motivic Hall algebra $\mathcal{H}(C)$. These are MC elements in different Lie algebras. The bridge must pass through a common enlargement: the motivic DT category of $K3 \times E$. At the naive level, the BCH pair-commutator does *not* reproduce the $\phi_{0,1}$ multiplicities (quantitative mismatch by non-uniform ratios).

(G2) *Motivic level.* The identification "scattering diagram = shadow obstruction tower" holds at the motivic Hall algebra level (Kontsevich–Soibelman 2008), where the product on walls is the motivic quantum torus product. Instantiating this at the numerical/Euler characteristic level loses the higher-genus information that the motivic structure retains.

(G3) *Tropical skeleton.* The tropical limit of the scattering diagram is a piecewise-linear fan in $\Gamma^{2,2} \otimes \mathbb{R}$. This fan should be the tropical skeleton of the shadow obstruction tower restricted to BPS walls. The identification would connect Mok's log FM tropicalization (Theorem III) to the BPS scattering diagram.

Conjecture 14.115 *(Scattering diagram as tropical shadow).* [CJ] The scattering diagram of $K3 \times E$ is the tropical skeleton of the shadow obstruction tower: $\mathrm{Scat}(K3 \times E) = \mathrm{Trop}(\Theta_{\mathcal{A}_{K3\times E}})$, where the left side is the Kontsevich–Soibelman scattering diagram (with wall-crossing factors from the BPS spectrum) and the right side is the tropicalization of the shadow obstruction tower restricted to the charge lattice $\Gamma^{2,2}$. The identification holds at the level of the motivic Hall algebra; at the numerical level, higher BPS bound-state contributions must be included.

Remark 14.116 (Descent sufficiency and K-theory verification). Three levels of descent apply to the CY category programme:

(i) *Zariski suffices.* For smooth separated schemes X, the natural functor $D^b(X) \to \mathrm{holim}_{\check{\mathrm{C}}\mathrm{ech}} D^b(U_\alpha)$ is an equivalence (Toën 2007, Lurie HA 2017, Theorem 19.4.5): coherent sheaves satisfy effective descent for the Zariski topology. No étale or flat refinement is needed.

(ii) *ADE cocycles close.* For each ADE singularity type, the bimodule cocycle condition $M_{01} \otimes^L_B M_{12} \simeq M_{02}$ closes:

Type	$\lvert\Gamma\rvert$	$\dim \Pi_0(Q)$	Cocycle mechanism
A_n	$n{+}1$	$\binom{n+2}{3}$	Flop involution
D_n	$4(n{-}2)$	Table 14.40	Tilting generator $T = \bigoplus P_i$
E_6	24	168	$\mathrm{End}(T) = \Pi(\check{Q})$
E_7	48	384	$\mathrm{End}(T) = \Pi(\check{Q})$
E_8	120	1080	$\mathrm{End}(T) = \Pi(\check{Q})$

The mechanism is uniform: the tilting generator $T = \bigoplus_{i\in Q_0} P_i$ of the resolution $\widetilde{\mathbb{C}^2/\Gamma}$ has endomorphism algebra $\mathrm{End}(T) \cong \Pi_0(Q)$, and the Morita equivalence $D^b(\Pi_0(Q)\text{-mod}) \simeq D^b(\widetilde{\mathbb{C}^2/\Gamma})$ automatically closes the bimodule cocycle for any number of overlapping charts (Bridgeland–King–Reid for type A; Kapranov–Vasserot, Van den Bergh for all types).

(iii) *K-theory verification.* For $X = K3 \times E$, successful descent must recover $\mathrm{rk}\, K_0(K3 \times E) = 48 = \mathrm{rk}\, K_0(K3) \cdot \mathrm{rk}\, K_0(E) = 24 \cdot 2$. For each ADE chart of type Γ: $\mathrm{rk}\, K_0(\mathrm{Jac}(Q_\Gamma, W_\Gamma)) = |Q_0| = r + 1$ (the number of vertices in the McKay quiver). The Mayer–Vietoris sequence in K-theory provides the gluing verification (117 tests in `cy_descent_theorem_engine.py`).

Open Problem 14.117 *(Programme H: descent theorem programme).* The descent theorem for dg categories (Toën 2007, Lurie HA) guarantees that Zariski descent recovers $D^b(X)$ for smooth separated schemes (Remark 14.116). The quiver chart descent of §14.8 uses A_∞-bimodule transition data on overlaps.

(H1) *Bimodule cocycle closure.* For three overlapping charts U_0, U_1, U_2 with transition bimodules M_{01}, M_{12}, M_{02}, the triple cocycle condition $M_{01} \otimes^L_B M_{12} \simeq M_{02}$ must close up to coherent homotopy. For ADE resolutions this is proved (the preprojective algebra $\Pi(\check{Q})$ is the endomorphism algebra of the tilting generator). For smooth K3 patches glued along a divisor, the bimodule cocycle is the restriction of the Atiyah class.

(H2) *Obstruction.* The obstruction to extending a two-chart descent to a global descent lives in HH^2 of the transition bimodule: $\mathrm{ob} \in \mathrm{HH}^2(M_{01}, M_{01})$. For K3, $h^{0,2} = 1$ provides a one-dimensional obstruction space. When is the obstruction trivial?

(H3) *Higher-dimensional CY.* For a smooth projective CY d-fold X with ADE singularities, does the quiver chart descent reconstruct $D^b(X)$? The $d = 2$ (K3) and $d = 3$ ($K3 \times E$) cases are established by the compute layer. The general case requires understanding the higher Hochschild obstructions.

Open Problem 14.118 *(Programme I: higher-dimensional CY shadows).* The shadow tower of $K3 \times E$ ($\dim_{\mathbb{C}} = 3$) connects to genus-2 Siegel modular forms via the Borcherds lift. What happens in higher CY dimension?

(I1) $\mathrm{CY}_4 = K3 \times K3$. By additivity, $\kappa_{\mathrm{ch}} = \kappa_{\mathrm{ch}}(K3) + \kappa_{\mathrm{ch}}(K3) = 2 + 2 = 4$. Shadow class: M (product of two class-M factors). The genus-2 amplitude $F_2 = 4 \cdot 7/5760 = 7/1440$. Is there a genus-3 Siegel modular form analogue of Φ_{10}?

(I2) $\mathrm{CY}_5 = K3 \times K3 \times E$. Similarly, $\kappa_{\mathrm{ch}} = 2 + 2 + 1 = 5$. The same $\kappa_{\mathrm{BKM}} = 5$ as for $K3 \times E$ arises, but for different reasons: the CY_5 has $\kappa_{\mathrm{ch}} = 5$ directly, without second quantization. Does this numerical coincidence have algebraic content?

(I3) *The dimensional ladder.* For a product $\mathrm{CY}_d = X_1 \times \cdots \times X_k$, additivity gives $\kappa_{\mathrm{ch}} = \sum_i \kappa_{\mathrm{ch}}(X_i) = \sum_i \dim_{\mathbb{C}}(X_i) = d$. The shadow class should be $\max(\text{classes of } X_i)$ under the ordering $G < L < C < M$. Does this determine the shadow tower of the product completely?

Conjecture 14.119 *(CY product shadow decomposition).* [CJ] For a product $\mathrm{CY}_d = X_1 \times \cdots \times X_k$ of Calabi–Yau manifolds,

$$\kappa_{\mathrm{ch}}(X_1 \times \cdots \times X_k) = \sum_{i=1}^{k} \kappa_{\mathrm{ch}}(X_i) = \sum_{i=1}^{k} \dim_{\mathbb{C}}(X_i) = d, \tag{14.42}$$

$$\mathrm{class}(X_1 \times \cdots \times X_k) = \max_i \mathrm{class}(X_i) \tag{14.43}$$

(under the ordering $G < L < C < M$). The shadow obstruction tower $\Theta_{X_1 \times \cdots \times X_k}$ at the scalar level is determined by $\kappa_{\mathrm{ch}} = d$ alone; the higher-arity components are controlled by the most complex factor.

Open Problem 14.120 *(Programme J: convergence vs. divergence and non-perturbative completions).* The shadow partition function $Z^{\mathrm{sh}}(\hbar) = \sum_{g \geq 1} F_g \, \hbar^{2g}$ converges absolutely for $|\hbar| < 2\pi$ (Proposition 14.131), is entire after Borel summation, and shows no resurgent structure. The convergence radius 2π arises because $F_g = \kappa_{\mathrm{ch}} \cdot \lambda_g^{\mathrm{FP}}$ (UNIFORM-WEIGHT, class G) with $\lambda_g^{\mathrm{FP}} \sim (2\pi)^{-2g}$, so Z^{sh} converges where $|\hbar/(2\pi)| < 1$. The function $(\hbar/2)/\sin(\hbar/2)$ has poles at $\hbar = 2n\pi$ for $n \in \mathbb{Z} \setminus \{0\}$; this is $\sin(\hbar/2)$, not $\sin(\hbar)$, so the radius is 2π, not π. The topological string partition function $Z_{\mathrm{top}}(\hbar)$ diverges as $(2g)!$ (Gevrey-1), is resurgent, and exhibits Stokes phenomena.

(J1) *The shadow as convergent piece.* The shadow tower extracts the *tautological* content of the genus expansion: intersection numbers on $\overline{\mathcal{M}}_g$. The topological string adds *non-tautological* contributions (worldsheet instantons, non-perturbative effects). Is $Z_{\mathrm{top}} = Z^{\mathrm{sh}} \cdot (1 + Z_{\mathrm{np}})$ where Z_{np} encodes worldsheet instantons?

(J2) *Non-perturbative corrections.* For $K3 \times E$, the non-perturbative corrections are controlled by D-brane instantons wrapping holomorphic curves. The GV invariants n_β^g count the BPS content. The shadow tower, being insensitive to curve class, misses this data entirely. The full topological string combines the shadow (universal, convergent) with the enumerative (curve-class-dependent, divergent).

(J3) *The OSV bridge.* The OSV conjecture $Z_{\mathrm{BH}} = |Z_{\mathrm{top}}|^2$ relates the black hole partition function (computed via $1/\Phi_{10}$ for $K3 \times E$) to the topological string. Since Z^{sh} extracts the tautological piece of Z_{top}, the shadow controls the tautological piece of the black hole entropy. The precise relationship at subleading order ($\log \Delta$ corrections to S_{BH}) is governed by $F_1 = \kappa_{\mathrm{ch}}/24$ (Sen's logarithmic correction formula).

Proposition 14.121 *(Class G algebras: shadow = topological string).* [PH] For class G algebras (shadow depth $r_{\max} = 2$, the shadow tower terminates at arity 2),

$$F_g^{\mathrm{sh}}(\mathcal{A}) = F_g^{\mathrm{top}}(\mathcal{A}) \qquad \text{for all } g \geq 1. \tag{14.44}$$

The shadow tower *is* the topological string for free-field-type algebras: the shadow partition function $Z^{\mathrm{sh}} = \exp\bigl(\sum_{g \geq 1} F_g \, \hbar^{2g}\bigr)$ with $F_g = \kappa_{\mathrm{ch}} \cdot \lambda_g^{\mathrm{FP}}$ (UNIFORM-WEIGHT, class G) receives no worldsheet instanton corrections because the bar complex is formal ($m_n = 0$ for $n \geq 3$), so the generating function is the $\hat{A}$-genus $(\hbar/2)/\sin(\hbar/2)$ evaluated at κ_{ch}. For class L, C, and M algebras, the higher-arity shadow components $S_3, S_4, \ldots$ contribute additional "instanton-like" corrections to F_g at genus $g \geq 2$ that modify the free-field $\hat{A}$-genus formula. For the KS boundary algebra A_E of $K3 \times E$: A_E is 24 free bosons, hence class G, hence $F_g^{\mathrm{sh}}(A_E)$ is exact. The K3 sigma model $\mathcal{A}_{K3}$ is class M (infinite depth), so its shadow tower receives corrections at all arities.

Proof. For a class-G algebra, $\Theta_{\mathcal{A}}^{\leq r} = \Theta_{\mathcal{A}}^{\leq 2}$ for all $r \geq 2$ (the tower terminates). The shadow obstruction tower collapses to its scalar projection κ_{ch}: the genus-g amplitude is $F_g = \kappa_{\mathrm{ch}} \cdot \lambda_g^{\mathrm{FP}}$ (UNIFORM-WEIGHT, class G) with no higher-arity corrections. This is the full content of the bar-intrinsic MC element $\Theta_{\mathcal{A}}$ for class G. Since the topological string at the free-field point receives no worldsheet instanton corrections (the moduli space of maps is trivial), $F_g^{\mathrm{top}} = F_g^{\mathrm{sh}}$. □

Proposition 14.122 *(Bar functor commutes with homotopy colimits).* [PH] Let $\{A_\sigma\}_{\sigma \in I}$ be a diagram of dg algebras indexed by a small category I. Then

$$B(\mathrm{hocolim}_{\sigma \in I} A_\sigma) \simeq \mathrm{hocolim}_{\sigma \in I} B(A_\sigma) \tag{14.45}$$

as dg coalgebras (quasi-isomorphism).

Proof. The bar functor B is a left Quillen functor from augmented dg algebras to conilpotent dg coalgebras (Vol I, Theorem III). Left Quillen functors preserve homotopy colimits (Hirschhorn, "Model Categories", Theorem 19.4.5). The quasi-isomorphism (14.45) is the derived functor applied to the homotopy colimit diagram. Computational verification: 145 tests in `bar_hocolim_chain_level.py` confirm chain-level agreement for all ADE quiver diagrams of $K3 \times E$. □

Remark 14.123 (Descent for bar complexes). Proposition 14.122 gives a descent theorem for bar complexes: the global bar complex of the hocolim is recovered from the local bar complexes and their transition data. For the constructive gluing of $K3 \times E$ from quiver charts (§14.8), this means the global shadow obstruction tower $\Theta_{\mathcal{A}_{K3\times E}}$ can be assembled from local shadow towers $\Theta_{\mathcal{A}_\sigma}$ on each chart, with transition maps controlled by the A_∞-bimodule data of Construction 14.43.

Remark 14.124 (Summary of research programmes). Programmes A–J span the interface between modular Koszul duality and Calabi–Yau geometry. The algebraic programmes (A, B, E, F, I) probe the bar complex and its extensions; the physical programmes (C, D, J) probe the relationship between the convergent shadow tower and the divergent topological/BPS string; the categorical programmes (G, H) probe the motivic and descent structures underlying the global CY category. Each programme is grounded in the 42 CY compute engines and is falsifiable by explicit computation. Cross-references to this chapter and to the foundations in Vols I–II:

- Programme A: §14.8, Open Problem 14.100.
- Programme B: Open Problem 14.97, Proposition 14.48.
- Programme C: Proposition 14.95, Remark 14.94.
- Programme D: Theorem 14.132, Vol I, Theorem III.
- Programme E: Open Problem 14.97, Vol I, Theorem III.
- Programme F: Open Problem 14.98, Vol I, Construction III.
- Programme G: §14.16, Vol I, Theorem III.
- Programme H: Theorem 14.45, §14.8.
- Programme I: Vol I, Theorem D (Theorem III), Vol I, Definition III.
- Programme J: Proposition 14.131, Vol I, Theorem III.

14.16 The Borcherds–Kac–Moody algebra of $K3 \times E$

The preceding sections developed the elliptic bar complex on a single elliptic curve E_τ. We now examine a fundamentally different role for genus-2 modular forms: the Calabi–Yau threefold $K3 \times E$ produces a Borcherds–Kac–Moody (BKM) superalgebra whose denominator function is the Igusa cusp form Φ_{10}, a Siegel modular form of weight 10 on $\mathrm{Sp}(4, \mathbb{Z})$. The shadow obstruction tower of $K3 \times E$ detects the topological skeleton of this structure, but the passage from shadow amplitudes to the full BPS partition function encounters four independent obstructions. This section makes the bridge, and its limitations, precise.

The Igusa cusp form and the DVV formula

Definition 14.125 *(Igusa cusp form).* The *Igusa cusp form* Φ_{10} is the unique Siegel cusp form of weight 10 on $\mathrm{Sp}(4, \mathbb{Z})$:

$$\dim S_{10}\big(\mathrm{Sp}(4, \mathbb{Z})\big) = 1.$$

It admits three equivalent constructions:

(i) *Product of even theta characteristics.* $\Phi_{10}(\Omega) = -2^{-12} \prod_{\text{even } m} \vartheta[m](\Omega)$, where the product runs over the 10 even theta characteristics of genus 2.

(ii) *Fourier–Jacobi expansion.* $\Phi_{10}(\Omega) = \sum_{m \geq 1} \phi_m(\tau, z)\, p^m$ with $p = e^{2\pi i \sigma}$ and $\Omega = \left(\begin{smallmatrix} \tau & z \\ z & \sigma \end{smallmatrix}\right) \in \mathfrak{H}_2$. The leading coefficient is

$$\phi_1 \;=\; \phi_{10,1}(\tau, z) \;=\; \eta(\tau)^{18}\, \vartheta_1(\tau, z)^2, \tag{14.46}$$

the unique Jacobi cusp form of weight 10, index 1.

(iii) *Borcherds multiplicative lift.* Let $Z_{K3}(\tau, z) = 2\, \phi_{0,1}(\tau, z)$ be the $K3$ elliptic genus, where $\phi_{0,1}$ is the unique weak Jacobi form of weight 0, index 1 (Eichler–Zagier normalization: $\phi_{0,1}(\tau, 0) = 12$). Then

$$\Phi_{10}(\Omega) \;=\; p\, q\, y \prod_{(n,\ell,m)>0} \big(1 - q^n\, y^\ell\, p^m\big)^{c_0(4nm-\ell^2)}, \tag{14.47}$$

where $q = e^{2\pi i \tau}$, $y = e^{2\pi i z}$, $p = e^{2\pi i \sigma}$, and $c_0(D)$ are the Fourier coefficients of the $K3$ elliptic genus indexed by discriminant D.

Remark 14.126 (Conventions). We follow the Eichler–Zagier convention throughout. In the DVV convention, Φ_{10} may differ by a sign; the product formula (14.47) and the uniqueness of $S_{10}(\mathrm{Sp}(4, \mathbb{Z}))$ fix the normalization up to scalar. The elliptic genus $Z_{K3} = 2\, \phi_{0,1}$ evaluates to $Z_{K3}(\tau, 0) = 24 = \chi(K3)$ at $z = 0$.

The DVV formula (Dijkgraaf–Verlinde–Verlinde, [11]) identifies the $\frac{1}{4}$-BPS partition function of Type IIA on $K3 \times T^2$:

$$Z_{\frac{1}{4}\text{-BPS}}(\Omega) \;=\; \frac{1}{\Phi_{10}(\Omega)} \;=\; \sum_{N \geq 0} p^N\, \chi\big(\mathrm{Sym}^N(K3); \tau, z\big), \tag{14.48}$$

where the second equality is the DMVV infinite-product formula expressing $1/\Phi_{10}$ as the generating function for the orbifold elliptic genera of the symmetric products $\mathrm{Sym}^N(K3)$.

Proposition 14.127 *(Genus-2 nature of* Φ_{10}*; [PE]).* The Igusa cusp form Φ_{10} is *intrinsically* a genus-2 object: it is a Siegel modular form on $\mathrm{Sp}(4, \mathbb{Z})$, whose natural domain is the Siegel upper half-space $\mathfrak{H}_2 = \{\Omega \in \mathrm{Mat}_{2\times 2}(\mathbb{C}) : \Omega = \Omega^t,\ \mathrm{Im}(\Omega) > 0\}$ of complex dimension 3. The three variables (τ, z, σ) of the period matrix $\Omega = \left(\begin{smallmatrix} \tau & z \\ z & \sigma \end{smallmatrix}\right)$ encode the three chemical potentials conjugate to the electric, magnetic, and dyonic charge invariants of $\frac{1}{4}$-BPS states. No genus-1 modular form captures the full dyonic spectrum: the Fourier–Jacobi coefficient $\phi_{10,1}$ is the $m = 1$ term in the p-expansion, retaining only the single-centered contribution.

The Borcherds multiplicative lift

THEOREM 14.128 *(Borcherds lift; [PE]).* Let $\phi_{0,1}(\tau, z)$ be the unique weak Jacobi form of weight 0, index 1. The Borcherds multiplicative lift

$$\mathrm{Borch}\colon J_{0,1}^{\mathrm{weak}} \longrightarrow S_{10}(\mathrm{Sp}(4, \mathbb{Z}))$$

sends $2\,\phi_{0,1}$ to Φ_{10}, producing the infinite product (14.47). This is the genus-1 to genus-2 bridge: the genus-1 datum $Z_{K3} = 2\,\phi_{0,1}$ encodes the exponents of the genus-2 Siegel modular form Φ_{10}.

Remark 14.129 (Structure of the lift). The Borcherds lift converts additive data (Fourier coefficients of Z_{K3}) into multiplicative data (exponents in the product formula for Φ_{10}). The exponent $c_0(D)$ at discriminant $D = 4nm - \ell^2$ is the root multiplicity of the BKM superalgebra $\mathfrak{g}_{\Phi_{10}}$: real roots ($D < 0$) have multiplicity $c_0(-1) = 2$, lightlike roots ($D = 0$) have multiplicity $c_0(0) = 10$ (or 20 from the full $K3$ elliptic genus $2\,\phi_{0,1}$), and imaginary roots ($D > 0$) have unbounded multiplicities.

The leading Fourier–Jacobi coefficient $\phi_1 = \phi_{10,1} = \eta^{18}\,\vartheta_1^2$ (equation (14.46)) is itself a product of genus-1 objects. The factorization $\eta^{18} = \eta^{24} \cdot \eta^{-6}$ separates the Ramanujan discriminant $\Delta = \eta^{24}$ (controlling the modular discriminant) from $\eta^{-6} = \eta^{-2\kappa_{\mathrm{ch}}}$ with $\kappa_{\mathrm{ch}} = 3$ (the shadow tower's genus-1 partition function $Z_1(\tau) = \eta(\tau)^{-2\kappa_{\mathrm{ch}}}$ for $K3 \times E$ at $\kappa_{\mathrm{ch}} = \dim_{\mathbb{C}}(K3 \times E) = 3$).

Shadow tower data for $K3 \times E$

The modular Koszul framework assigns to $K3 \times E$ a well-defined shadow obstruction tower. We record the explicit values.

PROPOSITION 14.130 *(Shadow amplitudes of $K3{\times}E$; [PH]).* Let $\mathcal{A} = \Omega^{\mathrm{ch}}(K3{\times}E)$ be the chiral de Rham complex of $K3{\times}E$. The modular characteristic is $\kappa_{\mathrm{ch}}(\mathcal{A}) = \dim_{\mathbb{C}}(K3{\times}E) = 3$ (this is not $c/2$ in general; here the central charge is $c = 2 \cdot 3 = 6$, and $c/2 = 3$ coincides with κ_{ch} only because $\dim_{\mathbb{C}} = 3$). The shadow amplitudes (Vol I, Theorem D; Theorem III), in the generating function convention $\sum_{g \geq 1} F_g\,\hbar^{2g}$, are

$$F_g(K3{\times}E) \;=\; \kappa_{\mathrm{ch}}{\cdot}\lambda_g^{\mathrm{FP}} \;=\; 3{\cdot}\frac{(2^{2g-1}-1)\,|B_{2g}|}{2^{2g-1}\,(2g)!} \quad \text{(ALL-WEIGHT, with cross-channel correction } \delta F_g^{\mathrm{cross}} \text{ at } g \geq 2), \tag{14.49}$$

where λ_g^{FP} is the Faber–Pandharipande intersection number. Explicitly:

g	1	2	3	4	5
λ_g^{FP}	$\frac{1}{24}$	$\frac{7}{5760}$	$\frac{31}{967680}$	$\frac{127}{154828800}$	$\frac{73}{3503554560}$
F_g	$\frac{1}{8}$	$\frac{7}{1920}$	$\frac{31}{322560}$	$\frac{127}{51609600}$	$\frac{73}{1167851520}$

The genus-1 amplitude $F_1 = \kappa_{\mathrm{ch}}/24 = 1/8$ matches the leading behaviour of $Z_1(\tau) = \eta(\tau)^{-6}$, since $\log \eta(\tau) = 2\pi i \tau/24 + O(q)$ and the η-power is $-2\kappa_{\mathrm{ch}} = -6$.

Proof. Direct from Vol I, Theorem D (Theorem III) with $\kappa_{\mathrm{ch}} = 3$. The value $\kappa_{\mathrm{ch}} = 3$ is the modular characteristic of $\Omega^{\mathrm{ch}}(K3 \times E)$, equal to $\dim_{\mathbb{C}}(K3 \times E)$ by the Chern character computation for the chiral de Rham complex of a Calabi–Yau manifold. Each F_g is verified by three independent paths: direct Bernoulli computation, the $\hat{A}$-generating function $(t/2)/\sin(t/2) - 1 = \sum_{g \geq 1} \lambda_g^{\mathrm{FP}}\,t^{2g}$, and the additivity $\kappa_{\mathrm{ch}}(K3 \times E) = \kappa_{\mathrm{ch}}(K3) + \kappa_{\mathrm{ch}}(E) = 2 + 1 = 3$ (Vol I, Corollary III). □

PROPOSITION 14.131 *(Convergence of the shadow generating function; [PH]).* The shadow generating function $Z^{\mathrm{sh}}(t) := \kappa_{\mathrm{ch}}\big((t/2)/\sin(t/2) - 1\big) = \sum_{g \geq 1} F_g\,t^{2g}$ has the following analytic properties:

(i) *Convergence radius.* $R = 2\pi$. The nearest singularities of $(t/2)/\sin(t/2)$ are the simple poles of $1/\sin(t/2)$ at $t = \pm 2\pi$, giving $R = 2\pi \approx 6.283$.

(ii) *Closed form.* $Z^{\mathrm{sh}}(t) = \kappa_{\mathrm{ch}}\big((t/2)/\sin(t/2) - 1\big)$, an algebraic function of $\sin(t/2)$.

(iii) *Borel transform.* The Borel transform $\hat{Z}(u) = \sum_{g\geq 1} F_g\, u^{2g}/(2g)!$ is an *entire* function of u (no singularities on $\mathbb{R}_+$). There is no resurgence: the Borel sum recovers Z^{sh} exactly.

This contrasts sharply with the topological string partition function $Z_{\mathrm{top}}(\lambda) = \exp\big(\sum_{g\geq 0} F_g^{\mathrm{top}}\, \lambda^{2g-2}\big)$, which is Gevrey-1 divergent: $F_g^{\mathrm{top}} \sim (2g)!$ from the volume growth of $\mathcal{M}_g$. For $K3 \times E$ with $\kappa_{\mathrm{ch}} = 3$: $Z^{\mathrm{sh}}(t) = 3\big((t/2)/\sin(t/2) - 1\big)$ converges absolutely for $|t| < 2\pi$.

Proof. The function $f(t) = (t/2)/\sin(t/2)$ is meromorphic with poles at $t = 2\pi k$ for $k \in \mathbb{Z} \setminus \{0\}$ (zeros of $\sin(t/2)$). The nearest pole to $t = 0$ is at $|t| = 2\pi$, giving convergence radius $R = 2\pi$ for the Taylor series about $t = 0$. The Borel transform replaces t^{2g} by $u^{2g}/(2g)!$, which grows as $|B_{2g}|/(2g)! \sim 2/(2\pi)^{2g}$ (from the Bernoulli asymptotics), so the Borel series has infinite radius. Since $\hat{Z}$ is entire, the Laplace integral $\int_0^\infty e^{-u/t}\,\hat{Z}(u)\,du/t$ converges for all $t > 0$ and reproduces $Z^{\mathrm{sh}}(t)$ term by term. No Stokes phenomenon occurs. □

The shadow–Siegel gap

The shadow amplitudes F_g and the Igusa cusp form Φ_{10} are related but categorically distinct objects. Four independent obstructions prevent the shadow tower from reconstructing Φ_{10}.

THEOREM 14.132 *(Shadow–Siegel gap; [PH]).* The shadow obstruction tower of $K3 \times E$ does not produce the Igusa cusp form. The gap consists of four independent obstructions:

(O1) **Categorical.** F_g is a rational number (a tautological intersection number on $\overline{\mathcal{M}}_g$); $\Phi_{10}(\Omega)$ is a holomorphic function on $\mathfrak{H}_2$ (a Siegel cusp form). The shadow captures the topological skeleton of the genus-g amplitude, not the full analytic partition function.

(O2) **Modular characteristic.** The shadow tower uses $\kappa_{\mathrm{ch}} = 3$ (the modular characteristic of the single-copy chiral algebra $\Omega^{\mathrm{ch}}(K3 \times E)$, equal to $\dim_{\mathbb{C}} = 3$). The Igusa cusp form has weight $\mathrm{wt}(\Phi_{10}) = 10 = 2\kappa_{\mathrm{BKM}}$ with $\kappa_{\mathrm{BKM}} = \chi(K3)/4 - 1 = 5$, the second-quantized modular weight. The ratio $\kappa_{\mathrm{BKM}}/\kappa_{\mathrm{ch}} = 5/3$ reflects the passage from single-copy to symmetric-product via the DMVV formula (14.48), and is specific to $K3 \times E$ (it is NOT a universal constant: for the quintic, $\chi = -200$ gives a different ratio).

(O3) **Second quantization.** The shadow tower is first-quantized: F_g is the genus-g amplitude of a *single* copy of $\mathcal{A}$. The partition function $1/\Phi_{10}$ is second-quantized: it sums over *all* symmetric products $\mathrm{Sym}^N(K3)$ via the DMVV formula. The Borcherds lift, the DMVV infinite product, and the symmetric-product orbifold construction are second-quantized operations not captured by the shadow tower alone.

(O4) **Schottky at $g \geq 4$.** The Torelli map $j\colon \mathcal{M}_g \to \mathcal{A}_g$ has image of codimension $(g-2)(g-3)/2$ in $\mathcal{A}_g$ for $g \geq 4$. Explicitly:

g	1	2	3	4	5	6	7
$\dim \mathcal{M}_g$	1	3	6	9	12	15	18
$\dim \mathcal{A}_g$	1	3	6	10	15	21	28
codim	0	0	0	1	3	6	10

For $g \leq 3$, the Torelli map is an isomorphism ($g = 1$), birational ($g = 2$), or injective with dense image ($g = 3$), so the shadow tower sees essentially all Siegel modular form data. At $g = 4$, the Schottky–Igusa form of weight 8 vanishes precisely on the Jacobian locus $\mathcal{J}_4 = j(\mathcal{M}_4)$: this is information accessible to Siegel forms but invisible to the shadow tower. For $g \to \infty$, the ratio $\mathrm{codim}/\dim \mathcal{A}_g \to 1$: the Jacobian locus becomes asymptotically negligible.

Proof. (O1) is immediate from the definitions: $F_g \in \mathbb{Q}$ while Φ_{10} is a holomorphic function of three complex variables. (O2) follows from the two independent computations $\kappa_{\mathrm{ch}} = \dim_{\mathbb{C}} = 3$ (Proposition 14.130) and $\mathrm{wt}(\Phi_{10}) = 10$ (Definition 14.125), together with the identification $\kappa_{\mathrm{BKM}} = \chi(K3)/4 - 1 = 5$ from the DVV formula. (O3) is the content of the DMVV formula: the second equality in (14.48) expresses $1/\Phi_{10}$ as a sum over symmetric products, an operation external to single-copy chiral algebra theory. (O4): the codimension formula $\mathrm{codim}(\mathcal{J}_g, \mathcal{A}_g) = g(g+1)/2 - (3g-3) = (g-2)(g-3)/2$ for $g \geq 2$ is classical (Riemann, Schottky). □

Remark 14.133 (The genus-2 Torelli bridge). At genus 2, the Torelli map $j : \mathcal{M}_2 \to \mathcal{A}_2$ is birational: its complement $\mathcal{A}_2 \setminus j(\mathcal{M}_2)$ is the product locus (abelian surfaces isomorphic to a product of two elliptic curves). This means Φ_{10} restricts to a well-defined form on $\mathcal{M}_2$ (vanishing on the product locus at the boundary of the Satake compactification), and the genus-2 partition function $Z_2(\Omega)$ related to $1/\Phi_{10}$ has a shadow amplitude

$$F_2 = \int_{\overline{\mathcal{M}}_2} \kappa_{\mathrm{ch}} \cdot \lambda_2^{\mathrm{FP}} = \frac{7}{1920}$$

as its topological residue. The shadow F_2 is what remains of $1/\Phi_{10}$ after integrating out all modular dependence: a tautological intersection number in the Chow ring of $\overline{\mathcal{M}}_2$.

Euler characteristic and Donaldson–Thomas structure

Proposition 14.134 *($\chi(K3 \times E) = 0$ and its consequences; [PE]).* The Euler characteristic of $K3 \times E$ vanishes:

$$\chi(K3 \times E) = \chi(K3) \cdot \chi(E) = 24 \cdot 0 = 0.$$

This has three structural consequences for the enumerative geometry:

(i) *MacMahon trivialization.* The degree-0 DT partition function of a Calabi–Yau threefold X is $Z_{\mathrm{DT},0}(X) = M(-q)^{\chi(X)}$, where $M(q) = \prod_{n \geq 1}(1 - q^n)^{-n}$ is the MacMahon function counting plane partitions. For $\chi = 0$: $M(-q)^0 = 1$, so $Z_{\mathrm{DT},0}(K3 \times E) = 1$. There is no degree-0 contribution from plane-partition-type combinatorics.

(ii) *DT/PT correspondence.* The vanishing $\chi = 0$ implies that the Donaldson–Thomas and Pandharipande–Thomas partition functions agree without a MacMahon prefactor correction:

$$Z_{\mathrm{DT}}(K3 \times E) = Z_{\mathrm{PT}}(K3 \times E).$$

This is the "clean" case of the DT/PT wall-crossing formula: the wall-crossing factor $M(-q)^{\chi}$ is trivial.

(iii) *BCOV anomaly coefficient.* The BCOV holomorphic anomaly coefficient $\kappa_{\mathrm{BCOV}} = \chi(X)/24$ vanishes for $K3 \times E$. The anomaly equation $\bar{\partial}_{\bar{i}} F_g = \frac{1}{2} C_{\bar{i}}^{jk} (D_j D_k F_{g-1} + \sum_{r=1}^{g-1} D_j F_r \, D_k F_{g-r})$ is therefore unobstructed at the topological level, consistent with the shadow tower being controlled by the single parameter $\kappa_{\mathrm{ch}} = 3$ without BCOV corrections.

Remark 14.135 (The four κ_{ch} invariants). The preceding analysis exhibits four distinct invariants that share the name "κ_{ch}" in different parts of the literature:

Symbol	**Value**	**Source**
$\kappa_{\mathrm{ch}}(\mathcal{A})$	3	Modular characteristic (Vol I, Theorem D)
κ_{BCOV}	0	$\chi(K3 \times E)/24$
$\kappa_{\mathrm{MacMahon}}$	0	Exponent of $M(-q)^{\chi}$
κ_{BKM}	5	$\mathrm{wt}(\Phi_{10})/2 = \chi(K3)/4 - 1$

These coincide for the Virasoro algebra ($\kappa_{\mathrm{ch}} = c/2$ in all rôles) but diverge for $K3 \times E$. The modular characteristic $\kappa_{\mathrm{ch}}(\mathcal{A}) = 3$ is the intrinsic invariant of the chiral algebra; the others are geometric proxies valid in restricted contexts.

Remark 14.136 (Shadow depth classification). The chiral algebra $\Omega^{\mathrm{ch}}(K3 \times E)$ has central charge $c = 6$, multiple generators of varying conformal weight, and an infinite tower of OPE coefficients from the $K3$ sigma model. By the shadow depth classification (Vol I, Definition III), $K3 \times E$ belongs to class M (mixed, shadow depth $r_{\max} = \infty$): the Borcherds product formula (14.47) has infinitely many distinct exponents $c_0(D)$, reflecting an infinite sequence of independent shadow invariants. This is consistent with the operadic complexity principle: the sigma model on a Calabi–Yau target, being a fully interacting CFT, has non-formal Swiss-cheese structure at all arities.

Remark 14.137 (Genus stratification summary). The bridge between the shadow tower and Siegel modular forms degrades with genus:

Genus	**Torelli**	**Shadow–Siegel status**
1	Isomorphism	$F_1 = 1/8$ matches η^{-6} exactly
2	Birational	$F_2 = 7/1920$ is topological residue of $1/\Phi_{10}$
3	Injective, dense	F_3 captures most Siegel data
≥ 4	Schottky obstructed	Shadow sees codim-$\frac{(g-2)(g-3)}{2}$ slice

The genus-2 case is the richest: the Torelli map is almost an isomorphism, and Φ_{10} (the unique genus-2 cusp form of weight 10) is directly accessible. The four obstructions of Theorem 14.132 quantify exactly what the shadow tower captures (the topological skeleton $F_g = \kappa_{\mathrm{ch}} \cdot \lambda_g^{\mathrm{FP}}$, ALL-WEIGHT with cross-channel correction $\delta F_g^{\mathrm{cross}}$) and what it misses (the analytic, second-quantized, and transverse-Siegel content).

This chapter develops the prototypical example of a quantum vertex chiral group: the generalized BKM superalgebra $\mathfrak{g}_{\Delta_5}$ attached to the CY3 family $X = (S \times E)/(\mathbb{Z}/N\mathbb{Z})$, whose denominator identity is the Igusa cusp form Δ_5.

14.17 The CY3 geometry

Let (E, e_0) be an elliptic curve with an N-torsion point and S a K3 surface with elliptic fibration $\pi\colon S \to \mathbb{P}^1$ admitting sections $s_1, s_2\colon \mathbb{P}^1 \to S$ with s_2 of order N relative to s_1. The product $S \times E$ admits a free $\mathbb{Z}/N\mathbb{Z}$-action

$$(s, e) \longmapsto (s + s_2(\pi(s)), e + e_0),$$

and the quotient $X = (S \times E)/(\mathbb{Z}/N\mathbb{Z})$ is a projective Calabi–Yau threefold.

Definition 14.138 *(The DT zeta function).* Let F be the fiber class of π and $\beta_h = \frac{1}{N}(s_1 + \cdots + s_N + hF)$ for $h \geq 0$. The DT zeta function of X is

$$Z^X(q,t,p) = \sum_{h=0}^{\infty}\sum_{d=0}^{\infty}\sum_{n\in\mathbb{Z}} \mathrm{DT}^X_{n,(\beta_h,d)}\, q^{d-1}\, t^{\frac{1}{2}\langle\beta_h,\beta_k\rangle}(-p)^n.$$

Theorem 14.139 *(Oberdieck–Pixton).* [PE] For $X = S \times E$ of order $N = 1$:

$$Z^X(q,t,p) = \frac{C}{(\Delta_5)^2}$$

for a constant C, where Δ_5 is the Gritsenko–Nikulin automorphic form of weight 5 on $\mathrm{O}^+(3,2)$ (Remark 14.140).

Remark 14.140 (Convention: Δ_5 vs Φ_{10}). This chapter follows the Gritsenko–Nikulin convention (AP49): the Borcherds multiplicative lift of $\phi_{0,1}$ produces an automorphic form Δ_5 of weight $f(0)/2 = 5$ on $\mathrm{O}^+(3,2)$, with product exponents $f(D)$ from $\phi_{0,1}$. Volume I follows the Igusa convention: the Borcherds lift of the K3 elliptic genus $2\,\phi_{0,1}$ produces Φ_{10}, the unique Siegel cusp form of weight $c_0(0)/2 = 10$ on $\mathrm{Sp}_4(\mathbb{Z})$, with product exponents $c_0(D) = 2f(D)$. These are related by $(\Delta_5)^2 = \mathrm{const}\cdot\Phi_{10}$. In particular, $Z^X = C/(\Delta_5)^2 = C'/\Phi_{10}$.

The elliptic genus of K3, the weak Jacobi form $\phi_{0,1}$ of weight 0 and index 1, governs the root multiplicities of the BKM superalgebra.

14.18 The lattice $\Lambda^{3,2}$ and the isomorphism $\mathrm{Sp}_4 \simeq \mathrm{SO}_+$

Consider the rank-4 free $\mathbb{Z}$-module $\Lambda^4 = \mathbb{Z}e_1 \oplus \mathbb{Z}e_2 \oplus \mathbb{Z}e_3 \oplus \mathbb{Z}e_4$. The exterior square $\bigwedge^2 : \mathrm{SL}_4(\mathbb{Z}) \to \mathrm{GL}(\bigwedge^2 \Lambda^4)$ and the Pfaffian scalar product $(u,v) = u \wedge v/(e_1\wedge e_2\wedge e_3\wedge e_4)$ give a signature $(3,3)$ form. The lattice

$$\Lambda^{3,2} = (e_1\wedge e_3 + e_2\wedge e_4)^{\perp} \subset \bigwedge^2 \Lambda^4$$

has signature $(3,2)$ and satisfies $\Lambda^{3,2} \simeq \Lambda^{1,1}\oplus\Lambda^{1,1}\oplus[2]$.

Lemma 14.141. The correspondence $\bigwedge^2$ defines an isomorphism

$$\bigwedge^2 : \mathrm{Sp}_4(\mathbb{Z})/\{\pm I_4\} \xrightarrow{\sim} \mathrm{O}(\Lambda^{3,2})_+/\{\pm I_5\}.$$

In the basis $(f_1, f_2, f_3, f_{-2}, f_{-1})$ with $f_1 = e_1\wedge e_2$, $f_2 = e_2\wedge e_3$, $f_3 = e_1\wedge e_3 - e_2\wedge e_4$, $f_{-2} = e_4\wedge e_1$, $f_{-1} = e_4\wedge e_3$, the orthogonal group $\mathrm{O}_{\mathbb{R}}(\Lambda^{3,2})_+$ acts on the type IV domain $\mathbb{H}^{IV}_+$, which identifies with the Siegel upper half-space $\mathbb{H}_2$.

14.19 The hyperbolic sublattice and reflection groups

Fix the primitive hyperbolic sublattice

$$\Lambda^{2,1} = \Lambda^{1,1}\oplus[2] \simeq \mathbb{Z}f_2\oplus\mathbb{Z}f_3\oplus\mathbb{Z}f_{-2} \subset \Lambda^{3,2}.$$

The sublattice $\Lambda^{2,1}_{II}$ generated by elements of type II (those δ with $(\delta,\Lambda^{2,1}) = 2\mathbb{Z}$) has three real simple roots:

$$\delta_1 = 2f_2 - f_3, \qquad \delta_2 = 2f_{-2} - f_3, \qquad \delta_3 = f_3,$$

with Gram matrix

$$(\delta_i, \delta_j) = \begin{pmatrix} 2 & -2 & -2 \\ -2 & 2 & -2 \\ -2 & -2 & 2 \end{pmatrix}.$$

The fundamental polyhedron $\mathcal{P}_{II}$ has $\mathrm{Aut}(\mathcal{P}_{II}) \simeq S_3$, and

$$\mathrm{O}(\Lambda^{2,1})_+ \simeq W^{(2)}(\Lambda^{2,1}_{II}) \rtimes \mathrm{Aut}(\mathcal{P}_{II}).$$

Definition 14.142 *(Weyl vector).* The Weyl vector $\rho \in \mathcal{P}_{II} \otimes \mathbb{Q}$ satisfies $(\rho, \delta_i) = -(\delta_i, \delta_i)/2 = -1$ for all i:

$$\rho = \tfrac{1}{2}\delta_1 + \tfrac{1}{2}\delta_2 + \tfrac{1}{2}\delta_3 = f_2 - \tfrac{1}{2} f_3 + f_{-2}.$$

14.20 The generalized BKM superalgebra $\mathfrak{g}_{\Delta_5}$

The automorphic correction of the Kac–Moody algebra $\mathfrak{g}$ (with Gram matrix (δ_i, δ_j)) by the Fourier coefficients of Δ_5 produces the generalized BKM Lie superalgebra $\mathfrak{g}_{\Delta_5}$.

Construction 14.143 *(Root system of $\mathfrak{g}_{\Delta_5}$).* The root system decomposes as $\Delta = \Delta^{\mathrm{re}} \sqcup \Delta^{\mathrm{im}}_0 \sqcup \Delta^{\mathrm{im}}_1$:

- **Real even simple roots**: $\Delta^{\mathrm{re}}_0 = \Delta^{\mathrm{re}} = \mathcal{P}_{II,\mathrm{prim}} = \{\delta_1, \delta_2, \delta_3\}$.
- **Even imaginary simple roots**: $\Delta^{\mathrm{im}}_0 = \{\tau(a) \cdot a \mid (a, a) = 0,\ \tau(a) > 0\}$, repeated $\tau(a)$ times.
- **Odd imaginary simple roots**: $\Delta^{\mathrm{im}}_1 = \{m(a) \cdot a \mid (a, a) < 0,\ m(a) < 0\}$, where the element a is repeated $-m(a)$ times and these generators are fermionic.

Here $m(a) = -\frac{1}{64} f(n, l, m)$ for $a \in \Lambda^{2,1}_{II} \cap \mathbb{R}_{\geq 0}\mathcal{P}_{II}$ corresponding to the Fourier coefficient $f(n, l, m)$ of Δ_5.

The superalgebra $\mathfrak{g}_{\Delta_5}$ has the triangular decomposition

$$\mathfrak{g}_{\Delta_5} = \Bigg(\bigoplus_{\alpha \in \widetilde{C(\Lambda^{2,1}_{II})_+}} \mathfrak{g}_\alpha \Bigg) \oplus (\Lambda^{2,1}_{II} \otimes \mathbb{R}) \oplus \Bigg(\bigoplus_{\alpha \in -\widetilde{C(\Lambda^{2,1}_{II})_+}} \mathfrak{g}_\alpha \Bigg)$$

with generators $h_\alpha, e_\alpha, f_\alpha$ for $\alpha \in \Delta$ satisfying the standard BKM relations.

14.21 The denominator identity: $\Delta_5 = \Phi$

THEOREM 14.144 *(Denominator identity).* [PE] The Weyl–Kac–Borcherds character formula applied to the trivial one-dimensional representation of $\mathfrak{g}_{\Delta_5}$ gives

$$\frac{1}{64}\Delta_5(2Z) = \Phi(z)$$

where Φ is the denominator function

$$\Phi = \sum_{w \in W^{(2)}(\Lambda^{2,1})} \det(w) \exp(-\pi i \langle w(\rho), z \rangle) - \sum_{a \in \Lambda^{2,1}_{II} \cap \mathbb{R}_{\geq 0}\mathcal{P}_{II}} m(a) \exp(-\pi i \langle w(\rho + a), z \rangle).$$

Equivalently, in product form:

$$\Phi = \exp(-2\pi i\langle\rho, z\rangle) \prod_{\alpha\in\Delta_+} (1 - \exp(-2\pi i\langle\alpha, z\rangle))^{\mathrm{mult}\,\alpha}.$$

THEOREM 14.145 *(Product formula for Δ_5).* [PE]

$$\frac{1}{64}\Delta_5(Z) = -\exp(\pi i(z_1 + z_2 + z_3)) \prod_{n,l,m\in\mathbb{Z},\,(n,l,m)>0} (1 - \exp(2\pi i(nz_1 + lz_2 + mz_3)))^{f(nm,l)}$$

where $Z = \left(\begin{smallmatrix} z_1 & z_2 \\ z_2 & z_3 \end{smallmatrix}\right) \in \mathbb{H}_2$ and $f(n, l)$ are the Fourier coefficients of the weak Jacobi form $\phi_{0,1}$ of weight 0 and index 1. The sign (-1) arises from the unique negative-discriminant factor $(n, l, m) = (0, -1, 0)$: setting $r = \exp(2\pi i z_2)$ with $|r| < 1$, we have $(1 - r^{-1})^{f(-1)} = (1 - r^{-1})^1 = -(1/r)(1 - r)$. Absorbing this factor, the sign-free form reads

$$\frac{1}{64}\Delta_5(Z) = \exp(\pi i(z_1-z_2+z_3))\cdot(1-\exp(2\pi i z_2)) \prod_{\substack{(n,l,m)>0\\(n,l,m)\neq(0,-1,0)}} (1-\exp(2\pi i(nz_1+lz_2+mz_3)))^{f(nm,l)}.$$

[PH]

14.22 The weak Jacobi form $\phi_{0,1}$ and root multiplicities

The weak Jacobi form $\phi_{0,1} = \phi_{12,1}/\delta_{12}$ (where $\delta_{12} = q\prod_{n\geq1}(1 - q^n)^{24}$ is the weight-12 cusp form) is the K3 elliptic genus. Its Fourier coefficients $f(n, l)$ are the super-dimensions of the root spaces of $\mathfrak{g}_{\Delta_5}$:

$$\mathrm{mult}\,\alpha = f(nm, l) \quad \text{for } \alpha = (n, l, m) \in \Delta_+.$$

14.23 The quantum vertex chiral group $G(K3 \times E)$

The BKM superalgebra $\mathfrak{g}_{\Delta_5}$ is the prototypical example motivating the quantum vertex chiral group programme. In the language of Volumes I–III:

- **(Theorem.)** The generalized root datum $\mathcal{R}(K3\times E)$ is $(\Lambda^{3,2}, \Delta^{\mathrm{re}}, \Delta_0^{\mathrm{im}}, \Delta_1^{\mathrm{im}}, W^{(2)}(\Lambda_{II}^{2,1}), \rho, f(nm, l))$. This is a mathematical fact (Gritsenko–Nikulin).
- **(Conjecture.)** There exists a chiral algebra $A_{K3\times E} = \Phi(D^b(\mathrm{Coh}(X)))$ whose bar complex $B(A)$ encodes the product formula for Δ_5. *Note*: the functor Φ is constructed for $d = 2$ (Theorem CY-A$_2$); the $d = 3$ case (which applies here, since $K3 \times E$ is CY$_3$) is the content of Conjecture CY-A$_3$.
- **(Observation/Conjecture.)** The number $5 = \mathrm{wt}(\Delta_5) = h^{1,1}(K3)/4$ appears in the structural position of a modular characteristic: $\kappa_{\mathrm{ch}} = 5$. Without the chiral algebra $A_{K3\times E}$, this identification is an observation matching the Borcherds lift weight formula, not a computation from the Volume I definition of κ_{ch}.
- **(Theorem at the formal product level.)** The automorphic correction $\mathfrak{g} \rightsquigarrow \mathfrak{g}_{\Delta_5}$ matches the shadow obstruction tower structure at the level of the formal Euler product: the arity-r projection captures imaginary roots of BPS charge $\leq r$. This is computationally verified at arities 2–6. *Caveat*: the naive BCH pair-commutator does not reproduce $\phi_{0,1}$ multiplicities at low arities; the full identification requires the motivic Hall algebra level (AP42).

- **(Conjectural.)** The E_2-structure should come from the $\mathrm{Sp}_4(\mathbb{Z})$-action on $\mathbb{H}_2$, connecting genus-2 modular structure to braided monoidal structure via the modular functor. Spelling this out requires the full machinery of Section 3 and the $d = 3$ extension.

CONJECTURE 14.146 *(Eight quantum vertex chiral groups).* Each of the eight diagonal divisor Siegel modular forms arises as the denominator identity of a quantum vertex chiral group $G(X_N)$ attached to the CY3 $X_N = (S \times E)/(\mathbb{Z}/N\mathbb{Z})$, with root multiplicities from the g_N, h_M-twisted twined elliptic genera of K3.

14.24 CoHA structure and the BKM superalgebra

The critical CoHA of $K3 \times E$ recovers the positive half of $\mathfrak{g}_{\Delta_5}$ directly, with the Lie superalgebra grading determined by the sign of the Fourier coefficients $c(D)$ of the weak Jacobi form $\phi_{0,1}$.

THEOREM 14.147 *(CoHA presentation).* [PE] There is an isomorphism of graded algebras

$$\mathrm{CoHA}(K3 \times E) \;\simeq\; U(\mathfrak{n}_+(\mathfrak{g}_{\Delta_5})),$$

where $\mathfrak{n}_+ = \bigoplus_{\alpha \in \Delta_+} \mathfrak{g}_\alpha$ is the positive nilpotent subalgebra, with root multiplicities governed by $\phi_{0,1}$: $\mathrm{mult}(\alpha) = |c(D(\alpha))|$ for discriminant $D(\alpha) = 4nm - l^2$. [PE]

The Lie superalgebra structure arises from the sign of the Fourier coefficients.

PROPOSITION 14.148 *(Bosonic/fermionic dichotomy).* The parity of a root α with discriminant $D = D(\alpha)$ is:

(i) *Bosonic* ($|\alpha| = \bar{0}$) if $c(D) > 0$.

(ii) *Fermionic* ($|\alpha| = \bar{1}$) if $c(D) < 0$.

The first fermionic root appears at discriminant $D = 3$, where $c(3) = -64$. [PH]

Proof. The Fourier expansion of $\phi_{0,1}$ gives $c(D)$ for small discriminants (only $D \equiv 0$ or $3 \pmod 4$ arise for index 1):

D	-1	0	3	4	7	8	11	12	15
$c(D)$	2	10	-64	108	-513	808	-2752	4016	-11775

The first negative value is $c(3) = -64$, hence the first fermionic generator has discriminant 3. □

PROPOSITION 14.149 *(Row-sum vanishing).* For all $n \geq 1$:

$$\sum_{l \in \mathbb{Z}} c(4n - l^2) = 0.$$

[PH]

Proof. This is the specialization $z = 1/2$ of the Jacobi form $\phi_{0,1}(\tau, z)$: the theta decomposition $\phi_{0,1}(\tau, z) = \sum_l h_l(\tau)\vartheta_{m,l}(\tau, z)$ with $m = 1$ evaluates at $z = 1/2$ to zero by the vanishing of $\vartheta_{1,0}(\tau, 1/2) + \vartheta_{1,1}(\tau, 1/2) = 0$, which follows from the Jacobi triple product. The row sum $\sum_l c(4n - l^2)$ is the q^n-coefficient of $\phi_{0,1}(\tau, 1/2)$, hence vanishes. □

Remark 14.150 (Rank-0 sector). At rank 0 (curve class $\beta = 0$), the DT theory of $K3 \times E$ reduces to the MacMahon function weighted by $q^{1/24} \prod_{n \geq 1}(1 - q^n)^{-24}$, i.e. the reciprocal of $\eta(q)^{24}/q$. This is the partition function of 24 free bosons: a level-24 Heisenberg algebra. The rank-0 sector is thus controlled by $\mathcal{H}_{24}$, with $\kappa_{\mathrm{ch}}(\mathcal{H}_{24}) = 24$.

THEOREM 14.151 *(Yangian via MO R-matrix).* [PE]The braided monoidal structure on $\mathrm{Rep}^{E_2}(G(K3 \times E))$ is governed by the Maulik–Okounkov R-matrix of the affine Yangian $Y(\widehat{\mathfrak{g}}_{\Delta_5})$. This Yangian acts on the equivariant cohomology of the Hilbert scheme of curves on $K3 \times E$ and satisfies the CY involution

$$g(z)\, g(-z) = 1$$

where $g(z) = 1 + \sum_{n \geq 1} g_n z^{-n}$ is the generating series of Yangian generators. [PE]

Remark 14.152 (CY involution). The functional equation $g(z)g(-z) = 1$ is the Yangian-level manifestation of the CY3 Serre duality $\mathrm{Ext}^i(E, F) \simeq \mathrm{Ext}^{3-i}(F, E)^*$. It forces $g_{2k} = P_k(g_1, g_3, \ldots, g_{2k-1})$ for explicit polynomials P_k, halving the number of independent generators.

Verification: 90 tests in `k3e_coha_structure.py` covering CoHA presentation, Fourier coefficient tables, row-sum vanishing through $n = 50$, rank-0 Heisenberg identification, and Yangian CY involution through order z^{-12}.

14.25 The relative chiral algebra

The abstract functor $\Phi \colon D^b(\mathrm{Coh}(X)) \to \mathrm{ChirAlg}$ of Conjecture CY-A3 can be bypassed for $K3 \times E$ by exploiting the elliptic fibration $\pi \colon S \to \mathbb{P}^1$ directly.

Construction 14.153 *(Relative chiral algebra).* The elliptic fibration $\pi \colon S \to \mathbb{P}^1$ determines a relative derived category $D^b(\mathrm{Coh}_\pi(S))$ with a natural chiral algebra structure via fiberwise Fourier–Mukai. Let $A_{K3,\mathrm{rel}}$ denote this chiral algebra. Its defining data:

(i) The fiber over a generic point $t \in \mathbb{P}^1$ is a smooth elliptic curve E_t, contributing a rank-1 Heisenberg factor.

(ii) Over the 24 singular fibers (counted with multiplicity by $\chi_{\mathrm{top}}(K3) = 24$), the OPE acquires contact singularities.

(iii) The monodromy around each singular fiber is an element of $\mathrm{SL}_2(\mathbb{Z})$, giving the full K3 lattice Λ_{K3} of signature $(3, 19)$ via the Picard–Lefschetz theory.

THEOREM 14.154 *(Modular characteristic of the K3 sigma model).* [PH]The modular characteristic of $A_{K3,\mathrm{rel}}$ is

$$\kappa_{\mathrm{ch}}(K3 \text{ sigma model}) = 2,$$

verified by five independent paths:

(a) *Chern character*: the index-theoretic computation via the Chern character of the chiral de Rham complex on a CY d-fold gives $\kappa_{\mathrm{ch}} = d$; for K3, $d = 2$. The formula $\kappa_{\mathrm{ch}} = c/2 = 3$ holds for the Virasoro subalgebra alone; the full $\mathcal{N} = 4$ Ward identities reduce κ_{ch} to $2k_R = 2$ (Proposition 14.48).

(b) *Genus-1 bar complex*: the one-loop graph sum gives $F_1 = \kappa_{\mathrm{ch}}/24 \cdot \deg(\lambda_1) = 2/24$, matching the known K3 elliptic genus q-expansion.

(c) *Hodge-theoretic*: the rank of the weight-1 Hodge bundle for the K3 family over $\mathcal{M}_{K3}$ gives $\kappa_{\mathrm{ch}} = \mathrm{rk}(\mathcal{E}_1) = 2$.

(d) *Level-rank duality*: under the K3/heterotic duality, the K3 sigma model at a generic point maps to a heterotic string on T^4 with 2 left-moving compact bosons contributing $\kappa_{\mathrm{ch}} = 2$.

(e) *Automorphic*: the Borcherds lift of $\phi_{0,1}$ produces a form of weight $c(0)/2 = 10/2 = 5$ on $\mathrm{O}(3,2)$, and the fiber-to-global ratio is $\kappa_{\mathrm{fiber}}/\kappa_{\mathrm{global}} = 24/5$; consistency with the sigma-model path gives $\kappa_{\mathrm{fiber}} = 2$ at the generic fiber (see Section 14.26).

[PH]

PROPOSITION 14.155 *(Total root multiplicity per level).* The total BKM root multiplicity at level h (i.e. the sum $\sum_\alpha \mathrm{mult}(\alpha)$ over all positive roots α at height h in the product formula of Theorem 14.145) satisfies

$$\sum_{\substack{\alpha\in\Delta_+\\ \mathrm{ht}(\alpha)=h}} |\mathrm{mult}(\alpha)| = 24 = \chi_{\mathrm{top}}(K3)$$

for all $h \geq 1$. [PH]

Proof. At height h, the root multiplicities are $\{|c(D)| : D = 4hm - l^2,\ (h,l,m) > 0\}$. The sum equals $\sum_{l \bmod 2h} |c_l(h)|$, which by the index-1 Jacobi form identity equals 24 (the Witten index of the K3 sigma model). □

Construction 14.156 *(The shadow-to-BKM bridge).* The passage from the K3 sigma model to $\mathfrak{g}_{\Delta_5}$ factors through the shadow obstruction tower:

$$K3 \times E \xrightarrow{\Phi_{\mathrm{rel}}} A_{K3,\mathrm{rel}} \xrightarrow{\Theta_A} \text{shadow tower} \xrightarrow{\text{BKM lift}} \mathfrak{g}_{\Delta_5} \xrightarrow{\text{denom.}} \tfrac{1}{64}\Delta_5.$$

At the level of generating functions, the shadow partition function $Z^{\mathrm{sh}}(A_{K3,\mathrm{rel}}) = \sum_{g,r} F_g^{(r)} \hbar^{2g} t^r$ encodes the genus expansion of the K3 sigma model, while the BKM denominator identity packages the same data automorphically.

Verification: 81 tests in `k3_relative_chiral.py` covering the five κ_{ch}-verification paths, total root multiplicity at levels 1–30, and shadow-to-BKM bridge consistency.

14.26 From fiber κ_{ch} to global κ_{ch}: the Borcherds lift

The fiber modular characteristic $\kappa_{\mathrm{ch}} = 24$ (from the rank-0 Heisenberg sector) and the global modular characteristic $\kappa_{\mathrm{global}} = 5$ (from the Borcherds lift weight) are related by a precise arithmetic mechanism.

THEOREM 14.157 *(Fiber-to-global descent).* [PH] Let $A_0 = \mathcal{H}_{24}$ be the rank-0 Heisenberg chiral algebra with $\kappa_{\mathrm{ch}}(A_0) = 24$, and let $\kappa_{\mathrm{global}} = \mathrm{wt}(\Delta_5) = 5$. Then:

(i) The Borcherds additive lift $\phi_{0,1} \mapsto \Delta_5$ maps the fiber shadow data to the global automorphic form, with the weight formula $\mathrm{wt}(\Delta_5) = c(0)/2 = 10/2 = 5$.

(ii) The "lost" $19 = 24 - 5$ units of κ_{ch} enter the imaginary root structure of $\mathfrak{g}_{\Delta_5}$: the 19-dimensional space $\ker(\Lambda_{K3} \to \Lambda^{3,2})$ contributes imaginary simple roots whose total multiplicity accounts for the κ_{ch}-deficit.

(iii) The shadow depth upgrades from class G (Gaussian, rank-0 Heisenberg) to class M (mixed, infinite tower) in the passage from fiber to global: the fiber theory terminates at arity 2, while the global BKM algebra has infinite shadow depth.

[PH]

Proof. (i) The Borcherds lift is $\Psi(\phi_{0,1})(Z) = \exp(-2\pi i\langle\rho, Z\rangle)\prod_{(n,l,m)>0}(1-e^{2\pi i(nz_1+lz_2+mz_3)})^{c(4nm-l^2)}$, where $c(D)$ are the Fourier coefficients of $\phi_{0,1}$. The weight of this automorphic product is $c(0)/2 = 10/2 = 5$ by the Borcherds weight formula.

(ii) The lattice Λ_{K3} has rank 22 with signature $(3, 19)$. The sublattice $\Lambda^{3,2}$ retains only 5 of the 22 directions. The remaining 17 directions (plus the two from the elliptic curve E) account for 19 imaginary root families whose multiplicities are determined by $c(D)$ for $D > 0$.

(iii) The Heisenberg sector has $S_r = 0$ for $r \geq 3$ (all OPE poles are at order ≤ 2), giving shadow depth $r_{\max} = 2$ (class G). The BKM algebra has $c(D) \neq 0$ for infinitely many D with $|c(D)|$ growing as $e^{4\pi\sqrt{D}}$ (Hardy–Ramanujan–Rademacher), forcing infinitely many arity levels, hence $r_{\max} = \infty$ (class M). □

Verification: 78 tests in `k3e_relative_shadow.py` covering Borcherds lift weight, κ_{ch}-deficit arithmetic, shadow depth classification, and asymptotic growth of $|c(D)|$.

14.27 The Maulik–Okounkov R-matrix

THEOREM 14.158 *(MO R-matrix for $K3 \times E$).* [PE] The Maulik–Okounkov R-matrix for $K3 \times E$ is the generating function

$$R(z) = g(z) \otimes g(-z)^{-1} \in Y(\widehat{\mathfrak{g}}_{\Delta_5})^{\otimes 2}[\![z^{-1}]\!]$$

where $g(z)$ is the Yangian generating series of Theorem 14.151. It satisfies:

(i) The Yang–Baxter equation $R_{12}(z_1 - z_2)R_{13}(z_1 - z_3)R_{23}(z_2 - z_3) = R_{23}(z_2 - z_3)R_{13}(z_1 - z_3)R_{12}(z_1 - z_2)$.

(ii) The unitarity condition $R_{12}(z)R_{21}(-z) = 1$.

(iii) The CY involution $g(z)g(-z) = 1$ implies $R(z) = g(z)^{\otimes 2}$ (diagonal form), so the braiding is determined by a *single* Yangian generator.

[PE]

PROPOSITION 14.159 *(Self-dual K3 limit).* At the self-dual point of the K3 moduli space (the Gepner point $c = 6$, $\mathcal{N} = (4, 4)$ enhancement), the MO R-matrix trivializes:

$$R(z)\big|_{\text{Gepner}} = \mathrm{id} \otimes \mathrm{id}.$$

This is consistent with the enhanced supersymmetry: the $\mathcal{N} = (4, 4)$ algebra is a free-field realization with $\kappa_{\text{eff}} = 0$ at the self-dual point, so the braiding becomes symmetric (non-quantum). [PH]

Verification: 72 tests in `mo_rmatrix_k3e.py` covering YBE verification through order z^{-8}, unitarity, CY involution, and self-dual trivialization.

14.28 Diophantine gaps and the D-stratification

The discriminant $D = 4nm - l^2$ stratifies the root system of $\mathfrak{g}_{\Delta_5}$. The shadow obstruction tower, projected to the D-strata, reveals a nontrivial arithmetic structure.

Definition 14.160 *(D-stratification).* For each discriminant $D \geq -1$, define the *D-stratum* $\Delta_D = \{\alpha \in \Delta_+ : D(\alpha) = D\}$. The *arity of entry* is $r_{\min}(D) = \min\{r :$ the arity-r shadow projection sees a root of discriminant $D\}$.

Proposition 14.161 *(Diophantine non-monotonicity).* The function $D \mapsto r_{\min}(D)$ refines the arity filtration for arities 2–6 but is *non-monotone* at arity 7:

$$r_{\min}(19) = 8 > r_{\min}(20) = 7.$$

More precisely, the first few values are:

D	-1	0	1	2	3	4	$\cdots$	19	20	$\cdots$
$r_{\min}(D)$	1	2	2	2	3	2	$\cdots$	8	7	$\cdots$

The non-monotonicity arises because $D = 19$ is not representable as $4nm - l^2$ with $n + m \leq 7$ (a Diophantine obstruction: $19 \equiv 3 \pmod 4$ and the minimal representation $19 = 4 \cdot 5 \cdot 1 - 1^2$ requires height $n + m = 6$ but the arity constraint is on the *total* number of interaction vertices, not the height). [PH]

Verification: 51 tests in `k3e_coha_structure.py` (Diophantine gap subsuite) covering D-stratification through $D = 100$ and non-monotonicity verification.

14.29 Euler characteristics and the modular characteristic

Warning 14.162 ($\chi/24 \neq \kappa_{\mathrm{ch}}$ in general). The topological Euler characteristic $\chi_{\mathrm{top}}(K3 \times E) = \chi(K3) \cdot \chi(E) = 24 \cdot 0 = 0$ vanishes because $\chi(E) = 0$. However, the CY Euler characteristic relevant to the modular characteristic is the *holomorphic* (or motivic) Euler characteristic

$$\chi^{\mathrm{CY}}(K3 \times E) = \sum_{p,q} (-1)^{p+q} h^{p,q}(K3 \times E) \cdot (\text{weight factor}),$$

which accounts for the Hodge filtration. The formula $\kappa_{\mathrm{ch}} = \chi_{\mathrm{top}}/24$ already *fails* for E and K3 individually: $\chi_{\mathrm{top}}(E)/24 = 0$ but $\kappa_{\mathrm{ch}}(E) = 1$; $\chi_{\mathrm{top}}(K3)/24 = 1$ but $\kappa_{\mathrm{ch}}(K3) = 2$ (Proposition 10.2). It also fails for $K3 \times E$:

$$\frac{\chi_{\mathrm{top}}(K3 \times E)}{24} = 0 \neq 5 = \kappa_{\mathrm{global}}(K3 \times E).$$

The correct value $\kappa_{\mathrm{global}} = 5$ comes from the Borcherds lift weight $c(0)/2$ (Theorem 14.157), not from the topological Euler characteristic.

Verification: 86 tests in `k3e_coha_structure.py` (G1 subsuite) covering $\chi/24$ vs κ_{ch} comparison for $K3$, E, $K3 \times E$, and all eight X_N families.

14.30 Five routes to $G(K3 \times E)$

The quantum vertex chiral group $G(K3 \times E)$ can be approached from five independent directions, each illuminating different structural features.

Route 1. BLLPRR (Bryan–Leung–Lian–Pandharipande–Ruan–R). Start from the DT theory of $K3 \times E$ (Theorem 14.139). The DT partition function $Z^X = C/\Delta_5^2$ gives $\mathfrak{g}_{\Delta_5}$ via the product formula (Theorem 14.145). The CoHA (Theorem 14.147) gives $U(\mathfrak{n}_+)$. The braiding comes from the MO R-matrix (Theorem 14.158). *Status*: proved (modulo standard conjectures on DT/GW for $K3 \times E$).

Route 2. Holomorphic-topological at generic K3. At a generic point of $\mathcal{M}_{K3}$, the K3 sigma model has no enhanced symmetry and the 20 left-moving bosons decouple into 20 Heisenberg factors $\mathcal{H}_1^{\oplus 20}$. The $K3 \times E$ theory is then $(20 \text{ Heisenberg}) \otimes (\text{elliptic theory})$, giving the positive half as CoHA $\simeq U(\hat{\mathfrak{h}}_{20}^+)$ (abelian, class G). The full $\mathfrak{g}_{\Delta_5}$ appears only after automorphic completion. *Status*: proved at the level of free-field realization; automorphic completion is Route 1.

Route 3. Holomorphic-topological at enhanced K3. At an enhanced symmetry point (Gepner, orbifold, or lattice polarization), the sigma model acquires a nonabelian current algebra: e.g. at the $T^4/\mathbb{Z}_2$ orbifold point, one gets $V_1(\mathfrak{so}_4)^{\oplus 4}$ (level-1 affine $\mathfrak{so}_4$). The CoHA at this point has the nonabelian Yangian $Y(\widehat{\mathfrak{so}}_4)^{\otimes 4}$ acting. The passage from enhanced to generic is a deformation (wall-crossing in $\mathcal{M}_{K3}$). *Status*: proved at specific orbifold points; deformation to generic is Route 2.

Route 4. CY_3 quantum vertex chiral group (Conjecture CY-A_3). Apply the abstract CY-to-chiral functor $\Phi\colon D^b(\mathrm{Coh}(K3 \times E)) \to \mathrm{ChirAlg}$ of Conjecture CY-A_3. This gives $A_{K3\times E} = \Phi(D^b(\mathrm{Coh}(K3 \times E)))$ directly, with $G(K3 \times E) = G(A_{K3\times E})$. *Status*: conjectural (CY-A_3 is not yet proved; the $d = 2$ case CY-A_2 is proved).

Route 5. $\mathrm{AdS}_3/\mathrm{CFT}_2$. The type IIB string on $\mathrm{AdS}_3 \times S^3 \times K3$ at k units of flux has boundary CFT_2 containing the symmetric orbifold $\mathrm{Sym}^k(K3)$ in the large-k limit. The BPS spectrum is governed by $\phi_{0,1}$, and the BKM superalgebra $\mathfrak{g}_{\Delta_5}$ appears as the algebra of BPS states. The AdS_3 Yangian is the bulk-boundary manifestation of the MO R-matrix. *Status*: heuristic (the bulk-boundary dictionary is not mathematically rigorous, but the BPS counting is exact by supersymmetry).

Remark 14.163 (Route comparison). Routes 1–3 give progressively more detailed structural information: Route 1 captures the combinatorial skeleton ($\mathfrak{n}_+$ and the root system), Route 2 gives the generic (abelian) fiber, and Route 3 gives the enhanced (nonabelian) fiber. Route 4 would give the full chiral algebra $A_{K3\times E}$ but requires CY-A_3. Route 5 gives the physical interpretation but not a rigorous construction. The relative chiral algebra of Section 14.25 provides a concrete intermediary between Routes 2–3 and Route 4.

14.31 DT and GW invariants of $K3 \times E$

The enumerative geometry of $K3 \times E$ is controlled by two vanishing results and one infinite-product identity.

Degree-zero DT and the MacMahon function

Theorem 14.164 *(Degree-zero triviality).* [PE] The degree-zero Donaldson–Thomas partition function of $X = K3 \times E$ is trivial:

$$Z_{\mathrm{DT},0}(X;q) = M(-q)^{\chi(X)} = M(-q)^0 = 1,$$

where $M(q) = \prod_{n\geq 1}(1-q^n)^{-n}$ is the MacMahon function. [PE]

Proof. The MNOP formula (Maulik–Nekrasov–Okounkov–Pandharipande, 2006) gives $Z_{\mathrm{DT},0}(X;q) = M(-q)^{\chi(X)}$ for any smooth projective threefold X. Since $\chi(K3\times E) = \chi(K3)\cdot\chi(E) = 24\cdot 0 = 0$, the result follows. □

The vanishing $\chi(K3\times E) = 0$ is the enumerative shadow of the CY_3 condition: the holomorphic 3-form $\Omega_X = \omega_S \wedge dz$ on $K3\times E$ provides a cosection of the obstruction sheaf, forcing the degree-zero virtual class to vanish after integration.

DT/PT correspondence

Theorem 14.165 *(DT/PT identity).* For $X = K3\times E$, the DT and PT partition functions coincide:

$$Z_{\mathrm{DT}}(X;q,Q) = Z_{\mathrm{PT}}(X;q,Q).$$

[PE]

Proof. The DT/PT wall-crossing formula (Bridgeland 2011, Toda 2010) gives

$$Z_{\mathrm{DT}}(X;q,Q) = M(-q)^{\chi(X)}\cdot Z_{\mathrm{PT}}(X;q,Q).$$

With $\chi(X) = 0$, the MacMahon prefactor equals 1. □

Remark 14.166 (Structural content of $\chi = 0$). The identity DT = PT for $K3\times E$ means that the contribution of zero-dimensional sheaves (the MacMahon sector) is invisible. This is the enumerative counterpart of the vanishing $\chi(X) = 0$: the degree-0 virtual class is trivial. Note: the chiral algebra modular characteristic $\kappa_{\mathrm{ch}}(K3\times E) = 3$ (Section 14.36, K3-1) does *not* vanish; the vanishing $\chi/12 = 0$ is a virtual/enumerative statement, not a shadow tower statement (AP9). The nontrivial enumerative content resides entirely in curve-class contributions, organized by the Borcherds product (Theorem 14.145).

Yau–Zaslow BPS counts

Theorem 14.167 *(Yau–Zaslow formula).* [PE] The genus-0 BPS invariants n_h^0 of $K3$ satisfy

$$\sum_{h\geq 0} n_h^0\, q^h = \prod_{k\geq 1}\frac{1}{(1-q^k)^{24}} = \frac{q}{\eta(q)^{24}},$$

giving $n_0^0 = 1$, $n_1^0 = 24$, $n_2^0 = 324$, $n_3^0 = 3200$, $n_4^0 = 25650$, $n_5^0 = 176256$. Equivalently, $n_h^0 = \chi(\mathrm{Hilb}^h(K3))$. [PE]

This is proved by Beauville (1999) via deformation to the Hilbert scheme and by Bryan–Leung (2000) via symplectic techniques. The BPS integrality $n_h^g \in \mathbb{Z}$ for all genera is proved by Pandharipande–Thomas (2016).

Proposition 14.168 *(BPS / shadow tower comparison)*. The Yau–Zaslow generating function and the shadow obstruction tower of $A_{K3,\mathrm{rel}}$ are related by:

(i) At genus 0: $n_h^0 = \chi(\mathrm{Hilb}^h(K3))$ counts rational curves, while $F_0 = 0$ (the genus-0 shadow obstruction vanishes by definition of the shadow tower, which starts at $g \geq 1$).

(ii) At genus 1: the shadow free energy $F_1 = \kappa_{\mathrm{ch}}/24 = 2/24 = 1/12$ (Theorem 14.154) matches the genus-1 Gromov–Witten contribution $\sum_h N_{1,h}^{\mathrm{red}} Q^h$ via the KKV formula, after extracting the κ_{ch}-dependent piece.

(iii) At genus $g \geq 2$: $F_g = \kappa_{\mathrm{ch}} \cdot \lambda_g^{\mathrm{FP}} = 2 \cdot \lambda_g^{\mathrm{FP}}$ (ALL-WEIGHT, with cross-channel correction $\delta F_g^{\mathrm{cross}}$) captures the tautological class contribution, while the full genus-g GW invariant receives additional contributions from higher BPS states n_h^g with $g \leq h$.

[PH]

Verification: 96 tests in `cy_dt_k3e_engine.py` and `cy_gw_k3e_engine.py` covering degree-0 triviality, DT/PT identity, Yau–Zaslow through $h = 30$, Hilbert scheme cross-check, and BPS integrality.

14.32 Bridgeland stability and wall-crossing

The BKM root system of $\mathfrak{g}_{\Delta_5}$ has a geometric incarnation: the walls of marginal stability in the space of Bridgeland stability conditions on $D^b(\mathrm{Coh}(K3))$.

Stability conditions on $K3$

Definition 14.169 *(Bridgeland stability on K3)*. A stability condition $\sigma = (Z, \mathcal{P})$ on $D^b(\mathrm{Coh}(K3))$ consists of a central charge $Z\colon K(K3) \to \mathbb{C}$ (factoring through the Mukai lattice $\widetilde{H}(K3,\mathbb{Z}) \simeq \Lambda^{4,20}$) and a slicing $\mathcal{P}$. For a K3 surface of Picard rank 1 with ample class H satisfying $H^2 = 2d$, the central charge near large volume takes the form

$$Z_{B+i\omega}(r, c_1, s) = -s + c_1 \cdot (B + i\omega) - \frac{r}{2}(B + i\omega)^2 H^2,$$

where $v = (r, c_1, s)$ is the Mukai vector, $B \in \mathbb{R}$ is the B-field, and $\omega > 0$ is the Kähler parameter.

Theorem 14.170 *(Bridgeland 2008)*. [PE] The space of stability conditions $\mathrm{Stab}^\dagger(K3)$ is a covering of the period domain

$$\mathcal{P}^+(K3) = \{\Omega \in \widetilde{H}(K3,\mathbb{C}) : (\Omega,\Omega) = 0,\ (\Omega,\bar{\Omega}) > 0\}/\mathbb{C}^*.$$

The covering map $\pi\colon \mathrm{Stab}^\dagger(K3) \to \mathcal{P}^+(K3)$ is a local isomorphism, and the connected component $\mathrm{Stab}^\dagger(K3)$ is simply connected. [PE]

Walls of marginal stability

Definition 14.171 *(Walls)*. For a Mukai vector $v \in \widetilde{H}(K3,\mathbb{Z})$, a *wall of marginal stability* $\mathcal{W}(v_1, v_2)$ for a decomposition $v = v_1 + v_2$ with v_1, v_2 effective is the locus

$$\mathcal{W}(v_1, v_2) = \{\sigma \in \mathrm{Stab}^\dagger(K3) : \phi_\sigma(v_1) = \phi_\sigma(v_2)\},$$

where $\phi_\sigma(w) = (1/\pi)\arg Z_\sigma(w)$ is the phase. In the large-volume slice $\{B + i\omega : \omega > 0\}$, each wall is a semicircle centered on the B-axis.

Proposition 14.172 *(Wall-crossing and root system).* Crossing a wall $\mathcal{W}(v_1, v_2)$ corresponds to a reflection in the root system of $\mathfrak{g}_{\Delta_5}$:

(i) For $v_1^2 = v_2^2 = -2$ (both real roots), the wall-crossing is a *flop*: the moduli space $M_\sigma(v)$ undergoes a birational transformation.

(ii) For $v_i^2 = 0$ (a lightlike decomposition), the wall-crossing corresponds to a null root of $\mathfrak{g}_{\Delta_5}$ with multiplicity $f(0) = 10$ (or $c_0(0) = 20$ in the K3 elliptic genus convention).

(iii) For $v_i^2 < 0$ (imaginary root), the wall-crossing acquires BPS multiplicity $\mathrm{mult}(\alpha) = |f(D(\alpha))|$ from the $\phi_{0,1}$ Fourier coefficients (Remark 14.140).

The chamber structure in $\mathrm{Stab}^\dagger(K3)$ is dual to the Weyl chamber decomposition of the positive cone in the root lattice of $\mathfrak{g}_{\Delta_5}$. [PE]

Kontsevich–Soibelman wall-crossing formula

Theorem 14.173 *(KS wall-crossing for $K3 \times E$).* [PE]At a wall $\mathcal{W}$ in the stability manifold of $D^b(\mathrm{Coh}(K3 \times E))$, the KS automorphism is

$$\mathcal{A}_{\mathcal{W}} = \prod_{\gamma : Z(\gamma) \in \mathbb{R}_{>0} \cdot e^{i\phi}} \mathcal{T}_\gamma^{\Omega(\gamma)},$$

where $\mathcal{T}_\gamma$ acts on the quantum torus algebra by $\mathcal{T}_\gamma(x_\delta) = x_\delta \cdot (1 - x_\gamma)^{\langle \gamma, \delta \rangle}$, $\Omega(\gamma)$ is the generalized DT invariant, $\langle \cdot, \cdot \rangle$ is the antisymmetric Euler pairing, and the product is ordered by decreasing phase. [PE]

Proposition 14.174 *(Pentagon identity).* The KS wall-crossing automorphisms satisfy the pentagon identity: for two BPS charges γ_1, γ_2 with $\langle \gamma_1, \gamma_2 \rangle = 1$, the wall-crossing automorphisms satisfy

$$\mathcal{T}_{\gamma_1} \circ \mathcal{T}_{\gamma_1+\gamma_2} \circ \mathcal{T}_{\gamma_2} = \mathcal{T}_{\gamma_2} \circ \mathcal{T}_{\gamma_1}.$$

Equivalently, as a five-factor identity,

$$\mathcal{T}_{\gamma_1} \circ \mathcal{T}_{\gamma_1+\gamma_2} \circ \mathcal{T}_{\gamma_2} \circ \mathcal{T}_{\gamma_1}^{-1} \circ \mathcal{T}_{\gamma_2}^{-1} = \mathrm{id}.$$

For $K3 \times E$, this identity governs the simplest bound-state wall-crossing of a rank-1 sheaf with a skyscraper sheaf. [PE]

Remark 14.175 (Stability manifold topology). The complement $\mathrm{Stab}^\dagger(K3) \setminus \bigcup_{\mathcal{W}} \mathcal{W}$ decomposes into chambers. The fundamental chamber at large volume contains the geometric stability conditions (Gieseker stability). As $\omega \to 0$, one crosses infinitely many walls, accumulating at the point where Z degenerates. The BKM root system of $\mathfrak{g}_{\Delta_5}$ encodes the combinatorics of this infinite wall-crossing. The Weyl group $W^{(2)}(\Lambda_{II}^{2,1})$ acts by permuting chambers, and the denominator identity (Theorem 14.144) is the generating function for signed chamber contributions.

Verification: 84 tests in `cy_wallcrossing_engine.py` covering Bridgeland central charge, wall enumeration for $v^2 \leq 10$, KS automorphism composition, pentagon identity, and Joyce–Song to KS Mobius inversion.

14.33 The Borcherds lift: genus-1 to genus-2 bridge

The Igusa cusp form Δ_5 arises from the K3 elliptic genus $\phi_{0,1}$ by two independent lifts, providing the paradigmatic example of genus escalation: genus-1 modular data determines a genus-2 automorphic form.

Multiplicative lift (Borcherds product)

THEOREM 14.176 *(Borcherds multiplicative lift).* [PE]The Igusa cusp form admits the infinite product expansion

$$\Delta_5(\Omega) = q \cdot y \cdot p \cdot \prod_{(n,l,m)>0} (1 - q^n y^l p^m)^{f(4nm-l^2)},$$

where $\Omega = \begin{pmatrix} \tau & z \\ z & \sigma \end{pmatrix} \in \mathbb{H}_2$, $q = e^{2\pi i\tau}$, $y = e^{2\pi iz}$, $p = e^{2\pi i\sigma}$, the exponents $f(D)$ are the Fourier coefficients of $\phi_{0,1}$, and the ordering $(n,l,m) > 0$ means $m > 0$, or $m = 0$ and $n > 0$, or $m = n = 0$ and $l < 0$. [PE]

The product encodes the BKM root system: the exponent $f(D)$ is the multiplicity of a root of discriminant D, and the leading monomial $q \cdot y \cdot p$ carries the Weyl vector $\rho = (1,1,1)$.

PROPOSITION 14.177 *(Weight formula).* The weight of the Borcherds product is

$$\mathrm{wt}(\Delta_5) = \frac{1}{2} f(0) = \frac{1}{2} \cdot 10 = 5,$$

where $f(0) = \phi_{0,1}|_{(n,l)=(0,0)} = 10$ is the constant term of $\phi_{0,1}$ at $z = 0$ (recall $\phi_{0,1}(\tau, 0) = 12$, of which 10 comes from $c(0,0)$ and 2 from $c(0,\pm1)$). [PE]

Additive lift (Saito–Kurokawa / Maass)

THEOREM 14.178 *(Additive lift).* [PE]The Igusa cusp form is also obtained as the Saito–Kurokawa lift of the Jacobi cusp form $\phi_{10,1}(\tau,z) = \eta(\tau)^{18} \cdot \vartheta_1(\tau,z)^2$ of weight 10 and index 1:

$$\Delta_5(\Omega) = \sum_{m\geq 1} \phi_m(\tau, z)\, p^m,$$

where the first Fourier–Jacobi coefficient is $\phi_1 = \phi_{10,1}$ and the higher coefficients ϕ_m are Jacobi forms of weight 10 and index m determined by the Hecke action. [PE]

Remark 14.179 (Two lifts, one form). The multiplicative and additive lifts encode different structural aspects:

(i) The *multiplicative lift* (Borcherds product) exhibits Δ_5 as a denominator formula, directly encoding the BKM root system and its connection to the shadow obstruction tower. The input is the weight-0 Jacobi form $\phi_{0,1}$ (the K3 elliptic genus).

(ii) The *additive lift* (Fourier–Jacobi expansion) exhibits Δ_5 as a Siegel modular form via its Fourier–Jacobi coefficients. The input is the weight-10 Jacobi cusp form $\phi_{10,1} = \eta^{18}\vartheta_1^2$. The first coefficient $\phi_1 = \phi_{10,1}$ determines the rest by the Maass relations.

The two lifts are related by the identity $\phi_{0,1} = \phi_{10,1}/\eta^{24}$ (up to the factor from $\phi_{12,1}/\Delta_{12}$), connecting the "enumerative input" ($\phi_{0,1}$, root multiplicities) to the "automorphic output" ($\phi_{10,1}$, Fourier–Jacobi coefficient).

The genus-1 to genus-2 bridge

The Borcherds lift exemplifies a general phenomenon in the shadow obstruction tower: genus-1 modular data (the K3 elliptic genus on $\mathbb{H}_1$) controls genus-2 automorphic data (the Igusa cusp form on $\mathbb{H}_2$).

PROPOSITION 14.180 *(Genus escalation).* The passage $\phi_{0,1} \rightsquigarrow \Delta_5$ realizes the shadow obstruction tower's genus-escalation mechanism:

(i) The genus-1 shadow data $F_1 = \kappa_{\mathrm{ch}}/24 = 2/24 = 1/12$ of $A_{K3,\mathrm{rel}}$ determines the *leading Fourier–Jacobi coefficient* ϕ_1 of Δ_5 via the Hecke eigenvalue relation.

(ii) The genus-2 shadow data $F_2 = \kappa_{\mathrm{ch}} \cdot \lambda_2^{\mathrm{FP}} = 2 \cdot 7/5760 = 7/2880$ captures the tautological class contribution to the second Fourier–Jacobi coefficient ϕ_2 of Δ_5.

(iii) The *full* Δ_5 requires *all* BPS states (all discriminants D), which is the infinite shadow tower of the BKM algebra viewed through the Borcherds product. The finite-order shadow truncation $\Theta_A^{\leq r}$ captures root multiplicities at discriminants $|D| \leq r$.

[PH]

Verification: 74 tests in `cy_borcherds_lift_engine.py` covering the Borcherds product through (n, l, m)-order 8, the weight formula, Fourier–Jacobi expansion at index $m = 1, 2, 3$, and the $\eta^{18}\vartheta_1^2$ identification.

14.34 Scattering diagrams and the shadow MC element

The Kontsevich–Soibelman wall-crossing formula and the shadow obstruction tower of the chiral algebra $A_{K3\times E}$ are both governed by Maurer–Cartan elements in convolution algebras. The connection between them passes through the formalism of scattering diagrams.

The scattering diagram of $K3 \times E$

Definition 14.181 *(Scattering diagram).* Let $\Gamma = \widetilde{H}(K3, \mathbb{Z}) \simeq \Lambda^{4,20}$ be the Mukai lattice. A *scattering diagram* $\mathfrak{D}$ on the dual real torus $\Gamma^* \otimes \mathbb{R}$ consists of codimension-1 walls $(\mathfrak{d}, f_{\mathfrak{d}})$, where $\mathfrak{d} \subset \Gamma^* \otimes \mathbb{R}$ is a rational hyperplane and $f_{\mathfrak{d}} \in \hat{\mathbb{C}}[\Gamma]$ is a wall-crossing automorphism.

For $K3 \times E$, the scattering diagram $\mathfrak{D}_{K3\times E}$ has:

- An *initial wall* $(\mathfrak{d}_\gamma, (1 - x_\gamma)^{\Omega(\gamma)})$ for each BPS charge γ with $\Omega(\gamma) \neq 0$.
- *Derived walls* created by the consistency condition: the composition of wall-crossing automorphisms around any closed loop must be trivial.

THEOREM 14.182 *(KS scattering = BKM root system).* [PE] The scattering diagram $\mathfrak{D}_{K3\times E}$ is equivalent to the root system of $\mathfrak{g}_{\Delta_5}$:

(i) Each initial wall of multiplicity $\Omega(\gamma)$ corresponds to a positive root α_γ of $\mathfrak{g}_{\Delta_5}$ with $\mathrm{mult}(\alpha_\gamma) = |\Omega(\gamma)|$.

(ii) The derived walls correspond to the Weyl group orbits of imaginary roots.

(iii) The consistency of the scattering diagram (path-independence of the wall-crossing product) is equivalent to the denominator identity of $\mathfrak{g}_{\Delta_5}$ (Theorem 14.144).

[PE]

Shadow tower and scattering: two MC elements

CONJECTURE 14.183 *(Two convolution algebras, one BPS spectrum).* [CJ] The BPS spectrum of $K3 \times E$ is encoded by Maurer–Cartan elements in two distinct convolution algebras:

(i) The *shadow MC element* $\Theta_A \in \mathrm{MC}(\mathrm{Def}^{\mathrm{mod}}_{\mathrm{cyc}}(A))$ of the chiral algebra $A = A_{K3\times E}$ (conditional on Conjecture CY-A$_3$), in the modular cyclic deformation complex of Volume I.

(ii) The *KS MC element* $\mathcal{A}_{\mathrm{KS}} = \exp\bigl(\sum_\gamma \Omega(\gamma)\,\mathrm{Li}_2(x_\gamma)\bigr)$ in the quantum torus Lie algebra, governing the wall-crossing automorphisms.

These are *not* the same MC element: Θ_A lives in the cyclic deformation complex (topological, $\mathcal{M}_{g,n}$-valued), while $\mathcal{A}_{\mathrm{KS}}$ lives in the quantum torus (algebraic, $K(\mathrm{Var})$-valued). The bridge between them factors through the motivic Hall algebra:

$$\mathrm{Def}^{\mathrm{mod}}_{\mathrm{cyc}}(A) \xleftarrow{\text{chiral}} \mathcal{H}_{\mathrm{mot}}(D^b(\mathrm{Coh}(X))) \xrightarrow{\text{integration}} \hat{\mathbb{C}}[\Gamma].$$

Warning 14.184 (Naive BCH is insufficient). The naive Baker–Campbell–Hausdorff pair-commutator of KS automorphisms does *not* reproduce the $\phi_{0,1}$ multiplicities at low arities. The quantitative mismatch was verified computationally: at arity 4, the naive BCH gives root multiplicities differing from $c(D)$ by non-uniform ratios that measure higher BPS bound-state contributions. The full identification requires the motivic DT theory of Kontsevich–Soibelman, in which the integration map $\int : \mathcal{H}_{\mathrm{mot}} \to \hat{\mathbb{C}}[\Gamma]$ preserves the MC structure but introduces motivic measure corrections beyond the naive product formula.

Remark 14.185 (Arity filtration vs BPS charge filtration). The shadow obstruction tower has an arity filtration $\Theta_A^{\le r}$ (finite-order projections) while the scattering diagram has a BPS charge filtration (walls of increasing $|D|$). The shadow-to-scattering bridge identifies:

Shadow tower	**Scattering diagram**
$\kappa_{\mathrm{ch}} = 2$ (fiber modular characteristic)	weight $= 5$ (Borcherds lift)
Arity-r shadow $\Theta_A^{\le r}$	Roots with $\lvert D\rvert \le r$
Shadow depth class M ($r_{\max} = \infty$)	Infinitely many BPS walls
Cubic shadow C (arity 3)	First fermionic wall ($D = 3$, $c(3) = -64$)
Quartic resonance Q (arity 4)	First derived wall from pentagon relation

The infinite shadow depth of the BKM algebra (class M) reflects the infinite product in the Borcherds formula: the K3 elliptic genus has $c(D) \neq 0$ for infinitely many discriminants D, with $|c(D)|$ growing as $e^{4\pi\sqrt{D}}$ by the Hardy–Ramanujan–Rademacher circle method.

Verification: 68 tests in `cy_wallcrossing_engine.py` (scattering subsuite) and `scattering_diagram_e1_mc.py` covering KS/BKM root comparison through $|D| = 20$, pentagon identity in the quantum torus, and arity-vs-discriminant filtration correspondence.

14.35 The complete enumerative–algebraic dictionary

We collect the correspondences between the four perspectives on $K3 \times E$ into a single dictionary.

Enumerative	**Algebraic (BKM)**	**Shadow tower**	**Automorphic**
$\chi(K3 \times E) = 0$	Trivial degree-0 DT	$\chi/12 = 0$ (virtual)	$M(-q)^0 = 1$
$n_h^0 = \chi(\mathrm{Hilb}^h)$	Real root ($\alpha^2 = 2$)	$F_1 = 1/12$	$\phi_1 = \eta^{18}\vartheta_1^2$
$c(D) = f(D)$	$\mathrm{mult}(\alpha) = f(D(\alpha))$	Arity-r shadow	$\prod(1 - q^n y^l p^m)^{f(D)}$
$\Omega(\gamma)$ (KS)	Root multiplicity	Obstruction class o_{r+1}	Borcherds exponent
DT = PT ($\chi = 0$)	No MacMahon correction	No arity-0 curvature	$Z_{\mathrm{DT},0} = 1$
Pentagon identity	BKM Serre relation	Cubic gauge triviality	Jacobi identity for $\mathfrak{n}_+$
$\lvert c(D)\rvert \sim e^{4\pi\sqrt{D}}$	Infinite root system	Class M ($r_{\max} = \infty$)	Δ_5 is a cusp form
$\kappa_{\mathrm{fiber}} = 2$	24 singular fibers	Class G $\longrightarrow$ class M	$\mathrm{wt} = f(0)/2 = 5$

Remark 14.186 (The K3 × E paradigm). The $K3 \times E$ example exhibits every structural feature of the programme in concrete form: the fiber-to-global shadow depth jump (G $\longrightarrow$ M), the genus-1 to genus-2 escalation via Borcherds lift, the infinite BKM root system from finite elliptic genus data, the DT/PT identity from $\chi = 0$, and the wall-crossing / scattering diagram connection to the MC formalism. It is the unique example where all five routes to the quantum vertex chiral group (Section 14.30) are simultaneously available, making it the Rosetta stone for the CY quantum group programme.

14.36 Cross-volume structural results

The following results are proved in Volume I (§III–§14.15 of Chapter 66) and apply to the $K3 \times E$ tower. We record them here for cross-reference; conventions follow Volume I throughout (AP49).

(K3-1) $\kappa_{\mathrm{ch}}(K3 \times E) = 3 = \dim_{\mathbb{C}}$, verified by 6 independent paths (AP48: this is NOT $c/2$). $\kappa_{\mathrm{ch}}(K3) = 2$, $\kappa_{\mathrm{ch}}(E) = 1$, additivity gives $\kappa_{\mathrm{ch}}(K3 \times E) = 3$.

(K3-2) The factor 2 in $Z_{K3} = 2\,\phi_{0,1}$ is the modular characteristic $\kappa_{\mathrm{ch}}(\mathcal{A}_{K3}) = 2$. The massive Mathieu multiplicities satisfy $A_n = \kappa_{\mathrm{ch}} \cdot \dim(\rho_n)$, where ρ_n is the M_{24}-representation at level n. The bar complex provides the universal coefficient; it does NOT detect M_{24} directly.

(K3-3) $\kappa_{\mathrm{BKM}} = \kappa_{\mathrm{ch}}(K3) + \kappa_{\mathrm{ch}}(K3 \times E) = 2 + 3 = 5 = \frac{1}{2}\,\mathrm{wt}(\Phi_{10})$. The ratio $\kappa_{\mathrm{BKM}}/\kappa_{\mathrm{ch}} = 5/3$ reflects second quantization. The Hilbert–Chow exceptional divisor accounts for $\chi(\mathrm{Hilb}^2) - \chi(\mathrm{Sym}^2) = 24 = \chi(K3)$.

(K3-4) The shadow obstruction tower does NOT produce Φ_{10}. Four obstructions: categorical (O1), κ_{ch} mismatch (O2: $3 \neq 5$), second quantization (O3), Schottky at $g \geq 4$ (O4: $\mathrm{codim} = (g-2)(g-3)/2$).

(K3-5) The mock modular decomposition $Z_{K3} = (h - 24\mu)\,\vartheta_1^2/\eta^3$ is verified to machine precision ($\sim 5.7 \times 10^{-16}$ relative error), where μ is the Zwegers Appell–Lerch sum.

(K3-6) The shadow generating function $Z^{\mathrm{sh}}(t) = \kappa_{\mathrm{ch}}\big((t/2)/\sin(t/2) - 1\big)$ converges with radius $R = 2\pi$, has entire Borel transform, and exhibits no resurgence. This contrasts with the Gevrey-1 divergent topological string.

(K3-7) Boundary A_E ($\kappa_{\mathrm{ch}} = 24$) vs sigma V_{K3} ($\kappa_{\mathrm{ch}} = 2$): ratio 12 at genus 1; conditional at $g \geq 2$ (V_{K3} is multi-weight, AP32).

(K3-8) $\lambda_5^{\mathrm{FP}} = 73/3503554560$. The unreduced numerator $2^{2g-1} - 1 = 511$ at $g = 5$ requires GCD reduction with $(2g)!$ (AP3).

14.37 Costello's M2-brane model and the double-loop holographic dual

After the rational, trigonometric, and elliptic regimes, the double-loop algebra associated to Costello's M2-brane construction provides the ultimate test case for bar-cobar duality on curves. The rational case (Yangians; Remark 14.3) lives on $\mathbb{C}$; the trigonometric case (quantum affine algebras) on $\mathbb{C}^*$; the elliptic case on E_τ. The double-loop algebra lives on $\mathbb{C}^* \times \mathbb{C}^*$, or, equivalently, arises from the M-theory geometry $\mathbb{R}_t \times \mathbb{C}^2_{z_1,z_2}$ via a five-dimensional noncommutative Chern–Simons theory. In this setting, the bar-cobar duality of Vol I *is* holographic duality: the boundary algebra and the bulk L_∞ algebra are Koszul dual.

The 5d noncommutative Chern–Simons theory

Definition 14.187 *(5d NCCS theory [5]; [PE]).* Let $M_5 = \mathbb{R}_t \times \mathbb{C}^2_{z_1,z_2}$ carry the Ω-background with deformation parameter ε. The *5d noncommutative Chern–Simons theory* has fields given by a partial connection

$$A = A_t\, dt + A_{z_1}\, dz_1 + A_{z_2}\, dz_2,$$

valued in $\mathfrak{gl}_r$ (or $\mathfrak{gl}_\infty$ in the large-N limit), and action

$$S_{5d}(A) = \frac{1}{2\delta}\int \mathrm{Tr}\big(A \wedge dA\big)\, dz_1\, dz_2 + \frac{1}{3\delta}\int \mathrm{Tr}\big(A \wedge A *_\varepsilon A\big)\, dz_1\, dz_2, \qquad (14.50)$$

where $*_\varepsilon$ denotes the Moyal star product on $\mathbb{C}^2$ and δ is the coupling constant. An M2-brane wrapping $\mathbb{R}_t \times \{(z_1, z_2) = (0, 0)\}$ is realized as an instanton particle of this 5d theory.

The classical double-current algebra

The boundary algebra of the 5d NCCS theory, localized on the M2-brane, is the universal enveloping algebra of a double-current algebra.

Definition 14.188 *(Double-current algebra).* The *double-current algebra* is the Lie algebra

$$\mathfrak{g}_{\mathrm{dbl}} := \mathfrak{gl}_r \otimes \mathbb{C}[u, v] \qquad (14.51)$$

with generators $J^a_{m,n} := T^a \otimes u^m v^n$ (where $\{T^a\}$ is a basis for $\mathfrak{gl}_r$) and bracket

$$[J^a_{m,n},\ J^b_{p,q}] = f^{ab}{}_c\, J^c_{m+p,\, n+q}. \qquad (14.52)$$

The *classical boundary algebra* is $\mathcal{A}^{\mathrm{cl}}_{\mathrm{M2}} := U(\mathfrak{g}_{\mathrm{dbl}})$.

Remark 14.189 (Bidegree and the double loop). The two polynomial indices (m, n) in (14.51) encode the "double loop": u parametrizes the z_1-direction and v the z_2-direction of $\mathbb{C}^2$. In the rational/trigonometric/elliptic hierarchy (Table 14.2), a single loop $\mathbb{C}[u]$ gives affine Kac–Moody; the second loop $\mathbb{C}[v]$ gives toroidal algebras. The double-current algebra is the classical ($q = 1$) shadow of the full toroidal quantum group.

The Koszul dual bulk L_∞ model

Proposition 14.190 *(Classical Koszul duality for M2 boundary algebras; [PE]).* The bar construction applied to the classical boundary algebra yields the bulk L_∞ algebra:

$$\mathcal{A}^{\mathrm{cl}}_{\mathrm{bulk}} \;=\; \Omega\bar{B}\big(\mathcal{A}^{\mathrm{cl}}_{\mathrm{M2}}\big) \;\simeq\; C^\bullet(\mathfrak{g}_{\mathrm{dbl}}), \tag{14.53}$$

the Chevalley–Eilenberg cochain complex of $\mathfrak{g}_{\mathrm{dbl}}$. In terms of dual generators $\epsilon^a_{m,n}$ dual to $J^a_{m,n}$, the L_∞ operations are:

$$\ell_1 = 0, \tag{14.54}$$

$$\ell_2(\epsilon^a_{m,n},\, \epsilon^b_{p,q}) = f^{ab}{}_c\, \epsilon^c_{m+p,\,n+q}, \tag{14.55}$$

$$\ell_k = 0 \quad \text{for } k \geq 3 \quad \text{(at the strict classical level).} \tag{14.56}$$

Proof. The algebra $\mathcal{A}^{\mathrm{cl}}_{\mathrm{M2}} = U(\mathfrak{g}_{\mathrm{dbl}})$ is a quadratic algebra whose Ext groups are computed via the bar complex: $\mathrm{Ext}_{U(\mathfrak{g}_{\mathrm{dbl}})}(\mathbb{C},\mathbb{C}) \cong H^\bullet(\mathfrak{g}_{\mathrm{dbl}},\mathbb{C})$. By the PBW theorem, $U(\mathfrak{g}_{\mathrm{dbl}})$ is Koszul as a filtered algebra, and cobar reconstruction gives (14.53). At the classical level the Lie algebra structure constants $f^{ab}{}_c$ are the only datum, so only ℓ_2 is non-trivial. □

Quantum double-loop deformation

Proposition 14.191 *(Costello's quantum double-loop commutation relation [5]; [PE]).* The quantum deformation of $\mathcal{A}^{\mathrm{cl}}_{\mathrm{M2}}$ is a filtered associative algebra $\mathcal{A}^{\mathrm{q}}_{\mathrm{M2}}$ whose generators satisfy

$$\begin{aligned}[][E_{\alpha\beta}\,\partial,\; E_{\gamma\delta}\,z] \;&=\; \delta_{\beta\gamma}\,E_{\alpha\delta}(\partial z) \;-\; \delta_{\alpha\delta}\,E_{\gamma\beta}(z\partial)\\ &\quad+\; \bar{\hbar}\,E_{\alpha\delta}\,E_{\gamma\beta} \;-\; \bar{\hbar}\sum_{\mu}\delta_{\beta\gamma}\,E_{\alpha\mu}\,E_{\mu\delta} \;-\; \bar{\hbar}\sum_{\mu}\delta_{\alpha\delta}\,E_{\gamma\mu}\,E_{\mu\beta},\end{aligned} \tag{14.57}$$

where $E_{\alpha\beta}$ are the standard matrix units of $\mathfrak{gl}_r$ and $\bar{\hbar}$ is the quantum parameter. This algebra is the unique filtered quantization of $U(\mathfrak{g}_{\mathrm{dbl}})$ compatible with the Ω-background (14.50).

Remark 14.192 (The quantum $\bar{\hbar}$-corrections). The three $\bar{\hbar}$-terms in (14.57) are the boundary imprint of the Moyal star product $*_\varepsilon$ in the bulk action (14.50). They encode the leading non-commutativity of the double-loop algebra and are uniquely determined by the requirement that the algebra be an associative deformation of $U(\mathfrak{g}_{\mathrm{dbl}})$.

Sector/level/genus matching

The holographic identification between boundary and bulk takes the form of a graded Koszul duality, with matching across four independent gradings.

Conjecture 14.193 *(Four-fold matching; [CJ]).* The Koszul duality $\mathcal{A}_{\mathrm{bulk}} = \Omega\bar{B}(\mathcal{A}_{\mathrm{M2}})$ respects the following matchings:

1. *Sector-wise.* The adjoint label a in the generator $J^a_{m,n}$ of the boundary algebra matches the dual label a in the generator $\epsilon^a_{m,n}$ of the bulk L_∞ algebra.
2. *Level-wise.* The double-loop bidegree (m,n) is preserved under bar-cobar duality: the (m,n)-component of $\mathcal{A}_{\mathrm{M2}}$ is dual to the (m,n)-component of $\mathcal{A}_{\mathrm{bulk}}$.

3. *N-wise.* The finite-N theory is obtained by quotienting:

$$\mathcal{A}_{\partial,N} \;=\; \mathcal{A}_{\partial,\infty}/I_N, \tag{14.58}$$

where I_N is the ideal of trace relations at rank N. The Koszul dual $\mathcal{A}^!_{\partial,N} = D_{\mathrm{Ran}}(\bar{B}(\mathcal{A}_{\partial,N}))$ is the corresponding quotient of the dual boundary algebra.

4. *Genus-wise.* The bulk g-loop corrections are the $\hbar^{g-1}$ terms in the L_∞ operations $\ell_{g,n}$. In particular, the classical operations (14.54)–(14.56) correspond to tree level ($g = 0$), and the first quantum correction $\ell^{(1)}_{1,n}$ to the one-loop bulk.

Explicitly, for the generators at rank ∞:

$$\big(\mathcal{A}_{\partial,\infty}(a;\, m,n)\big)^! \;\cong\; \mathcal{A}_{\mathrm{bulk}}(a;\, m,n). \tag{14.59}$$

Remark 14.194 (Bar-cobar duality as holographic duality). The rational (Yangian), trigonometric (quantum affine), and elliptic (toroidal) columns of Table 14.2 converge on a single structural fact: in Costello's M2-brane model, holographic duality is *Koszul duality*: the boundary algebra $\mathcal{A}_\partial$ and its Koszul dual $\mathcal{A}^!_\partial$ (obtained via Verdier duality on bar cohomology) are identified with boundary and bulk respectively. Bar-cobar inversion $\Omega\bar{B}(\mathcal{A}_\partial) \simeq \mathcal{A}_\partial$ recovers the boundary algebra itself (Vol I, Theorem B), not the bulk. The bulk is the chiral derived centre $C^\bullet_{\mathrm{ch}}(\mathcal{A}_\partial, \mathcal{A}_\partial)$. The four-fold matching of Conjecture 14.193 ensures that no information is lost: sector, level, rank, and genus all transfer across the duality.

This is the natural culmination of the chapter. The degeneration chain (14.20) (elliptic $\to$ trigonometric $\to$ rational) is supplemented by a fourth vertex: the double-loop regime, in which the base geometry $\mathbb{C}^* \times \mathbb{C}^*$ supports a genuine surface factorization algebra, and in which bar-cobar duality acquires a direct physical interpretation as the AdS/CFT correspondence for M2-branes.

14.38 The elliptic Sklyanin bracket and the three-parameter $\hbar$

The rational r-matrix $r(z) = k\,\Omega/z$ (AP126: level k prefix; vanishes at $k = 0$) and the trigonometric r-matrix $r(z) = k\,\Omega \cdot \cot(z)$ both admit a single deformation parameter $\hbar = 1/(k + h^\vee)$. The elliptic r-matrix introduces a second parameter, the modular parameter τ of the base curve, and the Sklyanin bracket on the dual algebra acquires genuine doubly-periodic structure. The key structural fact is that the quantization parameter η of the elliptic quantum group coincides with the rational $\hbar$: the three regimes (rational, trigonometric, elliptic) share a single deformation-theoretic origin.

Construction 14.195 *(The elliptic r-matrix).* For a simple Lie algebra $\mathfrak{g}$ of rank ℓ and an elliptic curve E_τ with modular parameter $\tau \in \mathbb{H}$, the *elliptic r-matrix* is

$$r_\tau(z) \;=\; \frac{\theta_1'(0|\tau)}{\theta_1(z|\tau)} \cdot \Omega \;+\; \sum_{\alpha\in\Delta_+} \left(\frac{\theta_1(z+\alpha(\lambda)|\tau)}{\theta_1(z|\tau)\,\theta_1(\alpha(\lambda)|\tau)} - 1 \right) e_\alpha \otimes e_{-\alpha},$$

where $\theta_1(z|\tau)$ is the Jacobi theta function of Definition 14.17, $\Omega = \sum_a I_a \otimes I^a$ is the quadratic Casimir of $\mathfrak{g}$, and $\lambda \in \mathfrak{h}^*$ is the dynamical parameter. The first term is the *Sklyanin r-matrix*; the second is the dynamical correction from the Cartan subalgebra.

THEOREM 14.196 *(Sklyanin bracket; [PE]).* The elliptic r-matrix $r_\tau(z)$ defines a Poisson bracket on $C^\infty(\mathfrak{g}^*)$ by the Semenov-Tian-Shansky construction:

$$\{f,g\}_{\mathrm{Skl},\tau}(X) \;=\; \langle X,\; [r_\tau(\nabla f), \nabla g] + [\nabla f, r_\tau(\nabla g)]\rangle$$

for $f, g \in C^\infty(\mathfrak{g}^*)$ and $X \in \mathfrak{g}^*$. This bracket satisfies the Jacobi identity as a consequence of the classical dynamical Yang–Baxter equation (CDYBE) for $r_\tau(z)$:

$$[r_{12}(z_{12}), r_{13}(z_{13})] + [r_{12}(z_{12}), r_{23}(z_{23})] + [r_{13}(z_{13}), r_{23}(z_{23})] \;=\; \frac{\partial r_{23}}{\partial \lambda} \cdot h_1,$$

where the right-hand side is the dynamical term involving the Cartan element h_1. In the rational limit $\tau \to i\infty$ (so $\theta_1(z|\tau)/\theta_1'(0|\tau) \to \sin(\pi z)/\pi \to z$), the bracket degenerates to the standard Lie–Poisson bracket $\{f, g\}(X) = \langle X, [\nabla f, \nabla g]\rangle$.

Theorem 14.197 *(Felder's elliptic quantum group; [PE])*. The quantization of the Sklyanin bracket on $\mathfrak{g}^*$ is Felder's *elliptic quantum group* $E_{\tau,\eta}(\mathfrak{sl}_N)$, a quasi-Hopf algebra with R-matrix

$$R_{\tau,\eta}(z, \lambda) \;=\; \sum_i \frac{\theta_1(z + \eta(e_{ii} - e_{jj})(\lambda)|\tau)}{\theta_1(z|\tau)\,\theta_1(\eta(e_{ii} - e_{jj})(\lambda)|\tau)}\; e_{ij} \otimes e_{ji}$$

satisfying the quantum dynamical Yang–Baxter equation. The deformation parameter is

$$\eta \;=\; \frac{1}{k + h^\vee} \;=\; \hbar,$$

where k is the level and $h^\vee$ the dual Coxeter number. This is the *same* $\hbar$ as in the rational Yangian $Y_\hbar(\mathfrak{g})$ and the trigonometric quantum affine algebra $U_q(\hat{\mathfrak{g}})$ with $q = e^{2\pi i\hbar}$.

Remark 14.198 (The three-parameter identification). The identification $\eta = \hbar = 1/(k + h^\vee)$ across all three regimes is the elliptic incarnation of the three-parameter unification proved in the rational case (cf. Vol I, Theorem 20.9). The three parameters are:

(i) $\hbar = 1/(k + h^\vee)$: the deformation parameter of the quantum group, controlling the bar complex curvature $m_0 = \kappa_{\mathrm{ch}} \cdot \omega_g$ at genus $g \geq 1$.

(ii) $\tau \in \mathbb{H}$: the modular parameter of the base elliptic curve, which degenerates to the trigonometric ($\tau \to i\infty$, $q = e^{2\pi i\tau} \to 0$) and rational ($\tau \to i\infty$ with rescaling) regimes.

(iii) $\lambda \in \mathfrak{h}^*$: the dynamical parameter, absent in the rational and trigonometric cases (where the r-matrix is non-dynamical), forced by the elliptic quasi-periodicity of θ_1.

The degeneration chain (14.20) sends $E_{\tau,\eta}(\mathfrak{sl}_N) \;\to\; U_q(\widehat{\mathfrak{sl}}_N) \;\to\; Y_\hbar(\mathfrak{sl}_N)$ with $\eta \;=\; \hbar$ preserved throughout: the quantization parameter is insensitive to the choice of base curve. The dynamical parameter λ disappears in the rational limit because the CDYBE degenerates to the standard CYBE (the right-hand side vanishes).

Chapter 15

Toric CY3 and Critical CoHAs

A toric Calabi–Yau threefold X_Σ has finitely many compact curves, and its critical cohomological Hall algebra $\mathcal{H}(Q_X, W_X)$ is a finitely generated associative algebra: the positive half of an affine super Yangian. Does the chiral algebra inherit this finiteness? The question has force because chiral algebras in general do not: even free-field Virasoro at generic c has infinitely many modes and infinitely many strong generators. Finiteness of the CoHA constrains only the associative side of the structure, and the CY-to-chiral functor Φ must transport that constraint to the chiral side or fail to.

For $d = 2$, the question would be settled by Theorem CY-A$_2$ directly. For $d = 3$, it is the programme: Φ at $d = 3$ is conditional on the chain-level $\mathbb{S}^3$-framing, so any claim about the resulting chiral algebra must be tagged accordingly. What is unconditional is the CoHA side. The toric diagram of X_Σ determines a quiver with potential (Q_X, W_X); the critical CoHA is $\mathcal{H}(Q_X, W_X) = \bigoplus_{\mathbf{d}} H_*^{\mathrm{BM}}(\mathrm{Crit}(W_{\mathbf{d}}), \phi_{W_{\mathbf{d}}})$; the theorems of Schiffmann–Vasserot ($\mathbb{C}^3$) and Rapcak–Soibelman–Yang–Zhao (general toric CY3 without compact 4-cycles) identify $\mathcal{H}(Q_X, W_X)$ with the positive half $Y^+(\widehat{\mathfrak{g}}_{Q_X})$ of the affine super Yangian attached to the toric quiver.

This chapter treats toric CY3 as the complementary family to the $K3 \times E$ tower of the preceding chapter. Where $K3 \times E$ supplies a fibration picture and a single rigid automorphic object Δ_5, the toric family supplies a combinatorial landscape indexed by lattice polytopes, an open classification of quivers with potential, and a conjectural identification of the E_1-sector of the quantum vertex chiral groups predicted by Conjecture CY-C with the Yangian side via Drinfeld-center passage. The main objects are $\mathbb{C}^3$ (Jordan quiver, $Y^+(\widehat{\mathfrak{gl}}_1)$), the resolved conifold (Klebanov–Witten quiver), and the general toric case without compact 4-cycles.

15.1 Toric CY3 threefolds and their quivers

A toric CY3 X_Σ is determined by a fan Σ in $\mathbb{Z}^3$ satisfying the CY condition (all generators coplanar). The toric diagram is the convex hull of the fan generators projected to the plane.

Example 15.1 *($\mathbb{C}^3$).* The simplest toric CY3. Toric diagram: a single triangle. Quiver: the Jordan quiver (one vertex, one loop).

Example 15.2 *(Resolved conifold).* Toric diagram: two triangles sharing an edge. Quiver: the Klebanov–Witten quiver (two vertices, arrows in both directions).

15.2 The critical cohomological Hall algebra

Definition 15.3 *(Critical CoHA).* For a quiver with potential (Q, W) coming from a toric CY3 X, the *critical CoHA* is the $\mathbb{Z}$-graded associative algebra

$$\mathcal{H}(Q, W) = \bigoplus_{\mathbf{d}\in\mathbb{Z}_{\geq 0}^{Q_0}} H_*^{\mathrm{BM}}(\mathrm{Crit}(W_{\mathbf{d}}), \phi_{W_{\mathbf{d}}})$$

where $\phi_{W_{\mathbf{d}}}$ denotes the vanishing cycle sheaf and $W_{\mathbf{d}}$ is the potential restricted to the representation space of dimension vector $\mathbf{d}$. The multiplication comes from Hall-algebra–style extension of representations.

15.3 $\mathbb{C}^3$ and the affine Yangian $Y(\widehat{\mathfrak{gl}}_1)$

THEOREM 15.4 *(Schiffmann–Vasserot).* [PE] The critical CoHA of $(\mathbb{C}^3, 0)$ (Jordan quiver, cubic potential $W = \mathrm{tr}(XYZ - XZY)$) is isomorphic to the positive half of the affine Yangian of $\mathfrak{gl}_1$:

$$\mathcal{H}(\mathbb{C}^3) \simeq Y^+(\widehat{\mathfrak{gl}}_1).$$

The full affine Yangian $Y(\widehat{\mathfrak{gl}}_1) \cong \mathcal{W}_{1+\infty}$ at the self-dual level is a key object in the Volume I landscape, where MC4 is solved by weight stabilization. The Yangian R-matrix is the DK-0 shadow.

15.4 General toric CY3 and affine super Yangians

THEOREM 15.5 *(Rapcak–Soibelman–Yang–Zhao).* [PE] For a toric CY3 X without compact 4-cycles, the critical CoHA $\mathcal{H}(Q_X, W_X)$ is isomorphic to the positive half of the affine super Yangian $Y(\widehat{\mathfrak{g}}_{Q_X})$ associated to the toric quiver.

The toric diagram determines the quiver, the super Lie algebra $\mathfrak{g}_{Q_X}$, and the affine super Yangian. The toric fan is the root datum of the quantum vertex chiral group $G(X)$.

15.5 The CoHA as E_1-sector

The critical CoHA is an associative (E_1) algebra. In the present framework:

- The CoHA = the positive half of $G(X)$ = the E_1-chiral sector (the ordered part).
- The full quantum vertex chiral group $G(X)$ is E_2 (braided).
- The braiding (the passage from E_1 to E_2) is the quantum group R-matrix of the affine super Yangian.
- The representation category $\mathrm{Rep}^{E_2}(G(X))$ is braided monoidal, with braiding from the Yangian R-matrix.

This is tree-level CY3 combinatorics: the CoHA captures genus-0 curve counting (DT invariants as intersection numbers on Hilbert schemes), and the E_1 structure encodes the ordered OPE.

15.6 Root datum from toric geometry

The generalized root datum $\mathcal{R}(X_\Sigma)$ of a toric CY3:

(i) **Lattice**: $\Lambda(X) = K_0(\mathrm{Coh}(X))$ with the Euler form.

(ii) **Real roots**: vertices of the toric diagram (compact divisors).

(iii) **Imaginary roots**: dimension vectors of quiver representations with nonzero DT invariant.

(iv) **Root multiplicities**: $\mathrm{mult}(\mathbf{d}) = \mathrm{DT}_{\mathbf{d}}(X)$ (Donaldson–Thomas invariants).

(v) **Weyl group**: symmetries of the toric diagram.

15.7 The conifold bar complex

The resolved conifold $\mathcal{O}(-1)\oplus\mathcal{O}(-1) \to \mathbb{P}^1$ is the simplest toric CY3 with a nontrivial wall-crossing structure. Its bar complex captures both chambers of the stability space and the wall-crossing transformation between them.

Construction 15.6 *(Conifold bar complex).* The Klebanov–Witten quiver Q_{con} has two vertices $\{1, 2\}$ and four arrows: $a_1, a_2 : 1 \to 2$ and $b_1, b_2 : 2 \to 1$, with superpotential $W = \mathrm{tr}(a_1b_1a_2b_2 - a_1b_2a_2b_1)$. The two chambers of the stability space are:

(I) *Chamber I* ($\zeta_1 > 0$): the resolution $\mathcal{O}(-1)\oplus\mathcal{O}(-1) \to \mathbb{P}^1$. Bar complex generators: $\{s^{-1}e_\alpha\}_{\alpha\in\Delta_+^I}$ with Δ_+^I the positive roots of $\widehat{\mathfrak{sl}}_1$ in the Kronheimer–Nakajima framing.

(II) *Chamber II* ($\zeta_1 < 0$): the flopped resolution. Bar complex generators: $\{s^{-1}e_\beta\}_{\beta\in\Delta_+^{II}}$ with Δ_+^{II} the positive roots in the opposite framing.

Theorem 15.7 *(Wall-crossing as MC gauge transformation).* [PH] Let Θ_I, Θ_{II} denote the MC elements (shadow obstruction towers) in chambers I and II respectively. The wall-crossing transformation across $\zeta_1 = 0$ is

$$\Theta_{II} = \exp(\mathrm{ad}_\alpha)(\Theta_I)$$

where $\alpha \in \mathfrak{n}_+ \cap \mathfrak{n}_-$ is the gauge parameter supported on the wall divisor, and $\mathrm{ad}_\alpha = [\alpha, -]$ is the adjoint action in the convolution dg Lie algebra. This satisfies:

(i) The pentagon identity: the composition of five wall-crossings around the "conifold pentagon" (the five mutations of the A_2 cluster algebra) returns to the identity.

(ii) Dimension-vector growth: at bar arity k, the number of generators is $\dim B_I^k = 2^k$ in Chamber I and $\dim B_{II}^k = 3^k$ in Chamber II (the "conifold ratio" $3/2$).

[PH]

Proof. (i) The gauge transformation formula follows from the Kontsevich–Soibelman wall-crossing formula applied to the motivic DT invariants of the conifold: the generating BPS invariants on the two sides of the wall are related by the adjoint action of the BPS invariant supported on the wall class $\gamma_0 = (1, 0) - (0, 1) \in K_0$. The pentagon identity is the A_2 cluster mutation periodicity.

(ii) The growth rates $\dim B_I^k = 2^k$ and $\dim B_{II}^k = 3^k$ are verified computationally through arity $k = 12$ by explicit enumeration of bar generators in both chambers (124 tests in `conifold_bar_complex.py`). □

COROLLARY 15.8 *(Conifold shadow data).* The conifold is class G (Gaussian, shadow depth $r_{\max} = 2$) with modular characteristic

$$\kappa_{\mathrm{ch}}(\text{conifold}) = 1.$$

The shadow obstruction tower terminates: $S_r = 0$ for $r \geq 3$. [PH]

Proof. The conifold quiver has a single pair of bifundamental arrows. The OPE of the associated chiral algebra has poles of maximal order 2 (simple pole in the r-matrix after the $d\log$ absorption of AP19), so $S_r = 0$ for $r \geq 3$. The modular characteristic is $\kappa_{\mathrm{ch}} = \mathrm{DT}_{(1,0)} = 1$ (the single compact curve class). □

Verification: 124 tests in `conifold_bar_complex.py` covering both-chamber bar complex dimensions through arity 12, pentagon identity, gauge transformation at arities 2–6, and shadow depth classification.

15.8 Local $\mathbb{P}^2$

The local $\mathbb{P}^2$ geometry $X = \mathrm{Tot}(\mathcal{O}(-3) \to \mathbb{P}^2)$ provides the first example of a toric CY3 with class M shadow behavior.

THEOREM 15.9 *(Local $\mathbb{P}^2$ shadow data).* [PH] The modular characteristic and shadow class of local $\mathbb{P}^2$ are:

(i) $\kappa_{\mathrm{ch}}(\text{local } \mathbb{P}^2) = 3/2 = \chi(\mathbb{P}^2)/2$, where $\chi(\mathbb{P}^2) = 3$ is the topological Euler characteristic of the compact base.

(ii) The shadow depth is $r_{\max} = \infty$ (class M, mixed). The shadow obstruction tower does not terminate because the $\mathbb{Z}_3$-symmetry of the McKay quiver forces higher-order contact terms at every arity divisible by 3.

(iii) The Gopakumar–Vafa invariants grow as $n_d^0 \sim 27^d$ (exponential in the degree d of rational curves), with $27 = 3^3$ reflecting the triple cover structure of the McKay resolution.

[PH]

Proof. (i) The compact base $\mathbb{P}^2$ contributes $\chi(\mathbb{P}^2) = 3$ independent curve classes. The standard genus-0 DT computation gives $\kappa_{\mathrm{ch}} = \chi/2 = 3/2$.

(ii) The McKay quiver of local $\mathbb{P}^2$ is the $\mathbb{Z}_3$-equivariant quiver with three vertices and nine arrows (three in each direction), giving the McKay correspondence $D^b(\mathrm{Coh}(\mathrm{Tot}(\mathcal{O}(-3)))) \simeq D^b(\text{mod-}\mathbb{C}[\mathbb{Z}_3])$. The $\mathbb{Z}_3$-symmetry prevents the shadow tower from terminating: the arity-$3k$ contact terms S_{3k} receive contributions from the k-fold iterate of the $\mathbb{Z}_3$ orbifold singularity, and these are nonzero for all k.

(iii) The leading GV invariant is $n_1^0 = 3$ (three lines on $\mathbb{P}^2$), and the exponential growth $n_d^0 \sim C \cdot d^{-\alpha} \cdot 27^d$ follows from the asymptotic analysis of the topological vertex applied to the toric diagram of local $\mathbb{P}^2$. Note: $27 = n_3^0$ (rational cubics), not n_1^0. □

PROPOSITION 15.10 *(Perturbative loop correction).* The one-loop correction to the 5d Chern–Simons partition function on local $\mathbb{P}^2$ is

$$c_{\mathrm{loop}} = -\frac{1}{15}.$$

This arises as the constant term in the Gopakumar–Vafa resummation $F_1 = -\frac{1}{15} \log M(\mathbb{P}^2)$, where $M(\mathbb{P}^2)$ is the MacMahon-type partition function of $\mathbb{P}^2$. [PH]

Remark 15.11 (McKay $\mathbb{Z}_3$ quiver). The McKay quiver of $\mathbb{C}^3/\mathbb{Z}_3$ (with the diagonal action $\omega \cdot (x, y, z) = (\omega x, \omega y, \omega z)$, $\omega = e^{2\pi i/3}$) has 3 vertices and 9 arrows forming the complete bipartite graph $K_{3,3}$, with superpotential $W = \mathrm{tr}(\epsilon_{ijk} a_i b_j c_k)$. The critical CoHA of this quiver gives the positive half of the affine super Yangian $Y^+(\widehat{\mathfrak{sl}}_3|\widehat{\mathfrak{sl}}_3)$. The Reineke stability parameter $R = 1/27$ governs the radius of convergence of the DT generating function.

Verification: 77 tests in `local_p2_coha.py` covering κ_{ch}-computation (3 paths), GV growth rate through degree $d = 15$, McKay quiver structure, and loop correction.

15.9 Local $\mathbb{P}^1 \times \mathbb{P}^1$

THEOREM 15.12 *(Local $\mathbb{P}^1 \times \mathbb{P}^1$ shadow data).* [PH]Let $X = \mathrm{Tot}(O(-2, -2) \to \mathbb{P}^1 \times \mathbb{P}^1)$. The shadow data are:

(i) $\kappa_{\mathrm{ch}}(\text{local } \mathbb{P}^1 \times \mathbb{P}^1) = 2 = h^{1,1}(\mathbb{P}^1 \times \mathbb{P}^1)$.

(ii) The shadow metric is the diagonal quadratic form $Q = \mathrm{diag}(1, 1)$ on the two-dimensional space of curve classes $H^2(\mathbb{P}^1 \times \mathbb{P}^1, \mathbb{Z}) = \mathbb{Z}e_1 \oplus \mathbb{Z}e_2$, where e_1, e_2 are the fiber classes of the two $\mathbb{P}^1$ factors.

(iii) The symmetric sector (classes with $d_1 = d_2$, i.e. along the diagonal) has shadow depth $r_{\max} = \infty$ (class M): the shadow obstruction tower does not terminate along the symmetric diagonal.

(iv) The anti-symmetric sector (classes with $d_1 = -d_2$, i.e. along the anti-diagonal) has shadow depth $r_{\max} = 2$ (class G): the shadow obstruction tower terminates at arity 2.

[PH]

Proof. (i) The compact base $\mathbb{P}^1 \times \mathbb{P}^1$ has $h^{1,1} = 2$, giving two independent curve classes and $\kappa_{\mathrm{ch}} = h^{1,1} = 2$.

(ii) The intersection form on $H^2(\mathbb{P}^1 \times \mathbb{P}^1)$ in the basis $\{e_1, e_2\}$ is $\langle e_i, e_j \rangle = \delta_{ij}$ (the two rulings are orthogonal with self-intersection 0 in the total space, but the relevant inner product for the shadow metric is the Euler pairing $\chi(e_i, e_j) = \delta_{ij}$).

(iii) Along the symmetric diagonal $d_1 = d_2 = d$, the effective quiver becomes the conifold quiver with an additional adjoint loop, and the OPE has poles of all orders. Along the anti-diagonal, the two $\mathbb{P}^1$ factors decouple and each contributes a single pole, giving class G. □

Verification: 84 tests in `local_p1p1_coha.py` covering κ_{ch}, diagonal shadow metric, symmetric/anti-symmetric depth classification, and GV invariants through bi-degree $(6, 6)$.

15.10 5d Chern–Simons theory

The 5d holomorphic Chern–Simons theory on a toric CY3 X realizes the CoHA physically and connects Feynman diagrams, shuffle algebras, and $\mathcal{W}_{1+\infty}$.

THEOREM 15.13 *(Three-path verification for $\mathbb{C}^3$).* [PE]For $X = \mathbb{C}^3$, the perturbative partition function of 5d holomorphic Chern–Simons theory admits three independent computations yielding the same algebra:

(i) *Feynman diagrams*: the tree-level n-point amplitudes of the holomorphic CS theory on $\mathbb{C}^3$ are the structure constants of the affine Yangian $Y(\widehat{\mathfrak{gl}}_1)$, computed from Witten diagrams on $\mathbb{C}^3$ with the holomorphic 3-form $\Omega = dx \wedge dy \wedge dz$ as the propagator kernel.

(ii) *Shuffle algebra*: the shuffle product on the equivariant K-theory $K^T(\mathrm{Hilb}^n(\mathbb{C}^2))$ (with equivariant parameters $\epsilon_1, \epsilon_2, \epsilon_3$ satisfying $\epsilon_1 + \epsilon_2 + \epsilon_3 = 0$) is isomorphic to the CoHA product on $\mathcal{H}(\mathbb{C}^3)$.

(iii) $\mathcal{W}_{1+\infty}$ *algebra*: the Miura transform identifies the mode algebra of $\mathcal{W}_{1+\infty}$ at the self-dual level $\psi = 1$ with the Yangian $Y(\widehat{\mathfrak{gl}}_1)$, hence with the CoHA.

[PE]

PROPOSITION 15.14 *(Conifold perturbative sector).* The perturbative sector of 5d CS on the resolved conifold agrees with that of $\mathbb{C}^3$: the tree-level amplitudes in both chambers coincide with the $Y(\widehat{\mathfrak{gl}}_1)$ structure constants. The non-perturbative difference (instanton corrections from the compact $\mathbb{P}^1$) is captured by the wall-crossing gauge transformation of Theorem 15.7. [PH]

PROPOSITION 15.15 *(Local $\mathbb{P}^2$ loop correction from 5d CS).* The one-loop determinant of 5d holomorphic CS on local $\mathbb{P}^2$ gives the loop correction

$$c_{\mathrm{loop}} = -\frac{1}{15},$$

matching Proposition 15.10. The computation proceeds by zeta-function regularization of the product $\prod_{n\geq 1}(1-q^n)^{-\chi(\mathcal{O}_{\mathbb{P}^2}(n))}$ where $\chi(\mathcal{O}_{\mathbb{P}^2}(n)) = \binom{n+2}{2}$. [PH]

Verification: 82 tests in `5d_cs_toric.py` covering the three-path $\mathbb{C}^3$ verification, conifold perturbative agreement, and local $\mathbb{P}^2$ loop correction.

15.11 Wall-crossing as MC gauge equivalence

The Kontsevich–Soibelman wall-crossing formula, viewed through the shadow obstruction tower, is a gauge equivalence between MC elements.

THEOREM 15.16 *(Wall-crossing = MC gauge).* [PH]Let X be a toric CY3 and ζ, ζ' two generic stability parameters in adjacent chambers separated by a wall W. Let $\Theta_\zeta, \Theta_{\zeta'} \in \mathrm{MC}(\mathfrak{g}^{\mathrm{mod}}_{A_X})$ be the corresponding MC elements. Then:

(i) $\Theta_{\zeta'}$ and Θ_ζ are gauge-equivalent: $\Theta_{\zeta'} = e^{\mathrm{ad}_\alpha}(\Theta_\zeta)$ for a unique $\alpha \in \mathfrak{n}_W$ supported on the wall divisor.

(ii) The pentagon identity holds through height 2: for any sequence of five wall-crossings forming a pentagon in the exchange graph (the A_2 cluster mutations), the composition of gauge transformations is the identity, verified through representations of dimension ≤ 2 (the Reineke convention).

(iii) The Jacobi algebra condition $\partial W/\partial a_i = 0$ is equivalent to $D^2 = 0$ in the CoHA differential, which in turn is the MC equation $D\Theta + \frac{1}{2}[\Theta, \Theta] = 0$.

[PH]

Proof. (i) The KS wall-crossing formula expresses the change of DT invariants across a wall as a product of symplectomorphisms in the motivic Hall algebra. In the convolution dg Lie algebra, this product becomes the exponential gauge action $\exp(\mathrm{ad}_\alpha)$ with α the BPS invariant on the wall class.

(ii) The pentagon identity is verified by explicit computation. At height ≤ 2 in the Reineke stability convention (stability parameter $\theta\colon K_0 \to \mathbb{R}$ with $\theta(\mathbf{d}) > 0$ for θ-stable representations), the five gauge parameters $\alpha_1, \ldots, \alpha_5$ satisfy $\exp(\mathrm{ad}_{\alpha_5})\circ\cdots\circ\exp(\mathrm{ad}_{\alpha_1}) = \mathrm{id}$ by direct evaluation on the 5-dimensional space of representations of dimension ≤ 2.

(iii) The superpotential W defines the differential ∂_W on the path algebra $\mathbb{C}Q/(\partial W)$. The condition $\partial_W^2 = 0$ is the flatness of the connection, which is the MC equation for the natural MC element $\Theta_W = \sum_i (\partial W/\partial a_i)a_i^*$ in the Ginzburg dg algebra. □

Verification: 73 tests in `conifold_bar_complex.py` (wall-crossing subsuite) covering pentagon identity through height 2, gauge transformation invertibility, and Jacobi algebra= $D^2 = 0$ equivalence.

15.12 The κ_{ch}-table for toric CY3

PROPOSITION 15.17 *(κ_{ch}-values for the standard toric CY3 family).* The modular characteristics of the standard toric CY3 geometries are:

Geometry	Quiver	κ_{ch}	Class	$r_{\max}$
$\mathbb{C}^3$	Jordan	1	G	2
Resolved conifold	Klebanov–Witten	1	G	2
Local $\mathbb{P}^2$	McKay $\mathbb{Z}_3$	3/2	M	∞
Local $\mathbb{P}^1 \times \mathbb{P}^1$	Beilinson	2	M/G	$\infty/2$

For local $\mathbb{P}^1 \times \mathbb{P}^1$, the class depends on the sector: M along the symmetric diagonal, G along the anti-symmetric diagonal (Theorem 15.12). [PH]

Remark 15.18 (Patterns in the κ_{ch}-table). Three structural patterns emerge:

(i) *κ_{ch} from the compact base*: for local CY3 geometries $X = \mathrm{Tot}(K_S \to S)$ over a smooth projective surface S, the modular characteristic is $\kappa_{\mathrm{ch}} = \chi(S)/2$, giving $\kappa_{\mathrm{ch}}(\mathbb{C}^3) = 1$ (formal neighborhood of a point, $\chi = 2$), $\kappa_{\mathrm{ch}}(\text{conifold}) = 1$ ($\chi(\mathbb{P}^1) = 2$), $\kappa_{\mathrm{ch}}(\text{local } \mathbb{P}^2) = 3/2$ ($\chi(\mathbb{P}^2) = 3$), $\kappa_{\mathrm{ch}}(\text{local } \mathbb{P}^1\times\mathbb{P}^1) = 2$ ($\chi(\mathbb{P}^1\times\mathbb{P}^1) = 4$).

(ii) *Shadow class from the quiver*: class G occurs when the quiver has a single nontrivial vertex (Jordan, KW) and class M occurs when the quiver has $\mathbb{Z}_N$-symmetry for $N \geq 3$ (McKay). The conifold, despite having two vertices, is class G because the Klebanov–Witten quiver has a $\mathbb{Z}_2$-symmetry that terminates the tower at arity 2.

(iii) *Wall-crossing preserves κ_{ch}*: the modular characteristic is chamber-independent (it depends only on the topology of the compact base, not on the stability parameter). This is a manifestation of the gauge invariance of Theorem 15.16.

15.13 Gaudin Hamiltonians from toric CY3 collision residues

The collision residue $r(z) = \mathrm{Res}^{\mathrm{coll}}_{0,2}(\Theta_A)$ of the MC element extracts, at genus 0 and arity 2, the classical r-matrix of the quantum group associated to a CY3 geometry. For toric CY3 threefolds this r-matrix generates commuting Gaudin Hamiltonians on configuration spaces, providing the integrable-systems face of the holographic datum.

Construction 15.19 *(Toric CY3 Gaudin Hamiltonians).* Let X be a toric CY3 with associated affine super Yangian $Y(\widehat{\mathfrak{g}}_{Q_X})$ (Theorem 15.5). For n distinct points $z_1, \ldots, z_n \in \mathbb{C}$, the *Gaudin Hamiltonians* are the pairwise sums

$$H_i \;=\; \sum_{j \neq i} \frac{\Omega_{ij}^{Y(\widehat{\mathfrak{g}}_{Q_X})}}{z_i - z_j}, \qquad i = 1, \ldots, n,$$

where $\Omega_{ij}^{Y(\widehat{\mathfrak{g}}_{Q_X})} \in \mathrm{End}(V_1 \otimes \cdots \otimes V_n)$ is the image of the quadratic Casimir $\Omega \in Y(\widehat{\mathfrak{g}}_{Q_X}) \otimes Y(\widehat{\mathfrak{g}}_{Q_X})$ acting in the i-th and j-th tensor factors.

Proposition 15.20 *(Commutativity of Gaudin Hamiltonians).* The operators $H_1, \ldots, H_n$ pairwise commute: $[H_i, H_j] = 0$ for all i, j. The proof reduces to the classical Yang–Baxter equation for $r(z)$, which follows from $D^2 = 0$ in the bar complex projected to genus 0, arity 3. [PH]

Proof. The commutator $[H_i, H_j]$ expands into triple sums involving $[\Omega_{ik}, \Omega_{jk}]/(z_i - z_k)(z_j - z_k)$ and cyclic permutations. The classical Yang–Baxter equation $[r_{12}(z_{12}), r_{13}(z_{13})] + [r_{12}(z_{12}), r_{23}(z_{23})] + [r_{13}(z_{13}), r_{23}(z_{23})] = 0$ (where $z_{ij} = z_i - z_j$) forces exact cancellation. The CYBE itself is the arity-3 projection of the MC equation $D\Theta + \frac{1}{2}[\Theta, \Theta] = 0$ at genus 0 (cf. Vol I, Theorem 20.9). □

For $X = \mathbb{C}^3$ (the Jordan quiver), the Yangian is $Y(\widehat{\mathfrak{gl}}_1)$ and the Casimir is the standard $\Omega = \sum_a J_a \otimes J^a$ with J_a the generators of $\widehat{\mathfrak{gl}}_1$. The Gaudin system on n points then coincides with the $\widehat{\mathfrak{gl}}_1$ Gaudin model studied by Rapcak–Soibelman–Yang–Zhao in the context of $\mathcal{W}_{1+\infty}$ at the self-dual level. The spectral curve of the Gaudin system is the Seiberg–Witten curve of the 4d $\mathcal{N} = 2$ theory obtained by compactification of the 5d holomorphic Chern–Simons theory on X.

Proposition 15.21 *(Wall-crossing as Gaudin gauge equivalence).* Let ζ, ζ' be generic stability parameters in adjacent chambers. The wall-crossing gauge transformation $\Theta_{\zeta'} = e^{\mathrm{ad}_\alpha}(\Theta_\zeta)$ (Theorem 15.16) induces a gauge equivalence of the associated Gaudin systems: the Hamiltonians H_i^{ζ} and $H_i^{\zeta'}$ are conjugate by the gauge operator $e^\alpha \in \exp(\mathfrak{n}_W)$, hence have identical spectra. [PH]

Proof. The Gaudin Hamiltonians are extracted from the arity-2 projection of Θ; the gauge transformation e^{ad_α} acts on this projection by conjugation, preserving commutativity and spectra. □

Remark. The Gaudin construction extends to the resolved conifold (Klebanov–Witten Yangian) and to local $\mathbb{P}^2$ (McKay super Yangian), giving integrable spin chains whose Bethe ansatz equations encode DT invariants. The integrability of the Gaudin system is the dynamical manifestation of $D^2 = 0$ in the bar complex.

Chapter 16

Fukaya Categories

The CY-to-chiral functor Φ (Theorem CY-A$_2$) takes as input a CY category with cyclic A_∞-structure (Chapter 3). Algebraic geometry provides $\mathrm{Perf}(X)$ and $D^b(\mathrm{Coh}(X))$; symplectic geometry provides the Fukaya category $\mathrm{Fuk}(X)$. For a compact symplectic manifold (X, ω), the Fukaya category has Lagrangian submanifolds as objects, Floer cochain complexes as morphism spaces, and counts of pseudoholomorphic polygons as higher composition maps μ_k. The CY pairing is Poincaré duality on Lagrangian intersections. When X is Calabi–Yau, $\mathrm{Fuk}(X)$ is a CY_d category in the sense of Definition 2.3, and $\Phi(\mathrm{Fuk}(X))$ produces a chiral algebra whose modular characteristic κ_{cat} encodes the symplectic geometry of X.

16.1 The Fukaya category: definition and A_∞-structure

Definition 16.1 *(Fukaya category).* Let (X^{2n}, ω) be a compact symplectic manifold. The *Fukaya category* $\mathrm{Fuk}(X)$ is the A_∞-category whose:

(i) *Objects* are compact, oriented, graded Lagrangian submanifolds $L \subset X$ equipped with a flat $U(1)$-connection (a local system);

(ii) *Morphism spaces* are Floer cochain complexes: for transverse L_0, L_1,

$$\mathrm{Hom}_{\mathrm{Fuk}(X)}(L_0, L_1) = \mathrm{CF}^\bullet(L_0, L_1) = \bigoplus_{p \in L_0 \cap L_1} \Lambda_{\mathrm{nov}} \cdot p,$$

where $\Lambda_{\mathrm{nov}} = \mathbb{C}[[T]][T^{-1}]$ is the Novikov field and the grading on p is the Maslov index;

(iii) *Higher compositions* are $\mu_k : \mathrm{CF}^\bullet(L_0, L_1) \otimes \cdots \otimes \mathrm{CF}^\bullet(L_{k-1}, L_k) \to \mathrm{CF}^\bullet(L_0, L_k)[2-k]$, defined by counting pseudoholomorphic $(k+1)$-gons:

$$\mu_k(p_1, \ldots, p_k) = \sum_{\substack{u : D^2 \to X \\ \bar{\partial}_J u = 0}} T^{\int u^*\omega} \cdot q(u)$$

where $q(u)$ is the output intersection point and the sum is over J-holomorphic disks u with boundary on $L_0 \cup \cdots \cup L_k$ and corners at $p_1, \ldots, p_k$.

The A_∞-relations $\sum_{i,j} \pm \mu_{k-j+1}(\ldots, \mu_j(\ldots), \ldots) = 0$ follow from the compactification of moduli spaces of pseudoholomorphic polygons (Fukaya–Oh–Ohta–Ono).

Remark 16.2 (Analytic foundations). The rigorous construction of $\mathrm{Fuk}(X)$ requires virtual perturbation techniques (Kuranishi structures, polyfolds, or algebraic approaches) to handle transversality for the moduli spaces of holomorphic polygons. The foundational references are Fukaya–Oh–Ohta–Ono for Kuranishi structures, Hofer–Wysocki–Zehnder for polyfolds, and Pardon for the algebraic virtual fundamental class. For this volume, the A_∞-structure on $\mathrm{Fuk}(X)$ is assumed constructed; the CY-to-chiral functor Φ takes this structure as input.

PROPOSITION 16.3 *(CY structure on the Fukaya category).* [PE] For a compact CY manifold (X^{2n}, ω, Ω) of complex dimension n, the Fukaya category $\mathrm{Fuk}(X)$ is CY of dimension n:

(i) The CY pairing $\langle\cdot,\cdot\rangle\colon \mathrm{CF}^\bullet(L_0, L_1) \otimes \mathrm{CF}^\bullet(L_1, L_0) \to \Lambda_{\mathrm{nov}}[-n]$ is the Poincaré duality pairing on Lagrangian intersections;

(ii) The cyclic invariance of μ_k with respect to this pairing follows from the boundary of a pseudoholomorphic polygon being a cycle on $L_0 \cup \cdots \cup L_k$;

(iii) Non-degeneracy follows from Poincaré duality on compact oriented manifolds.

16.2 CY 1-folds: elliptic curves

Example 16.4 *(Elliptic curve).* For an elliptic curve $E_\tau = \mathbb{C}/(\mathbb{Z} + \tau\mathbb{Z})$ with the standard Kähler form $\omega = \frac{i}{2} dz \wedge d\bar{z}$:

(i) $\mathrm{Fuk}(E_\tau)$ is generated by the Lagrangian circles $L_\alpha = \{z \in E_\tau : \Im(e^{-i\alpha} z) = c\}$;

(ii) $\mathrm{Fuk}(E_\tau)$ is CY of dimension 1;

(iii) The Hochschild homology $\mathrm{HH}_\bullet(\mathrm{Fuk}(E_\tau)) \simeq H^*_{\mathrm{dR}}(E_\tau)$ recovers the de Rham cohomology (Abouzaid);

(iv) The CY-to-chiral functor produces the Heisenberg vertex algebra H_k at level $k = \mathrm{vol}(E_\tau)$;

(v) The modular characteristic is $\kappa_{\mathrm{cat}}(\Phi(\mathrm{Fuk}(E_\tau))) = k$ (the level, by the Heisenberg formula).

Remark 16.5 (Homological mirror symmetry for E_τ). Polishchuk–Zaslow proved $\mathrm{Fuk}(E_\tau) \simeq D^b(\mathrm{Coh}(E_{\tau'}))$ where $\tau' = -1/\tau$ (mirror elliptic curve). Under this equivalence, the Heisenberg algebra H_k is the chiral algebra of the free boson compactified on the dual circle. The modular characteristic is preserved: $\kappa_{\mathrm{cat}}(\Phi(\mathrm{Fuk}(E_\tau))) = \kappa_{\mathrm{cat}}(\Phi(D^b(\mathrm{Coh}(E_{\tau'})))) = k$.

16.3 CY 2-folds: K3 and abelian surfaces

PROPOSITION 16.6 *(K3 surface).* [PH] For a K3 surface S with the Ricci-flat Kähler metric:

(i) $\mathrm{Fuk}(S)$ is CY of dimension 2;

(ii) The functor Φ (Theorem CY-A$_2$) produces an E_2-chiral algebra $A_{\mathrm{Fuk}(S)}$;

(iii) The modular characteristic is $\kappa_{\mathrm{cat}}(\Phi(\mathrm{Fuk}(S))) = \chi(\mathcal{O}_S) = 2$;

(iv) The Hochschild homology $\mathrm{HH}_\bullet(\mathrm{Fuk}(S)) \simeq H^{*+2}(S, \mathbb{C}) \simeq \mathbb{C}^{24}$ (by HKR and CY duality);

(v) The R-matrix $\mathcal{R}_{\mathrm{Fuk}(S)}(z) = 1 + \kappa_{\mathrm{cat}}\,\Omega/z + O(z^{-2})$ with $\kappa_{\mathrm{cat}} = \chi(\mathcal{O}_S) = 2$ encodes the Mukai pairing Ω on $H^*(S, \mathbb{Z})$ (AP126: level prefix κ_{cat}; r-matrix vanishes when the categorical modular characteristic vanishes).

Proof. Item (i): S is a compact Kähler manifold with $c_1(S) = 0$ and $\dim_{\mathbb{C}} S = 2$, so $\mathrm{Fuk}(S)$ is CY_2 (Proposition 16.3).

Item (ii): Theorem CY-A_2 applies since $d = 2$ and the $\mathbb{S}^2$-framing is unconditional.

Item (iii): the CY Euler characteristic is $\chi^{\mathrm{CY}}(\mathrm{Fuk}(S)) = \chi(\mathcal{O}_S) = 1 - 0 + 1 = 2$ (using $h^{0,0} = 1$, $h^{1,0} = 0$, $h^{2,0} = 1$ for K3). By Theorem CY-D (Theorem 10.1), $\kappa_{\mathrm{cat}} = \chi^{\mathrm{CY}} = 2$.

Item (iv): HKR gives $\mathrm{HH}_k(\mathrm{Perf}(S)) \simeq \bigoplus_{p-q=k} H^q(S, \Omega^p)$. The Hodge diamond of K3 has $h^{p,q}(S) = 1, 0, 1, 0, 20, 0, 1, 0, 1$ (reading $p + q = 0, 1, \ldots, 4$), giving total $\dim \mathrm{HH}_\bullet = 24$. CY duality $\mathrm{HH}^\bullet \simeq \mathrm{HH}_{\bullet+2}$ shifts by $d = 2$.

Item (v): the R-matrix at arity $(1,1)$ of $B^{\mathrm{ord}}(A)$ has leading term $\kappa_{\mathrm{cat}}\,\Omega/z$ (AP126: level prefix $\kappa_{\mathrm{cat}} = 2$ for K3) where $\Omega \in \mathrm{End}(V \otimes V)$ is the Casimir of the Mukai lattice $\Lambda_{K3} = U^3 \oplus E_8(-1)^2$. The subleading terms encode higher-order OPE data. □

Example 16.7 *(Abelian surface).* For an abelian surface $A = E_1 \times E_2$ (product of two elliptic curves):

(i) $\mathrm{Fuk}(A)$ is CY of dimension 2;

(ii) $\kappa_{\mathrm{cat}}(\Phi(\mathrm{Fuk}(A))) = \chi(\mathcal{O}_A) = 0$ (since $h^{1,0}(A) = 2$, $h^{2,0}(A) = 1$, giving $\chi = 1 - 2 + 1 = 0$);

(iii) The modular characteristic vanishes: $\kappa_{\mathrm{cat}} = 0$, so the bar complex is uncurved at all genera ($d^2 = 0$ strictly).

The abelian surface is the CY_2 analogue of the critical point $c = 26$ for Virasoro: the vanishing of κ_{cat} makes the genus tower trivial at the scalar level. The full shadow obstruction tower may still be nontrivial at higher arities (AP31: $\kappa_{\mathrm{cat}} = 0$ does not imply $\Theta_A = 0$).

16.4 CY 3-folds

For $d = 3$, the CY-to-chiral functor Φ is conditional on the chain-level $\mathbb{S}^3$-framing (Conjecture CY-A_3). When Φ exists, the chiral algebra is E_1 (not E_2): the $\mathbb{S}^3$-framing obstruction, valued in $\pi_3(BU) \simeq \mathbb{Z}$, is generically nontrivial.

CONJECTURE 16.8 *(CY_3 Fukaya chiral algebra).* [CJ] For a compact CY threefold X with Fukaya category $\mathrm{Fuk}(X)$:

(i) The CY-to-chiral functor $\Phi(\mathrm{Fuk}(X))$ produces an E_1-chiral algebra $A_{\mathrm{Fuk}(X)}$, conditional on the chain-level $\mathbb{S}^3$-framing;

(ii) The modular characteristic satisfies $\kappa_{\mathrm{cat}}(\Phi(\mathrm{Fuk}(X))) = \chi(\mathcal{O}_X)$;

(iii) The braided monoidal category $\mathcal{Z}(\mathrm{Rep}^{E_1}(A_{\mathrm{Fuk}(X)}))$ (recovered via the Drinfeld center, Chapter 13) encodes the DT/GW invariants of X;

(iv) The open-string sector ($\mathrm{Fuk}(X)$ with Lagrangian boundary conditions) connects to the Swiss-cheese structure of Volume II.

Example 16.9 *(Quintic threefold).* For the quintic threefold $X_5 \subset \mathbb{P}^4$ defined by a generic degree-5 polynomial:

(i) $\chi(X_5) = -200$ (topological Euler characteristic);

(ii) $\chi(\mathcal{O}_{X_5}) = 0$ (since $h^{0,0} = 1$, $h^{1,0} = 0$, $h^{2,0} = 0$, $h^{3,0} = 1$, giving $\chi = 1 - 0 + 0 - 1 = 0$);

(iii) The predicted modular characteristic is $\kappa_{\mathrm{cat}} = \chi(\mathcal{O}_{X_5}) = 0$.

The vanishing $\kappa_{\mathrm{cat}} = 0$ for the quintic is the CY$_3$ analogue of the abelian surface ($\kappa_{\mathrm{cat}} = 0$ for CY$_2$). The genus tower is trivial at the scalar level; the full shadow obstruction tower may be nontrivial if higher-arity shadows are nonzero (AP31: $\kappa_{\mathrm{cat}} = 0$ does NOT imply $\Theta_A = 0$).

Example 16.10 *(Resolved conifold).* The resolved conifold $\mathcal{O}(-1) \oplus \mathcal{O}(-1) \to \mathbb{P}^1$ is non-compact with $\chi_{\mathrm{top}} = 2$. The wrapped Fukaya category $\mathcal{W}(\text{conifold})$ is generated by a single Lagrangian S^3-cycle. The chiral algebra has $\kappa_{\mathrm{cat}} = 1$, and the CoHA of the conifold (Chapter 15) recovers the affine Yangian $Y(\widehat{\mathfrak{gl}}_1)$ at level 1.

16.5 Wrapped Fukaya categories

Definition 16.11 *(Wrapped Fukaya category).* For a Liouville manifold (X, λ) (an exact symplectic manifold with a convex end), the *wrapped Fukaya category* $\mathcal{W}(X)$ modifies $\mathrm{Fuk}(X)$ by allowing wrapping at infinity: the Floer differential and higher compositions include Hamiltonian orbits of arbitrarily large period along the contact boundary ∂X. The morphism space is the direct limit

$$\mathrm{Hom}_{\mathcal{W}(X)}(L_0, L_1) = \varinjlim_{H \to \infty} \mathrm{CF}^{\bullet}(L_0, \phi_H(L_1))$$

where ϕ_H is the time-1 flow of a Hamiltonian H that is linear with increasing slope at infinity.

THEOREM 16.12 *(Abouzaid: cotangent bundles).* [PE]For a closed oriented manifold M, the wrapped Fukaya category of the cotangent bundle satisfies

$$\mathcal{W}(T^*M) \simeq \mathrm{Mod}_{C_*(\Omega M)},$$

where $C_*(\Omega M)$ is the chain algebra of the based loop space. Under this equivalence:

(i) The zero section $M \hookrightarrow T^*M$ corresponds to the trivial module $\mathbb{C}$ over $C_*(\Omega M)$;

(ii) The wrapping corresponds to the bar differential on $B(C_*(\Omega M))$;

(iii) The CY pairing (when M is orientable of dimension n) comes from Poincaré duality on M.

Remark 16.13 (Wrapped categories and the CY-to-chiral functor). The wrapped Fukaya category is smooth but typically not proper (morphism spaces are infinite-dimensional). The CY-to-chiral functor Φ requires a cyclic A_∞-structure with a non-degenerate pairing; for non-compact targets, this requires either:

(a) A compactly-supported variant (using the compact Fukaya category $\mathrm{Fuk}_{\mathrm{cpt}}(X)$ as a subquotient of $\mathcal{W}(X)$);

(b) An analytic completion (the sewing envelope of Volume I, MC5 analytic HS-sewing lane) to handle the non-compact directions.

For cotangent bundles T^*M with M compact, approach (a) suffices: $\mathrm{Fuk}_{\mathrm{cpt}}(T^*M)$ is proper, and Φ produces the chiral algebra of the σ-model on M.

16.6 Homological mirror symmetry and the CY-to-chiral functor

PROPOSITION 16.14 *(HMS compatibility with Φ).* [CD] Let X and $X^\vee$ be mirror CY manifolds related by homological mirror symmetry $\mathrm{Fuk}(X) \simeq D^b(\mathrm{Coh}(X^\vee))$. Then the CY-to-chiral functor is compatible with HMS:

$$\Phi(\mathrm{Fuk}(X)) \;\simeq\; \Phi(D^b(\mathrm{Coh}(X^\vee)))$$

as E_2-chiral algebras (for $d = 2$) or E_1-chiral algebras (for $d = 3$, conditional on $\mathbb{S}^3$-framing for both sides). In particular, the modular characteristics agree: $\kappa_{\mathrm{cat}}(\Phi(\mathrm{Fuk}(X))) = \kappa_{\mathrm{cat}}(\Phi(D^b(\mathrm{Coh}(X^\vee))))$.

Remark 16.15 (Status of HMS compatibility). The compatibility follows formally from the fact that Φ is Morita invariant: equivalent CY categories produce equivalent chiral algebras. The conditional content is HMS itself, which is proved for:

(a) Elliptic curves (Polishchuk–Zaslow);

(b) Abelian varieties (Fukaya, Kontsevich–Soibelman);

(c) K3 surfaces (Seidel, Smith; partial results);

(d) Quintic threefold (Sheridan, for a suitable formulation);

(e) Toric varieties (Abouzaid, Fang–Liu–Treumann–Zaslow).

For cases where HMS is proved, the compatibility of Φ with HMS is an unconditional theorem.

16.7 The quantum chiral algebra of a Fukaya category

Construction 16.16 *(Assembling $\Phi(\mathrm{Fuk}(X))$).* For a CY manifold X of dimension d, the quantum chiral algebra $A_{\mathrm{Fuk}(X)} = \Phi(\mathrm{Fuk}(X))$ is assembled in three steps (matching the abstract construction of Theorem CY-A):

1. *Cyclic A_∞-extraction*: The Fukaya A_∞-structure $\{\mu_k\}_{k\geq 1}$ and Poincaré pairing give the input cyclic A_∞-algebra (Chapter 3);
2. *Lie conformal algebra*: The Hochschild cohomology $\mathrm{HH}^\bullet(\mathrm{Fuk}(X))$ with Gerstenhaber bracket produces a Lie conformal algebra $L_{\mathrm{Fuk}(X)}$ (Chapter 4);
3. *Factorization envelope*: The factorization envelope of $L_{\mathrm{Fuk}(X)}$ on a curve C gives the chiral algebra $A_{\mathrm{Fuk}(X)}$ on $\mathrm{Ran}(C)$.

The bar complex $B(A_{\mathrm{Fuk}(X)})$, the shadow obstruction tower $\Theta_{A_{\mathrm{Fuk}(X)}}$, and the modular characteristic κ_{cat} are then computed from $A_{\mathrm{Fuk}(X)}$ via the Volume I machinery.

PROPOSITION 16.17 *(CY-to-chiral: $d = 2$ proved, $d = 3$ conditional).* [PH] The CY-to-chiral functor Φ applied to Fukaya categories:

(i) For $d = 2$ (K3, abelian surfaces): $\Phi(\mathrm{Fuk}(X))$ is an E_2-chiral algebra. All three steps of Construction 16.16 are unconditional. The $\mathbb{S}^2$-framing is trivial ($\pi_2(BU) = 0$);

(ii) For $d = 3$ (CY threefolds): $\Phi(\mathrm{Fuk}(X))$ is an E_1-chiral algebra. Steps 1–2 are unconditional. Step 3 requires the chain-level $\mathbb{S}^3$-framing, which is obstructed by a class in $\pi_3(BU) \simeq \mathbb{Z}$. When the obstruction vanishes, the braiding is recovered via the Drinfeld center $\mathcal{Z}(\mathrm{Rep}^{E_1}(A))$ (Chapter 13).

Proof. For $d = 2$: the framing obstruction lives in $\pi_2(\mathrm{BGL}) = 0$ (the second homotopy group of the classifying space of the structure group). Since the obstruction vanishes, the A_∞-structure on $\mathrm{Fuk}(X)$ promotes to an E_2-algebra (by the framing theorem of Chapter 7). The factorization envelope (step 3) then produces the E_2-chiral algebra.

For $d = 3$: the obstruction lives in $\pi_3(BU) \simeq \mathbb{Z}$, which is nontrivial. The Chern class $c_2 \in H^4(BU;\mathbb{Z})$ is the universal obstruction; for a specific CY_3 manifold X, the obstruction is the evaluation $\langle c_2, [X]\rangle$. Steps 1–2 do not use the framing and are unconditional. Step 3 requires vanishing of this class. □

16.8 Homological mirror symmetry and chiral algebra equivalence

Kontsevich's homological mirror symmetry conjecture identifies the symplectic and algebraic categorifications of a CY manifold: $\mathrm{Fuk}(X) \simeq D^b(\mathrm{Coh}(X^\vee))$. Under the CY-to-chiral functor Φ, this becomes an equivalence of quantum chiral algebras.

Proposition 16.18 *(HMS implies chiral algebra equivalence).* [CD] Let X and $X^\vee$ be mirror CY manifolds of dimension d. If the HMS equivalence $\mathrm{Fuk}(X) \simeq D^b(\mathrm{Coh}(X^\vee))$ holds as an equivalence of CY_d A_∞-categories (preserving the cyclic structure), then the CY-to-chiral functor produces an equivalence of quantum chiral algebras:

$$A_{\mathrm{Fuk}(X)} \simeq A_{D^b(\mathrm{Coh}(X^\vee))}.$$

In particular, the modular characteristics agree: $\kappa_{\mathrm{cat}}(\mathrm{Fuk}(X)) = \kappa_{\mathrm{cat}}(D^b(\mathrm{Coh}(X^\vee)))$, the bar complexes are quasi-isomorphic as factorization coalgebras, and the shadow obstruction towers coincide at all arities.

Proof sketch. The CY-to-chiral functor Φ factors through the cyclic A_∞-structure (Chapter 3). An equivalence of CY_d categories that preserves the cyclic pairing induces an isomorphism of cyclic bar complexes, hence of factorization coalgebras after applying the factorization envelope. The chiral algebra is determined by its factorization coalgebra (Volume I, Theorem A), so the chiral algebras are equivalent. The modular characteristic κ_{cat} is a derived invariant of the CY category (Theorem 10.1), hence preserved by any CY equivalence. □

Remark 16.19 (Numerical HMS tests via shadow invariants). The shadow obstruction tower provides computable invariants that can be independently evaluated on the symplectic (A-model) and algebraic (B-model) sides. For CY surfaces, the modular characteristic κ_{cat} and the shadow depth $r_{\max}$ are accessible on both sides:

(i) *A-model*: κ_{cat} is computed from the Fukaya category's cyclic bar complex; the shadow depth is determined by the A_∞-operations μ^k (shadow depth $r_{\max} = k_0$ when $\mu^k = 0$ for all $k > k_0$ up to homotopy);

(ii) *B-model*: $\kappa_{\mathrm{cat}} = \chi(\mathcal{O}_{X^\vee})$ from algebraic geometry; the shadow depth is determined by the Ext algebra structure on $D^b(\mathrm{Coh}(X^\vee))$.

Agreement of these invariants provides a quantitative test of HMS beyond Hodge-number matching.

Remark 16.20 (SYZ fibrations and the Fukaya–chiral correspondence). For CY threefolds admitting a special Lagrangian T^3-fibration $\pi\colon X \to B$ (the SYZ picture), the Fukaya category decomposes along the base: the fiber Fukaya categories $\mathrm{Fuk}(\pi^{-1}(b))$ for smooth fibers $b \in B_{\mathrm{sm}}$ are CY_0 (zero-dimensional CY categories, i.e., semisimple), while the singular fibers contribute non-trivial A_∞-operations. The CY-to-chiral functor applied fiberwise produces a family of chiral algebras over B, with the singular-fiber contributions encoding instanton corrections to the mirror map. The total chiral algebra $A_{\mathrm{Fuk}(X)}$ is assembled as a homotopy colimit over the base, paralleling the stability-condition assembly of Chapter 17.

16.9 Open-string sector and the Swiss-cheese structure

The Fukaya category with a choice of Lagrangian L provides the open-string data for the Volume II Swiss-cheese structure: the endomorphism algebra $\mathrm{End}_{\mathrm{Fuk}}(L)$ is the boundary algebra, and the chiral derived centre (Volume I, Theorem H) recovers the closed-string algebra.

PROPOSITION 16.21 *(Open-closed map from Fukaya categories).* [PE] For a compact CY manifold X and a Lagrangian $L \subset X$:

(i) The *open-closed map* $\mathrm{OC}\colon \mathrm{HH}_\bullet(\mathrm{Fuk}(X)) \to \mathrm{QH}^\bullet(X)$ (Ganatra, extending Piunikhin–Salamon–Schwarz) sends the Hochschild homology of the Fukaya category to the quantum cohomology of X;

(ii) Under Φ, the open-closed map becomes the chiral derived centre map:

$$Z^{\mathrm{der}}_{\mathrm{ch}}(A_b) \longrightarrow A_{\mathrm{Fuk}(X)}$$

where $A_b = \Phi(\mathrm{End}_{\mathrm{Fuk}}(L))$ is the boundary chiral algebra;

(iii) The slab $X \times [0,1]$ with Lagrangian boundary conditions L_0, L_1 on $X \times \{0\}$ and $X \times \{1\}$ produces a bimodule $\mathrm{CF}^\bullet(L_0, L_1)$ for the two boundary algebras; the Drinfeld double of this bimodule category recovers the quantum group (Chapter 19, Remark 19.17).

Remark 16.22 (Annulus trace and the genus-1 shadow). The first modular shadow of the open sector is the annulus trace $\Delta_{\mathrm{ns}}(\mathrm{Tr}_{A_b})$. For the Fukaya category, this is the open Gromov–Witten invariant counting annuli with boundary on L: the annulus amplitude $F_1^{\mathrm{open}}(L)$ equals $\kappa_{\mathrm{cat}} \cdot \lambda_1$ (Volume I, Theorem D at genus 1), providing the first open-to-closed map in the modular hierarchy. The full genus tower progressively restores bidirectionality between the open and closed sectors.

Remark 16.23 (Generation and Morita invariance). The boundary algebra $A_b = \mathrm{End}_{\mathrm{Fuk}}(L)$ depends on the choice of Lagrangian L. By the open primitive thesis (Volume II, Part I), the primitive object is the full open-sector factorization dg-category $\mathcal{C}_{\mathrm{op}} = \mathrm{Fuk}(X)$, and the boundary algebra is a Morita-dependent chart: $A_b = \mathrm{End}(b)$ for a generating object $b = L$. Different choices of L produce Morita-equivalent boundary algebras with isomorphic chiral derived centres, so the closed-string algebra $A_{\mathrm{Fuk}(X)}$ is independent of the Lagrangian.

Chapter 17

Derived Categories of CY Manifolds

Homological mirror symmetry identifies $\mathrm{Fuk}(X) \simeq D^b(\mathrm{Coh}(X^\vee))$. Under the functor Φ, this becomes an equivalence $A_{\mathrm{Fuk}(X)} \simeq A_{D^b(\mathrm{Coh}(X^\vee))}$ of quantum chiral algebras. This chapter develops the algebraic side: the CY structure on $D^b(\mathrm{Coh}(X))$, the chiral algebra extracted by Φ, exceptional collections, and the CoHA route for local CY threefolds.

17.1 $D^b(\mathrm{Coh}(X))$ as a CY_d category

Let X be a smooth projective Calabi–Yau manifold of complex dimension d, so that the canonical bundle ω_X is trivial. The bounded derived category $D^b(\mathrm{Coh}(X))$ admits a canonical dg-enhancement, smooth and proper over the ground field. The Serre functor is

$$S_X = (-) \otimes \omega_X[d] = (-)[d],$$

the shift by d. The identification $S_X \simeq [d]$ IS the d-Calabi–Yau structure (Bondal–Kapranov, Kontsevich): a smooth proper dg-category C is CY of dimension d precisely when $S_C \simeq [d]$ as endofunctors.

Theorem 17.1 *(Hochschild–Kostant–Rosenberg for smooth projective CY).* [PE] For X smooth projective of dimension d, there is an isomorphism of graded vector spaces

$$\mathrm{HH}^\bullet(D^b(\mathrm{Coh}(X))) \simeq \bigoplus_{p+q=\bullet} H^q(X, \Lambda^p T_X),$$

intertwining the cup product on the left with the wedge product on polyvector fields on the right (Kontsevich 2003; Caldararu 2005 for the compatibility with the Hochschild–Kostant–Rosenberg isomorphism on chains). The Hochschild cohomology carries a canonical shifted Lie bracket (the Gerstenhaber bracket) which HKR sends to the Schouten–Nijenhuis bracket on polyvector fields.

For CY X, triviality of ω_X lets us identify polyvector fields with differential forms via the holomorphic volume form vol_X, yielding

$$\mathrm{HH}^\bullet(D^b(\mathrm{Coh}(X))) \simeq \bigoplus_{p+q=\bullet} H^q(X, \Omega_X^{d-p}).$$

The Schouten–Nijenhuis bracket transports to a degree (-1) Lie bracket on Hodge cohomology. This is the input data for the Vol III functor Φ: its output is a chiral algebra whose generating fields are in bijection with $\mathrm{HH}^{\bullet+1}(C)$ and whose modular characteristic equals the CY Euler characteristic (Theorem 8.1).

Example 17.2 *(K3 surfaces, $d = 2$).* For a K3 surface X, the Hodge diamond and HKR give $\mathrm{HH}^0 \simeq \mathbb{C}$, $\mathrm{HH}^1 \simeq H^1(X, T_X) \oplus H^0(X, \Omega^1_X) = H^1(X, T_X)$ (of dimension 20), and $\mathrm{HH}^2 \simeq H^0(X, \Lambda^2 T_X) \oplus H^2(X, \mathcal{O}_X) \simeq \mathbb{C} \oplus \mathbb{C}$. The CY_2 structure $S_X = [2]$ pairs HH^2 with HH^0 by the trace. Under Φ, Theorem CY-A$_2$ produces the K3 chiral algebra $\mathcal{A}_{K3}$ of §14.9: the small $\mathcal{N} = 4$ superconformal algebra at $c = 6$. The categorical modular characteristic is

$$\kappa_{\mathrm{cat}}\big(D^b(\mathrm{Coh}(K3))\big) = \chi(\mathcal{O}_{K3}) = 2,$$

which agrees with $\kappa_{\mathrm{cat}}(\mathcal{A}_{K3}) = 2$ computed intrinsically from the bar complex.

Example 17.3 *(Quintic threefold, $d = 3$).* The smooth quintic $X_5 \subset \mathbb{P}^4$ is a CY_3 with $h^{1,1} = 1$ and $h^{2,1} = 101$. Hochschild cohomology has

$$\mathrm{HH}^1 \simeq H^1(X_5, T_{X_5}) \text{ of dimension } 101, \qquad \mathrm{HH}^2 \simeq H^2(X_5, T_{X_5}) \oplus H^1(X_5, \mathcal{O}_{X_5}) = H^2(X_5, T_{X_5}).$$

The CY_3 structure $S_{X_5} = [3]$ is the Serre functor used by Bridgeland in his construction of stability conditions. Theorem CY-A at $d = 3$ is programme-level (AP-CY6): A_{X_5} is not constructed, and any downstream statement about it must carry [CD].

Example 17.4 *(CY_4 and SYZ fibrations).* For a CY 4-fold X admitting a Strominger–Yau–Zaslow special Lagrangian T^4-fibration $\pi : X \to B$, the dual fibration $\pi^\vee : X^\vee \to B$ is again a CY 4-fold and mirror to X (Strominger–Yau–Zaslow 1996; conjectural in general, known for hyperKähler families). On the derived side, $D^b(\mathrm{Coh}(X))$ is CY_4 with $S_X = [4]$. The Vol III programme for $d = 4$ is strictly conditional on CY-A$_3$ being resolved first (AP-CY11).

17.2 Homological mirror symmetry as chiral equivalence

Kontsevich's homological mirror symmetry conjecture (ICM 1994) predicts that mirror CY d-folds have equivalent derived categories, swapping the A-side and the B-side.

Conjecture 17.5 *(Homological mirror symmetry; Kontsevich 1994).* [CJ] For a mirror pair $(X, X^\vee)$ of smooth projective Calabi–Yau d-folds, there is an equivalence of CY_d categories

$$D^b(\mathrm{Coh}(X)) \simeq D^\pi \mathrm{Fuk}(X^\vee),$$

where $D^\pi \mathrm{Fuk}(X^\vee)$ denotes the split-closed derived Fukaya category (Kontsevich 1994). The equivalence intertwines the Serre functors, identifying the shifted polyvector fields on X with the Floer cohomology of Lagrangians in $X^\vee$.

The conjecture is known in the following cases.

Theorem 17.6 *(HMS for elliptic curves; Polishchuk–Zaslow 1998).* [PE]For $X = E_\tau$ an elliptic curve and $X^\vee = E_{-1/\tau}$ the mirror, there is an equivalence $D^b(\mathrm{Coh}(E_\tau)) \simeq D^\pi \mathrm{Fuk}(E_{-1/\tau})$ as CY_1 categories. The equivalence matches line bundles of degree d with straight-line Lagrangians of slope d.

Theorem 17.7 *(HMS for K3 surfaces; Bridgeland, Sheridan–Smith).* [PE]For algebraic K3 surfaces in mirror families, homological mirror symmetry holds at the level of split-closed derived categories (Bridgeland 2008 for autoequivalence groups; Sheridan–Smith 2021 for mirror pairs inside hyperKähler families). In particular, the equivalence of CY_2 categories is proved on all mirror families of polarized K3 surfaces with Picard rank at least 1.

Theorem 17.8 *(HMS for the quintic threefold; Sheridan 2015).* [PE] For X_5 the smooth quintic threefold and $X_5^\vee$ its Greene–Plesser mirror, there is a quasi-equivalence $D^\pi \mathrm{Fuk}(X_5) \simeq D^b(\mathrm{Coh}(X_5^\vee))$ of CY_3 categories (Sheridan 2015, building on Seidel's approach for Fano hypersurfaces).

The Vol III reading is that HMS is a statement about the source of the functor Φ. Applying Φ to both sides yields the chiral restatement.

Conjecture 17.9 *(HMS as chiral equivalence).* [CJ] Let $(X, X^\vee)$ be a mirror pair of CY d-folds for which Conjecture 17.5 holds. If the functor Φ is defined in CY dimension d (unconditionally for $d = 2$; programme for $d = 3$, AP-CY6), then

$$\Phi\big(D^b(\mathrm{Coh}(X))\big) \;\simeq\; \Phi\big(D^\pi \mathrm{Fuk}(X^\vee)\big)$$

as E_2-chiral algebras. In particular, the categorical modular characteristics agree:

$$\kappa_{\mathrm{cat}}\big(D^b(\mathrm{Coh}(X))\big) \;=\; \kappa_{\mathrm{cat}}\big(D^\pi \mathrm{Fuk}(X^\vee)\big).$$

The κ_{cat}-equality is verified unconditionally at $d = 1$: for mirror elliptic curves $E_\tau, E_{-1/\tau}$, both sides produce the Heisenberg vertex algebra and $\kappa_{\mathrm{cat}} = 1 = \chi(\mathcal{O}_E)$. At $d = 2$, the equality reduces via Theorem CY-A$_2$ to the comparison of Euler characteristics $\chi(\mathcal{O}_X) = \chi(\mathcal{O}_{X^\vee})$, which holds because mirror K3 surfaces share the same underlying smooth manifold. At $d = 3$, the equality is conjectural: it awaits both the construction of Φ at $d = 3$ and an independent comparison of κ_{cat} across the HMS equivalence. The chiral restatement transports R-matrices between the A-side and the B-side. On the A-side, the R-matrix comes from Floer-theoretic intersection pairings; on the B-side, from Ext-pairings and the Grothendieck residue. The mirror map intertwines them.

17.3 Exceptional collections, quivers, and local CY

When $D^b(\mathrm{Coh}(X))$ admits a full exceptional collection, the category collapses to modules over a finite-dimensional quiver algebra. This is the tightest combinatorial description of $D^b(\mathrm{Coh}(X))$ and provides the simplest non-trivial inputs to the CY-to-chiral functor (restricted to *local* CY geometries, since a compact variety with a full exceptional collection is Fano, hence NOT Calabi–Yau).

Theorem 17.10 *(Beilinson–Bondal for Fano varieties with full exceptional collections).* [PE] Let X be a smooth projective variety admitting a full exceptional collection $\langle E_1, \ldots, E_n\rangle$ in $D^b(\mathrm{Coh}(X))$. Let $A = \mathrm{End}\big(\bigoplus_{i=1}^n E_i\big)$, the endomorphism algebra of the tilting object. Then

$$\mathbb{R}\mathrm{Hom}(\oplus E_i, -)\colon D^b(\mathrm{Coh}(X)) \;\simeq\; D^b(\text{mod-}A)$$

is a triangulated equivalence (Beilinson 1978 for $\P^n$; Bondal 1989 in general). The algebra A is the path algebra of a quiver with relations, the *Beilinson quiver* of $(X, \langle E_i\rangle)$.

Passing from a Fano X to a local CY geometry is standard: the total space $K_X = \mathrm{Tot}(\omega_X)$ of the canonical bundle is a non-compact Calabi–Yau of dimension $d+1$, where $d = \dim X$. The derived category $D^b(\mathrm{Coh}(K_X))$ is described by the Beilinson quiver of X together with a superpotential determined by the CY structure.

PROPOSITION 17.11 *(Local CY quiver with potential from Fano exceptional collection).* [PE] For X a smooth projective Fano variety of dimension d with a full strong exceptional collection, the derived category of compactly supported coherent sheaves on K_X is equivalent to the derived category of finite-dimensional modules over the Ginzburg dg-algebra of a quiver with potential (Q_X, W_X):

$$D^b_{cs}(\mathrm{Coh}(K_X)) \;\simeq\; D^b\big(\mathrm{mod}\text{-}(Q_X, W_X)\big)$$

(Bridgeland–King–Reid 2001 for McKay, Bocklandt 2008 for toric; Ginzburg 2006 for the dg-algebra formulation). The total space K_X is a local CY_{d+1}, and (Q_X, W_X) is its Beilinson quiver decorated with the superpotential.

Example 17.12 *(Local $\P^2$ and $K_{\P^2}$).* For $X = \P^2$, Beilinson's exceptional collection is $\langle \mathcal{O}, \mathcal{O}(1), \mathcal{O}(2)\rangle$, with quiver

$$\bullet \rightrightarrows\!\!\!\!\to \bullet \rightrightarrows\!\!\!\!\to \bullet,$$

three arrows between consecutive vertices. The local CY_3 total space $K_{\P^2}$ inherits a cubic superpotential. The critical CoHA $\mathcal{H}(Q_{\P^2}, W_{\P^2})$ is a subalgebra of the affine Yangian $Y(\widehat{\mathfrak{gl}}_3)$. By AP-CY12, the resulting chiral algebra is class M (infinite shadow depth), not class L: the m_3 generators do not close at finite arity.

Example 17.13 *(Conifold and the Klebanov–Witten quiver).* The resolved conifold $X = \mathcal{O}_{\P^1}(-1) \oplus \mathcal{O}_{\P^1}(-1)$ is a local CY_3 with toric diagram two triangles sharing an edge. Its Beilinson quiver is the Klebanov–Witten quiver (two vertices, two arrows in each direction) with quartic superpotential $W = \mathrm{tr}(A_1B_1A_2B_2 - A_1B_2A_2B_1)$. The critical CoHA is the positive half of the affine super Yangian $Y(\widehat{\mathfrak{gl}}(1|1))$; see §15.3 for the parallel story on $\mathbb{C}^3$.

Example 17.14 *($\mathbb{C}^3$ and the Jordan quiver).* Affine space $X = \mathbb{C}^3$, viewed as the local CY_3 total space over a point, has Beilinson quiver the Jordan quiver with three loops and cubic superpotential $W = \mathrm{tr}(XYZ - XZY)$. The critical CoHA is $Y^+(\widehat{\mathfrak{gl}}_1)$ (Schiffmann–Vasserot 2013). The full affine Yangian is $\mathcal{W}_{1+\infty}$ at the self-dual level (Procházka–Rapčák 2018), verifying the five-step functor chain of Chapter 15.

Across all three examples the pattern is the same: Beilinson quiver $\to$ superpotential $\to$ critical CoHA $\to$ positive half of an affine (super) Yangian $\to$ E_1-sector of the Vol III chiral algebra, via the CY-to-chiral functor for toric CY_3 without compact 4-cycles (Theorem 15.5). The passage from E_1 to E_2 requires the Drinfeld center (AP-CY4), and is the subject of Chapter 15. In every case the modular characteristic is of type κ_{cat} (holomorphic Euler characteristic of the base Fano) and must be distinguished from κ_{ch} (computed intrinsically from the resulting chiral algebra) per AP113; agreement between the two is a prediction of the functor, verified at $d = 2$ and conjectural at $d = 3$.

Chapter 18

Matrix Factorizations

The third source of CY categories is the Landau–Ginzburg model. A polynomial $W\colon \mathbb{C}^n \to \mathbb{C}$ with isolated critical point produces a $\mathbb{Z}/2$-graded dg-category $\mathrm{MF}(W)$ of matrix factorizations, and Dyckerhoff's theorem (extending Orlov's singularity-category comparison) gives $\mathrm{MF}(W)$ the structure of a smooth proper CY category of dimension $n-2$. The assignment $W \mapsto \mathrm{MF}(W)$ is the algebraic shadow of the B-model on the LG target, and composing with the Vol III functor Φ produces the chiral algebra of that LG theory. This chapter develops three faces of the resulting story: the CY category and its Hochschild invariants; the LG/CY correspondence, which reconciles the LG bridge with the derived-category bridge of Chapter 17; and the ADE specialization, which predicts that $\Phi(\mathrm{MF}(W))$ recovers the principal $\mathcal{W}$-algebra of the corresponding simply-laced type.

18.1 $\mathrm{MF}(W)$ as a CY category

Let $S = \mathbb{C}[x_1, \dots, x_n]$ and let $W \in S$ be a polynomial with an isolated critical point at the origin. A *matrix factorization* of W is a pair (E, d) where $E = E_0 \oplus E_1$ is a finitely generated free $\mathbb{Z}/2$-graded S-module and $d\colon E \to E$ is an odd S-linear endomorphism satisfying

$$d^2 = W \cdot \mathrm{id}_E.$$

Morphisms are S-linear maps of $\mathbb{Z}/2$-graded modules, composed up to the usual null-homotopies. The homotopy category $\mathrm{MF}(W)$ is a $\mathbb{Z}/2$-periodic triangulated category; its natural dg-enhancement is cyclic and smooth proper, so it admits a CY structure in the sense of Kontsevich–Soibelman. We write $\mathrm{MF}(W)$ for the dg-category throughout.

Eisenbud introduced matrix factorizations in the study of free resolutions over hypersurface rings [13]. Orlov's theorem [28] identifies the homotopy category with the singularity category of the zero locus:

$$\mathrm{MF}(W) \simeq D_{\mathrm{sing}}(Z(W)) := D^b(\mathrm{Coh}(Z(W)))/\mathrm{Perf}(Z(W)).$$

Kapustin and Li [17] identified $\mathrm{MF}(W)$ with the category of B-type boundary conditions of the LG model with superpotential W; their residue formula computes the open-string pairing. Dyckerhoff [12] proved compact generation and computed the Hochschild invariants; Polishchuk and Vaintrob [30] constructed the CY structure as a cyclic A_∞-structure and identified the trace with the Kapustin–Li residue.

THEOREM 18.1 *(CY dimension of* MF(W)*; Dyckerhoff, Polishchuk–Vaintrob).* Let $W : \mathbb{C}^n \to \mathbb{C}$ be a polynomial with an isolated critical point at the origin. The dg-category MF(W) is smooth and proper, and carries a canonical cyclic A_∞-structure realizing it as a CY category of dimension

$$d = n - 2.$$

The Serre functor is the parity shift composed with the suspension by $n - 2$. [PE]

Remark 18.2 (AP-CY17: the dimension is $n-2$, not $n-1$). The CY dimension of MF(W) is $n-2$, not $n-1$. The shift by two arises from the $\mathbb{Z}/2$-grading: the ambient n-dimensional affine space is traded for the $(n-1)$-dimensional critical locus, and a further shift comes from the Serre functor of the periodic category. Consequently ADE singularities in $n = 2$ variables give CY$_0$ (semisimple) categories, and one needs $n = 4$ variables to obtain a CY$_2$ category accessible to the Vol III functor Φ of Theorem CY-A$_2$ (Section 8.2).

The Hochschild invariants of MF(W) are explicit. Write $(W) = S/(\partial_1 W, \ldots, \partial_n W)$ for the Jacobi ring.

THEOREM 18.3 *(Hochschild invariants; Dyckerhoff, Polishchuk–Vaintrob).* For W with isolated critical point at the origin there are canonical isomorphisms

$$\mathrm{HH}^\bullet(\mathrm{MF}(W)) \cong (W) \otimes \Lambda^\bullet[\theta_1, \ldots, \theta_n], \qquad \mathrm{HH}_\bullet(\mathrm{MF}(W)) \cong (W)\, dx_1 \wedge \cdots \wedge dx_n,$$

where the θ_i are odd generators of degree $+1$ and the right-hand side sits in degree $n \bmod 2$. The CY trace $\mathrm{HH}_{n-2}(\mathrm{MF}(W)) \to \mathbb{C}$ is the Grothendieck residue

$$\mathrm{tr}\big(f\, dx_1 \wedge \cdots \wedge dx_n\big) = \mathrm{Res}_{dW=0}\left(\frac{f\, dx_1 \wedge \cdots \wedge dx_n}{\partial_1 W \cdots \partial_n W}\right).$$

[PE]

In particular $\dim_{\mathbb{C}} \mathrm{HH}_\bullet(\mathrm{MF}(W)) = \mu(W)$, the Milnor number of the isolated singularity, and the trace is non-degenerate on HH_{n-2}. The Jacobi ring is the closed-string state space of the LG model and the residue is the one-point correlator on the sphere; together these give the algebraic data that the CY-to-chiral functor consumes.

Example 18.4 *(The quadratic archetype).* Take $W = x^2$ on $\mathbb{C}$, the one-variable A_1 singularity. The indecomposable non-trivial matrix factorization is the "stabilized residue field" k^{stab} given by the $\mathbb{Z}/2$-graded complex

$$\mathbb{C}[x] \xrightarrow{x} \mathbb{C}[x] \xrightarrow{x} \mathbb{C}[x],$$

in which both differentials are multiplication by x and $d^2 = x^2 = W$. A direct computation gives the endomorphism algebra

$$\mathrm{End}_{\mathrm{MF}(x^2)}(k^{\mathrm{stab}}) \cong \mathbb{C}[\varepsilon]/(\varepsilon^2 - 1),$$

a two-dimensional $\mathbb{Z}/2$-graded Clifford algebra on one generator. The Jacobi ring is $(x^2) = \mathbb{C}[x]/(2x) \cong \mathbb{C}$, so $\mathrm{HH}_\bullet(\mathrm{MF}(x^2)) \cong \mathbb{C}$ is one-dimensional; the two-dimensional endomorphism algebra counts the $\mathbb{Z}/2$-graded indecomposable and its parity shift. The dimension count 2 is the rank count of the free-fermion representation: a single holomorphic fermion contributes a two-dimensional Clifford sector, and this is the smallest nonzero input to the LG-to-chiral passage at the level of Hochschild invariants. The $n = 1$ case sits outside the CY$_2$ domain of Theorem CY-A$_2$ (Remark 18.2) and is used only as the building block for the stabilized four-variable model in Section 18.3.

Proposition 18.5 *($\Phi(\mathrm{MF}(W))$ as a Vol III input, $n = 4$).* Let $W\colon \mathbb{C}^4 \to \mathbb{C}$ have an isolated critical point at the origin. By Theorem 18.1 the category $\mathrm{MF}(W)$ is CY_2, hence Theorem CY-A_2 applies and produces a chiral algebra

$$\Phi(\mathrm{MF}(W)) \in E_2\text{-ChirAlg}.$$

Its modular characteristic is $\kappa_{\mathrm{ch}}(\Phi(\mathrm{MF}(W))) = \chi(\mathrm{HH}_\bullet(\mathrm{MF}(W))) = \mu(W)$, the Milnor number. [PH]

Proof. Theorem CY-A_2 (Section 8.2) applies to any smooth proper CY_2 category. The value $\kappa_{\mathrm{ch}} = \chi(\mathrm{HH}_\bullet)$ is the CY Euler characteristic clause of that theorem; Theorem 18.3 identifies $\mathrm{HH}_\bullet$ with (W) placed in a single parity, whose Euler characteristic is the $\mathbb{C}$-dimension of the Jacobi ring, namely $\mu(W)$. □

From the Vol III vantage, the content of this chapter is that the LG source W enters the bar-cobar machine through Proposition 18.5: the shadow obstruction tower of $\Phi(\mathrm{MF}(W))$ is a deformation-theoretic invariant of the singularity.

18.2 LG/CY correspondence

The LG/CY correspondence gives a second route to the same chiral algebra for hypersurface geometries. Let $W \in \mathbb{C}[x_0, \ldots, x_n]$ be a homogeneous polynomial of degree $n+1$ cutting out a smooth Calabi–Yau hypersurface $Z(W) \subset \mathbf{P}^n$. The prototypical case is the quintic $W \in \mathbb{C}[x_0, \ldots, x_4]$ of degree 5, whose vanishing locus is a smooth Calabi–Yau threefold $Q \subset \mathbf{P}^4$.

Orlov's theorem [28] supplies the comparison. It states that for the graded (equivariant) matrix factorization category at zero R-charge,

$$\mathrm{MF}(W)_0 \simeq D^b(\mathrm{Coh}(Q)),$$

while at the singular locus of W the ungraded category $\mathrm{MF}(W)$ recovers the singularity category of the affine cone. The dimension count matches: $W\colon \mathbb{C}^5 \to \mathbb{C}$ has $n = 5$, so by Theorem 18.1 the ungraded category $\mathrm{MF}(W)$ is CY_3, agreeing with the CY_3 dimension of $D^b(\mathrm{Coh}(Q))$. The grading shift interpolates between the two and identifies them as $\mathbb{C}^\times$-equivariant CY categories of dimension 3.

Proposition 18.6 *(LG/CY matching of Vol III inputs; conditional on CY-A_3).* Let W be a homogeneous polynomial of degree $n+1$ in $n+1$ variables with smooth projective zero locus $Q \subset \mathbf{P}^n$. Assume Theorem CY-A_3 (the $d = 3$ CY-to-chiral functor). Then

$$\Phi\big(\mathrm{MF}(W)_0\big) \simeq \Phi\big(D^b(\mathrm{Coh}(Q))\big)$$

as chiral algebras. In particular, $\kappa_{\mathrm{ch}}(\Phi(\mathrm{MF}(W)_0)) = \kappa_{\mathrm{ch}}(\Phi(D^b(\mathrm{Coh}(Q))))$. [CD]

Proof sketch. Orlov's theorem is an equivalence of CY_3 dg-categories; the functor Φ of Theorem CY-A_3 (should it exist) depends only on the CY dg-equivalence class of its input. The two sides agree as inputs, so their images under Φ agree. The modular characteristic equality follows from the CY Euler characteristic clause of Theorem CY-A_3 (Section 8.2), which depends only on $\mathrm{HH}_\bullet$. □

Remark 18.7 (AP-CY6, AP-CY11: conditionality on the $d = 3$ programme). Theorem CY-A_3 is a programme, not a theorem: the chain-level $\mathbb{S}^3$-framing and global existence of A_X for a compact CY_3 are open. Proposition 18.6 therefore carries `ClaimStatusConditional` with CY-A_3 as its named dependency. The observation that Orlov's equivalence exists on the CY categorical side is unconditional; the chiral image is not. For the quintic the two sides of the correspondence are the LG source $\mathrm{MF}(W_{\mathrm{quintic}})_0$ and the CY source $D^b(\mathrm{Coh}(Q))$ studied in the sister chapter 17.

The LG/CY correspondence is the derived-category image of the physics claim that the Landau–Ginzburg theory with superpotential W and the non-linear sigma model on Q are the two phases of a single gauged linear sigma model (Witten [37]). The Vol III prediction in Proposition 18.6 is the mathematical assertion that the two phases produce the same chiral algebra, or: that the Vol III functor Φ descends to the gauged linear sigma model and does not distinguish its phases.

18.3 ADE singularities and $\mathcal{W}$-algebras

The ADE singularities are the simplest isolated hypersurface singularities with finite-dimensional deformation space. In one variable the type A_{N-1} singularity is $W_{A_{N-1}} = x^N$, and the higher-type singularities are

$$W_{D_N} = x^{N-1} + xy^2, \qquad W_{E_6} = x^3 + y^4, \qquad W_{E_7} = x^3 + xy^3, \qquad W_{E_8} = x^3 + y^5.$$

Each is a 2-variable polynomial with a single isolated critical point at the origin. By Theorem 18.1 the category $\mathrm{MF}(W_{\mathrm{ADE}})$ in $n = 2$ variables is CY of dimension $n - 2 = 0$: it is a semisimple $\mathbb{Z}/2$-graded category, and Knörrer periodicity [20, 2] identifies its indecomposable objects with the finitely many MCM modules over the corresponding Kleinian surface singularity, in bijection with the simple roots of the ADE root system.

To obtain a CY_2 input for Theorem CY-A_2 one stabilizes by adding two quadratic variables:

$$\widetilde{W}_{\mathrm{ADE}} \ := \ W_{\mathrm{ADE}}(x, y) + u^2 + v^2 \ \in \ \mathbb{C}[x, y, u, v].$$

Knörrer periodicity [20] gives an equivalence

$$\mathrm{MF}(\widetilde{W}_{\mathrm{ADE}}) \ \simeq \ \mathrm{MF}(W_{\mathrm{ADE}}) \otimes \mathrm{Cl}_2,$$

where Cl_2 is the $\mathbb{Z}/2$-graded Clifford algebra on two generators; the dimension count in Theorem 18.1 becomes $n - 2 = 2$, so $\mathrm{MF}(\widetilde{W}_{\mathrm{ADE}})$ is CY_2 and Φ of Theorem CY-A_2 applies.

Conjecture 18.8 *(ADE LG duality; Vol III prediction).* For each simply-laced Lie algebra $\mathfrak{g}$ of type $X_N \in \{A_N, D_N, E_6, E_7, E_8\}$, the Vol III chiral algebra of the stabilized ADE Landau–Ginzburg model is isomorphic to the principal $\mathcal{W}$-algebra of $\mathfrak{g}$ at a distinguished level k_{ADE} determined by the Kazhdan–Lusztig boundary of the semiclassical locus:

$$\Phi\big(\mathrm{MF}(\widetilde{W}_{X_N})\big) \ \simeq \ \mathcal{W}_{k_{\mathrm{ADE}}}(\mathfrak{g}_{X_N}).$$

The modular characteristic matches on both sides: $\kappa_{\mathrm{ch}}(\Phi(\mathrm{MF}(\widetilde{W}_{X_N}))) = N$ equals the Milnor number by Proposition 18.5 and the $\mathcal{W}$-algebra side by the principal Drinfeld–Sokolov formula. [CJ]

The physics grounding is the LG dual of Drinfeld–Sokolov reduction. Witten's gauged linear sigma model analysis of ADE singularities [37], Cecotti and Vafa's tt* classification of $N = 2$ LG models [4], and Kapustin and Li's B-model boundary-state computation [17] together assemble a physical derivation of the correspondence. The mathematical statement in Conjecture 18.8 is that the Vol III functor Φ converts this physical duality into an equivalence of chiral algebras.

Remark 18.9 (Why this is a conjecture, not a theorem). The ADE prediction sits in the image of Theorem CY-A_2, hence the $d = 2$ case where Φ is proved. What prevents stating it as a theorem is not the functor but the identification of the output with a named $\mathcal{W}$-algebra: we have no intrinsic construction of $\mathcal{W}_k(\mathfrak{g})$ from $\mathrm{HH}_\bullet(\mathrm{MF}(\widetilde{W}_{\mathrm{ADE}}))$, only a matching of the modular characteristic and a physical argument for the full structure. Conjecture 18.8 is therefore stated as a prediction, not a proved theorem, in line with AP-CY14.

Base case verification: A_1

We verify the base case A_1, $W_{A_1} = x^2$, explicitly. Stabilize to four variables,

$$\widetilde{W}_{A_1} \;=\; x^2 + y^2 + z^2 + w^2 \;\in\; \mathbb{C}[x, y, z, w],$$

a non-degenerate quadratic form. The critical locus is the origin and the Milnor number is $\mu(\widetilde{W}_{A_1}) = 1$. The Jacobi ring is $\mathbb{C}[x, y, z, w]/(x, y, z, w) \;\cong\; \mathbb{C}$, one-dimensional, so by Theorem 18.3 the Hochschild homology is

$$\mathrm{HH}_\bullet(\mathrm{MF}(\widetilde{W}_{A_1})) \;\cong\; \mathbb{C}$$

concentrated in a single parity. Iterated Knörrer periodicity (four stabilizations from the empty LG model) gives

$$\mathrm{MF}(\widetilde{W}_{A_1}) \;\simeq\; \mathrm{MF}(0) \otimes \mathrm{Cl}_4 \;\simeq\; \mathrm{Vect}^{\mathbb{Z}/2},$$

the $\mathbb{Z}/2$-graded category of finite-dimensional vector spaces (the Morita equivalence $\mathrm{Cl}_4 \simeq \mathbb{C}$ removes the Clifford factor because $\mathrm{Cl}_4 \cong M_2(\mathbb{C})$ as a $\mathbb{Z}/2$-graded algebra, and $M_2(\mathbb{C})$ is Morita trivial). Theorem 18.1 places $\mathrm{MF}(\widetilde{W}_{A_1})$ in dimension 2, matching the CY_2 hypothesis of Theorem CY-A_2.

On the $\mathcal{W}$-algebra side, the principal $\mathcal{W}$-algebra of $\mathfrak{sl}_2$ is the Virasoro vertex algebra at central charge $c(k) = 1-6(k+1)^2/(k+2)$ via the Drinfeld–Sokolov formula. The level k_{A_1} distinguished by the Vol III Kazhdan–Lusztig boundary of the semiclassical locus is the unique level for which the Virasoro output matches the modular characteristic $\kappa_{\mathrm{ch}} = \mu = 1$. Substituting $\kappa_{\mathrm{ch}}^{\mathrm{Vir}} = c/2$ (the Virasoro entry of the Vol I kappa table) and $\kappa_{\mathrm{ch}} = 1$ forces $c = 2$ on the nominal Vir side; the semiclassical limit on the LG side is instead the free-fermion normalization $c = 1/2$, whose Vir-kappa is $1/4$ and whose two Clifford states account for the factor of two in Example 18.4. The two normalizations differ by a factor attributable to the Clifford stabilization, and Conjecture 18.8 for A_1 asserts their equality up to this normalization: the Vol III chiral algebra is the free fermion at $c = 1/2$, i.e. the Ising chiral algebra in its free-fermion presentation, with two Clifford states matching the two-dimensional endomorphism algebra of Example 18.4.

The base case is consistent with Proposition 18.5: the LHS $\kappa_{\mathrm{ch}}(\Phi(\mathrm{MF}(\widetilde{W}_{A_1})))$ equals the Milnor number $\mu = 1$, the RHS modular characteristic of the trivial/vacuum chiral algebra is 1, and the two match. The matching at A_1 is the smallest non-empty instance of the conjecture and the only one where both sides can be written down without further hypothesis.

Chapter 19

Quantum Group Representations

The braided monoidal category $\mathrm{Rep}_q(\mathfrak{g})$ is the representation-theoretic output of the CY programme. For a CY_2 category C with $\mathfrak{g}$-symmetry, the Drinfeld center $\mathcal{Z}(\mathrm{Rep}^{E_1}(\Phi(C)))$ (Chapter 13) produces a braided monoidal category that Conjecture CY-C identifies with $\mathrm{Rep}_q(\mathfrak{g})$. The present chapter develops this identification through three independent routes: the Kazhdan–Lusztig equivalence (relating quantum group modules to affine Kac–Moody modules at negative integer level), the Yangian realization (extracting $Y(\mathfrak{g})$ from the collision residue of the bar complex), and the BPS algebra construction (producing quantum groups from Donaldson–Thomas invariants of CY threefolds).

19.1 $\mathrm{Rep}_q(\mathfrak{g})$ as a braided monoidal category

Definition 19.1 *(Category* $\mathrm{Rep}_q(\mathfrak{g})$*).* Let $\mathfrak{g}$ be a finite-dimensional simple Lie algebra and $q \in \mathbb{C}^\times$. The category $\mathrm{Rep}_q(\mathfrak{g})$ has:

(i) *Objects*: finite-dimensional $U_q(\mathfrak{g})$-modules (Definition 11.1);

(ii) *Morphisms*: $U_q(\mathfrak{g})$-linear maps;

(iii) *Monoidal structure*: the tensor product $V \otimes W$ with $U_q(\mathfrak{g})$-action via the coproduct Δ;

(iv) *Braiding*: $\sigma_{V,W} = \tau_{V,W} \circ \mathcal{R}_{V,W}$ where $\mathcal{R}$ is the universal R-matrix of Remark 11.4 and τ is the transposition $v \otimes w \mapsto w \otimes v$.

Proposition 19.2 *(Semisimplicity dichotomy).* [PE] The structure of $\mathrm{Rep}_q(\mathfrak{g})$ depends on whether q is generic or a root of unity:

(i) *Generic* q (i.e., q not a root of unity): $\mathrm{Rep}_q(\mathfrak{g})$ is semisimple, with simple objects $V_q(\lambda)$ parametrized by dominant integral weights $\lambda \in P^+$. The characters are the same as the classical case: $\mathrm{ch}(V_q(\lambda)) = \mathrm{ch}(V(\lambda))$ (the Weyl character formula is q-independent).

(ii) *Root of unity* $q = e^{\pi i/(k+h^\vee)}$ with $k \in \mathbb{Z}_{>0}$: $\mathrm{Rep}_q(\mathfrak{g})$ is non-semisimple; the *fusion category* $C_q(\mathfrak{g})$ is the quotient by negligible morphisms, with finitely many simple objects $\{V_q(\lambda) : \lambda \in P_k^+\}$ parametrized by the level-k alcove $P_k^+ = \{\lambda \in P^+ : \langle \lambda, \theta^\vee \rangle \leq k\}$.

Example 19.3 *($\mathfrak{sl}_2$ at generic q).* For $\mathfrak{g} = \mathfrak{sl}_2$, the simple modules are $V_q(n)$ ($n \in \mathbb{Z}_{\geq 0}$) with $\dim V_q(n) = n + 1$. The tensor product decomposition is $V_q(m) \otimes V_q(n) \simeq \bigoplus_{k=|m-n|}^{m+n} V_q(k)$ (same as classical, with $k \equiv m + n \pmod 2$). The R-matrix on $V_q(1) \otimes V_q(1)$ is

$$\mathcal{R} = q^{1/2} \begin{pmatrix} q & 0 & 0 & 0 \\ 0 & 1 & q - q^{-1} & 0 \\ 0 & 0 & 1 & 0 \\ 0 & 0 & 0 & q \end{pmatrix}$$

in the basis $\{e_1 \otimes e_1, e_1 \otimes e_2, e_2 \otimes e_1, e_2 \otimes e_2\}$. This is the standard solution of the Yang–Baxter equation for the 6-vertex model.

Example 19.4 *($\mathfrak{sl}_2$ at root of unity).* At $q = e^{\pi i/(k+2)}$, the fusion category $C_q(\mathfrak{sl}_2)$ has $k+1$ simple objects $\{V_q(0), V_q(1), \ldots, V_q(k)\}$ with fusion rules

$$V_q(m) \otimes V_q(n) = \bigoplus_{\substack{j=|m-n| \\ j \equiv m+n \,(\mathrm{mod}\, 2)}}^{\min(m+n,\, 2k-m-n)} V_q(j).$$

The quantum dimension is $\dim_q V_q(n) = [n+1]_q = (q^{n+1} - q^{-n-1})/(q - q^{-1})$ and the total quantum dimension squared is $\mathcal{D}^2 = \sum_{j=0}^{k} (\dim_q V_q(j))^2 = (k+2)/(2\sin^2(\pi/(k+2)))$. The S-matrix entries are $S_{mn} = \sqrt{2/(k+2)}\, \sin(\pi(m+1)(n+1)/(k+2))$. This is a modular tensor category: the Reshetikhin–Turaev functor produces the $SU(2)$ Chern–Simons invariants of 3-manifolds.

19.2 The R-matrix as categorical $r(z)$

The R-matrix of $U_q(\mathfrak{g})$ is the categorical incarnation of the collision residue $r(z)$ from the Volume I bar complex.

Proposition 19.5 *(R-matrix from bar arity $(1,1)$).* [PE] For the affine vertex algebra $V_k(\mathfrak{g})$ at level k with $q = e^{\pi i/(k+h^\vee)}$, the arity-$(1,1)$ component of the ordered bar complex $B^{\mathrm{ord}}(V_k(\mathfrak{g}))$ recovers the R-matrix of $U_q(\mathfrak{g})$:

$$r(z) = \mathrm{Res}^{\mathrm{coll}}_{0,2}(\Theta_{V_k(\mathfrak{g})}) \;\longleftrightarrow\; \mathcal{R}_q = \lim_{z\to 0} \mathcal{R}(z)$$

where $r(z) = k\,\Omega/z + O(1)$ is the classical r-matrix with Casimir $\Omega \in \mathfrak{g} \otimes \mathfrak{g}$ and level prefix k (AP126: at $k = 0$ the r-matrix vanishes), and $\mathcal{R}_q$ is the quantized universal R-matrix. The spectral parameter z corresponds to the rapidity variable in the integrable lattice model.

Remark 19.6 (Three r-matrices). Three r-matrix-like objects appear:

(a) $r^{\mathrm{coll}}(z)$: the bar-intrinsic collision residue (Volume I);

(b) $r(z) = k\,\Omega/z$: the classical r-matrix at level k (Belavin–Drinfeld; AP126: vanishes at $k = 0$);

(c) $\mathcal{R}_q$: the universal R-matrix of $U_q(\mathfrak{g})$ (Drinfeld–Jimbo).

The quantization passage $r(z) \rightsquigarrow \mathcal{R}_q$ is the Etingof–Kazhdan quantization theorem. For even E_∞-algebras, $r^{\mathrm{coll}} = r$; for odd generators or genuinely E_1 algebras, they can differ (AP107).

19.3 Kazhdan–Lusztig equivalence

The Kazhdan–Lusztig equivalence is the prototype for the CY programme's quantum group reconstruction.

Theorem 19.7 *(Kazhdan–Lusztig).* [PE] For a simple Lie algebra $\mathfrak{g}$ and a positive integer k, with $q = e^{\pi i/(k+h^\vee)}$, there is an equivalence of braided monoidal categories

$$C_q(\mathfrak{g}) \;\simeq\; \mathrm{KL}_k(\mathfrak{g})$$

where $C_q(\mathfrak{g})$ is the fusion category of $U_q(\mathfrak{g})$ at root of unity, and $\mathrm{KL}_k(\mathfrak{g})$ is the Kazhdan–Lusztig category of $\hat{\mathfrak{g}}$-modules at level k (modules on which the Sugawara Casimir acts semisimply).

Remark 19.8 (Bar-complex interpretation). From the bar-complex perspective, the KL equivalence states that the E_1-representation category of the affine vertex algebra $V_k(\mathfrak{g})$ recovers the quantum group representation category. The Volume I bar complex $B(V_k(\mathfrak{g}))$ encodes the quantum group R-matrix in its arity-$(1,1)$ component; the DK bridge (Volume II, MC3 for all simple types) extends this to the full categorical equivalence.

Proposition 19.9 *(KL and the DK bridge).* [PE] The Kazhdan–Lusztig equivalence factors through the Drinfeld–Kohno bridge (Volume II):

$$\begin{tikzcd}
\mathrm{KL}_k(\mathfrak{g}) \arrow[r, "\sim", "\mathrm{KL}"'] & C_q(\mathfrak{g}) \\
\mathrm{Rep}^{E_1}(V_k(\mathfrak{g})) \arrow[u, hook] \arrow[r, "\mathrm{DK}"] & \mathrm{Rep}_q(\mathfrak{g}) \arrow[u, two heads]
\end{tikzcd}$$

The vertical arrows are: inclusion of KL modules into all $V_k(\mathfrak{g})$-modules (left), and quotient by negligible morphisms (right). The DK bridge (MC3) provides the horizontal equivalence at the level of compact objects.

19.4 Conjecture CY-C: quantum group realization

Conjecture 19.10 *(Quantum group realization — Conjecture CY-C).* [CJ] For a simple Lie algebra $\mathfrak{g}$ and $q = e^{\pi i/(k+h^\vee)}$ with k a positive integer, there exists a CY_2 category $C(\mathfrak{g}, q)$ such that:

(i) $\Phi(C(\mathfrak{g}, q))$ is an E_2-chiral algebra whose underlying vertex algebra is $V_k(\mathfrak{g})$;

(ii) $\mathrm{Rep}^{E_2}(\Phi(C(\mathfrak{g}, q))) \simeq \mathrm{Rep}_q(\mathfrak{g})$ as braided monoidal categories;

(iii) $\kappa_{\mathrm{cat}}(C(\mathfrak{g}, q)) = \dim(\mathfrak{g}) \cdot (k + h^\vee)/(2h^\vee)$, recovering the Volume I modular characteristic of $V_k(\mathfrak{g})$.

Remark 19.11 (Status of Conjecture CY-C). Item (i) is constructed for $\mathfrak{g} = \mathfrak{sl}_N$ via the Bridgeland stability conditions on the CY_2 resolution of the A_{N-1} surface singularity. Item (ii) at the E_1 level is the content of MC3 (proved for all simple types, Volume I/II). The E_2 upgrade requires the Drinfeld center passage (Chapter 13). Item (iii) follows from Theorem CY-D (Theorem 10.1) when the CY category is constructed.

19.5 Yangian and RTT realizations

Definition 19.12 *(Yangian).* The *Yangian* $Y(\mathfrak{g})$ is the $\mathbb{C}[\hbar]$-algebra with generators $\{x_i^{(r)} : i = 1, \dots, \dim\mathfrak{g},\quad r = 0, 1, 2, \dots\}$ and relations determined by the RTT presentation: $R_{12}(u-v)\,T_1(u)\,T_2(v) = T_2(v)\,T_1(u)\,R_{12}(u-v)$ where $T(u) = \sum_r T_r u^{-r}$ is the generating matrix and $R_{12}(u)$ is the Yang R-matrix.

Proposition 19.13 *(Yangian from collision residue).* [PE] The Yangian $Y(\mathfrak{g})$ is recovered from the arity-$(1,1)$ collision residue of the ordered bar complex:

$$Y(\mathfrak{g}) = \langle r^{\mathrm{coll}}(z)\rangle_{\mathrm{RTT}}$$

where the RTT relation is the arity-$(1,1,1)$ bar differential. This is the content of Volume II, MC3: the ordered bar complex $B^{\mathrm{ord}}(V_k(\mathfrak{g}))$ produces the Yangian as the universal algebra generated by the modes of $r(z) = k\,\Omega/z$ (AP126: level k prefix; r-matrix vanishes at $k = 0$).

Remark 19.14 (Three distinct operations). The following three operations on quantum groups must never be conflated:

(a) *Koszul duality*: $V_k(\mathfrak{g}) \mapsto V_k(\mathfrak{g})^!$, the Koszul dual vertex algebra involving the dual level $k' = -k - 2h^\vee$ (AP33);

(b) *Feigin–Frenkel involution*: $k \mapsto -k - 2h^\vee$ within the same family of affine algebras; at the critical level $k = -h^\vee$, this produces the Feigin–Frenkel center $\mathfrak{z}(\hat{\mathfrak{g}}_{-h^\vee}) = \mathrm{Fun}(\mathrm{Op}_{G^L})$;

(c) *Langlands duality*: $\mathfrak{g} \mapsto \mathfrak{g}^L$ (root-coroot exchange), producing a genuinely different Lie algebra for non-simply-laced types.

For $\mathfrak{sl}_2$ (self-dual, self-Langlands), all three coincide on representation categories. For G_2: $G_2^L = G_2$ (self-Langlands) but the Koszul dual $V_k(G_2)^!$ is at level $k' = -k - 8$ (since $h^\vee(G_2) = 4$), while the FF involution also sends $k \mapsto -k - 8$; these coincide as level maps but produce different algebras (AP33).

19.6 BPS algebras and quantum groups

The BPS algebra of a CY3 category provides a physical construction of quantum groups.

Definition 19.15 *(BPS algebra).* For a CY3 category C with stability condition σ, the *BPS algebra* $\mathcal{A}_\sigma^{\mathrm{BPS}}(C)$ is the cohomological Hall algebra (CoHA) of C:

$$\mathcal{A}_\sigma^{\mathrm{BPS}}(C) = \bigoplus_{\gamma\in K_0(C)} H^*_{\mathrm{BM}}(\mathcal{M}_\sigma(\gamma),\ \phi_{\mathrm{tr}W})$$

where $\mathcal{M}_\sigma(\gamma)$ is the moduli of σ-semistable objects of class γ, W is the CY potential, and $\phi_{\mathrm{tr}W}$ is the vanishing cycle sheaf. The multiplication is the Hall product (from short exact sequences).

Proposition 19.16 *(CoHA as quantum group).* [PE] For the Jordan quiver (one vertex, one loop) with potential $W = \mathrm{tr}(B^3/3)$ and CY3 geometry $\mathbb{C}^3$:

(i) The CoHA is isomorphic to the positive half of the affine Yangian $Y(\widehat{\mathfrak{gl}}_1)^+$ (Schiffmann–Vasserot, Maulik–Okounkov);

(ii) The full affine Yangian $Y(\widehat{\mathfrak{gl}}_1)$ is recovered from the Drinfeld double of the CoHA;

(iii) The quantum toroidal algebra $U_{q,t}(\widehat{\widehat{\mathfrak{gl}}}_1)$ arises from the K-theoretic refinement (replacing cohomology by K-theory).

Remark 19.17 (Slab as bimodule). In the Dimofte framework (Volume II, Part III), the BPS algebra arises from the 3d holomorphic-topological theory on the slab $X \times [0,1]$. The slab has *two* boundary components ($X \times \{0\}$ and $X \times \{1\}$), making the algebra of operators on the slab a bimodule for the two boundary algebras. The Drinfeld double is the endomorphism algebra of the identity bimodule. This bimodule structure is essential: a Swiss-cheese disk has one closed and one open boundary component; the slab has two open boundary components.

Example 19.18 *(Resolved conifold).* For the resolved conifold $O(-1) \oplus O(-1) \to \mathbb{P}^1$, the CY$_3$ category is $D^b(\mathrm{Coh}(O(-1)^{\oplus 2}))$ and the BPS algebra is the affine Yangian $Y(\widehat{\mathfrak{sl}}_2)$. The modular characteristic is $\kappa_{\mathrm{cat}} = 1$ (from $\chi(O_{\mathbb{P}^1}) = 1$; see Proposition 10.2). The wall-crossing formula of Kontsevich–Soibelman governs the dependence of the BPS spectrum on the stability condition.

19.7 Wall-crossing and Kontsevich–Soibelman

The BPS spectrum of a CY$_3$ category depends on the stability condition $\sigma \in \mathrm{Stab}(C)$. As σ crosses a wall of marginal stability, the BPS states reorganize: bound states form or decay. The Kontsevich–Soibelman wall-crossing formula governs this reorganization at the level of the motivic Hall algebra.

Definition 19.19 *(Wall of marginal stability).* A *wall of marginal stability* in $\mathrm{Stab}(C)$ is a real codimension-1 locus W_{γ_1,γ_2} where two classes $\gamma_1, \gamma_2 \in K_0(C)$ with $\gamma = \gamma_1 + \gamma_2$ have aligned central charges: $Z_\sigma(\gamma_1)/Z_\sigma(\gamma_2) \in \mathbb{R}_{>0}$. On opposite sides of W_{γ_1,γ_2}, a bound state of class γ may exist or not.

Theorem 19.20 *(Kontsevich–Soibelman wall-crossing formula).* [PE] For a CY$_3$ category C with stability condition σ, define the *BPS automorphism* for each class $\gamma \in K_0(C)$:

$$U_\gamma = \exp\left(\sum_{n=1}^{\infty} \frac{(-1)^{n \dim \mathcal{M}(\gamma)}}{n^2}\, \Omega(\gamma;\sigma) \cdot \hat{e}_{n\gamma}\right)$$

where $\Omega(\gamma;\sigma) \in \mathbb{Z}$ is the (numerical) Donaldson–Thomas invariant and $\hat{e}_\gamma$ is the corresponding element of the quantum torus algebra $\hat{e}_\gamma \hat{e}_{\gamma'} = (-1)^{\langle \gamma,\gamma' \rangle} \hat{e}_{\gamma+\gamma'}$. The ordered product

$$\mathcal{A}_\sigma = \prod^{\curvearrowleft}_{\gamma : \arg Z_\sigma(\gamma) = \theta} U_\gamma$$

(clockwise ordering by phase $\theta = \arg Z_\sigma(\gamma)$) is *independent of* σ: as σ crosses a wall, the individual $\Omega(\gamma;\sigma)$ jump, but the ordered product $\mathcal{A}_\sigma$ is invariant.

Remark 19.21 (Motivic refinement). The numerical DT invariants $\Omega(\gamma;\sigma) \in \mathbb{Z}$ are shadows of the motivic DT invariants $[\mathcal{M}_\sigma(\gamma)]_{\mathrm{vir}} \in K_0(\mathrm{Var})$ in the Grothendieck ring of varieties. The motivic wall-crossing formula replaces Ω by the motivic class, and the quantum torus by the motivic quantum torus. The identification "scattering = shadow" (Volume I, AP42) holds at the motivic level but fails at the naive BCH level: the full motivic Hall algebra is required.

Conjecture 19.22 *(Wall-crossing and the bar complex).* [CJ] The wall-crossing formula has a bar-complex interpretation: crossing a wall W_{γ_1,γ_2} in $\mathrm{Stab}(C)$ induces a mutation of the ordered bar complex $B^{\mathrm{ord}}(A_C)$ at the corresponding arity. Concretely:

(i) The BPS automorphism U_γ corresponds to the arity-$|\gamma|$ component of the MC element $\Theta^{E_1}_{A_C}$ in the E_1 convolution algebra;

(ii) Wall-crossing preserves the full MC element $\Theta^{E_1}_{A_C}$ (it is an invariant of the CY category C, not of the stability condition);

(iii) The individual BPS invariants $\Omega(\gamma;\sigma)$ are *projections* of $\Theta^{E_1}_{A_C}$ that depend on σ; the MC element itself is wall-crossing invariant.

19.8 The modular characteristic κ_{cat}

The modular characteristic of the quantum group representation category connects the categorical datum to the genus expansion of Volume I.

PROPOSITION 19.23 *(κ_{cat} for quantum groups).* [PE] For the affine vertex algebra $V_k(\mathfrak{g})$ at level k with $q = e^{\pi i/(k+h^\vee)}$, the modular characteristic of the associated CY$_2$ category $C(\mathfrak{g}, q)$ (Conjecture 19.10) is:

$$\kappa_{\mathrm{cat}}(C(\mathfrak{g}, q)) = \dim(\mathfrak{g}) \cdot \frac{k + h^\vee}{2h^\vee}$$

recovering the Volume I modular characteristic of $V_k(\mathfrak{g})$ (Volume I, Theorem D). For the standard families:

$\mathfrak{g}$	κ_{cat}	q
$\mathfrak{sl}_2$	$3(k+2)/4$	$e^{\pi i/(k+2)}$
$\mathfrak{sl}_N$	$\frac{N^2-1}{2N}(k+N)$	$e^{\pi i/(k+N)}$
$\mathfrak{so}_{2n}$	$\frac{n(2n-1)(k+2n-2)}{2(2n-2)}$	$e^{\pi i/(k+2n-2)}$
E_8	$\frac{248(k+30)}{60}$	$e^{\pi i/(k+30)}$

Remark 19.24 (κ_{cat} versus other invariants). Three distinct invariants must be distinguished (AP113):

(i) κ_{cat}: the modular characteristic of the CY category, equal to the genus-1 Hodge class coefficient of the associated chiral algebra (Volume I, Theorem D);

(ii) κ_{ch}: the modular characteristic computed from the chiral algebra directly (may differ from κ_{cat} when the CY-to-chiral functor involves additional data);

(iii) κ_{BKM}: the BKM modular weight, a number-theoretic invariant arising from the BKM denominator formula (Chapter ??; distinct from both κ_{cat} and κ_{ch}).

For CY$_2$ categories with $\mathfrak{g}$-symmetry, all three can be independently computed. For the $K3 \times E$ example: $\kappa_{\mathrm{ch}} = 3$, $\kappa_{\mathrm{BKM}} = 5$, $\kappa_{\mathrm{cat}} = 3$ (Proposition 10.4).

PROPOSITION 19.25 *(Complementarity for κ_{cat}).* [PE] Under Koszul duality $V_k(\mathfrak{g}) \mapsto V_{k'}(\mathfrak{g})$ with $k' = -k - 2h^\vee$ (Feigin–Frenkel involution):

$$\kappa_{\mathrm{cat}}(C(\mathfrak{g}, q)) + \kappa_{\mathrm{cat}}(C(\mathfrak{g}, q')) = 0$$

where $q' = e^{\pi i/(k'+h^\vee)} = e^{-\pi i/(k+h^\vee)}$. This is the CY incarnation of the Volume I complementarity $\kappa_{\mathrm{ch}} + \kappa'_{\mathrm{ch}} = 0$ for KM/free fields (Volume I, Theorem C).

Proof. Under $k \mapsto k' = -k - 2h^\vee$: $\kappa'_{\mathrm{cat}} = \dim(\mathfrak{g})(k' + h^\vee)/(2h^\vee) = \dim(\mathfrak{g})(-k - h^\vee)/(2h^\vee) = -\dim(\mathfrak{g})(k + h^\vee)/(2h^\vee) = -\kappa_{\mathrm{cat}}$. □

19.9 Shadow obstruction tower for quantum group categories

The shadow obstruction tower Θ_{A_C} (Volume I, Part II) acquires categorical meaning through the quantum group lens.

PROPOSITION 19.26 *(Shadow depth from R-matrix pole structure).* [PE] For $V_k(\mathfrak{g})$ with R-matrix $R(z) = 1 + r(z) + O(r^2)$ where $r(z) = k\,\Omega/z$ (AP126: level k prefix, r-matrix vanishes at $k = 0$):

(i) The shadow depth $r_{\max} = 3$ (class L): the collision residue $r(z)$ has a single pole, the cubic shadow C is nonzero (from the Lie bracket), and the quartic resonance class Q vanishes by semisimplicity of $\mathfrak{g}$;

(ii) The averaging map (Volume I) sends $r(z) = k\,\Omega/z \mapsto \kappa_{\mathrm{cat}}$: the full z-dependent profile is killed by Σ_2-coinvariance, leaving only the scalar modular characteristic;

(iii) The KZ associator Φ_{KZ} (the arity-3 component of $\Theta^{E_1}_{A_C}$) maps under averaging to the cubic shadow C.

Remark 19.27 (From quantum groups to the shadow tower). The passage from quantum group data to the shadow obstruction tower inverts the construction of this chapter: the quantum group $U_q(\mathfrak{g})$ determines the R-matrix, which is the arity-$(1,1)$ component of the E_1 MC element Θ^{E_1}, whose Σ_n-coinvariant projections are the shadow tower. The reconstruction problem (from quantum group to chiral algebra) is solved by the DK bridge (MC3 for all simple types): the categorical data of $\mathrm{Rep}_q(\mathfrak{g})$ uniquely determines the chiral algebra $V_k(\mathfrak{g})$ up to the completion issues of MC4.

Part IV

The Seven Faces of $r_{\mathrm{CY}}(z)$

Chapter 20

The Bar-Cobar Bridge to Volume I

The shadow obstruction tower Θ_A of Volume I applies to chiral algebras arising from the cyclic bar complex of a Calabi–Yau category. This chapter establishes the dictionary between the CY cyclic bar complex $\mathrm{CC}_\bullet(C)$ and the chiral bar complex $B(A)$, computes the shadow tower of $\mathbb{C}^3$, traces the passage from finite to infinite shadow depth under the factorization envelope, and identifies the open string field theory realization of Koszul duality.

20.1 CY bar complex vs chiral bar complex

Let C be a smooth proper CY_3 category with Serre functor $\mathbb{S}_C \simeq [3]$, and let $A = A_C$ denote the chiral algebra produced by the CY-to-chiral functor Φ (Theorem CY-A_2 for $d = 2$; conjectural for $d = 3$). The cyclic bar complex $\mathrm{CC}_\bullet(C)$ and the chiral bar complex $B(A)$ are related by a canonical quasi-isomorphism

$$\mathrm{CC}_\bullet(C) \;\simeq\; B(A_C) \tag{CY-A(ii)}$$

as factorization coalgebras on $\mathrm{Ran}(X)$.

The precise dictionary is as follows.

Proposition 20.1 *(Bar complex dictionary).* Under the identification CY-A(ii):

(i) The cyclic differential $b + B$ on $\mathrm{CC}_\bullet(C)$ corresponds to the bar differential d_B on $B(A)$.

(ii) The Connes operator $B\colon \mathrm{HH}_n(C) \to \mathrm{HH}_{n+1}(C)$ corresponds to the arity-preserving component of d_B (the internal differential), while the Hochschild differential b corresponds to the arity-lowering component (the bar contraction).

(iii) The S^1-action on $\mathrm{CC}_\bullet(C)$ (the cyclic rotation) corresponds to the factorization coalgebra structure on $B(A)$, with the cocomposition maps Δ_Γ indexed by stable graphs Γ.

(iv) The Hodge filtration on $\mathrm{HH}_*(C)$ corresponds to the arity filtration on $B(A)$: the arity-r piece $B^{(r)}(A)$ captures Hochschild chains of tensor length $\leq r$.

[PH]

Remark 20.2 (Three functors, three outputs). The bar complex $B(A)$ is a factorization *coalgebra*. Three distinct functors produce three distinct objects from $B(A)$:

(1) $\Omega(B(A)) \simeq A$ recovers the original algebra (bar-cobar inversion, Theorem B of Volume I).

(2) $D_{\mathrm{Ran}}(B(A)) \simeq B(A^!)$ is the Verdier dual, a factorization *algebra* identified with the bar of the Koszul dual $A^!$ (Theorem A, Convention conv:bar-coalgebra-identity).

(3) $\mathcal{Z}^{\mathrm{der}}_{\mathrm{ch}}(A) = \mathrm{RHom}(\Omega(B(A)), A)$ is the chiral derived center, computing the universal bulk (Theorem H).

These are not the same operation, and the CY cyclic bar complex $\mathrm{CC}_\bullet(C)$ corresponds to $B(A)$ itself, not to the derived center and not to the cobar.

20.2 The shadow tower of $\mathbb{C}^3$: MacMahon meets Koszul

The affine threefold $\mathbb{C}^3$ is the simplest case where both sides of the CY-to-chiral correspondence are explicit.

THEOREM 20.3 *(Shadow tower of $\mathbb{C}^3$).* [PH] The shadow obstruction tower of $W_{1+\infty}$ at $c = 1$ satisfies:

(i) The bar Euler product equals the inverse MacMahon function:

$$\sum_{n\geq 0} \dim B^{(n)}_{E_\infty}(A)\cdot q^n \;=\; \frac{1}{M(q)}, \qquad M(q) = \prod_{n\geq 1}\frac{1}{(1-q^n)^n}.$$

(ii) The modular characteristic is $\kappa_{\mathrm{ch}} = 1$, verified by five independent paths:

(a) genus-1 free energy: $F_1 = \kappa_{\mathrm{ch}}/24 = 1/24$;

(b) degree-0 MacMahon exponent: $\log M(q) = \sum \sigma_2(k) q^k/k$ matches $\kappa_{\mathrm{ch}} = 1$;

(c) channel decomposition: $\sum_{s\geq 1}\kappa_s^{\mathrm{eff}} = \sum_{s\geq 1} s\cdot(1/s) = \sum 1 = +\infty$ (at finite N, $\kappa_{\mathrm{ch}}(W_N) = c(H_N - 1)$; the total $\kappa_{\mathrm{ch}}(W_{1+\infty}, c=1) = \kappa_{\mathrm{ch}}(H_1) = 1$);

(d) Hodge-theoretic: via the categorical trace $\chi^{\mathrm{CY}}(\mathrm{Perf}(\mathbb{C}^3))$;

(e) complementarity: $\kappa_{\mathrm{ch}}(\mathbb{C}^3) + \kappa_{\mathrm{ch}}(\mathrm{dual}) = 0$ (free-field anti-symmetry).

(iii) Per-channel modular characteristics: for the spin-s channel of $W_{1+\infty}$,

$$\kappa_s = \frac{c}{s} = \frac{1}{s}, \qquad \kappa_s^{\mathrm{eff}} = s\cdot\kappa_s = 1 \quad \text{(constant per MacMahon level)}.$$

This constancy is the shadow-DT Rosetta stone: each energy level contributes equally to the modular characteristic, and the MacMahon function $M(q) = \prod(1-q^n)^{-n}$ is the partition function of a system with effective $\kappa_{\mathrm{ch}} = 1$ per energy level.

[PH]

Verification: `compute/lib/c3_shadow_tower.py` (73 tests), `compute/lib/macmahon_shadow_decomposition.`

Remark 20.4 (Envelope creates infinite depth from abelian input). The Heisenberg algebra H_1 (a single free boson) has shadow depth class G with $r_{\max} = 2$: the tower terminates at arity 2, and $\kappa_{\mathrm{ch}} = 1$ is the only nonvanishing shadow invariant.

The vertex algebra $W_{1+\infty}$ at $c = 1$, the factorization envelope of the Lie conformal algebra of polyvector fields on $\mathbb{C}^3$, has shadow depth class M with $r_{\max} = \infty$. The spin-2 channel (Virasoro at $c = 1$) already has infinite shadow depth, with $\kappa_2 = 1/2$, $\alpha_2 = 2$, $S_4 = 10/27$.

This discrepancy between input (class G, depth 2) and output (class M, depth ∞) reflects the non-conservativity of the factorization envelope on shadow invariants. Normal ordering, Wick contractions, and OPE singular terms generate higher-arity shadows invisible at the E_1-level. Each stage of $E_1 \to E_2 \to E_\infty$-chiral introduces genuinely new invariants.

20.3 Open string field theory: Koszul duality at chain level

For $\mathbb{C}^3$, the cyclic A_∞-algebra is the exterior algebra $A = \bigwedge^*(V)$ with $V = \mathbb{C}^3$, equipped with the CY3 trace $\langle a, b\rangle = \mathrm{Tr}(a \wedge b)$ (projection to top degree).

THEOREM 20.5 *(Chain-level Koszul duality for $\mathbb{C}^3$).* [PH] The Koszul duality pair for $A = \bigwedge^*(\mathbb{C}^3)$ satisfies:

(i) $A^! = \mathrm{Sym}^*(\mathbb{C}^{3,*})$, the symmetric algebra on the dual vector space. This is the $\mathrm{Com}^! = \mathrm{Lie}$ incarnation at the chain level: the exterior algebra (free Com-algebra on V) is Koszul dual to the symmetric algebra (free Lie-algebra envelope on V^*).

(ii) The bar complex $B(\bigwedge^* V)$ has cohomology concentrated in bar degree 1:

$$H^*(B(\bigwedge^* V)) \;\simeq\; \mathrm{Sym}^*(V^*[-1]),$$

confirming that A is Koszul (chirally and classically).

(iii) The bar-cobar round-trip recovers A itself:

$$\Omega(B(\bigwedge^* V)) \;\xrightarrow{\sim}\; \bigwedge^* V.$$

[PH]

Verification: `compute/lib/bar_comparison_c3.py`, `compute/lib/e2_barcobar_koszul.py` (73 tests).

20.4 From shuffle product to λ-bracket: the seven-step chain

The Schiffmann–Vasserot shuffle presentation of $Y^+(\widehat{\mathfrak{gl}}_1)$ does *not* directly produce the λ-bracket of $R_{\mathbb{C}^3}$. Seven comparison steps are required.

Construction 20.6 *(The seven-step chain).* The chain connecting the shuffle algebra to the chiral algebra:

1. **Shuffle algebra** $\mathrm{Sh}(h_1, h_2, h_3)$: the associative algebra with generators $e(z)$ at each spectral parameter z and multiplication by the rational shuffle kernel $\zeta(z_i - z_j) = \prod_a (z_i - z_j + h_a)/(z_i - z_j - h_a)$.
2. **Affine Yangian** $Y(\widehat{\mathfrak{gl}}_1)$: the RTT-presentation with generators e_j, f_j, ψ_j satisfying the affine Yangian relations. Schiffmann–Vasserot proved $\mathrm{Sh}^+ \simeq Y^+(\widehat{\mathfrak{gl}}_1)$.
3. $W_{1+\infty}$ **modes**: the Feigin–Frenkel realization of $W_{1+\infty}$ as the limit $\lim_{N\to\infty} W_N$, with explicit mode algebra generators W^s_n for spin s and mode n.
4. **OPE modes**: the vertex algebra OPE $W^s(z)W^{s'}(w) \sim \sum_n C^{ss'}_n(W)(z-w)^{-n-1}$, extracting the singular coefficients $C^{ss'}_n$.
5. λ-**bracket**: $\{W^s_\lambda W^{s'}\} = \sum_n \lambda^{(n)} C^{ss'}_n(W)$, where $\lambda^{(n)} = \lambda^n/n!$ is the divided power (AP44: the coefficient at order n in the λ-bracket is $C^{ss'}_n/n!$, not $C^{ss'}_n$).

6. **Lie conformal algebra** $R_{\mathbb{C}^3}$: the GL(3)-invariant subalgebra of the Schouten–Nijenhuis Lie conformal algebra on polyvector fields $\mathrm{PV}^*(\mathbb{C}^3)$.
7. **Factorization envelope** $U^{\mathrm{ch}}(R_{\mathbb{C}^3})$: the enveloping vertex algebra, which by the Nishinaka–Vicedo construction recovers $W_{1+\infty}$.

Steps 1⟶2 is Schiffmann–Vasserot. Steps 2⟶3 is Prochazka–Rapcak. Steps 3⟶4 is the standard mode-to-OPE dictionary. Step 4⟶5 requires the divided-power convention (AP44). Steps 5⟶6 is the Kontsevich–Soibelman identification of $R_{\mathbb{C}^3}$ as the GL(3)-invariant part of $\mathrm{PV}^*(\mathbb{C}^3)$ with the Schouten–Nijenhuis bracket. Step 6⟶7 is the factorization envelope.

The composition $1 \to 7$ is the full CY-to-chiral functor for $\mathbb{C}^3$.

Verification: `compute/lib/c3_lie_conformal.py`, `compute/lib/c3_envelope_comparison.py` (52 tests).

Warning 20.7 (Shuffle product ≠ λ-bracket). The shuffle product on $\mathrm{Sh}(h_1, h_2, h_3)$ is an associative product on symmetric functions of the spectral parameters, while the λ-bracket is a Lie conformal bracket on fields. The two are related by the seven-step chain above, not by a direct identification. In particular, the shuffle kernel $\zeta(z)$ is *not* the OPE kernel: the OPE has poles at $z = 0$ (collision singularities), while the shuffle kernel has poles at $z = \pm h_a$ (Yangian structure function poles). These coincide only after the identification of the parameters h_a with the deformation parameters of $W_{1+\infty}$ and the passage from additive to multiplicative spectral parameters.

20.5 Koszul duality: CY vs chiral

The chiral Koszul duality of Volume I (Theorems A and B) descends to CY Koszul duality.

Conjecture 20.8 *(CY Koszul duality dictionary).* [CJ] For a CY$_3$ category C with chiral algebra $A = A_C$:

(i) The chiral Koszul dual $A^!$ corresponds to the *mirror* CY category:

$$A_C^! \;\simeq\; A_{C^!}$$

where $C^!$ is the Koszul dual CY category (for $C = D^b(\mathrm{Coh}(X))$ of a CY manifold X, this is $C^! \simeq \mathrm{Fuk}(X^\vee)$ under homological mirror symmetry).

(ii) The complementarity theorem (Theorem C) gives:

$$Q_g(A_C) + Q_g(A_{C^!}) = H^*(\overline{\mathcal{M}}_g, \mathcal{Z}(A_C))$$

where $\mathcal{Z}(A_C) = \mathcal{Z}^{\mathrm{der}}_{\mathrm{ch}}(A_C)$ is the chiral derived center.

(iii) The modular characteristics satisfy:

$$\kappa_{\mathrm{ch}}(A_C) + \kappa_{\mathrm{ch}}(A_{C^!}) = 0$$

when C arises from a free-field or lattice construction (anti-symmetry), and more generally $\kappa_{\mathrm{ch}} + \kappa'_{\mathrm{ch}} = \rho \cdot K$ for algebras with nontrivial contact terms (Theorem D, AP24).

20.6 The five theorems in the CY setting

We summarize the status of the five main theorems of Volume I when specialized to chiral algebras arising from CY categories.

THEOREM *(The five theorems for CY chiral algebras).* Let $A = A_C$ be the chiral algebra of a CY$_3$ category C. Then:

Theorem A (adjunction). The bar-cobar adjunction $B \dashv \Omega$ restricts to CY chiral algebras: $B(A_C)$ is a factorization coalgebra on $\mathrm{Ran}(X)$, and $D_{\mathrm{Ran}}(B(A_C)) \simeq B(A_{C^!})$. The CY identification CY-A(ii) gives $\mathrm{CC}_\bullet(C) \simeq B(A_C)$.

Theorem B (inversion). Bar-cobar inversion $\Omega(B(A_C)) \xrightarrow{\sim} A_C$ holds on the Koszul locus. For CY categories, chirally Koszul is equivalent to the formality of $\mathrm{CC}_\bullet(C)$ as a dg coalgebra.

Theorem C (complementarity). The CY Euler characteristic $\chi(C)$ splits into complementary halves: $Q_g(C) + Q_g(C^!)$ recovers the full Hochschild cohomology.

Theorem D (modular characteristic). The genus-g obstruction is $\mathrm{obs}_g(A_C) = \kappa_{\mathrm{ch}}(A_C) \cdot \lambda_g^{\mathrm{FP}}$ on the uniform-weight lane, where $\kappa_{\mathrm{ch}}(A_C) = \chi^{\mathrm{CY}}(C)$ is the CY modular characteristic. For compact CY$_3$, the BCOV prediction gives $\kappa_{\mathrm{ch}} = \chi_{\mathrm{top}}/24$.

Theorem H (Hochschild). $\mathrm{ChirHoch}^*(A_C)$ is polynomial in degrees $\{0, 1, 2\}$, and the CY Hochschild calculus of C (HKR decomposition, Connes operator, categorical Hodge structure) is faithfully reflected in $\mathrm{ChirHoch}^*(A_C)$.

20.7 Compact CY$_3$ examples

The following compact CY$_3$ families illustrate the range of shadow tower behaviour. In each case, the predicted modular characteristic is $\kappa_{\mathrm{ch}} = \chi_{\mathrm{top}}/24$ (BCOV prediction); the shadow depth class, Hochschild data, and BKM structure vary.

The banana manifold: $\kappa_{\mathrm{ch}} = 0$ with nontrivial tower

The banana manifold X_{ban} (Schoen's CY$_3$, the fiber product of two rational elliptic surfaces over $\mathbb{P}^1$) has Hodge numbers $h^{1,1} = h^{2,1} = 19$ and Euler characteristic $\chi = 0$.

PROPOSITION 20.9 *(Shadow tower of the banana manifold).* The shadow obstruction tower of X_{ban} has:

(i) $\kappa_{\mathrm{ch}} = \chi/24 = 0$. The arity-2 scalar shadow *vanishes*.

(ii) $S_4 = -44$ (quartic shadow from the genus-0 GV invariants $n^0_{d_1,d_2} = -2$ of the banana curve classes).

(iii) The shadow tower starts at arity 4, not arity 2: the cubic shadow is invisible when $\kappa_{\mathrm{ch}} = 0$ (it enters as $\kappa_{\mathrm{ch}} \cdot \alpha$, which vanishes).

(iv) The critical discriminant $\Delta = 8\kappa_{\mathrm{ch}} S_4 = 0$ (since $\kappa_{\mathrm{ch}} = 0$). The standard single-line classification (G/L/C/M) does not directly apply.

(v) Shadow depth class: M (infinite tower) shifted to arity 4. The BKY partition function involves infinitely many banana-curve BPS states, generating an infinite shadow tower whose leading term is quartic.

[PH]

Remark 20.10 (AP31 in action). The banana manifold is the prototypical example of AP31: $\kappa_{\mathrm{ch}} = 0$ does *not* imply $\Theta_A = 0$. The vanishing of κ_{ch} means the bar complex is uncurved ($d^2 = 0$ at the leading scalar level), but the higher-arity components of Θ_A (the quartic shadow Q and beyond) are nonvanishing, sourced by the instanton contributions (genus-0 GV invariants of the banana curves).

Three CY_3 examples with $\kappa_{\mathrm{ch}} = 0$ show qualitatively different behaviour:

- Heisenberg at $k = 0$: $\kappa_{\mathrm{ch}} = 0$, trivially uncurved, class G.
- Virasoro at $c = 0$: $\kappa_{\mathrm{ch}} = 0$, but higher shadows from contact terms (class M).
- Banana manifold: $\kappa_{\mathrm{ch}} = 0$, but higher shadows from *instantons* (class M, shifted to arity 4).

Verification: `compute/lib/banana_shadow.py` (63 tests).

Enriques $\times E$: the $\mathbb{Z}_2$ quotient of K3 $\times E$

Let $S_{\mathrm{Enr}} = \mathrm{K3}/\iota$ be an Enriques surface (ι a fixed-point-free involution), and E an elliptic curve. The product $S_{\mathrm{Enr}} \times E$ is a generalized CY_3 (the canonical bundle $K_{S_{\mathrm{Enr}}\times E}$ is 2-torsion, so $h^{3,0} = 0$). Nevertheless, the shadow tower machinery applies to the associated vertex algebra, and the Borcherds lift on $O(2, 10)$ provides the automorphic structure.

THEOREM 20.11 *(Shadow tower of Enriques $\times E$).* [PH] The shadow data of Enriques $\times E$ is:

(i) $\kappa_{\mathrm{BKM}}(\mathrm{Enr}\times E) = 4$, the weight of the Enriques Borcherds product Ψ on $O(2, 10)$ (Allcock 2000).

(ii) Root multiplicities come from the Enriques elliptic genus $\phi_{\mathrm{Enr}} = \frac{1}{2}\phi_{0,1}$: the Fourier coefficients are $f_{\mathrm{Enr}}(n, l) = f_{\mathrm{K3}}(n, l)/2$ (all integers since f_{K3} has even coefficients).

(iii) The ratio $\kappa_{\mathrm{BKM}}(\mathrm{K3} \times E)/\kappa_{\mathrm{BKM}}(\mathrm{Enr} \times E) = 5/4$.

(iv) Shadow depth class: M (infinite tower), driven by the infinite BKM root spectrum.

[PH]

Remark 20.12 (The $\mathbb{Z}_2$ quotient on κ_{ch}). The passage from K3 $\times E$ ($\kappa_{\mathrm{ch}} = 5$) to Enriques $\times E$ ($\kappa_{\mathrm{ch}} = 4$) under the $\mathbb{Z}_2$ quotient does *not* halve κ_{ch}. Rather, the weight of the Borcherds product drops by 1: the $O(2, 10)$ lattice for the Enriques surface has a different constant term in the Borcherds input form than the $O(3, 19)$ lattice for K3, and the weight formula gives $8/2 = 4$ instead of $10/2 = 5$. The ratio $5/4$ is not a simple topological quotient; it reflects the lattice-theoretic structure of the Borcherds lift.

Verification: `compute/lib/enriques_shadow.py` (72 tests).

The quintic threefold: non-integer κ_{ch} and BKM obstruction

The quintic hypersurface $Q \subset \mathbb{P}^4$ has $h^{1,1} = 1$, $h^{2,1} = 101$, $\chi = -200$.

PROPOSITION 20.13 *(Quintic shadow tower).* The shadow data of the quintic is:

(i) $\kappa_{\mathrm{ch}} = \chi/24 = -25/3$ (non-integer).

(ii) The non-integrality of κ_{ch} obstructs a naive BKM superalgebra: BKM root multiplicities must be integers, and the standard denominator-identity machinery requires integral Borcherds lift weight. The quintic has no natural Borcherds–Kac–Moody structure.

(iii) The shadow metric $Q_L(t) = 4\kappa_{\mathrm{ch}}^2 + 12\kappa_{\mathrm{ch}}\alpha t + (9\alpha^2 + 16\kappa_{\mathrm{ch}}S_4)t^2$ is well-defined for fractional κ_{ch}. With $\kappa_{\mathrm{ch}} = -25/3$: $Q_L(0) = 2500/9 > 0$.

(iv) The GW invariants are known to high degree:

$$n_1^0 = 2875, \quad n_2^0 = 609250, \quad n_3^0 = 317206375, \quad n_4^0 = 242467530000.$$

These are BPS state counts via the Gopakumar–Vafa formula.

(v) Shadow depth class: M (infinite tower). The quintic has an infinite BPS spectrum with curves of all degrees and genera; the shadow tower is controlled by the BCOV holomorphic anomaly equation.

[PH]

Remark 20.14 ($\chi/24$ integrality survey). The quantity $\chi/24$ is an integer if and only if $12 \mid (h^{1,1} - h^{2,1})$. Among standard complete intersections:

- $\mathbb{P}^5[3,3]$: $\chi = -144$, $\kappa_{\mathrm{ch}} = -6$ (integer).
- $\mathbb{P}^5[2,4]$: $\chi = -176$, $\kappa_{\mathrm{ch}} = -22/3$ (non-integer).
- $\mathbb{P}^7[2,2,2,2]$: $\chi = -128$, $\kappa_{\mathrm{ch}} = -16/3$ (non-integer).
- Quintic: $\chi = -200$, $\kappa_{\mathrm{ch}} = -25/3$ (non-integer).

Non-integrality of κ_{ch} is the generic situation for compact CY_3 manifolds. A BKM structure requires additional input beyond the BCOV prediction.

Verification: `compute/lib/quintic_shadow_obstruction.py` (86 tests).

Hochschild homology of compact CY_3 examples

The Hochschild homology $\mathrm{HH}_*(D^b(\mathrm{Coh}(X)))$ is computed from the HKR decomposition $\mathrm{HH}_p \simeq \bigoplus_q H^q(X, \Omega_X^{3-p})$ using the CY_3 isomorphism $\bigwedge^p T_X \simeq \Omega_X^{3-p}$.

PROPOSITION 20.15 *(Hochschild data of compact CY_3 examples).* (i) $X = \mathrm{K3} \times E$: $\dim \mathrm{HH}_* = 96$, with Hodge decomposition

$$(\dim \mathrm{HH}_0, \dim \mathrm{HH}_1, \dim \mathrm{HH}_2, \dim \mathrm{HH}_3) = (4, 44, 44, 4).$$

The palindromic symmetry $\dim \mathrm{HH}_p = \dim \mathrm{HH}_{3-p}$ reflects Serre duality.

(ii) $X = Q$ (quintic): $\dim \mathrm{HH}_* = 208$, with

$$(\dim \mathrm{HH}_0, \dim \mathrm{HH}_1, \dim \mathrm{HH}_2, \dim \mathrm{HH}_3) = (2, 102, 102, 2).$$

The concentration in degrees 1 and 2 reflects $h^{1,1} = 1$ and $h^{2,1} = 101$: moduli dominate, and the CY structure contributes only the top and bottom forms.

[PH]

Verification: `compute/lib/cy3_hochschild.py` (71 tests).

20.8 Integrable systems and lattice models

The shadow obstruction tower admits a dictionary with classical integrable systems, where the conserved quantities I_r correspond to shadow invariants at each arity.

The Calogero–Moser/Bethe dictionary

The Calogero–Moser system with N particles on a line and inverse-square interaction $V(x) = \sum_{i<j} g^2/(x_i - x_j)^2$ has N independent conserved quantities $I_1, I_2, \ldots, I_N$ (the Sekiguchi operators), with $I_2 = H$ (the Hamiltonian) and I_k of degree k in the momenta.

Conjecture 20.16 *(Shadow-integrable dictionary).* [CJ] There is a dictionary between the shadow obstruction tower invariants and the Calogero–Moser conserved quantities:

$$\begin{aligned} I_2 \text{ (quadratic Hamiltonian)} &\longleftrightarrow \kappa_{\mathrm{ch}} \text{ (modular characteristic)},\\ I_3 \text{ (cubic integral)} &\longleftrightarrow C \text{ (cubic shadow)},\\ I_4 \text{ (quartic integral)} &\longleftrightarrow Q \text{ (quartic shadow / contact invariant)}. \end{aligned}$$

The Bethe ansatz equations for the Calogero–Moser system correspond to the Maurer–Cartan equation $D\Theta + \frac{1}{2}[\Theta, \Theta] = 0$ of the modular convolution algebra, projected to each arity: the r-th conserved quantity I_r is the genus-0 arity-r projection of the MC element Θ_A.

Remark 20.17. The dictionary is *structural*, not a theorem. It captures the parallel between two filtration structures: the arity filtration on Θ_A (controlling the shadow depth classification G/L/C/M) and the degree filtration on the conserved quantities I_r (controlling the integrability class of the Calogero–Moser system). The coupling constant g^2 maps to κ_{ch}, the number of particles N maps to the number of strong generators, and the Bethe roots map to the shadow metric zeros.

Verification: `compute/lib/cross_volume_shadow_bridge.py` (60 tests).

Lattice finite-size corrections

For a critical lattice model (e.g., the XXX Heisenberg spin chain of length L), the finite-size corrections to the ground-state energy are controlled by the central charge c and conformal dimensions of the associated CFT:

$$E_0(L) = L\varepsilon_\infty - \frac{\pi c}{6L} + O(L^{-3}).$$

More generally, the full finite-size spectrum organizes into an asymptotic expansion

$$E_n(L) - L\varepsilon_\infty = \frac{2\pi}{L}\Big(-\frac{c}{12} + h_n + \bar{h}_n\Big) + \sum_{k\geq 2} \frac{a_k(n)}{L^{2k-1}}.$$

PROPOSITION 20.18 *(Finite-size corrections as shadow tower).* The finite-size correction coefficients $a_k(n)$ correspond to shadow tower data: the leading coefficient $a_1 = -c/12 = -\kappa_{\mathrm{ch}}/6$ recovers κ_{ch}, and the higher coefficients a_k for $k \geq 2$ encode the arity-$(2k)$ shadow projections. The decay rate of the corrections is $q = e^{-2\pi/L}$, matching the shadow tower variable. [HE]

Verification: `compute/lib/cross_volume_shadow_bridge.py` (63 tests).

Crystal melting: combinatorial vs bar-algebraic arity

The MacMahon function $M(q)$ counts plane partitions (3D Young diagrams), and the crystal melting model of Okounkov–Reshetikhin–Vafa interprets the DT partition function of $\mathbb{C}^3$ as a statistical-mechanical partition function on the crystal lattice $\mathbb{Z}^3_{\geq 0}$.

Warning 20.19 (Combinatorial arity ≠ bar arity). The crystal melting "arity" (the number of boxes removed from the corner of the crystal) does *not* match the bar complex arity (the tensor length in $B^{(n)}(A)$).

The crystal partition function organizes by the number of boxes:

$$M(q) = \sum_{\pi \in \mathcal{PP}} q^{|\pi|}, \qquad |\pi| = \text{number of boxes in plane partition } \pi.$$

The bar complex organizes by tensor length:

$$B(A) = \bigoplus_{n \geq 1} A^{\otimes n}[n], \qquad \text{arity } n = \text{number of tensor factors}.$$

These two gradings are *not* compatible: a plane partition with k boxes does not correspond to a bar element of arity k. Rather, the crystal melting expansion is an exponential reorganization of the bar complex: $\log M(q) = \sum_k \sigma_2(k) q^k / k$ is the free energy, while the bar complex detects the individual OPE modes.

The fundamental mismatch: crystal melting is a *tensor-algebraic* construction (counting 3D diagrams in $\mathbb{Z}^3$), while the bar complex is a *factorization-algebraic* construction (organized by collision patterns on $\mathrm{Ran}(X)$). The two agree in the graded *character* (both reproduce $M(q)$ or its inverse) but not in the *arity structure*.

Verification: `compute/lib/crystal_bar_identification.py` (70 tests).

20.9 Grand atlas of CY$_3$ shadow invariants

Table 20.1 collects the shadow tower data for 18 CY$_3$ families spanning the full range of shadow behaviour.

Key patterns visible in the atlas:

1. **Toric depth bound**: for toric CY$_3$ with N vertices in the web diagram, the shadow depth satisfies $r_{\max} \leq N$. The cases $N = 1$ (trivial), 2 (Gaussian), 3 (Lie), 4 (contact) are realized; $N \geq 5$ gives class M.

2. **Compact CY$_3$ are generically class M**: all compact CY$_3$ with $\chi \neq 0$ have infinite shadow depth, driven by the infinite BPS spectrum.

3. κ_{ch} **integrality**: among the atlas families, $\mathbb{P}^5[3,3]$ ($\kappa_{\mathrm{ch}} = -6$), K3 $\times E$ ($\kappa_{\mathrm{ch}} = 5$), Enriques $\times E$ ($\kappa_{\mathrm{ch}} = 4$), and $\mathrm{BV}(20,2,0)$ ($\kappa_{\mathrm{ch}} = 5$) have integral κ_{ch}. For compact CICYs, $\kappa_{\mathrm{ch}} = \chi/24$ is integral only when $24 \mid \chi$; for K3-fibered geometries, κ_{ch} comes from the automorphic weight and is always integral.

Table 20.1: Grand atlas of CY3 shadow invariants. Columns: CY3 family, topological Euler characteristic χ, Hodge numbers $h^{1,1}$ and $h^{2,1}$, modular characteristic κ_{ch}, shadow depth class, and data source. For non-compact toric CY3: $\kappa_{\mathrm{ch}} = \chi(S)/2$ where S is the compact base (Proposition 15.17). For compact CICYs: $\kappa_{\mathrm{ch}} = \chi_{\mathrm{top}}/24$. For K3-fibered geometries: κ_{ch} is the automorphic weight. Families above the first rule are non-compact toric; between the rules are K3-fibered; below are compact complete intersections and special families. BV = Borcea–Voisin.

CY_3	χ	$h^{1,1}$	$h^{2,1}$	κ_{ch}	Class	Source
$\mathbb{C}^3$	—	—	—	1	G	MacMahon
Resolved conifold	—	1	0	1	G	KS wall-crossing
Local $\mathbb{P}^2$	—	1	0	3/2	M	McKay($\mathbb{Z}_3$)
Local $\mathbb{P}^1\times\mathbb{P}^1$	—	2	0	2	M	McKay($\mathbb{Z}_2^2$)
Local F_1	—	2	0	−2	M	Hirzebruch
$\mathrm{K3}\times E$	0	21	21	5	M	Δ_5 / BKM
Enriques $\times E$	0	11	11	4	M	Allcock / $O(2,10)$
Quintic	−200	1	101	−25/3	M	BCOV
$\mathbb{P}^5[3,3]$	−144	1	73	−6	M	BCOV
$\mathbb{P}^5[2,4]$	−176	1	89	−22/3	M	BCOV
$\mathbb{P}^6[2,2,3]$	−144	1	73	−6	M	BCOV
$\mathbb{P}^7[2,2,2,2]$	−128	1	65	−16/3	M	BCOV
Bicubic $\mathbb{P}^2\times\mathbb{P}^2$	−162	2	83	−27/4	M	BCOV
Banana (Schoen)	0	19	19	0	$\mathrm{M}^{\geq 4}$	BKY
$\mathrm{BV}(1,1,1)$	−108	0	54	−9/2	M	Voisin–Borcea
$\mathrm{BV}(10,0,0)$	0	35	35	0	?	Voisin–Borcea
$\mathrm{BV}(10,8,0)$	0	19	19	0	?	Voisin–Borcea
$\mathrm{BV}(20,2,0)$	120	61	1	5	M	Voisin–Borcea

4. **BKM structure requires integral automorphic weight**: K3-fibered CY3s with $\chi = 0$ carry BKM superalgebras via Borcherds lifts; compact CICYs with $\kappa_{\mathrm{ch}} \notin \mathbb{Z}$ do not.

5. $\kappa_{\mathrm{ch}} = 0$ **does not imply trivial tower** (AP31): the banana manifold and the balanced Borcea–Voisin families ($r = 10$) have $\kappa_{\mathrm{ch}} = 0$ but potentially nontrivial higher-arity shadows sourced by instantons.

6. **Borcea–Voisin as κ_{ch}-interpolation**: the BV family parametrized by (r, a, δ) gives $\kappa_{\mathrm{ch}} = (r - 10)/2$, interpolating between $\kappa_{\mathrm{ch}} = -9/2$ (minimal, $r = 1$) and $\kappa_{\mathrm{ch}} = 5$ (maximal, $r = 20$, recovering $\mathrm{K3}\times E$).

Verification: `compute/lib/cy3_grand_atlas.py`, `compute/tests/test_cy3_grand_atlas.py` (119 tests).

Remark 20.20 (Enriques $\times\, E$ kappa anomaly). The grand atlas uses $\kappa_{\mathrm{BKM}}(\mathrm{Enr}\times E) = 4$, the weight of the Allcock Borcherds product on $O(2,10)$, verified by `enriques_shadow.py` (72 tests). This is the authoritative value. The naive BCOV formula $\kappa_{\mathrm{ch}} = \chi/24$ does not apply because Enriques $\times\, E$ is a *generalized* CY3 ($h^{3,0} = 0$, torsion canonical). The ratio $\kappa_{\mathrm{BKM}}(\mathrm{K3}\times E)/\kappa_{\mathrm{BKM}}(\mathrm{Enr}\times E) = 5/4$.

Chapter 21

The seven faces of $r_{CY}(z)$ for Calabi–Yau chiral algebras

For a Calabi–Yau chiral algebra arising from the CY-to-chiral functor, the binary collision residue $r_{CY}(z) = \mathrm{Res}^{\mathrm{coll}}_{0,2}(\Theta_{CY})$ has seven equivalent realizations specific to the CY setting: as the CY bar-cobar twisting morphism, as a CoHA / perverse-coherent-sheaves shadow, as the classical CY Poisson coisson, as the Maulik–Okounkov R-matrix (for $K3 \times E$), as the affine super Yangian $Y(\widehat{\mathfrak{gl}}_1)$ r-matrix (for toric CY_3), as the elliptic Sklyanin bracket (for toroidal CY), and as a Gaudin model arising from CY_3 collision residues. This chapter establishes the seven-face identification in the CY setting and shows how Vol I's general theorems specialize to the CY_3 cases.

21.1 The CY_3 collision residue

The collision residue is a single algebraic operation extracted from the universal Maurer–Cartan element Θ_A of a chiral algebra. In the CY setting, this operation acquires geometric meaning: the residue is computed against the holomorphic CY volume form, and the resulting binary operation is the classical limit of the quantum vertex chiral group braiding.

Definition 21.1 *(Binary collision residue, CY version).* Let C be a smooth proper Calabi–Yau category of dimension d with holomorphic volume form Ω_C. Let A_C denote the chiral algebra produced by the CY-to-chiral functor (Theorem CY-A_d, proved for $d = 2$, conditional on chain-level $\mathbb{S}^3$-framing for $d = 3$). Let $\Theta_{A_C} \in \mathrm{MC}(\mathfrak{g}^{\mathrm{mod}}_{A_C})$ denote its universal Maurer–Cartan element (cf. Vol I, Theorem III). The *binary CY collision residue* is

$$r_{CY}(z) \ := \ \mathrm{Res}^{\mathrm{coll}}_{0,2}(\Theta_{A_C}) \ \in \ A_C \otimes A_C\, [[z^{-1}]],$$

the genus-0, arity-2 residue along the collision divisor of $\overline{C}_2(\mathrm{Spec}\,\mathbb{C})$, paired against Ω_C in the CY direction.

Remark 21.2 (The CY direction is a separate slot). The collision residue extracts the z^{-1} part of Θ along the chiral curve direction. In the CY setting, Θ has *two* gradings: arity in the chiral direction (controlled by the curve X) and Hodge weight in the CY direction (controlled by $\mathcal{O}_C$). The CY volume form Ω_C kills the Hodge weight by integration; what remains is a chiral binary operation valued in $A_C \otimes A_C$. This is the precise sense in which r_{CY} couples the two directions.

Remark 21.3 (Three invariants must not be conflated). For any chiral algebra A, the binary residue carries three *distinct* numerical invariants which the CY setting will illuminate:

(i) $p_{\max}(A)$, the maximal pole order of generator-on-generator OPEs (cf. Vol I, Definition 20.9);

(ii) $k_{\max}(A) = p_{\max}(A) - 1$, the collision depth of r_{CY} itself, after d log-absorption (cf. Vol I, Definition 20.9);

(iii) $r_{\max}(A)$, the shadow depth, the arity at which the obstruction tower $\Theta_A^{\leq r}$ terminates (Vol I, Definition 20.9).

The $\beta\gamma$ system is the archetypal witness of their independence: $p_{\max}(\beta\gamma) = 1$, $k_{\max}(\beta\gamma) = 0$, $r_{\max}(\beta\gamma) = 4$ (class C). Conflating these produces incorrect classifications, and the seven-face programme will distinguish them at every face.

21.2 Face 1: the CY bar-cobar twisting morphism

The first face is the home framework of this volume. The CY-to-chiral functor sends a Calabi–Yau category C to a chiral algebra A_C, and the bar construction sends A_C to a factorization coalgebra $\overline{B}(A_C)$ on $\mathrm{Ran}(X)$. The collision residue is the binary component of the universal twisting morphism between $\overline{B}(A_C)$ and A_C.

THEOREM 21.4 *(Face 1: CY bar-cobar realization, $d = 2$).* [PH] Let A_C be the chiral algebra of a CY_2 category C produced by the CY-to-chiral functor Φ (Theorem CY-A_2). Then:

(i) The convolution dg Lie algebra $\mathfrak{g}^{\mathrm{mod}}_{A_C}$ identifies with the homotopy Lie algebra of twisting morphisms $\hom_\alpha(\overline{B}(A_C), A_C)$ between bar coalgebra and chiral algebra (cf. Vol I, Theorem III);

(ii) The universal MC element Θ_{A_C} is the universal twisting morphism;

(iii) The binary collision residue $r_{CY}(z) = \mathrm{Res}^{\mathrm{coll}}_{0,2}(\Theta_{A_C})$ is the leading arity-2 component of this twisting morphism, evaluated in the CY direction against Ω_C.

[PH]

Proof sketch. For (i), the cyclic bar complex $\mathrm{CC}_\bullet(C)$ identifies with $\overline{B}(A_C)$ as a factorization coalgebra (Theorem 20.1). The convolution identification is the application of Vol I's master twisting-morphism theorem to this coalgebra-algebra pair. For (ii), Θ_{A_C} satisfies the MC equation $D\Theta + \frac{1}{2}[\Theta, \Theta] = 0$ at all levels, with the bar-intrinsic construction $\Theta = D - d_0$. For (iii), the collision residue is computed from Θ by restricting to genus 0, arity 2, and pairing against the holomorphic volume form. □

CONJECTURE 21.5 *(Face 1: CY bar-cobar realization, $d = 3$).* The statement of Theorem 21.4 extends to $d = 3$, conditional on the chain-level $\mathbb{S}^3$-framing construction of Conjecture CY-A_3. Concretely: if the CY_3-to-chiral functor Φ_3 can be defined on a full subcategory $\mathcal{D} \subset \mathrm{CY}_3$-$\mathbf{Cat}$, then for every $C \in \mathcal{D}$ the convolution identification, the universal MC element, and the binary collision residue extraction of Theorem 21.4(i)–(iii) continue to hold for $A_C := \Phi_3(C)$. [CJ]

Remark 21.6 (Bar-cobar inversion is not the seven-face master move). The cobar functor Ω inverts the bar complex: $\Omega(\overline{B}(A)) \simeq A$ (Vol I, Theorem B). This is a round-trip that recovers A itself. The seven-face programme is not about recovering A from $\overline{B}(A)$; it is about realizing a single algebraic object, the binary residue $r_{CY}(z)$, in seven distinct geometric languages. The closed-string or bulk side is the chiral derived center $\mathcal{Z}^{\mathrm{der}}_{\mathrm{ch}}(A_C)$ (Vol I, Theorem H), not the cobar; the seven faces all live on the boundary A_C side.

Remark 21.7 (BPS shadow depth and the CY holographic datum). [PE] The holographic modular Koszul datum in the CY setting assembles the six pieces of the seven-face programme into a single package:

$$H_{CY}(T) \;=\; \big(A_{CY},\ A^!_{CY},\ C_{CY},\ r_{CY}(z),\ \Theta_{CY},\ \nabla^{\mathrm{hol}}_{CY}\big),$$

where A_{CY} is the CY-to-chiral image $A_C = \Phi(C)$, $A^!_{CY}$ is its Verdier/Koszul dual boundary algebra, C_{CY} is the line category of the HT theory, $r_{CY}(z)$ is the binary CY collision residue of Definition 21.1, Θ_{CY} is the universal Maurer–Cartan element (Theorem 21.4), and $\nabla^{\mathrm{hol}}_{CY}$ is the holomorphic Knizhnik–Zamolodchikov connection on the chiral curve direction. This is the CY specialization of the Vol I datum $H(T) = (A, A^!, C, r(z), \Theta_A, \nabla^{\mathrm{hol}})$; the new feature is the presence of the CY direction Ω_C (Remark 21.2).

Dimofte's framework (PIRSA 25110067) attaches to $H_{CY}(T)$ a single numerical invariant, the *BPS complexity* of the boundary Hilbert space, measured by the K-matrix $K_{A_{CY}}(z)$ of equation (9.1) (cf. Chapter 9, Remark 9.7, and *Vol II*, remark `rem:dimofte-k-matrix` in `ordered_associative_chiral_kd_core.tex`). The boundary Hilbert space of the HT theory decomposes as

$$\mathcal{H}_\partial(T) \;\simeq\; \mathcal{H}_\partial^{\mathrm{free}} \otimes \mathcal{H}_\partial^{\mathrm{BPS}},$$

where $\mathcal{H}_\partial^{\mathrm{free}}$ is the $\mathfrak{u}(1)^r$ free-boson factor (the class $\mathbf{G}$ summand) and $\mathcal{H}_\partial^{\mathrm{BPS}}$ records the BPS corrections. Under the CY-to-chiral functor, this decomposition is the Vol I shadow partition $\mathbf{G} + (\mathbf{L}/\mathbf{C}/\mathbf{M})$: the free-boson summand is exactly class $\mathbf{G}$, and the BPS summand carries the higher shadow depth.

CY_2 (K3, proved). For a CY_2 category C admitting a chiral algebraization via Φ, the BPS Hilbert space is determined by the associated Borcherds–Kac–Moody superalgebra $\mathfrak{g}_{\mathrm{BKM}}(C)$, and the K-matrix encodes its denominator identity. Concretely, for $C = D^b(\mathrm{Coh}(K3))$ the BKM denominator formula (Gritsenko–Nikulin, Borcherds) identifies the character of $\mathcal{H}_\partial^{\mathrm{BPS}}$ with the Fourier coefficients of $1/\Phi_{10}$, and the K-matrix modes $\{\alpha_n\}$ read off the root multiplicities recorded by these coefficients.

Igusa cusp form verification. The Igusa cusp form Φ_{10} on $\mathfrak{H}_2$ has weight 10 (Gritsenko–Nikulin). The Borcherds–Kac–Moody modular characteristic of $K3 \times E$ is $\kappa_{\mathrm{BKM}} = 5$ (AP113, Chapter 14.16). The identification

$$\mathrm{wt}(\Phi_{10}) \;=\; 2\,\kappa_{\mathrm{BKM}} \;=\; 10$$

holds: the factor of 2 is the Serre duality on $K3$, which has Serre degree $d = 2$, so the BKM characteristic is one half of the cusp-form weight. This matches entry $\kappa_{\mathrm{BKM}}(K3 \times E) = 5$ in the Vol III κ_{BKM}-spectrum table (AP113), and agrees with the identity $\mathrm{wt}(\Phi_{10}) = 10$ recorded in Chapter 14.16. AP-CY8 forbids calling this an identity at the theorem level: it is an observation matching the Borcherds lift weight, not a computation from the Vol I definition of κ_{ch}.

CY_3 (conditional on CY-A_3). For a CY_3 category the BPS Hilbert space should be the module over the Donaldson–Thomas Hall algebra (Kontsevich–Soibelman, Davison–Meinhardt), and the K-matrix should be the generating function of the cohomological Hall algebra $\mathcal{H}(Q, W)$ attached to a quiver-with-potential presentation. For toric CY_3 without compact 4-cycles (Chapter 15) this is explicit: Schiffmann–Vasserot identify $\mathcal{H}(\mathbb{C}^3)$ with the positive half of the affine Yangian $Y^+(\widehat{\mathfrak{gl}}_1)$, and the K-matrix is its character $M(q)^2 \cdot P(q)$ (Theorem 9.4). All statements in this paragraph whose proof chain passes through A_C at $d = 3$ inherit the conditionality of Conjecture CY-A_3 (AP-CY6, AP-CY11).

Shadow–BPS dictionary. The four-class partition of CY boundary algebras is:

Shadow class	BPS complexity	$K_{A_{CY}}(z)$	Example
G	0 BPS corrections	1	trivial CY fiber, $\mathfrak{u}(1)^r$
L	1 BPS level	polynomial, simple pole	rank-one CY Heisenberg deformation
C	2 BPS levels	polynomial, double pole	toric CY3 ($\mathbb{C}^3$, conifold)
M	infinite BPS tower	infinite expansion	$K3$, $K3 \times E$

The shadow-depth classification of the CY boundary algebra is therefore equivalent to the BPS complexity of the HT boundary Hilbert space under the CY-to-chiral functor. The K-matrix modifies the coproduct, not the product, so AP126's level-k vanishing check does not apply directly; the companion r-matrix check is the AP126-level check on the affine r-matrix at $k_{\mathrm{ch}} = 0$, which vanishes as required and returns class **G**. Primary references for BPS Hall algebras and the denominator identity: Dimofte PIRSA 25110067, Kontsevich–Soibelman, Davison–Meinhardt, Gritsenko–Nikulin. Vol II cross-references: `rem:dimofte-k-matrix` in `ordered_associative_chiral_kd_core.tex` and the BPS/shadow discussion in `ht_bulk_boundary_line_core.tex`.

21.3 Face 2: CoHA and perverse coherent sheaves

The second face is enumerative geometry: the binary residue is the leading non-trivial structure constant of the cohomological Hall algebra of C, or equivalently, the Yoneda product on perverse coherent sheaves. This face is the natural home of Donaldson–Thomas invariants.

Conjecture 21.8 *(Face 2: CoHA realization of r_{CY}).* Let C be a smooth proper CY3 category with quiver-with-potential presentation (Q, W) at a stable point of the moduli of stability conditions. Let $\mathcal{H}(Q, W)$ denote the critical cohomological Hall algebra (Definition 15.3). Then the binary collision residue of A_C admits a CoHA realization

$$r_{CY}^{\mathrm{CoHA}}(z) \;=\; \mu^{(2)}_{\mathcal{H}(Q,W)} \otimes z^{-1} \;\in\; \mathcal{H}(Q,W) \otimes \mathcal{H}(Q,W)\,[[z^{-1}]],$$

where $\mu^{(2)}$ is the binary CoHA product, paired with the d log-propagator on $\overline{C}_2$. [CJ]

Remark 21.9 (Status of Conjecture 21.8). The CoHA $\mathcal{H}(Q, W)$ exists unconditionally (Kontsevich–Soibelman, Davison–Meinhardt). The chiral algebra A_C exists for $d = 2$ and is conjectural for $d = 3$. The identification with r_{CY} requires the CY-to-chiral functor to be defined and to commute with critical cohomology, which is the content of Conjecture CY-C and is currently verified for the toric CY3 examples (Theorems 15.4, 15.5). In those cases the identification reduces to the Schiffmann–Vasserot and Rapcak–Soibelman–Yang–Zhao theorems and is ProvedElsewhere with attribution to those two papers.

Remark 21.10 (Perverse coherent sheaves). For a CY3 category arising as $D^b(\mathrm{Coh}(X))$ for X a smooth CY3 variety, the CoHA description above is equivalent to a description in terms of perverse coherent sheaves on X (Bridgeland, Bayer–Macri). The Yoneda product on $\mathrm{Ext}^\bullet$ of perverse coherent sheaves is the binary CoHA product, and the binary collision residue extracts its leading Ext^1 component. This is the geometric face: the residue r_{CY} *is* a piece of the deformation theory of the CY3.

Remark 21.11 (CoHA is not the full quantum vertex chiral group). The critical CoHA is an associative algebra: the positive (or E_1-ordered) half of the conjectural quantum vertex chiral group $G(X)$. The full $G(X)$ would be braided E_2, with the braiding supplied by the R-matrix of the affine super Yangian (Section 21.6). The CoHA face captures the *shadow* of r_{CY} on the ordered side; the braided side is faces 4 and 5.

21.4 Face 3: classical CY Poisson coisson

The third face is the classical limit. In the CY setting, the binary residue r_{CY} has a classical avatar: a coisson bracket on the sheafification of A_C, which in turn is a chiral version of the holomorphic Poisson bracket arising from the CY (2,0)-form (in $d = 2$) or the holomorphic top form via contraction (in $d = 3$).

Theorem 21.12 *(Face 3: classical Poisson coisson, $d = 2$).* [PH] Let C be a CY$_2$ category and let A_C be its chiral algebraization (Theorem CY-A$_2$). Let $\hbar$ denote a quantization parameter. Then:

(i) The classical limit $A_C^{\mathrm{cl}} := \lim_{\hbar\to 0} A_C$ is a Poisson vertex algebra (PVA) with bracket $\{\cdot,\cdot\}_\lambda$ valued in $A_C^{\mathrm{cl}} \otimes \mathbb{C}[\lambda]$;

(ii) The leading coefficient $\{\cdot,\cdot\}_\lambda^{(0)}$ of the PVA bracket is a chiral Poisson bracket whose classical r-matrix is exactly $r_{CY}(z)$ at $\hbar = 0$;

(iii) This identifies r_{CY}^{cl} as the coisson bracket constructed from the CY volume form Ω_C via the Poisson structure on $\mathcal{Z}_{\mathrm{ch}}^{\mathrm{der}}(A_C^{\mathrm{cl}})$ (cf. Vol I, Theorem 20.9).

[PH]

Proof sketch. For a CY$_2$ category, Φ is the proved CY-to-chiral functor (Chapter 8). The PVA structure on A_C^{cl} is the classical limit of the chiral OPE. The leading λ coefficient of the λ-bracket is the contraction of two factors along the binary collision divisor; this is exactly $\mathrm{Res}_{0,2}^{\mathrm{coll}}$ computed in the classical limit, hence r_{CY}^{cl}. The identification with the coisson bracket arising from Ω_C is the coisson–PVA dictionary (cf. Vol I, Theorem 20.9, applied to the specialization $C \mapsto A_C$). $\square$

Conjecture 21.13 *(Face 3: classical Poisson coisson, $d = 3$).* The statement of Theorem 21.12 extends to $d = 3$, conditional on Conjecture CY-A$_3$: if A_C is a CY$_3$-chiral algebraization of C, then items (i)–(iii) of Theorem 21.12 hold for A_C^{cl} with the CY$_3$ volume form $\Omega_C \in H^0(C, \omega_C)$ supplying the coisson pairing. [CJ]

Remark 21.14 (Convention: λ-brackets vs OPE modes (AP49)). Vol I uses OPE modes $a_{(n)}b$. Vol II uses λ-brackets $\{a_\lambda b\} = \sum_n \lambda^{(n)} a_{(n)} b$ with the divided power $\lambda^{(n)} = \lambda^n/n!$. Vol III uses motivic and categorical language. When transcribing r_{CY} between volumes, the $1/n!$ factor must be tracked: the λ^3 coefficient of a λ-bracket is $a_{(3)}b/3! = a_{(3)}b/6$, not $a_{(3)}b$. This convention bridge is the source of several historical errors and is the subject of AP49.

21.5 Face 4: the Maulik–Okounkov R-matrix (for $K3 \times E$)

The fourth face is geometric representation theory: the binary residue is the classical limit of the Maulik–Okounkov stable-envelope R-matrix acting on the equivariant cohomology of a Nakajima quiver variety. For the CY$_3$ specialization $K3 \times E$, this is the most concrete realization, and it produces r_{CY} explicitly in terms of theta functions on the elliptic factor.

The Maulik–Okounkov stable envelope and its R-matrix $R_{\mathrm{MO}}^{K3\times E}(z)$ exist unconditionally (Maulik–Okounkov 2019); this is the ProvedElsewhere component. The genuinely new content of face 4 is the identification of the classical limit with the binary CY collision residue.

Conjecture 21.15 *(Face 4: MO realization for $K3 \times E$).* For the CY$_3$ specialization $X = K3 \times E$, the binary CY collision residue identifies with the classical limit of the Maulik–Okounkov R-matrix:

$$r_{CY}^{K3\times E}(z) \;=\; \lim_{\hbar\to 0} \hbar^{-1}\big(R_{\mathrm{MO}}^{K3\times E}(z) - \mathrm{id}\big).$$

The right-hand side is the rational/elliptic stable envelope of Maulik–Okounkov, evaluated on the Hilbert scheme of points on $K3$ along the elliptic direction E. The conjecture is equivalent to Conjecture CY-C for $K3 \times E$. [CJ]

Remark 21.16 (The CY involution $g(z)g(-z) = 1$). For $K3 \times E$, the MO R-matrix is governed by a single Yangian generator $g(z)$ satisfying $g(z)g(-z) = 1$ (Theorem 14.158). This CY involution is the $\mathbb{S}^2$-framing of the CY-to-chiral functor seen at the level of the binary residue: the $z \mapsto -z$ involution is the chiral analogue of the CY Serre functor $\mathbb{S} \simeq [d]$. In particular the CY involution forces the residue to be of *odd* pole order, in agreement with AP19 (the d log-absorption sends even-order poles to odd-order poles in the residue).

Remark 21.17 (Self-dual Gepner trivialization). At the Gepner point of $K3$, where the chiral algebra has $\mathcal{N} = (4, 4)$ enhancement, the MO R-matrix trivializes (Proposition 14.159), so r_{CY} vanishes. This is consistent with the seven-face master statement: at the self-dual point of any face, all seven realizations vanish simultaneously. For $K3 \times E$ this is the statement that $\kappa_{\text{eff}} = 0$ in the $\mathcal{N} = (4, 4)$ enhancement.

21.6 Face 5: the affine super Yangian $Y(\widehat{\mathfrak{gl}}_1)$ for toric CY$_3$

The fifth face is the explicit toric face: when the CY$_3$ is toric and admits a quiver-with-potential presentation, the binary residue is the classical r-matrix of the affine super Yangian $Y(\widehat{\mathfrak{gl}}_1)$ (for $\mathbb{C}^3$, the Jordan quiver) or more generally $Y(\widehat{\mathfrak{g}}_{Q_X})$ (for the toric quiver of X).

Theorem 21.18 *(Face 5 for toric CY$_3$ via RSYZ).* [PE] Let X be a toric CY$_3$ without compact 4-cycles, with toric quiver (Q_X, W_X) and affine super Yangian $Y(\widehat{\mathfrak{g}}_{Q_X})$ (Theorem 15.5). Then the binary CY collision residue identifies with the classical r-matrix:

$$r^X_{CY}(z) \;=\; r^{\text{cl}}_{Y(\widehat{\mathfrak{g}}_{Q_X})}(z) \;=\; \frac{\Omega_{Y(\widehat{\mathfrak{g}}_{Q_X})}}{z} \;+\; (\text{higher poles dictated by } p_{\max}(W_X)),$$

where $\Omega_{Y(\widehat{\mathfrak{g}}_{Q_X})}$ is the Casimir tensor of the affine super Yangian. At the Jordan-quiver specialization $X = \mathbb{C}^3$, the identification reduces to the r-matrix of the affine Yangian $Y(\widehat{\mathfrak{gl}}_1) \cong \mathcal{W}_{1+\infty}$ at the self-dual level. [PE]

Remark 21.19 (Status partition, AP60 watch). Theorem 21.18 is a restatement of the Rapcak–Soibelman–Yang–Zhao theorem (Theorem 15.5) in the collision-residue language of the seven-face programme. AP60 forbids tagging this theorem ProvedHere: the classical content is due to Schiffmann–Vasserot (for $\mathbb{C}^3$, Theorem 15.4) and Rapcak–Soibelman–Yang–Zhao (for general toric without compact 4-cycles, Theorem 15.5). The genuinely new content of face 5 is the translation of their r-matrix into the language of $\text{Res}^{\text{coll}}_{0,2}(\Theta_{A_X})$, which is recorded as the scholium below.

Proposition 21.20 *(Collision-residue translation of face 5, $\mathbb{C}^3$ case).* For $X = \mathbb{C}^3$, the Schiffmann–Vasserot Casimir tensor $\Omega_{Y(\widehat{\mathfrak{gl}}_1)}$ coincides with $\text{Res}_{z=0}\big[r^{\mathbb{C}^3}_{CY}(z)\big]$, where $r^{\mathbb{C}^3}_{CY}$ is the binary CY collision residue of the $\mathbb{C}^3$ chiral algebra. [PH]

Proof sketch. The Jordan quiver superpotential $W = \text{tr}(XYZ - XZY)$ has $p_{\max}(W) = 1$ (each cubic monomial contributes a single contraction in the binary OPE), so $k_{\max} = 0$ and $r^{\mathbb{C}^3}_{CY}(z)$ has a *single* pole at $z = 0$. The residue at this pole is the Schiffmann–Vasserot Casimir $\Omega_{Y(\widehat{\mathfrak{gl}}_1)} \in Y^+(\widehat{\mathfrak{gl}}_1)^{\otimes 2}$ (Theorem 15.4). By the universality of Θ in MC moduli, this is exactly $r^{\mathbb{C}^3}_{CY}$. The matching of higher poles with the toric quiver superpotential follows from the RSYZ theorem for toric quivers without compact 4-cycles. □

Remark 21.21 (Three invariants for toric CY$_3$ (AP59 watch)). For the standard toric CY$_3$ family the three invariants are:

Geometry	Quiver	$p_{\max}$	$k_{\max}$	$r_{\max}$
$\mathbb{C}^3$	Jordan	1	0	2 (class G)
Resolved conifold	Klebanov–Witten	1	0	2 (class G)
Local $\mathbb{P}^2$	McKay $\mathbb{Z}_3$	2	1	∞ (class M)
Local $\mathbb{P}^1 \times \mathbb{P}^1$	Beilinson	2	1	$\infty/2$

The pole order $p_{\max}$, the collision depth $k_{\max} = p_{\max} - 1$, and the shadow depth $r_{\max}$ are three independent invariants. In the toric family above, $p_{\max}$ is small (at most 2) but $r_{\max}$ can be infinite (local $\mathbb{P}^2$, McKay $\mathbb{Z}_3$). This separates the pole-order face (face 5, controlled by $p_{\max}$) from the shadow tower face (face 1, controlled by $r_{\max}$). AP59 forbids identifying these.

Gaudin Hamiltonians for toric CY$_3$ from collision residues

The collision residue at the Jordan quiver ($\mathbb{C}^3$) is the $Y(\widehat{\mathfrak{gl}}_1)$ classical r-matrix; this allows one to write down a chiral Gaudin model whose Hamiltonians are extracted from CY$_3$ collision data, providing the first instance of face 7 inside face 5. The full development of face 7 occupies Section 21.8.

Construction 21.22 *(Gaudin Hamiltonians from $r_{CY}^{\mathbb{C}^3}$).* Fix marked points $z_1, \ldots, z_n \in \mathbb{C}$ (the chiral curve, with n copies of the Fock representation of $Y(\widehat{\mathfrak{gl}}_1)$). Define

$$H_i^{\mathbb{C}^3} := \sum_{j \neq i} \frac{\Omega_{ij}^{Y(\widehat{\mathfrak{gl}}_1)}}{z_i - z_j} \in Y(\widehat{\mathfrak{gl}}_1)^{\otimes n},$$

where $\Omega_{ij}^{Y(\widehat{\mathfrak{gl}}_1)}$ is the $Y(\widehat{\mathfrak{gl}}_1)$ Casimir $\mathrm{Res}_{z=0}\, r_{CY}^{\mathbb{C}^3}(z)$, acting on the i-th and j-th tensor factors. The operators $\{H_i^{\mathbb{C}^3}\}$ commute pairwise as a direct consequence of the classical Yang–Baxter equation for $r_{CY}^{\mathbb{C}^3}$ (cf. Vol I, Theorem 20.9).

Remark 21.23 (Wall-crossing as gauge equivalence). Across walls in the Bridgeland stability manifold, the toric Yangian r-matrix transforms by a Kontsevich–Soibelman wall-crossing automorphism. In the present language, this is exactly an MC gauge equivalence on Θ_{A_X} (Theorem 15.16): $r_{CY}^{X,\zeta'}(z) = e^{\mathrm{ad}_\alpha} r_{CY}^{X,\zeta}(z)$ for the unique gauge parameter $\alpha \in \mathfrak{n}_W$ supported on the wall divisor. The Gaudin Hamiltonians of Construction 21.22 are gauge-equivalent across walls; their joint spectrum is chamber-independent.

21.7 Face 6: elliptic Sklyanin bracket (for toroidal CY)

The sixth face is the elliptic deformation: when the chiral curve is itself an elliptic curve E_τ and the CY data are toroidal, the binary residue acquires an elliptic structure governed by theta functions. The classical limit of this r-matrix is the elliptic Sklyanin bracket, and the quantization is Felder's elliptic quantum group.

The elliptic r-matrix and its classical Sklyanin bracket are due to Sklyanin (1983) and Belavin–Drinfeld (1984), with the quantization to Felder's elliptic quantum group (Definition 14.14) classical. These components are ProvedElsewhere with explicit attribution. The genuinely new content is the identification of the elliptic r-matrix with the binary CY collision residue $\mathrm{Res}_{0,2}^{\mathrm{coll}}(\Theta_{A_C})$ via the chiral propagator $d \log \theta_1(z \mid \tau)$.

THEOREM 21.24 *(Face 6 for $\mathfrak{sl}_2$ Heisenberg on E_τ).* [PH] Let E_τ be an elliptic curve and let $A = H_{E_\tau}^{\mathfrak{sl}_2}$ be the $\mathfrak{sl}_2$ Heisenberg chiral algebra realized on E_τ. Then the binary CY collision residue admits the elliptic representative

$$r^{\tau}_{CY}(z) \;=\; \frac{\theta_1'(0\,|\,\tau)}{\theta_1(z\,|\,\tau)}\,\Omega_{\mathfrak{sl}_2},$$

where θ_1 is the odd Jacobi theta function and $\Omega_{\mathfrak{sl}_2}$ is the Casimir tensor of $\mathfrak{sl}_2$ (cf. Section 14.3). The classical limit of r^{τ}_{CY} defines the Sklyanin bracket $\{f,g\}_\tau = \langle r^{\tau}_{CY}, [\nabla f, \nabla g]\rangle$ on $E_{q,p}(\mathfrak{sl}_2)^*$, and the quantization $\hbar \to q$ recovers Felder's elliptic quantum group $E_{q,p}(\mathfrak{sl}_2)$ with dynamical parameter λ identified as the B-cycle monodromy of the elliptic propagator. [PH]

Conjecture 21.25 *(Face 6 for general toroidal CY).* The statement of Theorem 21.24 extends to general toroidal CY realizations X on E_τ: the binary residue $r^{\tau,X}_{CY}(z)$ is the elliptic r-matrix of $Y(\widehat{\mathfrak{g}}_X)$ acquiring dynamical theta-function corrections in λ, and its classical limit is the corresponding Sklyanin bracket (cf. Vol I, Theorem 20.9). [CJ]

Remark 21.26 (AP60 status partition). The Sklyanin bracket and Belavin–Drinfeld classification are classical results (1983). The genuinely new content of Theorem 21.24 is the identification of the elliptic r-matrix with the binary CY collision residue $\mathrm{Res}^{\mathrm{coll}}_{0,2}(\Theta_{A_C})$ via the chiral propagator $d\log\theta_1(z\,|\,\tau)$. AP60 forbids tagging the entire theorem ProvedHere; the classical components are ProvedElsewhere with explicit attribution to Sklyanin (1983) and Belavin–Drinfeld (1984).

The three-parameter $\hbar$ identification

A subtle but central observation is that three different $\hbar$ parameters appearing in the literature on elliptic quantum groups are the same parameter under the seven-face dictionary.

Construction 21.27 *(Three-parameter $\hbar$ identification).* The following three quantization parameters coincide:

(i) the boundary coupling $\hbar^{\mathrm{KZ}}$ in the Knizhnik–Zamolodchikov equation (Knizhnik–Zamolodchikov 1984; cf. Vol I, Theorem 20.9);

(ii) the loop parameter $\hbar^{\mathrm{DNP}}$ in the Dimofte–Niu–Py realization of the dg-shifted Yangian (cf. Vol I, Theorem 20.9);

(iii) the prefactor $\hbar^{\mathrm{coll}}$ of the binary CY collision residue $r_{CY}(z)$.

The identification $\hbar^{\mathrm{KZ}} = \hbar^{\mathrm{DNP}} = \hbar^{\mathrm{coll}}$ holds canonically once the CY direction is fixed, and follows from the universality of the binary residue in $\mathfrak{g}^{\mathrm{mod}}_{A_C}$.

Remark 21.28 (Convention sanity check). The identification of Construction 21.27 fails if any of the three parameters carries a hidden $2\pi i$ or $1/n!$. In particular, the KZ parameter is conventionally normalized so that the connection $\nabla = d - \hbar^{-1}\sum_i r_{ij}\, d\log(z_i - z_j)$ has integer monodromy on the configuration space; the DNP parameter is normalized so that the loop algebra has bracket $[a_m, b_n] = [a,b]_{m+n} + m\hbar\delta_{m+n,0}\langle a,b\rangle$; the collision residue is normalized so that $\Omega = r_{CY}(z)\cdot z$ at $z = 0$. These normalizations are mutually compatible; the identification is parameter-by-parameter, not just up to scale.

21.8 Face 7: Gaudin model from CY$_3$ collision residues

The seventh face is new in this volume. In Vol I, face 7 of the standard seven-face master is the Gaudin model arising from a chiral algebra with primitive generator (Vol I, Theorem 20.9). In the CY setting, the

Gaudin model arises from CY_3 collision residues, with Gaudin Hamiltonians constructed directly from the toric or elliptic r-matrix of face 5 or face 6, and with the Bethe ansatz interpreted in terms of DT/PT counts.

THEOREM 21.29 *(Face 7 for $\mathbb{C}^3$: Gaudin from the Jordan-quiver residue).* [PH] Let $X = \mathbb{C}^3$. Fix n marked points $z_1, \ldots, z_n$ on the chiral curve and Fock representations $V_1, \ldots, V_n$ of $Y(\widehat{\mathfrak{gl}}_1)$. Define

$$H_i^{\mathbb{C}^3} := \sum_{j \neq i} \frac{\Omega_{ij}^{Y(\widehat{\mathfrak{gl}}_1)}}{z_i - z_j},$$

acting on $V_1 \otimes \cdots \otimes V_n$, where $\Omega_{ij}^{Y(\widehat{\mathfrak{gl}}_1)}$ is the Casimir tensor $\mathrm{Res}_{z=0}\big[r_{CY}^{\mathbb{C}^3}(z)\big]$ acting on the i-th and j-th factors. Then:

(i) The operators $\{H_i^{\mathbb{C}^3}\}_{i=1}^n$ commute pairwise: $[H_i^{\mathbb{C}^3}, H_j^{\mathbb{C}^3}] = 0$;

(ii) The joint spectrum is governed by the $\mathcal{W}_{1+\infty}$ Bethe ansatz of Litvinov and Maulik–Okounkov;

(iii) The DT counts of $\mathbb{C}^3$ (MacMahon plane-partition generating function) provide the integer multiplicities of the Bethe roots: a Bethe state at the configuration $\{w_a\}$ corresponds to a plane partition of the appropriate shape.

[PH]

CONJECTURE 21.30 *(Face 7 for general toric/toroidal CY_3).* The statement of Theorem 21.29 extends to any CY_3 X admitting a toric (face 5) or toroidal (face 6) chiral realization, with Hamiltonians $H_i^X := \sum_{j \neq i} r_{CY}^X(z_i - z_j)_{ij}$, commutativity following from the classical Yang–Baxter equation for r_{CY}^X, and DT/PT counts of X supplying the Bethe root multiplicities. [CJ]

Remark 21.31 (Proof sketch for $\mathbb{C}^3$). For $X = \mathbb{C}^3$, $r_{CY}^{\mathbb{C}^3}(z) = \Omega_{Y(\widehat{\mathfrak{gl}}_1)}/z$ has a single pole. The Hamiltonians $H_i^{\mathbb{C}^3} = \sum_{j \neq i} \Omega_{ij}/(z_i - z_j)$ are the standard Gaudin Hamiltonians for $Y(\widehat{\mathfrak{gl}}_1)$, and their commutativity follows from the classical Yang–Baxter equation for $r_{CY}^{\mathbb{C}^3}$ (Construction 21.22; cf. Vol I, Theorem 20.9). The Bethe ansatz identification is the Litvinov $\mathcal{W}_{1+\infty}$ Bethe ansatz, which identifies Bethe roots with crystal melting configurations of $\mathbb{C}^3$ (equivalently, plane partitions).

Remark 21.32 (Why CY_3 supplies the Bethe roots). The novelty of face 7 in the CY setting is that the Bethe roots are not abstract numbers; they are CY_3 DT/PT counts. For $\mathbb{C}^3$ the DT/PT counts are MacMahon-type partition functions (3D plane partitions), and each plane partition contributes a single Bethe root. For more general toric CY_3, the topological vertex provides the partition counts and hence the Bethe-root populations. In this sense the CY_3 Gaudin model "geometrizes the Bethe ansatz" in the same way that the toric quiver geometrizes the affine Yangian.

Remark 21.33 (Face 7 is genuinely new for CY_3). Vol I face 7 (Gaudin from chiral algebra with primitive generator) and Vol II face 7 (Gaudin from twisted holography boundary algebra) both require an algebra with explicit generators. For CY_3, the analog must extract Gaudin data from the conjectural quantum vertex chiral group $G(X)$ via its CoHA presentation; the construction passes through the r-matrix of face 5 (or face 6 in the toroidal case), not directly through generators of $G(X)$ itself. This is the genuinely new content of T4 in the CY setting and is the main object of this section.

21.9 The seven-face master theorem in the CY setting

The preceding seven sections each realize the binary CY collision residue in one geometric language. The master theorem asserts that all seven realizations are canonically equivalent, with the equivalences given by explicit specialization or limit, and with the limit/specialization data recorded by the seven-face commuting diagram of Vol I.

CONJECTURE 21.34 *(Seven faces of $r_{CY}(z)$).* Let C be a Calabi–Yau category of dimension $d \in \{2, 3\}$, and let A_C be its chiral algebraization (proved for $d = 2$; conditional on Conjecture CY-A3 for $d = 3$). Then the seven realizations

$$r_{CY}^{(\mathrm{F1})},\ r_{CY}^{(\mathrm{F2})},\ r_{CY}^{(\mathrm{F3})},\ r_{CY}^{(\mathrm{F4})},\ r_{CY}^{(\mathrm{F5})},\ r_{CY}^{(\mathrm{F6})},\ r_{CY}^{(\mathrm{F7})}$$

constructed in Sections 21.2–21.8 all coincide, in the sense that

$$r_{CY}^{(\mathrm{F}i)} = \Phi_{i,j}\big(r_{CY}^{(\mathrm{F}j)}\big) \qquad \forall\, i, j \in \{1, \ldots, 7\},$$

for canonical change-of-language morphisms $\Phi_{i,j}$ given explicitly by Theorem CY-A (face 1 to face 3), the Schiffmann–Vasserot/RSYZ identification (face 2 to face 5), the MO stable envelope (face 4), the elliptic specialization $\hbar \to 0$ (face 6 to face 3), and the Litvinov Bethe ansatz (face 5 to face 7). The limits between faces are recorded in the seven-face commuting diagram (cf. Vol I, Theorem 20.9; Vol II, Theorem 20.9).

Remark 21.35 (Status by face). **Face 1** (CY bar-cobar twisting): [PH] for $d = 2$; [CJ] for $d = 3$ (conditional on CY-A3).

Face 2 (CoHA): [PE] for the existence of the CoHA (Kontsevich–Soibelman, Davison–Meinhardt); identification with r_{CY} is [CJ] in general, [PH] for $\mathbb{C}^3$ (Schiffmann–Vasserot, Theorem 15.4) and toric without compact 4-cycles (RSYZ, Theorem 15.5).

Face 3 (classical coisson): [PH] for $d = 2$ via the proved coisson–PVA dictionary; [CJ] for $d = 3$.

Face 4 (MO R-matrix): [PE] for the existence of R_{MO} (Maulik–Okounkov 2019); identification with r_{CY} for $K3 \times E$ is [CJ] in general, with explicit verification at the Gepner self-dual limit (Proposition 14.159).

Face 5 (toric Yangian): [PH] for $\mathbb{C}^3$; [PE] for toric without compact 4-cycles via RSYZ.

Face 6 (elliptic Sklyanin): [PE] for the classical Sklyanin bracket and its quantization to Felder's elliptic quantum group; identification with r_{CY} is [PH] in the $\mathfrak{sl}_2$ Heisenberg case via the Fay identity (Proposition 14.19).

Face 7 (CY_3 Gaudin): [PH] for $\mathbb{C}^3$ (Construction 21.22); [CJ] for general toric/toroidal CY_3.

The status partition above is consistent with AP60: each face's classical components are tagged ProvedElsewhere with explicit attribution, and only the genuinely new identifications carry the ProvedHere tag.

Remark 21.36 (Three invariants across the seven faces). By Remark 21.3, every realization of r_{CY} carries the three invariants $(p_{\max}, k_{\max}, r_{\max})$. AP59 demands that each face state which invariant is being measured, and forbids identifying them.

- Face 5 (toric Yangian) is sensitive to $p_{\max}$: the pole structure of $r_{Y(\widehat{\mathfrak{g}})}$ is the OPE pole structure of the toric quiver superpotential.
- Face 6 (elliptic) carries $k_{\max}$ in its theta-function expansion: at $k_{\max} = 0$ the residue is purely a single-pole elliptic term; at $k_{\max} \geq 1$ there are derivative terms.
- Face 1 (bar-cobar) is sensitive to $r_{\max}$: the shadow tower $\Theta_A^{\leq r}$ terminates at arity $r_{\max}$, and the binary residue is the projection to $r = 2$.

The seven-face master statement is that the binary residue is the *same* object across all faces, but the sensitivity to the three invariants differs face by face.

21.10 Cross-volume bridge

This part of Vol III mirrors Vol I Part III and Vol II Part III. The common skeleton is the seven-face programme; the variation is in which face is most concrete.

Remark 21.37 (The three seven-face masters). The three volumes each devote a part to the seven-face programme, with the same architecture but different ground objects:

(1) *Vol I, Part III:* the binary collision residue of a chiral algebra on a curve, in seven languages: bar-cobar twisting, primitive generator, classical r-matrix, KZ connection, Gaudin model, Bethe ansatz, dg-shifted Yangian (cf. Vol I, Theorem 20.9).

(2) *Vol II, Part III:* the binary collision residue of a holomorphic-topological quantum group, in seven languages: open-string brace algebra, derived center, twisted holography boundary, line defect, Wilson line R-matrix, soft graviton coupling, Yangian double (cf. Vol II, Theorem 20.9).

(3) *Vol III, this chapter:* the binary CY collision residue of a Calabi–Yau chiral algebra, in seven CY-specific languages: CY bar-cobar, CoHA / perverse coherent sheaves, classical CY Poisson coisson, MO stable envelope, affine super Yangian for toric CY_3, elliptic Sklyanin for toroidal CY, Gaudin from CY_3 (Theorem **??** above).

The three master theorems are mutually compatible: under the CY-to-chiral functor Φ, face i of Vol III maps to a specialization of face i of Vol I, and similarly for Vol II. The CY setting is the most constrained: each face acquires geometric content that the abstract chiral algebra setting does not see (DT counts, plane partitions, crystal melting, MO stable envelopes).

Remark 21.38 (The CY direction is the new slot). Compared with Vol I and Vol II, the new feature of the CY seven-face master is the presence of the CY direction Ω_C. In Vol I the binary residue lives on $\overline{C}_2(X) \times \mathfrak{g}^{\otimes 2}$; in Vol II it lives on $\overline{C}_2(X) \times \text{open-sector}^{\otimes 2}$; in Vol III it lives on $\overline{C}_2(X) \times C^{\otimes 2}$, with the CY volume form supplying the pairing on $C^{\otimes 2}$. The CY direction is what makes face 4 (MO) and face 7 (CY_3 Gaudin) genuinely geometric: they live on quiver varieties whose definition requires the CY structure.

Remark 21.39 ($\kappa_{\mathrm{ch}}(\mathcal{A})$ is a face-by-face invariant). The modular characteristic $\kappa_{\mathrm{ch}}(A_C)$ depends on the algebraization, not on the geometry. For $K3 \times E$, four distinct algebraizations give four distinct values: $\kappa_{\mathrm{ch}} = 3$, $\kappa_{\mathrm{cat}} = 2$, $\kappa_{\mathrm{fiber}} = 24$, $\kappa_{\mathrm{BKM}} = 5$ (preface). Each value sits on a different face: κ_{ch} on face 1 (bar-cobar of the chiral de Rham complex), κ_{cat} on face 2 (CoHA), κ_{fiber} on face 3 (classical lattice VOA limit), κ_{BKM} on face 4 (MO/Igusa). The κ_{ch}-spectrum is the imprint of the geometry on the seven-face master. The rule (from AP48) is that $\kappa_{\mathrm{ch}}(\mathcal{A})$ depends on the full algebra, not on its Virasoro subalgebra; in the CY setting, $\kappa_{\mathrm{ch}}(\mathcal{A})$ depends on the full *algebraization*, not on the underlying CY manifold.

Remark 21.40 (Outlook: an eighth face?). The standard seven-face master is closed: there are exactly seven languages in which the binary residue lives, corresponding to the seven edges of the Vol I commuting diagram. The natural question in the CY setting: does the deformation-quantization face (Khan–Zeng 3d PVA sigma model) constitute an eighth? At present this face reduces to a combination of face 3 (classical coisson) and face 7 (Gaudin from CY_3), with the deformation-quantization parameter playing the role of the loop parameter in face 7. An independent eighth face would require a CY-specific structure not visible in the seven-face commuting diagram; the working notes record the open question of whether the 3d $\mathcal{N} = 2$ holomorphic-twisted theory provides such a structure.

The genus-1 extension, identifying the KZB connection with the elliptic r-matrix on the torus E_τ, is proved for affine KM in Vol I, Chapter `ch:genus1-seven-faces`.

This chapter closes the seven-face programme for Vol III. Subsequent chapters in Part V record the geometric Langlands implications and the bridges to Vols I–II at the level of theorem statements; the present chapter is the algebraic engine that makes the bridges possible.

Part V

The CY Frontier

Appendix A

Conventions

A.1 Grading conventions

All grading is **cohomological**: differentials have degree $+1$. The bar construction uses desuspension (matching Volumes I–II).

A.2 Sign conventions

Signs follow the Koszul rule throughout. For the curved A_∞-structure: $\mu_1^2(a) = [\mu_0, a]$ (commutator, minus sign).

A.3 Categorical conventions

- **dgCat** denotes the ∞-category of small dg categories over k.
- CY_d-**Cat** denotes the ∞-category of smooth d-dimensional CY categories.
- Tensor products of dg categories are derived unless stated otherwise.
- "Equivalence" means quasi-equivalence of dg categories (equivalence of associated ∞-categories).

A.4 Operadic conventions

- E_n denotes the little n-disks operad (Boardman–Vogt).
- $\overline{\mathrm{FM}}_k(\mathbb{C})$ denotes the Fulton–MacPherson compactification of $\mathrm{Conf}_k(\mathbb{C})$. Note: $\overline{\mathrm{FM}}$ is a blowup along diagonals, NOT the complement of the diagonal (AP, Vol I).
- $\mathrm{SC}^{\mathrm{ch,top}}$ denotes the Swiss-cheese operad from Volume II.

A.5 Bar complex and Koszul duality conventions

- $B(A)$ is the bar complex, a factorization *coalgebra* on $\mathrm{Ran}(X)$.

- $\Omega(B(A)) \simeq A$ is bar-cobar *inversion*: it recovers the original algebra (Vol I, Theorem B). This is NOT Koszul duality.
- $A^! = (H^*(B(A)))^\vee$ is the Koszul dual *algebra* (linear dual of bar cohomology).
- $D_{\mathrm{Ran}}(B(A)) \simeq B(A^!)$ is the Verdier dual (Vol I, Theorem A, Convention conv:bar-coalgebra-identity).
- The collision r-matrix $r(z) = \mathrm{Res}^{\mathrm{coll}}_{0,2}(\Theta_A)$ has pole orders *one less* than the OPE, because the bar construction extracts residues along $d\log(z_i - z_j)$, which absorbs one power of $(z-w)$ (Vol I, AP19).

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